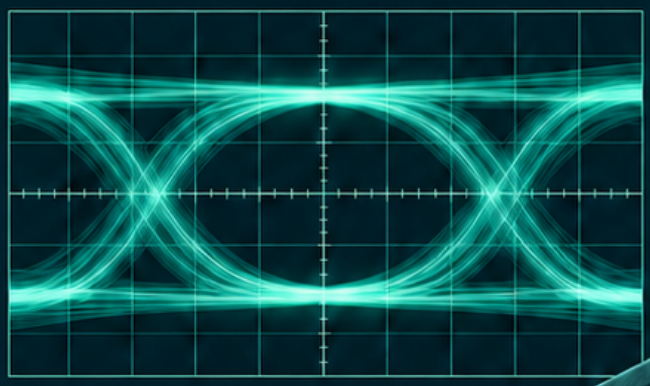
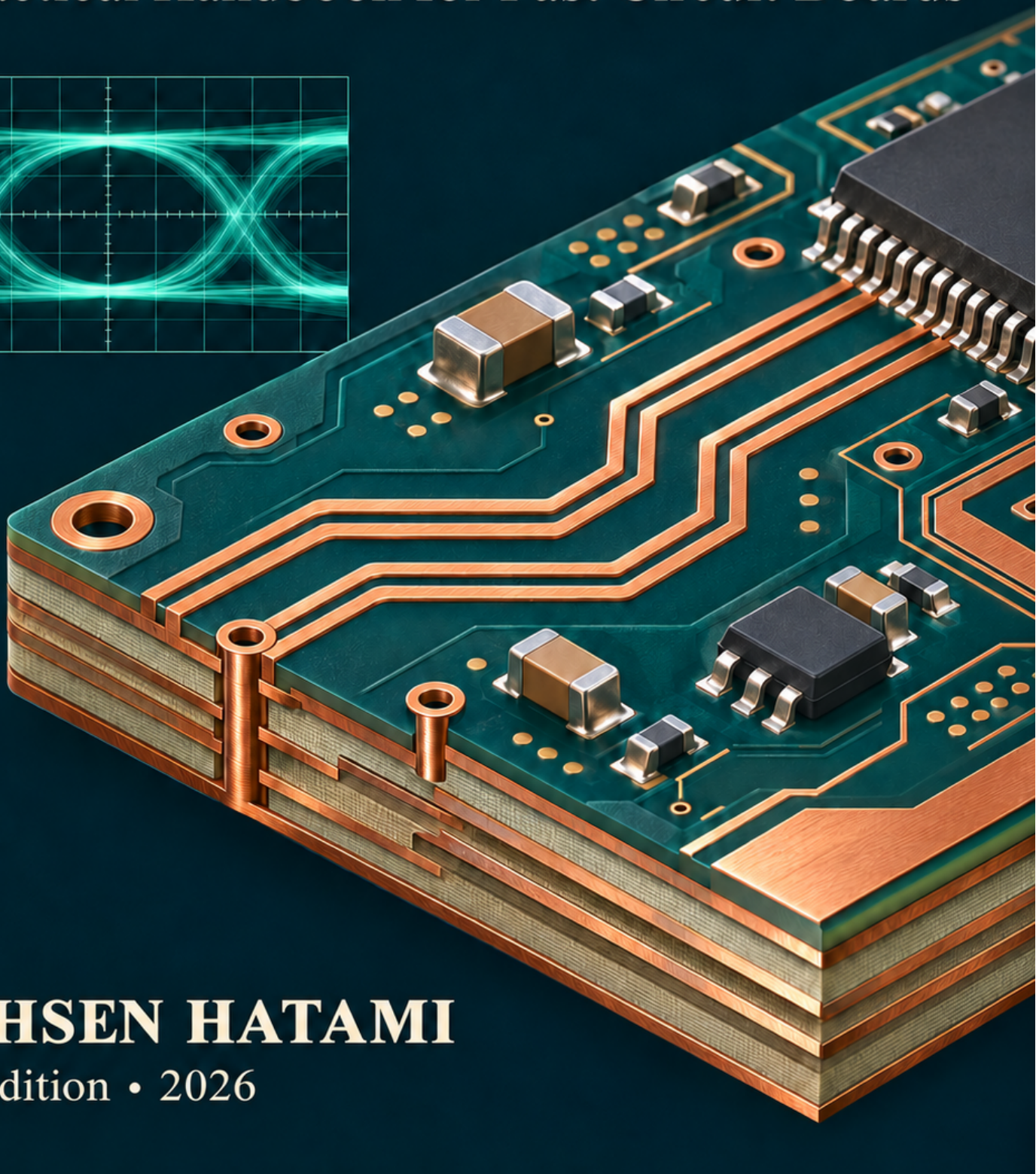
# High-Speed Digital Design

## A Practical Handbook for Fast Circuit Boards





**MOHSEN HATAMI**

First Edition • 2026

**High-Speed Digital Design: A Practical Handbook for Fast Circuit Boards**
First Edition, 2026

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Typeset in Times New Roman (or the metric-compatible TeX Gyre Termes) with XeLaTeX. All figures, schematics, and plots were generated natively with TikZ, CircuiTikZ, and PGFPlots from the equations in the text.

# Preface

## Why another book on high-speed design?

A modern circuit board is a microwave structure that happens to carry digital data. A single 10-layer board can host an LPDDR5 memory bus, several PCI Express lanes, a USB 3.x port, a gigabit Ethernet PHY, and MIPI camera and display links, each with edge rates of a few tens of picoseconds. At those speeds the copper traces, planes, vias, and connectors stop behaving like ideal wires and start behaving like transmission lines, antennas, resonant cavities, and coupled networks. Whether the product works on first power-up is decided long before the board is fabricated, by choices made in the stackup, the routing rules, and the power distribution network.

The classic texts on this subject—Johnson and Graham's *Black Magic* books [1], [2], Bogatin's *Signal and Power Integrity—Simplified* [3], Hall and Heck's *Advanced Signal Integrity* [4], Ott's *Electromagnetic Compatibility Engineering* [5], and the power-integrity work of Novak, Smith, and Bogatin [6], [7]—remain indispensable. This handbook does not replace them. Its purpose is narrower and more practical: to put, in one place and in one consistent notation, the equations, rules, budgets, and checklists that an engineer actually reaches for while designing and bringing up a dense multilayer board, and to be explicit about where each approximation stops being trustworthy.

## Who this book is for

The intended reader is a practicing hardware or PCB design engineer, or a graduate student, who is comfortable with circuit theory (phasors, Laplace transforms, RLC networks) and has some exposure to electromagnetics, but who wants a rigorous and usable treatment ofsignal integrity (SI), power integrity (PI), and electromagnetic compatibility (EMC) as they apply to real boards. No prior knowledge of microwave engineering is assumed; the needed transmission-line and field concepts are developed from first principles in Part I.

## How the book is organized

The book is divided into six parts.

**Part I — Foundations**
opens with the circuit board itself—its physical elements, what the fabricator controls, and why geometry and not just connectivity decides behavior—then defines what makes a design "high-speed," reviews the electromagnetic ideas a board designer needs (loop inductance, return current, skin depth), and develops transmission-line theory from the telegrapher's equations.

**Part II — Signal Integrity**
opens with a chapter that defines signal integrity—what it is, how it fails, what causes it, how it is quan-

tified and measured—and then covers reflections and termination, stackup and impedance control, loss, crosstalk, differential signaling, vias, and timing and jitter, ending with eye diagrams and equalization.

**Part III — Power Integrity**
opens with a chapter that defines power integrity and the power delivery network, introduces droop and target impedance, and contrasts PI with SI; a second chapter follows one load step through the network—current sharing, path inductance, resonance, first to third droops and the impedance as a response function; it then treats ground bounce and simultaneous switching noise, frequency-domain PDN design around a target impedance, and the closely related subject of planes and return paths.

**Part IV — EMI and EMC**
explains how boards radiate, how to keep them from doing so, and how the regulatory limits are structured.

**Part V — Interfaces in Practice**
applies the preceding theory to DDR4/DDR5 and LPDDR4/LPDDR5, PCI Express, USB and Type-C, Ethernet, MIPI CSI-2 and DSI, and clock distribution. Each chapter gives a specification summary table, impedance targets, loss and skew budgets, routing rules, and the failure modes most often seen in the lab.

**Part VI — Simulation, Measurement, and Bring-Up**
describes the pre- and post-layout simulation flow, the measurement instruments used to verify a design (TDR, VNA, oscilloscope, near-field probes), a structured board bring-up procedure, and the documentation required to release a board to fabrication.

Nine appendices collect a formula reference sheet, laminate property tables, design-rule checklists, unit conversions and rules of thumb, Python scripts for impedance, PDN target-impedance, and bounce-diagram calculations, a case study that applies the whole book to a 10-layer SoC board (with a requirements ledger), one-page interface quick-reference sheets, a glossary, and study paths with integrated exercises that cross chapter boundaries.

## How to use this book

Each chapter opens with an overview and a list of learning objectives, and closes with key takeaways and review problems. Answers to the numerical review problems are collected at the end of the book. Throughout the text, four kinds of callout box are used consistently:

**Design Rule: Example**

A green box states a design rule: a recommendation that is safe to apply on most boards, together with the conditions under which it holds.

**Rule of Thumb: Example**

A blue box gives a rule of thumb: a quick estimate for back-of-the-envelope work. Its accuracy limits are always stated.

**Common Pitfall: Example**

A red box describes a common pitfall: a mistake that appears regularly in design reviews and failure analysis.

**Specification Note**

A purple box flags a value or requirement that depends on a particular revision of an industry specification. Always confirm such values against the current document.

Most technical sections follow the same pattern: the problem, the model and its assumptions, a derivation, a worked calculation, a check of where the approximation fails, and the design decision it supports. Where a quick estimate is compared with a fuller model, a small three-column table at the end of the worked example gives the quick estimate, the full-model result and their difference.

Every figure caption ends with an *evidence label*—[Analytical], [Simulation], [Extraction], [Measured], [Conceptual] or [Schematic]—so that a curve computed from an equation is never mistaken for a measurement (Section 1.5). Review problems include three recurring types, marked in the text: *approximation failure*, *inverse design*, and *imperfect data*. Appendix I suggests study paths through the book and integrated exercises.

Worked examples use realistic geometries and materials—for instance, a 4 mil trace on 3.5 mil prepreg with $D_k = 3.8$ at 1 GHz—and carry the arithmetic through to a number, so that the reader can check a calculator or field-solver result against them.

## A note on accuracy

High-speed design is full of formulas that are easy to remember and easy to misuse. The closed-form microstrip and stripline impedance equations, the 3W crosstalk rule, the $0.35/t_r$ bandwidth estimate, and the "six inches per nanosecond" propagation delay are all approximations with limited ranges of validity. Wherever such an expression appears, this book states the range in which it is accurate and what to use outside it. When a value is taken from an industry specification, the specification is cited; when an exact value could not be verified, the text says so rather than guessing. Final design decisions should always be confirmed with a field solver, with the fabricator's stackup calculations, and with the silicon vendor's current design guide.

## About this revision

This revision incorporates an external technical review. Besides the new opening chapter, the lattice-diagram tutorial, the LPDDR5 timing groups, and the sections on pulse response, PDN transients, shunt-through measurement and TDR resolution, it corrects several results of the first draft: the trapezoid spectrum now uses the full ramp duration rather than the 10–90 % rise time; capacitor counts and anti-resonances are computed from complete R–L–C branches; the far-end-crosstalk, split-termination, DFE, ground-bounce and AC-coupling discussions have been restated with their correct limits; impedance formulas are checked against their validity ranges; and tolerance budgets keep worst-case and statistical results apart. Where a correction changed a number, the examples, problem answers, appendices and scripts were updated together.

## Acknowledgments

This handbook owes a great deal to the authors cited above and to the many engineers whose application notes, conference papers, and design reviews have shaped the practice of signal and power integrity. Errors that remain are the author's own; corrections and suggestions are welcome.

Mohsen Hatami

Binghamton, New York
2026

# Contents

**Preface** v

**List of Figures** xix

**List of Tables** xxvii

**List of Symbols and Abbreviations** xxxi

**I Foundations** 1

**1 The Printed Circuit Board as an Electrical System** 3
1.1 Anatomy of a multilayer board . . . 3
1.2 Connectivity and geometry . . . 4
1.3 Three views of the same fields . . . 7
1.4 Representations of a design . . . 7
1.5 Notation and conventions . . . 8
Further Reading . . . 9
Review Problems . . . 9

**2 What Makes a Design "High-Speed"** 11
2.1 Edges, not clocks . . . 11
2.2 Rise time and bandwidth . . . 12
2.3 Spectrum of a trapezoidal waveform . . . 13
2.4 Propagation delay . . . 15
2.5 Lumped versus distributed behavior . . . 16
2.6 Data rate, unit interval, and Nyquist frequency . . . 17
2.7 A checklist for deciding what matters . . . 17
Further Reading . . . 18
Review Problems . . . 18

**3 Electromagnetic Fundamentals for Board Designers** 21
3.1 Every signal is a current loop . . . 21
3.2 Capacitance . . . 21
3.3 Inductance . . . 22
3.4 Return current distribution . . . 24
3.5 Skin effect . . . 25
3.6 Summary of field quantities . . . 27

Further Reading . . . 28
Review Problems . . . 28

**4 Transmission Line Theory** **29**
4.1 The distributed RLGC model . . . 29
4.2 The telegrapher's equations . . . 30
4.3 Lossless lines . . . 30
4.4 Low-loss lines and attenuation . . . 31
4.5 Frequency dependence of the characteristic impedance . . . 32
4.6 Input impedance of a terminated line . . . 33
4.7 TEM, quasi-TEM, and dispersion . . . 35
Further Reading . . . 36
Review Problems . . . 36

**II Signal Integrity** **37**

**5 What Is Signal Integrity?** **39**
5.1 Definition . . . 39
5.2 What happens when signal integrity is poor . . . 40
5.3 What causes signal-integrity problems . . . 42
5.4 Important signal-integrity parameters . . . 42
5.5 How to improve signal integrity . . . 44
5.6 How to measure signal integrity . . . 44
5.7 A practical example: PCI Express . . . 45
5.8 Signal integrity and power integrity . . . 46
Further Reading . . . 46
Review Problems . . . 47

**6 Reflections and Termination** **49**
6.1 The reflection coefficient . . . 49
6.2 Lattice diagrams . . . 50
6.3 Reactive loads and discontinuities . . . 53
6.4 Termination strategies . . . 55
Further Reading . . . 58
Review Problems . . . 58

**7 PCB Stackup and Impedance Control** **59**
7.1 Transmission-line geometries . . . 59
7.2 Closed-form impedance approximations . . . 60
7.3 Stackup design principles . . . 63
7.4 Laminate materials . . . 65
7.5 Glass-weave skew . . . 65
7.6 Copper foil and roughness . . . 66
7.7 Impedance tolerance . . . 67
Further Reading . . . 68
Review Problems . . . 68

**8 Losses** **69**
8.1 Loss as a low-pass filter . . . 69
8.2 Conductor loss . . . 69
8.3 Dielectric loss . . . 71
8.4 Total insertion loss . . . 72
Further Reading . . . 74
Review Problems . . . 74

**9 Crosstalk** **75**
9.1 Coupling mechanisms . . . 75
9.2 Near-end crosstalk . . . 76
9.3 Far-end crosstalk . . . 77
9.4 Termination of the victim . . . 79
9.5 Spacing and the 3W rule . . . 79
9.6 Guard traces . . . 80
9.7 Other coupling paths . . . 80
9.8 Crosstalk budgets . . . 80
Further Reading . . . 81
Review Problems . . . 81

**10 Differential Signaling** **83**
10.1 Modes of a coupled pair . . . 83
10.2 Odd, even, differential, and common-mode impedance . . . 83
10.3 Terminating both modes . . . 84
10.4 Skew and mode conversion . . . 85
10.5 Length matching and serpentines . . . 86
Further Reading . . . 88
Review Problems . . . 88

**11 Vias and Discontinuities** **89**
11.1 Anatomy of a via . . . 89
11.2 Lumped via models . . . 89
11.3 The via as a transmission line . . . 90
11.4 Anti-pads . . . 91
11.5 Via stubs and back-drilling . . . 96
11.6 Blind, buried, and micro-vias . . . 98
11.7 Return and stitching vias . . . 99
11.8 BGA and connector breakout . . . 99
Further Reading . . . 100
Review Problems . . . 101

**12 Timing, Jitter, and Eye Diagrams** **103**
12.1 Clocking architectures . . . 103
12.2 Jitter . . . 105
12.3 Eye diagrams, masks, and bathtub curves . . . 106
12.4 Equalization . . . 108
Further Reading . . . 111

Review Problems . . . 111

**III Power Integrity** **113**

**13 What Is Power Integrity?** **115**
13.1 Definition . . . 115
13.2 What causes power-integrity problems . . . 116
13.3 The power delivery network . . . 116
13.4 Calculating voltage droop . . . 117
13.5 PDN impedance and target impedance . . . 119
13.6 How to improve power integrity . . . 119
13.7 How to measure power integrity . . . 120
13.8 Signal integrity versus power integrity . . . 121
Further Reading . . . 123
Review Problems . . . 123

**14 Transient Current, Inductance, and the PDN Response** **125**
14.1 The circuit between the regulator and the chip . . . 125
14.2 A load step: why fast current changes matter . . . 126
14.3 Local capacitors and the current shortfall . . . 127
14.4 How path inductance sets the droop . . . 128
14.5 From the time domain to PDN impedance . . . 129
14.6 Target impedance and the voltage budget . . . 130
14.7 Resonance, ringing and anti-resonance . . . 131
14.8 First, second and third droops . . . 131
14.9 The inductive voltage is a derivative . . . 131
14.10 Impedance is the response function . . . 133
14.11 Practical ways to improve the response . . . 133
Further Reading . . . 135
Review Problems . . . 136

**15 Ground Bounce and Simultaneous Switching Noise** **137**
15.1 The mechanism . . . 137
15.2 Estimating the switching current . . . 138
15.3 Parallel pins and mutual inductance . . . 139
15.4 Effects on timing and logic . . . 140
15.5 Mitigation . . . 141
Further Reading . . . 142
Review Problems . . . 142

**16 Power Distribution Network Design** **143**
16.1 The target-impedance method . . . 143
16.2 Frequency regions of a PDN . . . 144
16.3 Capacitor models . . . 146
16.4 Anti-resonance . . . 148
16.5 Planes and spreading inductance . . . 150

16.6 From impedance to a load transient . . . 150
16.7 DC IR drop . . . 151
Further Reading . . . 152
Review Problems . . . 152

**17 Planes and Return Paths** **155**
17.1 Crossing a plane split . . . 155
17.2 Reference-plane changes . . . 156
17.3 Plane cavity resonance . . . 157
17.4 Mixed-signal partitioning . . . 158
Further Reading . . . 159
Review Problems . . . 159

**IV EMI and EMC** **161**

**18 Emission Mechanisms and EMC Design** **163**
18.1 Differential-mode radiation . . . 163
18.2 Common-mode radiation . . . 165
18.3 Slot and edge radiation . . . 165
18.4 Techniques for reducing emissions . . . 166
18.5 Regulatory overview . . . 167
Further Reading . . . 168
Review Problems . . . 168

**V Interfaces in Practice** **171**

**19 DDR4/DDR5 and LPDDR4/LPDDR5 Memory Interfaces** **173**
19.1 Interface overview . . . 173
19.2 Signaling and termination . . . 175
19.3 Topologies . . . 176
19.4 Routing and matching budgets . . . 178
19.5 Common failure modes . . . 179
Further Reading . . . 180
Review Problems . . . 180

**20 PCI Express Gen3 to Gen6** **181**
20.1 Electrical overview . . . 181
20.2 Channel loss budgeting . . . 182
20.3 AC coupling . . . 183
20.4 Reference clocks . . . 184
20.5 Routing rules . . . 184
20.6 Gen6 and PAM4 . . . 184
Further Reading . . . 186
Review Problems . . . 186

**21 USB 2.0, USB 3.x, USB4, and Type-C** **189**
21.1 Generations and signaling . . . 189
21.2 The Type-C connector . . . 190
21.3 Routing the USB 2.0 pair . . . 190
21.4 High-speed lanes . . . 191
21.5 VBUS and power . . . 192
Further Reading . . . 193
Review Problems . . . 193

**22 Ethernet: 1000BASE-T, 10GBASE-KR, SGMII, and RGMII** **195**
22.1 1000BASE-T . . . 195
22.2 10GBASE-KR . . . 197
22.3 MAC-to-PHY interfaces . . . 197
Further Reading . . . 199
Review Problems . . . 199

**23 MIPI CSI-2 and DSI: D-PHY and C-PHY** **201**
23.1 D-PHY . . . 201
23.2 C-PHY . . . 202
23.3 Boards, connectors, and flex cables . . . 203
Further Reading . . . 204
Review Problems . . . 204

**24 Clock Distribution** **205**
24.1 Clock sources . . . 205
24.2 Clock tree and output standards . . . 206
24.3 Jitter of clocks . . . 207
24.4 PLLs and loop filters . . . 208
24.5 Clock routing . . . 209
Further Reading . . . 210
Review Problems . . . 210

**VI Simulation, Measurement, and Bring-Up** **211**

**25 Modeling and Simulation** **213**
25.1 Buffer models . . . 213
25.2 Interconnect models and field solvers . . . 214
25.3 S-parameters . . . 214
25.4 A pre- and post-layout flow . . . 217
Further Reading . . . 219
Review Problems . . . 219

**26 Measurement: TDR, VNA, Oscilloscopes, and Near-Field Probing** **221**
26.1 Time-domain reflectometry . . . 221
26.2 Vector network analysis . . . 223
26.3 Measuring milliohm PDN impedance . . . 224

26.4 Oscilloscopes and probing . . . 225
26.5 Near-field probing . . . 226
Further Reading . . . 227
Review Problems . . . 227

**27 Board Bring-Up and Debug** **229**
27.1 The bring-up flow . . . 229
27.2 Before power . . . 229
27.3 First power . . . 231
27.4 JTAG and boundary scan . . . 232
27.5 Interface bring-up and margining . . . 232
27.6 Root-cause method . . . 232
Further Reading . . . 234
Review Problems . . . 234

**28 Design for Manufacturing and Release** **237**
28.1 Fabrication drawing and notes . . . 237
28.2 Impedance coupons . . . 238
28.3 Tolerances . . . 239
28.4 Design for manufacturing and test . . . 240
28.5 The release package . . . 240
Further Reading . . . 241
Review Problems . . . 241

**Appendices** **245**

**A Formula Reference Sheet** **245**
Edges, bandwidth, and delay . . . 245
Electromagnetics . . . 245
Transmission lines and reflections . . . 245
Impedance approximations . . . 246
Loss, crosstalk, skew, vias . . . 246
Timing, jitter, and clocks . . . 246
Power integrity and EMC . . . 247
Measurement . . . 247

**B Material Property Tables** **249**
B.1 Laminate classes . . . 249
B.2 Frequency dependence of a causal model . . . 249
B.3 Glass styles . . . 249
B.4 Copper . . . 250
B.5 Surface finishes . . . 250

**C Design-Rule Checklists** **251**
C.1 Schematic review . . . 251
C.2 Stackup . . . 251
C.3 Routing . . . 251
C.4 Power distribution . . . 252
C.5 EMI/EMC . . . 252
C.6 Release . . . 252
C.7 Simulation . . . 252
C.8 Bring-up . . . 253

**D Unit Conversions and Rules of Thumb** **255**
D.1 Physical constants . . . 255
D.2 Length and copper . . . 255
D.3 Decibels . . . 255
D.4 Rules of thumb and their limits . . . 255

**E Python Scripts** **257**
E.1 Microstrip and stripline impedance . . . 257
E.2 PDN target impedance and capacitor sizing . . . 260
E.3 Reflection and bounce-diagram calculation . . . 262

**F Case Study: A 10-Layer SoC Single-Board Computer** **265**
F.1 Requirements . . . 265
F.2 Floorplan . . . 266
F.3 Stackup and impedance plan . . . 266
F.4 Loss budgets . . . 267
F.5 Timing and matching . . . 268
F.6 Power distribution . . . 268
F.7 EMC and protection . . . 269
F.8 Bring-up plan . . . 269

**G Interface Quick-Reference Sheets** **271**
G.1 DDR4 / DDR5 / LPDDR4 / LPDDR5 . . . 272
G.2 PCI Express . . . 273
G.3 USB and Type-C . . . 274
G.4 Ethernet . . . 275
G.5 MIPI CSI-2 / DSI . . . 276
G.6 Clocks . . . 277

**H Glossary** **279**

**I Study Paths and Integrated Exercises** **283**
I.1 Study paths . . . 283
I.2 Exercise A: from pulse response to receiver margin . . . 283
I.3 Exercise B: from a rail tolerance to a validated PDN . . . 284
I.4 Exercise C: from a coupon report to a production decision . . . 285

**Answers to Review Problems** **287**

**Bibliography** **295**

**Index** **299**

# List of Figures


1.1 Conceptual cross-section of a four-layer board with one component. The signal current in the L1 trace (⊗, into the page) returns in the L2 plane directly beneath it (⊙). Not to scale. . 4
1.2 Foil construction of a ten-layer board: four double-sided cores and five prepreg plies, with copper foil on the outer layers. . . . . 5
1.3 Via types in a ten-layer board: a through via with anti-pads in the planes it does not connect, a back-drilled via leaving a residual stub, a buried via, a blind laser microvia and stacked microvias. Not to scale. . . . . 6
1.4 High-frequency return current in the reference plane (top view). (a) With an intact plane it flows beneath the trace. (b) A slot forces a detour; the shaded area is added loop area. . . . 6

2.1 An ideal step compared with a physical digital edge. The rise time $t_r$ is the interval between the 10 % and 90 % crossings; the edge shown has $t_r = 100\,\text{ps}$. . . . . 12
2.2 A linear ramp with $t_r = 0.5\,\text{ns}$ (10–90 %) has a full duration $T_{\text{ramp}} = 0.625\,\text{ns}$. . . . . 14
2.3 Harmonic amplitudes of a 100 MHz, 50 % duty-cycle, 3.3 V clock with $t_r = 0.5\,\text{ns}$ ($T_{\text{ramp}} = 0.625\,\text{ns}$), computed from Eq. (2.6), with the asymptotic envelope of Eq. (2.8). Even harmonics vanish for a 50 % duty cycle. . . . . 15
2.4 Critical length versus rise time from Eq. (2.11). Lines shorter than $\ell_{\text{crit}}$ may be modeled as lumped elements; longer lines need transmission-line treatment and, usually, termination. Vertical lines mark representative edge rates. . . . . 16

3.1 The loop formed by a signal and its return current. (a) With an adjacent reference plane, the loop area, and therefore the loop inductance, is set by the dielectric height $H$. (b) A distant return conductor encloses a much larger area and has much higher loop inductance. . . . . 23
3.2 Return current density in the reference plane under a microstrip, from Eq. (3.10). A thinner dielectric concentrates the return current more tightly beneath the trace. The shaded band, $\pm 3H$ for $H = 4$ mil, carries 79.5 % of the return current. . . . . 25
3.3 Top view of a U-shaped trace over a plane. (a) At low frequency the return current takes the shortest, lowest-resistance path across the plane. (b) At high frequency it follows directly beneath the trace, the lowest-inductance path. . . . . 26
3.4 Skin depth of copper versus frequency, from Eq. (3.12). Above about 4 MHz the skin depth is smaller than the thickness of 1 oz foil; above a few gigahertz it is comparable to the surface roughness of the foil. . . . . 27

4.1 Circuit models of an interconnect. (a) An electrically short line is a lumped inductance and capacitance. (b) A long line is a ladder of RLGC cells; in the limit $\Delta z \to 0$ it becomes the distributed model described by the telegrapher's equations. . . . . 30

4.2 Snapshots of a 0.5 V step propagating along a lossless line at $v_p = 6\,\text{in/ns}$. The edge moves without distortion and occupies a rise length $\ell_r = t_r/t_{pd}$. . . . . 32
4.3 Magnitude of $Z_0$ for a lossy 50 Ω stripline ($R'_{dc} = 0.1\,\Omega/\text{in}$, skin-effect $R'$, $\tan\delta = 0.02$). Below about 2 MHz the line is an RC network; above, $Z_0$ is essentially $\sqrt{L'/C'}$. . . . . 33
4.4 Input impedance magnitude of a lossless 50 Ω line with $T_D = 0.51$ ns (3 in of stripline) for several loads, from Eq. (4.16). Mismatched loads produce an input impedance that oscillates with period $1/(2T_D)$; open and shorted lines resonate at odd multiples of the quarter-wave frequency. . . . . 34
4.5 Ratio of the exact equivalent reactance of a lossless open or shorted stub to its lumped value (left) and the resulting lumped-model error (right), from Eq. (4.17). The marker is the stub of Worked Example 4.3. Ideal uniform line; no pad capacitance. . . . . 35

5.1 An 8 Gb/s NRZ pattern at the transmitter and after a 10.75 in channel with a via region, a connector and an 8 in trace (the reference channel of this chapter: loss $0.10\,\text{dB/in}/\sqrt{\text{GHz}}$ plus $0.12\,\text{dB/(in}\cdot\text{GHz)}$, 170 ps/in). . . . . 40
5.2 Signal-integrity impairments, each shown alone. (a) Step seen at the source of a line with a 75 Ω segment. (b) Under-terminated source ($R_S = 15\,\Omega$) driving an open line. (c) Aggressor edge and victim NEXT and FEXT. (d) A 250 ps pulse before and after the reference channel. (e) A data pattern through a one-pole channel: the isolated 1 never reaches full level. (f) Twelve edges with random timing. . . . . 41
5.3 A high-speed channel from transmitter to receiver, with the impairments each element mainly contributes. . . . . 42
5.4 Anatomy of an eye diagram. Noise and ISI thicken the rails and reduce the eye height; jitter spreads the crossings and reduces the eye width. The receiver samples at the center of the opening. . . . . 43
5.5 Three views of the same reference channel: (a) TDR impedance profile with a 30 ps edge, showing the low-impedance via region and the high-impedance connector; (b) insertion and return loss; (c) 8 Gb/s PRBS7 eye at the receiver, no equalization. . . . . 45

6.1 Lattice diagram for a 3.3 V driver with $R_S = 10\,\Omega$ driving an open-circuited 50 Ω line with $T_D = 1$ ns. Wave amplitudes are shown on the diagonals; cumulative voltages at the source (left) and load (right) are shown in boxes. The load voltage rings about its final value of 3.3 V. 50
6.2 Load and source voltages of Worked Example 6.1 built from Table 6.1. The load rings with period $4T_D$; the source changes only at even multiples of $T_D$. . . . . 52
6.3 Receiver voltage for a 1 V source driving an open 50 Ω line ($T_D = 1$ ns, $t_r = 150$ ps) through three source resistances, computed by superposing the waves of the lattice diagram. . . . . 52
6.4 Normalized response of the same 4 pF capacitance at the end of a matched-source 50 Ω line and as a shunt in the middle of a matched line, for an ideal incident step. The added 10–90 % rise times are $\ln 9\,\tau$. . . . . 54
6.5 A 50 Ω line with matched source ($T_D = 1$ ns) driving a capacitive load, for an ideal 1 V source step. The load charges with $\tau = Z_0 C$; the reflection seen at the source is a dip that recovers with the same time constant. . . . . 54
6.6 Termination schemes for single-ended and differential lines. . . . . 55

7.1 Cross-sections of PCB transmission lines with their dimensional parameters. . . . . 60

7.2 Characteristic impedance versus trace width for a microstrip ($H = 3.5$ mil, $D_k = 3.8$) and a symmetric stripline ($B = 10.6$ mil, $D_k = 3.8$). The two microstrip formulas differ by 10–20 %, partly because of thickness and partly because of the approximations themselves. . 62
7.3 An example 10-layer stackup (62.6 mil over copper). Four signal layers each have an adjacent ground plane; two power planes are coupled to ground through 3 mil cores; the stackup is symmetric about its center. . . . 64
7.4 Glass-weave skew: when a differential pair runs parallel to the weave, one trace may lie over glass and the other over resin. Rotating the route relative to the weave averages the two. . . . 66
7.5 Distribution of the IPC-2141 microstrip impedance of Worked Example 7.4 for $W = 4$ mil $\pm$ 0.5 mil, $H = 3.5$ mil $\pm$ 10 %, $D_k = 3.8 \pm 5$ % (independent uniform inputs, 100 000 samples, fixed seed). Dashed lines: exact corners of the same formula; dotted: $\pm$10 %. . . . 67

8.1 Roughness correction factor versus frequency. The Hammerstad model (Eq. (8.4)) saturates at 2; the Huray model (Eq. (8.5)) rises further, toward its own finite asymptote 1 + 1.5 SR, and better matches measurements of rough foil above about 10 GHz. . . . 71
8.2 Conductor and dielectric attenuation of a 50 Ω, 5 mil stripline in standard FR-4 ($D_k = 4.2$, $D_f = 0.020$, smooth copper). Dielectric loss dominates above about 2.3 GHz. . . . 72
8.3 Insertion loss of a 10 in, 50 Ω, 5 mil stripline from Eq. (8.9), for standard FR-4 with standard foil and for a low-loss laminate with HVLP foil. The smooth-copper curve shows how much of the low-loss laminate's loss is due to roughness. . . . 73

9.1 Coupled aggressor and victim lines. Distributed mutual capacitance $C_m$ and mutual inductance $L_m$ produce a backward wave at the victim's near end (NEXT) and a forward wave at its far end (FEXT). . . . 76
9.2 Far-end crosstalk from the modal model of Eq. (9.7) (solid) and its derivative approximation (dashed), for a 1 V Gaussian edge with $t_r = 100$ ps. Left: waveforms for $\Delta T/t_r = 0.2$ and 2. Right: peak magnitude versus modal delay difference. Matched, lossless, symmetric pair. . 77
9.3 Crosstalk on a matched victim from a 1 V, 100 ps aggressor edge over 6 in of coupled microstrip ($C_m/C = 0.04$, $L_m/L = 0.06$). NEXT is a plateau of duration $2T_D$; FEXT is a negative pulse at $T_D$. . . . 78
9.4 Relative coupling versus center-to-center spacing normalized to dielectric height. Microstrip coupling falls as $1/[1+(D/H)^2]$; stripline coupling falls approximately exponentially. Curves are normalized and show trends only; absolute values require a field solver. . . . 79

10.1 Electric field patterns of an edge-coupled microstrip pair. (a) In the odd mode, field lines run strongly from one trace to the other, and the symmetry plane is a virtual ground. (b) In the even mode, both traces are at the same potential and fields are pushed toward the plane. . . . 84
10.2 Mode conversion by intra-pair skew. Left: the modal voltage ratio of Eq. (10.4) and the power-wave ratio $|S_{CD21}|$ of Eq. (10.5), which is 6.02 dB higher. Right: differential and converted power fractions for 5 ps of skew; they sum to one in this lossless model. . . . 86
10.3 Length-matching geometries. (a) A serpentine for matching a single-ended net to a bus: segment spacing $s$ of at least $4H$ limits self-coupling. (b) Intra-pair compensation: small bumps on the inner trace placed immediately next to the bend that created the mismatch. . . 87

11.1 Cross-section of a through via carrying a signal from L1 to L3 in a 10-layer board. (a) As drilled, the barrel below L3 forms an open stub. (b) After back-drilling with an oversized bit from the bottom, only a short residual stub remains. . . . 90

11.2 A signal via passing through three ground planes. (a) Cross-section: the field (arrows) runs from the barrel to the plane edges at each anti-pad. (b) Top view of one plane with an illustrative 0.30 mm drill, 0.60 mm pad and 0.90 mm anti-pad: the radial clearance between pad and plane is 0.15 mm. Not to scale. . . . . 91
11.3 Effect of anti-pad diameter for a 0.30 mm drill and 0.60 mm pads in a 1.6 mm board with $\varepsilon_r = 3.8$. Left: coaxial-limit impedance from Eq. (11.4). Right: Johnson–Graham pad capacitance from Eq. (11.2). Trends only; final dimensions come from a 3-D solver. . . . . 93
11.4 Simulated TDR (left, 25 ps Gaussian edge) and return loss (right) of a 1.6 mm via between 50 Ω lines for anti-pad diameters of 0.70, 0.90, 1.20 and 1.55 mm, modeled as a uniform line with the coaxial-limit impedance of Eq. (11.4) and 10.4 ps delay. No pads or stub. . . . . 94
11.5 Top view of a plane under a differential via pair. (a) A shared obround anti-pad with symmetric ground return vias. (b) Oversized anti-pads that merge into a slot under a trace on the adjacent layer, interrupting its return path. . . . . 95
11.6 One leg of a PCIe via transition from L1 to L3 in a ten-layer board: via pad, functional pad on L3, anti-pads in the ground planes, non-functional pads removed, back-drilled barrel with a short residual stub, and ground stitching vias. Not to scale. . . . . 96
11.7 Insertion loss of a 50 Ω through path loaded by an open via stub ($Z_{stub} = 45\,\Omega$, $\varepsilon_r = 3.8$), computed from $S_{21} = 2/(2 + Z_0 Y_{stub})$ with $Y_{stub} = \tanh(\gamma \ell)/Z_{stub}$. The quarter-wave notch moves down in frequency as the stub lengthens. . . . . 97
11.8 Left: ideal shunt open-stub transmission for residual stubs of 15, 23.5, 31.5 and 50 mil ($Z_{main} = 50\,\Omega$, $Z_{stub} = 45\,\Omega$, $\varepsilon_{eff} = 3.8$). Right: quarter-wave frequency versus residual stub length for $\varepsilon_{eff} = 3.8$ and $\varepsilon_{eff} = 3.99$ (5 %), with the 48 GHz target. Transmission-line calculation, not a 3-D via extraction. . . . . 98
11.9 Dog-bone breakout under a BGA. A differential pair is assigned to adjacent balls, its vias are placed symmetrically with nearby ground return vias, and it escapes between ball rows on an inner layer. . . . . 100

12.1 Common-clock timing. Data launched $t_{co}$ after the driver's clock edge arrives at the receiver $t_{flight}$ later and must be stable from $t_{su}$ before to $t_h$ after the receiver's next clock edge. . . . . 104
12.2 Decomposition of total jitter into random and deterministic components. . . . . 106
12.3 Synthetic NRZ eye diagram: a PRBS7 sequence through a band-limited channel with random jitter and noise, overlaid over two unit intervals. Eye height and eye width are marked; a receiver mask would occupy the open region at the center. . . . . 107
12.4 Dual-Dirac bathtub curve. The eye width at $10^{-12}$ is the distance between the two walls of the bathtub at that BER. . . . . 107
12.5 Sampled pulse response of Worked Example 12.5, normalized to the main cursor, before and after an ideal three-tap DFE (correct prior decisions, exact taps; no noise or adaptation). The precursor remains. . . . . 109
12.6 Channel loss, CTLE, and two-tap FFE responses for a 32 GT/s link. The equalized response is flatter than the raw channel; the residual slope is left to the DFE. . . . . 110

13.1 A 1.0 V rail responding to a 10 A load step applied at 3 µs and released at 8 µs. Ripple comes from the regulator; droop and ringing from the PDN's inductance and capacitance; the settled offset is the IR drop. Model: 1.5 mΩ and 1 nH feed, 200 µF at the load, 500 kHz, ±2 mV ripple; no regulator loop. . . . . 117

13.2 The power delivery network from regulator to die. The frequency bands are typical orders of magnitude, not limits; Figure 16.2 shows a computed example. . . . 117
13.3 Rail deviation for a 10 A, 100 ns load ramp through 1 mΩ and 1 nH: without load capacitance, with 1 µF (ESR 2 mΩ) and with 22 µF (ESR 1 mΩ). The small capacitor forms a lightly damped resonance and makes the droop worse. . . . 118
13.4 PDN impedance of two designs against a 5 mΩ target. Design A meets it over the shaded range; Design B has an anti-resonance peak near 20 MHz. Lumped models of VRM, bulk, MLCC banks and planes. . . . 120
13.5 Three ways to measure power integrity: (a) oscilloscope with a rail probe, (b) VNA shunt-through impedance, (c) injected load step with voltage and current measured together. Probe grounds, cable shields and isolation are omitted. . . . 121
13.6 How power-integrity noise becomes a signal-integrity problem. . . . 122

14.1 Simplified one-rail PDN: regulator, effective path resistance and inductance of the supply-and-return loop, a local capacitor, and the chip's current demand. . . . 126
14.2 Calculated 2–12 A load-step response of the circuit of Figure 14.1 with the values of Table 14.1 and $L_{\text{path}} = 1$ nH. The path current lags the demand, overshoots to 15.5 A and rings; the capacitor current is the difference. . . . 127
14.3 Chip voltage for the load step of Table 14.1 with 1 nH and 5 nH path inductance. The 950 mV line is an illustrative limit, not a device specification. Ideal source, ideal capacitor, no regulator loop. . . . 128
14.4 Impedance of a simple PDN: path (2 mΩ, 1 nH), a 100 µF capacitor (ESR 2.5 mΩ, ESL 0.4 nH) and a 1 µF capacitor (ESR 3 mΩ, ESL 0.25 nH), against a 5 mΩ target. Labels mark the local peaks. . . . 130
14.5 Die-voltage response of a three-stage ladder to a 10 A, 1 ns load step (log time axis): die 200 nF with package 12.7 pH; board MLCCs 100 µF behind 253 pH; bulk 2 mF behind an inductive regulator model (14 nH, 0.5 mΩ). The droops are 73 mV at 2.8 ns, 19 mV at 0.29 µs and 26 mV at 6.3 µs. . . . 132
14.6 Current (top) and voltage across 1 nH (bottom). (a) A 0–5 A pulse with 10 ns linear edges gives ±0.5 V rectangles. (b) The same pulse with raised-cosine edges of the same full duration gives bell-shaped pulses peaking at 0.79 V. (c) A 2.5 A, 10 MHz sinusoid gives 0.16 V, leading by 90°. . . . 132
14.7 Periodic steady-state rail deviation for a 0–10 A, 50 % square-wave load with 15 ns edges applied to the PDN of Figure 14.4, at 6.8 MHz and 3.4 MHz, from the Fourier series and Eq. (14.11). . . . 134
14.8 The end-to-end relationship: the chip's current spectrum excites the PDN impedance, producing the supply disturbance and its consequences. . . . 134

15.1 Equivalent circuit of ground bounce. $N$ switching outputs discharge their loads through the shared package ground inductance $L_{\text{gnd}}$. A quiet driver held low follows the die ground and transmits the bounce onto its own line. . . . 138
15.2 Ground bounce on a quiet output for $N$ simultaneously switching outputs (Worked Example 15.1 parameters, raised-cosine current edges). The pulse scales with $N$ and is followed by ringing of the package inductance with the on-die capacitance. . . . 139

16.1 Lumped equivalent circuit of a PDN from the regulator to the die. Each stage has a capacitance limited by its own series inductance and resistance. . . . 145

16.2 PDN impedance computed from the lumped model of Figure 16.1, seen at the board (planes and capacitors) and at the die (through the package). The board meets the 5 mΩ target up to about 90 MHz but has its own anti-resonance between MLCC inductance and plane capacitance near 200 MHz; seen from the die, the package inductance hides that peak, and the package inductance and on-die capacitance instead form an anti-resonance near 60 MHz that exceeds the target. . . . 145
16.3 Impedance of four mounted MLCCs from Eq. (16.2). Each is capacitive below its self-resonant frequency (SRF), resistive at it, and inductive above it. Above about 30 MHz the curves converge, because mounted inductances are similar. . . . 146
16.4 Left: impedance of a 100 nF, 10 mΩ, 0.5 nH capacitor and its inductive-only approximation. Right: pointwise number of ideal, independent capacitors needed for $Z_{\text{target}} = 2.4\,\text{m}\Omega$, from Eq. (16.4) and from the shortcut. Shared inductance is ignored. . . . 147
16.5 Decoupling-capacitor mounting patterns, from highest to lowest mounted inductance. Short connections, vias at the sides of the pads, and multiple via pairs reduce the loop area. . . . 148
16.6 Exact impedance of the two branches of Worked Example 16.3 and of their parallel combination. The cross marks the prediction of the $1/\sqrt{L_1 C_2}$ shortcut. . . . 149
16.7 Impedance (left) and response to a 1 A, 20 ns load step (right) of the network of Eq. (16.10) for two feed resistances. Linear model; no regulator loop or capacitor ESR. . . . 151

17.1 A trace crossing a gap between two plane regions. (a) The return current detours to the end of the gap, enclosing a large loop. (b) A stitching capacitor next to the crossing restores a short return path at high frequency. . . . 156
17.2 Return current at a layer transition from L1 (referenced to L2 GND) to L8 (referenced to L7 GND). A ground via adjacent to the signal via carries the return current between the two ground planes. . . . 157
17.3 Voltage distribution of the first three modes of a 10 in × 8 in plane pair with $\varepsilon_r = 4.2$, from Eq. (17.1). Red and blue denote opposite polarity. . . . 157

18.1 Emission mechanisms. (a) Differential-mode current in a small loop radiates as a magnetic dipole. (b) Common-mode current, driven by a noise voltage between the board's ground and an attached cable, flows in the same direction on all of the cable's conductors and radiates as a monopole. . . . 164
18.2 Envelope of differential-mode emission at 3 m from a 4 in trace carrying the harmonics of a 100 MHz clock ($t_r = 0.5$ ns, 10 mA harmonic envelope), for two return-path geometries, from Eq. (18.1), compared with the FCC Class B limit. Only the envelope is shown; actual emissions occur at discrete harmonics. Free-space field, RMS; a test site with a ground plane can read up to about 6 dB higher. . . . 164
18.3 Effect of −0.5 % down-spread SSC on the fundamental of a 100 MHz clock, as seen by a receiver with a 120 kHz resolution bandwidth (schematic). The peak falls by about 6 dB, the ratio of the 500 kHz spread to the resolution bandwidth expressed in decibels. . . . 166

19.1 Logical LPDDR5/5X timing groups of one channel: CK with command/address, WCK with write data, and RDQS with read data where used. Conceptual architecture, not a pinout, termination schematic, clock-ratio specification or training sequence. . . . 175
19.2 Pseudo-open-drain DQ signaling: the receiver's on-die termination pulls up to $V_{\text{DDQ}}$; the receiver compares the line against an internally generated, trained $V_{\text{REF}}$. . . . 176

19.3 Command/address and clock topologies. (a) Fly-by: a daisy chain with short stubs, terminated to $V_{TT}$, giving clean signals but staggered arrival times. (b) Balanced T: simultaneous arrival but larger reflections. . . . 177
19.4 Write leveling. The controller delays DQS for a byte lane until its rising edge coincides with the CK rising edge at that lane's DRAM, as reported by the DRAM sampling CK with DQS. 178

20.1 A PCIe channel from transmitter to receiver, through packages, system-board traces, AC-coupling capacitors, the CEM connector, and an add-in card. The reference clock is distributed to both ends in a common-clock architecture. . . . 182
20.2 An illustrative insertion-loss allocation at 8 GHz for a PCIe Gen4 channel with a 28 dB budget. Actual allocations come from the silicon vendor and the CEM specification. . . . 182
20.3 Synthetic NRZ and PAM4 eyes at the same symbol rate and swing: 1600 random symbols, one-pole channel ($\tau = 0.1$ UI), additive Gaussian noise $\sigma = 0.025$, 80 overlaid two-UI windows, fixed seed. Dotted lines: PAM4 decision thresholds. . . . 185
20.4 NRZ and PAM4 at the same peak-to-peak swing. PAM4 carries two bits per symbol with Gray-coded levels; each eye has one-third of the NRZ amplitude. . . . 186

21.1 Type-C receptacle contact map (front view). Row B is rotated by 180 degrees relative to row A, making the connector reversible. . . . 190
21.2 Host-side signal path for SuperSpeed or USB4 lanes to a Type-C receptacle. The CC logic determines orientation and configures the mux. . . . 191

22.1 One pair of a 1000BASE-T interface: PHY, isolation transformer with center taps, common-mode choke (CMC), RJ45, and Bob Smith termination of the cable-side center tap to chassis ground. . . . 196
22.2 Layout zones of an Ethernet port: signal ground under the PHY, an isolation gap under the magnetics, and chassis ground under the connector. . . . 197
22.3 RGMII transmit timing. Data change at the source clock edges; delaying the clock by about 1.5–2 ns centers each edge in the 4 ns data eye. . . . 198

23.1 D-PHY lane voltages during a transition from the low-power stop state (LP-11) through LP-01 and LP-00 into high-speed data and back (schematic). . . . 202
23.2 A C-PHY trio. Three wires each take one of three levels in every symbol; three differential receivers recover the symbol, and the guaranteed transitions carry the clock. . . . 202

24.1 A clock tree: oscillator, PLL or jitter cleaner, and fanout buffers with point-to-point outputs to each load. . . . 206
24.2 Representative phase noise of a 100 MHz oscillator. Integrating Eq. (24.2) over the shaded band gives about 114 fs RMS. . . . 207
24.3 Charge-pump PLL with a second-order passive loop filter ($R$, $C_1$, $C_2$). . . . 208

25.1 Port numbering for a four-port differential channel used in this book, and the corresponding mixed-mode ports. . . . 215
25.2 Mixed-mode S-parameters of a representative 10 in differential channel: insertion loss, return loss from a 93 Ω section ($T_D = 0.6$ ns) between 85 Ω ports, and mode conversion from 3 ps of intra-pair skew. . . . 216
25.3 Pre- and post-layout signal- and power-integrity simulation flow. . . . 218

26.1 A TDR impedance profile: a capacitive via (dip), an inductive connector (bump), a 60 Ω trace section, and an open end. . . . 222
26.2 Simulated TDR of a 60 Ω segment with 20 ps one-way delay between matched 50 Ω lines, for Gaussian-equivalent system edges of 10, 35 and 100 ps. Multiple reflections in the segment are included; no loss or fixture. With the 100 ps edge, the 60 Ω plateau becomes a low bump. 223
26.3 VNA measurement chain. Calibration places the reference planes at the cable ends; de-embedding removes the test fixtures to isolate the DUT. . . . 224
26.4 Left: ideal two-port shunt-through connection; cable shields, calibration standards and ground loops are omitted but matter in the laboratory. Right: $|S_{21}|$ versus a resistive DUT impedance from Eq. (26.4). . . . 225
26.5 Step response through a passive probe with 10 pF tip capacitance and a 25 Ω source: a 150 nH ground lead rings at 130 MHz; a 10 nH spring tip resonates near 500 MHz and is well damped. 226

27.1 Board bring-up flow, from inspection to link qualification. . . . 230
27.2 An example power-up sequence: I/O, auxiliary, core, and memory rails ramp in order, and reset is released after the last rail is stable. The required order and delays come from each device's data sheet. . . . 231

28.1 Organization of a fabrication drawing: outline and dimensions, stackup and impedance table, drill table, fabrication notes, and title block (schematic). . . . 238
28.2 An impedance coupon with single-ended and differential test lines, ground launch vias (green) next to the probe pads, and layer identification. . . . 239

F.1 Block diagram of the case-study board. . . . 265
F.2 Floorplan of the case-study board (schematic). The memory is placed beside the SoC's DDR ball field, high-speed connectors along one edge, and the Ethernet magnetics and RJ45 in a corner chassis zone. . . . 267

# List of Tables


1.1 One physical cause, three consequences. . . . 7
1.2 Representations of a board and what each captures. . . . 8
1.3 Physical elements of a board and the chapters that treat them. . . . 9

2.1 Criteria for deciding when an interconnect is electrically long. . . . 17
2.2 Unit interval and Nyquist frequency of representative interfaces (NRZ unless noted). Edge rates are implementation dependent and are not listed. . . . 18

3.1 Electromagnetic quantities and closed-form estimates used in board design. . . . 27

4.1 Special cases of the input impedance of a lossless line. . . . 33

5.1 Signal-integrity parameters. . . . 43
5.2 Summary: the signal-integrity view of a design. . . . 46

6.1 Arrival table for Worked Example 6.1 ($\Gamma_S = -2/3$, $\Gamma_L = +1$). . . . 51
6.2 Comparison of termination schemes. . . . 57

7.1 Closed-form impedance formulas and their limits. . . . 63
7.2 Representative 8-, 10-, and 12-layer stackups (S = signal, G = ground, P = power). Thicknesses are illustrative and must be confirmed with the fabricator. . . . 65
7.3 Laminate classes. Ranges are representative of published data at about 1–10 GHz and vary with resin content; consult the specific product data sheet. . . . 65

8.1 Loss mechanisms of PCB transmission lines. . . . 74

9.1 Crosstalk characteristics and mitigations. . . . 81

10.1 Typical skew targets expressed as a fraction of a UI. Interface specifications and controller design guides take precedence. . . . 88

11.1 Features around vias and pads. . . . 95
11.2 Via technologies. . . . 99

12.1 Gaussian multipliers for total jitter, $2Q_{\mathrm{BER}}$ (for a transition density of one). . . . 106
12.2 Equalizer types. . . . 110

13.1 Techniques that improve power integrity. . . . 120
13.2 Signal integrity and power integrity side by side. . . . 122
13.3 Summary: the power-integrity view of a design. . . . 122

14.1 Illustrative load-step parameters. . . . 126
14.2 Results of the lumped simulation, rounded to the nearest millivolt. . . . 128
14.3 Reactance of the path inductance alone (not the impedance of the full PDN). . . . 129
14.4 Current waveform and ideal inductor voltage $V_L = L\ \mathrm{d}I/\mathrm{d}t$. . . . 132
14.5 Problems and engineering approaches. . . . 134
14.6 Equation reference for this chapter. . . . 135

15.1 Techniques for controlling simultaneous switching noise. . . . 141

16.1 Illustrative voltage allocation for a 0.8 V, $\pm 3$ % rail (requirement values chosen for the example). 144
16.2 PDN elements and their effective frequency ranges (typical). . . . 152

17.1 Return-path situations and remedies. . . . 159

18.1 Emission mechanisms and design controls. . . . 167

19.1 Representative characteristics of DDR4, DDR5, LPDDR4/4X, and LPDDR5/5X. Values reflect the base JEDEC releases and common practice; see the specification note below. . . . 174
19.2 Timing groups of the four DRAM families (conceptual; consult the controller documentation for the implemented groups). . . . 174
19.3 Typical DDR routing parameters. Controller layout guides take precedence. . . . 179

20.1 PCIe generations. Loss budgets are commonly cited end-to-end (bump-to-bump) values at Nyquist and are given for orientation only. . . . 181
20.2 Typical PCIe board routing practice. . . . 184

21.1 USB physical layers (representative; see the specification note). . . . 189
21.2 USB layout summary. . . . 192

22.1 Ethernet interfaces on boards. . . . 199

23.1 D-PHY and C-PHY compared. . . . 204

24.1 Common clock output standards and their terminations. . . . 207

25.1 Field-solver classes and their uses. . . . 214

26.1 Measurement instruments and their uses. . . . 227

27.1 A minimal margining matrix and log template (illustrative). . . . 233
27.2 Common high-speed symptoms and likely causes. . . . 233

28.1 Key DFM and DFT checks. . . . 240
28.2 Contents of a PCB release package. . . . 241

B.1 Representative laminate properties at about 1–10 GHz. Ranges, not guaranteed values; consult product data sheets. . . . 249
B.2 $D_k$ / $D_f$ versus frequency from a Djordjevic–Sarkar fit at 1 GHz. . . . 249
B.3 Common E-glass styles in prepreg (approximate; pressed thickness depends on resin content and adjacent copper). . . . 250

B.4 Copper foil weights and properties. . . . 250
B.5 PCB surface finishes and their high-speed relevance. . . . 250

D.1 Physical constants. . . . 255
D.2 Length conversions. . . . 255
D.3 Decibel conversions. . . . 256
D.4 Rules of thumb used in this book, with their limits. . . . 256

F.1 Interface requirements of the case-study board (illustrative). . . . 266
F.2 Requirements ledger for the case-study board (excerpt; values illustrative). . . . 266
F.3 Summary of the case-study design decisions. . . . 269

I.1 Core path and advanced modules. . . . 283
I.2 Design inputs for Exercise A. . . . 284
I.3 Rail budget for Exercise B. . . . 285

# List of Symbols and Abbreviations

Symbols are listed with the SI unit in which they are normally expressed in this book. PCB dimensions are frequently given in mils (1 mil = 0.001 in = 25.4 µm); conversions appear in Appendix D.

## Symbols

| Symbol | Meaning | Unit |
|---|---|---|
| $\alpha$ | Attenuation constant (total, conductor $\alpha_c$, dielectric $\alpha_d$) | Np/m, dB/in |
| $\beta$ | Phase constant | rad/m |
| $\gamma$ | Propagation constant, $\gamma = \alpha + \mathrm{j}\beta$ | 1/m |
| $\Gamma$ | Reflection coefficient ($\Gamma_\mathrm{S}$ at source, $\Gamma_\mathrm{L}$ at load) | — |
| $\delta_\mathrm{s}$ | Skin depth | m |
| $\varepsilon_0$ | Permittivity of free space, $8.854 \times 10^{-12}$ F/m | F/m |
| $\varepsilon_\mathrm{r}$, $D_\mathrm{k}$ | Relative permittivity (dielectric constant) | — |
| $\varepsilon_\mathrm{eff}$ | Effective relative permittivity of a non-homogeneous line | — |
| $\mu_0$ | Permeability of free space, $4\pi \times 10^{-7}$ | H/m |
| $\rho$ | Resistivity (copper: $1.72 \times 10^{-8}\ \Omega \cdot$ m at 20 °C) | $\Omega \cdot$ m |
| $\sigma$ | Conductivity; also standard deviation of random jitter | S/m, s |
| $\tan\delta$, $D_\mathrm{f}$ | Dielectric loss tangent (dissipation factor) | — |
| $\omega$ | Angular frequency, $2\pi f$ | rad/s |
| $C$, $C'$ | Capacitance; per-unit-length capacitance | F, F/m |
| $C_m$ | Mutual capacitance per unit length | F/m |
| $f_\mathrm{clk}$ | Clock frequency | Hz |
| $f_\mathrm{knee}$ | Knee frequency, $\approx 0.5/t_\mathrm{r}$ | Hz |
| $f_\mathrm{Nyq}$ | Nyquist frequency of a data stream, $1/(2\,\mathrm{UI})$ | Hz |
| $f_\mathrm{SRF}$ | Self-resonant frequency of a capacitor | Hz |
| $G$, $G'$ | Conductance; per-unit-length shunt conductance | S, S/m |
| $H$ | Dielectric height between trace and reference plane | mil, µm |
| $I$, $i(t)$ | Current | A |
| $K_b$, $K_f$ | Backward (near-end) and forward (far-end) crosstalk coefficients | — |
| $\ell$ | Physical length of an interconnect | m, in |
| $\ell_\mathrm{crit}$ | Critical length beyond which a line must be treated as distributed | m, in |
| $L$, $L'$ | Inductance; per-unit-length series inductance | H, H/m |
| $L_\mathrm{loop}$ | Loop inductance | H |
| $L_m$ | Mutual inductance per unit length | H/m |

*continued on next page*

| Symbol | Meaning | Unit |
|---|---|---|
| $R$, $R'$ | Resistance; per-unit-length series resistance | Ω, Ω/m |
| $S$ | Edge-to-edge spacing between traces | mil, µm |
| $S_{ij}$ | Scattering parameter, port $j$ to port $i$ | — (dB) |
| $T$, $t$ | Conductor thickness | mil, µm |
| $T_{\mathrm{D}}$ | One-way delay of a line, $\ell/v_{\mathrm{p}}$ | s |
| $t_{\mathrm{pd}}$ | Propagation delay per unit length, $1/v_{\mathrm{p}}$ | ps/in |
| $t_{\mathrm{r}}$, $t_{\mathrm{f}}$ | Rise and fall times (10–90 % unless stated) | s |
| $t_{\mathrm{su}}$, $t_{\mathrm{h}}$ | Setup and hold times | s |
| $t_{\mathrm{flight}}$ | Flight time from driver to receiver threshold | s |
| $v_{\mathrm{p}}$ | Propagation velocity | m/s, in/ns |
| $V_{\mathrm{DD}}$ | Supply voltage | V |
| $V_{\mathrm{REF}}$ | Reference voltage for single-ended receivers | V |
| $W$ | Trace width | mil, µm |
| $Z_0$ | Characteristic impedance | Ω |
| $Z_{\mathrm{diff}}$, $Z_{\mathrm{cm}}$ | Differential and common-mode impedance | Ω |
| $Z_{\mathrm{odd}}$, $Z_{\mathrm{even}}$ | Odd- and even-mode impedance | Ω |
| $Z_{\mathrm{in}}$ | Input impedance | Ω |
| $Z_{\mathrm{S}}$, $Z_{\mathrm{L}}$ | Source and load impedance | Ω |
| $Z_{\mathrm{target}}$ | PDN target impedance, $V_{\mathrm{ripple}}/I_{\mathrm{transient}}$ | Ω |

## Abbreviations

| Abbreviation | Meaning |
|---|---|
| AC | Alternating current; also AC coupling capacitor |
| BER | Bit error ratio |
| BGA | Ball grid array |
| BOM | Bill of materials |
| CDR | Clock and data recovery |
| CM | Common mode |
| CTLE | Continuous-time linear equalizer |
| CPWG | Coplanar waveguide with ground |
| DC | Direct current |
| DDJ | Data-dependent jitter |
| DDR | Double data rate (synchronous DRAM) |
| DFE | Decision-feedback equalizer |
| DFM | Design for manufacturing |
| DFT | Design for test |
| DJ | Deterministic jitter |
| DM | Differential mode |
| DUT | Device under test |
| DVT | Design verification testing |

*continued on next page*

| Abbreviation | Meaning |
|---|---|
| EMC | Electromagnetic compatibility |
| EMI | Electromagnetic interference |
| ESD | Electrostatic discharge |
| ESL | Equivalent series inductance |
| ESR | Equivalent series resistance |
| FEXT | Far-end crosstalk |
| FFE | Feed-forward equalizer |
| IBIS | I/O Buffer Information Specification |
| IBIS-AMI | IBIS Algorithmic Modeling Interface |
| IL | Insertion loss |
| ISI | Inter-symbol interference |
| JTAG | Joint Test Action Group (IEEE 1149.1 boundary scan) |
| LPDDR | Low-power double data rate DRAM |
| MIPI | Mobile Industry Processor Interface (Alliance) |
| NEXT | Near-end crosstalk |
| NRZ | Non-return-to-zero signaling |
| ODT | On-die termination |
| PAM4 | Four-level pulse amplitude modulation |
| PCB | Printed circuit board |
| PCIe | PCI Express |
| PDN | Power distribution network |
| PI | Power integrity |
| PJ | Periodic jitter |
| PLL | Phase-locked loop |
| PRBS | Pseudo-random bit sequence |
| RJ | Random jitter |
| RL | Return loss |
| RMS | Root mean square |
| SERDES | Serializer/deserializer |
| SI | Signal integrity |
| SoC | System on chip |
| SRF | Self-resonant frequency |
| SSC | Spread-spectrum clocking |
| SSN | Simultaneous switching noise |
| TDR | Time-domain reflectometry |
| TDT | Time-domain transmission |
| TIE | Time interval error |
| UI | Unit interval (one bit period) |
| VNA | Vector network analyzer |
| VRM | Voltage regulator module |

Part I

# Foundations

Chapter 1

# The Printed Circuit Board as an Electrical System

**Chapter Overview**

A schematic says which pins are connected; a circuit board decides *how*. This chapter describes the physical board—copper, dielectric, planes, vias, soldermask and legend—and what the fabricator and assembler add to the designer's data. It then separates two views that the rest of the book keeps combining: *connectivity*, which the schematic captures, and *geometry*, which determines delay, impedance, loss, coupling and radiation. It ends with the representations an engineer works with (schematic, layout, stackup, models and measurements), a practical sequence for a new design, the notation used throughout the book, and how the evidence behind each figure is labeled.

**Learning Objectives**

1. Name the physical elements of a multilayer board and what each contributes electrically.
2. Distinguish what the designer specifies from what the fabricator and assembler control.
3. Explain why connectivity alone does not describe a high-speed net, and estimate when geometry matters.
4. Relate signal integrity, power integrity and EMC as three views of the same currents and fields.
5. Use the book's notation for delay and per-unit-length quantities, and its evidence labels.

## 1.1 Anatomy of a multilayer board

A rigid board is a laminate of copper and dielectric layers (Figure 1.1).

**Traces and pads**
: Etched copper on signal layers. Traces carry signals between pads; pads accept component terminals. Their width, thickness and distance from the nearest plane set impedance, resistance and coupling (Chapter 7).

**Planes**
: Large copper areas, normally ground or a supply. A plane is the *return conductor* of every trace that runs next to it, and a pair of planes on thin dielectric is a distributed capacitor for the power distribution network (Chapters 16 and 17).

**Dielectric**
: Glass-reinforced resin: *cores* (cured laminate with copper on both sides) and *prepreg* (uncured sheets that bond the stack during lamination). Its permittivity sets propagation delay and impedance; its loss tangent sets dielectric loss (Chapter 8).

**Vias**

Plated holes that connect layers. Through vias span the board; blind and buried vias and microvias span only some layers (Figure 1.3). Every via is a short, non-uniform transmission line with a possible stub (Chapter 11).

**Soldermask and legend**

The polymer coating that protects outer copper and defines solder openings, and the printed component markings. Soldermask is a dielectric: it lowers the impedance and raises the delay of outer-layer traces.

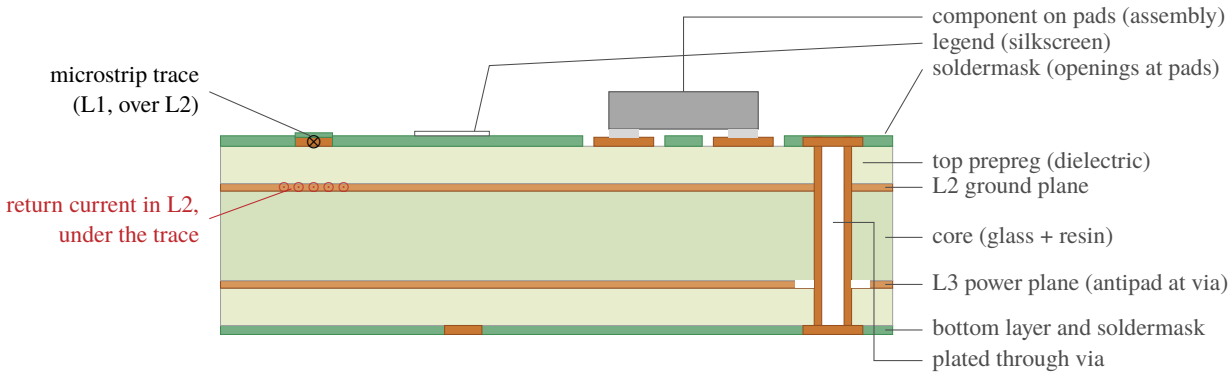



**Figure 1.1.** Conceptual cross-section of a four-layer board with one component. The signal current in the L1 trace (⊗, into the page) returns in the L2 plane directly beneath it (⊙). Not to scale. [Conceptual]

### 1.1.1 What the fabricator and the assembler add

The designer delivers data: copper artwork for each layer, drill files, a stackup with target impedances, and fabrication notes (Chapter 28). The fabricator turns it into a board by choosing the actual cores and prepregs, pressing them (Figure 1.2), drilling, plating and etching. Several electrically important quantities are therefore set by the fabricator's process rather than by the drawing: final dielectric thicknesses after resin flow, etched trace width and its trapezoidal profile, plated copper thickness, the copper surface roughness, and the actual permittivity of the chosen glass and resin. A fabricator permitted to "adjust widths to meet impedance" will change the artwork to compensate. The assembler then adds components, solder joints and, through the reflow profile, further thermal stress. Every model of a board is a model of a fabricated, assembled object, and its inputs carry the tolerances of these processes.

## 1.2 Connectivity and geometry

A netlist records that pin A connects to pin B. For a slow signal that is the whole story: the copper between them is a wire, and any route that connects them works. For a fast signal the route is a component with its own delay, impedance, loss and coupling, and two layouts with identical connectivity can behave entirely differently. The relevant comparison is between the signal's edge time and the time the signal takes to cross the route, introduced in Chapter 2.

**Worked Example 1.1: The same net, two electrical objects**

A 3 in stripline route in a laminate with $D_k = 3.8$ carries (a) a DDR data signal with 100 ps edges and (b) a slow control signal with 5 ns edges. Compare the one-way delay with each edge time.

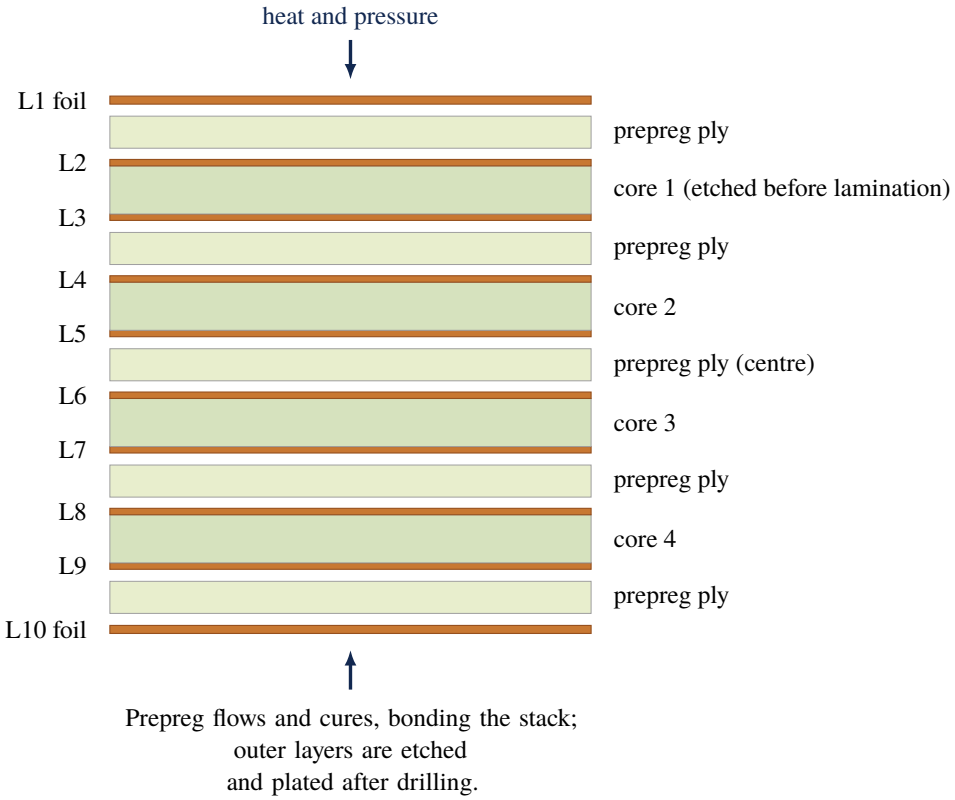



**Figure 1.2.** Foil construction of a ten-layer board: four double-sided cores and five prepreg plies, with copper foil on the outer layers. [Conceptual]

**Solution.** The delay per inch is $84.73\sqrt{3.8} = 165$ ps/in (Section 1.5), so $T_D = 496$ ps. For (a), $T_D/t_r = 5$: the edge occupies a fifth of the route, the route is a transmission line, and its impedance and termination matter (Chapters 4 and 6). For (b), $T_D/t_r = 0.1$: the whole route charges almost uniformly and behaves as a small lumped capacitance. The schematic is the same; the electrical object is not.

### 1.2.1 Return current is part of the circuit

Current flows in loops. The signal current in a trace is matched by an equal and opposite return current, and at high frequency that current is not free to take any path: the fields that carry the signal are concentrated between the trace and its nearest plane, so the return current flows in that plane, concentrated beneath the trace (Figure 1.4a). It does not "choose" a path; the distribution of current is the one that satisfies the field equations for that geometry, and at high frequency that is the distribution of lowest loop inductance (Chapter 3). When the plane is interrupted by a slot, a split or a cluster of merged anti-pads, the return current must flow around the interruption (Figure 1.4b). The loop grows, its inductance rises, the trace sees an impedance discontinuity, neighboring signals that share the detour couple to each other, and the enlarged loop radiates.

**Design Rule: Route every fast signal over an unbroken reference**

Give every high-speed trace an adjacent, continuous reference plane for its whole length, and wherever it changes layer or reference, provide a return path at the transition (a stitching via or capacitor) within a short distance of the signal via. Check plane layers for slots created by merged anti-pads before release.

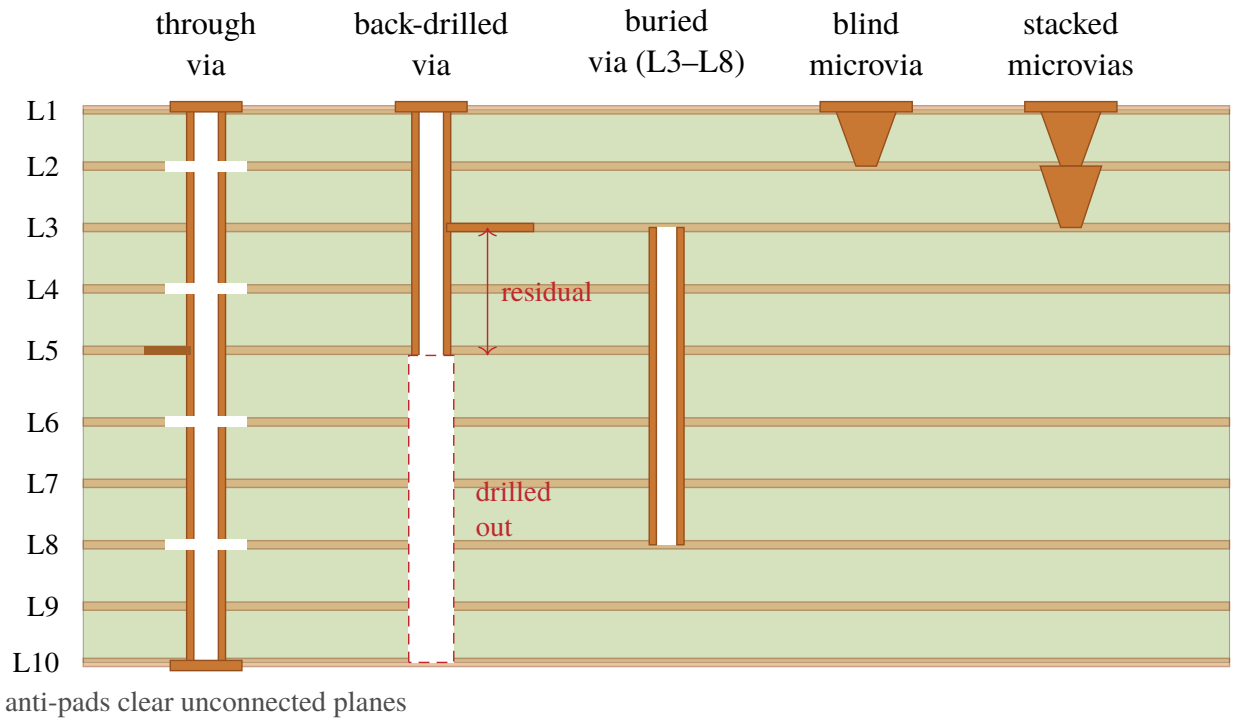



**Figure 1.3.** Via types in a ten-layer board: a through via with anti-pads in the planes it does not connect, a back-drilled via leaving a residual stub, a buried via, a blind laser microvia and stacked microvias. Not to scale. [Conceptual]

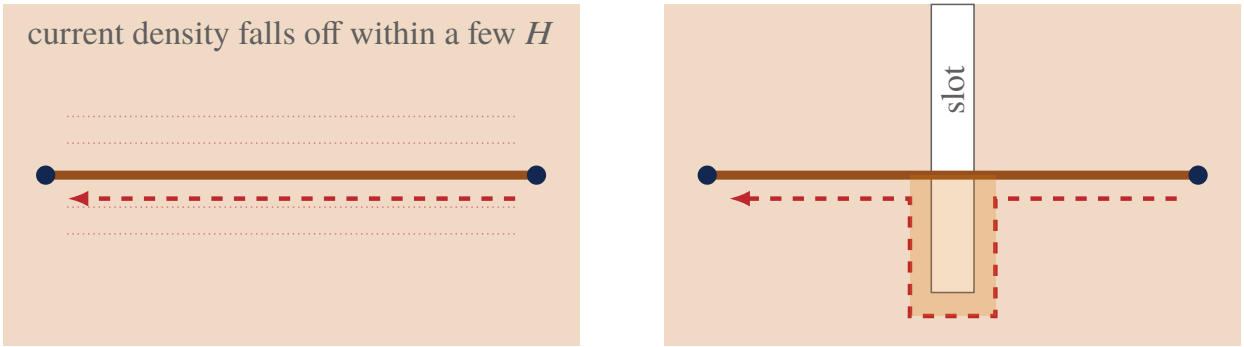



**Figure 1.4.** High-frequency return current in the reference plane (top view). (a) With an intact plane it flows beneath the trace. (b) A slot forces a detour; the shaded area is added loop area. [Conceptual]

**Worked Example 1.2: Resistance and capacitance the schematic does not show**

Estimate (a) the DC resistance of a 6 in, 5 mil-wide inner-layer trace in 0.5 oz copper (17.4 µm), and (b) the capacitance of a 4 in × 4 in power–ground plane pair separated by 3 mil of dielectric with $D_k = 4.0$.

**Solution.** (a) $R = \rho\ell/(Wt) = 1.724\times10^{-8}\times0.1524/(127\times10^{-6}\times17.4\times10^{-6}) = 1.19\ \Omega$, about 0.2 Ω/in. Negligible beside 50 Ω at DC, but significant for a supply trace carrying amperes, and multiplied by the skin effect at high frequency (Chapter 8). (b) $C = \varepsilon_0 D_k A/h = 8.854\times10^{-12}\times4\times0.01032/(76.2\times10^{-6}) = 4.8$ nF: a capacitor with almost no series inductance, which the PDN uses above the range of discrete capacitors (Chapter 16).

## 1.3 Three views of the same fields

Signal integrity, power integrity and electromagnetic compatibility are often taught as separate subjects, but they describe the same currents. A signal's return current flows in a plane that also distributes power; a slot that forces a return detour raises the signal's discontinuity (SI), couples noise into the plane cavity (PI) and creates a radiating loop (EMC). A decoupling capacitor that lowers PDN impedance also provides a return path for signals changing reference between a power and a ground plane. Simultaneous switching current is a PI problem whose symptom is jitter on signals. Table 1.1 lists the most common interactions; treating any one of them in isolation is how many designs fail.

**Table 1.1.** One physical cause, three consequences.

| Cause | Signal integrity | Power integrity | EMC |
|---|---|---|---|
| Slot or split under a trace | Discontinuity, crosstalk via shared detour | Cavity coupling | Loop and slot radiation |
| Reference change without stitching | Return-path discontinuity | Noise injected between planes | Plane-pair resonance excited |
| Shared ground vias or pins | Ground bounce on quiet lines | Higher loop inductance | Common-mode voltage on cables |
| Fast edges | Reflections, ringing | Larger $\mathrm{d}i/\mathrm{d}t$ | Higher harmonic content |
| Low-loss, low-ESR PDN | — | Sharp anti-resonances | Resonant emission |

**Common Pitfall: Fixing one view and breaking another**

Cutting a ground plane to "isolate" noisy circuitry (an EMC or PI intention) and routing fast signals across the cut (an SI disaster), or adding small capacitors to lower PDN impedance at one frequency and creating an anti-resonance at another. Every change to copper, planes or decoupling should be reviewed against all three views.

## 1.4 Representations of a design

An engineer never works with the board itself until the end; the work is done on representations, each of which keeps some information and discards the rest (Table 1.2). Most design errors happen at the boundaries: a constraint that exists in a simulation but not in the layout rules, a stackup assumed in a model but never sent to the fabricator, or a measurement taken at a different reference plane from the model it is compared with.

**Table 1.2.** Representations of a board and what each captures.

| Representation | Captures | Omits |
|---|---|---|
| Schematic | Connectivity, components, intent | Geometry, return paths, delay |
| Layout database | Copper geometry, layers, vias, placement | Material properties as fabricated |
| Stackup and fabrication notes | Layer order, thicknesses, materials, impedance targets | Etch profile, resin flow, roughness as built |
| Simulation model | Chosen physics over a chosen bandwidth | Everything outside its assumptions |
| Measurement | The real object at one reference plane | Conditions not exercised; fixture effects |

### 1.4.1 A practical sequence

For each new board the following sequence turns the representations into one design. It recurs throughout the book and in the case study of Appendix F.

1. *Classify the nets.* From the edge rates and data rates (Chapter 2), decide which nets are electrically long and which are sensitive or aggressive.
2. *Choose the stackup.* Assign a reference plane to every signal layer, pair power with ground, and choose materials for the fastest link (Chapters 7 and 8).
3. *Write constraints.* Turn budgets—loss, skew, crosstalk, target impedance—into layout rules with the evidence that justifies each.
4. *Place and route* with the constraints, keeping return paths intact.
5. *Verify* the routed design by extraction and simulation at the corners (Chapter 25).
6. *Correlate* on the first boards: measure at defined reference planes and reconcile the results with the models (Chapters 26 and 27).

**Rule of Thumb: When a board is "high-speed"**

If any edge on the board is faster than about six times the delay of the longest route it drives, or any link runs above a few hundred megabits per second, plan the stackup and constraints before placement. *Limits:* a screening rule only; the quantitative tests are in Chapter 2.

## 1.5 Notation and conventions

Delay and per-unit-length quantities are easy to confuse, and confusing them is the most common source of factor-of-two errors in this subject. The book uses the notation of the box below throughout; the full list of symbols is in the front matter.

**Delay and per-unit-length notation**

| | | |
|---|---|---|
| $t_{pd}$ | delay per unit length (ps/in) | $t_{pd} = \sqrt{\varepsilon_{eff}}/c \approx 84.73\,\text{ps/in} \times \sqrt{\varepsilon_{eff}}$ |
| $T_D$ | one-way delay of a line of length $\ell$ | $T_D = \ell\, t_{pd}$ |
| $2T_D$ | round-trip delay (what a TDR displays) | |
| $L'$, $C'$ | inductance and capacitance per unit length | $Z_0 = \sqrt{L'/C'}$, $t_{pd} = \sqrt{L'C'}$ |
| $L_{tot}$, $C_{tot}$ | totals for length $\ell$ | $L_{tot} = Z_0 T_D$, $C_{tot} = T_D/Z_0$ |
| $t_r$ | 10–90 % rise time (unless stated) | $T_{ramp} = t_r/0.8$ for a linear edge |

**Evidence labels.** Every figure caption ends with a bracketed label stating what kind of evidence the figure is: [Analytical] (evaluated from the equations in the text), [Simulation] (a numerical model, with its assumptions in the caption), [Extraction] (a field-solver result), [Measured] (laboratory data) or [Conceptual] or [Schematic] (a drawing that explains an idea and is not to scale). An analytical curve is exactly as good as its model; it is not a measurement.

**Table 1.3.** Physical elements of a board and the chapters that treat them.

| Element | Electrical role | Chapters |
|---|---|---|
| Trace | Transmission line; resistance; coupling | 4–10 |
| Plane | Return conductor; PDN capacitance; cavity | 3, 16, 17 |
| Dielectric | Delay, impedance, loss, skew | 7, 8 |
| Via | Discontinuity, stub, return transition | 11 |
| Soldermask | Outer-layer impedance and delay shift | 7 |
| Component and package | Load, source, SSN path | 15, 19–24 |

**Key Takeaways**

- A board is copper and dielectric whose geometry, not just connectivity, sets delay, impedance, loss, coupling and radiation.
- The fabricator controls final dielectric thickness, etched width, copper thickness and roughness; models must carry those tolerances.
- A route is lumped when its delay is small against the edge and a transmission line when it is not; the same net can be either.
- Return current flows in the adjacent plane, beneath the trace; interruptions enlarge the loop and harm SI, PI and EMC together.
- Keep $t_{pd}$, $T_D$ and $2T_D$ distinct, and read every figure's evidence label.

## Further Reading

Ritchey [8] explains board construction from the fabricator's side; Bogatin [3] and Ott [5] develop the return-current view used here.

## Review Problems

**P1.1** Name three board properties that the fabricator, not the layout, determines, and one way each affects impedance.

**P1.2** A 2 in microstrip with $t_{pd} = 145\,\text{ps/in}$ is driven by edges of 1 ns and of 80 ps. Classify the route in each case.

**P1.3** Find the DC resistance of a 10 in, 8 mil-wide trace in 1 oz (34.8 µm) copper.

**P1.4** Find the total capacitance and inductance of a 50 Ω line with $T_D = 0.5\,\text{ns}$.

**P1.5** A TDR shows a discontinuity 1.2 ns after the launch on a line with $t_{pd} = 170\,\text{ps/in}$. How far from the launch is it?

**P1.6** For each of the following, name the SI, PI and EMC consequence: a ground-plane slot under a clock trace; a clock trace that changes from a ground reference to a power-plane reference with no stitching capacitor.

Chapter 2

# What Makes a Design "High-Speed"

**Chapter Overview**

A design is "high-speed" not because its clock is fast but because its *edges* are fast compared with the time it takes a signal to travel across the interconnect. This chapter makes that statement quantitative. It relates rise time to bandwidth, introduces the knee frequency, shows how the spectrum of a trapezoidal waveform is shaped by its pulse width and its edge rate, and derives the critical length beyond which a trace must be treated as a transmission line rather than a lumped wire.

**Learning Objectives**

1. Define 10–90 % rise time and relate it to the $-3$ dB bandwidth of a single-pole system.
2. Estimate the significant spectral content of a digital edge using the knee frequency $f_{\text{knee}} \approx 0.5/t_{\text{r}}$.
3. Sketch the spectral envelope of a trapezoidal pulse train and locate its two corner frequencies.
4. Compute propagation delay from the effective dielectric constant and decide whether an interconnect is lumped or distributed.
5. Combine the rise times of cascaded components and specify the bandwidth of a measurement system.

## 2.1 Edges, not clocks

Consider two boards. The first carries a 25 MHz microcontroller bus driven by a modern 1.8 V CMOS output stage whose edges switch in 300 ps. The second carries a 25 MHz bus driven by a decades-old TTL part whose edges take 5 ns. The clocks are identical, yet the first board may ring, cross-talk, and radiate, while the second is well behaved on almost any layout. The difference is the *rise time*.

Every physical edge is a smooth transition between two logic levels (Figure 2.1). By convention the rise time $t_{\text{r}}$ is measured between the 10 % and 90 % points of the swing; some data sheets use 20 % to 80 %, which yields a shorter number for the same edge, so the convention must always be stated. The fall time $t_{\text{f}}$ is defined analogously. Throughout this book, $t_{\text{r}}$ means the 10–90 % rise time unless noted otherwise.

Semiconductor scaling has pushed rise times down relentlessly, because the transistors in an output stage switch as fast as the process allows unless the designer deliberately slows them. A 100 MHz system built today can easily have 200 ps edges—faster than the edges of a 1 GHz system of a generation ago. As Johnson and Graham put it in their classic treatment, the clock frequency sets how often edges happen, but the edge rate sets how hard each edge stresses the interconnect [1].

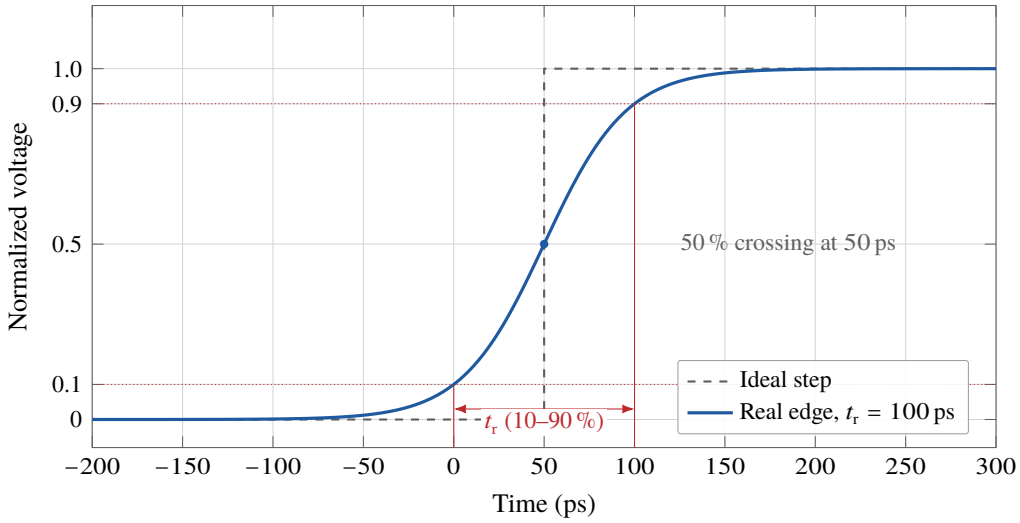



**Figure 2.1.** An ideal step compared with a physical digital edge. The rise time $t_r$ is the interval between the 10 % and 90 % crossings; the edge shown has $t_r = 100$ ps.

**Common Pitfall: Judging speed by the clock**

Treating a “slow” 10 MHz SPI or I$^2$C bus as immune to signal-integrity problems. If the driver is a fast CMOS output, the edge contains energy into the gigahertz range and long traces will ring. Always look up the driver’s edge rate (or its IBIS model) before deciding that a net is not critical.

## 2.2 Rise time and bandwidth

### 2.2.1 The single-pole system

The simplest model of a band-limited edge is the step response of a first-order RC low-pass filter with time constant $\tau = RC$:

$$v(t) = V\left(1 - \mathrm{e}^{-t/\tau}\right), \qquad t \geq 0. \tag{2.1}$$

Setting $v = 0.1V$ and $v = 0.9V$ gives $t_{10} = \tau \ln(10/9)$ and $t_{90} = \tau \ln 10$, so

$$t_r = t_{90} - t_{10} = \tau \ln 9 \approx 2.197\,\tau. \tag{2.2}$$

The same filter has a $-3$ dB bandwidth $f_{3\mathrm{dB}} = 1/(2\pi\tau)$. Eliminating $\tau$ between the two results yields the most widely used relation in high-speed design:

$$f_{3\mathrm{dB}} = \frac{\ln 9}{2\pi\, t_r} \approx \frac{0.35}{t_r}. \tag{2.3}$$

Equation (2.3) is exact only for a single-pole response. A Gaussian response (typical of cascaded stages and of many oscilloscopes) gives a constant of about 0.34; oscilloscopes with a flatter, “brick-wall” response can have a constant of 0.4 to 0.45. For engineering estimates the spread does not matter, but for precise measurement work (Chapter 26) the instrument’s actual step response should be used.

### 2.2.2 The knee frequency

A real digital waveform is not a sinusoid, and its $-3$ dB bandwidth does not capture how far up in frequency its energy extends. Johnson and Graham introduced the *knee frequency* as a practical upper limit of the

significant spectral content of an edge [1]:

$$f_{\text{knee}} \approx \frac{0.5}{t_{\text{r}}}. \tag{2.4}$$

Below $f_{\text{knee}}$ the spectrum of a digital pulse train falls at roughly $-20\,\text{dB/decade}$; above it the spectrum falls much faster. The practical meaning is that an interconnect whose response is reasonably flat up to $f_{\text{knee}}$ will pass the edge with little distortion, while features of the interconnect that only matter above $f_{\text{knee}}$ will have little effect on it.

**Rule of Thumb: Knee frequency**

$f_{\text{knee}} \approx 0.5/t_{\text{r}}$ and $f_{3\text{dB}} \approx 0.35/t_{\text{r}}$. A 100 ps edge has significant content to about 5 GHz. *Limits:* $f_{\text{knee}}$ is a heuristic boundary, not a cutoff. Energy above it is reduced, not absent, and it still matters for EMI (Chapter 18) and for resonances such as via stubs (Chapter 11).

**Worked Example 2.1: Bandwidth of a DDR4 data edge**

A DDR4 data driver produces edges with $t_{\text{r}} = 100$ ps. Estimate the $-3$ dB bandwidth of the edge and its knee frequency.

**Solution.** From Eq. (2.3), $f_{3\text{dB}} = 0.35/100\,\text{ps} = 3.5\,\text{GHz}$. From Eq. (2.4), $f_{\text{knee}} = 0.5/100\,\text{ps} = 5\,\text{GHz}$. Even though DDR4-3200 toggles its data at a fundamental of only 1.6 GHz, models of the channel (package, vias, connectors) must remain accurate to at least 5 GHz.

### 2.2.3 Cascaded rise times

When an edge passes through several band-limited stages—a driver, a package, a channel, a probe, and an oscilloscope—each stage slows it. If each stage has an approximately Gaussian response, the overall rise time is the root sum of squares of the individual rise times:

$$t_{r,\text{total}} \approx \sqrt{t_{r,1}^2 + t_{r,2}^2 + \cdots + t_{r,n}^2}. \tag{2.5}$$

The relation is exact for Gaussian stages and a good approximation for most other well-damped responses. It fails for stages that ring or overshoot.

**Worked Example 2.2: Oscilloscope bandwidth**

A 4 GHz oscilloscope is used to measure an edge whose true rise time is 100 ps. What rise time is displayed, and what scope bandwidth would keep the error below 5 %?

**Solution.** The scope's own rise time is $0.35/4\,\text{GHz} = 87.5$ ps. By Eq. (2.5) the displayed rise time is $\sqrt{100^2 + 87.5^2}\,\text{ps} = 132.9$ ps, an error of 33 %. For an error below 5 % we need $\sqrt{1 + x^2} \le 1.05$, where $x$ is the ratio of scope to signal rise time; thus $x \le 0.32$, the scope rise time must be at most 32 ps, and its bandwidth at least $0.35/32\,\text{ps} \approx 11\,\text{GHz}$.

## 2.3 Spectrum of a trapezoidal waveform

A clock or a repetitive data pattern is well approximated by a trapezoidal pulse train with period $T$, amplitude $A$, pulse width $\tau$ (measured at the 50 % points), and equal linear rise and fall ramps of full duration $T_{\text{ramp}}$. Its Fourier series has one-sided harmonic amplitudes at $f_n = n/T$ given by

$$|c_n| = 2A\frac{\tau}{T}\left|\frac{\sin(n\pi\tau/T)}{n\pi\tau/T}\right|\left|\frac{\sin(n\pi T_{\text{ramp}}/T)}{n\pi T_{\text{ramp}}/T}\right|. \tag{2.6}$$

### 2.3.1 Edge duration is part of the model

Equation (2.6) is derived for a ramp that goes from 0 % to 100 % in $T_{\text{ramp}}$, whereas data sheets and oscilloscopes quote the 10–90 % rise time $t_r$. For a linear ramp $v(t) = At/T_{\text{ramp}}$ the 10 % and 90 % crossings occur at $0.1T_{\text{ramp}}$ and $0.9T_{\text{ramp}}$, so

$$t_r = 0.8\,T_{\text{ramp}} \qquad \Longleftrightarrow \qquad T_{\text{ramp}} = \frac{t_r}{0.8} = 1.25\,t_r \tag{2.7}$$

(Figure 2.2). The conversion applies only to a linear ramp. A single-pole exponential edge has no finite full duration, and a Gaussian edge has none either; for those shapes use the measured or modeled waveform directly rather than a trapezoid.

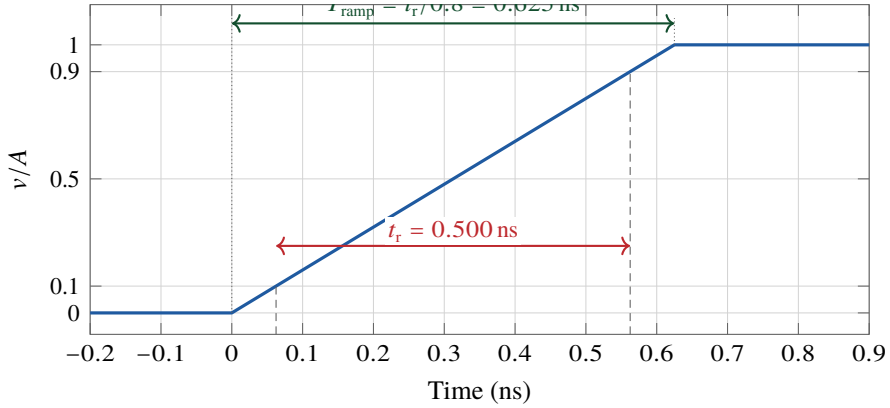



**Figure 2.2.** A linear ramp with $t_r = 0.5$ ns (10–90 %) has a full duration $T_{\text{ramp}} = 0.625$ ns. [Analytical]

The two $\sin(x)/x$ factors produce two corner frequencies in the spectral envelope. Replacing each $\sin(x)/x$ factor by its upper bound, $\min(1, 1/x)$, gives the asymptotic envelope

$$|c_n|_{\text{env}} = \begin{cases} 2A\tau/T, & f < f_1 = \dfrac{1}{\pi\tau}, \\ 2A\tau/T \cdot \dfrac{f_1}{f}, & f_1 \le f < f_2 = \dfrac{1}{\pi T_{\text{ramp}}} = \dfrac{0.8}{\pi t_r}, \\ 2A\tau/T \cdot \dfrac{f_1 f_2}{f^2}, & f \ge f_2. \end{cases} \tag{2.8}$$

The envelope is flat up to $f_1$, falls at −20 dB/decade between $f_1$ and $f_2$, and falls at −40 dB/decade above $f_2$ (Figure 2.3). The first corner depends only on the pulse width, the second only on the edge duration. Two conclusions follow directly:

1. Slowing the edges moves $f_2$ down and reduces *all* harmonics above it; this is the cheapest EMI fix available (Chapter 18) [5].
2. $f_2 = 0.8/(\pi t_r) \approx 0.255/t_r$ lies well below $f_{\text{knee}} = 0.5/t_r$. The knee frequency is a separate heuristic for the bandwidth an interconnect model must cover; it is not a corner of the trapezoid spectrum.

**Common Pitfall: Mixing edge definitions**

Inserting a 10–90 % rise time directly into the trapezoid formula overstates $f_2$ by 25 % and over-predicts every harmonic above it. State the edge definition beside every spectral estimate, and convert with Eq. (2.7) only when the edge really is close to linear.

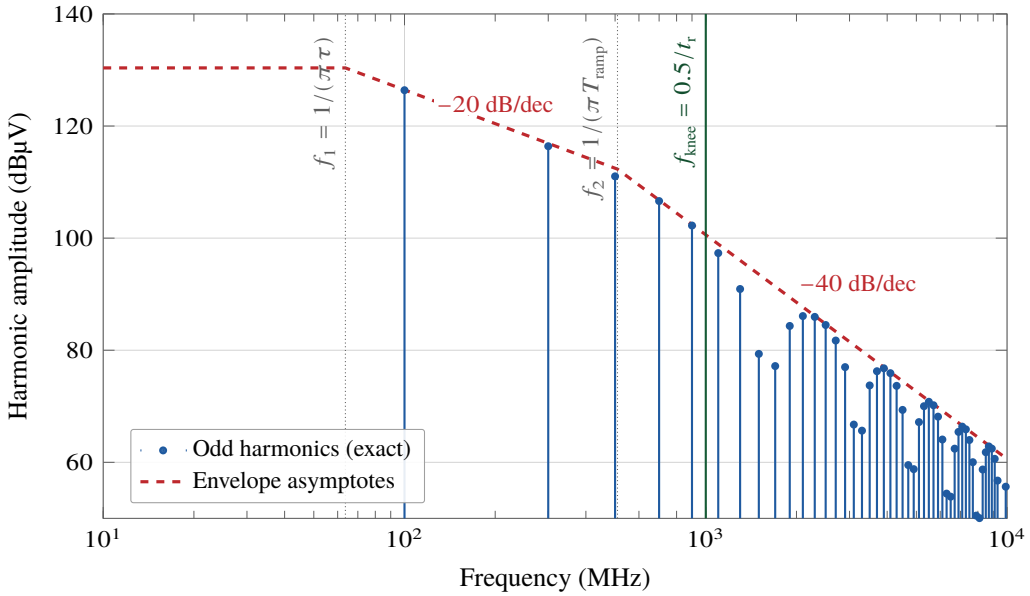



**Figure 2.3.** Harmonic amplitudes of a 100 MHz, 50 % duty-cycle, 3.3 V clock with $t_r = 0.5$ ns ($T_{\text{ramp}} = 0.625$ ns), computed from Eq. (2.6), with the asymptotic envelope of Eq. (2.8). Even harmonics vanish for a 50 % duty cycle. [Analytical]

**Worked Example 2.3: Harmonic content of a clock**

For the clock of Figure 2.3 ($T = 10$ ns, $\tau = 5$ ns, $t_r = 0.5$ ns), find the corner frequencies and the amplitude of the 11th harmonic relative to the fundamental.

**Solution.** $f_1 = 1/(\pi \cdot 5\,\text{ns}) = 63.7$ MHz. The ramp duration is $T_{\text{ramp}} = 0.5\,\text{ns}/0.8 = 0.625$ ns (Eq. (2.7)), so $f_2 = 1/(\pi \cdot 0.625\,\text{ns}) = 509$ MHz. With $\tau/T = 0.5$ and $T_{\text{ramp}}/T = 0.0625$, Eq. (2.6) gives for $n = 1$ the product $0.6366 \times 0.9936 = 0.6325$ and for $n = 11$ the product $0.05787 \times 0.3850 = 0.02228$. The ratio is 0.0352, or −29.1 dB. The 1.1 GHz harmonic is still only about 29 dB below the fundamental—more than enough to fail a radiated emissions test if it finds an antenna.

*Check against the shortcut.* Using $t_r$ in place of $T_{\text{ramp}}$ gives $f_2 = 637$ MHz and −25.6 dB: a 3.5 dB over-estimate of this harmonic, from the edge definition alone.

## 2.4 Propagation delay

A signal on a board travels as an electromagnetic wave guided by the trace and its reference plane. In a homogeneous dielectric of relative permittivity $\varepsilon_r$ the wave velocity is

$$v_p = \frac{c}{\sqrt{\varepsilon_r}}, \qquad c = 2.998 \times 10^8\ \text{m/s} = 11.80\ \text{in/ns}, \tag{2.9}$$

and the delay per unit length is

$$t_{pd} = \frac{1}{v_p} = \frac{\sqrt{\varepsilon_r}}{c} \approx 84.7\ \text{ps/in} \times \sqrt{\varepsilon_r}. \tag{2.10}$$

A stripline is embedded in a homogeneous dielectric, so its $\varepsilon_r$ is simply the laminate $D_k$. A microstrip has field lines partly in air and partly in the dielectric, so an *effective dielectric constant* $\varepsilon_{\text{eff}}$, lying between 1 and $D_k$, replaces $\varepsilon_r$; it is computed in Chapter 7. Soldermask raises $\varepsilon_{\text{eff}}$ of a microstrip slightly.

**Rule of Thumb: Six inches per nanosecond**

In FR-4 stripline ($D_k \approx 3.6$ to 4.2) signals travel at roughly 5.8–6.2 in/ns, i.e. 160–175 ps/in. Outer-layer microstrip is faster, typically 140–150 ps/in. *Limits:* $D_k$ varies with resin content, glass style, and frequency (Chapters 7 and 8); for timing budgets tighter than about 10 %, use the fabricator's $D_k$ at the frequency of interest.

The one-way delay of a line of length $\ell$ is $T_D = \ell\, t_{pd}$. A useful companion quantity is the *spatial extent of the edge*, or rise length, $\ell_r = t_r/t_{pd}$: the physical distance occupied by the rising edge as it travels down the line. A 100 ps edge in FR-4 stripline is only about 0.6 in long.

## 2.5 Lumped versus distributed behavior

If the edge is long compared with the interconnect, the whole trace sees essentially the same voltage at every instant, and it behaves like a lumped capacitor (plus, perhaps, a lumped inductor). If the edge is short compared with the trace, different parts of the trace are at different voltages at the same instant; the trace must then be analyzed as a transmission line (Chapter 4), and impedance discontinuities produce reflections (Chapter 6).

The boundary between the two regimes is expressed as a *critical length*

$$\ell_{crit} = \frac{t_r}{k\, t_{pd}}, \tag{2.11}$$

where $k$ is a safety factor. Johnson and Graham treat a line as lumped when its delay is less than about one-sixth of the rise time [1], i.e. $k = 6$; other authors use values from 4 to 10 [3], [4]. The choice depends on how much ringing the receiver can tolerate. Table 2.1 summarizes common criteria, and Figure 2.4 plots Eq. (2.11) for typical board materials.

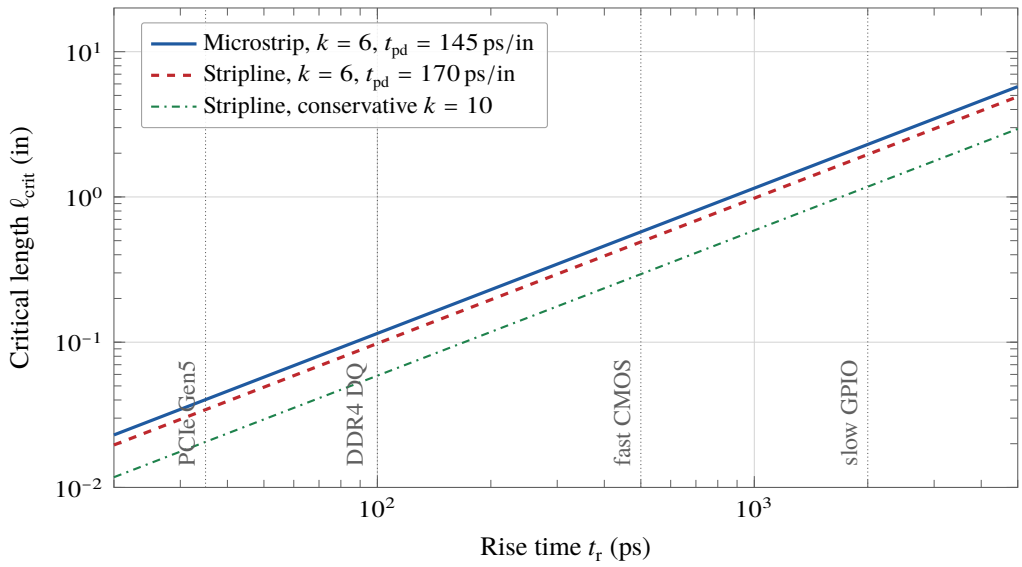



**Figure 2.4.** Critical length versus rise time from Eq. (2.11). Lines shorter than $\ell_{crit}$ may be modeled as lumped elements; longer lines need transmission-line treatment and, usually, termination. Vertical lines mark representative edge rates.

**Table 2.1.** Criteria for deciding when an interconnect is electrically long.

| Condition | Interpretation | Typical use |
|---|---|---|
| $T_D < t_r/10$ | Line is negligible; treat as lumped $C$ | Conservative, analog edges |
| $T_D < t_r/6$ | Lumped model adequate; ringing small | General digital ($k = 6$) |
| $T_D < t_r/4$ | Reflections overlap the edge; mild ringing | Short stubs, breakouts |
| $T_D < t_r/2$ | Round-trip delay shorter than edge; termination often optional | Low-speed buses |
| $T_D \geq t_r/2$ | Reflections arrive after the edge; termination needed | Most high-speed nets |

**Worked Example 2.4: Is a DDR4 stub lumped?**

A DDR4 data line uses $t_r = 100\,\text{ps}$ edges and is routed in stripline with $D_k = 3.8$. What is the critical length with $k = 6$, and is a 150 mil via-and-breakout stub electrically short?

**Solution.** From Eq. (2.10), $t_{pd} = 84.7\,\text{ps/in} \times \sqrt{3.8} = 165.2\,\text{ps/in}$. Then $\ell_{crit} = 100\,\text{ps}/(6 \times 165.2\,\text{ps/in}) = 0.101\,\text{in}$, about 100 mil or 2.6 mm. The 150 mil stub exceeds this; its effect on the edge will be visible as a reflection, and it should be either shortened or included in simulation. The rise length is $t_r/t_{pd} = 0.61\,\text{in}$.

**Design Rule: When to treat a net as a transmission line**

Compute $\ell_{crit} = t_r/(6t_{pd})$ using the fastest edge the driver can produce (the "fast/strong" IBIS corner), not the nominal one. Any net longer than $\ell_{crit}$ needs a controlled impedance and an explicit termination strategy; any stub, branch, or via longer than $\ell_{crit}$ must be included in the SI model.

## 2.6 Data rate, unit interval, and Nyquist frequency

Serial and memory interfaces are described by their data rate rather than by a clock. For non-return-to-zero (NRZ) signaling at $R_b$ bits per second, the *unit interval* is $\text{UI} = 1/R_b$ and the highest fundamental frequency, produced by an alternating 1010 pattern, is the Nyquist frequency

$$f_{Nyq} = \frac{R_b}{2} = \frac{1}{2\,\text{UI}}. \tag{2.12}$$

Channel loss budgets for serial links are specified at $f_{Nyq}$ (Chapter 20). Edge rates in such links are usually a sizeable fraction of a UI, so their knee frequency lies at two to three times $f_{Nyq}$; this is why channel models are routinely required to two or three times the Nyquist frequency. For multi-level signaling such as PAM4, the symbol rate (baud), not the bit rate, sets the UI. Table 2.2 lists the interfaces covered in Part V.

## 2.7 A checklist for deciding what matters

The concepts of this chapter combine into a quick triage that should be run on every net class at the start of a design:

1. Find the fastest edge rate ($t_r$) of each driver from its IBIS model or data sheet.
2. Compute $f_{knee} = 0.5/t_r$. All models of the channel (vias, connectors, packages) must be valid to at least $f_{knee}$.

**Table 2.2.** Unit interval and Nyquist frequency of representative interfaces (NRZ unless noted). Edge rates are implementation dependent and are not listed.

| Interface | Rate (GT/s) | UI (ps) | $f_{\mathrm{Nyq}}$ (GHz) |
|---|---|---|---|
| 1000BASE-T (per pair, PAM5) | 0.125 | 8000 | 0.0625 |
| MIPI D-PHY (one lane) | 2.5 | 400 | 1.25 |
| DDR4-3200 | 3.2 | 312.5 | 1.6 |
| LPDDR5-6400 | 6.4 | 156.25 | 3.2 |
| PCIe Gen3 | 8.0 | 125 | 4.0 |
| USB 3.2 Gen 2 | 10.0 | 100 | 5.0 |
| 10GBASE-KR | 10.3125 | 96.97 | 5.156 |
| PCIe Gen4 | 16.0 | 62.5 | 8.0 |
| PCIe Gen5 | 32.0 | 31.25 | 16.0 |
| PCIe Gen6 (PAM4, 32 GBd) | 64.0 | 31.25 | 16.0 |

3. Compute $\ell_{\mathrm{crit}}$ from Eq. (2.11) for the intended routing layer. Nets longer than $\ell_{\mathrm{crit}}$ need controlled impedance and termination; stubs longer than $\ell_{\mathrm{crit}}$ need simulation.
4. For serial links, compute $f_{\mathrm{Nyq}}$ and the loss budget at $f_{\mathrm{Nyq}}$ (Chapters 8 and 20).
5. Estimate the harmonic envelope (Eq. (2.8)) of clocks and periodic buses; these dominate radiated emissions.

**Key Takeaways**

- Edge rate, not clock frequency, determines whether a design is high-speed.
- $f_{3\mathrm{dB}} \approx 0.35/t_{\mathrm{r}}$ for a single-pole edge; the knee frequency $f_{\mathrm{knee}} \approx 0.5/t_{\mathrm{r}}$ bounds its significant spectral content.
- The spectrum of a trapezoidal waveform is flat to $1/(\pi\tau)$, falls at $-20\,\mathrm{dB/decade}$ to $1/(\pi T_{\mathrm{ramp}})$, then at $-40\,\mathrm{dB/decade}$; for a linear edge $T_{\mathrm{ramp}} = t_{\mathrm{r}}/0.8$.
- Signals travel at about 6 in/ns in FR-4 stripline; $t_{\mathrm{pd}} \approx 84.7\,\mathrm{ps/in} \times \sqrt{\varepsilon_{\mathrm{r}}}$.
- A line longer than $\ell_{\mathrm{crit}} = t_{\mathrm{r}}/(6t_{\mathrm{pd}})$ is a transmission line. With a 100 ps edge, that is about 0.1 in.
- Rise times of cascaded Gaussian-like stages add in root-sum-square fashion; measurement bandwidth must be several times the signal bandwidth.

## Further Reading

Johnson and Graham [1] remain the classic source on edge rates, knee frequency, and lumped versus distributed behavior; Bogatin [3] develops the same ideas with many quantitative rules of thumb, and Dally and Poulton [9] place them in the context of complete digital systems.

## Review Problems

**P2.1** A PCIe Gen5 transmitter produces edges with $t_{\mathrm{r}} = 35\,\mathrm{ps}$. Estimate the $-3\,\mathrm{dB}$ bandwidth and the knee frequency.

**P2.2** A 50 MHz clock with $t_r = 2$ ns travels 4 in on a microstrip with $t_{pd} = 145$ ps/in. Find $T_D$ and $\ell_{crit}$ ($k = 6$). Is the line lumped?

**P2.3** An edge with a true rise time of 60 ps is measured on an 8 GHz oscilloscope. What rise time is displayed?

**P2.4** Find the per-inch delay of a stripline in a laminate with $D_k = 3.4$, and the delay of a 7 in route.

**P2.5** A 25 MHz, 50 % duty-cycle clock has linear edges with $t_r = 1$ ns (10–90 %). Find the two envelope corner frequencies and the knee frequency.

**P2.6** A link runs at 16 GT/s NRZ with edges equal to 40 % of a UI. Find the UI, the Nyquist frequency, and the knee frequency.

Chapter 3

# Electromagnetic Fundamentals for Board Designers

**Chapter Overview**

Every signal-integrity, power-integrity, and EMI problem on a board can be traced back to a few electromagnetic quantities: capacitance, inductance (and in particular *loop* inductance), the path taken by return current, and the way current crowds toward conductor surfaces at high frequency. This chapter develops these ideas with the minimum of field theory needed to apply them, and gives the closed-form estimates used throughout the rest of the book.

**Learning Objectives**

1. Distinguish self, mutual, partial, and loop inductance, and compute loop inductance for common board structures.
2. Estimate parallel-plate and trace capacitance and the displacement current they carry.
3. Explain why high-frequency return current flows directly beneath a signal trace, and quantify how it spreads.
4. Compute skin depth and the resulting increase in conductor resistance.
5. Identify the frequency at which the return path changes from the lowest-resistance to the lowest-inductance route.

## 3.1 Every signal is a current loop

A signal does not "go" from a driver to a receiver along a single trace. The driver pushes current into the signal conductor and an equal current returns to the driver through some other conductor—a reference plane, a neighboring trace, a cable shield. Kirchhoff's current law makes this unavoidable, and at high frequency the return path is as much a part of the signal as the trace itself. Most of the failures examined in this book (ground bounce, crosstalk, emission from plane splits, reference-plane changes) are failures of the return path, not of the signal trace [3], [5].

The two conductors and the space between them form a loop. The loop stores energy in two forms: in the electric field between the conductors (described by capacitance) and in the magnetic field enclosed by the loop (described by inductance). Both depend only on geometry and materials.

## 3.2 Capacitance

### 3.2.1 Parallel-plate capacitance

Two conducting planes of area $A$ separated by a dielectric of thickness $h$ and relative permittivity $\varepsilon_r$ have capacitance

$$C = \frac{\varepsilon_0 \, \varepsilon_r A}{h}, \qquad \varepsilon_0 = 8.854\,\text{pF/m} = 0.2249\,\text{pF/in}. \tag{3.1}$$

Equation (3.1) ignores fringing fields at the edges, which is accurate when the lateral dimensions are much larger than $h$—true for power and ground planes, which is where it is most often used.

**Worked Example 3.1: Capacitance of a power–ground plane pair**

A 4 in × 4 in region of a 0.8 V core plane is separated from the adjacent ground plane by a 3 mil core with $D_k = 4.2$. Find the plane capacitance.

**Solution.** With $A = 16\,\text{in}^2$ and $h = 0.003$ in, Eq. (3.1) gives $C = 0.2249\,\text{pF/in} \times 4.2 \times 16/0.003 = 5.04\,\text{nF}$. This capacitance has extremely low inductance and dominates the PDN impedance in the hundreds-of-megahertz range (Chapter 16).

### 3.2.2 Capacitance of a trace

A trace over a plane has a capacitance per unit length $C'$ that includes the parallel-plate region under the trace plus fringing fields from its edges and top. For typical 50 Ω lines the fringing contribution is comparable to the parallel-plate term, so Eq. (3.1) applied to the trace width alone underestimates $C'$ by roughly a factor of two. Accurate values come from a 2-D field solver or from the impedance relations of Chapter 4: a 50 Ω line in FR-4 has $C' \approx 3.3\,\text{pF/in}$.

### 3.2.3 Displacement current

When the voltage across a capacitance changes, a displacement current

$$i = C\frac{dv}{dt} \tag{3.2}$$

flows through the dielectric. This is how a signal's return current crosses from one plane to another at a layer transition (Chapter 17) and how a fast edge couples capacitively into a neighbor (Chapter 9). A 1 V edge with $t_r = 100\,\text{ps}$ across 1 pF of stray capacitance drives 10 mA.

## 3.3 Inductance

### 3.3.1 Definitions

Inductance is the number of magnetic field lines (the flux $\Phi$) linking a current path, per ampere of current:

$$L = \frac{\Phi}{I}. \tag{3.3}$$

A changing current changes the flux, which by Faraday's law induces a voltage

$$v = L\frac{di}{dt}. \tag{3.4}$$

Several kinds of inductance must be distinguished [10]:

**Loop inductance**
: The flux through a *closed* current path per ampere. Only loop inductance is physically measurable, and only loop inductance appears in Faraday's law.

**Partial self inductance**
A bookkeeping quantity assigned to one *segment* of a loop, defined so that the loop inductance equals the sum of the partial self inductances of its segments minus the partial mutual inductances between them. It allows the inductance of a via or a pin to be quoted independently of the rest of its loop.

**Mutual inductance**
The flux from one current that links a second conductor, per ampere of the first current. It is the mechanism of inductive crosstalk (Chapter 9) and of ground bounce (Chapter 15).

For a loop made of an outgoing conductor (1) and a returning conductor (2) carrying equal and opposite currents,

$$L_{\text{loop}} = L_{11} + L_{22} - 2L_{12}, \tag{3.5}$$

where $L_{11}$ and $L_{22}$ are partial self inductances and $L_{12}$ is the partial mutual inductance. Bringing the return conductor closer increases $L_{12}$ and reduces $L_{\text{loop}}$. This single equation explains why every high-speed design rule about return paths says the same thing: keep the return conductor as close as possible to the signal conductor (Figure 3.1).

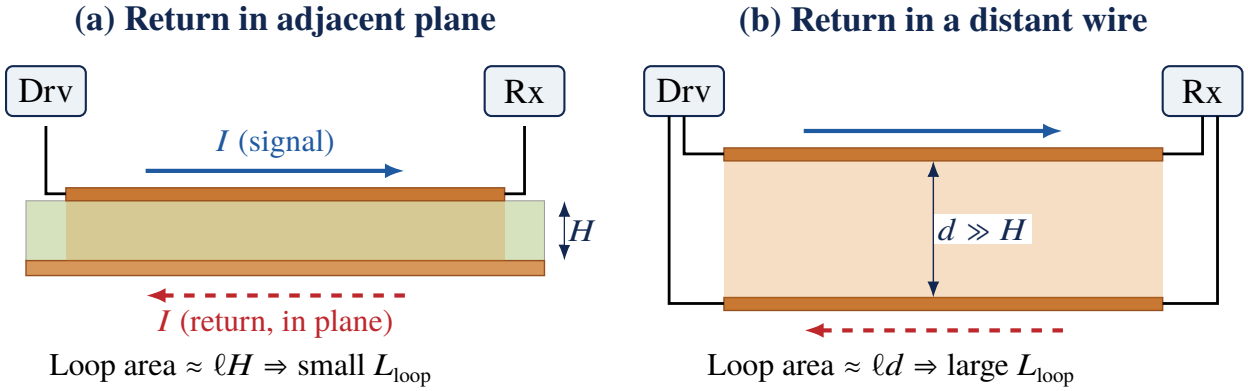



**Figure 3.1.** The loop formed by a signal and its return current. (a) With an adjacent reference plane, the loop area, and therefore the loop inductance, is set by the dielectric height $H$. (b) A distant return conductor encloses a much larger area and has much higher loop inductance.

### 3.3.2 Useful closed-form estimates

For a straight round wire of length $\ell$ and radius $r$, with $\ell \gg r$, the high-frequency partial self inductance is approximately

$$L_{\text{partial}} \approx \frac{\mu_0 \ell}{2\pi}\left[\ln\frac{2\ell}{r} - 1\right], \tag{3.6}$$

and the partial mutual inductance between two parallel wires of length $\ell$ separated by $s$ (with $\ell \gg s$) is

$$M \approx \frac{\mu_0 \ell}{2\pi}\left[\ln\frac{2\ell}{s} - 1\right]. \tag{3.7}$$

Substituting Eqs. (3.6) and (3.7) into Eq. (3.5) gives the loop inductance of a two-wire line,

$$L_{\text{loop}} \approx \frac{\mu_0 \ell}{\pi}\ln\frac{s}{r}, \qquad s \gg r, \tag{3.8}$$

which also describes a pair of vias carrying equal and opposite current—the dominant inductance of a decoupling capacitor's mounting (Chapter 16). Note that $\mu_0/(2\pi) = 0.2\,\text{nH/mm} = 5.08\,\text{nH/in}$.

**Rule of Thumb: Inductance of a wire**

The partial self inductance of a thin wire or trace is roughly 1 nH/mm (25 nH/in). *Limits:* Eq. (3.6) shows the value depends logarithmically on $\ell/r$; a 1 in wire of 0.5 mm radius has 18 nH, and thicker or shorter conductors have less. Use it only for order-of-magnitude estimates.

For two parallel planes separated by $h$, current spreading uniformly across a square region of any size sees the same inductance, the *sheet inductance*:

$$L_{\square} = \mu_0 h \approx 32\,\text{pH per square per mil of } h. \tag{3.9}$$

Thin dielectrics between power and ground planes therefore reduce both the spreading inductance and the impedance of the PDN.

**Worked Example 3.2: Loop inductance of a via pair**

A ground via and a power via pass through a 62 mil board. Each has a drill diameter of 10 mil ($r = 5$ mil) and they are 50 mil apart. Estimate the loop inductance if current enters on the top layer and leaves on the bottom layer.

**Solution.** From Eq. (3.8) with $\ell = 62\,\text{mil} = 1.575\,\text{mm}$: $L_{\text{loop}} = (\mu_0/\pi)\ell \ln(s/r) = 0.4\,\text{nH/mm} \times 1.575\,\text{mm} \times \ln 10 = 1.45\,\text{nH}$. If the same via pair connected a capacitor on the top layer to planes only 5 mil below, the inductance would scale with height, to about 0.12 nH. This is why decoupling capacitors should be mounted on the side of the board closest to the power–ground plane pair.

### 3.3.3 Internal and external inductance

Part of a conductor's inductance comes from the magnetic field *inside* the conductor. At low frequency, when current is uniform across the cross-section, this internal inductance is $\mu_0/(8\pi) = 50\,\text{nH/m}$ for a round wire. As frequency rises and current is pushed to the surface (Section 3.5), the internal field and the internal inductance shrink. The total inductance of a trace therefore decreases slightly with frequency, which is one reason a lossy line's characteristic impedance is frequency dependent (Chapter 4).

## 3.4 Return current distribution

### 3.4.1 Current density beneath a microstrip

At high frequency the return current in a plane beneath a trace concentrates directly under the trace, because that distribution minimizes the loop inductance and therefore the total impedance of the loop. For a trace at height $H$ above an infinite plane, image theory gives the current density in the plane as a function of lateral distance $x$ from the trace centerline [1]:

$$J(x) = \frac{I}{\pi H}\,\frac{1}{1 + (x/H)^2}. \tag{3.10}$$

Integrating Eq. (3.10) from $-D$ to $D$ gives the fraction of the return current within $\pm D$ of the centerline:

$$\frac{I(|x| < D)}{I} = \frac{2}{\pi}\arctan\frac{D}{H}. \tag{3.11}$$

Half the return current lies within $\pm H$, 79.5 % within $\pm 3H$, and 87.4 % within $\pm 5H$ (Figure 3.2). Equation (3.10) treats the trace as a thin wire; for real traces of width $W$ comparable to $H$, the peak is broader and lower, but the long tails, and the conclusion, are the same.

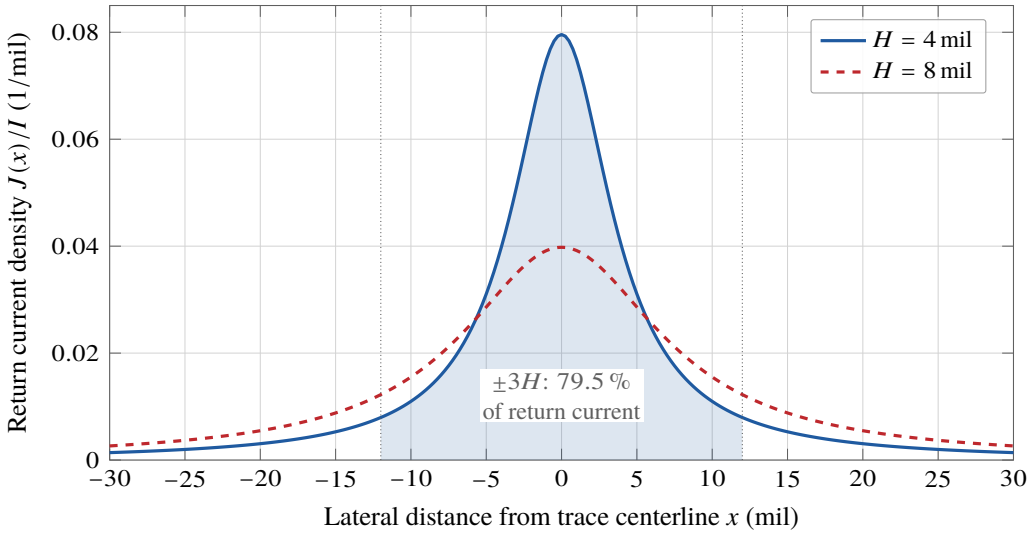



**Figure 3.2.** Return current density in the reference plane under a microstrip, from Eq. (3.10). A thinner dielectric concentrates the return current more tightly beneath the trace. The shaded band, $\pm 3H$ for $H = 4$ mil, carries 79.5 % of the return current.

**Design Rule: Keep the plane intact under the trace**

Treat the plane region within $\pm 3H$ to $\pm 5H$ of every high-speed trace as part of the signal path. Slots, anti-pad clusters, and plane splits inside this band disturb the return current and raise loop inductance, crosstalk, and emissions (Chapter 17).

### 3.4.2 Lowest resistance versus lowest inductance

The return current divides itself among the available paths to minimize total impedance $R + j\omega L$. At low frequency the resistive term dominates and the current spreads across the plane along the lowest-resistance route—roughly a straight line between the load and the driver. At high frequency the inductive term dominates and the current follows the lowest-inductance route, directly beneath the signal trace (Figure 3.3). The transition occurs where $\omega L \approx R$ for the loop, typically between a few kilohertz and a few hundred kilohertz for board-level loops. Every digital edge has most of its energy far above this transition, so for signal integrity the return current always follows the trace.

**Common Pitfall: Routing "ground" as if it were a wire**

Assuming that because all ground nets are connected, the return path will take care of itself. A slot cut in the plane under a trace forces the high-frequency return current around the slot. The added loop area shows up as inductance in series with the signal, as crosstalk to every other trace whose return shares the detour, and as a slot antenna.

## 3.5 Skin effect

### 3.5.1 Skin depth

A time-varying current inside a conductor generates a time-varying magnetic field, which induces eddy currents that oppose current flow in the interior. The result is that current density decays exponentially

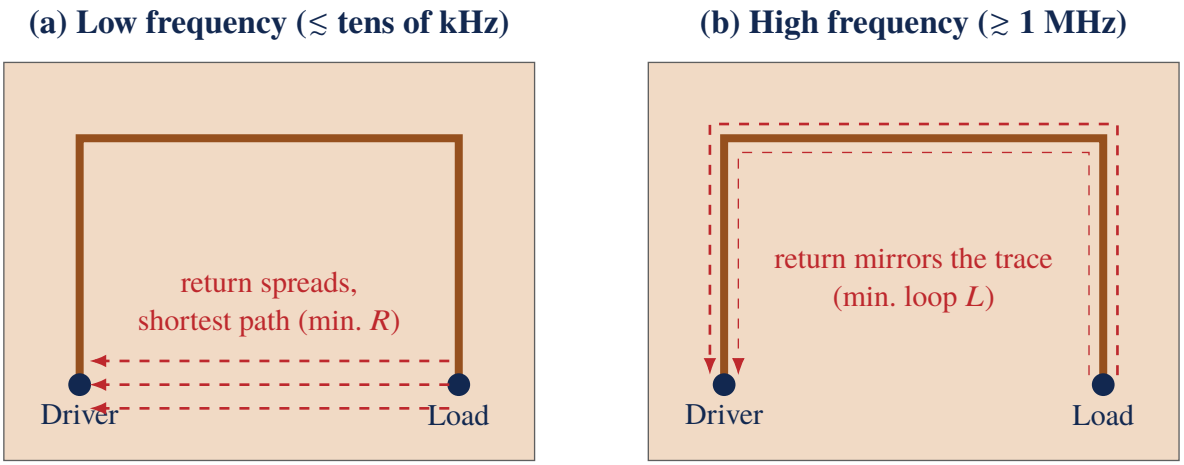



**Figure 3.3.** Top view of a U-shaped trace over a plane. (a) At low frequency the return current takes the shortest, lowest-resistance path across the plane. (b) At high frequency it follows directly beneath the trace, the lowest-inductance path.

with depth below the surface, with characteristic length—the *skin depth*—

$$\delta_s = \sqrt{\frac{\rho}{\pi f \mu_0 \mu_r}} = \frac{1}{\sqrt{\pi f \mu_0 \mu_r \sigma}}, \tag{3.12}$$

where $\rho = 1/\sigma$ is the resistivity. For annealed copper ($\rho = 1.724 \times 10^{-8}\ \Omega \cdot \text{m}$ at 20 °C, $\mu_r = 1$),

$$\delta_s \approx \frac{66.1\ \mu\text{m}}{\sqrt{f/\text{MHz}}}. \tag{3.13}$$

At 1 GHz the skin depth is 2.09 µm, about one-seventeenth of the thickness of 1 oz copper (Figure 3.4). Resistivity rises with temperature by about 0.39 %/°C, so the skin depth of a hot board is slightly larger. Ferromagnetic plating layers such as electroless nickel ($\mu_r \gg 1$) have a much smaller skin depth, which is one reason ENIG finish adds loss on outer-layer traces at high frequency.

### 3.5.2 AC resistance

When $\delta_s$ is much less than the conductor thickness, current flows in a layer roughly one skin depth thick on the surfaces facing the return current. For a stripline, with current on both the top and bottom faces of a trace of width $W$, the resistance per unit length is approximately

$$R'_{ac} \approx \frac{\rho}{2W\delta_s} \qquad \text{(signal conductor only)}, \tag{3.14}$$

compared with the DC value $R'_{dc} = \rho/(WT)$. Because $\delta_s \propto f^{-1/2}$, $R'_{ac} \propto \sqrt{f}$. The return planes carry the same current in a similar skin layer, so the total loop resistance is larger than Eq. (3.14); for a stripline it is close to twice the signal-conductor value. Edge effects and surface roughness raise it further (Chapter 8).

**Worked Example 3.3: AC resistance of a stripline trace**

A 5 mil wide, 1 oz (34.8 µm) stripline trace is 1 in long. Compare its DC resistance with its signal-conductor AC resistance at 1 GHz.

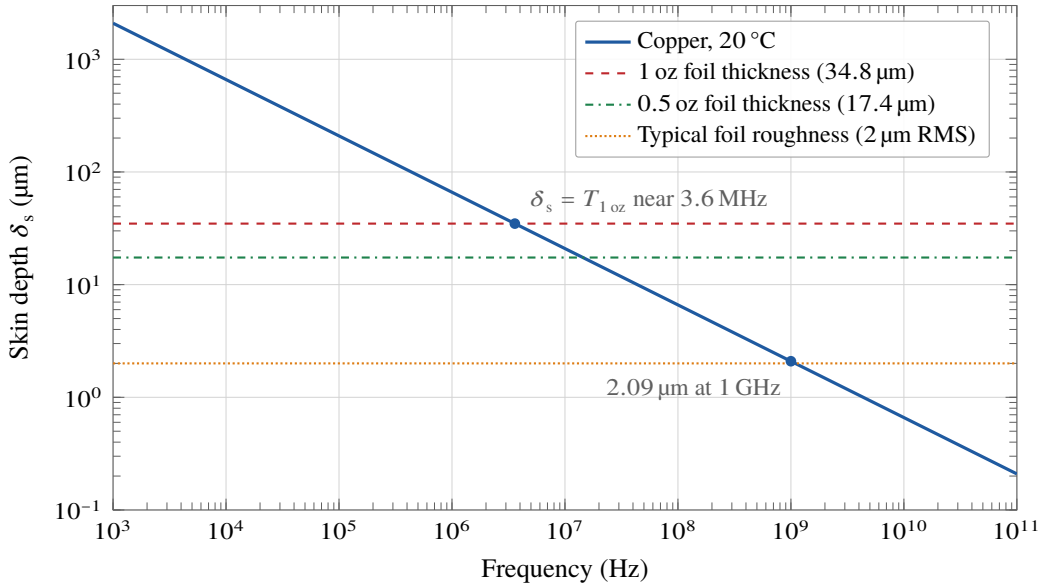



**Figure 3.4.** Skin depth of copper versus frequency, from Eq. (3.12). Above about 4 MHz the skin depth is smaller than the thickness of 1 oz foil; above a few gigahertz it is comparable to the surface roughness of the foil.

**Solution.** $W = 127\,\mu\text{m}$ and $\ell = 25.4\,\text{mm}$. DC: $R = \rho\ell/(WT) = 1.724 \times 10^{-8} \times 0.0254/(127 \times 10^{-6} \times 34.8 \times 10^{-6}) = 0.099\,\Omega$. AC: with $\delta_s = 2.09\,\mu\text{m}$, Eq. (3.14) gives $R = \rho\ell/(2W\delta_s) = 0.825\,\Omega$, about 8.3 times the DC value. Adding a comparable return-plane contribution, the loop resistance is about $1.6\,\Omega/\text{in}$ at 1 GHz.

## 3.6 Summary of field quantities

**Table 3.1.** Electromagnetic quantities and closed-form estimates used in board design.

| Quantity | Estimate | Validity |
|---|---|---|
| Plane capacitance | $C = \varepsilon_0\varepsilon_r A/h$; $0.2249\,\text{pF/in} \times \varepsilon_r A/h$ | Lateral size $\gg h$ |
| Sheet inductance | $L_\square = \mu_0 h$; 32 pH per square per mil | Plane pairs |
| Wire partial $L$ | $(\mu_0\ell/2\pi)[\ln(2\ell/r) - 1]$ | $\ell \gg r$ |
| Two-wire loop $L$ | $(\mu_0\ell/\pi)\ln(s/r)$ | $s \gg r$, $\ell \gg s$ |
| Return current spread | $J(x) \propto 1/(1 + (x/H)^2)$; 80 % within $\pm 3H$ | Thin trace over plane |
| Skin depth (Cu) | $66.1\,\mu\text{m}/\sqrt{f/\text{MHz}}$ | $\mu_r = 1$, 20 °C |
| Stripline $R_{ac}$ | $\rho/(2W\delta_s)$ per length (signal only) | $\delta_s \ll T$, smooth Cu |

**Key Takeaways**

- Every signal is a current loop; the return path is part of the signal.

- Only loop inductance is physical: $L_{\text{loop}} = L_{11} + L_{22} - 2L_{12}$. Bringing the return closer raises mutual inductance and lowers $L_{\text{loop}}$.
- Plane pairs have sheet inductance $\mu_0 h$ (32 pH per square per mil) and capacitance $\varepsilon_0 \varepsilon_r A/h$; thin dielectrics help both.
- High-frequency return current flows beneath the trace, with about 80 % within $\pm 3H$.
- Above a few kilohertz to a few hundred kilohertz the return current follows the trace; below, it takes the lowest-resistance path.
- Skin depth in copper is 2.09 µm at 1 GHz and scales as $f^{-1/2}$, so AC resistance grows as $\sqrt{f}$.

## Further Reading

Paul [10] gives a rigorous treatment of loop and partial inductance; Ott [5] and Johnson and Graham [1] discuss return current distribution and its consequences for layout; Ramo, Whinnery, and Van Duzer [11] cover the underlying field theory, including skin effect.

## Review Problems

**P3.1** Compute the skin depth of copper at 100 MHz and at 10 GHz.

**P3.2** Find the capacitance of a 6 in × 4 in plane pair separated by 2 mil of dielectric with $D_k = 4.0$.

**P3.3** What fraction of the return current under a thin microstrip flows within $\pm 2H$ of the centerline? Within $\pm 5H$?

**P3.4** Find the sheet inductance of a plane pair separated by 4 mil.

**P3.5** Two vias, each of radius 4 mil, are 40 mil apart and carry opposite currents over a height of 40 mil. Estimate their loop inductance.

**P3.6** A current of 20 mA switches in 100 ps through a loop inductance of 2 nH. What voltage is induced?

**P3.7** Find the DC resistance of a 10 in long, 4 mil wide trace in 0.5 oz (17.4 µm) copper.

Chapter 4

# Transmission Line Theory

**Chapter Overview**

When an interconnect is longer than its critical length, it must be analyzed as a transmission line. This chapter derives the telegrapher's equations from the distributed RLGC model, obtains the characteristic impedance and propagation constant, specializes them to lossless and low-loss lines, and develops the input-impedance formula that governs stubs, resonances, and matching. These results are the foundation of every later chapter on signal integrity.

**Learning Objectives**

1. Derive the telegrapher's equations from a distributed RLGC ladder.
2. Compute characteristic impedance, propagation velocity, and per-unit-length $L'$ and $C'$ from one another.
3. Separate conductor and dielectric attenuation in a low-loss line and express them in dB/in.
4. Use the input-impedance formula to analyze shorted, open, matched, and quarter-wave lines.
5. Explain why the characteristic impedance of a real line varies with frequency.

## 4.1 The distributed RLGC model

A pair of conductors with uniform cross-section—a trace and its plane, or the two traces of a differential pair—can be described by four per-unit-length parameters, and modeled by the circuits of Figure 4.1:

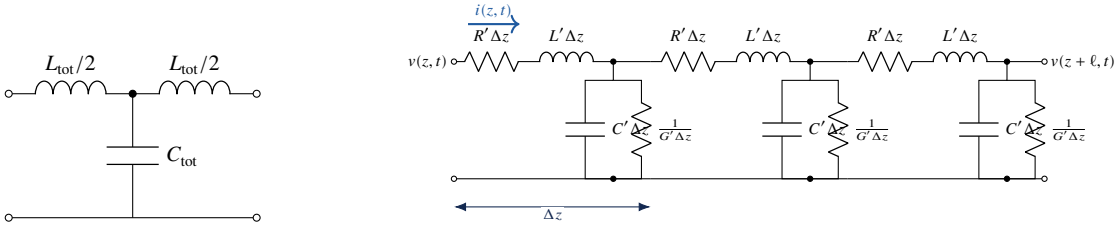
$R'$ ($\Omega$/m)
: series resistance of both conductors, rising as $\sqrt{f}$ above the skin-effect onset (Chapter 3);

$L'$ (H/m)
: series loop inductance;

$G'$ (S/m)
: shunt conductance of the dielectric, $G' = \omega C' \tan\delta$, which rises in proportion to frequency;

$C'$ (F/m)
: shunt capacitance between the conductors.

A line of length $\ell$ that is electrically short can be modeled by lumped totals $L'\ell$ and $C'\ell$ (Figure 4.1a). A long line is modeled by a cascade of infinitesimal cells of length $\Delta z$ (Figure 4.1b). In practice a ladder of $N$ lumped cells is accurate up to a frequency at which each cell is shorter than about one-tenth of a wavelength, i.e. $N \gtrsim 10\, T_{\mathrm{D}} f_{\max}$; equivalently, each cell's delay must be short compared with $t_{\mathrm{r}}$.




**(a)** Lumped T model ($T_{\text{D}} \ll t_{\text{r}}$) **(b)** Distributed RLGC ladder ($\Delta z \to 0$)

**Figure 4.1.** Circuit models of an interconnect. (a) An electrically short line is a lumped inductance and capacitance. (b) A long line is a ladder of RLGC cells; in the limit $\Delta z \to 0$ it becomes the distributed model described by the telegrapher's equations.

## 4.2 The telegrapher's equations

Applying Kirchhoff's voltage law around one cell of Figure 4.1b and Kirchhoff's current law at its shunt node gives, in the limit $\Delta z \to 0$,

$$\frac{\partial v(z,t)}{\partial z} = -R'\, i(z,t) - L' \frac{\partial i(z,t)}{\partial t}, \tag{4.1}$$

$$\frac{\partial i(z,t)}{\partial z} = -G'\, v(z,t) - C' \frac{\partial v(z,t)}{\partial t}. \tag{4.2}$$

For sinusoidal steady state with phasors $V(z)$ and $I(z)$ at angular frequency $\omega$, these become

$$\frac{\mathrm{d}V}{\mathrm{d}z} = -(R' + \mathrm{j}\omega L')\, I, \qquad \frac{\mathrm{d}I}{\mathrm{d}z} = -(G' + \mathrm{j}\omega C')\, V. \tag{4.3}$$

Differentiating the first and substituting the second gives the wave equation

$$\frac{\mathrm{d}^2 V}{\mathrm{d}z^2} = \gamma^2 V, \qquad \gamma = \alpha + \mathrm{j}\beta = \sqrt{(R' + \mathrm{j}\omega L')(G' + \mathrm{j}\omega C')}, \tag{4.4}$$

where $\gamma$ is the *propagation constant*, $\alpha$ the *attenuation constant* (Np/m), and $\beta$ the *phase constant* (rad/m). The general solution is a sum of forward and backward waves,

$$V(z) = V^{+}\mathrm{e}^{-\gamma z} + V^{-}\mathrm{e}^{+\gamma z}, \qquad I(z) = \frac{1}{Z_0}\left(V^{+}\mathrm{e}^{-\gamma z} - V^{-}\mathrm{e}^{+\gamma z}\right), \tag{4.5}$$

and substituting Eq. (4.5) back into Eq. (4.3) identifies the ratio of voltage to current in either wave as the *characteristic impedance*

$$Z_0 = \sqrt{\frac{R' + \mathrm{j}\omega L'}{G' + \mathrm{j}\omega C'}}. \tag{4.6}$$

Equation (4.6) is a property of the cross-section only; it does not depend on the length of the line or on what is connected to it. It is the impedance a driver sees looking into the line until the first reflection returns, and it is the impedance a wave "sees" as it travels. A line terminated in $Z_0$ looks infinitely long.

## 4.3 Lossless lines

When $R' \ll \omega L'$ and $G' \ll \omega C'$, which holds for board traces above a few megahertz (Section 4.5), the line is effectively lossless and

$$Z_0 = \sqrt{\frac{L'}{C'}}, \tag{4.7}$$

$$\gamma = \mathrm{j}\beta = \mathrm{j}\omega\sqrt{L'C'}, \qquad v_{\text{p}} = \frac{\omega}{\beta} = \frac{1}{\sqrt{L'C'}}. \tag{4.8}$$

For a TEM line in a homogeneous dielectric, $L'C' = \mu_0\varepsilon_0\varepsilon_r$, so $v_p = c/\sqrt{\varepsilon_r}$, recovering Eq. (2.9). Combining Eqs. (4.7) and (4.8) yields relations that let any two of $\{Z_0, t_{pd}, L', C'\}$ determine the other two:

$$C' = \frac{t_{pd}}{Z_0} = \frac{1}{v_p Z_0}, \qquad L' = Z_0\, t_{pd} = \frac{Z_0}{v_p}. \tag{4.9}$$

Multiplying by the length gives the totals for a line of one-way delay $T_D = \ell\, t_{pd}$:

$$C_{tot} = C'\ell = \frac{T_D}{Z_0}, \qquad L_{tot} = L'\ell = Z_0\, T_D. \tag{4.10}$$

These are the values a lumped model would use; Section 4.6 shows how long they remain valid.

**Three ways to the same delay.** $T_D$ is one-way and $t_{pd}$ is per unit length; the two are the most frequently confused quantities in this subject (Section 1.5). Any of three routes gives $T_D$, and using two of them is a useful check:

$$T_D = \ell\, t_{pd} = \frac{\ell\sqrt{\varepsilon_{eff}}}{c} \approx 84.73\ \text{ps/in} \times \ell_{in}\sqrt{\varepsilon_{eff}} = \ell\sqrt{L'C'}. \tag{4.11}$$

For 3 in of stripline with $D_k = 4.0$: $t_{pd} = 169.5$ ps/in gives 508 ps; $3 \times 84.73 \times 2 = 508$ ps; and with the $L'$ and $C'$ of Worked Example 4.1, $3\sqrt{8.47\times10^{-9} \times 3.39\times10^{-12}} = 508$ ps. A TDR would display this feature at $2T_D = 1.02$ ns.

**Worked Example 4.1: RLGC parameters of a stripline**

A 50 Ω stripline is built in a laminate with $D_k = 4.0$. Find $t_{pd}$, $C'$, and $L'$.

**Solution.** $t_{pd} = 84.73\ \text{ps/in} \times \sqrt{4.0} = 169.5$ ps/in. From Eq. (4.9), $C' = 169.5\ \text{ps/in}/50\ \Omega = 3.39$ pF/in and $L' = 50\ \Omega \times 169.5$ ps/in $= 8.47$ nH/in. These values—roughly 3.4 pF and 8.5 nH per inch—are worth memorizing for 50 Ω FR-4 lines.

**Rule of Thumb: Per-inch values of a 50 Ω line**

In FR-4, a 50 Ω line has about 3.4 pF/in and 8.5 nH/in; delay is about 170 ps/in in stripline. *Limits:* microstrip has the same $Z_0$ with lower $C'$ and $t_{pd}$ (because $\varepsilon_{eff} < D_k$) but similar $L'$; scale with Eq. (4.9) for other impedances.

Figure 4.2 shows how a step launched into a matched lossless line propagates: the waveform moves at $v_p$ without changing shape, occupying a length $\ell_r = t_r/t_{pd}$ along the line.

## 4.4 Low-loss lines and attenuation

Board traces at gigahertz frequencies satisfy $R' \ll \omega L'$ and $G' \ll \omega C'$ but the losses are not negligible over long routes. Expanding Eq. (4.4) to first order in $R'/(\omega L')$ and $G'/(\omega C')$ gives

$$\alpha \approx \frac{R'}{2Z_0} + \frac{G'Z_0}{2} = \alpha_c + \alpha_d, \qquad \beta \approx \omega\sqrt{L'C'}, \tag{4.12}$$

with $Z_0$ taken as the lossless value. The two terms are the *conductor loss* $\alpha_c$ and the *dielectric loss* $\alpha_d$. Substituting $G' = \omega C' \tan\delta$ and $\sqrt{L'C'} = \sqrt{\varepsilon_r}/c$ into $\alpha_d$:

$$\alpha_d = \frac{\omega\sqrt{L'C'}\,\tan\delta}{2} = \frac{\pi f\sqrt{\varepsilon_r}\,\tan\delta}{c} \quad \text{(Np/m)}. \tag{4.13}$$

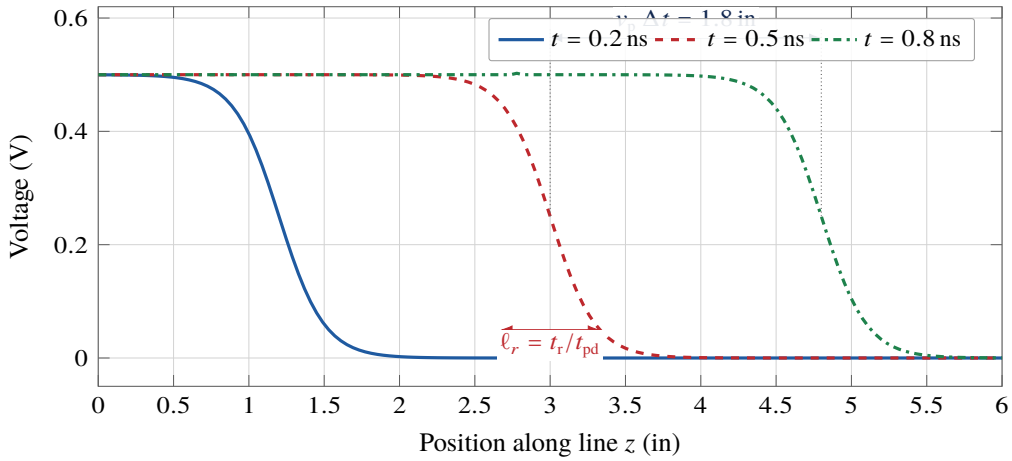



**Figure 4.2.** Snapshots of a 0.5 V step propagating along a lossless line at $v_p = 6$ in/ns. The edge moves without distortion and occupies a rise length $\ell_r = t_r/t_{pd}$.

Converting with 1 Np = 8.686 dB and 1 m = 39.37 in gives the practical form

$$\alpha_d\,[\text{dB/in}] \approx 2.31\, f_{\text{GHz}}\, D_f \sqrt{D_k}, \tag{4.14}$$

since $8.686 \times \pi \times 10^9/(2.998 \times 10^8 \times 39.37) = 2.31$. The substitution $\sqrt{L'C'} = \sqrt{\varepsilon_r}/c$ holds for a TEM line in a *homogeneous* dielectric, such as stripline; there, dielectric loss per inch is independent of geometry and cannot be reduced by changing trace width. In a microstrip, part of the field is in air, and $\sqrt{D_k}$ is replaced by an effective value that depends on the cross-section (Section 8.3). Conductor loss, by contrast, decreases with wider traces. These loss mechanisms, and the effect of copper roughness, are treated in detail in Chapter 8.

A voltage wave decays as $e^{-\alpha \ell}$, so the insertion loss of a matched line is $8.686\,\alpha\ell$ decibels. Because $\alpha_c \propto \sqrt{f}$ and $\alpha_d \propto f$, the line acts as a low-pass filter whose roll-off steepens with frequency, smearing fast edges and causing inter-symbol interference (Chapter 12).

**Worked Example 4.2: Attenuation at 5 GHz**

A 50 Ω stripline in a laminate with $D_k = 3.8$ and $D_f = 0.020$ has a loop resistance of 1.6 Ω/in at 1 GHz. Estimate $\alpha_c$, $\alpha_d$, and the insertion loss of a 10 in route at 5 GHz.

**Solution.** Resistance scales as $\sqrt{f}$, so $R' = 1.6\,\Omega/\text{in} \times \sqrt{5} = 3.58\,\Omega/\text{in}$ and $\alpha_c = R'/(2Z_0) = 0.0358\,\text{Np/in} = 0.31\,\text{dB/in}$. From Eq. (4.14), $\alpha_d = 2.31 \times 5 \times 0.02 \times \sqrt{3.8} = 0.45\,\text{dB/in}$. The total is 0.76 dB/in, so a 10 in route loses about 7.6 dB at 5 GHz. Dielectric loss already dominates.

## 4.5 Frequency dependence of the characteristic impedance

At low frequency the series resistance dominates the series impedance and the shunt capacitance dominates the shunt admittance, so Eq. (4.6) gives $|Z_0| \approx \sqrt{R'/(\omega C')}$, which rises without bound as frequency falls. In this *RC region* the line behaves like a diffusive RC network, as on-chip wires do. Above the frequency at which $\omega L' = R'$, $Z_0$ settles to the familiar $\sqrt{L'/C'}$ (Figure 4.3). For board traces this transition occurs at a few megahertz, far below the spectral content of any fast edge, which is why the lossless value is used for all signal-integrity work. At very high frequency, the drop of internal inductance and the rise of $R'$ make $Z_0$ slightly complex, by a few percent at most for typical traces.

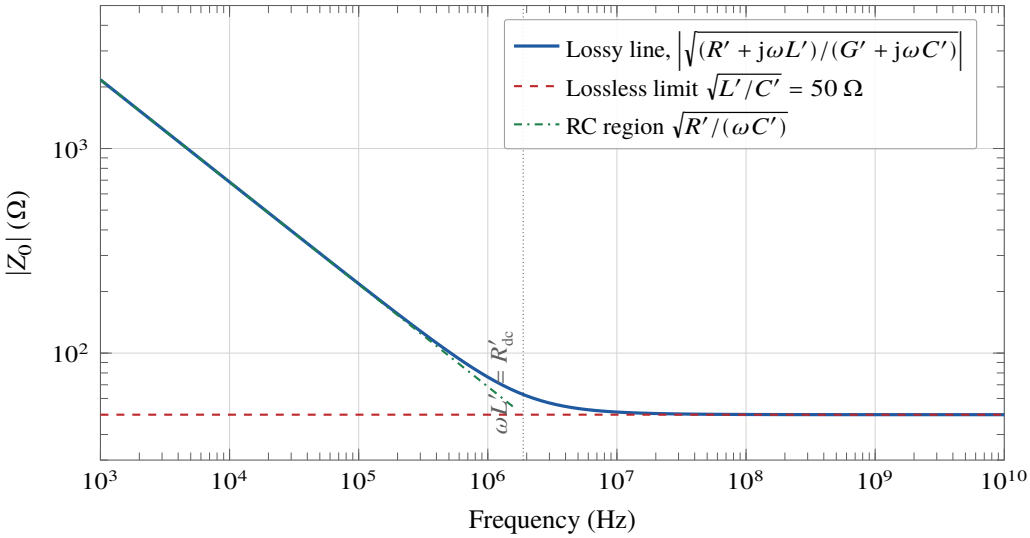



**Figure 4.3.** Magnitude of $Z_0$ for a lossy 50 Ω stripline ($R'_{dc} = 0.1\ \Omega/\text{in}$, skin-effect $R'$, $\tan\delta = 0.02$). Below about 2 MHz the line is an RC network; above, $Z_0$ is essentially $\sqrt{L'/C'}$.

## 4.6 Input impedance of a terminated line

Consider a line of length $\ell$ and characteristic impedance $Z_0$ terminated at $z = \ell$ by a load $Z_L$. Applying $V(\ell)/I(\ell) = Z_L$ to Eq. (4.5) and evaluating $V(0)/I(0)$ gives the impedance seen at the input:

$$Z_{in} = Z_0 \frac{Z_L + Z_0 \tanh(\gamma\ell)}{Z_0 + Z_L \tanh(\gamma\ell)}. \tag{4.15}$$

For a lossless line, $\tanh(j\beta\ell) = j\tan(\beta\ell)$:

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan\theta}{Z_0 + jZ_L \tan\theta}, \qquad \theta = \beta\ell = 2\pi f\, T_D. \tag{4.16}$$

Several special cases appear repeatedly in board design (Table 4.1 and Figure 4.4).

**Table 4.1.** Special cases of the input impedance of a lossless line.

| Load | Length | $Z_{in}$ | Board-level relevance |
|---|---|---|---|
| $Z_L = Z_0$ | any | $Z_0$ | Matched line; no reflection |
| Short | $\theta \ll 1$ | $j\omega(L'\ell)$ | Short shorted stub is an inductor |
| Open | $\theta \ll 1$ | $1/[j\omega(C'\ell)]$ | Short open stub is a capacitor |
| Open | $\lambda/4$ ($\theta = \pi/2$) | 0 | Via-stub notch (Chapter 11) |
| Short | $\lambda/4$ | $\infty$ | Shorted stub blocks this frequency |
| $Z_L$ | $\lambda/4$ | $Z_0^2/Z_L$ | Quarter-wave transformer |
| $Z_L$ | $\lambda/2$ | $Z_L$ | Impedance repeats every half wavelength |

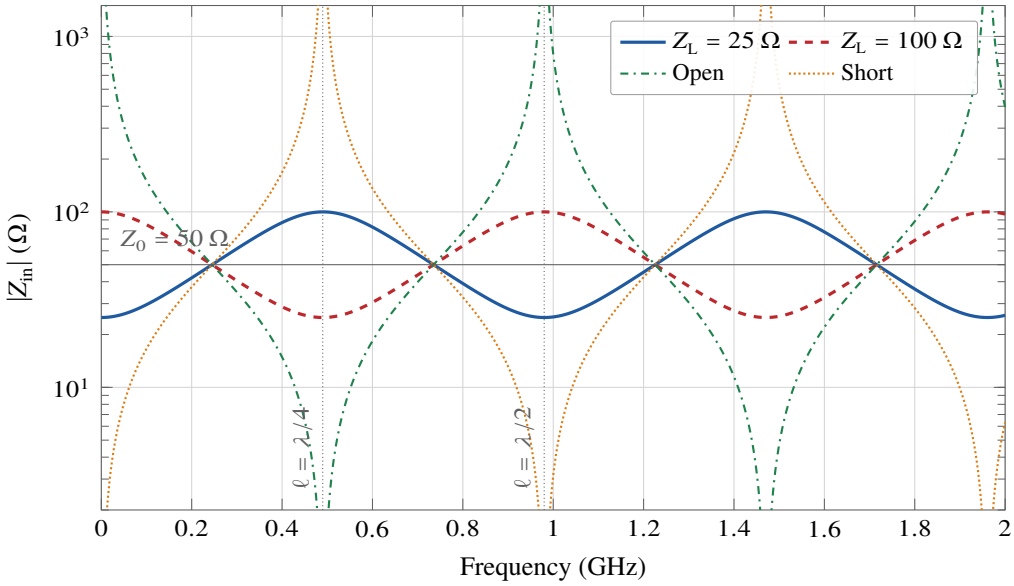



**Figure 4.4.** Input impedance magnitude of a lossless 50 Ω line with $T_D = 0.51$ ns (3 in of stripline) for several loads, from Eq. (4.16). Mismatched loads produce an input impedance that oscillates with period $1/(2T_D)$; open and shorted lines resonate at odd multiples of the quarter-wave frequency.

**Worked Example 4.3: An open stub is not just a capacitor**

A 0.5 in open-circuited stub of 50 Ω stripline ($t_{pd} = 170$ ps/in, $C' = 3.4$ pF/in) hangs off a signal net. Compare its input impedance at 2 GHz with that of its lumped capacitance.

**Solution.** $T_D = 85$ ps, so $\theta = 2\pi \times 2\,\text{GHz} \times 85\,\text{ps} = 1.068$ rad (61.2°). With $Z_L \to \infty$, Eq. (4.16) gives $Z_{in} = -jZ_0 \cot\theta = -j27.5\,\Omega$, equivalent to a capacitance of $1/(\omega \cdot 27.5\,\Omega) = 2.90$ pF. The lumped estimate, $C'\ell = T_D/Z_0 = 1.70$ pF, gives $1/(\omega C'\ell) = -j46.8\,\Omega$, too high by 70 %: at this frequency the stub is already approaching resonance. At $f = 1/(4T_D) = 2.94$ GHz it becomes a short circuit across the signal.

| Quick estimate | Full model | Difference |
|---|---|---|
| Lumped: −j46.8 Ω | Exact: −j27.5 Ω | 70 % |

### 4.6.1 When does a stub become distributed?

The example generalizes. For an open stub, $Z_{in} = -jZ_0 \cot\theta$, so the equivalent capacitance at one frequency is $C_{eff} = \tan\theta/(\omega Z_0)$; for a shorted stub, $Z_{in} = jZ_0 \tan\theta$ and $L_{eff} = Z_0 \tan\theta/\omega$. Dividing by the lumped totals of Eq. (4.10), both give the same ratio and the same relative error of the lumped model:

$$\frac{C_{eff}}{C'\ell} = \frac{L_{eff}}{L'\ell} = \frac{\tan\theta}{\theta}, \qquad e = 1 - \frac{\theta}{\tan\theta} \approx \frac{\theta^2}{3} \quad (\theta \ll 1). \tag{4.17}$$

The equivalent values are frequency dependent; they describe the stub at one frequency and are not physical components valid at all frequencies. Setting $e = 5$ % gives $\theta \approx \sqrt{0.15} = 0.39$ rad, a stub about $\lambda/16$ long at the highest frequency of interest; the common $\lambda/10$ criterion already implies a 14 % error (Figure 4.5).

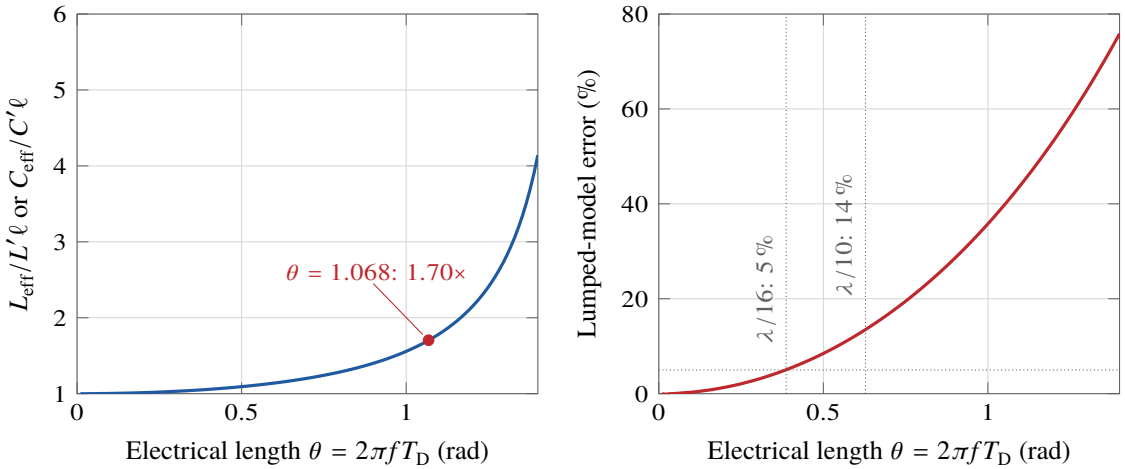



**Figure 4.5.** Ratio of the exact equivalent reactance of a lossless open or shorted stub to its lumped value (left) and the resulting lumped-model error (right), from Eq. (4.17). The marker is the stub of Worked Example 4.3. Ideal uniform line; no pad capacitance. [Analytical]

**Common Pitfall: Using lumped stub models above $\ell_{\text{crit}}$**

Representing a via stub, test-point branch, or unused connector pin by a lumped capacitor in simulation. Once the stub is longer than about one-tenth of a wavelength at the highest frequency of interest, its impedance departs from the lumped value and eventually resonates. Model stubs as transmission lines.

## 4.7 TEM, quasi-TEM, and dispersion

The telegrapher's equations assume a transverse electromagnetic (TEM) mode, in which electric and magnetic fields lie entirely in the cross-sectional plane. A stripline in a homogeneous dielectric supports a true TEM mode. A microstrip does not, because its fields straddle two media (air and laminate); its dominant mode is *quasi-TEM*, and $\varepsilon_{\text{eff}}$ rises slowly with frequency as the field concentrates in the dielectric. For typical board microstrips the effect is a few percent below 20 GHz and is usually absorbed in a frequency-dependent $\varepsilon_{\text{eff}}$ [12].

A second source of dispersion is the frequency dependence of $D_{\text{k}}$ itself. Causality requires that a lossy dielectric's permittivity decrease with frequency (the Kramers–Kronig relations), so a constant-$D_{\text{k}}$, constant-$D_{\text{f}}$ model is not physically realizable. Wideband models such as the Djordjevic–Sarkar model enforce causality [13] (Chapter 8).

**Design Rule: Model lines with frequency-dependent RLGC**

For any simulation of edges faster than about 200 ps or data rates above about 2 Gb/s, use a causal, frequency-dependent RLGC (or S-parameter) model of each line from a field solver, with the laminate's $D_{\text{k}}$ and $D_{\text{f}}$ specified at a frequency near the Nyquist frequency. Constant-parameter lossless models are acceptable only for topology exploration.

**Key Takeaways**

- A uniform line is described by $R'$, $L'$, $G'$, $C'$; the telegrapher's equations follow from Kirchhoff's laws on a ladder of cells.

- $Z_0 = \sqrt{(R' + j\omega L')/(G' + j\omega C')} \to \sqrt{L'/C'}$ above a few megahertz; $v_p = 1/\sqrt{L'C'}$.
- $C' = t_{pd}/Z_0$ and $L' = Z_0 t_{pd}$: about 3.4 pF/in and 8.5 nH/in for 50 Ω FR-4.
- Attenuation is $\alpha_c + \alpha_d = R'/(2Z_0) + G'Z_0/2$; dielectric loss is $\approx 2.31 f_{GHz} D_f \sqrt{D_k}$ dB/in, independent of geometry in stripline.
- $Z_{in}$ follows Eq. (4.16); open stubs become shorts at a quarter wavelength, and impedances repeat every half wavelength.

## Further Reading

Pozar [12] and Ramo et al. [11] derive transmission-line theory from Maxwell's equations; Paul [14] extends it to multiconductor lines; Hall and Heck [4] apply it to lossy digital interconnects.

## Review Problems

**P4.1** A line has $L' = 9.0$ nH/in and $C' = 3.6$ pF/in. Find $Z_0$, $t_{pd}$, and the equivalent $\varepsilon_r$.

**P4.2** A 45 Ω stripline uses a laminate with $D_k = 3.5$. Find $C'$ and $L'$.

**P4.3** Compute the dielectric attenuation at 8 GHz of a stripline in a low-loss laminate with $D_k = 3.6$ and $D_f = 0.004$.

**P4.4** What characteristic impedance is needed for a quarter-wave transformer that matches a 100 Ω load to a 50 Ω source?

**P4.5** A 1 in shorted stub of 50 Ω line ($t_{pd} = 170$ ps/in) is driven at 1 GHz. Find $Z_{in}$ and its equivalent inductance, and compare with $L'\ell$.

**P4.6** A line has $R'_{dc} = 0.2$ Ω/in and $L' = 8$ nH/in. Above what frequency does it behave as an LC line?

**P4.7** *(Approximation failure.)* Using the small-angle form of Eq. (4.17), find the largest stub delay for which a lumped model is within 5 % at 8 GHz. Check it with the exact expression.

**P4.8** *(Inverse design.)* A 45 Ω line must present no more than 0.5 pF total capacitance as a lumped stub. What is its maximum delay and length at 170 ps/in?

Part II

# Signal Integrity

Chapter 5

# What Is Signal Integrity?

**Chapter Overview**

This chapter opens the signal-integrity part of the book by answering a simple question precisely: what is signal integrity, and how do we know when we have it? It defines the term, shows what a signal looks like when integrity is lost, names the physical causes, lists the parameters engineers use to quantify it, and introduces the three measurements—oscilloscope, vector network analyzer and time-domain reflectometer—that reveal it. A PCI Express example ties the pieces together, and the chapter closes by separating signal integrity from power integrity. Every topic introduced here is developed in detail in the chapters that follow.

**Learning Objectives**

1. Define signal integrity and state what "good enough" means for a receiver.
2. Recognize reflection, ringing, overshoot, crosstalk, attenuation, ISI and jitter on a waveform.
3. Trace each impairment to its physical cause on the board.
4. Read the main SI parameters—impedance, rise time, eye opening, jitter, insertion and return loss, crosstalk, skew—and know how each is measured.
5. Distinguish signal integrity from power integrity and explain how they interact.

## 5.1 Definition

*Signal integrity* (SI) is the ability of an electrical signal to travel through an electronic system or PCB interconnect while keeping its intended shape, timing, amplitude and quality. In simple terms, signal integrity means that the signal received by a component is sufficiently similar to the signal transmitted by the source.

"Sufficiently similar" has a precise meaning. A digital receiver does not need a perfect copy of the transmitted waveform; it needs to decide, at each sampling instant, whether the symbol is a 0 or a 1 (or, for PAM4, which of four levels was sent), and to do so with a bit-error ratio below its requirement. A signal therefore has integrity when, at the receiver, it is on the correct side of the decision threshold by at least the receiver's voltage sensitivity, at a time that lies within the receiver's timing window, with margin left for noise and variation:

$$\text{SI margin} = \text{what the receiver gets} - \text{what the receiver needs}, \tag{5.1}$$

evaluated both in voltage and in time. Everything in the following chapters—impedance control, termination, loss budgets, crosstalk limits, via design, equalization—exists to keep this margin positive.

Figure 5.1 shows the idea on a real channel model. The transmitted waveform is a clean sequence of fast edges. After 10.75 in of board, two vias and a connector, the received waveform is slower, smaller, and

pattern dependent: an isolated bit no longer reaches the full level, because energy from earlier bits is still arriving. The receiver can still decide correctly here, but the margin is visibly smaller than at the transmitter.

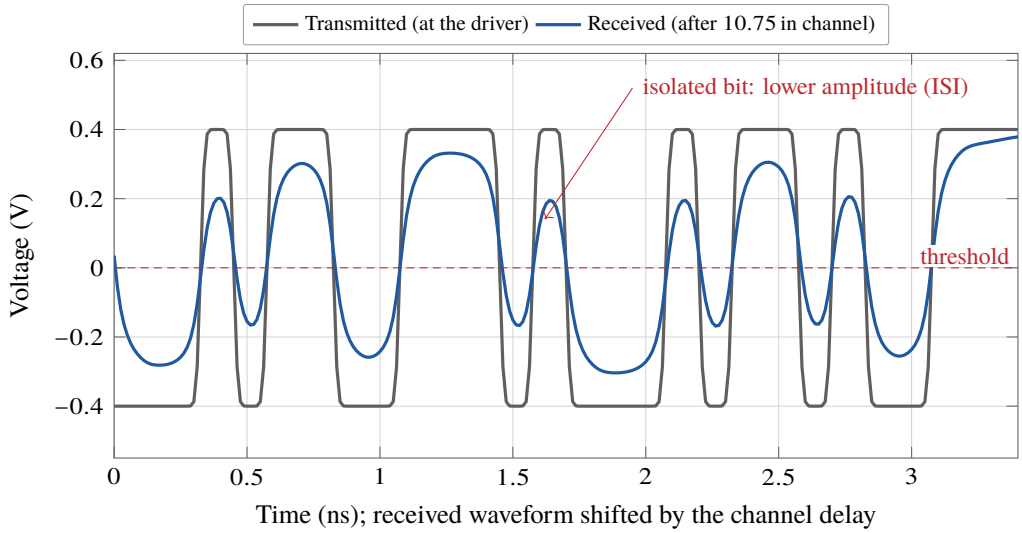



**Figure 5.1.** An 8 Gb/s NRZ pattern at the transmitter and after a 10.75 in channel with a via region, a connector and an 8 in trace (the reference channel of this chapter: loss 0.10 dB/in/$\sqrt{\text{GHz}}$ plus 0.12 dB/(in · GHz), 170 ps/in). [Simulation (frequency-domain channel model)]

Signal integrity is particularly important in high-speed digital designs such as PCI Express, DDR3, DDR4, DDR5 and LPDDR memory, USB, Ethernet, MIPI camera and display links, and SerDes (serializer and deserializer) interfaces in general, all of which are covered in Part V. As Chapter 2 shows, what makes a design "high-speed" is not the clock frequency but the edge rate relative to the size of the interconnect.

## 5.2 What happens when signal integrity is poor

When a signal travels through a PCB trace, several electrical effects can distort it. Figure 5.2 shows each one in isolation; on a real board they occur together.

**Reflection**
Part of the signal travels back toward the source because of an impedance mismatch (Chapter 6). The reflected fraction is the reflection coefficient $\Gamma = (Z_2 - Z_1)/(Z_2 + Z_1)$.

**Ringing**
Oscillation after a transition, caused by a reflection bouncing back and forth between two mismatched ends, or by a resonance of parasitic inductance and capacitance.

**Overshoot and undershoot**
The voltage exceeds the final high level, or falls below the low level, during the ringing. Excessive overshoot stresses input protection and gate oxides; undershoot below ground can forward-bias input diodes.

**Crosstalk**
An unwanted signal coupled from a neighboring trace through mutual capacitance and inductance (Chapter 9). It appears at the near end (NEXT) and the far end (FEXT) of the victim.

**Attenuation**
The signal amplitude decreases along the line because of conductor and dielectric losses, which grow with frequency (Chapter 8). Fast edges lose their high-frequency content first, so edges also slow down.

**Inter-symbol interference (ISI)**

Previous bits affect the interpretation of later bits. Loss spreads each bit's energy into the following bit periods, and reflections return energy several bit periods later (Chapter 12).

**Jitter**

Uncertainty in the timing of the signal transitions. Some jitter comes from the source (PLL, supply noise); some is created by the channel, because ISI and crosstalk move the threshold crossing depending on the data pattern (Section 12.2).

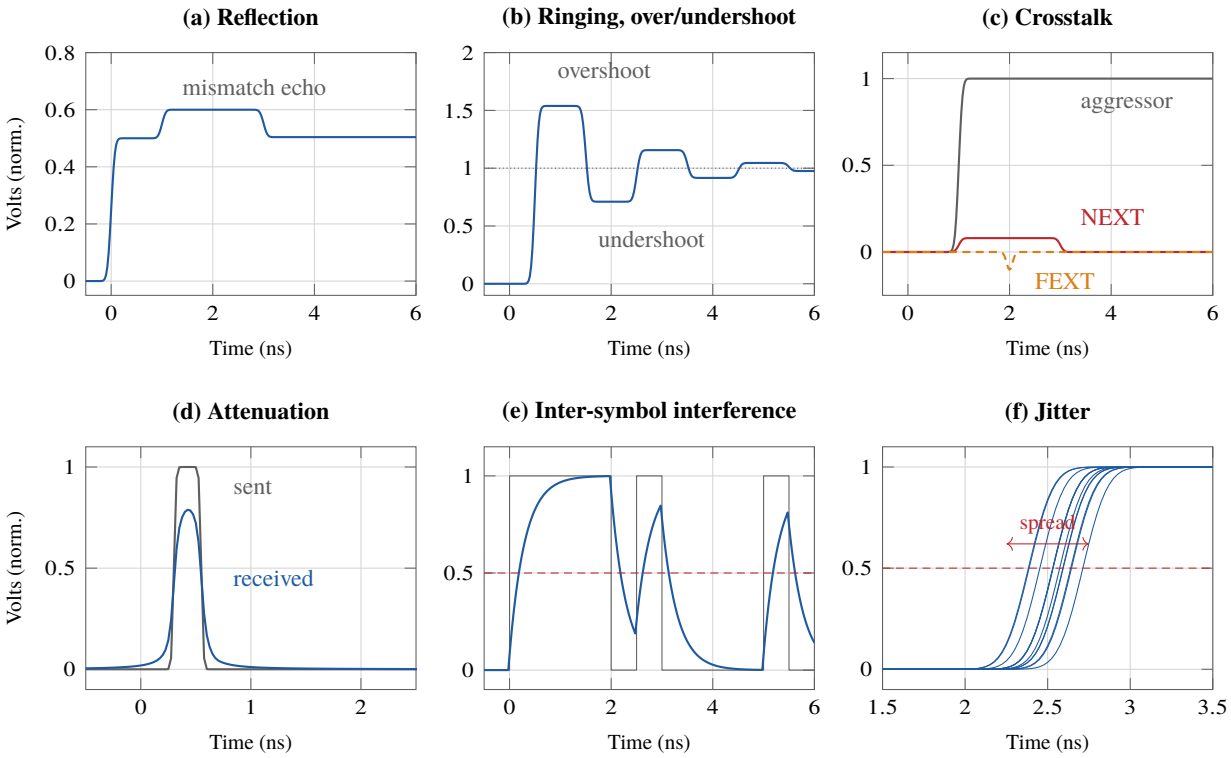



**Figure 5.2.** Signal-integrity impairments, each shown alone. (a) Step seen at the source of a line with a 75 Ω segment. (b) Under-terminated source ($R_S$ = 15 Ω) driving an open line. (c) Aggressor edge and victim NEXT and FEXT. (d) A 250 ps pulse before and after the reference channel. (e) A data pattern through a one-pole channel: the isolated 1 never reaches full level. (f) Twelve edges with random timing. [Simulation (analytical models)]

**Worked Example 5.1: How big is a reflection?**

A 50 Ω trace enters a connector region whose impedance is 60 Ω. The driver launches an 800 mV step. How large is the reflected step, and what return loss does this discontinuity have by itself?

**Solution.** $\Gamma = (60 - 50)/(60 + 50) = 0.091$, so 9.1 % of the incident step, 73 mV, travels back toward the driver, and the transmitted step is $1 + \Gamma = 1.091$ times the incident. Expressed as return loss, $-20\log_{10} 0.091 = 20.8$ dB. A single 10 Ω error is modest; several of them along a channel add echoes that arrive at different times and appear as ISI and pattern-dependent jitter at the receiver (Chapter 6).

## 5.3 What causes signal-integrity problems

The impairments above have a small number of physical causes, all of them decided during stackup design, placement and routing:

1. incorrect trace impedance, from wrong width, dielectric thickness or material (Chapter 7);
2. poor termination, or none, so that reflections are not absorbed (Chapter 6);
3. long traces, which add loss and delay (Chapter 8);
4. improper routing of differential pairs: skew, uneven spacing, asymmetry (Chapter 10);
5. inadequate ground reference planes, which break the return path (Chapter 17);
6. poor connector and via design, with stubs and impedance discontinuities (Chapter 11);
7. excessive capacitive and inductive loading: pads, test points, multiple loads (Chapter 6);
8. crosstalk between adjacent high-speed signals routed too close or too long in parallel (Chapter 9).

Figure 5.3 places these causes along a typical channel. The channel is everything between the two silicon dies: packages, vias, traces, connectors and coupling capacitors, together with the reference planes that carry the return current. Each element is a source of one or more impairments, and the receiver sees their sum.

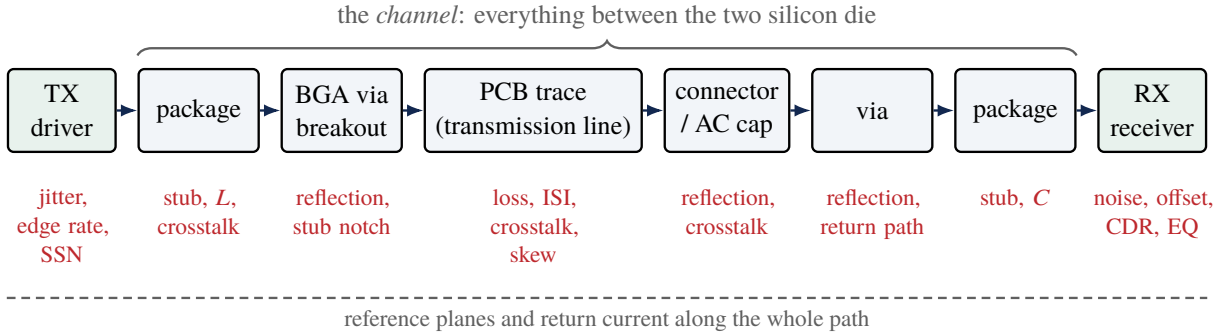



**Figure 5.3.** A high-speed channel from transmitter to receiver, with the impairments each element mainly contributes. [Conceptual]

**Common Pitfall: Treating the schematic as the channel**

Reviewing a high-speed interface only at the schematic level. The schematic shows that the pins are connected; it does not show the via stub, the plane split under the trace, or the 3 in of parallel routing next to a clock. Signal integrity is decided in the stackup and the layout, and must be reviewed there (Chapter 1).

## 5.4 Important signal-integrity parameters

Engineers quantify signal integrity with a handful of parameters (Table 5.1). Some describe the interconnect alone (impedance, insertion and return loss, crosstalk, skew); others describe the signal that results (rise time, eye opening, jitter).

The eye diagram deserves a closer look, because it combines all the others. An oscilloscope overlays many bit periods of a long random pattern on one display; the result looks like an eye (Figure 5.4). Its vertical opening, the *eye height*, is the voltage margin left after loss, ISI, reflections, crosstalk and noise; its horizontal opening, the *eye width*, is the timing margin left after jitter. A receiver specification usually states a minimum eye height and width, or a mask the eye must not touch, at a stated bit-error ratio.

**Table 5.1.** Signal-integrity parameters.

| Parameter | What it tells you | Measured with | Chapter |
|---|---|---|---|
| Impedance | Electrical characteristic of the transmission line; mismatches cause reflections | TDR, field solver | 4, 7 |
| Rise/fall time | How quickly the signal transitions; sets the bandwidth that matters | Oscilloscope | 2 |
| Eye opening | Signal quality: remaining noise and timing margin at the receiver | Oscilloscope, simulation | 12 |
| Jitter | Timing variation of the transitions | Oscilloscope, BERT | 12 |
| Insertion loss | Signal attenuation versus frequency, $\lvert S_{21}\rvert$ | VNA | 8 |
| Return loss | Amount of reflected signal versus frequency, $\lvert S_{11}\rvert$ | VNA | 6 |
| Crosstalk | Coupling between neighboring signals (NEXT, FEXT) | VNA, TDT, scope | 9 |
| Skew | Timing difference between signals (intra-pair, lane-to-lane, data-to-strobe) | TDT, scope, VNA phase | 10, 12 |

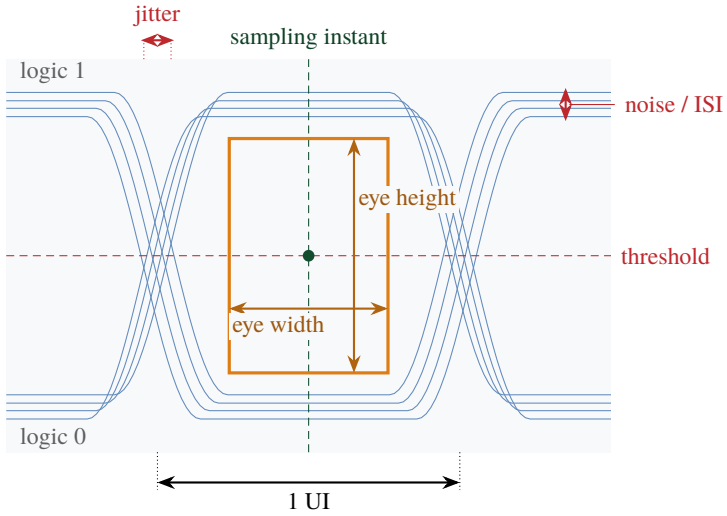



**Figure 5.4.** Anatomy of an eye diagram. Noise and ISI thicken the rails and reduce the eye height; jitter spreads the crossings and reduces the eye width. The receiver samples at the center of the opening. [Conceptual]

**Worked Example 5.2: From loss to eye height**

The reference channel of Figure 5.1 carries 8 Gb/s NRZ with an 800 mV peak-to-peak launch. Its insertion loss at the 4 GHz Nyquist frequency is 7.6 dB. Estimate the amplitude of the Nyquist-frequency component at the receiver, and compare with the simulated eye height.

**Solution.** $10^{-7.6/20} = 0.42$, so a 4 GHz square-wave pattern (alternating bits) arrives with about $0.42 \times 800\,\text{mV} = 330\,\text{mV}$ peak to peak. The simulated eye for a PRBS7 pattern (Figure 5.5c) opens to about 300 mV: slightly less, because long runs and isolated bits suffer different amounts of ISI, and because the connector and via reflections add small echoes. The Nyquist loss is a good first indicator of eye closure, but the eye itself is the quantity the receiver cares about.

| Quick estimate | Full model | Difference |
|---|---|---|
| $0.42 \times 800\,\text{mV} = 330\,\text{mV}$ | Simulated eye: 300 mV | 10 % |

**Rule of Thumb: Loss at Nyquist and eye closure**

For an unequalized NRZ link, an insertion loss of about 10 dB at the Nyquist frequency already reduces the eye height to roughly a quarter of the launched swing or less; links with more loss than that depend on equalization (Section 12.4). *Limits:* a rough guide only; the shape of the loss curve, reflections and crosstalk matter as much as the single Nyquist number.

## 5.5 How to improve signal integrity

Signal integrity is designed in, mostly before routing starts. The main techniques are:

**Impedance matching**
: Maintain controlled impedance—for example 50 Ω single-ended, or 85, 90 and 100 Ω differential depending on the interface—along the whole path, including breakouts and vias (Chapter 7).

**Termination**
: Use appropriate series, parallel or on-die termination so that reflections are absorbed at one end (Chapter 6).

**Length matching**
: Control trace lengths within a group and the skew within each differential pair (Chapters 10 and 12).

**Ground planes**
: Give every high-speed signal a continuous reference plane, and stitch the return path wherever the signal changes layer (Chapter 17).

**Via optimization**
: Minimize unnecessary vias, back-drill or avoid stubs, and tune anti-pads (Chapter 11).

**Trace spacing**
: Increase spacing between aggressors and victims, and limit parallel run length, to reduce crosstalk (Chapter 9).

**Stackup design**
: Select dielectric thickness, material and copper geometry to meet impedance and loss targets (Chapters 7 and 8).

**Design Rule: Budget before you route**

For every high-speed interface, write down the loss, return-loss, crosstalk and skew budgets before layout starts, allocate them among package, board, connector and cable, and turn the board's share into routing constraints. A budget that exists only after routing is a measurement, not a design.

## 5.6 How to measure signal integrity

Three instruments, each answering a different question, are used to verify signal integrity (Figure 5.5); Chapter 26 treats them in depth.

1. *Oscilloscope.* Measures the signal waveform: overshoot, undershoot, rise time, jitter and eye diagrams. It answers *does the link work, and with how much margin?*
2. *Vector network analyzer (VNA).* Measures S-parameters of the passive interconnect, including insertion loss ($S_{21}$) and return loss ($S_{11}$) versus frequency. It answers *how much does the channel attenuate and reflect?*
3. *Time-domain reflectometer (TDR).* Launches a fast step and records the reflections, giving impedance versus position along traces, vias and connectors. It answers *where is the discontinuity?*

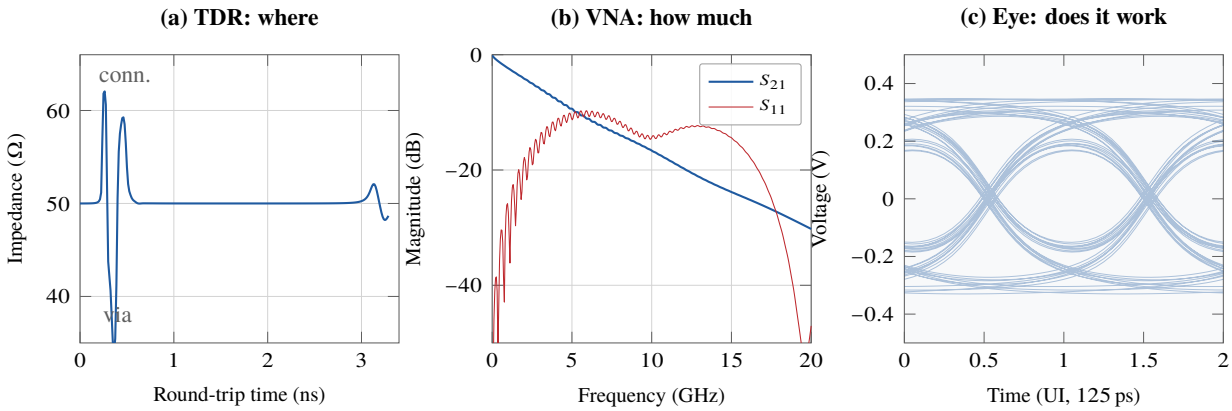



**Figure 5.5.** Three views of the same reference channel: (a) TDR impedance profile with a 30 ps edge, showing the low-impedance via region and the high-impedance connector; (b) insertion and return loss; (c) 8 Gb/s PRBS7 eye at the receiver, no equalization. [Simulation (frequency-domain channel model)]

## 5.7 A practical example: PCI Express

Suppose you are designing a PCI Express interface between an SoC and an external device, such as an M.2 solid-state drive. The signal-integrity work consists of the following checks, each developed later in the book:

- differential impedance, typically 85 Ω for PCIe (Chapter 20);
- insertion loss across the whole channel, against the generation's budget;
- return loss caused by impedance discontinuities at breakouts, vias, AC capacitors and the connector;
- differential-pair length matching and intra-pair skew;
- eye opening at the receiver, after the receiver's equalization;
- jitter, including the reference-clock contribution, and equalization settings;
- connector and via effects, especially via stubs (Chapter 11).

The ultimate goal is to ensure that the receiver can correctly distinguish logic 0 from logic 1 at the required data rate.

**Worked Example 5.3: A first loss check for a PCIe Gen4 lane**

A PCIe Gen4 lane (16 GT/s, Nyquist 8 GHz) runs 7 in on a low-loss laminate that measures about 0.75 dB/in at 8 GHz, through two via transitions (about 0.5 dB each) and an M.2 connector (about 1.5 dB). The channel budget for Gen4 is approximately 28 dB at 8 GHz (Table 20.1), shared with the two packages. If the packages are allocated 6 dB each (illustrative), is the board within its share?

**Solution.** Board share = $28 - 2 \times 6 = 16$ dB. Board loss = $7 \times 0.75 + 2 \times 0.5 + 1.5 = 7.75$ dB, leaving about 8 dB of margin for manufacturing variation, temperature, the AC capacitors and the device side. Loss is not the only check: the same channel must also meet return loss, crosstalk and skew limits, and the final judgement is the simulated eye with the reference receiver (Chapter 25).

## 5.8 Signal integrity and power integrity

A concise distinction, often asked in hardware-engineering interviews: *signal integrity focuses on the quality of signals traveling through interconnects, while power integrity focuses on maintaining stable power and ground voltages across the PCB.* The two are not independent. Every signal's return current flows in planes that also distribute power; simultaneous switching draws current spikes from the power network (Chapter 15); and supply noise moves driver delays and PLL phase, so it appears as jitter on signals. Power integrity is introduced in Chapter 13.

**Table 5.2.** Summary: the signal-integrity view of a design.

| Question | Answered by | Chapter |
|---|---|---|
| Is the interconnect electrically long? | $T_D$ versus $t_r$ | 2 |
| Will it reflect? | Impedance profile, termination | 6, 7 |
| How much is lost? | Insertion loss at Nyquist and its slope | 8 |
| What couples in? | NEXT, FEXT, spacing, parallel length | 9 |
| Is the pair balanced? | Skew, mode conversion | 10 |
| Do vias and connectors behave? | Stubs, anti-pads, return vias | 11 |
| Does the receiver have margin? | Eye height and width, jitter, BER | 12 |

**Key Takeaways**

- Signal integrity is the ability of a signal to arrive with enough voltage and timing margin for the receiver to decide correctly at the required error ratio.
- Poor SI appears as reflection, ringing, overshoot and undershoot, crosstalk, attenuation, ISI and jitter; on a board they act together.
- Its causes are physical: impedance, termination, length, pair routing, reference planes, vias and connectors, loading and spacing.
- Impedance, rise time, eye opening, jitter, insertion and return loss, crosstalk and skew quantify it; the eye diagram combines them.
- The oscilloscope shows whether the link works, the VNA how much the channel attenuates and reflects, the TDR where the discontinuities are.
- SI is about the signal path; PI is about the supply path; they share the same copper and interact.

## Further Reading

Bogatin [3] is an accessible introduction to the whole subject; Hall and Heck [4] cover the advanced material on channels and equalization.

## Review Problems

**P5.1** State in one sentence what "good signal integrity" means for a digital receiver.

**P5.2** A 50 Ω line meets a 40 Ω via region. Find the reflection coefficient, the reflected voltage for a 1 V incident step, and the return loss of this discontinuity.

**P5.3** Match each observation to the most likely impairment: (a) an isolated 1 after a long run of 0s barely crosses the threshold; (b) a quiet line shows a pulse when its neighbor switches; (c) the high level briefly reaches 1.4 V on a 1.0 V signal.

**P5.4** A channel has 12 dB of insertion loss at Nyquist. What fraction of the Nyquist-frequency amplitude arrives? Is this link likely to need equalization?

**P5.5** Which instrument would you use to find where along a 6 in trace an impedance discontinuity lies, and which to find how much it reflects at 8 GHz?

**P5.6** *(Inverse design.)* A Gen4 board share of 16 dB must cover two vias (0.5 dB each), a connector (1.5 dB) and 3 dB reserved for variation. What is the longest trace on a laminate with 0.9 dB/in at 8 GHz?

**P5.7** Give one way in which a power-integrity problem appears as a signal-integrity symptom.

Chapter 6

# Reflections and Termination

**Chapter Overview**

Whenever a wave meets a change in impedance, part of it reflects. On a board, the ends of every net—the driver's output impedance and the receiver's input impedance—are almost never equal to the line's characteristic impedance, so reflections are the default behavior of an unterminated line. This chapter derives the reflection coefficient, shows how to track multiple reflections with a lattice diagram, explains ringing and staircase waveforms, analyzes reactive loads, and compares the termination strategies available to the designer.

**Learning Objectives**

1. Derive reflection and transmission coefficients at an impedance discontinuity.
2. Construct a lattice (bounce) diagram and compute source and load voltages at any time.
3. Predict ringing for under-terminated lines and staircase steps for over-terminated lines.
4. Estimate the time constant and rise-time degradation caused by a capacitive load.
5. Choose and size series, parallel, Thevenin, AC, diode, and on-die terminations for a given topology.

## 6.1 The reflection coefficient

Let a wave of voltage $V_i$ traveling on a line of impedance $Z_0$ arrive at a load $Z_\mathrm{L}$ (which may be a resistor, the input of another line, or the input of a receiver). At the junction, the voltage and current must be continuous. If a reflected wave $V_r$ travels back and a transmitted wave $V_t$ is delivered to the load,

$$V_i + V_r = V_t, \qquad \frac{V_i}{Z_0} - \frac{V_r}{Z_0} = \frac{V_t}{Z_\mathrm{L}}. \tag{6.1}$$

Solving gives the reflection and transmission coefficients

$$\Gamma = \frac{V_r}{V_i} = \frac{Z_\mathrm{L} - Z_0}{Z_\mathrm{L} + Z_0}, \qquad T = \frac{V_t}{V_i} = 1 + \Gamma = \frac{2Z_\mathrm{L}}{Z_\mathrm{L} + Z_0}. \tag{6.2}$$

$\Gamma$ ranges from $-1$ (short circuit) through 0 (matched) to $+1$ (open circuit). A CMOS receiver input is, to first order, an open circuit with a few picofarads of capacitance, so a wave arriving at an unterminated receiver doubles. $T$ can exceed one; this is not a violation of energy conservation, since the current, not the voltage, is reduced.

**Rule of Thumb: Small mismatches**

For a small impedance change $\Delta Z$, $\Gamma \approx \Delta Z/(2Z_0)$: a 10 % impedance error reflects about 5 % of the incident voltage. *Limits:* valid for $|\Delta Z| \lesssim 0.2Z_0$; beyond that use Eq. (6.2).

### 6.1.1 Launch voltage

When a driver with Thevenin voltage $V_S$ and output resistance $R_S$ first switches, it sees only the line's characteristic impedance, because no reflection has yet returned. The voltage launched into the line is therefore set by a resistive divider:

$$V_{\text{launch}} = V_S \, \frac{Z_0}{Z_0 + R_S}. \tag{6.3}$$

When the reflection from the load returns to the source, it meets the source reflection coefficient $\Gamma_S = (R_S - Z_0)/(R_S + Z_0)$ and reflects again.

## 6.2 Lattice diagrams

A *lattice diagram* tracks each traveling wave on a distance–time plot. Each wave is drawn as a diagonal line from one end to the other, labeled with its amplitude; at each end the amplitude is multiplied by that end's reflection coefficient. The voltage at either end at any time is the sum of all waves that have arrived there, counting both incident and reflected components.

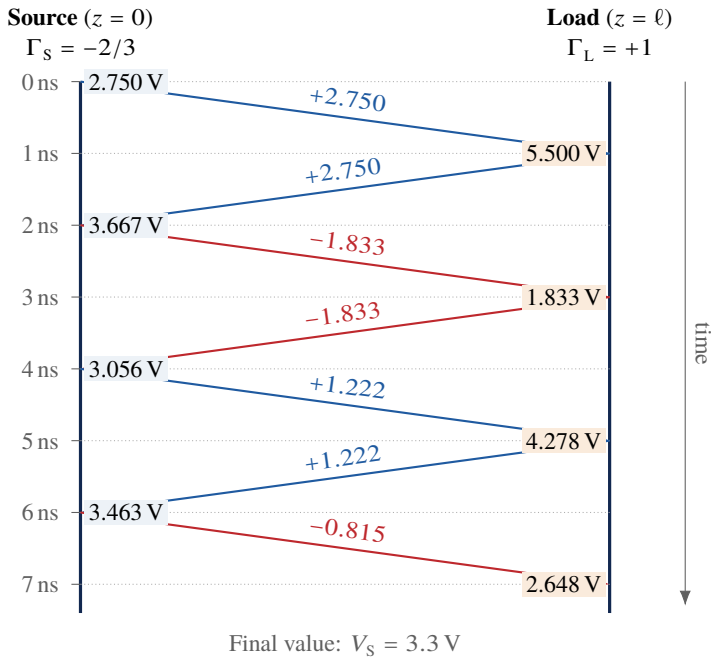



**Figure 6.1.** Lattice diagram for a 3.3 V driver with $R_S = 10\,\Omega$ driving an open-circuited $50\,\Omega$ line with $T_D = 1$ ns. Wave amplitudes are shown on the diagonals; cumulative voltages at the source (left) and load (right) are shown in boxes. The load voltage rings about its final value of 3.3 V.

**Worked Example 6.1: Bounce diagram of an under-terminated line**

A 3.3 V CMOS driver with $R_S = 10\,\Omega$ drives a $50\,\Omega$ line with $T_D = 1$ ns into a high-impedance receiver. Find the receiver voltage at 1, 3, 5 and 7 ns.

**Solution.** From Eq. (6.3), $V_{\text{launch}} = 3.3 \times 50/60 = 2.750$ V. $\Gamma_L = +1$ and $\Gamma_S = (10 - 50)/(10 + 50) = -2/3$. The waves arriving at the load are +2.750, −1.833, +1.222, and −0.815 V; each doubles on reflection. The load voltages are therefore 5.500, $5.500 - 3.667 = 1.833$, $1.833 + 2.444 = 4.278$, and $4.278 - 1.630 = 2.648$ V (Figure 6.1). The first overshoot is 2.2 V above the supply—well beyond

the absolute-maximum rating of most receivers—and the first undershoot dips to 1.83 V, close to the switching threshold. The ringing period is $4T_D = 4\,\text{ns}$.

### 6.2.1 A lattice diagram, step by step

The example compresses a procedure that is worth making explicit, because the same bookkeeping underlies every reflection simulator.

**Set up.** Draw the source at the left and the load at the right; time runs downward. Mark $\Gamma_S$ and $\Gamma_L$ at the two ends and the launch voltage of Eq. (6.3) at $t = 0$. Each diagonal takes one $T_D$ to cross.

**Propagate.** A wave of amplitude $V^+$ arriving at an end with reflection coefficient $\Gamma$ creates a reflected wave $\Gamma V^+$ that starts back immediately. The node voltage jumps by the sum of the arriving and reflected waves:

$$V_{\text{node}}^{\text{new}} = V_{\text{node}}^{\text{prev}} + (1 + \Gamma)\, V_{\text{arr}}^{+}. \tag{6.4}$$

The voltage between arrivals is constant (for an ideal step on a lossless line).

**Tabulate.** Table 6.1 lists the first arrivals for the 3.3 V, 10 Ω, 50 Ω, open-load case. Odd multiples of $T_D$ are load events and even multiples are source events.

**Table 6.1.** Arrival table for Worked Example 6.1 ($\Gamma_S = -2/3$, $\Gamma_L = +1$).

| $t/T_D$ | Node | Arriving wave | $V_{\text{arr}}^{+}$ (V) | $(1+\Gamma)$ | $V_{\text{node}}$ (V) |
|---|---|---|---|---|---|
| 0 | Source | launch | 2.750 | – | 2.750 |
| 1 | Load | 1st incident | 2.750 | 2 | 5.500 |
| 2 | Source | reflection of 1 | 2.750 | 0.333 | 3.667 |
| 3 | Load | 2nd incident | −1.833 | 2 | 1.833 |
| 4 | Source | reflection of 3 | −1.833 | 0.333 | 3.056 |
| 5 | Load | 3rd incident | 1.222 | 2 | 4.278 |
| 6 | Source | reflection of 5 | 1.222 | 0.333 | 3.463 |
| 7 | Load | 4th incident | −0.815 | 2 | 2.648 |

**Close the form.** Each round trip multiplies the wave by $\Gamma_S\Gamma_L = -2/3$, so the $(n+1)$th wave to reach the load is $2.75(-2/3)^n$, and summing Eq. (6.4) as a geometric series gives

$$V_L((2n+1)T_D) = 2V_{\text{launch}} \sum_{k=0}^{n} \left(-\tfrac{2}{3}\right)^k = 3.3\left[1 - \left(-\tfrac{2}{3}\right)^{n+1}\right] \text{V}. \tag{6.5}$$

**Check.** Three checks catch almost every bookkeeping error: (i) for $n \to \infty$ the load voltage tends to 3.3 V, the DC value with no current flowing; (ii) $n = 0$ gives 5.5 V, twice the launch, because an open end forces zero current, so the reflected wave must cancel the incident current and therefore adds an equal voltage; (iii) with $R_S = Z_0$, $\Gamma_S = 0$ and the series stops after its first term. Figure 6.2 plots both node voltages.

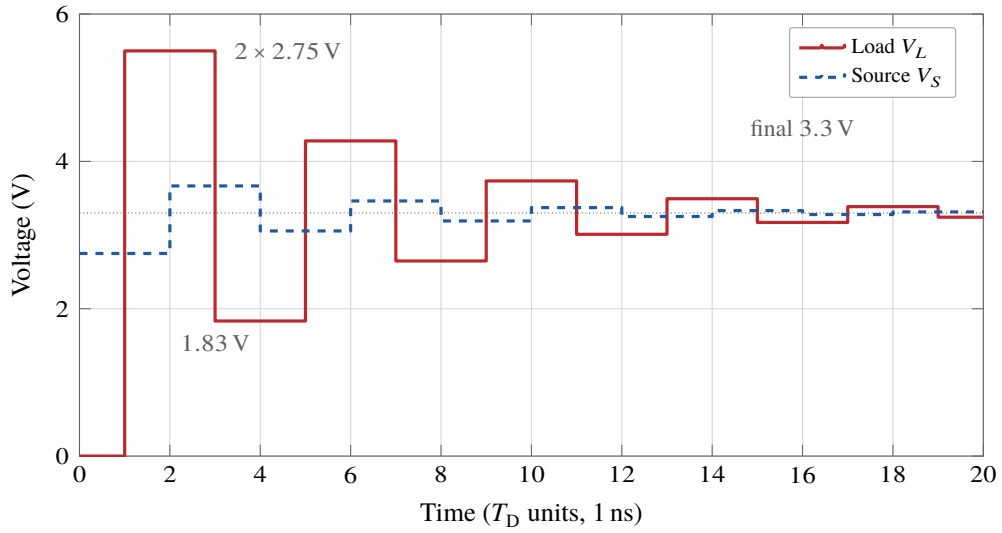



**Figure 6.2.** Load and source voltages of Worked Example 6.1 built from Table 6.1. The load rings with period $4T_D$; the source changes only at even multiples of $T_D$. [Analytical]

### 6.2.2 Under-, matched, and over-termination

The three regimes of source termination are compared in Figure 6.3 for an open-circuited load:

- **Under-terminated** ($R_S < Z_0$, $\Gamma_S < 0$): the load voltage overshoots and rings with period $4T_D$, decaying geometrically by $|\Gamma_S|$ per round trip.
- **Matched** ($R_S = Z_0$, $\Gamma_S = 0$): half the swing is launched, it doubles at the open end, and the single reflection is absorbed at the source. The load sees a clean step after one delay.
- **Over-terminated** ($R_S > Z_0$, $\Gamma_S > 0$): the load voltage approaches its final value in a staircase of steps spaced $2T_D$ apart, delaying the threshold crossing.

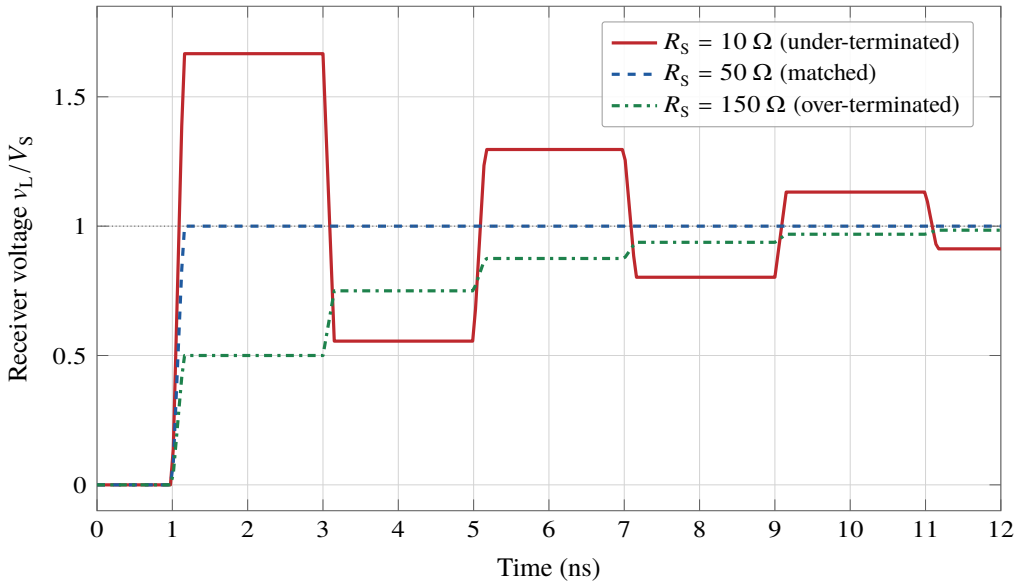



**Figure 6.3.** Receiver voltage for a 1 V source driving an open 50 Ω line ($T_D = 1$ ns, $t_r = 150$ ps) through three source resistances, computed by superposing the waves of the lattice diagram.

**Common Pitfall: Non-monotonic edges on clocks**

An undershoot or a staircase step that lingers near the receiver threshold produces a non-monotonic edge. On a data line it costs timing margin; on a clock or strobe it can double-clock a flip-flop, causing intermittent failures that are notoriously difficult to reproduce. Clocks and strobes must be terminated so that every edge is monotonic through the threshold region.

## 6.3 Reactive loads and discontinuities

Receivers, connectors, vias, and test pads present capacitance; bond wires, connector pins, and vias in thick boards present series inductance. Neither is a simple resistive $\Gamma$, because their impedance changes during the edge.

### 6.3.1 Capacitive load at the end of a line

Consider a matched-source line terminated by a capacitor $C$. The incident wave $V_i$ initially sees a short circuit ($\Gamma = -1$), and the capacitor then charges through the line's impedance. The Thevenin equivalent of the line at the load is a source $2V_i$ in series with $Z_0$, so the load voltage is

$$v_L(t') = 2V_i\left(1 - \mathrm{e}^{-t'/\tau}\right), \qquad \tau = Z_0 C, \tag{6.6}$$

where $t'$ is measured from the wave's arrival. The reflected wave is $v_r = v_L - V_i$, which starts at $-V_i$ and ends at $+V_i$; seen from the source, a capacitive load produces a dip before the voltage settles (Figure 6.5). An exponential rises from 10 % to 90 % in $\ln 9\,\tau \approx 2.2\tau$, so the capacitor adds about $2.2\tau$ to an ideal step; for a finite incident edge the two combine approximately by Eq. (2.5), and exactly by convolution.

### 6.3.2 Capacitance in the middle of a line

A capacitor $C$ at a point along a uniform line sees $Z_0$ in both directions, i.e. $Z_0/2$ in total, so the time constant halves:

$$\tau_{\mathrm{mid}} = \frac{Z_0 C}{2}. \tag{6.7}$$

The factor of two is not a different capacitor model but a different boundary condition: at the end of a line the arriving wave has a Thevenin equivalent $2V_i$ behind $Z_0$, whereas mid-line the capacitor is fed by $V_i$ behind $Z_0 \parallel Z_0$ (Figure 6.4). The reflected wave is a negative pulse of area $-V_i\tau_{\mathrm{mid}}$; the transmitted wave is slowed by about $2.2\tau_{\mathrm{mid}}$.

This is the mechanism by which via pads, test points, and connector pins degrade a fast edge, and it is the basis of TDR interpretation in Chapter 26.

**Worked Example 6.2: Rise-time degradation from receiver capacitance**

A series-terminated 50 Ω line delivers a 300 ps edge to a receiver whose input capacitance, including its pad and package, is 5 pF. Estimate the rise time at the receiver.

**Solution.** $\tau = Z_0 C = 250$ ps, which alone contributes $2.2\tau = 550$ ps. Combining by root sum of squares, $\sqrt{300^2 + 550^2} \approx 626$ ps. The receiver capacitance more than doubles the rise time. Reducing $Z_0$ to 40 Ω would reduce $\tau$ to 200 ps.

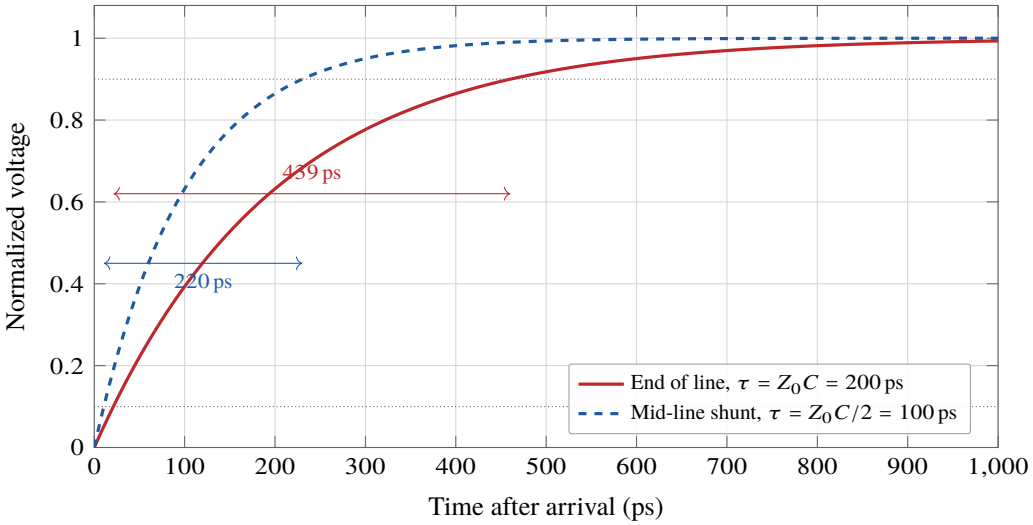



**Figure 6.4.** Normalized response of the same 4 pF capacitance at the end of a matched-source 50 Ω line and as a shunt in the middle of a matched line, for an ideal incident step. The added 10–90 % rise times are $\ln 9\,\tau$. [Analytical]

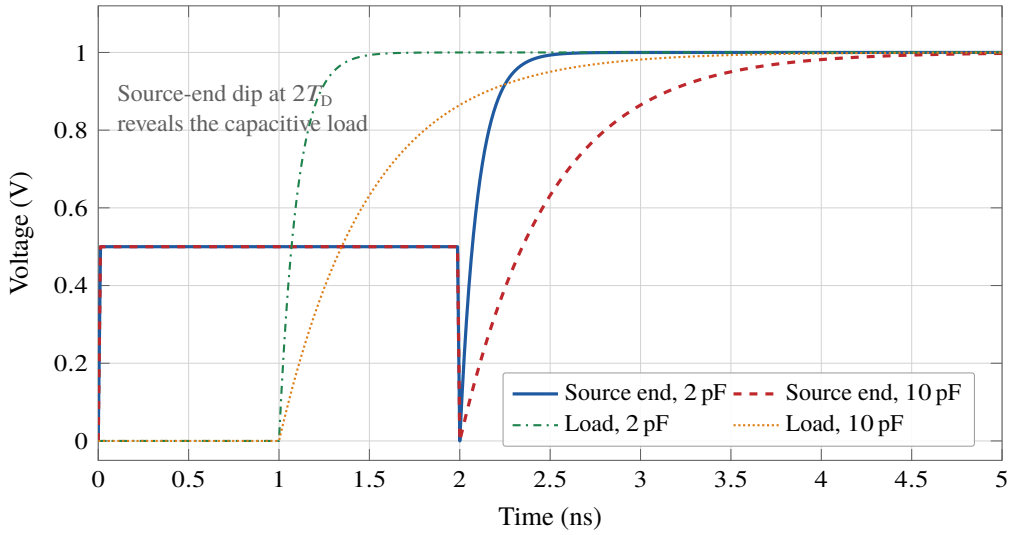


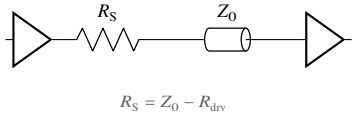

**Figure 6.5.** A 50 Ω line with matched source ($T_D = 1$ ns) driving a capacitive load, for an ideal 1 V source step. The load charges with $\tau = Z_0C$; the reflection seen at the source is a dip that recovers with the same time constant.

### 6.3.3 Series inductance

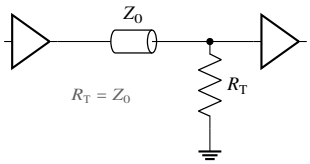
A series inductance $L$ in a line (a long via, a bond wire, a connector pin) looks initially like an open circuit and then relaxes to a short. The time constant is $\tau = L/(2Z_0)$, and the reflection is a positive pulse of area $V_i\tau$. Inductive and capacitive discontinuities of similar size partly cancel, which is the principle behind compensating a large connector pad with a reduced-width neck or an anti-pad cut-out.

## 6.4 Termination strategies

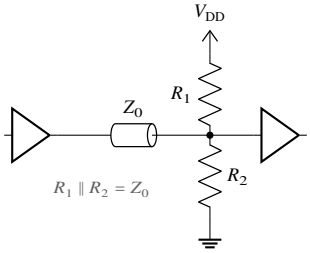
Termination places a resistance (or a resistance and capacitance) at one end of the line to make the reflection coefficient at that end approximately zero. The common schemes are drawn in Figure 6.6 and compared in Table 6.2.



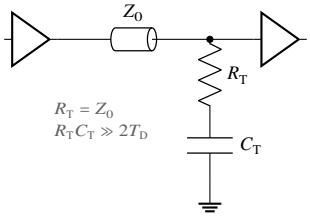

**(a)** Series (source) termination



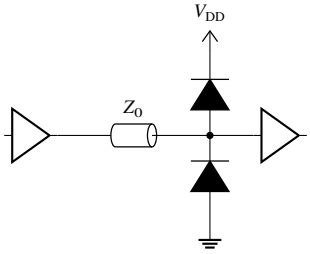

**(b)** Parallel (end) termination to ground or $V_{TT}$



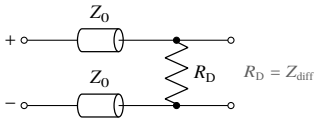

**(c)** Thevenin (split) termination




**(d)** AC (RC) termination




**(e)** Diode (clamp) termination




**(f)** Differential termination

**Figure 6.6.** Termination schemes for single-ended and differential lines.

### 6.4.1 Series (source) termination

A resistor $R_S$ in series with the driver makes the total source resistance equal to $Z_0$: $R_S = Z_0 - R_{\text{drv}}$. Half the swing is launched, the open receiver doubles it, and the reflection is absorbed at the source. Series termination dissipates no static power and needs one component, which is why it is the default for point-to-point CMOS nets. Its limitations are:

- intermediate points along the line see a half-amplitude plateau for $2T_D$, so it is unsuitable for multi-drop nets whose loads are distributed along the line;

- the driver's output resistance differs between high and low states and varies with process, voltage, and temperature, so the match is never exact;
- the full edge reaches the receiver only after $T_{\mathrm{D}}$, and the first transition is slowed by the receiver capacitance as in Worked Example 6.2.

**Worked Example 6.3: Choosing a series resistor**

An IBIS model shows a driver output resistance of 18 Ω pulling high and 22 Ω pulling low. Choose a series resistor for a 50 Ω line and find the worst-case source reflection coefficient.

**Solution.** The average driver resistance is 20 Ω, so $R_S = 30\ \Omega$ (an E24 value). The total source resistance is 48 Ω or 52 Ω, giving $\Gamma_{\mathrm{S}} = -0.020$ or $+0.020$. A $\pm 10\,\%$ variation of $Z_0$ widens this to about $\pm 0.07$, which is still well controlled.

### 6.4.2 Parallel (end) termination

A resistor $R_T = Z_0$ from the receiver end of the line to a termination voltage absorbs the incident wave. The full swing arrives in one pass, and every load along the line sees a clean edge, so parallel termination suits multi-drop and bidirectional buses. The cost is static power: with termination to ground, a logic high of $V_H$ dissipates $V_H^2/R_T$ in the resistor and requires the driver to source the full current. Terminating to a mid-supply voltage $V_{\mathrm{TT}} = V_{\mathrm{DD}}/2$, as DDR command/address buses do, halves the swing across the resistor and reduces power by a factor of four, at the cost of a dedicated $V_{\mathrm{TT}}$ regulator that must both source and sink current (Chapter 19).

### 6.4.3 Thevenin termination

Two resistors from the line to $V_{\mathrm{DD}}$ and ground provide an equivalent termination resistance $R_1 \parallel R_2 = Z_0$ and an equivalent voltage $V_{\mathrm{th}} = V_{\mathrm{DD}} R_2/(R_1 + R_2)$ without a separate supply. Their drawback is that current flows from $V_{\mathrm{DD}}$ to ground through $R_1 + R_2$ even when the line is idle.

**Worked Example 6.4: Static power of Thevenin termination**

A 1.8 V, 32-bit bus on 50 Ω lines uses Thevenin terminations $R_1 = R_2 = 100\ \Omega$. Find the static current and power.

**Solution.** $R_1 \parallel R_2 = 50\ \Omega$ and $V_{\mathrm{th}} = 0.9$ V. Each terminator draws $1.8/200 = 9$ mA from the supply, 16.2 mW per line, or 518 mW for 32 lines, before any signal switches. A 0.9 V $V_{\mathrm{TT}}$ termination with 50 Ω resistors has zero idle current when the bus is tri-stated.

### 6.4.4 AC termination

A resistor $R_T = Z_0$ in series with a capacitor $C_T$ terminates the line at high frequency while blocking DC current. The capacitor must be large enough that it does not charge appreciably during the reflection interval, typically $R_T C_T$ of at least three times the round-trip delay $2T_{\mathrm{D}}$, but small enough that its charging does not distort data patterns: a long run of identical bits charges $C_T$ and shifts the effective termination voltage. AC termination is therefore best suited to clocks and other DC-balanced signals.

**Worked Example 6.5: Sizing an AC termination**

A 50 Ω clock line has $T_{\mathrm{D}} = 1$ ns. Size an AC termination.

**Solution.** Require $R_T C_T \geq 3 \times 2T_D = 6\,\text{ns}$, so $C_T \geq 6\,\text{ns}/50\,\Omega = 120\,\text{pF}$. Choose $C_T = 150\,\text{pF}$ with $R_T = 49.9\,\Omega$. For a 100 MHz clock (5 ns half-period), the time constant of 7.5 ns is comparable to the half period, so the capacitor's voltage settles near mid-supply and the clock edges are unaffected.

### 6.4.5 Diode termination

Schottky diodes from the line to $V_{DD}$ and ground clamp overshoot and undershoot to about one diode drop beyond the rails. They do not match the line and so do not eliminate ringing, but they limit its amplitude and protect receivers. Most CMOS inputs contain such clamps intrinsically (as ESD protection), which is why an unterminated line often rings less severely than a lossless lattice diagram predicts.

### 6.4.6 On-die termination and impedance calibration

Memory and SERDES devices integrate termination resistors on the die, switchable in value and enabled only when the device is receiving. On-die termination (ODT) places the termination exactly at the receiver, eliminating the stub between an external resistor and the die, and allows the controller to tune the effective termination for each rank and topology. Drivers likewise calibrate their output resistance against an external precision resistor (the ZQ resistor in DDR memories) to remove process variation. ODT and calibrated drivers have made discrete terminations rare on modern memory and serial interfaces (Chapters 19 and 20).

### 6.4.7 Differential termination

A single resistor $R_D = Z_{\text{diff}} = 2Z_{\text{odd}}$ across a differential pair terminates the differential mode but leaves the common mode unterminated. A *split* termination—two resistors of $Z_{\text{diff}}/2 = Z_{\text{odd}}$ with their midpoint bypassed to ground by a capacitor—gives the common mode a defined, low-impedance path at high frequency, but it does not match it. The modal half-circuits make this clear (Chapter 10): in the odd mode the midpoint is a virtual ground, so each line sees $Z_{\text{odd}}$ (matched); in the even mode no current crosses between the lines, so each line sees $Z_{\text{odd}}$ to ground (through the capacitor) where it should see $Z_{\text{even}}$. For a coupled pair $Z_{\text{even}} > Z_{\text{odd}}$, so the even mode is under-terminated and reflects with $\Gamma_e = (Z_{\text{odd}} - Z_{\text{even}})/(Z_{\text{odd}} + Z_{\text{even}})$.

Both modes are matched by a *T network*: a resistor $R_1 = Z_{\text{odd}}$ from each line to a common node, and $R_2 = (Z_{\text{even}} - Z_{\text{odd}})/2$ from that node to ground (or to AC ground). In the odd mode the node is a virtual ground and each line sees $R_1 = Z_{\text{odd}}$; in the even mode each line sees $R_1 + 2R_2 = Z_{\text{even}}$. The equivalent $\pi$ network uses $Z_{\text{even}}$ from each line to ground and $2Z_{\text{even}}Z_{\text{odd}}/(Z_{\text{even}} - Z_{\text{odd}})$ across the pair. For a loosely coupled pair $Z_{\text{even}} \approx Z_{\text{odd}}$, $R_2 \to 0$, and the split termination is then nearly correct.

**Table 6.2.** Comparison of termination schemes.

| Scheme | Static power | Parts | Advantages | Limitations |
|---|---|---|---|---|
| Series | None | 1 | Simple; low power; slows edges slightly | Point-to-point only; depends on driver $R$ |
| Parallel | High | 1 | Full swing in one pass; multi-drop | $V^2/R$ power; strong driver needed |
| Thevenin | High | 2 | No extra supply; sets idle level | Idle current through $R_1 + R_2$ |
| $V_{TT}$ | Moderate | 1 + VRM | Low power; symmetric swing | Needs source/sink regulator |
| AC (RC) | None (DC) | 2 | Matches at HF; no DC current | Pattern-dependent baseline; clocks only |
| Diode | None | 2 | Clamps over/undershoot | Does not match; ringing remains |
| ODT | When active | 0 | At the die; tunable | Device must support it |

**Design Rule: Choosing a termination**

For point-to-point single-ended nets longer than $\ell_{\text{crit}}$, use series termination at the driver unless the driver's own output impedance already matches the line. For multi-drop or bidirectional buses, use parallel (or $V_{TT}$) termination at the far end, or on-die termination where available. Place discrete terminators within $\ell_{\text{crit}}$ of the pin they terminate, and verify the choice in simulation at the fast and slow IBIS corners.

**Key Takeaways**

- $\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$ and $T = 1 + \Gamma$; an open receiver doubles the arriving wave.
- The launched voltage is $V_S Z_0/(Z_0 + R_S)$; subsequent reflections can be tracked with a lattice diagram.
- Under-termination rings with period $4T_D$; over-termination produces a staircase with steps every $2T_D$.
- A capacitive load has time constant $Z_0 C$ at the end of a line and $Z_0 C/2$ in its middle; it slows edges by about $\ln 9\ \tau = 2.2\tau$.
- A split differential termination matches the odd mode only; a T network with $R_2 = (Z_{\text{even}} - Z_{\text{odd}})/2$ matches both.
- Series termination is the low-power default for point-to-point nets; parallel, $V_{TT}$, or on-die termination serves multi-drop buses.

## Further Reading

Johnson and Graham [1] and Hall, Hall, and McCall [15] treat reflections and termination in depth; Brooks [16] gives a practitioner's view with many examples.

## Review Problems

**P6.1** Find $\Gamma$ and $T$ for a 50 Ω line terminated in 75 Ω, and in 25 Ω.

**P6.2** A 1.2 V driver with $R_S = 25\ \Omega$ drives a 50 Ω line. Find the launched voltage and $\Gamma_S$.

**P6.3** A 1 V source with $R_S = 25\ \Omega$ drives an open 50 Ω line with $T_D = 1$ ns. Find the load voltage at 1, 3 and 5 ns.

**P6.4** A driver has an output resistance of 12 Ω. Choose an E24 series resistor for a 50 Ω line.

**P6.5** A 4 pF connector sits midway along a 50 Ω line. Find the time constant and the added 10–90 % rise time of the transmitted edge.

**P6.6** Size the capacitor of an AC termination for a 50 Ω line with $T_D = 0.8$ ns, using $RC \geq 3 \times 2T_D$.

**P6.7** A pair has $Z_{\text{odd}} = 45\ \Omega$ and $Z_{\text{even}} = 60\ \Omega$. Design a T network that matches both modes, and find the even-mode reflection coefficient of a split termination of two 45 Ω resistors.

**P6.8** *(Approximation failure.)* In the lattice example, the driver's output resistance is actually 10 Ω only near the rails and rises to 25 Ω in mid-swing. Which entries of Table 6.1 remain trustworthy, and what model would you use instead?

Chapter 7

# PCB Stackup and Impedance Control

**Chapter Overview**

The stackup is the single most consequential decision in a high-speed board design. It fixes the characteristic impedance that every trace can achieve, the return path of every signal layer, the plane capacitance available to the PDN, the amount of crosstalk between layers, and much of the cost. This chapter describes the standard transmission-line geometries, gives the closed-form impedance approximations of IPC-2141 and Hammerstad–Jensen together with their accuracy limits, presents practical 8-, 10-, and 12-layer stackups, and discusses the material effects—laminate grade, glass weave, and copper roughness—that set the ultimate performance of the board.

**Learning Objectives**

1. Identify microstrip, embedded microstrip, stripline, asymmetric stripline, edge-coupled pairs, and coplanar waveguide, and their dimensional parameters.
2. Estimate $Z_0$ and $Z_{\text{diff}}$ with closed-form formulas, and state where those formulas fail.
3. Construct a symmetric multilayer stackup that gives every signal layer an adjacent ground reference.
4. Select a laminate class from loss and cost requirements.
5. Explain glass-weave skew and copper roughness and mitigate them.
6. Perform a tolerance analysis of characteristic impedance.

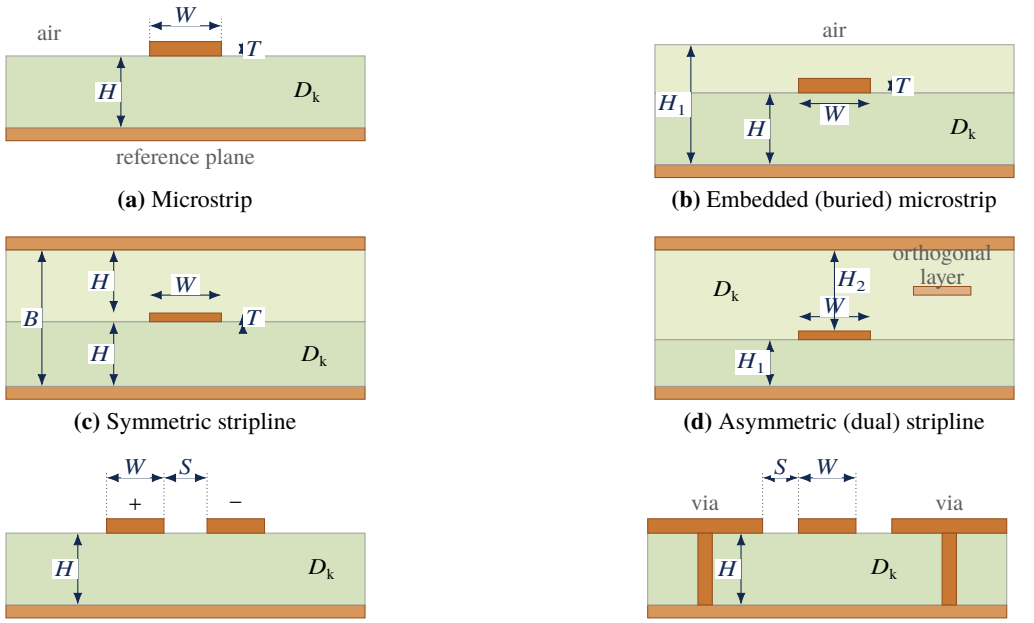
## 7.1 Transmission-line geometries

Figure 7.1 shows the geometries used on printed boards. Their dimensions are named consistently throughout this book: $W$ is trace width, $T$ trace thickness, $H$ dielectric height from trace to plane, $S$ edge-to-edge spacing, and $B$ the plane-to-plane separation of a stripline.

**Microstrip**
: A trace on an outer layer above one reference plane. Its field is partly in air, so $\varepsilon_{\text{eff}} < D_{\text{k}}$ and signals are faster than in stripline. It is accessible for rework and probing but exposed to EMI and to soldermask variation, and its far-end crosstalk does not cancel (Chapter 9).

**Embedded microstrip**
: A microstrip covered by additional dielectric (prepreg or a thick soldermask). Coverage raises $\varepsilon_{\text{eff}}$ and lowers $Z_0$.

**Symmetric stripline**
: A trace centered between two planes in a homogeneous dielectric. It is a true TEM line with no far-end crosstalk in the ideal case and excellent shielding.




**(a)** Microstrip

**(b)** Embedded (buried) microstrip

**(c)** Symmetric stripline

**(d)** Asymmetric (dual) stripline

**(e)** Edge-coupled differential microstrip

**(f)** Grounded coplanar waveguide (CPWG)

**Figure 7.1.** Cross-sections of PCB transmission lines with their dimensional parameters.

**Asymmetric (dual) stripline**
A trace closer to one plane than the other, usually because two signal layers share the space between two planes. Routing on the two layers is kept orthogonal to limit coupling.

**Edge-coupled pair**
Two traces side by side forming a differential pair, on either microstrip or stripline layers (Chapter 10).

**Grounded coplanar waveguide**
A trace with ground copper on both sides on the same layer, stitched to the plane below. It is common in RF and in very high-speed launches, where it controls fields near connectors.

## 7.2 Closed-form impedance approximations

Exact impedances require a field solution of the cross-section. For quick estimates, closed-form approximations are available. They differ in accuracy, and none of them should be used for final dimensions.

### 7.2.1 Microstrip

The IPC-2141 approximation [17], based on earlier work by Kaupp, is

$$Z_0 \approx \frac{87}{\sqrt{\varepsilon_r + 1.41}} \ln \frac{5.98H}{0.8W + T}, \qquad 0.1 < \frac{W}{H} < 2.0, \quad 1 < \varepsilon_r < 15. \tag{7.1}$$

It includes thickness but becomes poor outside its range and is typically in error by 5–10 % even within it. A widely reproduced *simplified* closed-form set for a zero-thickness strip with $u = W/H$, derived from Wheeler's and Hammerstad's work and given in most microwave texts [12], [18], is For $u \le 1$:

$$\varepsilon_{\text{eff}} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left[\left(1 + \frac{12}{u}\right)^{-1/2} + 0.04\,(1-u)^2\right], \qquad Z_0 = \frac{60}{\sqrt{\varepsilon_{\text{eff}}}} \ln\left(\frac{8}{u} + \frac{u}{4}\right); \tag{7.2}$$

for $u \geq 1$:

$$\varepsilon_{\mathrm{eff}} = \frac{\varepsilon_{\mathrm{r}}+1}{2} + \frac{\varepsilon_{\mathrm{r}}-1}{2}\left(1+\frac{12}{u}\right)^{-1/2}, \qquad Z_0 = \frac{120\pi}{\sqrt{\varepsilon_{\mathrm{eff}}}\,[u + 1.393 + 0.667\ln(u+1.444)]}. \tag{7.3}$$

Finite thickness is commonly included by replacing $W$ with an effective width $W + \Delta W$, where $\Delta W \approx (T/\pi)\,[1 + \ln(2H/T)]$ for $W/H > 1/(2\pi)$ [18], [19].

Equations (7.2) and (7.3) are *not* the full Hammerstad–Jensen model. The full model [20], [21] uses more elaborate fitted expressions for $\varepsilon_{\mathrm{eff}}(u, \varepsilon_{\mathrm{r}})$ and $Z_0(u)$, its own thickness correction, and (in later work) frequency dispersion; it is implemented in Listing E.1 as microstrip_hj_full. Published accuracy figures for any of these formulas compare them with a quasi-static solution of the *same idealized* geometry—rectangular strip, homogeneous substrate, infinite ground, no soldermask—and say nothing about how well that geometry represents an etched, plated, mask-coated board trace. Judge them against a field solver run on the fabricator's cross-section, not against each other.

### 7.2.2 Embedded microstrip

IPC-2141 treats a microstrip covered to total height $H_1$ above the plane by using the microstrip logarithm with a reduced effective permittivity:

$$Z_0 \approx \frac{60}{\sqrt{\varepsilon'_{\mathrm{r}}}} \ln\frac{5.98H}{0.8W+T}, \qquad \varepsilon'_{\mathrm{r}} = \varepsilon_{\mathrm{r}}\left[1 - \mathrm{e}^{-1.55H_1/H}\right]. \tag{7.4}$$

### 7.2.3 Stripline

For a symmetric stripline with plane separation $B = 2H + T$,

$$Z_0 \approx \frac{60}{\sqrt{\varepsilon_{\mathrm{r}}}} \ln\frac{4B}{0.67\pi(0.8W+T)} = \frac{60}{\sqrt{\varepsilon_{\mathrm{r}}}} \ln\frac{1.9B}{0.8W+T}, \qquad \frac{W}{B-T} < 0.35,\ \frac{T}{B} < 0.25. \tag{7.5}$$

Common 50 Ω stripline geometries often violate the width limit, so a second estimate valid over a wider range is useful. Wheeler's formula [18] for a strip of finite thickness, centered between planes, is

$$Z_0 \approx \frac{30}{\sqrt{\varepsilon_{\mathrm{r}}}} \ln\left\{1 + \frac{4}{\pi}\frac{B-T}{W'}\left[\frac{8}{\pi}\frac{B-T}{W'} + \sqrt{\left(\frac{8}{\pi}\frac{B-T}{W'}\right)^2 + 6.27}\right]\right\}, \tag{7.6}$$

where $W' = W + \Delta W$ and $\Delta W = \frac{T}{\pi}\left\{1 - \frac{1}{2}\ln\left[\left(\frac{T}{2B-T}\right)^2 + \left(\frac{0.0796\,T}{W+1.1T}\right)^m\right]\right\}$ with $m = 2/\left[1 + \frac{2}{3}T/(B-T)\right]$. For $T = 0$ it agrees with the exact conformal-mapping result for a zero-thickness strip to better than 0.1 % over the widths used on boards (checked numerically in Listing E.1); with thickness it remains an approximation. Asymmetric stripline is not well served by any simple formula; IPC-2141 gives an approximation, but a field solver should be used.

### 7.2.4 Edge-coupled differential pairs

IPC-2141 relates differential impedance to the single-ended $Z_0$ of either trace alone:

$$Z_{\mathrm{diff}} \approx 2Z_0\left[1 - 0.48\,\mathrm{e}^{-0.96S/H}\right] \qquad \text{(microstrip)}, \tag{7.7}$$

$$Z_{\mathrm{diff}} \approx 2Z_0\left[1 - 0.347\,\mathrm{e}^{-2.9S/B}\right] \qquad \text{(stripline)}. \tag{7.8}$$

Figure 7.2 compares the microstrip and stripline formulas over a range of widths. The differential formulas are the least reliable of those here; they ignore the effect of coupling on each trace's own capacitance and can be in error by more than 10 % for tightly coupled pairs.

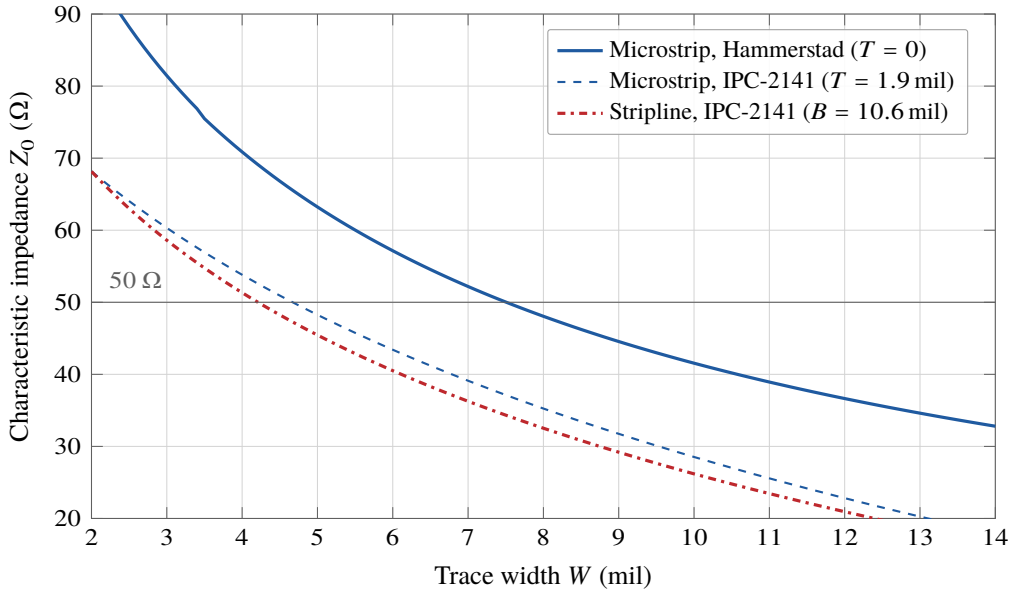



**Figure 7.2.** Characteristic impedance versus trace width for a microstrip ($H$ = 3.5 mil, $D_k$ = 3.8) and a symmetric stripline ($B$ = 10.6 mil, $D_k$ = 3.8). The two microstrip formulas differ by 10–20 %, partly because of thickness and partly because of the approximations themselves.

**Worked Example 7.1: Microstrip on 3.5 mil prepreg**

A 4 mil trace of 0.5 oz foil plated to $T$ = 1.9 mil sits on 3.5 mil prepreg with $D_k$ = 3.8 at 1 GHz. Estimate $Z_0$.

**Solution.** IPC-2141, Eq. (7.1): $Z_0 = \frac{87}{\sqrt{5.21}} \ln \frac{5.98\times3.5}{0.8\times4+1.9} = 38.12 \times \ln 4.104 = 53.8\,\Omega$. The simplified set with $T = 0$, $u = 1.143$: $\varepsilon_{\text{eff}} = 2.81$ and $Z_0 = 70.9\,\Omega$. This is not a like-for-like comparison, because the plated strip is more than half as thick as the dielectric. With Wheeler's thickness correction $\Delta W = 1.39$ mil, $u = 1.541$, $\varepsilon_{\text{eff}} = 2.87$, and $Z_0 = 60.7\,\Omega$; the full Hammerstad–Jensen model with its own thickness correction gives $Z_0 = 61.4\,\Omega$ and $\varepsilon_{\text{eff}} = 2.57$ (Listing E.1). The three thickness-including estimates span 53.8–61.4 Ω; IPC-2141 is the outlier here, and soldermask (not included) would lower the true value by a few ohms.

| Quick estimate | Full model | Difference |
|---|---|---|
| IPC-2141: 53.8 Ω | Full H–J, $T$ incl.: 61.4 Ω | −12 % |

The only reliable way to settle this is a 2-D field solver with the fabricator's actual dimensions, including the trapezoidal etch profile.

**Common Pitfall: Treating calculator results as final**

Releasing a design with trace widths taken from a closed-form calculator. The formulas above disagree with one another by 5–15 %, they ignore soldermask, etch trapezoid, resin-rich regions, and frequency-dependent $D_k$, and they are often used outside their validity range. Use them to choose a starting point; take final widths from the fabricator's field-solver stackup.

**Table 7.1.** Closed-form impedance formulas and their limits.

| Geometry | Formula | Range and typical accuracy |
|---|---|---|
| Microstrip | IPC-2141, Eq. (7.1) | $0.1 < W/H < 2$; 5–10 % |
| Microstrip | Simplified set, Eqs. (7.2) and (7.3) | All $u$, $T = 0$ (or Wheeler $\Delta W$); idealized geometry |
| Microstrip | Full Hammerstad–Jensen (Listing E.1) | Wide range, incl. $T$; idealized geometry |
| Embedded microstrip | IPC-2141, Eq. (7.4) | $H_1 > H$; rough |
| Stripline | IPC-2141, Eq. (7.5) | $W/(B-T) < 0.35$, $T/B < 0.25$; 5–10 % |
| Stripline | Wheeler, Eq. (7.6) | Wide range; exact limit for $T = 0$ |
| Differential pair | Eqs. (7.7) and (7.8) | Loose coupling; > 10 % error when tight |

**Worked Example 7.2: Stripline width for 50 Ω**

A stripline layer has 5 mil of dielectric above and below a 0.6 mil (0.5 oz) trace, $D_k = 3.8$. Estimate the width for 50 Ω and the differential impedance of a pair of such traces spaced $S = 6$ mil.

**Solution.** $B = 2 \times 5 + 0.6 = 10.6$ mil. Solving Eq. (7.5) for $W$: $\ln[1.9B/(0.8W + T)] = 50\sqrt{3.8}/60 = 1.624$, so $0.8W + T = 20.14/5.075 = 3.97$ mil and $W = 4.2$ mil. *Validity check:* $W/(B-T) = 4.2/10.0 = 0.42$, beyond the formula's limit of 0.35, so this result is an extrapolation. Wheeler's Eq. (7.6) gives 51.2 Ω at 4.2 mil and 50 Ω at $W = 4.4$ mil.

| Quick estimate | Full model | Difference |
|---|---|---|
| IPC-2141 (extrapolated): 4.2 mil | Wheeler: 4.4 mil | −5 % in width |

With Eq. (7.8), $Z_{\text{diff}} \approx 100[1 - 0.347\,e^{-2.9\times 6/10.6}] = 93$ Ω. To reach 100 Ω differential the traces would be narrowed or spaced further apart; the final geometry should again come from a field solver.

## 7.3 Stackup design principles

A good high-speed stackup obeys a small number of principles, which follow directly from the return-path physics of Chapter 3:

1. **Every signal layer is adjacent to a solid reference plane**, preferably ground. The reference defines $Z_0$ and carries the return current.
2. **Power planes are paired closely with ground planes** (2–4 mil of dielectric) to create low-inductance plane capacitance (Chapter 16).
3. **The stackup is symmetric about its center** in copper distribution and dielectric thickness, so that the board does not bow or twist during lamination and reflow.
4. **Adjacent signal layers are avoided** or, where two signal layers share a stripline cavity, routed orthogonally and spaced as far apart as the thickness budget permits.
5. **Fast signals change layers only between layers that share a reference**, or stitching vias are placed next to the signal via (Chapters 11 and 17).

### 7.3.1 Example stackups

Figure 7.3 shows a 10-layer stackup suitable for a board carrying LPDDR, PCIe, USB, and MIPI. Its routing resources are two microstrip layers (L1, L10) for breakout and short escapes and two well-shielded stripline layers (L3, L8) for long high-speed routes. Table 7.2 compares it with typical 8- and 12-layer alternatives.

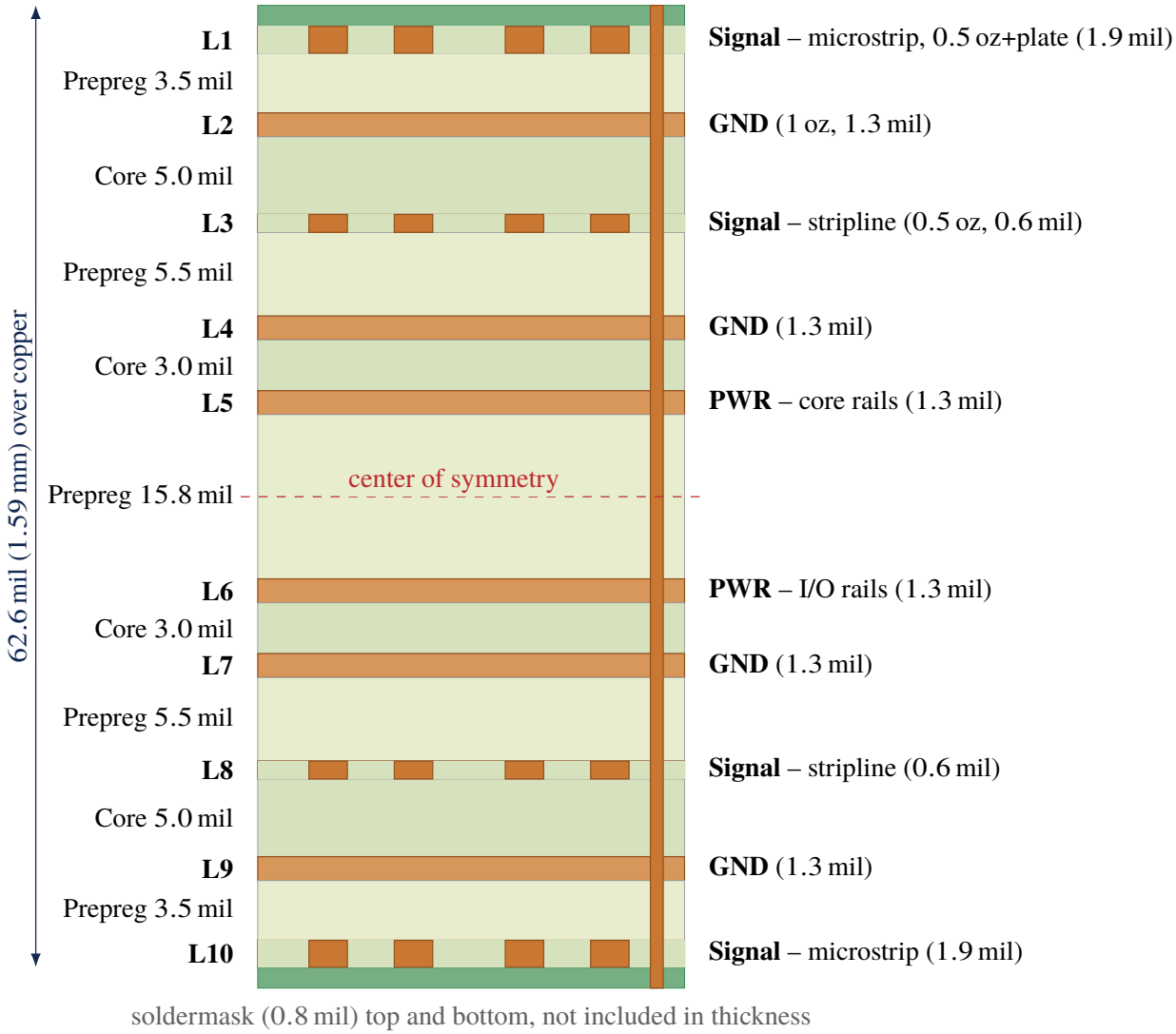



**Figure 7.3.** An example 10-layer stackup (62.6 mil over copper). Four signal layers each have an adjacent ground plane; two power planes are coupled to ground through 3 mil cores; the stackup is symmetric about its center.

The dielectric thicknesses given are representative; the fabricator will adjust them to the glass styles and resin contents it stocks.

**Design Rule: Stackup checklist**

Before routing: (1) every signal layer has an adjacent ground plane; (2) each power plane has a ground plane within 4 mil; (3) the stackup is symmetric; (4) the fabricator has confirmed dielectric thicknesses, $D_k$ at the target frequency, trace widths, and impedance tolerance; (5) via-stub lengths for the thickest transition are acceptable at the fastest interface's Nyquist frequency.

**Table 7.2.** Representative 8-, 10-, and 12-layer stackups (S = signal, G = ground, P = power). Thicknesses are illustrative and must be confirmed with the fabricator.

| Layers | Assignment (top to bottom) | Thickness | Notes |
|---|---|---|---|
| 8 | S–G–S–P–G–S–G–S | 62 mil | L3 and L6 are asymmetric striplines referenced mainly to L2 and L7; P/G pair at center (L4–L5) on a thin core. |
| 10 | S–G–S–G–P–P–G–S–G–S | 62.6 mil | Figure 7.3. Symmetric striplines L3, L8; two power planes each coupled to ground. |
| 12 | S–G–S–G–S–P–G–S–G–S–G–S | 78–93 mil | Six signal layers; L5 and L8 are inner striplines; one P/G pair at the center. Thicker board raises via-stub length (Chapter 11). |

## 7.4 Laminate materials

Board laminates are composites of woven E-glass and a resin system. The glass has a $D_k$ of about 6, the resin roughly 3; the composite's $D_k$ depends on the resin content, which differs between glass styles (thin, open styles such as 106 and 1080 are resin-rich; heavier styles such as 2116 and 7628 are glass-rich). Laminates are grouped by their dissipation factor, which sets the dielectric loss (Eq. (4.14)). Representative ranges are given in Table 7.3; vendor data for specific products appear in Appendix B.

**Table 7.3.** Laminate classes. Ranges are representative of published data at about 1–10 GHz and vary with resin content; consult the specific product data sheet.

| Class | $D_k$ | $D_f$ | Typical use |
|---|---|---|---|
| Standard FR-4 | 4.0–4.6 | 0.015–0.022 | Digital to ~ 3 Gb/s, short links |
| Mid-loss | 3.7–4.2 | 0.008–0.012 | DDR4, PCIe Gen3, 10 Gb/s short reach |
| Low-loss | 3.4–3.8 | 0.004–0.007 | PCIe Gen4, USB4, 25 Gb/s |
| Very low / ultra-low loss | 3.0–3.6 | 0.001–0.003 | PCIe Gen5/6, 56 Gb/s+ |
| PTFE and hydrocarbon RF | 2.2–3.5 | 0.001–0.004 | RF, microwave, radar |

Other laminate properties matter for reliability as much as for signal integrity: the glass-transition temperature $T_g$, the decomposition temperature $T_d$, the out-of-plane coefficient of thermal expansion (which stresses plated vias), and resistance to conductive anodic filament growth. Lead-free assembly requires laminates qualified for its higher reflow temperatures; IPC-4101 slash sheets classify these properties [22].

## 7.5 Glass-weave skew

Because glass and resin have different permittivities, the local $D_k$ seen by a trace depends on whether it lies over a glass bundle or over a resin-rich gap (Figure 7.4). The two traces of a differential pair can see different average $D_k$ along their length, producing intra-pair skew that converts differential signal into common mode (Chapter 10). The effect grows with route length and with the openness of the weave.

**Worked Example 7.3: Worst-case weave skew**

Suppose that over a 5 in parallel run one trace of a pair sees an average $D_k$ of 4.0 and the other 3.6 (an extreme case). Find the skew and compare it with a PCIe Gen4 unit interval.

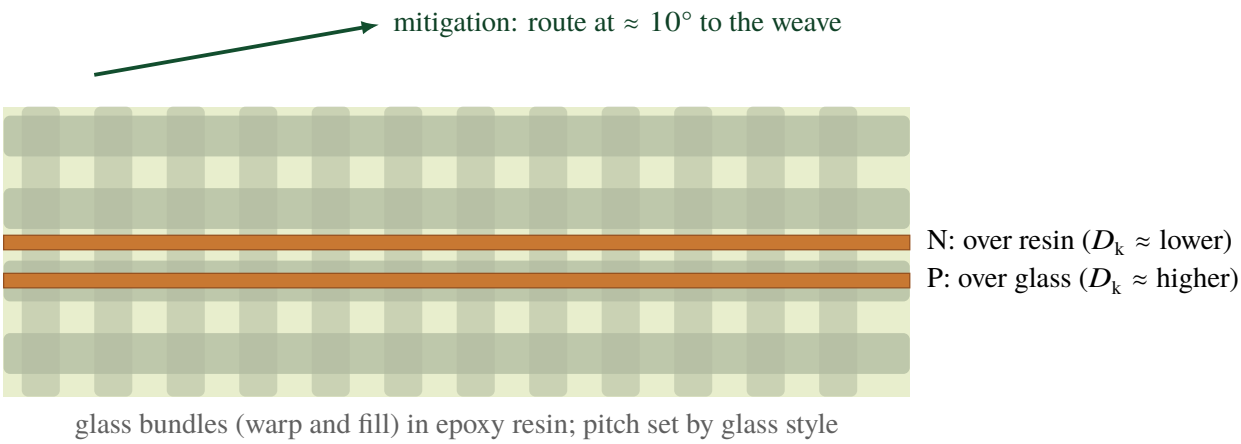



**Figure 7.4.** Glass-weave skew: when a differential pair runs parallel to the weave, one trace may lie over glass and the other over resin. Rotating the route relative to the weave averages the two.

**Solution.** $t_{pd} = 84.73\,\text{ps/in}\sqrt{D_k}$ gives 169.5 ps/in and 160.8 ps/in. The difference, 8.7 ps/in, accumulates to 43 ps over 5 in—70 % of the 62.5 ps UI at 16 GT/s. Realistic local $D_k$ variations are usually smaller, but even 5–10 ps is significant at Gen4 and above.

Mitigations, roughly in order of cost, are:

- route long pairs at an angle (often 10° to 15°) to the weave, or in a shallow zig-zag, so that each trace samples glass and resin equally;
- rotate the artwork on the panel by a few degrees;
- use spread or flat glass styles (e.g. 1067, 1078, 1035, 3313), in which the bundles are flattened to close the resin gaps;
- use two plies of prepreg of different styles so that the bundles do not align.

## 7.6 Copper foil and roughness

Copper foil is roughened on the side bonded to the laminate to improve adhesion. At gigahertz frequencies the skin depth (Figure 3.4) becomes comparable to this roughness and current is forced to follow the tooth profile, raising conductor loss by up to a factor of two or more (Chapter 8). Foils are sold in roughness grades: standard (STD), reverse-treated (RTF), very low profile (VLP), and hyper or ultra low profile (HVLP, ULP), with progressively smaller surface profiles. Lower-profile foils cost more and have lower peel strength. Oxide or oxide-alternative treatments applied to inner-layer cores before lamination also add roughness, which should be included in loss models.

**Rule of Thumb: When roughness matters**

Copper roughness begins to affect loss noticeably once the skin depth falls below the RMS roughness, about 1 GHz for standard foil and several gigahertz for VLP/HVLP. *Limits:* the threshold is gradual, and the size of the effect depends on the tooth shape, not only on the RMS value (Section 8.2.2).

## 7.7 Impedance tolerance

Fabricators typically guarantee impedance to $\pm 10\,\%$, and to $\pm 5\,\%$ or $\pm 7\,\%$ at extra cost. The variation comes from etched width, dielectric thickness after lamination, copper thickness, and $D_k$. Its sensitivity can be estimated from any of the closed-form formulas.

**Worked Example 7.4: Tolerance budget of a microstrip**

For the microstrip of Worked Example 7.1 ($Z_0 \approx 53.8\,\Omega$ by IPC-2141), estimate the impedance spread for $H \pm 10\,\%$, $W \pm 0.5$ mil, and $D_k \pm 5\,\%$.

**Solution.** From Eq. (7.1), $\partial Z_0/\partial H = 38.12/H$, so $\Delta H = 0.35$ mil gives $3.8\,\Omega$. $\partial Z_0/\partial W = -38.12 \times 0.8/(0.8W + T) = -5.98\,\Omega$/mil, so $\Delta W = 0.5$ mil gives $3.0\,\Omega$. $\partial \ln Z_0/\partial \varepsilon_r = -1/[2(\varepsilon_r + 1.41)]$, so $\Delta\varepsilon_r = 0.19$ gives $1.8\,\%$, or $1.0\,\Omega$.

These three numbers can be combined in three different ways, and they answer three different questions:

- *Worst case.* Evaluating the formula at the corners of the box gives 46.1–61.7 Ω, close to the linear sum $\pm(3.8 + 3.0 + 1.0) = \pm 7.8\,\Omega$. This is a bound—for this formula and these ranges only.
- *RSS of the bounds.* $\sqrt{3.8^2 + 3.0^2 + 1.0^2} = 4.9\,\Omega$ is neither a bound nor a standard deviation; it acquires a meaning only once a probability model is attached.
- *Statistical.* If each input is independent and uniformly distributed over its range, its standard deviation is (half-range)/$\sqrt{3}$, so $\sigma_Z \approx 4.9/\sqrt{3} = 2.8\,\Omega$. A Monte Carlo run of the same formula (Figure 7.5) confirms $\sigma_Z = 2.85\,\Omega$, and about 5 % of samples fall outside $\pm 10\,\%$.

The result is dominated by dielectric thickness. None of these numbers is a fabrication yield: real processes have correlated inputs (a fabricator adjusts etch width to the measured dielectric thickness), non-uniform distributions, copper-thickness variation and model error. Use process data and field-solver sensitivities before predicting production yield.

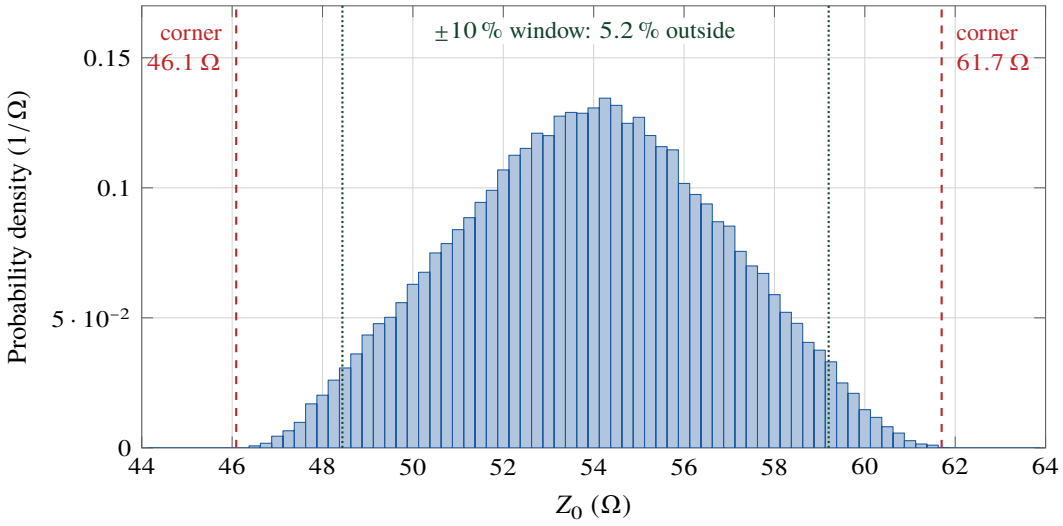



**Figure 7.5.** Distribution of the IPC-2141 microstrip impedance of Worked Example 7.4 for $W = 4$ mil $\pm$ 0.5 mil, $H = 3.5$ mil $\pm\,10\,\%$, $D_k = 3.8 \pm 5\,\%$ (independent uniform inputs, 100 000 samples, fixed seed). Dashed lines: exact corners of the same formula; dotted: $\pm 10\,\%$. [Analytical (Monte Carlo)]

**Common Pitfall: RSS presented as a guarantee**

Quoting the root-sum-square of worst-case contributions as if it bounded the result. RSS combines standard deviations of independent variables; it is a statistical estimate whose coverage depends on the distributions. Keep worst-case corners and statistical estimates in separate columns of a tolerance budget.

**Key Takeaways**

- Microstrip is fast and accessible; stripline is shielded, homogeneous, and free of ideal far-end crosstalk.
- Closed-form impedance formulas (IPC-2141, the simplified and full Hammerstad–Jensen sets, Wheeler) describe idealized geometries, disagree with each other by several percent, and must be checked against their validity ranges; use a field solver for final dimensions.
- Give every signal layer an adjacent ground, pair power with ground on thin dielectric, and keep the stackup symmetric.
- Choose laminate by $D_f$ at the Nyquist frequency of the fastest link.
- Glass-weave skew and copper roughness become first-order effects above roughly 10 Gb/s.
- Impedance tolerance of $\pm 10\,\%$ is dominated by dielectric thickness and etched width. Worst-case corners, RSS and Monte Carlo answer different questions; none of them is a yield prediction without process data.

## Further Reading

IPC-2141A [17] documents the closed-form formulas and coupon practice; Wadell [18] collects formulas for many geometries; Wheeler [19] and Hammerstad and Jensen [20] are the primary sources for microstrip; Ritchey [8] discusses stackup design from a fabrication perspective.

## Review Problems

**P7.1** Using IPC-2141, find the width of a 50 Ω microstrip with $H = 3.5$ mil, $T = 1.9$ mil, and $D_k = 3.8$.

**P7.2** Using Hammerstad–Jensen with $T = 0$, estimate the width of a 50 Ω microstrip with $H = 3.5$ mil and $D_k = 3.8$. Why does it differ from Problem P7.1?

**P7.3** A 50 Ω microstrip pair has $H = 3.5$ mil and $S = 5$ mil. Estimate $Z_{diff}$.

**P7.4** Find $Z_0$ of a stripline with $W = 4$ mil, $T = 0.6$ mil, $B = 10.6$ mil, and $D_k = 3.8$. Is the formula within its range?

**P7.5** An embedded microstrip ($W = 4$ mil, $T = 1.9$ mil, $H = 3.5$ mil, $D_k = 3.8$) is covered to $H_1 = 7$ mil. Estimate $\varepsilon_r'$ and $Z_0$.

**P7.6** A pair runs 8 in with one trace seeing $D_k = 3.75$ and the other $D_k = 3.65$. Find the skew.

**P7.7** *(Inverse design.)* A fabricator guarantees $H = 3.5$ mil $\pm\ 5\,\%$ and $D_k \pm 3\,\%$ for the microstrip of Worked Example 7.4. Using the linear sensitivities of that example, what etch tolerance on $W$ keeps the worst-case linear sum within $\pm 5.4\ \Omega$ ($\pm 10\,\%$)?

**P7.8** *(Imperfect data.)* Ten coupons from one panel measure 52.1, 52.8, 53.0, 51.9, 52.5, 52.2, 53.1, 52.7, 52.4 and 52.6 Ω. Can you infer the production spread of $\pm 10\,\%$ boards from them?

Chapter 8

# Losses

**Chapter Overview**

Above a few gigabits per second, the dominant signal-integrity impairment of a long board trace is not reflection or crosstalk but loss. Loss attenuates the high-frequency content of each bit, spreads its energy into neighboring bit periods, and closes the eye. This chapter quantifies the two loss mechanisms of a transmission line—conductor loss, including skin effect and surface roughness, and dielectric loss—and shows how to assemble them into an insertion-loss budget for a channel.

**Learning Objectives**

1. Derive the practical formulas for conductor and dielectric attenuation in dB/in.
2. Identify the crossover frequency at which dielectric loss overtakes conductor loss.
3. Apply the Hammerstad and Huray roughness models and state their regimes of validity.
4. Explain why a causal dielectric model is required and describe the Djordjevic–Sarkar model.
5. Build an insertion-loss budget and use it to select a laminate and a trace width.

## 8.1 Loss as a low-pass filter

From Eq. (4.12), the attenuation constant of a low-loss line is the sum of a conductor term $\alpha_c = R'/(2Z_0)$ and a dielectric term $\alpha_d = G'Z_0/2$. A matched line of length $\ell$ has insertion loss

$$\mathrm{IL}(f) = -20\log_{10}|S_{21}| = 8.686\,[\alpha_c(f) + \alpha_d(f)]\ell \quad \text{(dB)}. \tag{8.1}$$

The equality holds only when the line is matched to the measurement reference impedance. A mismatched line also loses transmitted power to reflection, so measured $|S_{21}|$ contains both dissipation (attenuation) and mismatch loss; the two are separated by checking $|S_{11}|^2 + |S_{21}|^2$, which equals 1 only for a lossless network, or by fitting $\alpha$ to lines of two lengths (Chapter 25). A ripple of period $1/(2T_{\mathrm{D}})$ on $|S_{21}|$ is the signature of mismatch, not of dielectric loss. Because both terms rise with frequency, the line is a low-pass filter. A digital bit stream has a broad spectrum; its high-frequency components, which form the sharp edges, are attenuated more than its low-frequency components, which form long runs of identical bits. The received waveform therefore has slow edges and pattern-dependent levels—*inter-symbol interference* (ISI)—which is analyzed with eye diagrams in Chapter 12.

## 8.2 Conductor loss

### 8.2.1 Smooth-copper conductor loss

Above the skin-effect onset, the signal conductor of a stripline carries current in a layer one skin depth thick on both its top and bottom faces, and each return plane carries the same current in a similar layer. From Eq. (3.14), doubling to include the return path, and inserting the skin depth of copper,

$$R'_{\text{loop}} \approx 2 \cdot \frac{\rho}{2W\delta_s} \approx \frac{8.25\sqrt{f_{\text{GHz}}}}{W_{\text{mil}}}\ \Omega/\text{in}. \tag{8.2}$$

Dividing by $2Z_0$ and converting nepers to decibels gives the widely used estimate [3]

$$\alpha_c \approx \frac{36\sqrt{f_{\text{GHz}}}}{W_{\text{mil}}\, Z_0}\ \text{dB/in} \qquad \text{(smooth copper, stripline)}. \tag{8.3}$$

Three conclusions follow. Conductor loss grows as $\sqrt{f}$; it is inversely proportional to trace width; and, for a fixed width, it is lower at higher impedance. Wider traces reduce loss only if the dielectric is thickened to keep $Z_0$ constant, which is why long-reach backplanes use thick stackups with wide traces. Equation (8.3) neglects the current crowding at trace edges and the slightly different geometry of microstrip; for microstrip the estimate is typically optimistic by 10–30 %.

**Rule of Thumb: Conductor loss**

A 50 Ω, 5 mil stripline with smooth copper loses about $0.14\sqrt{f_{\text{GHz}}}$ dB/in—about 0.3 dB/in at 5 GHz before roughness. *Limits:* smooth copper only; multiply by $K_{\text{rough}}$ (Section 8.2.2), which can approach 2 or more.

### 8.2.2 Surface roughness

When the skin depth becomes comparable to the copper's surface profile, current must follow the peaks and valleys of the surface, lengthening its path and increasing loss. The effect is described by a multiplicative factor on the smooth-copper conductor loss, $\alpha_{c,\text{rough}} = K_{\text{rough}}\,\alpha_c$.

The *Hammerstad* model, originally fitted to measurements of periodic grooves [20], is

$$K_H = 1 + \frac{2}{\pi}\arctan\left[1.4\left(\frac{\Delta}{\delta_s}\right)^2\right], \tag{8.4}$$

where $\Delta$ is the RMS surface roughness. It is simple and widely implemented, but it saturates at $K_H = 2$, whereas measured losses of rough foils at tens of gigahertz exceed twice the smooth value.

The *Huray* ("snowball") model represents the surface as stacks of copper spheres of radius $a$ on a flat base, and computes their absorption and scattering of the incident field [23], [24]. For a single sphere size it reduces to

$$K_{\text{Huray}} = \frac{A_{\text{matte}}}{A_{\text{flat}}} + \frac{3}{2}\,\frac{N\,4\pi a^2/A_{\text{hex}}}{1 + \dfrac{\delta_s}{a} + \dfrac{\delta_s^2}{2a^2}}, \tag{8.5}$$

where $N$ is the number of spheres on a hexagonal tile of area $A_{\text{hex}}$. The dimensionless ratio $\text{SR} = N4\pi a^2/A_{\text{hex}}$ is the *surface ratio*; with $A_{\text{matte}}/A_{\text{flat}} \approx 1$, $K$ rises from 1 at low frequency to $1 + 1.5\,\text{SR}$ at high frequency. Its parameters are best obtained from SEM images of the foil or by fitting measured insertion loss; typical sphere radii are a few tenths of a micrometre. Figure 8.1 compares the two models.

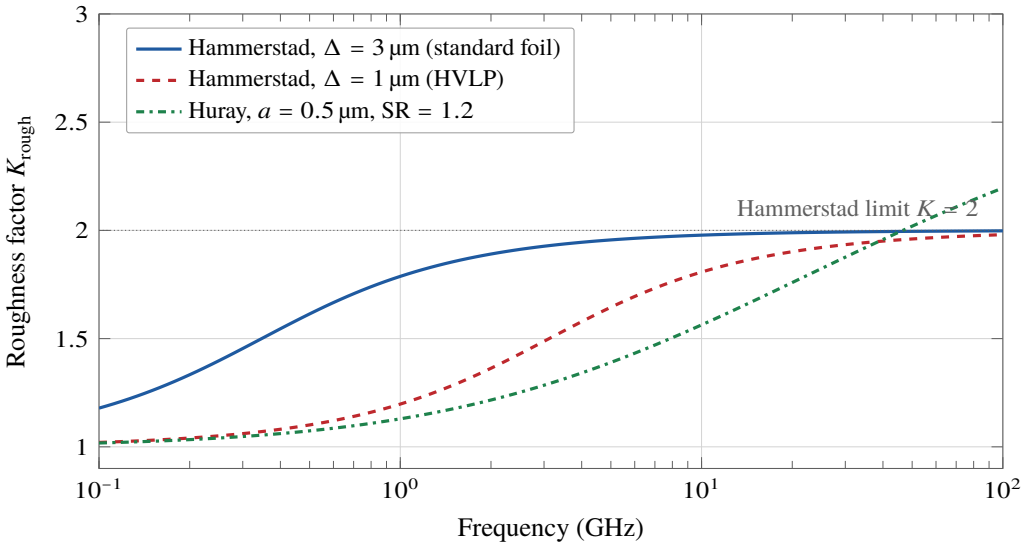



**Figure 8.1.** Roughness correction factor versus frequency. The Hammerstad model (Eq. (8.4)) saturates at 2; the Huray model (Eq. (8.5)) rises further, toward its own finite asymptote 1 + 1.5 SR, and better matches measurements of rough foil above about 10 GHz. [Analytical]

**Common Pitfall: Using datasheet roughness without context**

Entering a foil's catalog $R_z$ (a peak-to-valley height) as the RMS roughness $\Delta$ of the Hammerstad model, or ignoring the oxide treatment on inner cores. Both errors can misstate conductor loss by tens of percent. Use the fabricator's measured or fitted roughness parameters for the specific foil and bonding treatment, and say which model they belong to.

## 8.3 Dielectric loss

In an alternating field, the polarization of the dielectric lags the field, dissipating energy. The complex permittivity is written $\varepsilon = \varepsilon_0\varepsilon_r(1 - j\tan\delta)$, and the loss appears as a shunt conductance $G' = \omega C' \tan\delta$. From Eqs. (4.13) and (4.14),

$$\alpha_d\,[\mathrm{Np/m}] = \frac{\pi f\sqrt{D_k}\,D_f}{c}, \qquad \alpha_d\,[\mathrm{dB/in}] \approx 2.31 f_{\mathrm{GHz}}\,D_f\sqrt{D_k}. \tag{8.6}$$

The two forms differ only by units; writing "=" between them would mix nepers per meter with decibels per inch. Dielectric loss is proportional to frequency, so it eventually dominates. In a homogeneous TEM line (stripline) it is independent of trace width and impedance and can be reduced only by choosing a laminate with lower $D_f$ (Table 7.3). In microstrip, part of the field is in air; $\sqrt{D_k}$ is then replaced by $qD_k/\sqrt{\varepsilon_{\mathrm{eff}}}$, where $q$ is the filling factor (the fraction of electric energy in the dielectric), so $\alpha_d$ depends on the cross-section.

### 8.3.1 Crossover frequency

Equating Eqs. (8.3) and (8.6) gives the frequency at which dielectric loss overtakes smooth-copper conductor loss:

$$f_\times\,[\mathrm{GHz}] = \left(\frac{36}{2.31\,W_{\mathrm{mil}}Z_0D_f\sqrt{D_k}}\right)^2. \tag{8.7}$$

For a 5 mil, 50 Ω stripline in standard FR-4 ($D_f = 0.02$, $D_k = 4.2$) the crossover is at 2.3 GHz (Figure 8.2); in a laminate with $D_f = 0.004$ it moves to about 70 GHz, and conductor loss—with its roughness—dominates everywhere of practical interest.

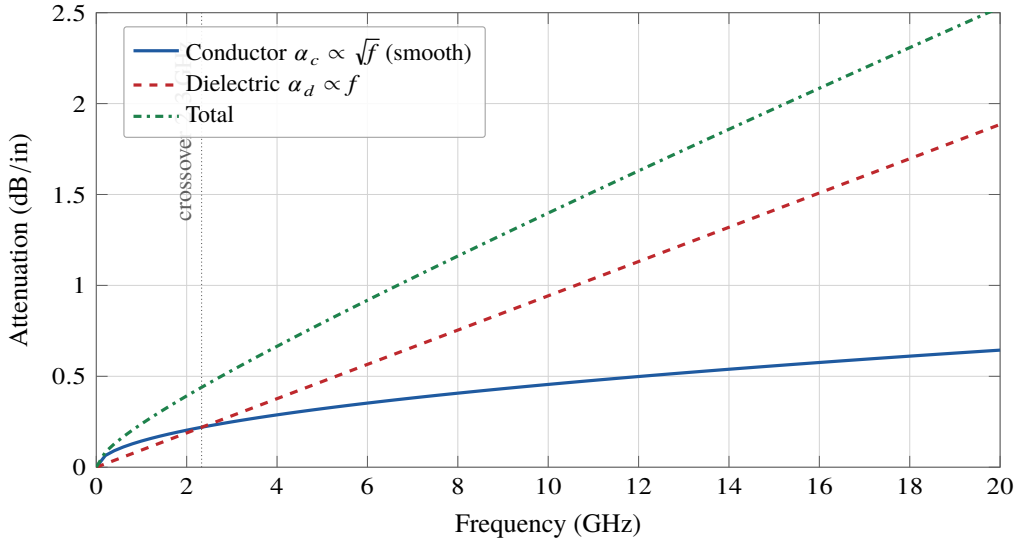



**Figure 8.2.** Conductor and dielectric attenuation of a 50 Ω, 5 mil stripline in standard FR-4 ($D_k = 4.2$, $D_f = 0.020$, smooth copper). Dielectric loss dominates above about 2.3 GHz.

### 8.3.2 Causal dielectric models

Data sheets often give a single $D_k$ and $D_f$ at one frequency. A model with both held constant over frequency is non-causal: a time-domain simulation built on it produces responses that begin before the stimulus. The Kramers–Kronig relations require that a lossy dielectric's $D_k$ decrease as frequency rises. The wideband Debye, or Djordjevic–Sarkar, model satisfies this with a distribution of relaxation frequencies between $\omega_1 = 10^{m_1}$ and $\omega_2 = 10^{m_2}$ [13]:

$$\varepsilon_r(\omega) = \varepsilon_\infty + \frac{\Delta\varepsilon}{m_2 - m_1} \log_{10} \frac{\omega_2 + j\omega}{\omega_1 + j\omega}. \tag{8.8}$$

Fitting $\varepsilon_\infty$ and $\Delta\varepsilon$ to one $(D_k, D_f)$ pair yields a model in which $D_f$ is nearly constant across several decades and $D_k$ falls slowly (typically a few percent per decade). Most field solvers and SI simulators implement this model; it should be selected whenever a laminate is entered from a data sheet.

**Design Rule: Specify materials at the right frequency**

Enter $D_k$ and $D_f$ at a frequency near the Nyquist frequency of the fastest interface on the layer, using a causal (Djordjevic–Sarkar or equivalent) model. Ask the fabricator for the $D_k$ of the specific glass style and resin content of each dielectric layer, not a generic value for the laminate family.

## 8.4 Total insertion loss

Combining the pieces,

$$\mathrm{IL}\,[\mathrm{dB}] \approx \ell_{in}\left[\frac{36\sqrt{f_{GHz}}}{W_{mil} Z_0} K_{rough}(f) + 2.31 f_{GHz} D_f \sqrt{D_k}\right] + \mathrm{IL}_{vias} + \mathrm{IL}_{conn} + \mathrm{IL}_{pkg}. \tag{8.9}$$

Figure 8.3 evaluates the line terms for a 10 in route in standard FR-4 and in a low-loss laminate. The two differ by a factor of almost three at 16 GHz. Table 8.1 summarizes the loss mechanisms.

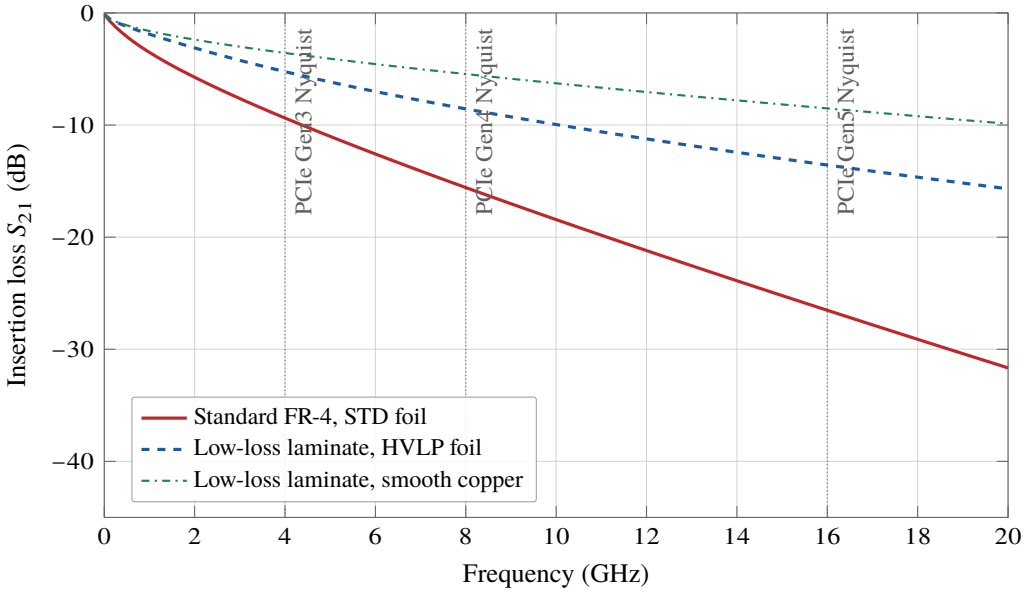



**Figure 8.3.** Insertion loss of a 10 in, 50 Ω, 5 mil stripline from Eq. (8.9), for standard FR-4 with standard foil and for a low-loss laminate with HVLP foil. The smooth-copper curve shows how much of the low-loss laminate's loss is due to roughness.

**Worked Example 8.1: Loss of an FR-4 channel at 4 GHz**

Estimate the line loss at 4 GHz (PCIe Gen3 Nyquist) of a 10 in, 50 Ω, 5 mil stripline in standard FR-4 ($D_k = 4.2$, $D_f = 0.020$) with 3 µm RMS roughness.

**Solution.** Smooth conductor loss: $36 \times 2/(5 \times 50) = 0.288$ dB/in. Skin depth: $2.09/\sqrt{4} = 1.045$ µm, so $1.4(\Delta/\delta_s)^2 = 11.5$ and $K_H = 1 + (2/\pi)\arctan 11.5 = 1.94$; rough conductor loss 0.560 dB/in. Dielectric loss: $2.31 \times 4 \times 0.02 \times \sqrt{4.2} = 0.379$ dB/in. The total is 0.94 dB/in, or 9.4 dB for 10 in—about 40 % of a typical Gen3 channel budget, before vias, connectors, and packages.

**Worked Example 8.2: Choosing a laminate for a Gen4 route**

A PCIe Gen4 design allocates 12 dB at 8 GHz to a 10 in, 50 Ω, 5 mil stripline with HVLP foil ($\Delta = 1$ µm). What $D_f$ is required if $D_k \approx 3.7$?

**Solution.** The allowance is 1.2 dB/in. Smooth conductor loss is $36\sqrt{8}/250 = 0.407$ dB/in; with $\delta_s = 0.739$ µm, $K_H = 1.76$ and the rough value is 0.718 dB/in. The dielectric may contribute at most $1.2 - 0.718 = 0.482$ dB/in, so $D_f \le 0.482/(2.31 \times 8 \times \sqrt{3.7}) = 0.0136$. A mid-loss laminate ($D_f \approx 0.010$) meets the budget with some margin; standard FR-4 does not.

**Design Rule: Insertion-loss budgeting**

Allocate the channel's total loss at the Nyquist frequency among packages, connectors, vias, and line in a written budget before choosing the laminate. Reserve at least 1–2 dB for manufacturing variation (etch,

**Table 8.1.** Loss mechanisms of PCB transmission lines.

| Mechanism | Frequency dependence | Controlled by |
|---|---|---|
| DC resistance | constant | Copper weight, width |
| Skin effect | $\propto \sqrt{f}$ | Width, impedance (thicker dielectric) |
| Surface roughness | factor $1 \to 2$ (Hammerstad) or more (Huray) | Foil grade, oxide treatment |
| Dielectric loss | $\propto f$ | Laminate $D_\mathrm{f}$ only |
| Radiation | rises steeply with $f$ | Shielding (stripline), launch design |
| Via and connector loss | resonant, $\propto f$ to $f^2$ | Back-drill, anti-pads, connector choice |

$D_\mathrm{k}$, roughness) and 5–10 % of the line loss for temperature, since both copper resistivity and laminate $D_\mathrm{f}$ rise when hot.

**Key Takeaways**

- Insertion loss is $8.686(\alpha_c + \alpha_d)\ell$ plus discrete contributions from vias, connectors, and packages.
- Smooth stripline conductor loss is about $36\sqrt{f_\mathrm{GHz}}/(WZ_0)$ dB/in; roughness multiplies it by up to 2 (Hammerstad) or more (Huray).
- Dielectric loss is $2.31 f_\mathrm{GHz} D_\mathrm{f} \sqrt{D_\mathrm{k}}$ dB/in in stripline, independent of geometry there; it dominates FR-4 above a few gigahertz. In microstrip it depends on the filling factor.
- Laminate data must be modeled causally; constant-$D_\mathrm{k}$/$D_\mathrm{f}$ models give non-physical time-domain results.
- Write the loss budget first, then choose laminate, foil, and width.

## Further Reading

Hall and Heck [4] model conductor and dielectric loss and roughness in detail; Huray [23], [24] develops the snowball roughness model; Djordjevic et al. [13] introduce the causal wideband dielectric model; Johnson and Graham's second book [2] treats lossy-line behavior.

## Review Problems

**P8.1** Estimate the smooth-copper conductor loss at 10 GHz of a 4 mil, 50 Ω stripline.

**P8.2** Find the dielectric loss at 10 GHz in a laminate with $D_\mathrm{k} = 3.4$ and $D_\mathrm{f} = 0.003$.

**P8.3** At what frequency does dielectric loss equal smooth conductor loss for a 5 mil, 50 Ω stripline with $D_\mathrm{f} = 0.02$ and $D_\mathrm{k} = 4.2$?

**P8.4** Evaluate the Hammerstad roughness factor at 5 GHz for $\Delta = 2\,\mu\mathrm{m}$.

**P8.5** What is the high-frequency limit of the Huray factor for a surface ratio SR = 1.2 and $A_\mathrm{matte}/A_\mathrm{flat} = 1$?

**P8.6** Estimate the line loss at 16 GHz of a 6 in, 50 Ω, 5 mil stripline in a laminate with $D_\mathrm{k} = 3.5$, $D_\mathrm{f} = 0.004$, and HVLP foil ($\Delta = 1\,\mu\mathrm{m}$, Hammerstad).

Chapter 9

# Crosstalk

**Chapter Overview**

Crosstalk is the unwanted transfer of signal from one net (the *aggressor*) to another (the *victim*) through their mutual capacitance and mutual inductance. On dense boards it is a leading source of noise and timing jitter. This chapter derives the near-end and far-end crosstalk of coupled transmission lines, explains saturation length and the cancellation of far-end crosstalk in stripline, examines the popular "3W rule" and its limits, and gives layout guidelines for controlling coupling.

**Learning Objectives**

1. Express crosstalk in terms of the ratios $C_m/C$ and $L_m/L$.
2. Compute near-end crosstalk (NEXT) and its saturation length.
3. Compute far-end crosstalk (FEXT) and explain why it vanishes in a homogeneous stripline.
4. Evaluate the 3W rule and choose spacing from $S/H$ rather than $S/W$.
5. Decide when guard traces help and how to stitch them.
6. Build a crosstalk budget across several aggressors.

## 9.1 Coupling mechanisms

Two parallel traces over a common plane share electric and magnetic fields. Their coupling is described by a mutual capacitance per unit length $C_m$ and a mutual inductance per unit length $L_m$ (Figure 9.1). A changing voltage on the aggressor injects current $C_m\ \mathrm{d}V_a/\mathrm{d}t$ into the victim; a changing aggressor current induces a series voltage $L_m\ \mathrm{d}I_a/\mathrm{d}t$ in the victim. Both are distributed along the coupled length, and their contributions travel in both directions along the victim.

The capacitive and inductive contributions add in the backward direction and subtract in the forward direction. To first order in the coupling (weak coupling, both lines terminated in $Z_0$), the standard results are [3], [4], [14]

$$K_b = \frac{1}{4}\left(\frac{C_m}{C} + \frac{L_m}{L}\right), \tag{9.1}$$

$$K_f = \frac{1}{2v_\mathrm{p}}\left(\frac{C_m}{C} - \frac{L_m}{L}\right), \tag{9.2}$$

where $C$ and $L$ are the per-unit-length self capacitance and inductance of a line in the presence of the other. $K_b$ is dimensionless; $K_f$ has units of time per length.

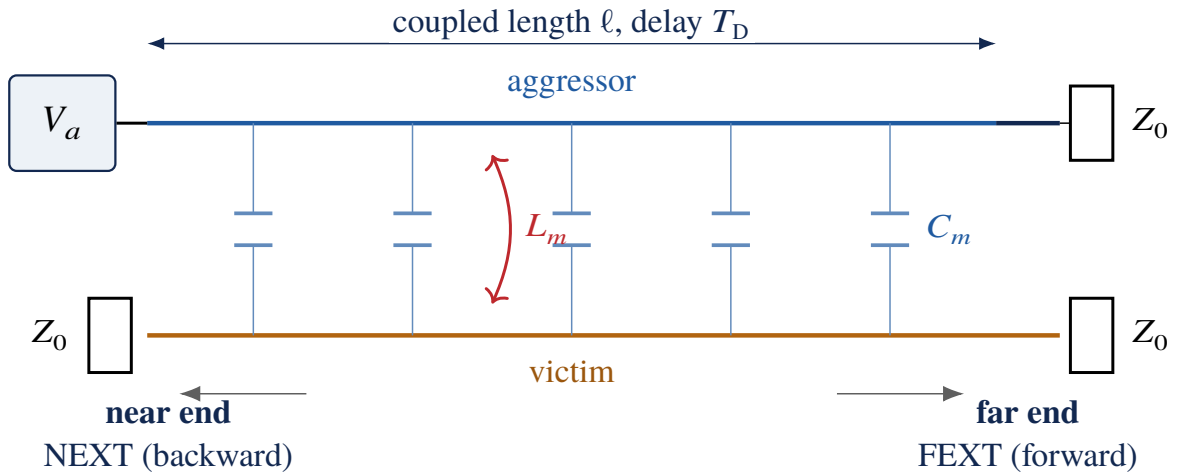



**Figure 9.1.** Coupled aggressor and victim lines. Distributed mutual capacitance $C_m$ and mutual inductance $L_m$ produce a backward wave at the victim's near end (NEXT) and a forward wave at its far end (FEXT).

**Reading the coefficients from a field solver.** A 2-D solver reports per-unit-length matrices for the pair,

$$\mathbf{L} = \begin{bmatrix} L_{11} & L_{12} \\ L_{12} & L_{22} \end{bmatrix}, \qquad \mathbf{C} = \begin{bmatrix} C_{11} & C_{12} \\ C_{12} & C_{22} \end{bmatrix}. \tag{9.3}$$

$\mathbf{L}$ is straightforward: $L = L_{11}$ and $L_m = L_{12} > 0$. The capacitance matrix is usually the *Maxwell* matrix, whose diagonal $C_{11}$ is the *total* capacitance of line 1 (to ground plus to line 2) and whose off-diagonal term is *negative*, $C_{12} = -C_m$. Hence $C = C_{11}$ and $C_m = -C_{12}$. Some tools report the "SPICE" (branch) form instead, with positive mutual and a diagonal that excludes it; mixing the two conventions flips the sign of the capacitive term in Eq. (9.2) and can turn a cancelling stripline FEXT into a large one. Check the convention by confirming that $C_m/C = L_m/L$ for a homogeneous stripline.

## 9.2 Near-end crosstalk

As the aggressor edge travels along the coupled region, it continuously sends a backward wave to the victim's near end. The near-end voltage builds up while the edge enters the coupled region, then holds at a plateau for as long as the edge is still traveling inside it, and finally decays. The plateau lasts $2T_D$ (the time for the edge to reach the far end plus the time for the last backward contribution to return). If $2T_D \geq t_r$, NEXT reaches its full value

$$V_{NEXT} = K_b V_a, \qquad 2T_D \geq t_r; \tag{9.4}$$

otherwise the plateau is never reached and the peak is reduced in proportion, $V_{NEXT} \approx K_b V_a \cdot 2T_D/t_r$. The coupled length at which NEXT saturates is the *saturation length*

$$\ell_{sat} = \frac{t_r\, v_p}{2}. \tag{9.5}$$

For a 100 ps edge in FR-4 stripline, $\ell_{sat}$ is only 0.3 in. Beyond saturation, making the parallel run longer does not increase NEXT amplitude; it only lengthens the plateau.

## 9.3 Far-end crosstalk

The forward contributions travel with the aggressor edge and arrive at the far end together, after $T_{\mathrm{D}}$. Their sum is proportional to the coupled length and to the derivative of the aggressor waveform:

$$V_{\mathrm{FEXT}}(t) = K_f \, \ell \, \frac{\mathrm{d}V_a(t - T_{\mathrm{D}})}{\mathrm{d}t} \approx K_f \, \ell \, \frac{V_a}{t_{\mathrm{r}}} \quad \text{(peak, linear ramp).} \tag{9.6}$$

FEXT is a pulse of width $\approx t_{\mathrm{r}}$ centered on the arrival of the aggressor edge (Figure 9.3). Unlike NEXT, it grows with coupled length and with edge rate—but only while it is small.

### 9.3.1 When the derivative approximation fails

Equation (9.6) is a first-order expansion. The exact behavior follows from the modal picture of Chapter 10: a step on the aggressor of a symmetric, matched pair launches equal even- and odd-mode waves, which travel with delays $T_e$ and $T_o$. The victim far-end voltage is half their difference,

$$V_{\mathrm{FEXT}}(t) = \frac{1}{2}\left[a(t - T_e) - a(t - T_o)\right] \approx -\frac{\Delta T}{2}\,\frac{\mathrm{d}a}{\mathrm{d}t}, \qquad \Delta T = T_e - T_o, \tag{9.7}$$

where $a(t)$ is the aggressor waveform. Comparing with Eq. (9.6) identifies $K_f \ell = -\Delta T/2$. The approximation holds while $|\Delta T| \ll t_{\mathrm{r}}$. When the modes separate by a large fraction of the edge, the two delayed copies no longer overlap: the victim sees a pulse of width $\Delta T$ whose height approaches half the aggressor swing and then stops growing (Figure 9.2). In a real bus, loss, reflections and unequal terminations modify this ideal limit, but the conclusion is general: FEXT does not grow without bound.

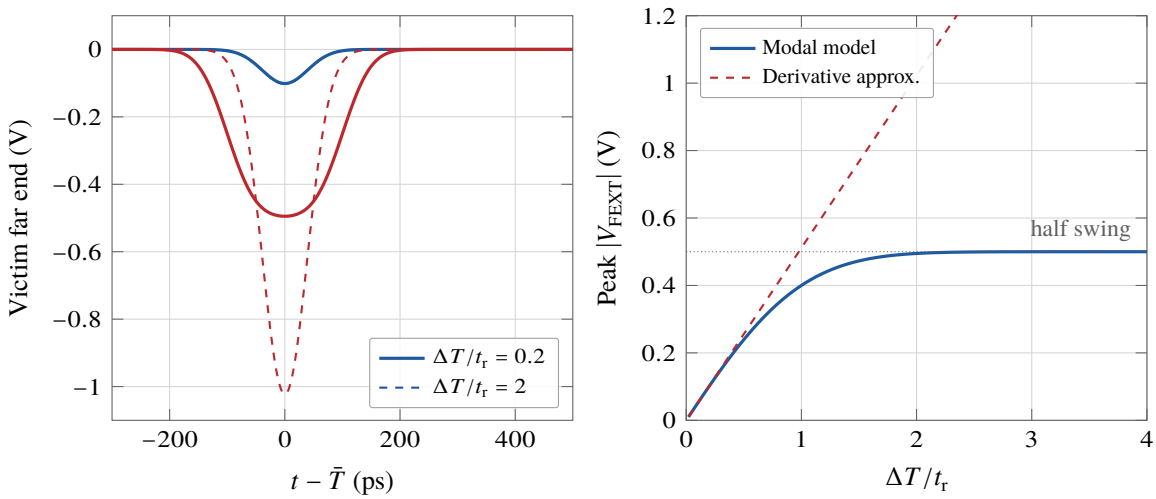



**Figure 9.2.** Far-end crosstalk from the modal model of Eq. (9.7) (solid) and its derivative approximation (dashed), for a 1 V Gaussian edge with $t_{\mathrm{r}} = 100\,\mathrm{ps}$. Left: waveforms for $\Delta T/t_{\mathrm{r}} = 0.2$ and 2. Right: peak magnitude versus modal delay difference. Matched, lossless, symmetric pair. [Analytical]

### 9.3.2 Stripline versus microstrip

In a homogeneous dielectric the capacitive and inductive coupling ratios are equal, $C_m/C = L_m/L$, so $K_f = 0$ and *FEXT cancels*. In microstrip, part of the electric field—and therefore of the mutual capacitance—lies in air, whereas the magnetic field is unaffected by the dielectric; thus $L_m/L > C_m/C$ and FEXT is negative

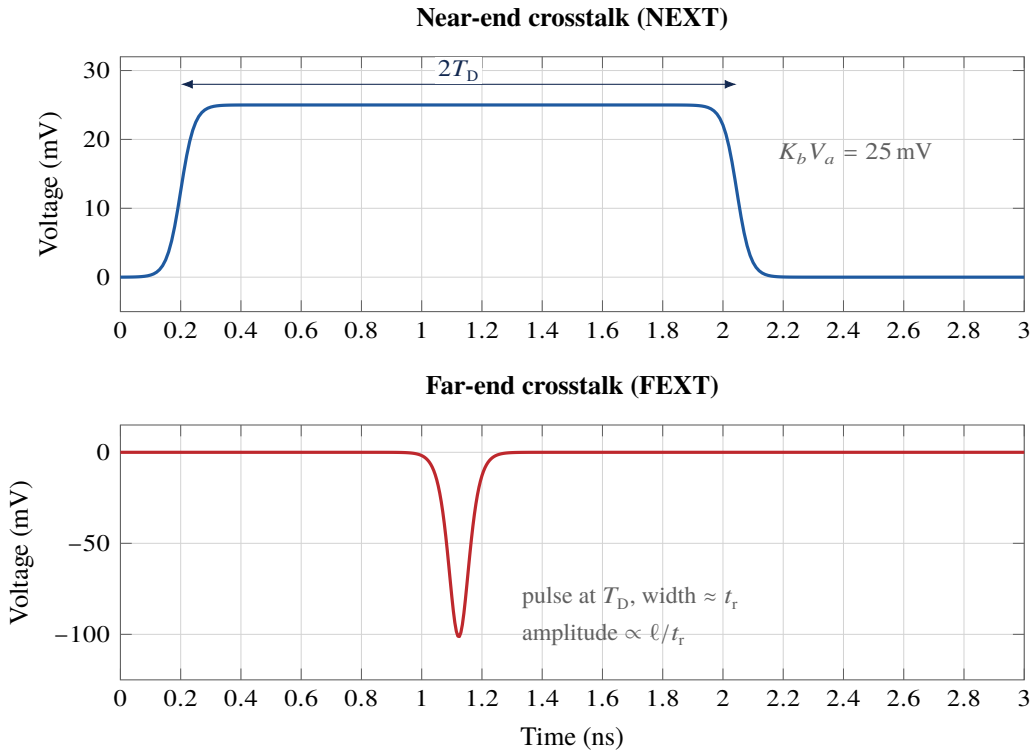



**Figure 9.3.** Crosstalk on a matched victim from a 1 V, 100 ps aggressor edge over 6 in of coupled microstrip ($C_m/C = 0.04$, $L_m/L = 0.06$). NEXT is a plateau of duration $2T_D$; FEXT is a negative pulse at $T_D$.

(the victim pulse is inverted relative to the aggressor edge). Embedded microstrip and stripline with unequal dielectrics (for example a resin-rich prepreg above and a glass-rich core below) have partial cancellation.

**Worked Example 9.1: Crosstalk on a microstrip bus**

Two 50 Ω microstrips are coupled over 6 in with $C_m/C = 0.04$ and $L_m/L = 0.06$; $v_p = 6.5$ in/ns. The aggressor is a 1 V edge with $t_r = 100$ ps. Find NEXT and FEXT.

**Solution.** $K_b = 0.25(0.04 + 0.06) = 0.025$. $T_D = 6/6.5 = 0.923$ ns, so $2T_D \gg t_r$ and NEXT saturates at 25 mV. $K_f \ell = (6/(2 \times 6.5))(0.04 - 0.06) = -9.23$ ps; with a linear ramp, $V_{FEXT} = -9.23/100 \times 1\text{ V} = -92$ mV. A smooth edge with the same 10–90 % time has a higher peak slope, about −100 mV in Figure 9.3. FEXT, not NEXT, dominates on long microstrip buses. *Validity:* $\Delta T = -2K_f \ell = 18.5$ ps, so $\Delta T/t_r = 0.18$ and the derivative approximation is adequate (Figure 9.2).

**Worked Example 9.2: Saturation length**

DDR data lines run in parallel stripline ($v_p = 6$ in/ns) with $t_r = 100$ ps. How long must the parallel run be before NEXT saturates, and what is NEXT for a 0.1 in coupled breakout?

**Solution.** $\ell_{sat} = 0.1\text{ ns} \times 6\text{ in/ns}/2 = 0.3$ in. A 0.1 in section has $2T_D = 33$ ps, so NEXT reaches $33/100 = 33$ % of its saturated value. Short breakout regions where traces must run close together therefore contribute much less NEXT than long parallel routes.

## 9.4 Termination of the victim

The expressions above assume that both ends of the victim are terminated in $Z_0$. A victim driven by a low-impedance driver at its near end ($\Gamma \approx -1$) reflects the NEXT, inverted, toward the far end, where it adds to FEXT one round trip later. A victim with a high-impedance receiver at its far end ($\Gamma \approx +1$) doubles the FEXT there, and reflects it back to the near end. A source-terminated victim absorbs the NEXT at the driver. These interactions are why crosstalk should be simulated with the actual drivers and receivers, and why series-terminated nets usually suffer less crosstalk than unterminated ones.

## 9.5 Spacing and the 3W rule

The coupling coefficients fall rapidly with separation. Johnson and Graham showed that, for microstrip, coupling decreases roughly as $1/[1 + (D/H)^2]$, where $D$ is the center-to-center spacing [1]. In stripline, the second plane confines the fields and coupling falls faster, approximately exponentially with $D/H$ (Figure 9.4).

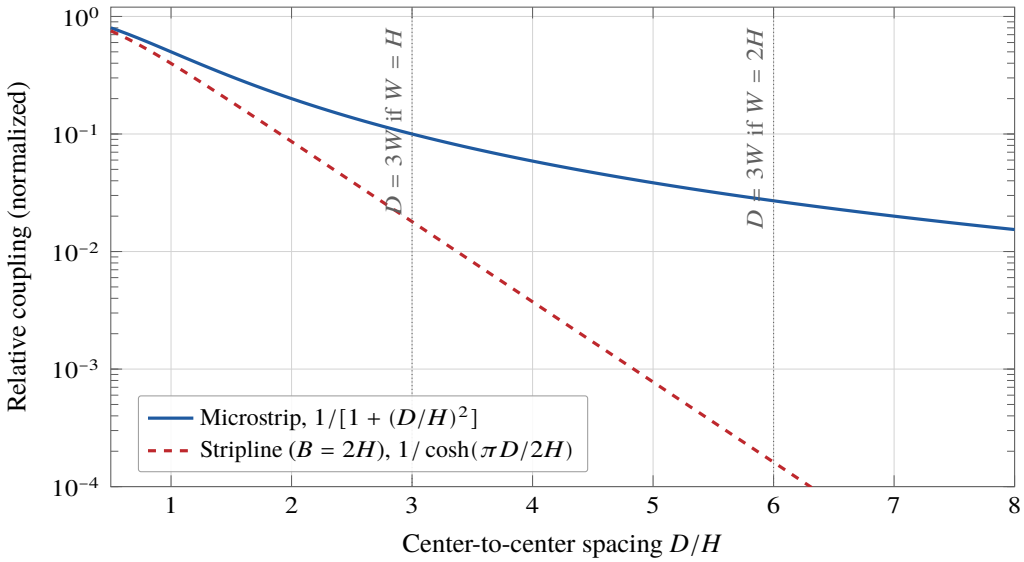



**Figure 9.4.** Relative coupling versus center-to-center spacing normalized to dielectric height. Microstrip coupling falls as $1/[1+(D/H)^2]$; stripline coupling falls approximately exponentially. Curves are normalized and show trends only; absolute values require a field solver.

The *3W rule* states that the center-to-center spacing of traces should be at least three times the trace width ($S \geq 2W$ edge to edge). Its physical basis is the fall-off of fringing fields with distance—but that distance is measured in units of the dielectric height $H$, not of $W$. The rule is reasonable only when $W$ and $H$ are comparable.

**Rule of Thumb: The 3W rule and its limits**

$D \geq 3W$ keeps microstrip NEXT to a few percent *when* $W \approx 2H$, as for a 50 Ω microstrip on FR-4. *Limits:* for a narrow trace over a thick dielectric (high impedance, $W < H$), 3W may leave the traces strongly coupled; for a wide trace over a thin dielectric, it is overly conservative. Specify spacing as a multiple of $H$ (typically $S \geq 2H$ to $3H$ edge to edge), and confirm with a field solver or crosstalk simulation.

**Worked Example 9.3: Is 3W enough?**

Compare the relative coupling at $D = 3W$ for (a) a microstrip with $W = 7$ mil, $H = 3.5$ mil, and (b) a microstrip with $W = 4$ mil, $H = 4$ mil.

**Solution.** (a) $D = 21$ mil $= 6H$, so $1/[1+36] = 0.027$. (b) $D = 12$ mil $= 3H$, so $1/[1+9] = 0.10$. The same "3W" spacing gives almost four times more coupling in case (b). Spacing (b) to $D = 6H = 24$ mil would restore the margin of case (a).

## 9.6 Guard traces

A grounded trace between aggressor and victim intercepts some of the coupled field. In stripline, where coupling is already small at moderate spacings, a guard trace adds little beyond what the extra space it occupies would provide. In microstrip it can reduce coupling by a factor of two or more—but only if it is well grounded. An unstitched or singly grounded guard is itself a resonant transmission line that can *increase* crosstalk at its resonant frequencies.

**Design Rule: Guard traces**

Use a guard trace only when spacing alone cannot meet the crosstalk budget. Ground it at both ends and stitch it to the reference plane with vias spaced no more than about one-tenth of the rise length, $s \le t_r v_p/10$. That spacing puts the half-wave resonance of each stitched segment, $v_p/(2s)$, at ten times the knee frequency. (The rule "$\lambda/10$ at $f_{knee}$" is often quoted too; since $\lambda(f_{knee}) = 2t_r v_p$, it allows twice the spacing, and its first segment resonance is only five times $f_{knee}$.) Never leave a guard trace floating or grounded at one end.

## 9.7 Other coupling paths

**Broadside coupling**
: Traces on adjacent signal layers that run parallel couple strongly because they face each other across a thin dielectric. In dual-stripline stackups, route adjacent layers orthogonally and avoid long parallel overlaps.

**Via coupling**
: Closely spaced vias, especially in BGA fields and connector pin fields, couple through the cavity between planes. Return vias adjacent to each signal via reduce this coupling (Chapter 11).

**Common return-path coupling**
: Signals that share a return path through a connector or package pin interact through the shared inductance; this is ground bounce (Chapter 15).

## 9.8 Crosstalk budgets

A victim usually has several aggressors. If their switching is uncorrelated, their peak contributions add worst-case only when all switch together; a conservative budget sums the peak values of the nearest aggressors on each side and adds a root-sum-square allowance for more distant ones. Crosstalk that arrives during the victim's own transition shifts its threshold crossing time and appears as timing jitter rather than as amplitude noise.

**Table 9.1.** Crosstalk characteristics and mitigations.

| | NEXT (backward) | FEXT (forward) |
|---|---|---|
| Coefficient | $K_b = \frac{1}{4}(C_m/C + L_m/L)$ | $K_f = \frac{1}{2v_p}(C_m/C - L_m/L)$ |
| Waveform | Plateau of length $2T_D$ | Pulse of width $\approx t_r$ at $T_D$ |
| Length dependence | Saturates at $\ell_{sat} = t_r v_p/2$ | $\propto \ell$ while $\lvert\Delta T\rvert \ll t_r$; bounded beyond |
| Edge dependence | Independent of $t_r$ once saturated | $\propto 1/t_r$ (small-$\Delta T$ regime) |
| Stripline | Present | Cancels in homogeneous dielectric |
| Mitigation | Spacing, source termination | Spacing, stripline routing, shorter parallel runs |

**Common Pitfall: Analyzing one aggressor**

Simulating a victim with only one neighbor. Buses have aggressors on both sides, and in dual-stripline stackups possibly above or below. Simulate at least the two nearest neighbors on each side switching in the worst-case pattern, with realistic driver strengths and terminations.

**Key Takeaways**

- Crosstalk arises from $C_m$ and $L_m$; it is characterized by $C_m/C$ and $L_m/L$.
- NEXT = $K_b V_a$ with $K_b = \frac{1}{4}(C_m/C + L_m/L)$; it saturates once the coupled length exceeds $t_r v_p/2$.
- FEXT grows with length and edge rate while $\lvert\Delta T\rvert \ll t_r$; beyond that it is bounded by modal separation and loss. It is negative in microstrip and cancels in a homogeneous stripline.
- Spacing should be specified relative to dielectric height $H$; the 3W rule is valid only when $W \approx 2H$.
- Guard traces help only if stitched densely to the plane.

## Further Reading

Paul [14] gives the multiconductor theory behind crosstalk; Hall and Heck [4] and Bogatin [3] derive NEXT and FEXT for practical geometries.

## Review Problems

**P9.1** Find $K_b$ for $C_m/C = 0.05$ and $L_m/L = 0.08$.

**P9.2** Find the NEXT saturation length for $t_r = 50\,\text{ps}$ and $v_p = 6\,\text{in/ns}$.

**P9.3** With the values of Problems P9.1 and P9.2, what NEXT does a 1 V aggressor produce over a 0.1 in coupled length?

**P9.4** Two microstrips are coupled over 4 in with $C_m/C = 0.04$, $L_m/L = 0.07$, $v_p = 6.5\,\text{in/ns}$. Estimate the FEXT from a 1 V, 80 ps edge.

**P9.5** In a stripline with $C_m/C = L_m/L = 0.05$, what are NEXT and FEXT for a 1 V aggressor?

**P9.6** By what factor does microstrip coupling fall when $D/H$ is increased from 3 to 6?

Chapter 10

# Differential Signaling

**Chapter Overview**

Nearly every interface above a gigabit per second uses differential pairs: PCIe, USB, Ethernet, MIPI, DDR clocks and strobes, and SERDES links of every kind. A pair carries information in the voltage difference between two conductors, which gives it immunity to common noise, low emissions, and a well-defined return path even when the ground reference is imperfect. Those advantages are realized only if the pair is symmetric. This chapter introduces odd- and even-mode analysis, defines differential and common-mode impedance, explains how skew converts differential signal into common-mode noise, and gives practical rules for length matching and serpentine design.

**Learning Objectives**

1. Decompose arbitrary signals on a pair into differential and common modes.
2. Compute odd-, even-, differential-, and common-mode impedances from per-unit-length parameters.
3. Design terminations that absorb both modes.
4. Quantify mode conversion caused by intra-pair skew.
5. Specify intra-pair and inter-pair skew budgets and implement length matching with serpentines.

## 10.1 Modes of a coupled pair

Any pair of voltages $(V_P, V_N)$ on the two traces can be written as a differential and a common-mode component:

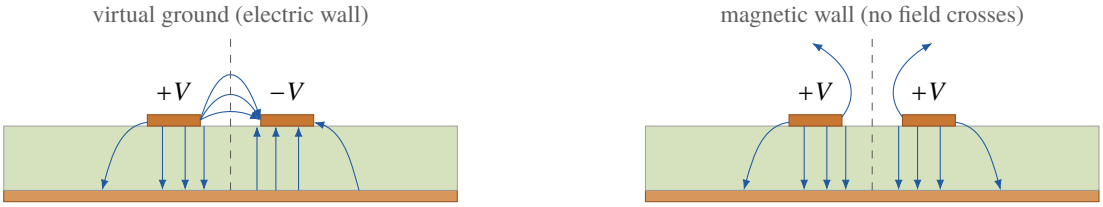
$$V_d = V_P - V_N, \qquad V_c = \frac{V_P + V_N}{2}, \tag{10.1}$$

with the inverse $V_P = V_c + V_d/2$ and $V_N = V_c - V_d/2$. A receiver that responds only to $V_d$ rejects any noise that appears equally on both traces. Driving the pair with equal and opposite signals ($V_c = 0$) excites the *odd mode*; equal signals ($V_d = 0$) excite the *even mode*. Their field patterns are sketched in Figure 10.1.

## 10.2 Odd, even, differential, and common-mode impedance

Let each trace have self inductance $L_{11}$ and capacitance to ground $C_g$, with mutual inductance $L_{12}$ and mutual capacitance $C_m$ between traces (per unit length). In the odd mode the mutual inductance opposes the self inductance and the mutual capacitance is effectively doubled (it is charged from both sides); in the even mode the mutual inductance aids and the mutual capacitance carries no charge. Hence

$$Z_{\text{odd}} = \sqrt{\frac{L_{11} - L_{12}}{C_g + 2C_m}}, \qquad Z_{\text{even}} = \sqrt{\frac{L_{11} + L_{12}}{C_g}}. \tag{10.2}$$




**(a)** Odd mode (differential): $Z_{odd} = Z_{diff}/2$ **(b)** Even mode (common): $Z_{even} = 2Z_{cm}$

**Figure 10.1.** Electric field patterns of an edge-coupled microstrip pair. (a) In the odd mode, field lines run strongly from one trace to the other, and the symmetry plane is a virtual ground. (b) In the even mode, both traces are at the same potential and fields are pushed toward the plane.

Each mode also has its own velocity, $v_p = 1/\sqrt{LC}$ with the corresponding $L$ and $C$; the two are equal only in a homogeneous dielectric.

The impedance seen by a differential signal is the series combination of the two odd-mode lines, and the impedance seen by a common-mode signal is their parallel combination:

$$Z_{diff} = 2Z_{odd}, \qquad Z_{cm} = \frac{Z_{even}}{2}. \tag{10.3}$$

For uncoupled traces, $Z_{odd} = Z_{even} = Z_0$; tighter coupling lowers $Z_{odd}$ and raises $Z_{even}$. A pair is designed for its target $Z_{diff}$ (85 Ω, 90 Ω, or 100 Ω, depending on the interface); $Z_{cm}$ is usually uncontrolled but matters for EMI and common-mode noise.

**Worked Example 10.1: Modal impedances of a coupled pair**

A field solver reports, per inch, $L_{11} = 8.5\,\text{nH}$, $L_{12} = 1.0\,\text{nH}$, $C_g = 3.0\,\text{pF}$, and $C_m = 0.3\,\text{pF}$. Find $Z_{odd}$, $Z_{even}$, $Z_{diff}$, and $Z_{cm}$.

**Solution.** From Eq. (10.2), $Z_{odd} = \sqrt{7.5\times10^{-9}/3.6\times10^{-12}} = 45.6\,\Omega$ and $Z_{even} = \sqrt{9.5\times10^{-9}/3.0\times10^{-12}} = 56.3\,\Omega$. Then $Z_{diff} = 91.3\,\Omega$ and $Z_{cm} = 28.1\,\Omega$.

### 10.2.1 Tight versus loose coupling

A common misconception is that the two traces of a pair must be tightly coupled for the pair to work. In fact the return current of a differential signal over a solid plane flows mostly in the plane under each trace, whatever the spacing; coupling only changes $Z_{odd}$. Loose coupling ($S \gtrsim 2W$) makes $Z_{diff}$ insensitive to spacing variation, eases routing around obstacles, and allows narrower traces for the same impedance. Tight coupling ($S \approx W$) gives denser routing, better rejection of noise that couples equally into both traces, and smaller loop area if the plane is interrupted. Most designs choose $S$ between $W$ and $2W$ and keep it constant along the route.

**Design Rule: Pair geometry**

Hold the pair's spacing constant along its length; split only where pins force it, and keep the uncoupled length shorter than about one-tenth of the rise length. Route both traces on the same layer, over the same reference plane, with the same number of vias placed symmetrically.

## 10.3 Terminating both modes

A single resistor $Z_{diff}$ across the pair terminates the odd mode but presents an open circuit to the even mode, which therefore reflects. Two networks terminate both modes:

**T network**

A resistor $R_1 = Z_{\text{odd}}$ from each trace to a center node, and $R_2 = (Z_{\text{even}} - Z_{\text{odd}})/2$ from that node to ground (or to AC ground through a capacitor).

**Π network**

A resistor $R_1 = Z_{\text{even}}$ from each trace to ground, and $R_2 = 2Z_{\text{even}}Z_{\text{odd}}/(Z_{\text{even}} - Z_{\text{odd}})$ across the pair.

For the pair of Worked Example 10.1, the T network uses $R_1 = 45.6\,\Omega$ and $R_2 = 5.3\,\Omega$. In practice the split termination—two resistors of $Z_{\text{diff}}/2$ with a capacitor to ground at the midpoint—is the common compromise; on-die terminations of modern receivers often implement it internally. It terminates the even mode in $Z_{\text{odd}}$ rather than $Z_{\text{even}}$: for this pair, $\Gamma_e = (45.6 - 56.2)/(45.6 + 56.2) = -0.10$. That is usually acceptable, but it is a mismatch, not a match.

## 10.4 Skew and mode conversion

### 10.4.1 Intra-pair skew

If one trace of a pair is delayed by $\Delta t$ relative to the other, a purely differential signal acquires a common-mode component. Writing the arriving voltages as $V_P = \frac{V_d}{2}e^{-j\omega t_1}$ and $V_N = -\frac{V_d}{2}e^{-j\omega(t_1+\Delta t)}$ and applying Eq. (10.1),

$$\left|\frac{V_c}{V_d}\right| = \frac{1}{2}\left|\sin(\pi f \Delta t)\right|, \qquad \left|\frac{V_{d,\text{out}}}{V_d}\right| = \left|\cos(\pi f \Delta t)\right|. \tag{10.4}$$

Skew therefore does two kinds of damage. It reduces the differential amplitude, slows the differential edge, and closes the eye horizontally; and it creates common-mode energy, which radiates from cables and couples into other circuits (Figure 10.2). In time-domain terms, the differential edge is smeared over $\Delta t$, and a common-mode pulse of width $\Delta t$ appears at every transition.

### 10.4.2 Volts, waves and watts

Equation (10.4) is a *voltage* ratio between modal voltages defined by Eq. (10.1). Mixed-mode S-parameters (Chapter 25) are ratios of *power waves*, normalized to reference impedances of $2Z_0$ for the differential mode and $Z_0/2$ for the common mode (100 Ω and 25 Ω for 50 Ω single-ended ports). With $a \propto V/\sqrt{Z_{\text{ref}}}$,

$$|S_{\text{CD21}}| = \left|\frac{V_c}{V_d}\right|\sqrt{\frac{2Z_0}{Z_0/2}} = 2\left|\frac{V_c}{V_d}\right| = \left|\sin(\pi f \Delta t)\right|, \tag{10.5}$$

so the two legitimate numbers differ by a factor of two, or 6.02 dB. The converted *power* fraction is $|S_{\text{CD21}}|^2$, and for this lossless model $|S_{\text{DD21}}|^2 + |S_{\text{CD21}}|^2 = \cos^2 + \sin^2 = 1$ (Figure 10.2). Always state which ratio a mode-conversion number is.

Sources of intra-pair skew include unequal trace lengths (especially at bends, where the outer trace is longer), asymmetric via placement, pin-escape asymmetry at BGAs and connectors, and glass-weave effects (Chapter 7). In mixed-mode S-parameters (Chapter 25), the resulting conversion appears as $S_{\text{CD21}}$ and $S_{\text{DC21}}$.

**Worked Example 10.2: Skew budget for a 10 Gb/s pair**

A USB 3.2 Gen 2 pair runs at 10 Gb/s (UI = 100 ps). If 5 % of a UI is allocated to intra-pair skew, what length mismatch is allowed in stripline with $t_{\text{pd}} = 160$ ps/in, and how large is the resulting common-mode voltage at the Nyquist frequency?

**Solution.** $\Delta t = 5$ ps, corresponding to $5/160 = 0.031$ in $= 31$ mil. At $f_{\text{Nyq}} = 5$ GHz, Eq. (10.4) gives $0.5\sin(\pi \times 5\times10^{9}\times5\times10^{-12}) = 0.039$, i.e. −28.1 dB: a 800 mV differential swing produces about 31 mV

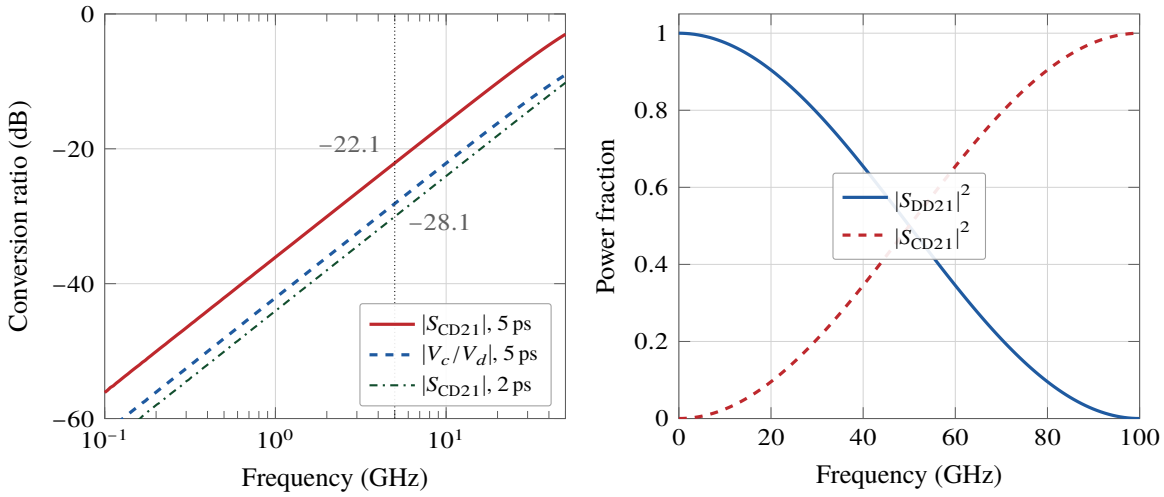



**Figure 10.2.** Mode conversion by intra-pair skew. Left: the modal voltage ratio of Eq. (10.4) and the power-wave ratio $|S_{CD21}|$ of Eq. (10.5), which is 6.02 dB higher. Right: differential and converted power fractions for 5 ps of skew; they sum to one in this lossless model. [Analytical]

of common-mode voltage at 5 GHz. The same skew gives $|S_{CD21}| = 0.078$ (−22.1 dB) by Eq. (10.5), and a converted power fraction of 0.62 %.

### 10.4.3 Inter-pair skew

Skew between different pairs matters only when the receiver must align them. Serial lanes with embedded clocks (PCIe, USB, Ethernet SERDES) deskew lanes in the receiver's logic, and their inter-pair skew tolerance is large—many nanoseconds in some specifications. Source-synchronous interfaces (DDR strobes and data, MIPI D-PHY clock and data lanes, RGMII) sample data with a forwarded clock, so their lane-to-clock skew budget is a fraction of a UI and must be matched carefully (Chapters 12, 19 and 23).

## 10.5 Length matching and serpentines

### 10.5.1 Where skew comes from at bends

When a pair turns, the outer trace travels further. For a pair of centerline pitch $p = W + S$ turning through an angle $\theta$ along a circular arc, the length difference is

$$\Delta\ell = \theta\, p, \tag{10.6}$$

or $(\pi/2)p$ for a 90° turn. Turns in opposite directions cancel. Because the skew of a bend appears *at the bend*, the compensation should be applied close to it, so that the pair is never misaligned for more than a small fraction of the rise length. A total length match achieved by a single bump at the far end of a long route still lets the signals travel misaligned—and converting mode—for most of the route.

**Worked Example 10.3: Skew from a bend**

A pair with $W = S = 5$ mil makes a single 90° turn in stripline ($t_{pd} = 165$ ps/in). Find the skew.

**Solution.** $p = 10\,\text{mil}$, so $\Delta\ell = (\pi/2) \times 10 = 15.7\,\text{mil}$, and $\Delta t = 0.0157 \times 165 = 2.6\,\text{ps}$. Three uncompensated turns in the same direction would use up the entire 5 %-UI skew budget of a 16 GT/s link.

### 10.5.2 Serpentine design

Serpentines (meanders) add length to the shorter trace. Their segments couple to one another, which makes the signal arrive slightly *earlier* than the straightened length predicts, because part of the edge propagates by crosstalk across the meander. The geometry rules of Figure 10.3 keep this error small.

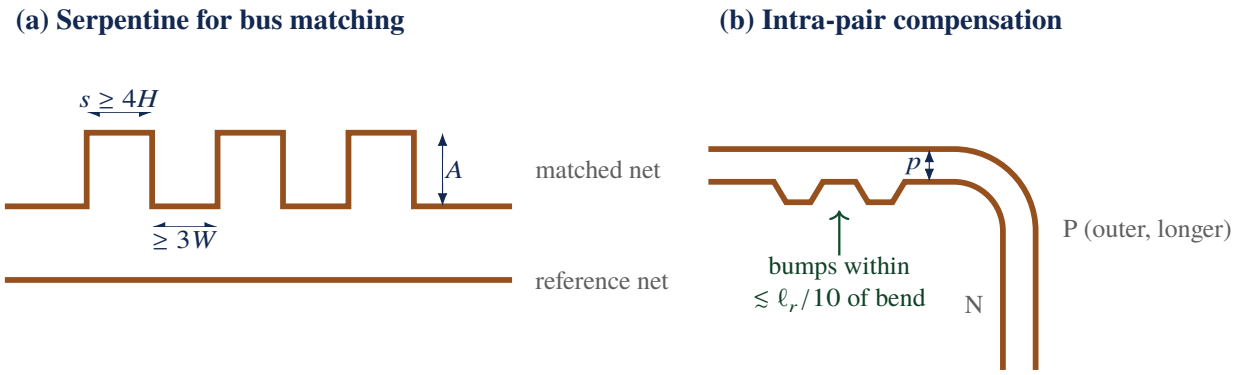



**Figure 10.3.** Length-matching geometries. (a) A serpentine for matching a single-ended net to a bus: segment spacing $s$ of at least $4H$ limits self-coupling. (b) Intra-pair compensation: small bumps on the inner trace placed immediately next to the bend that created the mismatch.

**Rule of Thumb: Serpentine dimensions**

Space meander segments by at least $3W$ to $4H$; make the parallel segment length at least 1.5 times the spacing and no longer than needed; prefer many low bumps to a few tall ones on differential pairs, keeping the bump height below about twice the pair spacing so that $Z_{\text{diff}}$ is disturbed only briefly. *Limits:* the coupling error grows with edge rate; above about 10 Gb/s, verify serpentine delay with a 3-D or 2.5-D field solver.

**Common Pitfall: Matching length instead of delay**

Matching physical length on layers with different propagation delay. A microstrip segment is faster than an equal stripline segment, and a via adds delay that layout tools often ignore. Matching must be done in units of time; modern layout tools can do this if each layer's $t_{\text{pd}}$ and the via delays are entered.

**Key Takeaways**

- Signals on a pair decompose into differential ($V_P - V_N$) and common ($\frac{1}{2}(V_P + V_N)$) modes.
- $Z_{\text{diff}} = 2Z_{\text{odd}}$ and $Z_{\text{cm}} = Z_{\text{even}}/2$; coupling lowers $Z_{\text{odd}}$ and raises $Z_{\text{even}}$.
- Loose or tight coupling both work over a solid plane; consistency of spacing matters more than its value.
- Intra-pair skew $\Delta t$ converts differential signal to common mode in the ratio $\frac{1}{2}\,|\sin \pi f\,\Delta t|$.

**Table 10.1.** Typical skew targets expressed as a fraction of a UI. Interface specifications and controller design guides take precedence.

| Skew type | Basis | Typical design target |
|---|---|---|
| Intra-pair (serial) | Mode conversion, differential eye | 2–5 % of UI |
| Inter-pair (embedded clock) | Receiver deskew range | Loose; see specification |
| Clock to data (source-synchronous) | Setup/hold budget | 2–10 % of UI after training |
| Within a DDR byte lane | DQ/DQS timing budget | a few picoseconds to tens of picoseconds |

- Compensate skew close to where it arises, and match delay, not length.

## Further Reading

Bogatin [3] and Hall and Heck [4] cover odd and even modes, mixed-mode S-parameters, and mode conversion; Johnson and Graham [2] discuss differential signaling and skew.

## Review Problems

**P10.1** A pair has $Z_{\text{odd}} = 48\ \Omega$ and $Z_{\text{even}} = 60\ \Omega$. Find $Z_{\text{diff}}$ and $Z_{\text{cm}}$.

**P10.2** Design a T termination for the pair of Problem P10.1.

**P10.3** What skew results from a 20 mil length mismatch in stripline with $t_{\text{pd}} = 170\ \text{ps/in}$?

**P10.4** Find the common-mode-to-differential ratio at 8 GHz for 4 ps of intra-pair skew.

**P10.5** A pair with pitch $p = 12$ mil makes four 90° turns in the same direction. Find the length and time skew ($t_{\text{pd}} = 170\ \text{ps/in}$).

**P10.6** Allocate 3 % of the UI of an 8 GT/s link to intra-pair skew. What length mismatch is allowed at 170 ps/in?

Chapter 11

# Vias and Discontinuities

**Chapter Overview**

A uniform transmission line is easy to design; the trouble lies in the places where it stops being uniform. Vias, pads, connector pins, BGA escapes, and reference-plane transitions all present local excess capacitance or inductance, and the unused portion of a through via forms a resonant stub. Above roughly 10 Gb/s, these discontinuities often dominate channel performance. This chapter models vias as lumped elements and as short transmission lines, explains stub resonance and back-drilling, surveys blind, buried, and micro-via technologies, and gives layout rules for anti-pads, return vias, and BGA and connector breakout.

**Learning Objectives**

1. Estimate the inductance and capacitance of a plated through via.
2. Treat a via as a short transmission line and estimate its impedance.
3. Compute via-stub resonant frequency and decide when back-drilling or blind vias are required.
4. Compare through, blind, buried, and micro-via structures.
5. Place return (stitching) vias and size anti-pads for controlled transitions.
6. Plan BGA and connector breakout for high-speed pairs.

## 11.1 Anatomy of a via

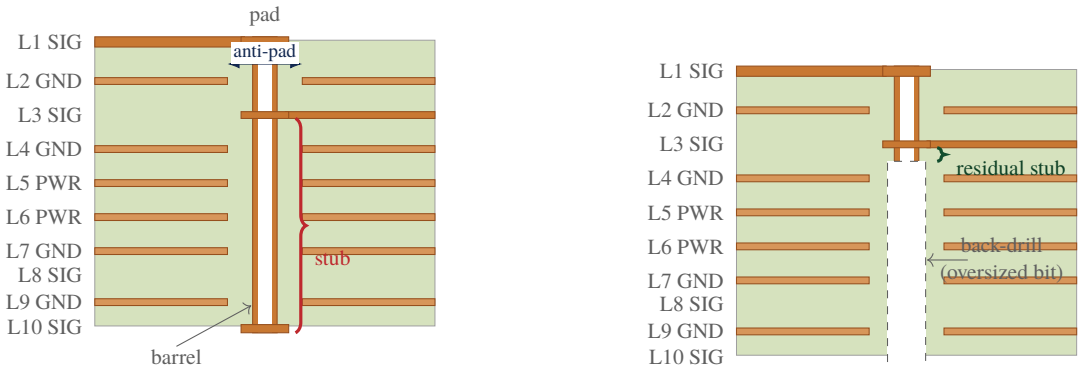
A plated through-hole (PTH) via consists of a drilled and plated *barrel*, *pads* on the layers where traces connect, and *anti-pads*—clearance holes—in the planes it passes through (Figure 11.1). Non-functional pads on unconnected layers are usually removed for high-speed vias, since they add capacitance without benefit.

## 11.2 Lumped via models

When a via is short compared with the rise length, it can be represented by a series inductance and a shunt capacitance. Johnson and Graham give two widely used estimates [1]. The partial inductance of a via of length $h$ and barrel diameter $d$ (both in inches) is

$$L_{\text{via}} \approx 5.08\, h \left[ \ln \frac{4h}{d} + 1 \right] \quad \text{nH}, \tag{11.1}$$

an empirical estimate. It is *not* the partial self-inductance of Eq. (3.6): with $\mu_0/2\pi = 5.08\,\text{nH/in}$ and $2\ell/r = 4h/d$, that expression gives $5.08\,h[\ln(4h/d) - 1]$, with $-1$ where Johnson and Graham have $+1$. For a short via the two differ by nearly a factor of two. Neither is the loop inductance the signal actually




**(a)** Through via, L1 → L3, with a long stub

**(b)** Back-drilled from the bottom

**Figure 11.1.** Cross-section of a through via carrying a signal from L1 to L3 in a 10-layer board. (a) As drilled, the barrel below L3 forms an open stub. (b) After back-drilling with an oversized bit from the bottom, only a short residual stub remains.

sees, which depends on where the return current flows (Section 11.7); treat the spread as the uncertainty of a lumped via model. The capacitance of a via pad of diameter $D_1$ in a board of thickness $T$ with anti-pad diameter $D_2$ is

$$C_{\text{via}} \approx \frac{1.41\,\varepsilon_r T D_1}{D_2 - D_1} \quad \text{pF}. \tag{11.2}$$

Both are rough: Eq. (11.1) is a partial inductance whose loop value depends on where the return current flows, and Eq. (11.2) lumps the barrel and pad capacitances into one empirical term.

**Worked Example 11.1: Lumped model of a through via**

Estimate the inductance and capacitance of a via in a 62 mil board with a 10 mil drill, 20 mil pads, 32 mil anti-pads, and $\varepsilon_r = 4.4$, and the time constant it presents to a 50 Ω line.

**Solution.** Equation (11.1): $L = 5.08 \times 0.062[\ln(24.8) + 1] = 1.33$ nH (the partial self-inductance with $-1$ would give 0.70 nH). Equation (11.2): $C = 1.41 \times 4.4 \times 0.062 \times 0.020 / 0.012 = 0.64$ pF. As a mid-line capacitance the via has $\tau = Z_0 C/2 = 16$ ps; as a series inductance, $\tau = L/(2Z_0) = 13$ ps. For a 1 ns edge it is invisible; for a 30 ps edge it is a significant discontinuity, and its capacitive and inductive parts must be balanced.

## 11.3 The via as a transmission line

A via surrounded by return vias is a short coaxial-like transmission line. If the return vias form a ring of effective diameter $D_{\text{ret}}$ around a barrel of diameter $d$, its impedance is approximately

$$Z_{\text{via}} \approx \frac{60}{\sqrt{\varepsilon_r}} \ln \frac{D_{\text{ret}}}{d}, \tag{11.3}$$

modified by the pads and anti-pads it passes, which add capacitance and lower the impedance. A typical signal via with a 10 mil drill and return vias at about 40 mil diameter has $Z_{\text{via}} \approx 43$ Ω in $D_k = 3.8$ material, close enough to 50 Ω that careful anti-pad sizing can tune it. Its delay is about $h\sqrt{\varepsilon_r}/c$, roughly 10 ps through a 62 mil board.

## 11.4 Anti-pads

An *anti-pad* (also written antipad) is a clearance area around a via or plated through-hole in a copper plane, where copper is intentionally removed. Its first purpose is to prevent an unwanted electrical connection: a signal via must pass through ground and power planes without touching them. Its second purpose, which matters as much in high-speed design, is to control the capacitance—and therefore the impedance—of the via. Anti-pads are particularly important in PCIe, DDR, USB, Ethernet and other SerDes designs. An anti-pad is not simply a manufacturing clearance; it is a signal-integrity design parameter that sets the electromagnetic behavior of the via.

### 11.4.1 What an anti-pad looks like

Imagine a signal via passing through a board with several ground planes (Figure 11.2). The structure has four elements:

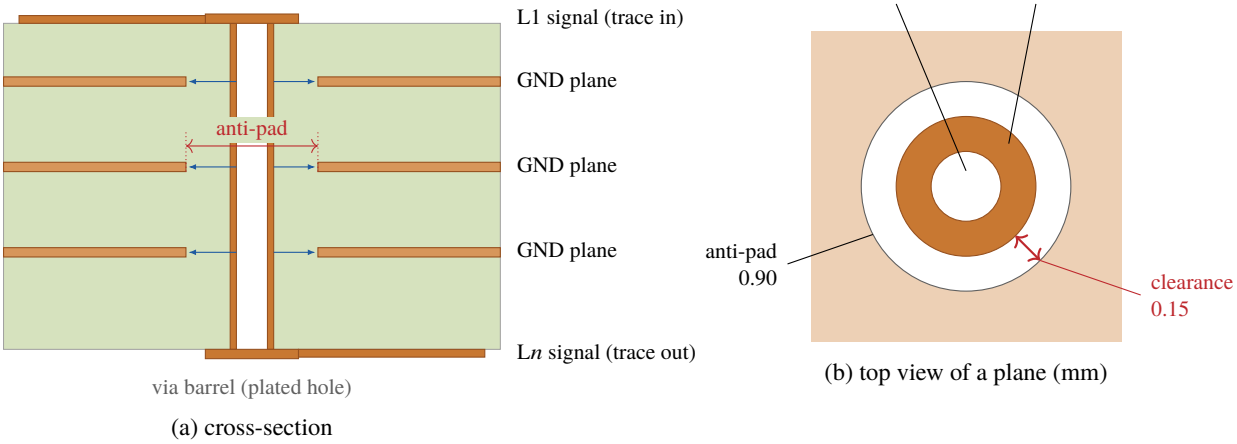
**Signal via**
carries the high-speed signal between layers.

**Via barrel**
the plated hole that forms the conductor; its *pads* are the copper rings where traces connect (functional pads) or where no trace connects (non-functional pads).

**Ground plane**
provides the return-current path for the signal on each side of the transition.

**Anti-pad**
the copper-free clearance surrounding the via on every plane it passes through without connecting.

The electric field of the via terminates on the plane edges at the anti-pad boundaries (Figure 11.2a), so the anti-pad diameter fixes how close the return conductor is to the barrel.




(a) cross-section

(b) top view of a plane (mm)

**Figure 11.2.** A signal via passing through three ground planes. (a) Cross-section: the field (arrows) runs from the barrel to the plane edges at each anti-pad. (b) Top view of one plane with an illustrative 0.30 mm drill, 0.60 mm pad and 0.90 mm anti-pad: the radial clearance between pad and plane is 0.15 mm. Not to scale. [Conceptual]

### 11.4.2 Why the anti-pad matters for signal integrity

A via is a short, non-uniform transmission line with parasitic inductance (from the barrel and its current loop) and parasitic capacitance (from the barrel and pads to the surrounding planes). The anti-pad determines much of that capacitance:

- *If the anti-pad is too small,* the plane edges come close to the barrel and pads, the capacitance increases, the via impedance falls below that of the line, and the impedance step reflects part of the signal.
- *If the anti-pad is too large,* the capacitance decreases and the via impedance rises, but the return path moves away from the signal, the plane loses copper, adjacent anti-pads may merge into slots, and neighboring traces may lose their reference over the void (Figure 11.5b). The larger opening also increases coupling between vias that share the plane cavity.

Anti-pad dimensions therefore have an optimum, and it is found by balancing the via's capacitance against its inductance. In the dense-plane limit—planes closely spaced compared with the anti-pad—the field around the barrel is nearly coaxial, with the anti-pad edge as the outer conductor. Then Eq. (11.3) applies with $D_{\text{ret}}$ replaced by the anti-pad diameter $D_{\text{ap}}$:

$$Z_{\text{via}} \approx \frac{60}{\sqrt{\varepsilon_{\text{r}}}} \ln \frac{D_{\text{ap}}}{d}. \tag{11.4}$$

Because pads add capacitance and the gaps between planes let the field fringe, the real impedance differs, but the trend is right: impedance rises logarithmically with anti-pad size (Figure 11.3), while the pad capacitance of Eq. (11.2) falls as $1/(D_{\text{ap}} - D_1)$.

In the lumped picture of Section 11.2, the via is an inductance $L_{\text{via}}$ and a capacitance $C_{\text{via}}$, with an equivalent impedance and a characteristic frequency

$$Z_{\text{via}} \approx \sqrt{\frac{L_{\text{via}}}{C_{\text{via}}}}, \qquad f_0 = \frac{1}{2\pi\sqrt{L_{\text{via}} C_{\text{via}}}}. \tag{11.5}$$

Changing the anti-pad changes $C_{\text{via}}$, and therefore both. For an isolated L–C pair, $f_0$ is a resonance. For a via embedded in a matched channel, $L$ and $C$ are spread along the barrel, and $f_0 = 1/(2\pi\sqrt{LC})$ is better read as the frequency at which the via's electrical length reaches about one radian—where a lumped model already has an error of tens of percent (Section 4.6.1). Above roughly $f_0/3$, model the via as a transmission line or with a 3-D field solver.

**Worked Example 11.2: Clearance, capacitance and impedance of one anti-pad**

A via has a 0.30 mm drill and 0.60 mm pads, and passes through planes with 0.90 mm anti-pads in a 1.6 mm board with $\varepsilon_{\text{r}} = 3.8$. Find the radial clearance, the coaxial-limit impedance, and the lumped estimates of Eqs. (11.1), (11.2) and (11.5). What anti-pad would the coaxial limit need for 50 Ω?

**Solution.** The radial copper clearance between pad and plane is

$$\text{clearance} = \frac{D_{\text{ap}} - D_{\text{pad}}}{2} = \frac{0.90 - 0.60}{2} = 0.15\,\text{mm},$$

and between the barrel and the plane, where non-functional pads are removed, $(0.90 - 0.30)/2 = 0.30$ mm. Equation (11.4): $Z_{\text{via}} = 30.8 \ln(0.90/0.30) = 33.8\,\Omega$, well below 50 Ω: the via is capacitive. For 50 Ω the anti-pad would have to be $0.30\, e^{50/30.8} = 1.52$ mm, more than most pitches allow. With $h = 63$ mil and $d = 11.8$ mil, Eq. (11.1) gives $L_{\text{via}} = 1.30$ nH and Eq. (11.2) $C_{\text{via}} = 0.68$ pF; then $\sqrt{L/C} = 44\,\Omega$ and $f_0 = 5.4$ GHz. Enlarging the anti-pad to 1.20 mm halves the capacitance to 0.34 pF.

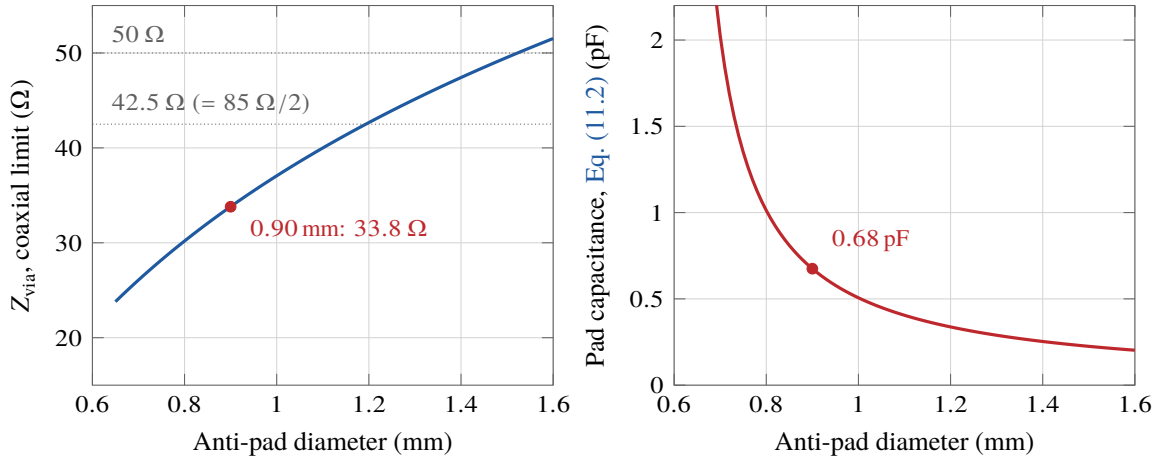



**Figure 11.3.** Effect of anti-pad diameter for a 0.30 mm drill and 0.60 mm pads in a 1.6 mm board with $\varepsilon_r = 3.8$. Left: coaxial-limit impedance from Eq. (11.4). Right: Johnson–Graham pad capacitance from Eq. (11.2). Trends only; final dimensions come from a 3-D solver. [Analytical]

The two estimates of impedance disagree (34 Ω and 44 Ω), and the lumped $\sqrt{LC} = 30$ ps is about three times the barrel's physical delay of 10 ps. Both are symptoms of the same fact: the formulas are trend indicators. These dimensions are illustrative, not a recommended PCIe rule.

| Quick estimate | Full model | Difference |
|---|---|---|
| Coaxial limit: 34 Ω | Lumped $\sqrt{L/C}$: 44 Ω | 30 % |

Figure 11.4 shows what the impedance trend means for a signal. A 25 ps TDR edge sees a capacitive dip at the via whose depth shrinks as the anti-pad grows, and the return loss at 16 GHz improves from about 6 dB to better than 17 dB between the 0.70 mm and 1.20 mm anti-pads. The via is electrically short (10 ps), so the 26 Ω section of the smallest anti-pad reads about 31 Ω on the TDR: the resolution limit of Section 26.1.2 understates the local mismatch, and the return loss reveals it.

### 11.4.3 Selecting anti-pad dimensions

There is no universal anti-pad dimension. It depends on:

1. the via drill diameter and pad size (and whether non-functional pads are kept);
2. the stackup: which layers are planes and how far apart they are;
3. the dielectric thickness and material;
4. the signal frequency and data rate, which set how much mismatch is tolerable;
5. the target impedance: 50 Ω single-ended, or an odd-mode impedance of 42.5 Ω, 45 Ω or 50 Ω for 85, 90 and 100 Ω differential;
6. the distance between reference planes, which decides how coaxial the field is;
7. the via stub length and geometry, and any back-drilling.

**Non-functional pads.** Pads on layers where no trace connects add capacitance without purpose. Removing them ("NFP removal") raises via impedance and is common for high-speed vias; some fabricators prefer to keep them on thick boards for drilling and plating reliability, so the choice belongs in the fabrication notes.

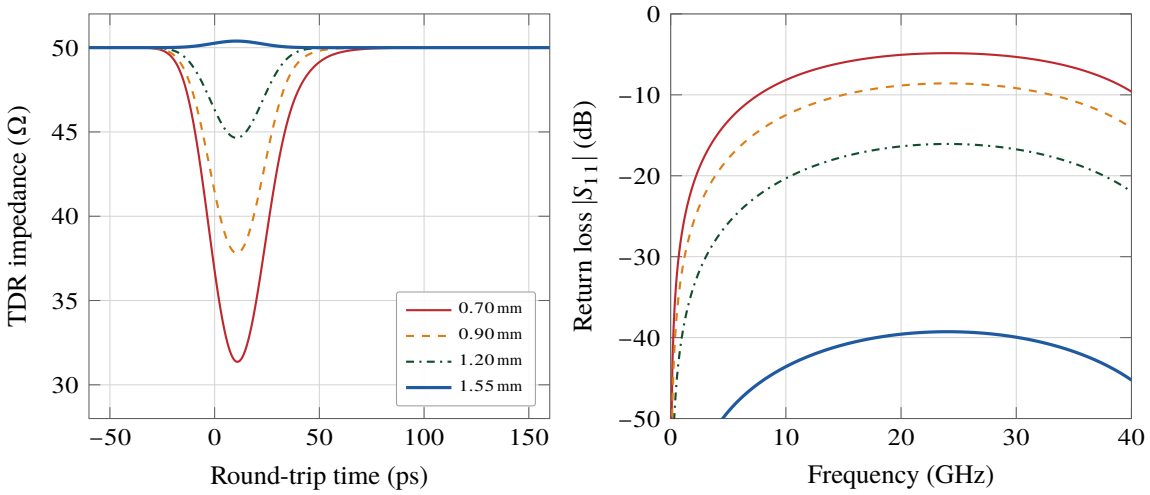



**Figure 11.4.** Simulated TDR (left, 25 ps Gaussian edge) and return loss (right) of a 1.6 mm via between 50 Ω lines for anti-pad diameters of 0.70, 0.90, 1.20 and 1.55 mm, modeled as a uniform line with the coaxial-limit impedance of Eq. (11.4) and 10.4 ps delay. No pads or stub. [Simulation (transmission-line model)]

**Layer-specific anti-pads.** Anti-pads need not be the same on every plane. Planes adjacent to the entry and exit layers carry most of the return current and are often given the carefully tuned value, while planes in the middle of a long barrel can be larger. On back-drilled layers the anti-pad must clear the back-drill bit, which is larger than the plated hole, plus its positional tolerance.

**Differential via pairs.** A pair is normally given a single obround ("stadium") anti-pad around both vias, with a ground return via on each side, symmetric (Figure 11.5a). The shared opening removes the plane copper between the two vias, which raises the pair's differential impedance and keeps both legs symmetric. As an upper bound, two parallel cylinders of diameter $d$ at pitch $p$ with no planes have

$$Z_{\text{diff}} = \frac{120}{\sqrt{\varepsilon_{\text{r}}}} \operatorname{arcosh} \frac{p}{d}, \tag{11.6}$$

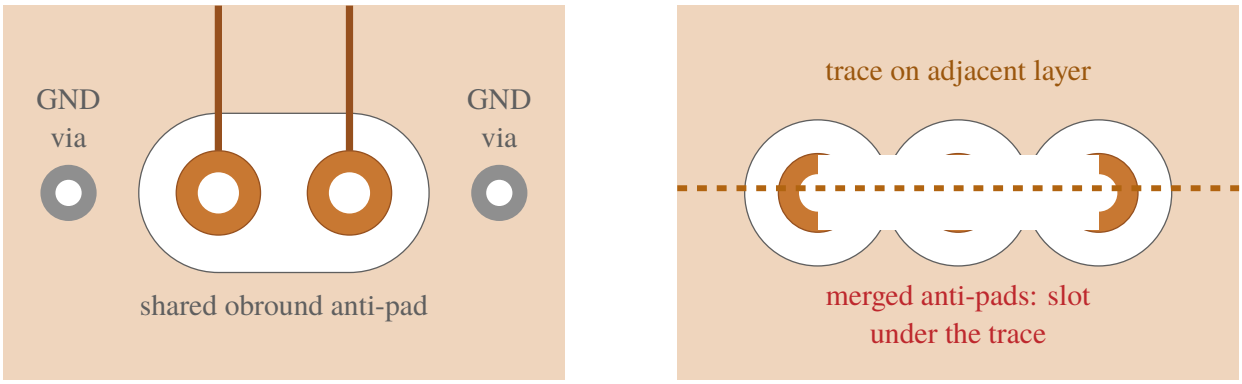
about 115 Ω for 0.30 mm vias at 1.0 mm pitch in $\varepsilon_{\text{r}} = 3.8$; planes, pads and anti-pad edges pull the real value down toward the 85–100 Ω targets.

**Plane integrity.** Every anti-pad removes plane copper. Under a BGA, rows of anti-pads leave only narrow webs between vias; if they merge, the plane is slotted for every trace that crosses it, and the power distribution network loses capacitance and gains spreading inductance and resistance (Chapter 16).

**Worked Example 11.3: How large can the anti-pads be under a BGA?**

Signal vias on a 1.0 mm grid pass through a ground plane. The fabricator requires a plane web of at least 0.15 mm between anti-pads. What is the largest round anti-pad, and what coaxial-limit impedance does it give with a 0.30 mm drill in $\varepsilon_{\text{r}} = 3.8$?

**Solution.** Along a row the anti-pad may be at most $1.0 - 0.15 = 0.85$ mm in diameter (diagonal neighbors at 1.41 mm are not the limit). The 0.90 mm anti-pad of Worked Example 11.2 would leave only a 0.10 mm web. At 0.85 mm, $Z_{\text{via}} = 30.8 \ln(0.85/0.30) = 32\ \Omega$. Higher impedance must then come from other means: removing non-functional pads, obround anti-pads for pairs aligned with the rows, smaller drills (micro-vias), or accepting the discontinuity and confirming it in the channel simulation.




(a) good: symmetric return vias

(b) bad: void under a neighbor

**Figure 11.5.** Top view of a plane under a differential via pair. (a) A shared obround anti-pad with symmetric ground return vias. (b) Oversized anti-pads that merge into a slot under a trace on the adjacent layer, interrupting its return path. [Conceptual]

### 11.4.4 Anti-pad, thermal relief and soldermask

Three plane and surface features are easily confused (Table 11.1).

**Table 11.1.** Features around vias and pads.

| Feature | Purpose | High-speed relevance |
|---|---|---|
| Anti-pad | Removes copper around a via on a plane it must not touch | Sets via capacitance and impedance; affects return path and plane integrity |
| Thermal relief | Connects a pad to a plane through narrow spokes, so it can be soldered | Adds inductance; avoid on high-current and decoupling connections |
| Soldermask | Insulating coating over outer copper; defines solder openings | Lowers outer-layer impedance slightly; tents or plugs vias |
| Via pad | Copper ring on a layer where the via connects | Adds capacitance; size per drill and annular-ring rules |
| Non-functional pad | Pad on a layer with no connection | Adds capacitance; often removed for high-speed vias |

### 11.4.5 A PCIe example

Suppose a PCIe Gen4 differential pair transitions from L1 to L3 of a ten-layer board (Figure 11.6). At the transition:

- the signal vias introduce parasitic inductance and capacitance;
- the anti-pad geometry sets the via impedance;
- ground stitching vias next to the pair maintain the return-current path between the reference planes (Section 11.7);
- the via stub below L3 resonates and reflects unless it is back-drilled (Section 11.5).

Optimize the anti-pad dimensions with a 3-D electromagnetic field solver, and verify the whole channel, including this transition, with S-parameters (Chapter 25).

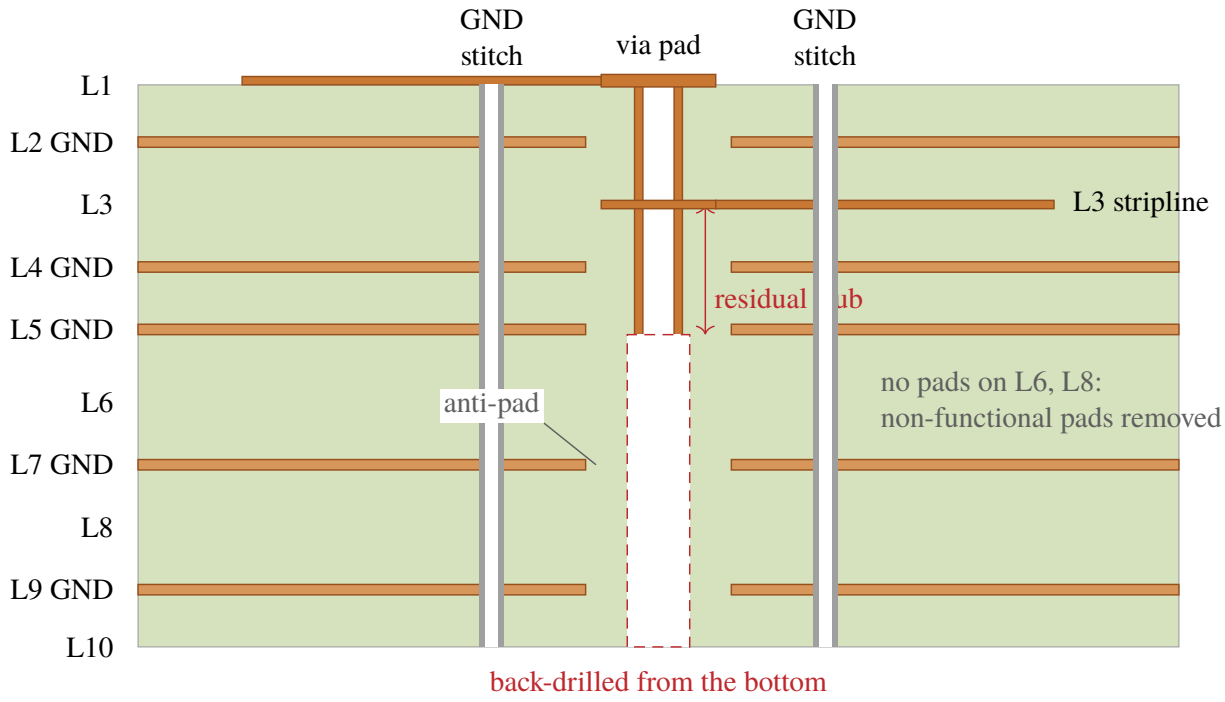



**Figure 11.6.** One leg of a PCIe via transition from L1 to L3 in a ten-layer board: via pad, functional pad on L3, anti-pads in the ground planes, non-functional pads removed, back-drilled barrel with a short residual stub, and ground stitching vias. Not to scale. [Conceptual]

**Design Rule: Anti-pad checklist for high-speed vias**

1. Start from the target: single-ended or odd-mode impedance of the connected lines.
2. Decide non-functional pads (remove where the fabricator allows).
3. Use one obround anti-pad per differential pair, with a symmetric ground via on each side.
4. Size anti-pads with a 3-D solver for the actual stackup, drill, pads and stub; tune the planes next to the entry and exit layers first.
5. Check plane webs between neighboring anti-pads (BGA rows, connector fields) against the fabricator's minimum, and look for merged voids.
6. Check that no trace on an adjacent layer crosses an anti-pad void.
7. On back-drilled layers, clear the back-drill diameter plus tolerance.
8. Verify the transition in the channel S-parameters and TDR at the link's edge rate.

**Common Pitfall: Anti-pads from the library default**

Using the CAD library's default plane clearance, chosen for manufacturing, on multi-gigabit vias. The result is usually a capacitive via, and the mismatch is hidden in a slow TDR. Treat the anti-pad as a designed dimension, recorded per via type in the stackup and constraint set.

## 11.5 Via stubs and back-drilling

When a through via connects two layers near the top of a board, the barrel below the lower connection is an open-circuited transmission line hanging off the signal path. From Table 4.1, an open stub becomes a short

circuit when its length equals a quarter wavelength, which occurs at

$$f_{\lambda/4} = \frac{c}{4\,\ell_{\text{stub}}\sqrt{\varepsilon_{\text{eff}}}} \approx \frac{11.8\,\text{GHz}\cdot\text{in}}{4\,\ell_{\text{stub}}\sqrt{\varepsilon_{\text{eff}}}}. \tag{11.7}$$

Near this frequency, the stub shunts the signal to ground and $S_{21}$ shows a deep notch (Figure 11.7). Below the notch, the stub behaves as a capacitance that increases loss and reflections. Pads and anti-pads add capacitance along the stub. That loading *slows* the wave—it raises the effective permittivity seen by the stub mode—and adds end capacitance, so the true resonance is often 10–30 % below the estimate of Eq. (11.7).

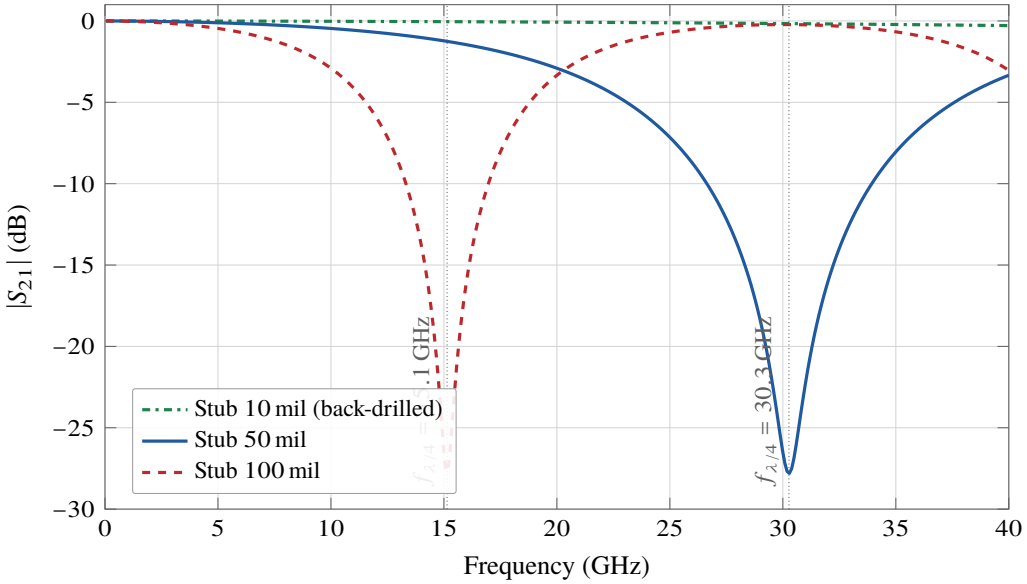



**Figure 11.7.** Insertion loss of a 50 Ω through path loaded by an open via stub ($Z_{\text{stub}} = 45\,\Omega$, $\varepsilon_{\text{r}} = 3.8$), computed from $S_{21} = 2/(2 + Z_0 Y_{\text{stub}})$ with $Y_{\text{stub}} = \tanh(\gamma\ell)/Z_{\text{stub}}$. The quarter-wave notch moves down in frequency as the stub lengthens.

**Rule of Thumb: Stub length**

Keep the quarter-wave resonance of every via stub at least three times above the Nyquist frequency of the signal it carries. *Limits:* this is a common design heuristic, not a specification requirement; the actual penalty depends on the stub's loading, the channel loss, and the receiver's equalization. Verify with a channel simulation that includes the 3-D via model.

**Worked Example 11.4: Does a Gen5 via need back-drilling?**

A PCIe Gen5 pair ($f_{\text{Nyq}} = 16\,\text{GHz}$) transitions from L1 to L3 in the 62.6 mil stackup of Figure 7.3, leaving a stub of about 50 mil. With $\varepsilon_{\text{eff}} \approx 3.8$, is the stub acceptable?

**Solution.** Equation (11.7) gives $f_{\lambda/4} = 11.8/(4 \times 0.050 \times 1.949) = 30.3\,\text{GHz}$, below the target $3f_{\text{Nyq}} = 48\,\text{GHz}$. Meeting the target requires $\ell_{\text{stub}} \le 11.8/(4 \times 1.949 \times 48) = 0.0315\,\text{in}$, about 32 mil. Options: back-drill from the bottom, route the pair on L8 so that the via spans most of the board (stub from L8 to L10 only), or use a blind via.

### 11.5.1 Design the tolerance, not just the nominal stub

Worked Example 11.4 produced a *maximum* residual stub of 31.5 mil. A back-drill does not land exactly on target: if the process can leave up to 8 mil more than intended, the nominal target must be at most $31.5 - 8 \approx 23.5$ mil. The permittivity matters too: from Eq. (11.7), a 5 % rise in $\varepsilon_{\text{eff}}$ lowers the notch by a factor $1/\sqrt{1.05} = 0.976$, about 2.4 %. Figure 11.8 shows both effects for an ideal 45 Ω stub on a 50 Ω path. The "$3 \times f_{\text{Nyq}}$" criterion is itself a heuristic; the allocation shown is illustrative, not a compliance requirement.

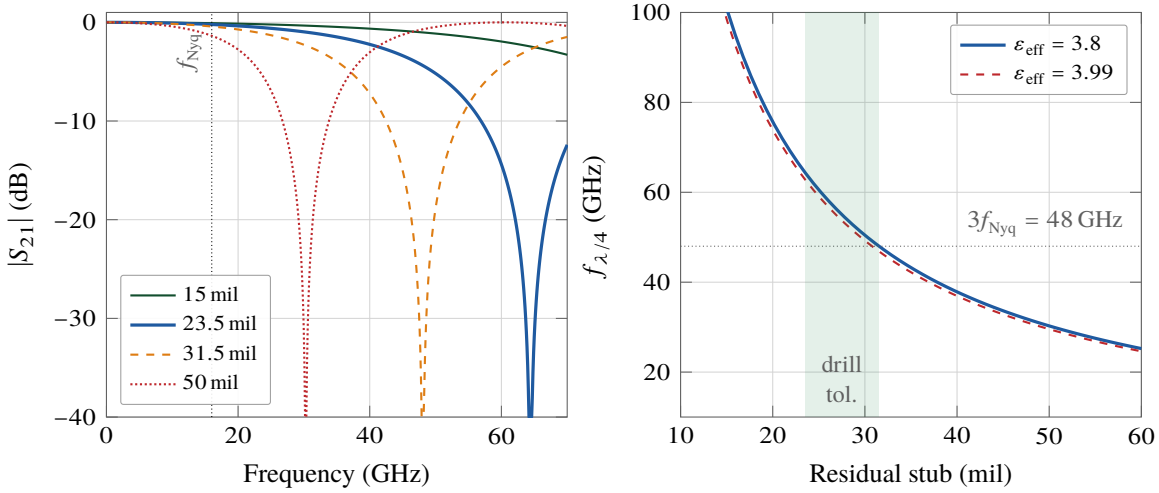



**Figure 11.8.** Left: ideal shunt open-stub transmission for residual stubs of 15, 23.5, 31.5 and 50 mil ($Z_{\text{main}} = 50\ \Omega$, $Z_{\text{stub}} = 45\ \Omega$, $\varepsilon_{\text{eff}} = 3.8$). Right: quarter-wave frequency versus residual stub length for $\varepsilon_{\text{eff}} = 3.8$ and $\varepsilon_{\text{eff}} = 3.99$ (5 %), with the 48 GHz target. Transmission-line calculation, not a 3-D via extraction. [Analytical]

Back-drilling removes the unused barrel with a drill slightly larger than the original hole, from the side opposite the connection. The drill depth must stop short of the connected layer, so a residual stub remains; its length depends on the fabricator's depth-control capability and is typically specified as a maximum in the fabrication notes.[1] Back-drilled holes must be identified on a separate drill layer with their target depth or the "must-not-cut" layer (Chapter 28).

## 11.6 Blind, buried, and micro-vias

**Blind vias**
: connect an outer layer to one or more inner layers without passing through the board; they have no stub on the far side. Mechanically drilled blind vias require sequential lamination or controlled-depth drilling.

**Buried vias**
: connect inner layers only and are drilled and plated in a sub-lamination before the outer layers are added.

**Micro-vias**
: are laser-drilled blind vias that span one thin dielectric layer, with typical aspect ratios (depth to diameter) of about 0.75:1 to 1:1. They form the basis of high-density interconnect (HDI) and enable fine-pitch BGA escape. Stacking micro-vias directly on top of one another saves space but has known reliability concerns under thermal cycling; staggered micro-vias are preferred where space allows, and stacked structures should be qualified with the fabricator.

[1] Residual stub lengths of roughly 6–12 mil are commonly quoted, but capability varies by fabricator and board thickness. Confirm the achievable value before relying on it in a loss budget.

HDI construction removes stubs, shortens vias (reducing their inductance), and frees routing channels under BGAs, at the cost of extra lamination cycles. It is standard practice for LPDDR and fine-pitch SoC packages with pitches of 0.5 mm and below.

**Table 11.2.** Via technologies.

| Type | Advantages | Limitations | Relative cost |
|---|---|---|---|
| Through (PTH) | Simple, robust, cheapest | Stubs; large anti-pads; blocks routing on all layers | 1 |
| Back-drilled PTH | Short residual stub | Extra drill step; depth tolerance; larger hole in planes | 1.1–1.3 |
| Blind / buried (mechanical) | No stub; routing freed below | Sequential lamination | 1.3–2 |
| Micro-via (HDI) | Very short, low $L$; fine pitch | Laser process; stack reliability; cost per layer | 1.5–3 |
| Via-in-pad (filled, capped) | Shortest BGA connection | Fill and planarize step | +10–20 % |

## 11.7 Return and stitching vias

When a signal changes layers, its return current must also change planes (Chapter 17). If the two reference planes are both ground, a ground via next to the signal via provides a short, low-inductance path for the return current. Without it, the return current spreads through the plane cavity as displacement current, exciting cavity resonances, raising crosstalk between all vias that share the cavity, and adding inductance to the signal path.

**Design Rule: Return vias**

Place at least one ground via within about one via pitch (typically 30–50 mil) of every high-speed signal via that changes reference plane, and two—one on each side, symmetric—for each differential via pair. If the reference planes are at different potentials (ground and power), place a stitching capacitor as close as possible to the transition, and prefer stackups that avoid such transitions for fast signals.

## 11.8 BGA and connector breakout

Under a BGA, signals must escape from the ball field through the gaps between vias. On coarse pitches (1.0 mm and 0.8 mm), "dog-bone" fan-out places a via diagonally offset from each ball, connected by a short trace (Figure 11.9). On finer pitches, via-in-pad with filled and capped vias (VIPPO) or micro-vias is required. For high-speed pairs:

- assign pairs to adjacent balls, and place their vias symmetrically so that the two legs of the pair have equal length and see the same neighbors;
- keep a ground via next to each signal via pair;
- neck traces down only where the gap requires it and for the shortest possible length—the reduced width raises impedance;
- route the fastest signals on the layers that give the shortest via stubs.

Connector pin fields follow the same principles. Press-fit connectors require plated holes of a specific finished size, which forces long barrels and stubs; their footprints are usually back-drilled for links above about 10 Gb/s.

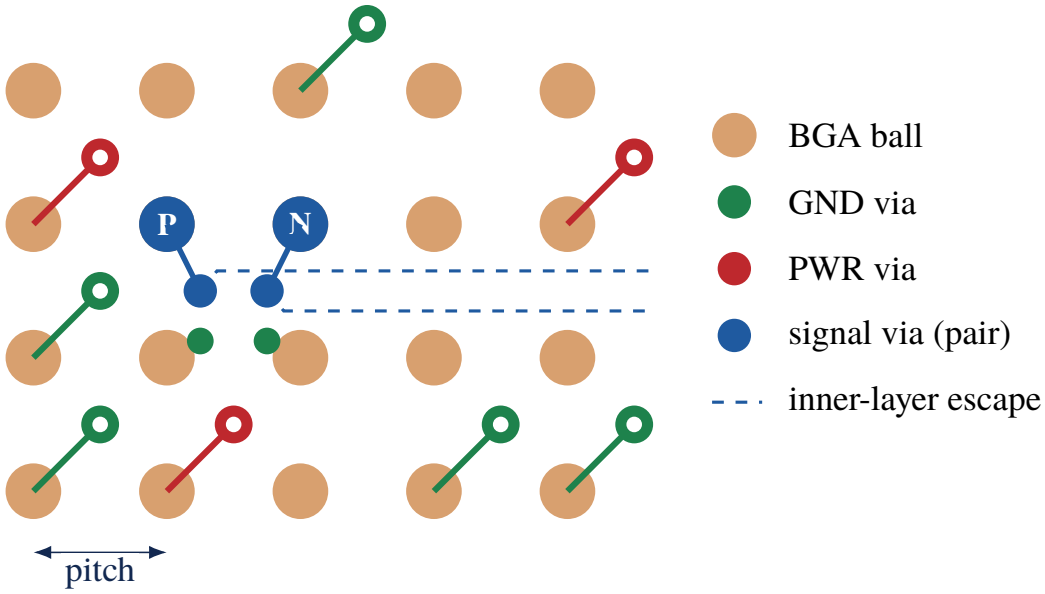



**Figure 11.9.** Dog-bone breakout under a BGA. A differential pair is assigned to adjacent balls, its vias are placed symmetrically with nearby ground return vias, and it escapes between ball rows on an inner layer.

**Common Pitfall: Swiss-cheese planes under BGAs**

A dense field of through vias with large anti-pads can leave the reference plane under the BGA as a mesh of narrow copper webs, or remove it entirely between rows. Traces escaping on adjacent layers then lose their return path exactly where they are most crowded. Check plane integrity under every BGA, and reduce anti-pad size or use back-drilling or HDI where the plane is compromised.

**Key Takeaways**

- A through via is roughly a nanohenry and a fraction of a picofarad; at edges of tens of picoseconds it must be modeled as a transmission line.
- Return vias set the via's impedance and provide its return path; place them adjacent to every signal via that changes reference.
- An open via stub resonates at $c/(4\ell\sqrt{\varepsilon_{\text{eff}}})$; keep that frequency at least three times the Nyquist frequency.
- Back-drilling, routing on deep layers, blind vias, and HDI remove stubs.
- The anti-pad sets via capacitance and impedance: too small makes the via capacitive, too large weakens the return path and the plane. Size it with a field solver, per via type.
- Under BGAs, symmetric pair escapes and intact planes matter more than shortest routes.

## Further Reading

Johnson and Graham [1] give the classic via estimates; Hall and Heck [4] model vias and stubs as transmission-line structures; Ritchey [8] addresses via fabrication and back-drilling practice.

## Review Problems

**P11.1** Estimate the partial inductance of a via 93 mil long with a 12 mil barrel.

**P11.2** Estimate the pad capacitance of a via in a 62 mil board with 18 mil pads, 30 mil anti-pads, and $\varepsilon_{\mathrm{r}} = 3.8$.

**P11.3** Find the quarter-wave resonance of a 70 mil stub with $\varepsilon_{\mathrm{eff}} = 3.6$.

**P11.4** What is the longest stub that keeps $f_{\lambda/4} \geq 3 f_{\mathrm{Nyq}}$ for a 16 GT/s link with $\varepsilon_{\mathrm{eff}} = 3.8$?

**P11.5** Estimate the impedance of a via with an 8 mil barrel and an effective return-via diameter of 32 mil in a laminate with $\varepsilon_{\mathrm{r}} = 4.0$.

**P11.6** A micro-via spans a 3 mil dielectric and the fabricator allows a maximum aspect ratio of 0.8:1. What is the minimum laser-drill diameter?

**P11.7** A via has a 0.25 mm drill, 0.50 mm pads and 0.80 mm anti-pads in $\varepsilon_{\mathrm{r}} = 3.6$. Find the radial pad-to-plane clearance and the coaxial-limit impedance.

**P11.8** *(Inverse design.)* For the via of the previous problem, what anti-pad diameter gives 50 Ω in the coaxial limit? Would it fit on a 1.0 mm grid with a 0.15 mm minimum web?

Chapter 12

# Timing, Jitter, and Eye Diagrams

**Chapter Overview**

Every impairment studied so far—reflections, loss, crosstalk, skew—ends up as a reduction of timing or voltage margin at the receiver's sampling point. This chapter turns those impairments into a pass/fail answer. It sets up timing budgets for the three clocking architectures used on boards (common-clock, source-synchronous, and embedded-clock), decomposes jitter into its random and deterministic components, relates jitter to bit-error ratio through the dual-Dirac model and bathtub curves, and explains how eye diagrams, masks, and equalization (CTLE, FFE, DFE) are used to qualify high-speed links.

**Learning Objectives**

1. Write setup and hold budgets for common-clock interfaces.
2. Explain why source-synchronous and embedded-clock interfaces scale to higher data rates.
3. Decompose total jitter into RJ, DJ, PJ, DDJ, and BUJ, and compute TJ at a target BER.
4. Read eye diagrams and bathtub curves and relate them to masks.
5. Describe the operation and limits of CTLE, FFE, and DFE equalizers.

## 12.1 Clocking architectures

### 12.1.1 Common-clock timing

In a common-clock interface, driver and receiver are clocked from the same source. Data launched by one clock edge must arrive and settle at the receiver before the *next* edge, by at least the setup time, and must not change until after the hold time (Figure 12.1). The budgets are

$$\text{setup margin} = T_{\text{clk}} - \left(t_{\text{co,max}} + t_{\text{flight,max}} + t_{\text{su}} + t_{\text{skew}} + t_{\text{jit}}\right) \geq 0, \tag{12.1}$$

$$\text{hold margin} = t_{\text{co,min}} + t_{\text{flight,min}} - t_{\text{h}} - t_{\text{skew}} \geq 0, \tag{12.2}$$

where $t_{\text{co}}$ is the driver's clock-to-output delay, $t_{\text{skew}}$ the worst-case difference in clock arrival between the two devices, and $t_{\text{jit}}$ the cycle-to-cycle clock jitter. The flight time includes the board delay plus the extra delay caused by loading and edge degradation at the receiver threshold.

**Worked Example 12.1: Common-clock budget**

A 133 MHz bus ($T_{\text{clk}} = 7.5$ ns) has $t_{\text{co}} = 1.0$–$4.0$ ns, flight time 0.6–1.2 ns, $t_{\text{su}} = 1.5$ ns, $t_{\text{h}} = 0.5$ ns, clock skew 0.3 ns, and jitter 0.2 ns. Find the margins and the maximum clock frequency.

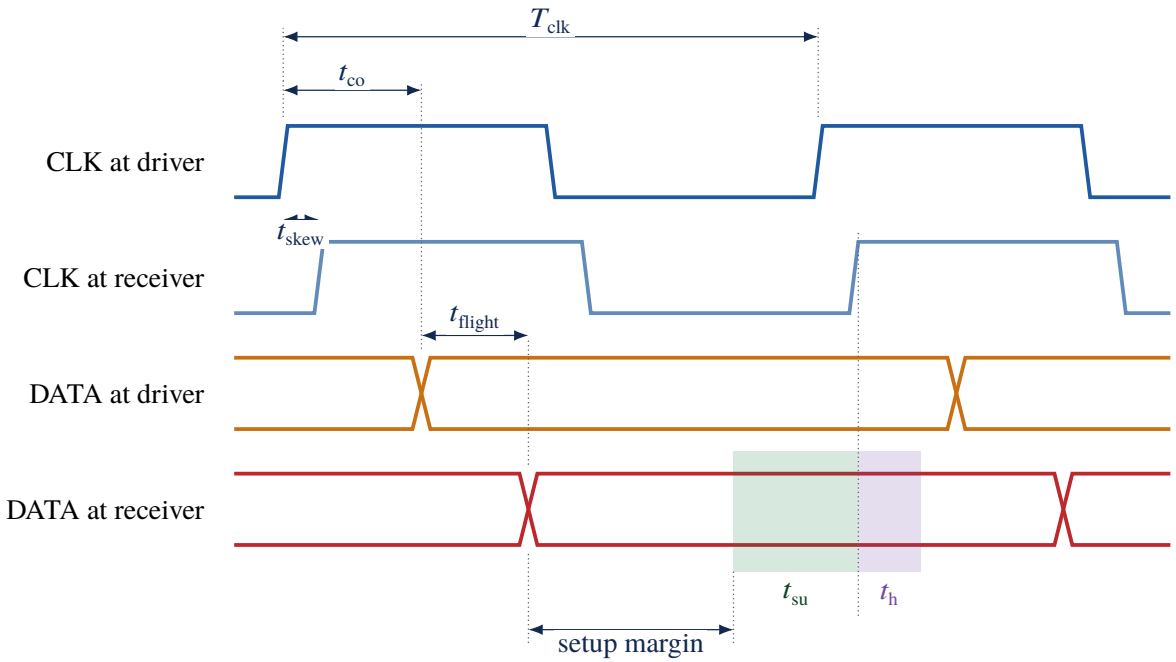



**Figure 12.1.** Common-clock timing. Data launched $t_{co}$ after the driver's clock edge arrives at the receiver $t_{flight}$ later and must be stable from $t_{su}$ before to $t_h$ after the receiver's next clock edge.

**Solution.** Setup: $7.5 - (4.0 + 1.2 + 1.5 + 0.3 + 0.2) = 0.3$ ns. Hold: $1.0 + 0.6 - 0.5 - 0.3 = 0.8$ ns. The minimum period is 7.2 ns, so $f_{max} = 139$ MHz. The driver's $t_{co}$ spread alone consumes 3 ns—which is why common-clock interfaces rarely exceed a few hundred megahertz.

### 12.1.2 Source-synchronous timing

A source-synchronous interface forwards a clock or strobe with the data, routed alongside it. Because strobe and data experience the same driver delay and nearly the same flight time, the absolute $t_{co}$ and flight times cancel; only their *differences* enter the budget. DDR memory (DQS with DQ), MIPI D-PHY, and RGMII work this way. The governing quantity is the skew between data and strobe, which is why these interfaces impose tight length matching within a byte lane or lane group (Chapters 19 and 23). The strobe is either *center-aligned* with the data eye (DDR writes) or *edge-aligned* (DDR reads), in which case the receiver delays it by a quarter of a clock period.

**Signed skew.** Define the data-to-strobe skew at the receiver as $\Delta = t_{DQ} - t_{DQS}$, with its sign. Late data ($\Delta > 0$) eats setup margin; early data ($\Delta < 0$) eats hold margin. A budget written as "$\pm|\Delta|_{max}$" charges the worst magnitude against *both* sides and wastes margin whenever the skew is biased in one direction, as package and via delays often are. Carry the minimum and maximum signed skew separately:

$$M_{setup} = \tfrac{1}{2}\mathrm{UI} - t_{su} - \Delta_{max} - t_{jit}, \qquad M_{hold} = \tfrac{1}{2}\mathrm{UI} - t_h + \Delta_{min} - t_{jit}, \tag{12.3}$$

for a strobe centered in the data eye. For example, with UI = 312.5 ps, illustrative $t_{su} = t_h = 60$ ps, jitter 30 ps, and a signed skew range of −5–25 ps, the setup margin is $156 - 60 - 25 - 30 = 41$ ps and the hold margin $156 - 60 - 5 - 30 = 61$ ps. A symmetric ±25 ps budget would report 41 ps on both sides and hide 20 ps of real hold margin—margin that a trained interface can redistribute by shifting the strobe (Chapter 19).

### 12.1.3 Embedded-clock timing

Serial links (PCIe, USB, Ethernet SERDES) send no separate clock. A clock-and-data-recovery (CDR) circuit in the receiver locks a local oscillator to the data transitions and samples each bit at the center of the recovered eye. The CDR tracks slow phase variations (low-frequency jitter and spread-spectrum modulation) and rejects only jitter faster than its loop bandwidth. The timing budget becomes a jitter budget, and line coding (8b/10b, 128b/130b, scrambling) guarantees enough transitions for the CDR to remain locked. This architecture removes flight-time and skew limitations altogether, which is why every multi-gigabit interface uses it.

**Jitter transfer.** A first-order CDR tracks input phase with a low-pass transfer function $H(s) = \omega_c/(s+\omega_c)$, where $f_c = \omega_c/2\pi$ is the CDR bandwidth. The sampling error that matters is the part it does *not* track,

$$\frac{\phi_{\text{err}}}{\phi_{\text{in}}}(s) = 1 - H(s) = \frac{s}{s+\omega_c}, \tag{12.4}$$

a high-pass response: jitter far below $f_c$ (spread-spectrum modulation at about 33 kHz, wander) is tracked and costs almost nothing, while jitter well above $f_c$ passes fully into the sampling error. For $f_c = 10\,\text{MHz}$, 1 MHz jitter is attenuated to $|\text{j}0.1/(1+\text{j}0.1)| = 0.0995$, about $-20\,\text{dB}$. This is why serial standards specify jitter with a defined *reference CDR* or filter: a jitter number without its filtering is incomplete. Real CDRs are usually second order, with peaking near $f_c$ that the standards also bound.

## 12.2 Jitter

Jitter is the deviation of signal transitions from their ideal times. It is characterized by its *time interval error* (TIE): the sequence of differences between each actual transition and the corresponding ideal one. Figure 12.2 shows the standard decomposition [25], [26].

**Random jitter (RJ)**
: Gaussian, unbounded jitter from thermal and shot noise in oscillators and buffers. It is specified by its RMS value $\sigma$. Because it is unbounded, its peak-to-peak value grows with the number of observed transitions.

**Deterministic jitter (DJ)**
: Bounded jitter with identifiable causes. Its components are *periodic jitter* (PJ) from power-supply ripple, spread spectrum, or coupled clocks; *data-dependent jitter* (DDJ), comprising inter-symbol interference (ISI) from channel loss and reflections and duty-cycle distortion (DCD); and *bounded uncorrelated jitter* (BUJ), typically crosstalk from asynchronous aggressors.

### 12.2.1 Total jitter at a bit-error ratio

Because RJ is unbounded, total jitter is meaningful only at a stated *bit-error ratio* (BER). The *dual-Dirac* model represents DJ by two delta functions separated by $\text{DJ}_{\delta\delta}$, each convolved with the Gaussian RJ. The total jitter at a target BER is then

$$\text{TJ}(\text{BER}) = \text{DJ}_{\delta\delta} + 2Q_{\text{BER}}\,\sigma_{\text{RJ}}, \tag{12.5}$$

where $Q_{\text{BER}}$ is the number of standard deviations at which the Gaussian tail equals the BER (adjusted for transition density). Common values are listed in Table 12.1. $\text{DJ}_{\delta\delta}$ is a model parameter fitted to the tails of the distribution and is generally smaller than the peak-to-peak DJ that would be measured directly.

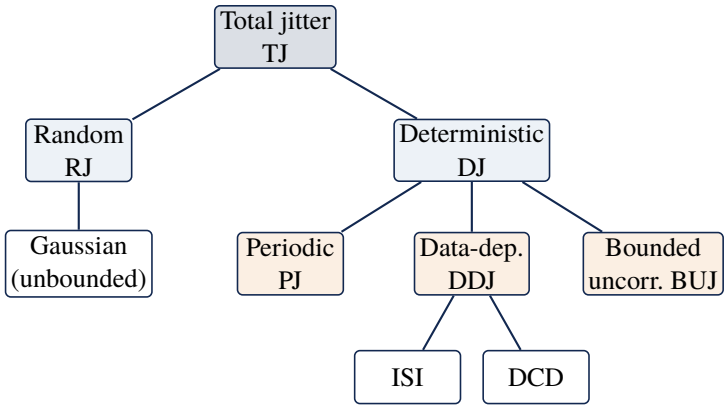



**Figure 12.2.** Decomposition of total jitter into random and deterministic components.

**Table 12.1.** Gaussian multipliers for total jitter, $2Q_{\mathrm{BER}}$ (for a transition density of one).

| BER | $10^{-9}$ | $10^{-12}$ | $10^{-15}$ |
|---|---|---|---|
| $Q_{\mathrm{BER}}$ | 5.998 | 7.034 | 7.941 |
| $2Q_{\mathrm{BER}}$ | 11.996 | 14.069 | 15.883 |

**Worked Example 12.2: Total jitter at $10^{-12}$**

A 16 GT/s receiver sees $\mathrm{DJ}_{\delta\delta} = 10$ ps and $\sigma_{\mathrm{RJ}} = 1$ ps. Find TJ at $10^{-12}$ and the remaining horizontal eye opening.

**Solution.** $\mathrm{TJ} = 10 + 14.07 \times 1 = 24.1$ ps. The UI is 62.5 ps, so the eye opening is $62.5 - 24.1 = 38.4$ ps, or 0.61 UI. Each additional picosecond of RJ costs 14 ps of eye width—RJ is expensive.

**Common Pitfall: Adding jitter components linearly or in RSS indiscriminately**

Adding RMS random jitter values to peak-to-peak deterministic values, or combining DJ terms by root sum of squares. Independent RJ sources add as RSS ($\sigma_{\mathrm{tot}}^2 = \sum \sigma_i^2$); DJ components combine by convolution of their distributions, which for budgeting purposes is approximately linear addition of $\mathrm{DJ}_{\delta\delta}$ values; and the two are combined only through Eq. (12.5).

## 12.3 Eye diagrams, masks, and bathtub curves

An *eye diagram* overlays many unit intervals of a waveform, each triggered by the recovered clock (Figure 12.3). The vertical opening at the sampling instant is the *eye height*, the horizontal opening at the decision threshold the *eye width*. Loss and reflections close the eye through ISI, crosstalk adds vertical noise, and jitter blurs the crossings. Interface specifications define a *eye mask*—a forbidden region inside the eye (and often above and below it)—that the waveform must not enter at a specified BER.

A *bathtub curve* plots BER against sampling phase across the UI (Figure 12.4). Near each crossing the BER is high; toward the center of the eye it falls steeply, at a rate determined by RJ. The width of the bathtub at a given BER is the eye width at that BER. Because measuring BER down to $10^{-12}$ directly takes a long time, the tails of the bathtub are usually extrapolated from measurements at higher BER using the dual-Dirac model.

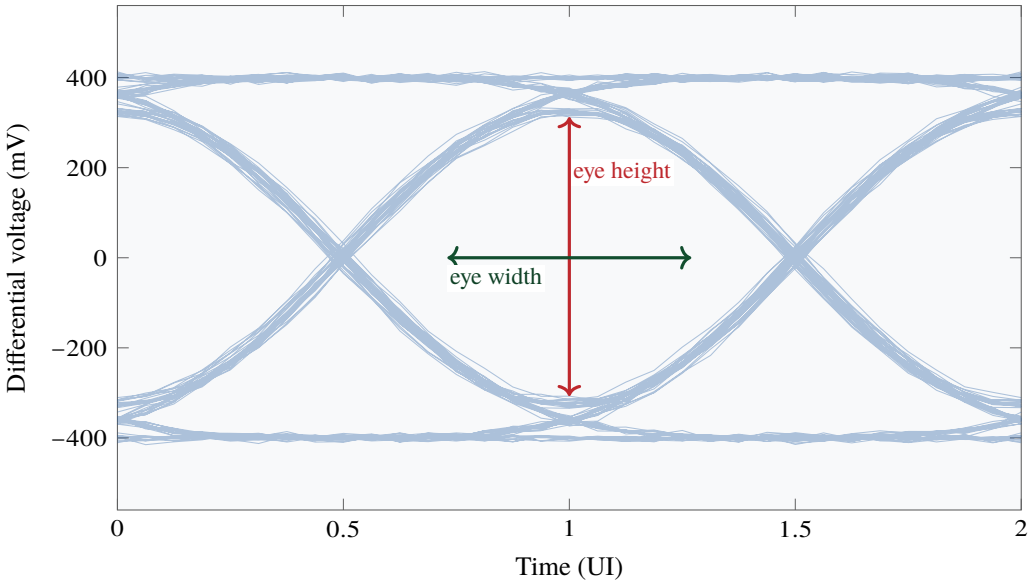
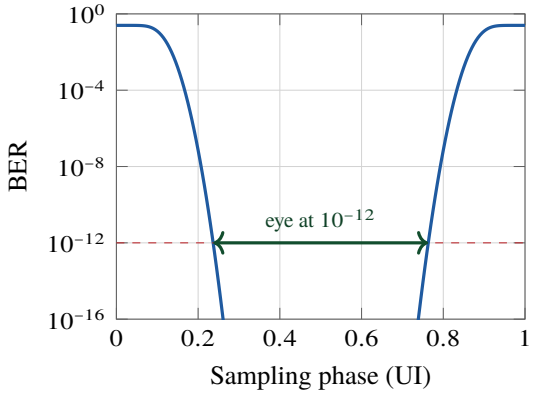



**Figure 12.3.** Synthetic NRZ eye diagram: a PRBS7 sequence through a band-limited channel with random jitter and noise, overlaid over two unit intervals. Eye height and eye width are marked; a receiver mask would occupy the open region at the center.




**(a)** Bathtub curve

| Parameter | Value |
|---|---|
| $\mathrm{DJ}_{\delta\delta}$ | 0.20 UI |
| $\sigma_{\mathrm{RJ}}$ | 0.020 UI |
| Transition density | 0.5 |
| Eye at $10^{-12}$ | 0.53 UI |

**(b)** Model parameters

**Figure 12.4.** Dual-Dirac bathtub curve. The eye width at $10^{-12}$ is the distance between the two walls of the bathtub at that BER.

**Worked Example 12.3: Measuring BER**

How many bits must be received without error to claim, at 95 % confidence, that BER $\le 10^{-12}$? How long does that take at 16 Gb/s?

**Solution.** For zero observed errors, the confidence level is $1 - \mathrm{e}^{-N\cdot\mathrm{BER}}$, so $N = -\ln(0.05)/10^{-12} = 3.0 \times 10^{12}$ bits. At 16 Gb/s this takes 187 s. Verifying $10^{-15}$ the same way would take more than two days, which is why extrapolation is used.

## 12.4 Equalization

When channel loss exceeds what an unequalized receiver can tolerate (roughly 10–15 dB at Nyquist for NRZ), the transmitter and receiver apply equalization to restore a flatter overall response (Figure 12.6).

### 12.4.1 Continuous-time linear equalizer (CTLE)

A CTLE is an analog high-pass (peaking) filter in the receiver front end, with a transfer function of the form

$$H_{\mathrm{CTLE}}(s) = A_{\mathrm{dc}} \frac{1 + s/\omega_z}{(1 + s/\omega_{p1})(1 + s/\omega_{p2})}, \tag{12.6}$$

which boosts frequencies between the zero $\omega_z$ and the first pole $\omega_{p1}$ by up to $20\log_{10}(\omega_{p1}/\omega_z)$ relative to DC. It is inexpensive and acts on the whole signal, but it amplifies high-frequency noise and crosstalk together with the signal.

### 12.4.2 Feed-forward equalizer (FFE) and pre-emphasis

An FFE is a finite-impulse-response filter, usually in the transmitter, that outputs a weighted sum of the current and neighboring bits: $y_n = \sum_k c_k\, x_{n-k}$. A two-tap FFE with a negative post-cursor tap $c_1$ reduces the amplitude of repeated bits relative to transitions (*de-emphasis*). Its frequency response is

$$|H_{\mathrm{FFE}}(f)| = \sqrt{c_0^2 + c_1^2 + 2c_0c_1\cos(2\pi f\ \mathrm{UI})}, \tag{12.7}$$

rising from $|c_0 + c_1|$ at DC to $|c_0 - c_1|$ at Nyquist. The de-emphasis ratio, in decibels, is

$$\mathrm{DE} = 20\log_{10}\left|\frac{c_0 - c_1}{c_0 + c_1}\right|.$$

Because the transmitter's peak swing is limited, de-emphasis works by attenuating low frequencies rather than boosting high ones. PCIe defines sets of pre-cursor and post-cursor coefficients (presets) that the link negotiates during training.

**Worked Example 12.4: De-emphasis of a two-tap FFE**

A transmitter uses $c_0 = 0.75$ and $c_1 = -0.25$. Find the DC and Nyquist gains and the de-emphasis.

**Solution.** DC: $c_0 + c_1 = 0.5$ ($-6.0$ dB). Nyquist: $c_0 - c_1 = 1.0$ (0 dB). De-emphasis is 6.0 dB: transitions are driven at full swing, repeated bits at half swing.

### 12.4.3 Pulse response and cursors

Equalization is easiest to reason about from the channel's *pulse response*: the received waveform for a single isolated one-UI pulse, sampled once per UI at the receiver's sampling phase. The largest sample $h_0$ is the *main cursor*; samples before it are *precursors* $h_{-1}, h_{-2}, \ldots$ and those after it are *postcursors* $h_1, h_2, \ldots$. For a linear channel and symbols $a_n = \pm 1$, the sampled signal is the superposition

$$r_n = \sum_k h_k\, a_{n-k} = h_0 a_n + \underbrace{\sum_{k \neq 0} h_k\, a_{n-k}}_{\text{ISI}}. \tag{12.8}$$

The worst-case (peak-distortion) eye half-height at a zero threshold is $h_0 - \sum_{k\neq 0}|h_k|$. This is the bridge between the interconnect chapters, which produce $h_k$, and the receiver, which must survive them. Loss produces a smooth tail of postcursors; a reflection produces an isolated echo at a delay equal to its round-trip time.

### 12.4.4 Decision-feedback equalizer (DFE)

A DFE subtracts from the received sample a weighted sum of previously *decided* symbols,

$$y_n = r_n - \sum_{k=1}^{N} b_k \, \hat{a}_{n-k}, \tag{12.9}$$

and with $b_k = h_k$ and correct decisions it removes postcursors 1 to $N$ exactly—whatever caused them. That includes echoes from reflections, provided they fall inside the tap span (some receivers add "floating" taps that can be placed on a distant echo). Because the DFE feeds back clean decisions rather than the noisy signal, it does not amplify noise or crosstalk. It cannot cancel precursor ISI or postcursors beyond its span, and a wrong decision subtracts the wrong correction: an error on a tap of weight $b_k$ injects an error of $2b_k$ (error propagation). DFEs with one to many taps are standard in receivers above about 10 Gb/s; DDR5 receivers also include a small DFE.

**Worked Example 12.5: What a three-tap DFE buys**

A sampled NRZ channel has $h_{-1} = -0.08$, $h_0 = 1$, and postcursors 0.25, 0.12, 0.05 (Figure 12.5). Noise at the slicer has $\sigma = 0.03$. Compare the worst-case half-height without equalization and with an ideal three-tap DFE.

**Solution.** Without DFE: $1 - (0.08 + 0.25 + 0.12 + 0.05) = 0.50$, or $16.7\sigma$. With correct decisions and exact taps the three postcursors vanish and only the precursor remains: $1 - 0.08 = 0.92$, or $30.7\sigma$. These are noiseless amplitude distances for a static sampling phase, not BER values.

*A plausible failure.* Add a reflection echo of 0.15 at $k = 8$ (500 ps at 16 GBd). It lies outside the tap span, so the half-height falls to $1 - 0.08 - 0.15 = 0.77$. The remedies compete: more (or floating) taps in the receiver, or a board change that removes the echo at its source.

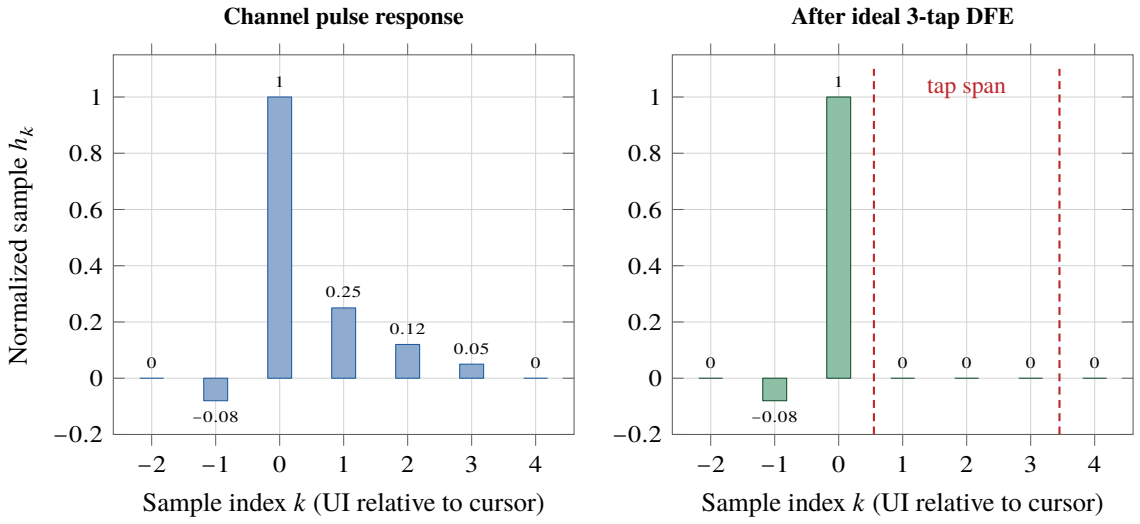



**Figure 12.5.** Sampled pulse response of Worked Example 12.5, normalized to the main cursor, before and after an ideal three-tap DFE (correct prior decisions, exact taps; no noise or adaptation). The precursor remains. [Analytical]

### 12.4.5 Statistical and bit-by-bit analysis

Two methods turn a pulse response into a margin. *Statistical* ("peak distortion" or convolution-based) analysis combines the ISI distribution of Eq. (12.8) with Gaussian noise and jitter models to predict the eye contour at very low BER, such as $10^{-12}$ or below, in seconds; it assumes linearity and time invariance, and its tails are only as good as its noise and jitter models. *Bit-by-bit* (time-domain) simulation runs millions of symbols through the channel and the receiver's adaptive equalization and CDR, and captures nonlinear and adaptive behavior, but it cannot reach $10^{-12}$ directly and extrapolates its tails. Use both: bit-by-bit to show the adaptation converges, statistical to estimate the low-BER margin. A finite eye overlay of a few thousand symbols demonstrates neither.

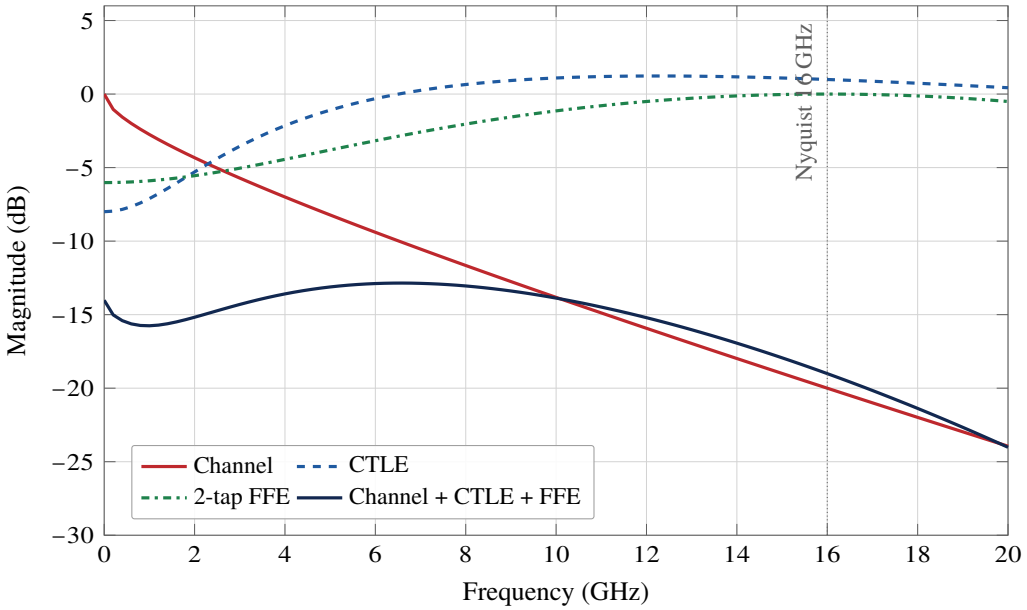



**Figure 12.6.** Channel loss, CTLE, and two-tap FFE responses for a 32 GT/s link. The equalized response is flatter than the raw channel; the residual slope is left to the DFE.

**Table 12.2.** Equalizer types.

| Type | Location | ISI cancelled | Limitation |
|---|---|---|---|
| CTLE | Rx (analog) | Pre- and post-cursor (broadband) | Boosts noise and crosstalk |
| FFE / de-emphasis | Tx (or Rx) | Pre- and post-cursor | Reduces signal amplitude |
| DFE | Rx (feedback) | Post-cursors within tap span (incl. echoes) | Error propagation; tap span |

**Design Rule: Design the channel for the equalizer you have**

Linear equalizers (CTLE, FFE) compensate loss that varies smoothly with frequency; they cannot undo the sharp notches of via stubs or crosstalk, and boosting near a notch amplifies noise. A DFE can cancel reflection echoes, but only those within its tap span, and only postcursors. Keep return loss and crosstalk within specification, place unavoidable discontinuities so that their echoes fall early, and leave the equalizer to handle the slope.

**Rule of Thumb: The cost of random jitter**

At a BER of $10^{-12}$, each picosecond of RMS random jitter consumes about 14 ps of horizontal eye opening. *Limits:* assumes Gaussian RJ and the dual-Dirac model; real tails may be non-Gaussian, and the multiplier changes with BER and transition density (Table 12.1).

**Key Takeaways**

- Common-clock interfaces are limited by $t_{co}$ spread and flight time; source-synchronous interfaces by data-to-strobe skew; embedded-clock links by jitter.
- Total jitter at a BER is $\mathrm{DJ}_{\delta\delta} + 2Q_{\mathrm{BER}}\sigma_{\mathrm{RJ}}$, with $2Q = 14.07$ at $10^{-12}$.
- Eye diagrams and masks summarize amplitude and timing margin; bathtub curves give eye width versus BER.
- The sampled pulse response (cursors) links the channel to the receiver; worst-case half-height is $h_0 - \sum |h_k|$.
- CTLE and FFE flatten the channel response at the cost of noise or amplitude; DFE cancels postcursors within its tap span—including reflection echoes—without amplifying noise, but not precursors.
- Carry signed skew in source-synchronous budgets, and quote jitter with its CDR filtering.

## Further Reading

Li [25] and Derickson and Müller [26] treat jitter decomposition, BER, and measurement thoroughly; Hall and Heck [4] and Dally and Poulton [9] cover timing budgets and equalization.

## Review Problems

**P12.1** A 200 MHz common-clock bus has $t_{co} = 0.8$–$2.8$ ns, flight time 0.4–0.9 ns, $t_{su} = 1.0$ ns, $t_h = 0.4$ ns, skew 0.2 ns, and jitter 0.15 ns. Find setup and hold margins.

**P12.2** Find TJ at $10^{-12}$ for $\mathrm{DJ}_{\delta\delta} = 12$ ps and $\sigma_{\mathrm{RJ}} = 0.8$ ps.

**P12.3** How long must a 16 Gb/s link run error-free to demonstrate BER $\leq 10^{-12}$ at 95 % confidence?

**P12.4** Find the de-emphasis of a two-tap FFE with $c_0 = 0.8$, $c_1 = -0.2$.

**P12.5** A CTLE has a zero at 2 GHz and its first pole at 8 GHz. What is its maximum peaking relative to DC, ignoring the second pole?

**P12.6** In Worked Example 12.5, the previous decision feeding the first DFE tap is wrong. What error does the DFE inject into the current sample, and is the current decision still safe without noise?

**P12.7** *(Inverse design.)* A receiver needs a worst-case half-height of 0.6 after an ideal two-tap DFE. The channel has $h_{-1} = -0.1$ and postcursors 0.3, 0.15 and $h_3$. What is the largest acceptable $h_3$?

**P12.8** A CDR has a first-order bandwidth of 4 MHz. By how much is 40 MHz sinusoidal jitter attenuated in the sampling error? And 400 kHz?

Part III

# Power Integrity

Chapter 13

# What Is Power Integrity?

**Chapter Overview**

This chapter opens the power-integrity part of the book. It defines power integrity, shows what a supply rail looks like when it is lost, explains the power delivery network (PDN) from the regulator to the transistors, and introduces the two calculations every PI engineer starts with: voltage droop from resistance and inductance, and target impedance. It then lists the techniques that improve PI, the measurements that verify it, and how power integrity and signal integrity interact. The following chapters develop each topic: simultaneous switching noise (Chapter 15), PDN design (Chapter 16) and planes and return paths (Chapter 17).

**Learning Objectives**

1. Define power integrity and the PDN.
2. Recognize voltage droop, ripple, ground bounce, simultaneous switching noise, PDN resonance and IR drop on a rail waveform.
3. Estimate droop from $IR$ and $L\ \mathrm{d}I/\mathrm{d}t$, and explain why decoupling capacitance helps only when it is placed and sized correctly.
4. Compute a target impedance and read a PDN impedance plot against it.
5. Choose the right instrument to measure a rail, and relate PI to SI.

## 13.1 Definition

*Power integrity* (PI) is the ability of a board's power delivery network to provide stable, clean and sufficient voltage and current to every component under all operating conditions. In simple terms, power integrity ensures that integrated circuits receive the correct supply voltage without excessive noise, ripple, voltage drop or transient disturbance.

As with signal integrity, "correct" has a precise meaning: every IC specifies a supply voltage and a tolerance, measured at a stated point (the package balls or the die), for example $0.8\,\mathrm{V}\ \pm 3\,\%$. A rail has integrity when, at that point and under every load condition the device can create, the voltage stays inside that window with margin. Power integrity is especially important in high-performance systems built around CPUs, FPGAs, GPUs and high-speed SoCs—an application processor such as the Rockchip RK3588 is a typical example—together with their DDR memory and PCIe and other SerDes interfaces, where core rails below 1 V carry many amperes that change in nanoseconds.

## 13.2 What causes power-integrity problems

Consider a processor core operating at 1.0 V. When it switches suddenly from low to high activity, its current consumption rises by several amperes within nanoseconds to microseconds. The power delivery network must respond quickly to this transient demand; if it cannot, the supply voltage drops. Figure 13.1 shows the resulting waveform and labels the common problems.

**Voltage droop**
: The supply voltage decreases when the load current increases, most sharply in the first moments after the step, before the regulator can respond.

**Voltage ripple**
: Periodic variation of the supply voltage, usually at the switching frequency of the regulator and its harmonics.

**Ground bounce**
: The ground voltage of a chip changes with respect to board ground because of transient current through package and via inductance (Chapter 15).

**Simultaneous switching noise (SSN)**
: Many digital outputs switching at once draw a large $\mathrm{d}I / \mathrm{d}t$ through shared power and ground paths, creating noise on both (Chapter 15).

**PDN resonance**
: Inductance and capacitance in the network form resonant peaks; a load current at or near such a frequency produces ringing and enlarged droop (Section 16.4).

**IR drop**
: Voltage lost in the resistance of traces, planes, vias and connectors between the regulator and the load; it persists as long as the current flows (Section 16.7).

## 13.3 The power delivery network

The PDN is the complete path that delivers power from the voltage regulator to the transistors on the die, *and* the return path back through ground (Figure 13.2). A typical PDN consists of:

1. a *voltage regulator module* (VRM), which generates the required supply voltage and holds its average value through its control loop;
2. *bulk capacitors*, which supply energy for relatively slow load transients, faster than the regulator loop can follow;
3. *decoupling capacitors* (MLCCs) near the load, which supply local transient current and lower the impedance at higher frequencies;
4. the *PCB power and ground planes*, which distribute current across the board and form a low-inductance capacitor of their own;
5. *package and on-die capacitance*, which maintains voltage stability closest to the silicon at the highest frequencies.

Each stage serves a band of frequencies and refills the next, faster stage. Because every connection between stages has resistance and inductance, a stage cannot help above the frequency at which its connection inductance dominates—which is why the smallest, fastest capacitance must sit closest to the load.

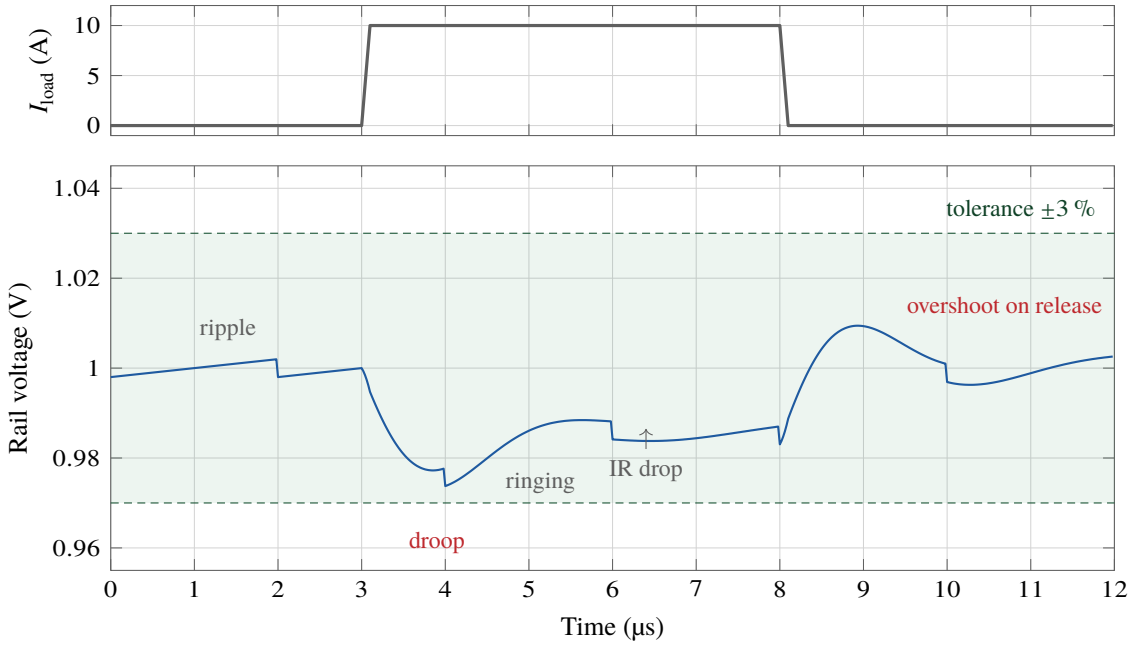



**Figure 13.1.** A 1.0 V rail responding to a 10 A load step applied at 3 µs and released at 8 µs. Ripple comes from the regulator; droop and ringing from the PDN's inductance and capacitance; the settled offset is the IR drop. Model: 1.5 mΩ and 1 nH feed, 200 µF at the load, 500 kHz, ±2 mV ripple; no regulator loop. [Simulation (lumped network)]

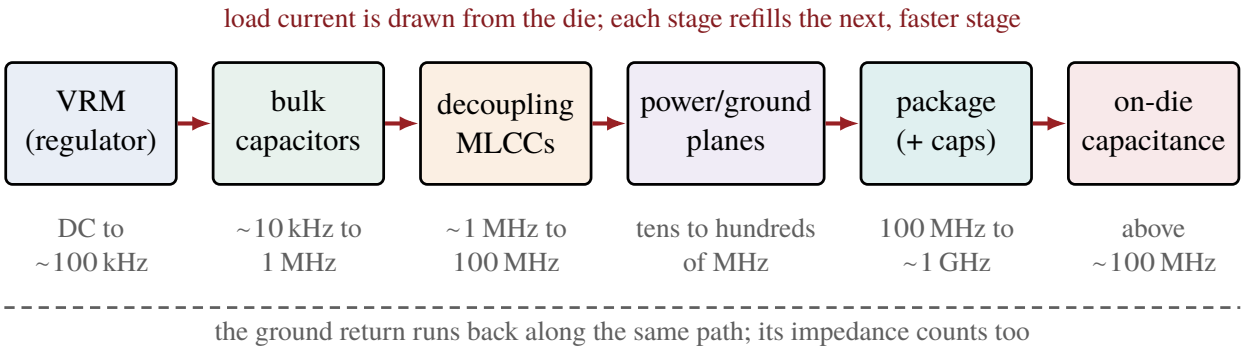



**Figure 13.2.** The power delivery network from regulator to die. The frequency bands are typical orders of magnitude, not limits; Figure 16.2 shows a computed example. [Conceptual]

## 13.4 Calculating voltage droop

Two equations describe most of the voltage disturbance on a rail. The resistive part,

$$\Delta V_R = I\,R, \tag{13.1}$$

is the IR drop, present as long as the current flows. During fast current transients, inductance also matters:

$$V_L = L\,\frac{\mathrm{d}I}{\mathrm{d}t}. \tag{13.2}$$

**Worked Example 13.1: Inductive droop**

A processor increases its current consumption by 10 A in 100 ns. The effective PDN inductance between the load and the nearest capacitance that could supply the current is 1 nH, and the path resistance is 1 mΩ. Estimate the disturbance.

**Solution.** Equation (13.2): $V_L = 1\,\text{nH} \times 10\,\text{A}/100\,\text{ns} = 100\,\text{mV}$ during the ramp, plus an IR drop of 10 mV that remains afterward (Figure 13.3, solid curve). On a 1.0 V rail with a ±3 % tolerance, 110 mV is almost four times the allowed deviation. The inductive term alone is a simplified estimate: it assumes that all the current must flow through the inductance, with no capacitance at the load.

Capacitance at the load changes the picture, because during the transient the capacitor supplies the current and the inductive feed refills it more slowly. The inductance and capacitance then form a resonant circuit with characteristic impedance

$$Z_c = \sqrt{L/C}, \tag{13.3}$$

and for a fast step the droop is roughly $\Delta I \times Z_c$. With 22 µF at the load, $Z_c = 6.7\,\text{m}\Omega$ and the droop falls to about 62 mV (Figure 13.3). With only 1 µF, $Z_c = 32\,\text{m}\Omega$: the circuit resonates at $1/(2\pi\sqrt{LC}) = 5\,\text{MHz}$, the 100 ns ramp excites it, and the droop *grows* to almost 200 mV. Adding capacitance does not automatically improve power integrity; the capacitance must be large enough, and its connection inductance small enough, that $\sqrt{L/C}$ is below the impedance the rail can tolerate.

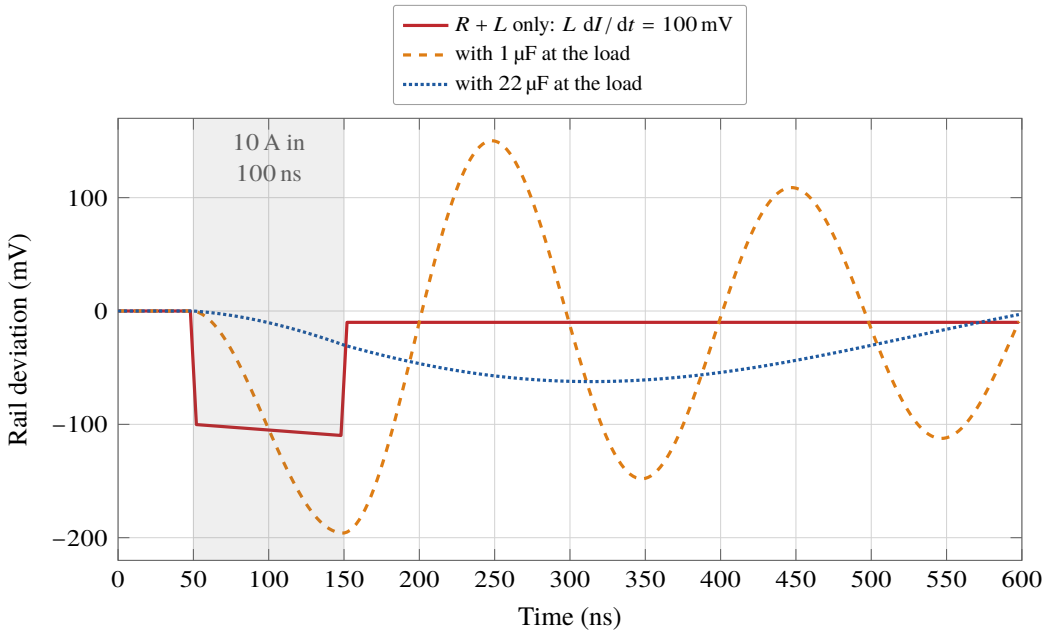



**Figure 13.3.** Rail deviation for a 10 A, 100 ns load ramp through 1 mΩ and 1 nH: without load capacitance, with 1 µF (ESR 2 mΩ) and with 22 µF (ESR 1 mΩ). The small capacitor forms a lightly damped resonance and makes the droop worse. [Simulation (lumped network)]

## 13.5 PDN impedance and target impedance

PDN impedance is the most important concept in power-integrity engineering. It describes how much the voltage at the load changes for a given change of current, at each frequency:

$$Z_{\mathrm{PDN}}(f) = \frac{\Delta V(f)}{\Delta I(f)}. \tag{13.4}$$

If the PDN impedance is low at every frequency the load current contains, the voltage cannot move much, whatever the current does. This leads to the *target impedance*, the largest PDN impedance that keeps the worst-case current change within the allowed voltage deviation:

$$Z_{\mathrm{target}} = \frac{\Delta V_{\mathrm{max}}}{\Delta I}. \tag{13.5}$$

**Worked Example 13.2: A target impedance**

A processor allows a maximum voltage deviation of 50 mV and experiences a transient current change of 10 A. Find the target impedance.

**Solution.** $Z_{\mathrm{target}} = 50\,\mathrm{mV}/10\,\mathrm{A} = 5\,\mathrm{m\Omega}$. The PDN, seen from the load, should ideally stay below 5 mΩ over the frequency range the load current contains. Two refinements follow in Chapter 16: the 50 mV must first be shared with the regulator's set-point error and the DC IR drop, so the dynamic share is smaller (Table 16.1); and meeting the impedance target in the frequency domain should be confirmed with a load-step simulation (Section 16.6).

Figure 13.4 compares two PDN designs against this target. Design A uses several staggered capacitor banks whose impedances overlap, so the total stays below 5 mΩ across the range the load excites. Design B uses fewer values with very low ESR; between them an anti-resonance peak rises to about 50 mΩ, ten times the target, near 20 MHz. A load current with energy at that frequency will produce ten times the allowed noise even though Design B has plenty of capacitance.

**Rule of Thumb: Which frequencies matter**

The load current of a step with rise time $t_r$ has significant content up to about $0.35/t_r$; a 100 ns step reaches about 3.5 MHz, a 1 ns step about 350 MHz. Above the frequency at which the package and on-die capacitance take over, the board's impedance is no longer visible to the die. *Limits:* periodic load patterns concentrate energy at specific frequencies and must be checked against resonances individually.

## 13.6 How to improve power integrity

Table 13.1 summarizes the main techniques and their limits.

**Common Pitfall: More capacitors is always better**

Adding capacitors without checking the impedance curve. Poor selection and placement can create anti-resonance peaks (Figure 13.4) or lightly damped resonances (Figure 13.3) that make the rail worse at exactly the frequencies where the load draws current. Every change to the decoupling should be checked with a PDN simulation.

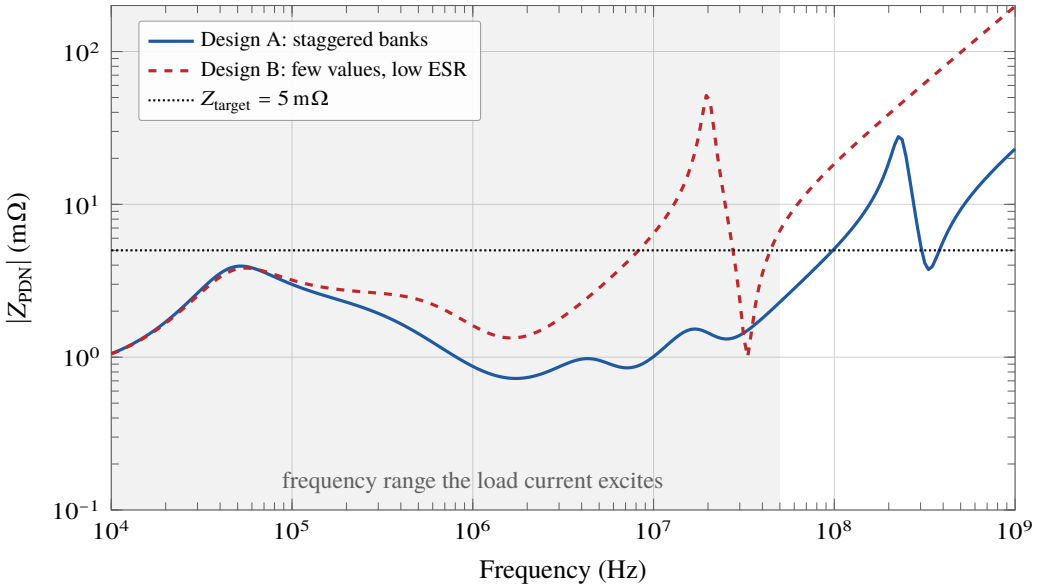



**Figure 13.4.** PDN impedance of two designs against a 5 mΩ target. Design A meets it over the shaded range; Design B has an anti-resonance peak near 20 MHz. Lumped models of VRM, bulk, MLCC banks and planes. [Analytical]

**Table 13.1.** Techniques that improve power integrity.

| Technique | Purpose | Watch out for |
|---|---|---|
| Multiple decoupling capacitors | Lower impedance across frequency | Anti-resonances between values |
| Proper capacitor placement | Minimize mounting inductance | Shared vias limit the gain |
| Low-ESR capacitors | Lower resistive impedance | Very low ESR sharpens peaks |
| Low-ESL capacitors | Better high-frequency response | Mounting dominates ESL |
| Solid power and ground planes | Low spreading impedance, plane capacitance | Splits and slots under signals |
| Multiple vias | Lower connection inductance and resistance | Via fields perforate planes |
| Proper VRM selection | Fast transient response, low ripple | Loop bandwidth versus stability |
| PDN simulation | Find impedance peaks and voltage drops | Models need biased capacitance and real ESL |

**Design Rule: Decouple from the die outward**

Start from the device's requirement at its pins, place the highest-frequency capacitors closest to the power balls with the shortest via connection, then add larger values farther out, and confirm with an impedance sweep and a load-step simulation that the target is met and no peak is exposed (Chapter 16).

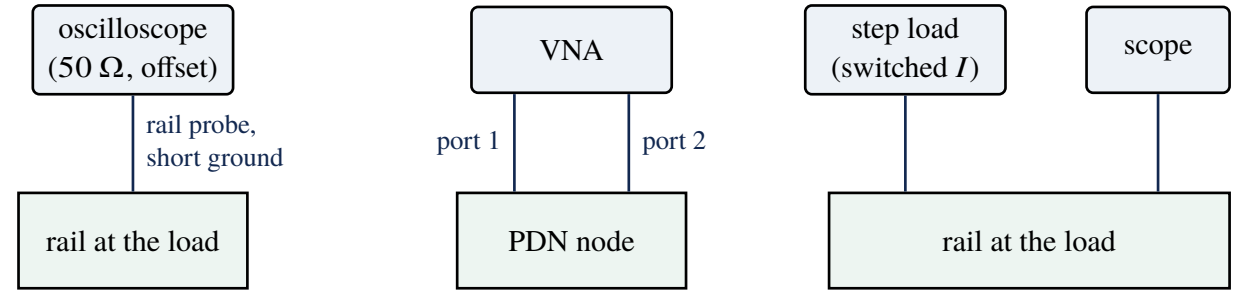
## 13.7 How to measure power integrity

Three measurements cover most needs (Figure 13.5); Chapter 26 gives the details.

1. *Oscilloscope.* Measures voltage ripple, transient droop, overshoot and ground bounce in the time domain. A dedicated rail probe with DC offset, a 50 Ω input and a very short ground connection is

needed: a standard 10 × probe with a long ground lead adds ringing and pickup larger than the millivolt signals being measured.

2. *Vector network analyzer.* Measures PDN impedance over frequency. For milliohm impedances the two-port shunt-through method is used (Section 26.3).
3. *Injected disturbance.* A controlled load step, or a small AC current injected by a signal generator and current driver, is applied to the rail while the voltage and current responses are recorded; the ratio gives the impedance, and the time response shows droop and settling directly.




(a) time domain: ripple, droop, overshoot

(b) frequency domain: $|Z(f)|$ shunt-through

(c) injected current: $\Delta V / \Delta I$ and transient

**Figure 13.5.** Three ways to measure power integrity: (a) oscilloscope with a rail probe, (b) VNA shunt-through impedance, (c) injected load step with voltage and current measured together. Probe grounds, cable shields and isolation are omitted. [Conceptual]

**Worked Example 13.3: Regulator ripple**

A buck regulator switching at 500 kHz has an inductor ripple current of 3 A peak to peak and 400 µF of output capacitance with a total ESR of 2 mΩ. Estimate the output ripple.

**Solution.** A common estimate for a buck converter adds the ESR and capacitive terms, $\Delta V \approx \Delta I_L \left(\mathrm{ESR} + \frac{1}{8 f_{sw} C}\right) = 3 \times (0.002 + 0.000625) = 7.9\,\mathrm{mV}$ peak to peak. ESL adds spikes at the switching edges that this estimate omits; they are often the larger part of the measured ripple and are seen only with a short probe ground and adequate bandwidth.

## 13.8 Signal integrity versus power integrity

The two disciplines use parallel concepts (Table 13.2), and the distinction matters particularly when designing carrier boards and high-speed embedded systems.

Poor power integrity also causes signal-integrity problems (Figure 13.6). Supply noise on a transceiver's PLL and VCO supply modulates the clock phase, increasing jitter and closing the PCIe eye; noise on output-driver supplies changes their delay (power-supply-induced jitter, Chapter 15); and ground bounce shifts the reference against which signals are compared. In the other direction, signal return currents that cross plane splits or change reference planes inject noise into the PDN (Chapter 17).

**Key Takeaways**

- Power integrity is the ability of the PDN to keep every supply inside its tolerance, at the point the device specifies, under all operating conditions.

**Table 13.2.** Signal integrity and power integrity side by side.

| Signal integrity (SI) | Power integrity (PI) |
|---|---|
| Focuses on signal quality | Focuses on power quality |
| Controlled impedance | Target impedance |
| Reflections | Voltage droop |
| Crosstalk | Ground bounce |
| Eye diagrams | PDN impedance plots |
| Jitter | Power-supply noise |
| Transmission lines | Power delivery network |

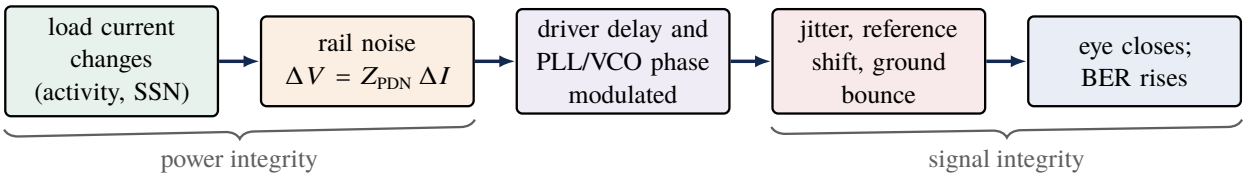



**Figure 13.6.** How power-integrity noise becomes a signal-integrity problem. [Conceptual]

**Table 13.3.** Summary: the power-integrity view of a design.

| Question | Answered by | Chapter |
|---|---|---|
| How much can the rail move? | Tolerance and its allocation | 16 |
| How large is the current step? | Device data, activity patterns | 16 |
| What impedance is allowed? | $Z_{\mathrm{target}} = \Delta V_{\mathrm{max}}/\Delta I$ | 16 |
| Are there resonances? | Impedance sweep of the whole network | 16 |
| How much DC drop? | Plane and via resistance, IR-drop simulation | 16 |
| How noisy are I/O supplies and grounds? | SSN, package inductance | 15 |
| Do return paths disturb the PDN? | Plane continuity, stitching | 17 |

- Droop, ripple, ground bounce, SSN, PDN resonance and IR drop are the common problems; they add on the same rail.
- The PDN runs from the VRM through bulk and decoupling capacitors, planes and package to the die; each stage serves a band of frequencies.
- Droop has a resistive part $IR$ and an inductive part $L\ \mathrm{d}I/\mathrm{d}t$; local capacitance helps only if $\sqrt{L/C}$ is small enough.
- Keep $|Z_{\mathrm{PDN}}|$ below $Z_{\mathrm{target}} = \Delta V_{\mathrm{max}}/\Delta I$ over the frequencies the load excites, and watch for anti-resonances.
- Measure rails with a rail probe, a VNA shunt-through setup or an injected load step; poor PI shows up as SI problems, especially jitter.

## Further Reading

Smith and Bogatin [7] develop the target-impedance method in depth; Swaminathan and Engin [27] cover PDN modeling and power–signal interaction; Novak and Miller [6] cover frequency-domain PDN characterization.

## Review Problems

**P13.1** Define power integrity in one sentence, and state where the supply voltage must be judged.

**P13.2** A load current rises by 5 A in 50 ns through 0.5 nH and 2 mΩ, with no local capacitance. Estimate the inductive and resistive drops.

**P13.3** Find the target impedance for a 0.9 V rail with a $\pm 2$ % tolerance, all of it assigned to dynamic noise, and a 6 A transient.

**P13.4** In Worked Example 13.1, what local capacitance gives $\sqrt{L/C} = 5\,\mathrm{m\Omega}$? Estimate the resulting droop for a fast step.

**P13.5** *(Approximation failure.)* A designer adds a single 1 µF capacitor behind 1 nH of mounting and feed inductance to "help" a 10 A, 100 ns load step. Why can this make the droop worse?

**P13.6** Which measurement would you use to find (a) the switching ripple of a regulator, (b) a 20 MHz anti-resonance in the PDN, (c) the droop during a real workload?

**P13.7** Explain how supply noise on a SerDes PLL can reduce a PCIe eye opening.

Chapter 14

# Transient Current, Inductance, and the PDN Response

**Chapter Overview**

When a chip suddenly draws more current, the path from the voltage regulator to the chip cannot change its current instantly. Its inductance resists the change, local capacitors supply the initial shortfall, and the chip's supply voltage droops and may ring. How much depends on the frequency-dependent impedance of the complete power delivery network. This chapter follows one load step from start to finish: the circuit between regulator and chip, how current is shared between the regulator path and the capacitors, how path inductance sets the droop, how the time-domain picture becomes a PDN impedance, why resonances matter, and how real PDNs show several droops on different time scales. It closes with two points that are often misunderstood: the voltage across an inductance is the *derivative* of its current, and a load transient *excites* the PDN impedance rather than changing it.

**Learning Objectives**

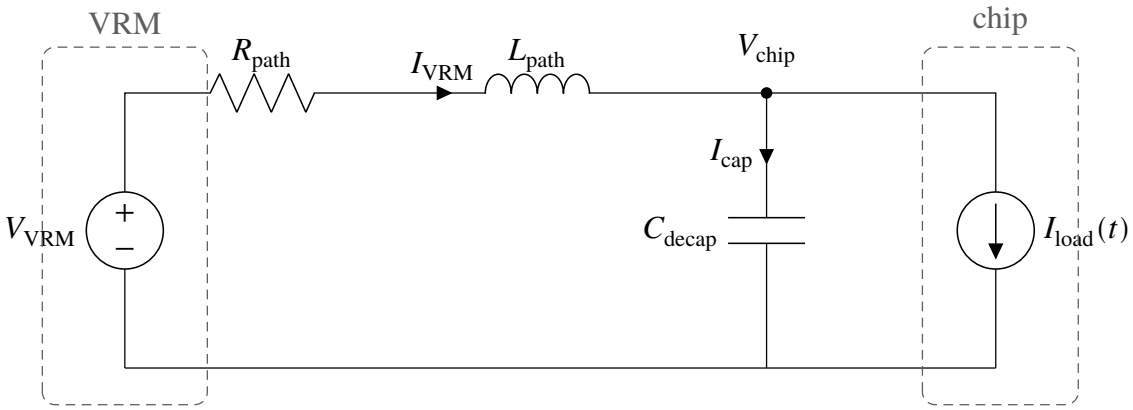
1. Write the voltage equation of the regulator-to-chip path and distinguish load current from regulator-path current.
2. Compute $L\ \mathrm{d}I/\mathrm{d}t$ for a load step and explain why it is not, by itself, the chip's droop.
3. Use charge balance to estimate capacitor droop, and the characteristic impedance, resonant frequency and damping of the path–capacitor pair.
4. Build a simple PDN impedance from branch impedances and read peaks against a target.
5. Recognize first, second and third droops and the PDN element behind each.
6. Predict the waveform of $L\ \mathrm{d}I/\mathrm{d}t$ for common current shapes, and use $\Delta V(f) = -Z(f)\,\Delta I(f)$ to reason about periodic loads.

## 14.1 The circuit between the regulator and the chip

The power delivery path includes the voltage regulator, board copper, vias, package connections, decoupling capacitors and the ground return path, all with parasitic resistance and inductance. The simplest useful model lumps them into a path resistance $R_{\text{path}}$ and inductance $L_{\text{path}}$, with one local capacitor at the chip (Figure 14.1). The resistance and inductance belong to the whole *loop*—supply and return together (Chapter 3)—not to the power copper alone.

If $I_{\text{VRM}}$ is the current in the path, the chip voltage is

$$V_{\text{chip}}(t) = V_{\text{VRM}}(t) - R_{\text{path}}\, I_{\text{VRM}}(t) - L_{\text{path}} \frac{\mathrm{d}I_{\text{VRM}}(t)}{\mathrm{d}t}, \tag{14.1}$$




return path (ground); its $R$ and $L$ are lumped into $R_{path}$ and $L_{path}$

**Figure 14.1.** Simplified one-rail PDN: regulator, effective path resistance and inductance of the supply-and-return loop, a local capacitor, and the chip's current demand. [Schematic]

and Kirchhoff's current law at the chip node gives

$$I_{load}(t) = I_{VRM}(t) + I_{cap}(t), \qquad I_{cap} = -C\frac{dV_{chip}}{dt}. \tag{14.2}$$

The distinction between the two currents is the key to the whole chapter: $I_{load}$, the current the chip demands, is *not* necessarily the current that flows immediately through the path inductance. Local capacitors can supply most of a fast transient.

## 14.2 A load step: why fast current changes matter

Assume a processor increases its current from 2 A to 12 A in 100 ns, with the illustrative values of Table 14.1.

**Table 14.1.** Illustrative load-step parameters.

| Parameter | Value | Unit |
|---|---|---|
| VRM output voltage | 1.0 | V |
| Initial chip current | 2 | A |
| Final chip current | 12 | A |
| Current rise time | 100 | ns |
| Path inductance | 1 | nH (and 5 nH for comparison) |
| Path resistance | 2 | mΩ |
| Local capacitance (ideal) | 100 | µF |

**Worked Example 14.1: The inductive voltage of a load step**

Find the current change, the average slew rate and the voltage across the path inductance if the whole change flows through it.

**Solution.** $\Delta I = 12 - 2 = 10\,\text{A}$ and $\Delta I/\Delta t = 10\,\text{A}/100\,\text{ns} = 10^8\,\text{A/s}$, so

$$V_L = L\frac{dI}{dt} = 1\,\text{nH} \times 10^8\,\text{A/s} = 100\,\text{mV}.$$

Across 5 nH the same ramp would develop 500 mV. These are the voltages across the inductance *if* the full 10 A change passes through it in 100 ns. They are not automatically the chip's droop: the local capacitor changes the current that flows in the inductance.

## 14.3 Local capacitors and the current shortfall

The chip wants its new current immediately, but the path current cannot rise without a voltage across the path inductance, and that voltage appears only as the chip voltage falls. Local capacitors make up the difference (Figure 14.2):

1. the chip switches into a higher-current state;
2. local capacitors provide most of the initial extra current;
3. the capacitors discharge, lowering the chip's supply voltage;
4. the path current rises and may temporarily *overshoot* the new load current;
5. the regulator eventually carries the sustained load and recharges the capacitors.

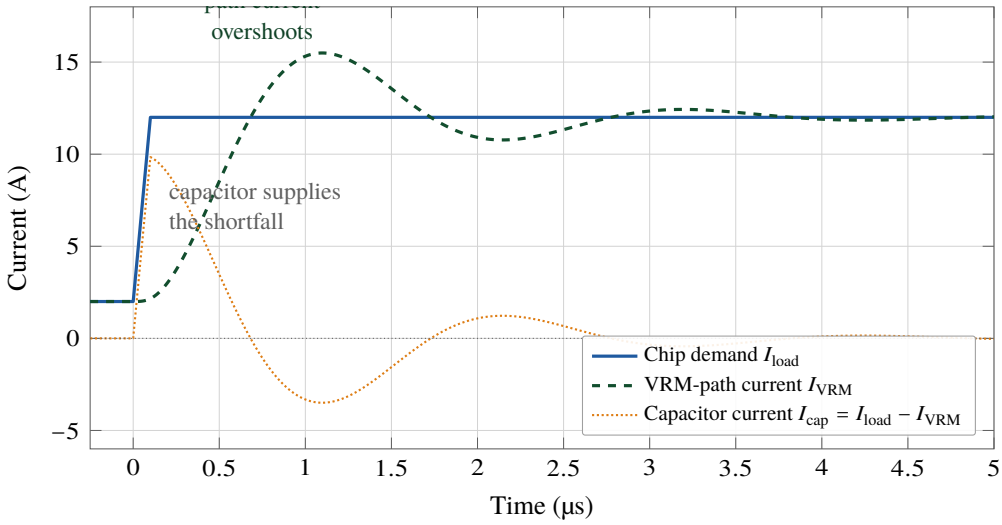



**Figure 14.2.** Calculated 2–12 A load-step response of the circuit of Figure 14.1 with the values of Table 14.1 and $L_{path}$ = 1 nH. The path current lags the demand, overshoots to 15.5 A and rings; the capacitor current is the difference. [Simulation (lumped network)]

Integrating Eq. (14.2) gives the capacitor's voltage change in terms of the charge it has delivered:

$$\Delta V_C = \frac{1}{C}\int (I_{load} - I_{VRM})\ \mathrm{d}t. \tag{14.3}$$

The chip voltage keeps falling as long as the load current exceeds the path current, so *the minimum voltage occurs where the two currents cross*.

**Worked Example 14.2: Charge-balance estimates of droop**

(a) A 200 µF capacitor supplies a constant 10 A shortfall for 1 µs. Estimate its droop. (b) In Figure 14.2, the shortfall integrates to 3.68 µC before the currents cross at 0.685 µs. What droop does Eq. (14.3) predict, and how does it compare with the simulated waveform?

**Solution.** (a) $\Delta V_C = \Delta I\,\Delta t/C = 10\,\text{A} \times 1\,\mu\text{s}/200\,\mu\text{F} = 50\,\text{mV}$. The estimate ignores the capacitor's ESR and ESL and the growing path current, so it is an upper bound for a stiff capacitor and an underestimate for one with significant ESR. (b) $3.68\,\mu\text{C}/100\,\mu\text{F} = 36.8\,\text{mV}$, which is exactly the simulated drop from 996–959 mV (Figure 14.3): for an ideal capacitor, charge balance *is* the droop.

## 14.4 How path inductance sets the droop

Compare two otherwise identical circuits with 1 nH and 5 nH of path inductance (Figure 14.3 and Table 14.2). Before the step both sit at $1.0 - 2 \times 0.002 = 996\,\text{mV}$, and after it they settle at $1.0 - 12 \times 0.002 = 976\,\text{mV}$. With more inductance the path current reacts more slowly, the capacitor supports the load for longer and delivers more charge, and the voltage falls further before recovering.

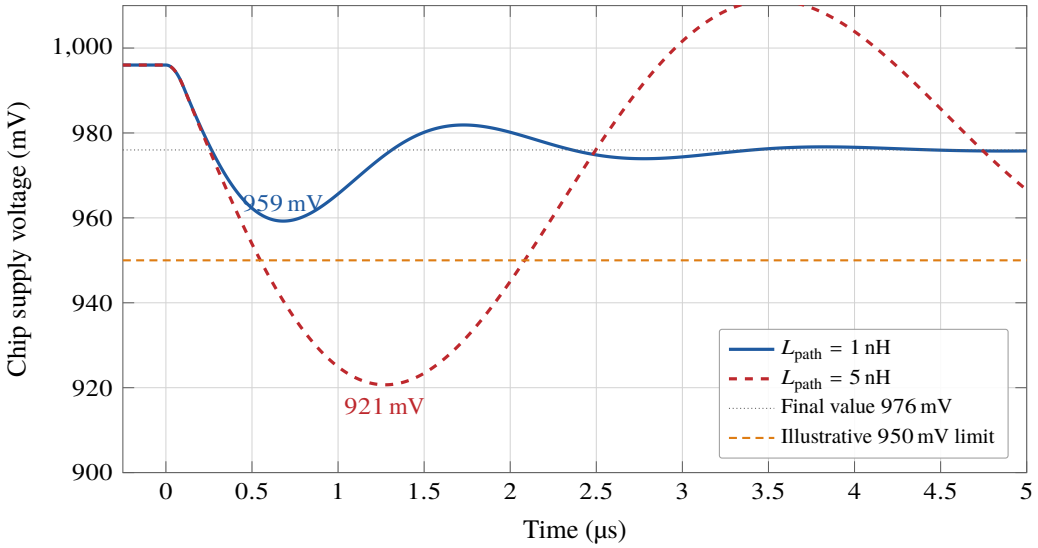



**Figure 14.3.** Chip voltage for the load step of Table 14.1 with 1 nH and 5 nH path inductance. The 950 mV line is an illustrative limit, not a device specification. Ideal source, ideal capacitor, no regulator loop. [Simulation (lumped network)]

**Table 14.2.** Results of the lumped simulation, rounded to the nearest millivolt.

| Result | 1 nH path | 5 nH path |
|---|---|---|
| Initial chip voltage | 996 mV | 996 mV |
| Minimum voltage | 959 mV | 921 mV |
| Droop from initial voltage | 37 mV | 75 mV |
| Characteristic impedance $\sqrt{L/C}$ | 3.2 mΩ | 7.1 mΩ |
| Resonant frequency | 503 kHz | 225 kHz |
| Damping ratio $\zeta$ | 0.32 | 0.14 |
| Qualitative response | smaller disturbance | larger droop, more ringing |

The last three rows of Table 14.2 explain the difference. The path inductance and local capacitance form a series-resonant loop with

$$f_0 = \frac{1}{2\pi\sqrt{LC}}, \qquad Z_c = \sqrt{\frac{L}{C}}, \qquad \zeta = \frac{R}{2}\sqrt{\frac{C}{L}}, \tag{14.4}$$

where $R$ is the total series resistance of the loop. For a step much faster than $1/f_0$, the dynamic part of the droop is of the order of $\Delta I\, Z_c$, reduced by damping; quintupling $L$ raises $Z_c$ by $\sqrt{5}$ and lowers $\zeta$ by the same factor, and the response rings longer. In a real system the regulator's control loop, the capacitor's ESR and ESL, the package inductance and on-die capacitance all change the result.

## 14.5 From the time domain to PDN impedance

The impedance of an ideal inductor is $Z_L = \mathrm{j}\omega L = \mathrm{j}2\pi f L$, so its magnitude grows in proportion to frequency (Table 14.3). High-frequency components of the load current therefore cannot be supplied through a distant regulator; they must come from capacitors close to the chip.

**Table 14.3.** Reactance of the path inductance alone (not the impedance of the full PDN).

| Frequency | $\lvert Z_L\rvert$ at 1 nH | $\lvert Z_L\rvert$ at 5 nH |
|---|---|---|
| 100 kHz | 0.63 mΩ | 3.14 mΩ |
| 1 MHz | 6.28 mΩ | 31.4 mΩ |
| 10 MHz | 62.8 mΩ | 314 mΩ |

In a basic model the path and each capacitor are branches in parallel, as seen from the chip:

$$Z_{\mathrm{path}}(\omega) = Z_{\mathrm{VRM}}(\omega) + R_{\mathrm{path}} + \mathrm{j}\omega L_{\mathrm{path}}, \tag{14.5}$$

$$Z_{\mathrm{cap}}(\omega) = R_{\mathrm{ESR}} + \mathrm{j}\omega L_{\mathrm{ESL}} + \frac{1}{\mathrm{j}\omega C}, \tag{14.6}$$

$$Z_{\mathrm{PDN}}(\omega) \approx Z_{\mathrm{path}} \parallel Z_{\mathrm{cap},1} \parallel Z_{\mathrm{cap},2} \parallel \cdots \tag{14.7}$$

Real PDNs add many capacitors, plane modes, package parasitics and on-die behavior, but the structure is the same.

**The regulator's own impedance.** $Z_{\mathrm{VRM}}$ is not zero. Within its control-loop bandwidth (typically tens to a few hundred kilohertz for a multiphase buck converter) the regulator holds its output close to the set point, and its output impedance is small and nearly resistive—deliberately so when a *load line* (adaptive voltage positioning) is specified. Above the loop bandwidth the regulator can no longer respond and its output looks inductive; the bulk capacitors must take over there. Remote sensing compensates the DC drop of the path but not its inductance.

Figure 14.4 evaluates Eq. (14.7) for the path of Table 14.1, a 100 µF capacitor (ESR 2.5 mΩ, ESL 0.4 nH) and a 1 µF capacitor (ESR 3 mΩ, ESL 0.25 nH). Two features stand out. The 0.5 MHz resonance of the path with the large capacitor is so damped by the ESR that its peak, 2.8 mΩ, stays below the target. But between the inductive side (the path in parallel with the large capacitor's ESL, 0.29 nH) and the 1 µF capacitor there is an anti-resonance near 6.8 MHz, where the impedance rises to 35.5 mΩ—seven times the target.

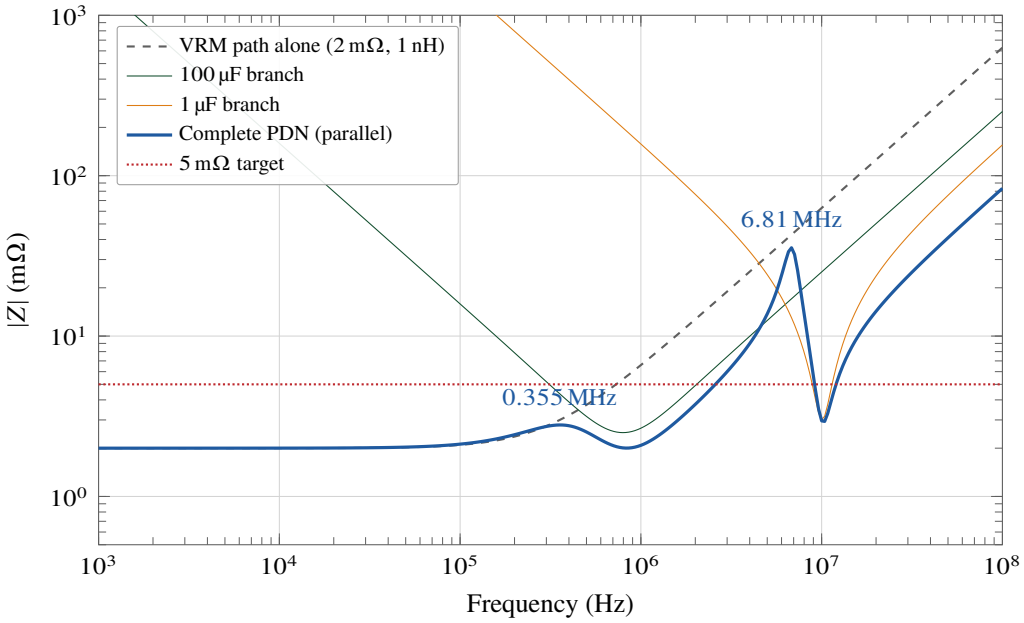



**Figure 14.4.** Impedance of a simple PDN: path (2 mΩ, 1 nH), a 100 µF capacitor (ESR 2.5 mΩ, ESL 0.4 nH) and a 1 µF capacitor (ESR 3 mΩ, ESL 0.25 nH), against a 5 mΩ target. Labels mark the local peaks. [Analytical]

**Worked Example 14.3: Locating the anti-resonance**

Estimate the frequency and height of the anti-resonance in Figure 14.4.

**Solution.** Above a few hundred kilohertz the path and the 100 µF branch are both inductive, so they combine as $1 \parallel 0.4 = 0.29$ nH with resistance $2 \parallel 2.5 = 1.1$ mΩ. Adding the small capacitor's own ESL, Eq. (16.6) gives $f = 1/(2\pi\sqrt{(0.29 + 0.25)\,\text{nH} \times 1\,\mu\text{F}}) = 6.9$ MHz. There the inductive reactance is $X = 2\pi f \times 0.29\,\text{nH} = 12.2\,\text{m}\Omega$, and Eq. (16.7) gives $|Z| \approx X^2/(1.1 + 3)\,\text{m}\Omega = 36\,\text{m}\Omega$. The exact sweep gives 6.81 MHz and 35.5 mΩ.

| Quick estimate | Full model | Difference |
|---|---|---|
| Estimate: 6.9 MHz, 36 mΩ | Exact: 6.81 MHz, 35.5 mΩ | 1 %, 1 % |

## 14.6 Target impedance and the voltage budget

For a first-order estimate, the target impedance is

$$Z_{\text{target}} = \frac{\Delta V_{\text{allowed}}}{\Delta I_{\text{transient}}}, \tag{14.8}$$

so a rail that may deviate by 50 mV for a 10 A load change needs $Z_{\text{target}} = 5$ mΩ (Chapter 13). The design goal is low PDN impedance over the frequencies present in the chip's current, which is not the same as low DC resistance.

**Common Pitfall: Target impedance as a guarantee**

Treating "$|Z| < Z_{\text{target}}$ everywhere" as proof that every load step meets the voltage specification. It is a first-pass guideline. The voltage budget must also cover DC IR drop, regulator tolerance and ripple

(Table 16.1), and the frequency-domain result must be confirmed with a time-domain load step that uses the real current profile (Section 16.6).

## 14.7 Resonance, ringing and anti-resonance

Inductance and capacitance exchange energy. For a simple *LC* loop the undamped natural frequency is that of Eq. (14.4); for 1 nH and 100 µF, $f_0 \approx 503$ kHz, which is why the voltage in Figure 14.3 oscillates with a period of about 2 µs instead of returning smoothly to its final value. With several capacitor values and their parasitic inductances, *anti-resonances* appear as well: narrow frequency ranges where the PDN impedance peaks (Figure 14.4). Damping—ESR chosen near the target, controlled-ESR parts, or lossy plane structures—and careful selection and placement of capacitors reduce the peaks (Section 16.4).

## 14.8 First, second and third droops

A real PDN is a ladder of stages (Figure 13.2), and each inductance–capacitance pair along it produces its own droop on its own time scale. Seen from the die after a fast load step, they appear in sequence (Figure 14.5):

**First droop**
: nanoseconds after the step: the on-die capacitance discharges while the package inductance limits the current from the board. It is usually the deepest and fastest, with a resonance from tens to a few hundred megahertz, and only on-die and package capacitance can address it.

**Second droop**
: hundreds of nanoseconds to about a microsecond: the board's ceramic capacitors discharge while board spreading and via inductance limits the current from the bulk capacitors; resonance around 1 MHz.

**Third droop**
: microseconds to tens of microseconds: the bulk capacitors discharge until the regulator's control loop responds.

After the third droop the voltage settles at the DC level set by the IR drop and the regulator's load line.

**Rule of Thumb: Who fixes which droop**

Board capacitors cannot fix the first droop: the package inductance stands between them and the die. The board designer controls the second droop (MLCC count, mounting inductance, plane spreading) and shares the third with the regulator design (bulk capacitance, loop bandwidth). *Limits:* the boundaries between the droops shift with the design; identify them from the impedance seen at the die.

## 14.9 The inductive voltage is a derivative

A useful first intuition is that "high-frequency transient current creates voltage on parasitic inductance, which then disturbs the power rail." Two details make it precise. First, $V_L$ need not look like a ramp: it is proportional to the *derivative* of the current, so its shape follows the current's rise, fall and curvature. Second, a load transient does not ordinarily change a linear PDN's impedance; it excites the existing impedance (Section 14.10).

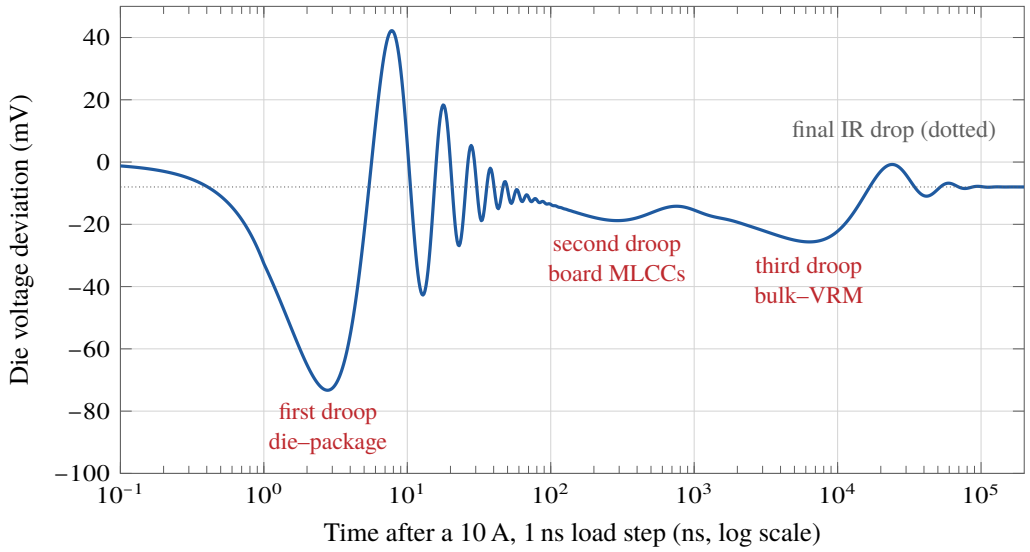



**Figure 14.5.** Die-voltage response of a three-stage ladder to a 10 A, 1 ns load step (log time axis): die 200 nF with package 12.7 pH; board MLCCs 100 µF behind 253 pH; bulk 2 mF behind an inductive regulator model (14 nH, 0.5 mΩ). The droops are 73 mV at 2.8 ns, 19 mV at 0.29 µs and 26 mV at 6.3 µs. [Simulation (lumped ladder)]

**Table 14.4.** Current waveform and ideal inductor voltage $V_L = L\ \mathrm{d}I/\mathrm{d}t$.

| Current waveform | Inductor voltage |
|---|---|
| Constant current | Zero |
| Linear ramp | Constant during the ramp |
| Pulse with finite linear edges | Positive and negative rectangular pulses at the edges |
| Pulse with smooth edges | Bell-shaped pulses, higher peak for the same edge duration |
| Sinusoid | Sinusoid, leading the current by 90° |

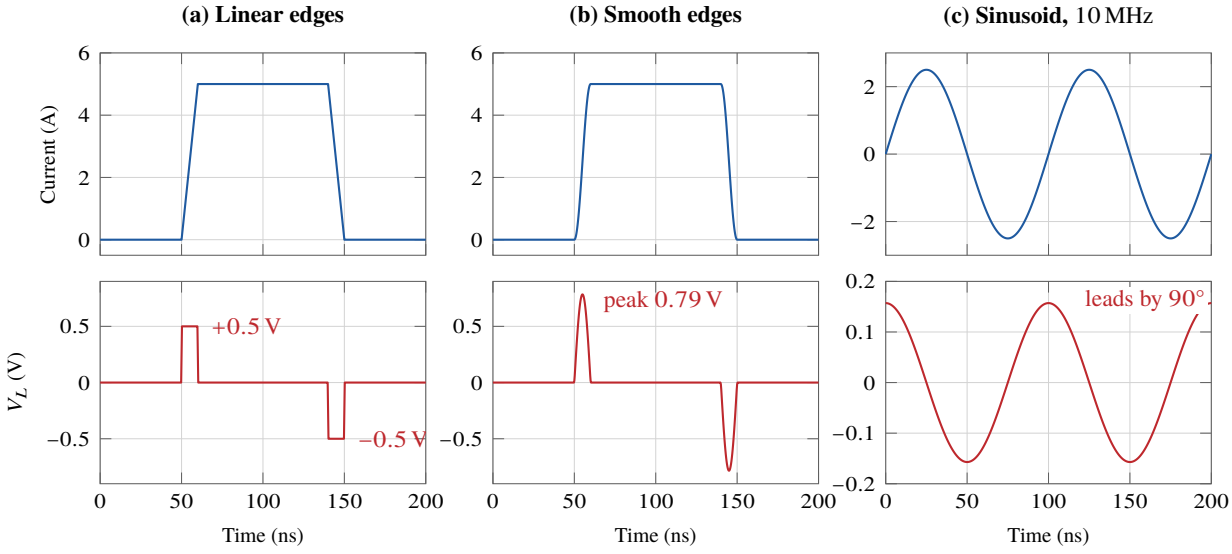



**Figure 14.6.** Current (top) and voltage across 1 nH (bottom). (a) A 0–5 A pulse with 10 ns linear edges gives ±0.5 V rectangles. (b) The same pulse with raised-cosine edges of the same full duration gives bell-shaped pulses peaking at 0.79 V. (c) A 2.5 A, 10 MHz sinusoid gives 0.16 V, leading by 90°. [Analytical]

**Worked Example 14.4: Inductive voltage of a current pulse**

A 0–5 A current pulse with 10 ns linear edges flows through 1 nH. Find the inductive voltage during each part of the pulse, and repeat for raised-cosine edges of the same full duration.

**Solution.** Rising edge: $V_L = 1\,\text{nH} \times 5\,\text{A}/10\,\text{ns} = +0.5\,\text{V}$; flat top: zero; falling edge: $-0.5\,\text{V}$ (Figure 14.6a). A linear ramp therefore produces a *constant* voltage during the ramp, not a voltage ramp. For a raised-cosine edge the peak slope is $\pi/2$ times the average, so the peak is $\frac{\pi}{2} \times 0.5\,\text{V} = 0.79\,\text{V}$ (Eq. (15.2) uses the same fact). These are voltages across the inductance, not necessarily the rail droop.

**Pulses contain many frequencies.** A transient can be written as a sum of sinusoids. For one component,

$$I(t) = I_p \sin(2\pi f t) \quad \Longrightarrow \quad V_L(t) = 2\pi f L\, I_p \cos(2\pi f t), \tag{14.9}$$

so for the same amplitude the inductive voltage grows with frequency and inductance, and a faster edge, which carries more high-frequency content, produces more of it. One nuance: raising a pulse train's repetition rate does not by itself raise the inductive peak at each edge, if each pulse keeps the same amplitude and edge time. Edge rate, spectral content and the circuit's response all matter.

**When the voltage subtracts or adds.** Without local capacitance, Eq. (14.1) with $I_{\text{VRM}} = I_{\text{load}} = I$ reads

$$V_{\text{chip}}(t) = V_{\text{VRM}}(t) - R\,I(t) - L\frac{\mathrm{d}I(t)}{\mathrm{d}t}. \tag{14.10}$$

With the current defined as flowing from the regulator to the chip, a rising current makes the inductive term subtract from the supply (droop); a falling current reverses its sign and produces overshoot, as on load release in Figure 13.1. In a real network the local capacitors and the several inductances share the transient and change the exact waveform.

## 14.10 Impedance is the response function

For an approximately linear PDN, define $\Delta I_{\text{load}}$ as positive for an increase in chip current. Then, in the frequency domain,

$$\Delta V_{\text{chip}}(f) = -Z_{\text{PDN}}(f)\,\Delta I_{\text{load}}(f). \tag{14.11}$$

The impedance is the network's *response function*; the transient current is the *excitation*. If the current spectrum has a strong component near a high-impedance peak, such as an anti-resonance, the voltage disturbance is large. The transient does not create a new inductance or change a linear network's impedance—it reveals it. (Nonlinear and operating-point-dependent effects, such as the regulator's slew limits or capacitance that varies with bias, are separate considerations.)

Figure 14.7 demonstrates this with the PDN of Figure 14.4. The same 0–10 A square-wave load is applied at two repetition rates. At 6.8 MHz its fundamental sits on the anti-resonance and the rail swings 433 mV peak to peak. At 3.4 MHz a 50 % square wave has no even harmonics, so no component lands on the 6.8 MHz peak, and the swing is 220 mV—half as much, from the same current amplitude.

## 14.11 Practical ways to improve the response

Table 14.5 maps each problem to its engineering approach.

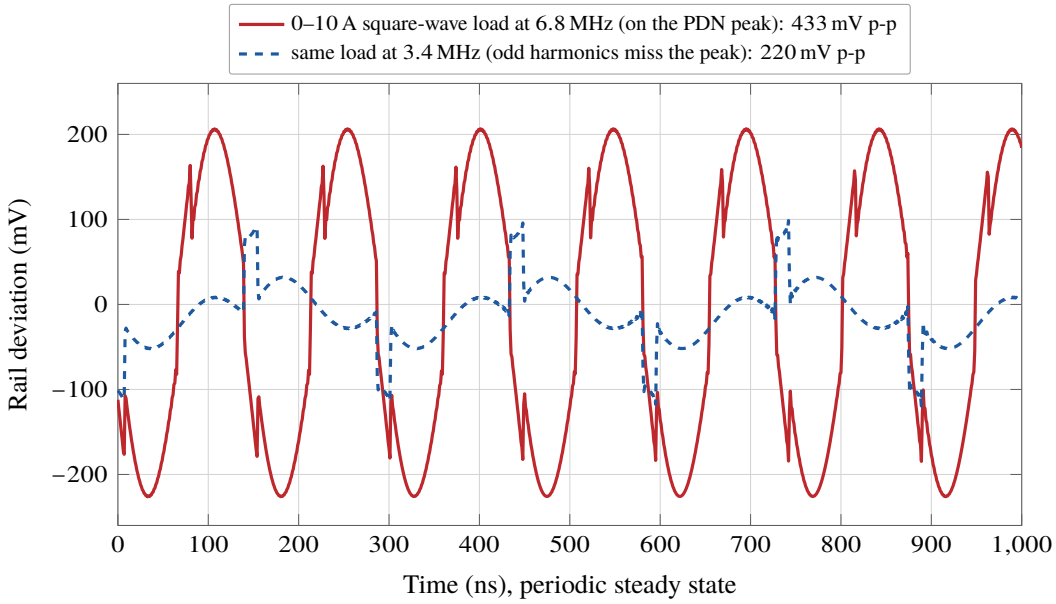



**Figure 14.7.** Periodic steady-state rail deviation for a 0–10 A, 50 % square-wave load with 15 ns edges applied to the PDN of Figure 14.4, at 6.8 MHz and 3.4 MHz, from the Fourier series and Eq. (14.11). [Analytical]

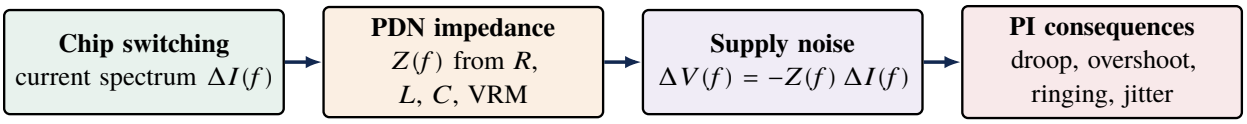



**Figure 14.8.** The end-to-end relationship: the chip's current spectrum excites the PDN impedance, producing the supply disturbance and its consequences. [Conceptual]

**Table 14.5.** Problems and engineering approaches.

| Problem | Engineering approach |
|---|---|
| High regulator-to-chip inductance | Closely coupled power/ground planes; short, wide current paths; via arrays with power and ground vias interleaved |
| Large transient current | Enough effective local decoupling capacitance (biased value) |
| Very fast current edges | Minimize capacitor connection and return-loop inductance; rely on package and die capacitance for the first droop |
| Excessive voltage droop | More local energy storage; faster regulator transient response |
| PDN impedance peaks | Optimize capacitor ESR, ESL, value, placement and damping |
| Excessive DC IR drop | Reduce copper resistance; use remote sensing where appropriate |
| Ringing | Control $LC$ resonances and provide damping |

**Design Rule: Minimize loop inductance, not path length**

Inductance belongs to the complete supply-and-return loop. Place power and ground vias in adjacent pairs so their opposing currents cancel flux, keep power and ground planes on thin dielectric next to each other (spreading inductance is about 32 pH per square per mil of separation, Chapter 3), and mount each capacitor with its vias at the pads. Reducing loop area can matter more than shortening the power path alone.

**The complete sequence.** A load step therefore unfolds in five stages: the chip current rises; inductance limits how fast the path current can follow; capacitors supply the difference and the voltage droops; the regulator catches up and restores the voltage; and damping and control dynamics decide any overshoot or ringing. High transient current alone does not determine the droop: the amplitude and speed of the current change, together with the full frequency-dependent PDN impedance, do. At very high frequencies a local capacitor's own ESL, its connection geometry and the package inductance become the limitation—the subject of Chapter 16.

**Table 14.6.** Equation reference for this chapter.

| Relationship | Equation |
|---|---|
| Inductor voltage | $V_L = L\,\mathrm{d}I/\mathrm{d}t$ |
| Node-current balance | $I_{\mathrm{VRM}} - I_{\mathrm{load}} = C\,\mathrm{d}V_{\mathrm{chip}}/\mathrm{d}t$ |
| Capacitor droop | $\Delta V_C = \frac{1}{C}\int (I_{\mathrm{load}} - I_{\mathrm{VRM}})\,\mathrm{d}t$ |
| Inductive reactance | $\lvert Z_L\rvert = 2\pi f L$ |
| Loop resonance, impedance, damping | $f_0 = 1/(2\pi\sqrt{LC})$, $Z_c = \sqrt{L/C}$, $\zeta = \frac{R}{2}\sqrt{C/L}$ |
| PDN voltage disturbance | $\Delta V_{\mathrm{chip}}(f) = -Z_{\mathrm{PDN}}(f)\,\Delta I_{\mathrm{load}}(f)$ |
| First-pass target impedance | $Z_{\mathrm{target}} = \Delta V_{\mathrm{allowed}}/\Delta I_{\mathrm{transient}}$ |

**Key Takeaways**

- The chip's demand $I_{\mathrm{load}}$ and the regulator-path current $I_{\mathrm{VRM}}$ differ during a transient; local capacitors supply the difference.
- $L\,\mathrm{d}I/\mathrm{d}t$ is the voltage across the inductance if all the current flows through it, not the chip's droop; the minimum voltage occurs where $I_{\mathrm{VRM}}$ catches up with $I_{\mathrm{load}}$, and charge balance gives its depth.
- More path inductance raises $\sqrt{L/C}$, lowers damping, and deepens and lengthens the droop.
- The PDN impedance is the parallel combination of the path and capacitor branches; anti-resonances between them can exceed the target many times.
- Real PDNs show first, second and third droops from die/package, board and bulk/VRM.
- $V_L$ follows the derivative of the current, and a transient excites the PDN impedance rather than changing it: $\Delta V(f) = -Z(f)\,\Delta I(f)$.

## Further Reading

Smith and Bogatin [7] develop the load-step and impedance views in detail, including first, second and third droops; Swaminathan and Engin [27] treat PDN modeling across die, package and board.

## Review Problems

**P14.1** A load current rises by 8 A in 40 ns. What voltage develops across 0.5 nH and across 2 nH if all of it flows there?

**P14.2** A 470 µF capacitor bank supplies a 6 A shortfall for 2 µs. Estimate its droop with Eq. (14.3).

**P14.3** For 2 nH of path inductance, 220 µF of local capacitance and 3 mΩ of loop resistance, find $f_0$, $Z_c$ and $\zeta$.

**P14.4** Sketch $V_L$ across 1 nH for a current that rises linearly from 0 to 4 A in 20 ns, stays for 50 ns, and falls in 10 ns.

**P14.5** *(Approximation failure.)* A colleague computes $L\,\mathrm{d}I/\mathrm{d}t = 500\,\mathrm{mV}$ for the 5 nH case of Table 14.1 and concludes the rail droops 500 mV. What is wrong, and what is the simulated droop?

**P14.6** *(Inverse design.)* With the 100 µF local capacitor of Table 14.1, what path inductance gives $Z_c = 2\,\mathrm{m\Omega}$?

**P14.7** A processor's workload toggles a 6 A current at 2.3 MHz with a 50 % duty cycle. Which harmonic of this load falls near the 6.8 MHz peak of Figure 14.4, and why does it matter?

**P14.8** *(Imperfect data.)* A scope capture of a rail shows a 60 mV dip lasting 3 ns after a load step. Which droop is this likely to be, and which board change would help most?

Chapter 15

# Ground Bounce and Simultaneous Switching Noise

**Chapter Overview**

When many outputs of a chip switch at the same time, their combined current flows through the shared inductance of the package's power and ground connections. The resulting voltage, $L\,\mathrm{d}i/\mathrm{d}t$, appears between the die's internal ground (or power) and the board's, and it disturbs every other circuit on the die that references the same rail. This chapter develops the equivalent circuit of ground bounce and simultaneous switching noise (SSN), shows how mutual inductance limits the benefit of adding ground pins, and surveys the package, board, and signaling techniques used to control it. It is the bridge between signal integrity and the power-integrity chapters that follow.

**Learning Objectives**

1. Derive the ground-bounce voltage from the shared return inductance and the switching current.
2. Estimate switching current and its rate of change for drivers loaded by transmission lines.
3. Compute the effective inductance of parallel ground pins, including mutual inductance.
4. Explain how SSN converts into timing jitter and false switching.
5. Select mitigations: pin assignment, package choice, slew control, on-die decoupling, differential signaling, and data-bus inversion.

## 15.1 The mechanism

Consider $N$ output buffers on one die driving loads high-to-low at the same time (Figure 15.1). Each pulls current from its load through the die's ground network, which connects to the board through package pins or balls with total effective inductance $L_{\mathrm{gnd}}$. The total current changes at a rate $N\,\mathrm{d}i/\mathrm{d}t$, so the die ground moves relative to the board ground by

$$V_{\mathrm{gb}} = L_{\mathrm{gnd}}\,N\frac{\mathrm{d}i}{\mathrm{d}t}. \tag{15.1}$$

A quiet driver on the same die, holding its output low, connects its output to the die ground through its on-resistance; its output therefore follows the die ground and launches a glitch of roughly $V_{\mathrm{gb}}$ onto its line. Inputs referencing the die ground see the same shift as a change in threshold. The identical mechanism on the power side—current drawn through the package's power inductance when outputs switch low-to-high—is called *power bounce*, and the two together are simultaneous switching noise [3], [27].

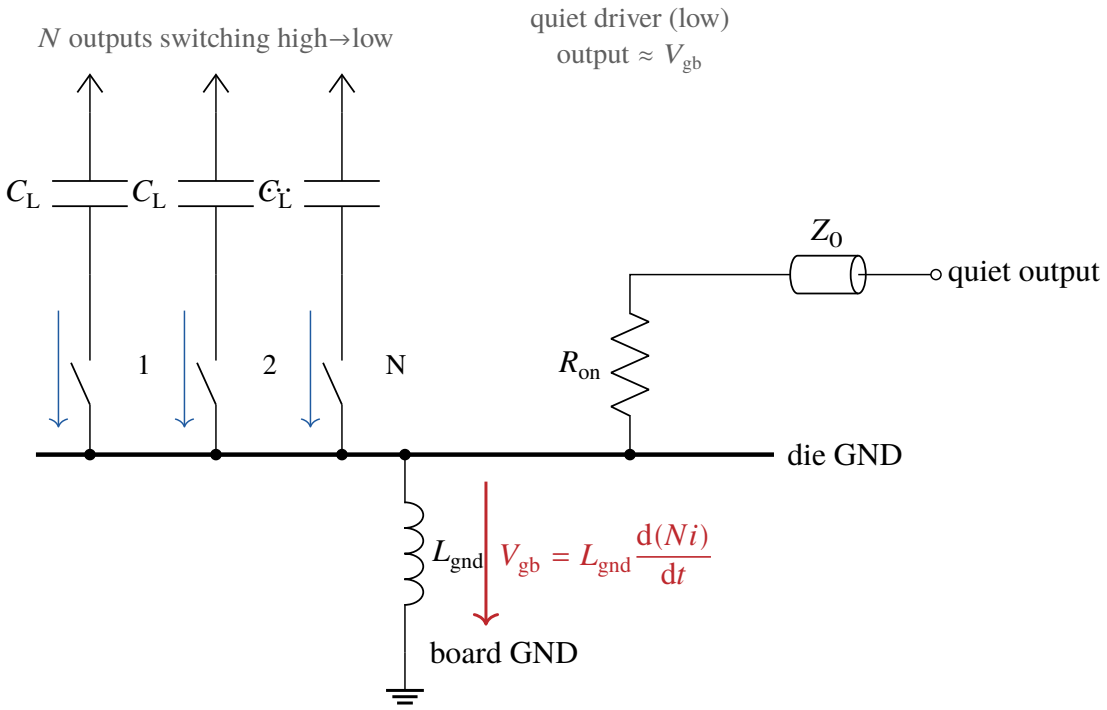



**Figure 15.1.** Equivalent circuit of ground bounce. $N$ switching outputs discharge their loads through the shared package ground inductance $L_{gnd}$. A quiet driver held low follows the die ground and transmits the bounce onto its own line.

## 15.2 Estimating the switching current

The peak $\mathrm{d}i/\mathrm{d}t$ depends on the shape and on the definition of the edge, so both must be stated. Let $T_e$ be the *full* (0–100 %) duration of the current edge and $t_r$ its 10–90 % time.

**Line-driven outputs.** A driver of output resistance $R_S$ launching into a line of impedance $Z_0$ draws a current *step* $\Delta I = \Delta V/(R_S + Z_0)$ whose shape follows the voltage edge. A linear ramp has $\mathrm{d}i/\mathrm{d}t = \Delta I/T_e = 0.8\,\Delta I/t_r$. A raised-cosine edge, $i = \frac{1}{2}\Delta I[1 - \cos(\pi t/T_e)]$, has peak slope $\frac{\pi}{2}\Delta I/T_e$; since its 10–90 % time is $0.590\,T_e$, that is $0.93\,\Delta I/t_r$. For a fixed load, the bounce scales as $N\Delta I/t_r$: halving the edge time doubles it.

**Lumped capacitive loads.** A driver charging a lumped capacitance $C_L$ draws $i = C_L\,\mathrm{d}v/\mathrm{d}t$: a current *pulse* that rises and falls within the edge. For a raised-cosine voltage edge of full duration $T_e$, $i = C_L\Delta V\,\frac{\pi}{2T_e}\sin(\pi t/T_e)$, whose peak derivative is

$$\left.\frac{\mathrm{d}i}{\mathrm{d}t}\right|_{\max} = \frac{\pi^2}{2}\,\frac{C_L\,\Delta V}{T_e^2}, \tag{15.2}$$

so the bounce scales as $1/T_e^2$: halving the edge time quadruples it. (A strictly linear voltage ramp would give a rectangular current with infinite slope at its corners; real edges are smooth, which is why the raised cosine is used.)

**Worked Example 15.1: Ground bounce of a 16-bit bus**

Sixteen outputs switch low together, each driving a 50 Ω line with a 1.2 V swing through 40 Ω of driver resistance, with current edges of full duration $T_e = 200$ ps. The die returns through eight ground balls, each with 1 nH of partial inductance; ignore mutual coupling. Estimate $V_{gb}$.

**Solution.** $\Delta I = 1.2/90 = 13.3$ mA per output. $L_{gnd} = 1\,\text{nH}/8 = 0.125$ nH. With a linear ramp, $V_{gb} = 0.125 \times 10^{-9} \times 16 \times 0.0133/(0.2 \times 10^{-9}) = 133$ mV; with a raised-cosine edge of the same full duration the peak is $\pi/2$ larger, about 209 mV (Figure 15.2). That is 17 % of the swing, injected onto every quiet line in the bank. The two edges have different 10–90 % times (160 ps and 118 ps), which is why a bounce estimate must name the edge definition it uses.

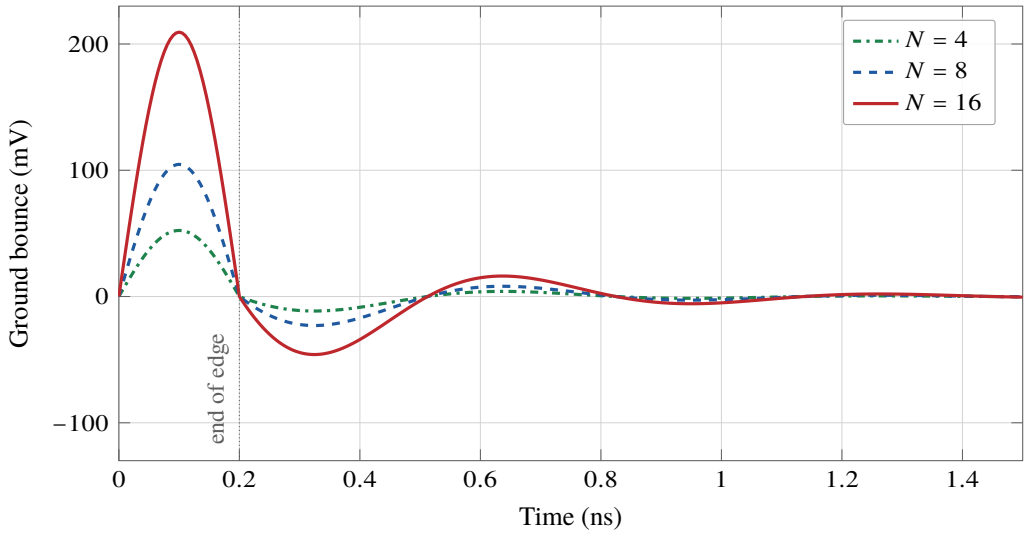



**Figure 15.2.** Ground bounce on a quiet output for $N$ simultaneously switching outputs (Worked Example 15.1 parameters, raised-cosine current edges). The pulse scales with $N$ and is followed by ringing of the package inductance with the on-die capacitance.

## 15.3 Parallel pins and mutual inductance

Adding ground pins in parallel reduces $L_{gnd}$, but not in simple proportion. If $N_g$ ground pins each have partial self inductance $L$ and every pair has partial mutual inductance $M$ (currents in the same direction), the effective inductance is

$$L_{\text{eff}} = \frac{L + (N_g - 1)M}{N_g} \xrightarrow{N_g \to \infty} M. \tag{15.3}$$

Closely spaced ground pins carrying current in the same direction share flux and add little. Interleaving power and ground pins helps twice: the opposite currents in neighboring power and ground pins have *negative* mutual inductance, reducing the loop inductance of each power–ground pair (Eq. (3.5)).

**Worked Example 15.2: Diminishing returns from extra ground pins**

For the bus of Worked Example 15.1, how many independent ground pins would keep the linear-ramp bounce below 50 mV? Repeat if adjacent ground pins have $M = 0.3$ nH.

**Solution.** Required $L_{\text{eff}} \le 0.05 \times 0.2 \times 10^{-9}/(16 \times 0.0133) = 47\,\text{pH}$. Without mutual inductance, $N_g \ge 1/0.047 \approx 22$ pins. With $M = 0.3\,\text{nH}$, Eq. (15.3) cannot go below 300 pH however many pins are added (eight pins give 388 pH). The budget can only be met by spreading the pins, interleaving power and ground, slowing edges, or reducing $N$.

**Common Pitfall: Clustering ground pins**

Assigning all ground balls of a BGA to a contiguous block in the corner and the signals to another. The clustered ground balls are strongly mutually coupled and far from the signals they return, so both $L_{\text{eff}}$ and the signal loop inductance are high. Distribute power and ground balls among the signal balls, ideally with a return ball adjacent to every few signals.

## 15.4 Effects on timing and logic

SSN harms a design in three ways:

1. **False switching.** A bounce exceeding the noise margin of a quiet output's receiver can be interpreted as a logic transition, especially on clocks and asynchronous resets.
2. **Push-out and jitter.** The switching outputs themselves are slowed, because the bounce reduces the effective drive voltage while they switch. The delay change depends on how many neighbors switch in the same direction, so it is pattern dependent: SSN appears as data-dependent jitter.
3. **Reference disturbance.** Single-ended receivers compare the input with a reference ($V_{\text{REF}}$) or with their own ground. A bouncing ground or reference shifts the threshold and closes the eye vertically.

### 15.4.1 From die bounce to the quiet line

The bounce is a voltage between die ground and board ground; what reaches a receiver depends on the path. A quiet driver held low connects its output to the die ground through its on-resistance $R_{\text{on}}$, and the line presents $Z_0$, so the wave launched onto the quiet line is

$$V_{\text{quiet}} \approx V_{\text{gb}} \frac{Z_0}{R_{\text{on}} + Z_0}, \tag{15.4}$$

which then propagates and reflects like any other wave: at an unterminated receiver it nearly doubles. For the bus of Worked Example 15.1 with $R_{\text{on}} = 40\,\Omega$, the 209 mV bounce launches 116 mV and can reach about 230 mV at an open receiver. Because Eq. (15.4) is linear, the quiet-line noise can be estimated from simulated or measured die-ground noise without repeating the full simultaneous-switching simulation.

### 15.4.2 Power-supply-induced jitter

Supply noise also moves edges. A driver's delay depends on its supply voltage; over a small range the dependence is linear, $\Delta t \approx k_{\text{PSIJ}}\,\Delta V$, with a sensitivity $k_{\text{PSIJ}}$ (picoseconds per millivolt) obtained from a transistor-level or IBIS simulation at the operating corner. Supply noise at frequencies below the CDR or training bandwidth is tracked out; noise above it appears as jitter. The PDN target of Chapter 16 therefore has a timing as well as an amplitude consequence, and the jitter budget of Chapter 12 should contain a PSIJ entry derived from the rail noise spectrum, not a guessed constant.

## 15.5 Mitigation

Table 15.1 lists the main remedies. Most act on one of the three factors of Eq. (15.1): the inductance, the number of simultaneously switching outputs, or the current slope.

**Table 15.1.** Techniques for controlling simultaneous switching noise.

| Technique | Mechanism | Factor reduced |
|---|---|---|
| More, well-distributed power/ground pins | Parallel paths; negative mutual between P/G | $L$ |
| Flip-chip or low-inductance package | Short connections, many bumps | $L$ |
| On-die and on-package decoupling | Local charge reservoir; current loop closes on die | Effective $L$ |
| Board vias adjacent to P/G balls | Short, low-inductance escape to planes | $L$ |
| Slew-rate control | Longer $t_r$ | $\mathrm{d}i/\mathrm{d}t$ |
| Weaker drive / series termination | Smaller $\Delta I$ | $\mathrm{d}i/\mathrm{d}t$ |
| Staggered output timing | Spreads switching in time | $N$ |
| Data-bus inversion (DBI) | Limits number of lines switching or driving low | $N$ |
| Differential signaling | Constant total current | $N\ \mathrm{d}i/\mathrm{d}t$ |

### 15.5.1 Data-bus inversion

DDR4 and later memories can transmit a byte either as is or inverted, flagged by an extra DBI signal, choosing whichever form minimizes the number of lines driven low (for pseudo-open-drain I/O, where only low outputs draw current) or the number of transitions. In the worst case, a byte then has at most four of its eight data lines drawing current, halving the worst-case SSN for one extra pin per byte [28].

**Rule of Thumb: Edge rate versus bounce**

Ground bounce scales as $1/t_r$: doubling the rise time halves the bounce, a 6 dB improvement, at no cost if the timing budget allows it. *Limits:* slower edges increase the uncertainty of threshold crossing under noise, and some interfaces specify minimum slew rates; check both before selecting a weaker drive or slew setting.

**Design Rule: Board-level SSN control**

Connect each power and ground ball of a high-I/O-count device to its plane with its own via, placed as close to the ball as the breakout allows; do not share one via among several balls. Place the plane pair that serves the I/O supply close to the device's side of the board, and provide decoupling whose mounted inductance is comparable to the package's own (Chapter 16).

**Key Takeaways**

- Ground bounce is $L_{\mathrm{gnd}} N\ \mathrm{d}i/\mathrm{d}t$ across the shared package return; quiet outputs and inputs on the same rail inherit it.
- The switching current of a driver into a line is $\Delta V/(R_S + Z_0)$; its slope is about $\Delta I/t_r$, or $\pi/2$ times that for smooth edges.
- Parallel ground pins approach the mutual inductance $M$ as a floor; interleave power and ground to exploit negative mutual inductance.

- SSN causes false switching, pattern-dependent delay (jitter), and threshold shifts.
- Mitigate with distributed pins, low-inductance packages, slew control, on-die decoupling, DBI, and differential signaling.

## Further Reading

Swaminathan and Engin [27] and Bogatin [3] analyze simultaneous switching noise and its package dependence.

## Review Problems

**P15.1** Eight outputs each switch 20 mA in 250 ps through a shared 0.5 nH ground inductance. Estimate the ground bounce for a linear current ramp of full duration 250 ps.

**P15.2** Four independent ground pins each have 1.2 nH. What is their effective inductance?

**P15.3** Repeat Problem P15.2 with 0.2 nH of mutual inductance between every pair.

**P15.4** With $L_{\text{eff}} = 0.1\,\text{nH}$, $\Delta I = 10\,\text{mA}$ per output, and linear current edges of full duration 150 ps, how many outputs may switch together for a 100 mV bounce limit?

**P15.5** A package ground inductance of 0.2 nH resonates with 10 nF of on-die capacitance. Find the resonant frequency.

**P15.6** By how many decibels is ground bounce reduced when the edge time is increased from 150 ps to 300 ps for line-driven outputs? For outputs driving lumped capacitive loads?

Chapter 16

# Power Distribution Network Design

**Chapter Overview**

The power distribution network (PDN) delivers current from the voltage regulator to every transistor on every die, and it must do so while keeping the supply within a tolerance band of a few percent, at frequencies from DC to beyond a gigahertz. The frequency-domain target-impedance method turns this time-domain requirement into a single impedance limit that each part of the PDN—regulator, bulk capacitors, ceramic capacitors, planes, package, and die—must meet in its own frequency band. This chapter develops the method, models each element, explains anti-resonance, and gives practical rules for selecting and placing decoupling capacitors and for budgeting DC IR drop.

**Learning Objectives**

1. Compute a PDN target impedance from ripple tolerance and transient current.
2. Identify which PDN element controls the impedance in each frequency band.
3. Model a capacitor by $C$, ESR, and mounted ESL, and compute its self-resonant frequency.
4. Predict and damp anti-resonances between capacitors, planes, and the package.
5. Choose capacitor quantity, values, placement, and via patterns.
6. Estimate DC IR drop in planes and size copper accordingly.

## 16.1 The target-impedance method

A load draws a current that varies in time. If the PDN looks like an impedance $Z(f)$ from the load's terminals, the supply noise is $V(f) = Z(f)I(f)$. Requiring $|Z(f)|$ to be below a constant value over the whole bandwidth of the current guarantees—for a resistive, flat PDN—that the voltage excursion stays within the allowed ripple for any current waveform within the stated amplitude [7], [29]:

$$Z_{\text{target}} = \frac{\Delta V_{\text{ripple}}}{\Delta I_{\text{transient}}} = \frac{V_{\text{DD}} \times (\text{ripple fraction})}{\Delta I_{\text{transient}}}. \tag{16.1}$$

The transient current is the largest expected change in load current, not the maximum current. When it is unknown, half the maximum current is a common assumption.

**Worked Example 16.1: Target impedance of a core rail**

An SoC core rail runs at 0.8 V with a $\pm 3\,\%$ tolerance and a maximum current of 20 A. Assume a 50 % transient.

**Solution.** $\Delta V = 0.03 \times 0.8 = 24\,\text{mV}$ and $\Delta I = 10\,\text{A}$, so $Z_{\text{target}} = 2.4\,\text{m}\Omega$. The tolerance must also cover the regulator's DC set-point accuracy and IR drop, so in practice only part of the 24 mV is available for dynamic noise (Table 16.1).

### 16.1.1 Allocating the voltage budget

A rail tolerance is shared by several independent error sources, and only one of them is the dynamic noise that $Z_{\text{target}}$ addresses. Table 16.1 shows an illustrative allocation of the $\pm 24\,\text{mV}$ of Worked Example 16.1. With 14 mV reserved for the specified 10 A step, the dynamic target becomes $1.4\,\text{m}\Omega$, not $2.4\,\text{m}\Omega$. The allocation must also say *where* the voltage is observed (board pads or die bumps) and must reserve a share for measurement uncertainty if it is to become an acceptance threshold. For continuity, the examples that follow use the unallocated $2.4\,\text{m}\Omega$.

**Table 16.1.** Illustrative voltage allocation for a 0.8 V, ±3 % rail (requirement values chosen for the example).

| Contribution | Allocation (mV) | Controlled by |
|---|---|---|
| Regulator set point, ripple, temperature | 4 | Regulator choice, remote sense |
| DC path (IR) drop, uncompensated | 6 | Copper, plane geometry |
| Dynamic response to 10 A step | 14 | PDN impedance ($Z_{\text{target}} = 1.4\,\text{m}\Omega$) |
| Total | 24 | ±3 % of 0.8 V |

**Common Pitfall: Treating the target as a hard flat line**

The target-impedance method assumes a flat, resistive PDN. A PDN that meets $Z_{\text{target}}$ everywhere but has high-$Q$ anti-resonance peaks just below it can still ring when the load current contains energy at those frequencies, as clock-gated loads do. Conversely, a narrow peak above $Z_{\text{target}}$ at a frequency where the load draws no current may be harmless. Use $Z_{\text{target}}$ as a design guide and confirm with a time-domain simulation of the expected current profile.

## 16.2 Frequency regions of a PDN

Each element of the PDN is effective only over a limited frequency range (Figures 16.1 and 16.2):

**Voltage regulator (DC to tens of kilohertz)**
The regulator's feedback loop holds the output impedance low up to roughly its loop bandwidth, above which its output impedance rises inductively.

**Bulk capacitors (kilohertz to about a megahertz)**
Electrolytic, polymer, or large ceramic capacitors hand off from the regulator. Their ESL of a nanohenry or more limits them above a megahertz.

**Ceramic capacitors (about 1 MHz to tens of megahertz)**
Banks of MLCCs, whose effectiveness is limited by *mounted* inductance.

**Planes (tens to hundreds of megahertz)**
The power–ground plane pair is a distributed capacitor with very low inductance, bounded at high frequency by its cavity resonances (Chapter 17).

**Package and die (above about 100 MHz)**

The package's inductance isolates the die from the board at high frequency; the die's own capacitance (gate and dedicated on-die decoupling) supplies the fastest current.

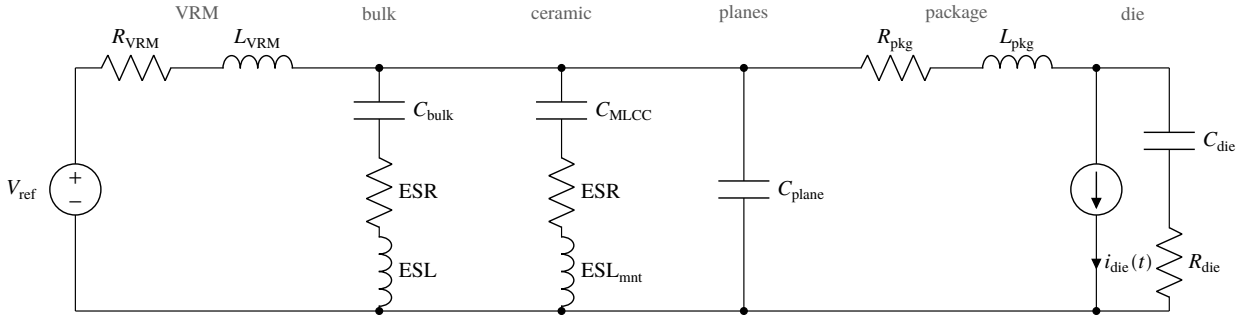



**Figure 16.1.** Lumped equivalent circuit of a PDN from the regulator to the die. Each stage has a capacitance limited by its own series inductance and resistance.

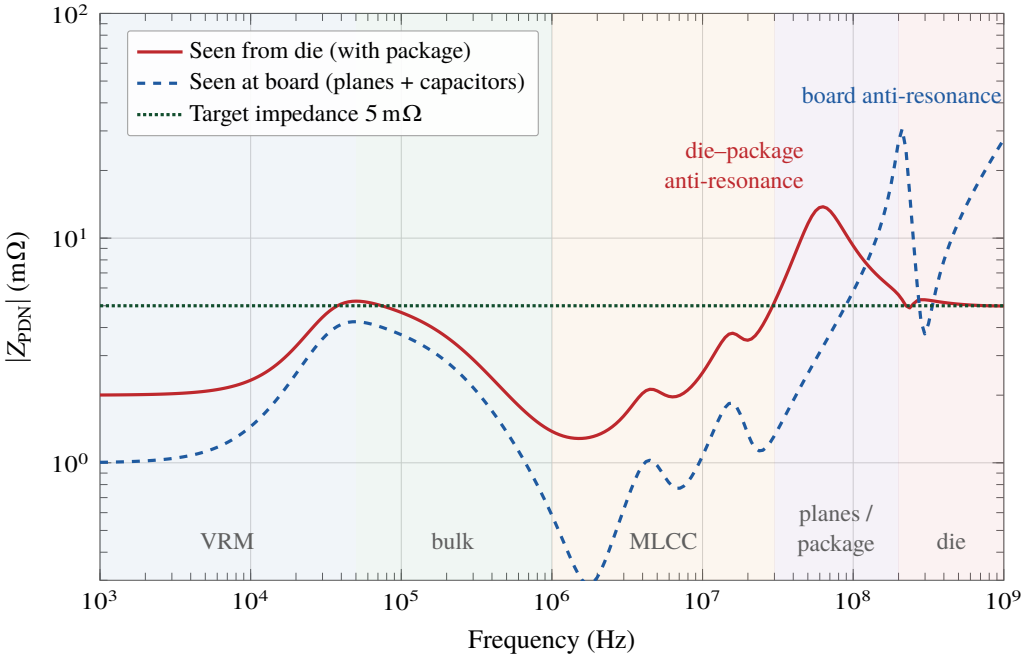



**Figure 16.2.** PDN impedance computed from the lumped model of Figure 16.1, seen at the board (planes and capacitors) and at the die (through the package). The board meets the 5 mΩ target up to about 90 MHz but has its own anti-resonance between MLCC inductance and plane capacitance near 200 MHz; seen from the die, the package inductance hides that peak, and the package inductance and on-die capacitance instead form an anti-resonance near 60 MHz that exceeds the target.

Figure 16.2 illustrates the most important limitation of board-level decoupling: above a few tens of megahertz, the impedance seen by the die is dominated by the package inductance and the die capacitance, not by anything on the board. The resulting die–package anti-resonance, typically between 30–200 MHz, is often the dominant PDN noise of a modern processor. The board can damp it—by presenting a low, resistive impedance at that frequency—but cannot move it.

## 16.3 Capacitor models

A real capacitor is a series RLC circuit:

$$Z_C(f) = \mathrm{ESR} + \mathrm{j}\omega\,\mathrm{ESL} + \frac{1}{\mathrm{j}\omega C}, \qquad |Z_C| = \sqrt{\mathrm{ESR}^2 + \left(\omega\mathrm{ESL} - \frac{1}{\omega C}\right)^2}, \tag{16.2}$$

with self-resonant frequency

$$f_{\mathrm{SRF}} = \frac{1}{2\pi\sqrt{\mathrm{ESL}\cdot C}}. \tag{16.3}$$

Below $f_{\mathrm{SRF}}$ the capacitor is capacitive; at $f_{\mathrm{SRF}}$ its impedance equals its ESR; above it, the capacitor is an inductor (Figure 16.3). Above its SRF a capacitor still helps, but only through its inductance, and that inductance is set mostly by how it is mounted, not by its capacitance value or case size. MLCC capacitance also falls with DC bias and temperature, especially in high-$k$ dielectrics (X5R, X7R); a 10 µF part at its rated voltage may provide only a third to a half of its nominal value.

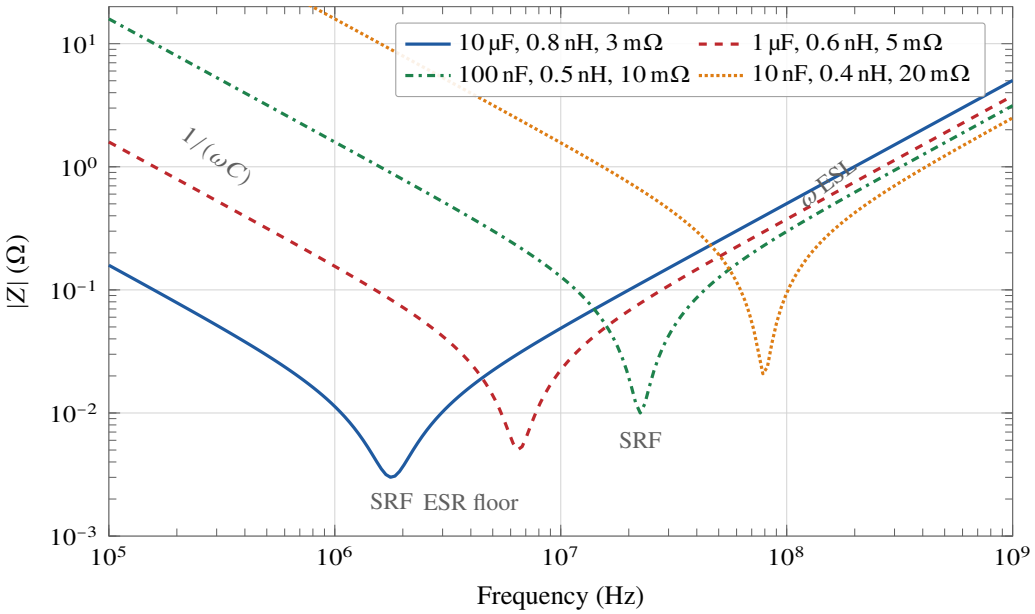



**Figure 16.3.** Impedance of four mounted MLCCs from Eq. (16.2). Each is capacitive below its self-resonant frequency (SRF), resistive at it, and inductive above it. Above about 30 MHz the curves converge, because mounted inductances are similar.

**Worked Example 16.2: Usefulness of a 100 nF capacitor**

A 100 nF 0402 MLCC has a mounted inductance of 0.5 nH. Find its SRF, its impedance at 100 MHz, and the number needed to reach 2.4 mΩ at 20 MHz. Take ESR = 10 mΩ.

**Solution.** $f_{\mathrm{SRF}} = 1/(2\pi\sqrt{0.5\times10^{-9}\times10^{-7}}) = 22.5$ MHz. At 100 MHz, $|Z| = 0.30$ Ω, close to $\omega\mathrm{ESL} = 0.31$ Ω—more than a hundred times the target.

At 20 MHz the capacitor is *below* its SRF, so both reactances matter: $X_L = \omega\mathrm{ESL} = 62.83$ mΩ and $X_C = 1/(\omega C) = 79.58$ mΩ. From Eq. (16.2), $|Z_C| = \sqrt{10^2 + (62.83 - 79.58)^2} = 19.50$ mΩ. For $N$

identical, independent capacitors the bank impedance is $Z_C/N$, so the pointwise requirement is

$$N(f) \geq \left\lceil \frac{|Z_C(f)|}{Z_{\text{target}}} \right\rceil, \tag{16.4}$$

which gives $N = \lceil 19.50/2.4 \rceil = 9$. The common shortcut $N = \omega\text{ESL}/Z_{\text{target}}$ assumes the capacitor is purely inductive and gives $\lceil 26.18 \rceil = 27$, three times too many at this frequency (Figure 16.4).

| Quick estimate | Full model | Difference |
|---|---|---|
| $\omega\text{ESL}/Z_{\text{target}}$: 27 | Full R–L–C: 9 | 200 % |

Equation (16.4) is a lower bound at one frequency in an idealized model. It ignores the mounting inductance the capacitors share (vias, spreading, the package), which prevents the $1/N$ improvement from continuing indefinitely, and it does not show that the bank meets the target at other frequencies. A sweep over frequency and a transient check (Section 16.6) are still required. Above about 30 MHz, the planes and the package must take over.

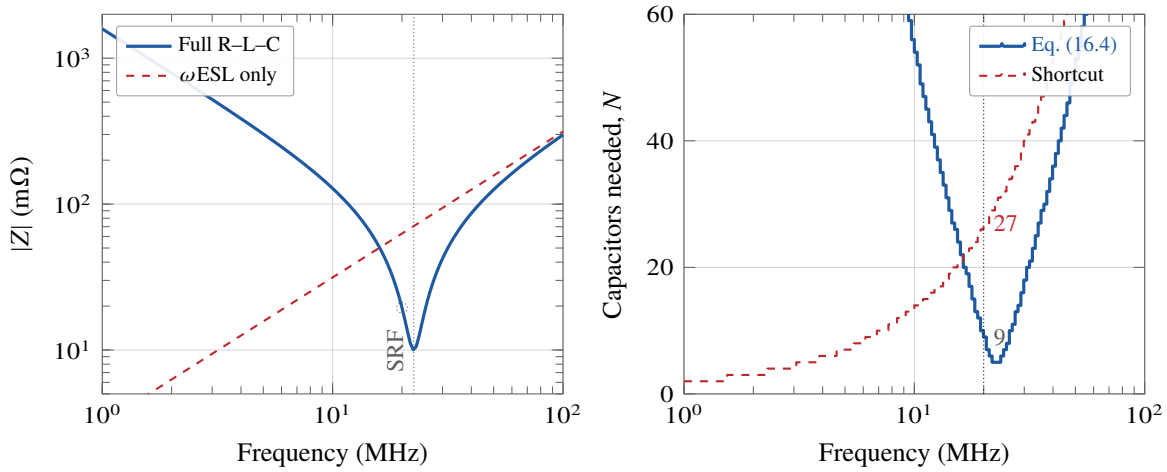



**Figure 16.4.** Left: impedance of a 100 nF, 10 mΩ, 0.5 nH capacitor and its inductive-only approximation. Right: pointwise number of ideal, independent capacitors needed for $Z_{\text{target}} = 2.4\,\text{m}\Omega$, from Eq. (16.4) and from the shortcut. Shared inductance is ignored. [Analytical]

### 16.3.1 Mounted inductance

The mounted ESL of a capacitor is the loop inductance of the path from the capacitor's terminals through its pads, mounting traces, and vias, to the plane pair, plus the capacitor's intrinsic inductance. The via portion follows Eq. (3.8): it is proportional to the depth of the plane pair below the capacitor and logarithmic in via spacing. Figure 16.5 ranks common mounting patterns. Placing vias at the sides of the pads, as close together as possible, roughly halves the inductance of end-placed vias; long traces between pad and via can double it.

**Design Rule: Mounting decoupling capacitors**

Connect every decoupling capacitor directly to its plane pair with vias at the sides of (or in) its pads; never route a trace from the pad to a distant via. Put the plane pair that serves a rail as close as possible to the

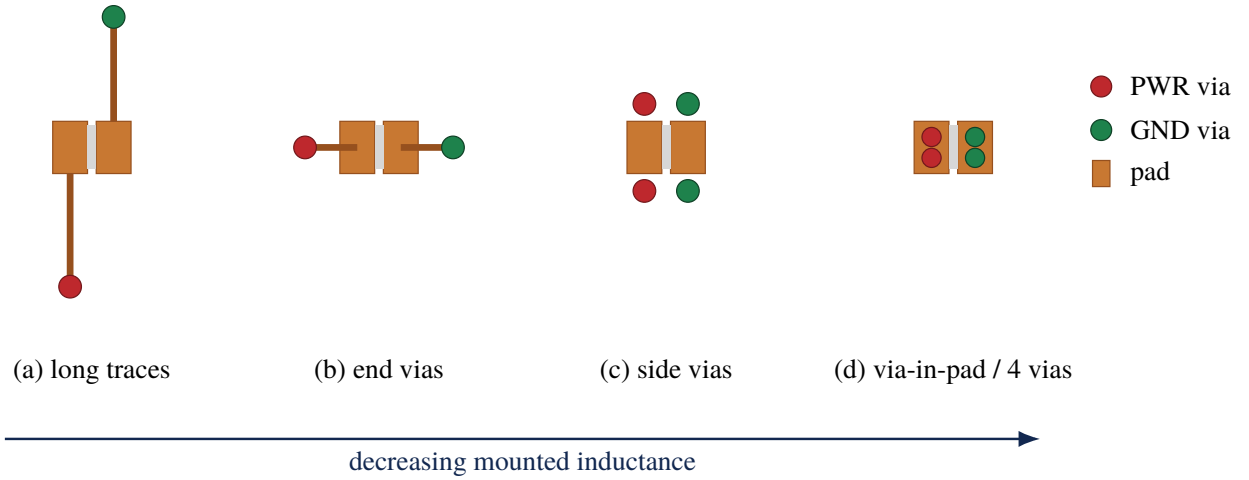



**Figure 16.5.** Decoupling-capacitor mounting patterns, from highest to lowest mounted inductance. Short connections, vias at the sides of the pads, and multiple via pairs reduce the loop area.

surface on which its capacitors are mounted. Prefer reverse-geometry (0204, 0306) or multi-terminal capacitors for the highest-frequency decoupling.

## 16.4 Anti-resonance

When two capacitors with different SRFs are placed in parallel, there is a frequency between their SRFs at which the larger capacitor is already inductive and the smaller is still capacitive. The two form a parallel LC tank, and the impedance peaks.

For two series branches $Z_k = R_k + \mathrm{j}(\omega L_k - 1/\omega C_k)$, $k = 1, 2$, the parallel impedance is

$$Z_{\parallel} = \frac{Z_1 Z_2}{Z_1 + Z_2}. \tag{16.5}$$

Without resistance the denominator vanishes when the two reactances are equal and opposite,

$$\omega L_1 - \frac{1}{\omega C_1} = -\left[\omega L_2 - \frac{1}{\omega C_2}\right],$$

that is, at

$$f_{\mathrm{ar}} = \frac{1}{2\pi}\sqrt{\frac{C_1 + C_2}{(L_1 + L_2)\, C_1 C_2}} \xrightarrow{C_1 \gg C_2} \frac{1}{2\pi\sqrt{(L_1 + L_2)\, C_2}}. \tag{16.6}$$

At that frequency the branches are $R_1 + \mathrm{j}X$ and $R_2 - \mathrm{j}X$, where $X = \omega L_1 - 1/(\omega C_1)$; if $X \gg R_1, R_2$, Eq. (16.5) gives

$$|Z|_{\mathrm{peak}} \approx \frac{X^2}{R_1 + R_2}. \tag{16.7}$$

The peak is high when ESRs are low—which is why the very low ESR of modern MLCCs, together with the low loss of plane capacitance, makes anti-resonance a real problem. A frequently quoted simplification keeps only the large capacitor's ESL and the small capacitor's capacitance, $f \approx 1/(2\pi\sqrt{L_1 C_2})$; it omits the small capacitor's own ESL, which is of the same order as $L_1$, and is not accurate enough to place a peak.

**Worked Example 16.3: Anti-resonance of two capacitors**

A 100 nF MLCC (ESL 0.5 nH, ESR 20 mΩ) is paralleled with a 10 nF MLCC (ESL 0.4 nH, ESR 30 mΩ). Estimate the anti-resonance.

**Solution.** Equation (16.6) with both inductances: $f_{ar} = \frac{1}{2\pi}\sqrt{110 \times 10^{-9}/(0.9 \times 10^{-9} \times 10^{-15})} = 55.6\,\text{MHz}$. There $X = \omega L_1 - 1/(\omega C_1) = 174.8 - 28.6 = 146\,\text{m}\Omega$, and Eq. (16.7) gives $|Z|_{peak} \approx 0.146^2/0.050 = 0.43\,\Omega$. A sweep of the exact Eq. (16.5) (Figure 16.6, Listing E.2) puts the maximum at 55.16 MHz and 0.443 Ω, about three times the 0.146 Ω that the 100 nF capacitor alone has at that frequency. Adding the small capacitor made the PDN worse there.

| Quick estimate | Full model | Difference |
|---|---|---|
| $1/\sqrt{L_1 C_2}$: 71.2 MHz, 1.0 Ω | Exact: 55.16 MHz, 0.443 Ω | 29 %, ×2.3 |

The shortcut misplaces the peak by 16 MHz and overstates it by more than a factor of two, because it ignores the 0.4 nH of the small capacitor. The qualitative lesson survives: a capacitor can create a harmful peak.

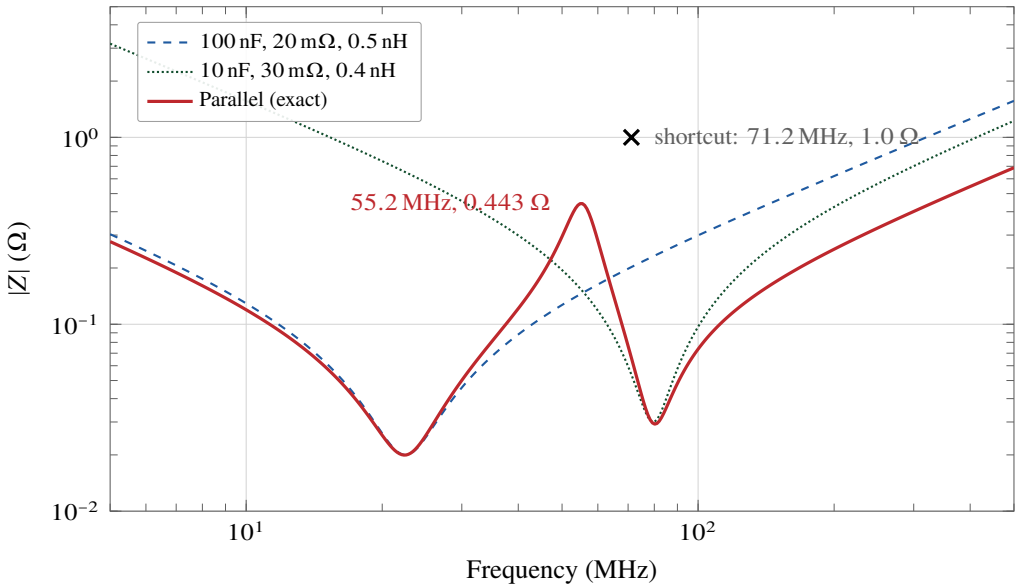



**Figure 16.6.** Exact impedance of the two branches of Worked Example 16.3 and of their parallel combination. The cross marks the prediction of the $1/\sqrt{L_1 C_2}$ shortcut. [Analytical]

Anti-resonances are controlled in three ways: by spacing capacitor values closely (several values per decade, or many capacitors of the same value, so that no deep inductive-capacitive pairing occurs); by increasing ESR (controlled-ESR capacitors, or simply fewer, larger capacitors whose inductive region is damped by ESR); and by designing for a flat impedance with the "frequency-domain target impedance" method, choosing each stage's ESR close to $Z_{target}$ so that the parallel combination is resistive [6], [30]. Raising ESR lowers a peak but also raises the impedance floor between peaks; there is no universal best sequence of capacitor values. The choice comes from a sweep of the actual complex impedance with biased capacitance, mounted ESL, ESR and the rest of the network.

**Rule of Thumb: Fewer values, more of each**

For board-level MLCC decoupling, two or three values per rail spread a decade apart (for example 10 µF, 1 µF, 0.1 µF), in quantities that make their banks' impedances overlap, usually perform better than a "one of every value" strategy. *Limits:* this assumes mounted ESL dominates above about 10 MHz; simulate the PDN impedance with the actual mounting and plane model before release.

## 16.5 Planes and spreading inductance

A plane pair provides capacitance $C = \varepsilon_0 D_k A/h$ (Eq. (3.1)) and, for current flowing from a capacitor's vias to a device's vias, a *spreading inductance*. For current spreading radially from a via of radius $r_v$ to a ring at distance $\rho$ between planes separated by $h$, the inductance is

$$L_{\text{spread}} \approx \frac{\mu_0 h}{2\pi} \ln \frac{\rho}{r_v}. \tag{16.8}$$

It is proportional to $h$ and only logarithmic in distance: for $h = 3$ mil, $r_v = 5$ mil, and $\rho = 250$ mil, $L_{\text{spread}} \approx 60$ pH. Thin plane dielectrics therefore matter more than placing capacitors within a fraction of an inch of the load—although placement still matters for the via portion of the loop and for avoiding shared vias.

**Design Rule: Plane pairs for low-voltage, high-current rails**

Give each high-current, low-voltage rail a dedicated plane (or wide pour) adjacent to a ground plane with 2–4 mil of dielectric. Place the pair near the surface carrying the device and its decoupling.

## 16.6 From impedance to a load transient

A target impedance is a frequency-domain proxy for a time-domain requirement: the rail voltage must stay inside its allocation while the load current changes. For a linear PDN with driving-point impedance $Z(s)$ at the observation port, the voltage deviation is

$$\Delta V(s) = Z(s)\, \Delta I(s), \tag{16.9}$$

so the shape of the current step decides which part of $Z$ is excited. The smallest network that shows the essential behavior is a feed $R + sL$ (regulator and bulk path) shunted by a capacitance $C$ at the load:

$$Z(s) = \frac{(R + sL) \cdot \frac{1}{sC}}{R + sL + \frac{1}{sC}} = \frac{R + sL}{1 + sRC + s^2 LC}. \tag{16.10}$$

It has a peak near $f_0 = 1/(2\pi\sqrt{LC})$ and a DC value $R$.

**Worked Example 16.4: Damping against DC drop**

A rail is fed through $L = 0.5$ nH and either $R = 2$ mΩ or 8 mΩ, with $C = 10$ µF at the load. The load current rises by 1 A in 20 ns and stays there. Compare the responses.

**Solution.** $f_0 = 1/(2\pi\sqrt{0.5 \times 10^{-9} \times 10^{-5}}) = 2.25$ MHz and the characteristic impedance $\sqrt{L/C} = 7.1$ mΩ. With $R = 2$ mΩ the circuit is underdamped ($Q = \sqrt{L/C}/R \approx 3.5$): the voltage dips to −7.5 mV, rings at 2.25 MHz, and settles at the 2 mV DC drop (Figure 16.7). With $R = 8$ mΩ ($Q \approx 0.9$) the ringing almost disappears, but the response still overshoots to −9.6 mV before settling at 8 mV: the "damped"

design has the *larger* worst-case deviation. Neither number is visible from a single $|Z|$ value; both follow from Eq. (16.9).

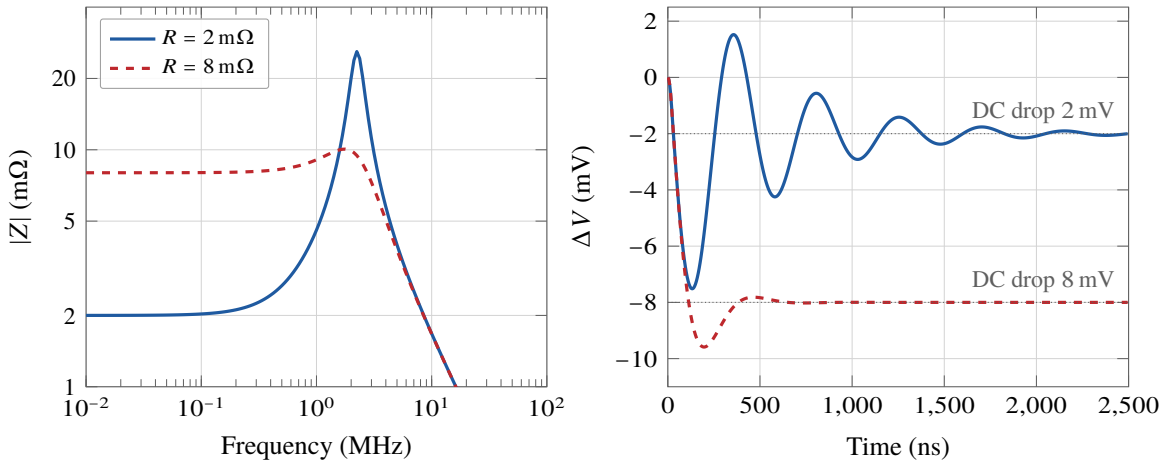



**Figure 16.7.** Impedance (left) and response to a 1 A, 20 ns load step (right) of the network of Eq. (16.10) for two feed resistances. Linear model; no regulator loop or capacitor ESR. [Simulation (analytical network)]

Two consequences matter in practice. First, a periodic load—a clock-gated block toggling at a fixed rate—can excite a narrow resonance coherently and produce a far larger deviation than an isolated step; the worst-case current pattern is part of the specification. Second, damping without DC loss comes from lossy shunt branches (controlled-ESR capacitors, an RC snubber) or from the regulator's output impedance, not from raising the series resistance.

**Design Rule: Close the PDN loop in the time domain**

After meeting $Z_{\text{target}}$ in the frequency domain, simulate the rail at the observation port with the specified current profile—amplitude, edge rate, repetition rate and dwell time—and with capacitance, ESL and ESR at their corners. Compare peak droop, overshoot and settling with the dynamic allocation of Table 16.1.

## 16.7 DC IR drop

At DC, the PDN must also deliver the load current with an acceptable voltage drop. A plane's resistance is conveniently expressed per square:

$$R_{\square} = \frac{\rho}{t}, \tag{16.11}$$

which for 1 oz copper (35 µm) is 0.49 mΩ per square at 20 °C. The resistance of a path is $R_{\square}$ times the number of squares along it. Via fields, anti-pad clusters, and neck-downs where the plane passes between BGAs crowd the current and raise the effective resistance; IR-drop simulation tools solve the plane geometry directly. Current density in vias and narrow necks must also be checked against IPC-2152 thermal limits [31].

**Worked Example 16.5: IR drop in a core plane**

A 0.8 V, 15 A rail flows through a 1 oz plane region three squares long from the regulator to the SoC. Estimate the drop.

**Solution.** $R = 3 \times 0.49\,\text{m}\Omega = 1.48\,\text{m}\Omega$, so $\Delta V = 15 \times 1.48 = 22\,\text{mV}$, 2.8 % of the rail—nearly the entire 3 % tolerance. Remedies: use 2 oz copper or two paralleled planes, move the regulator closer, or use the regulator's remote-sense input at the SoC so that the regulator compensates the static drop.

**Table 16.2.** PDN elements and their effective frequency ranges (typical).

| Element | Effective range | Limited by |
|---|---|---|
| Regulator | DC to ~10–100 kHz | Loop bandwidth; output inductance |
| Bulk (polymer, large MLCC) | 1 kHz to ~1 MHz | ESL (≳1 nH), ESR |
| MLCC bank | 1 MHz to ~50 MHz | Mounted ESL, DC-bias derating |
| Plane pair | ~20 MHz to cavity resonance | Area, dielectric thickness |
| Package capacitors | ~50 MHz to 300 MHz | Package inductance |
| On-die capacitance | > 100 MHz | Die area, leakage |

**Key Takeaways**

- $Z_{\text{target}} = \Delta V_{\text{ripple}} / \Delta I_{\text{transient}}$; a few milliohms is typical for low-voltage, high-current rails.
- Each PDN element—regulator, bulk, MLCC, planes, package, die—controls the impedance in its own band.
- Above its SRF a capacitor is an inductor; mounted ESL, not capacitance, sets high-frequency performance.
- Size capacitor banks from the full R–L–C branch impedance, $N \geq \lceil |Z_C| / Z_{\text{target}} \rceil$; the inductive-only shortcut is wrong below the SRF.
- Anti-resonances arise between parallel branches; place them with both ESLs (Eq. (16.6)), and control them with value selection, ESR, and simulation.
- Allocate the rail tolerance among set point, DC drop and dynamic noise, and confirm the result with a load-transient simulation.
- The die–package anti-resonance is set by the package and die; the board can only damp it.
- Budget DC IR drop with $R_{\square} = 0.49\,\text{m}\Omega$ per square for 1 oz copper, and use remote sense.

## Further Reading

Smith and Bogatin [7] is a complete treatment of frequency-domain PDN design; Smith et al. [29] introduced the target-impedance method; Novak and Miller [6] and Novak [30] cover PDN measurement and design methodologies.

## Review Problems

**P16.1** Find $Z_{\text{target}}$ for a 1.0 V rail with ±5 % tolerance and a 5 A transient.

**P16.2** Find the SRF of a 1 µF capacitor with a mounted ESL of 0.6 nH.

**P16.3** How many capacitors, each with 0.4 nH mounted ESL, are needed to keep the PDN below 5 mΩ at 30 MHz? Assume the capacitors are well above their SRF at that frequency.

**P16.4** An 8 A current flows through five squares of 0.5 oz (17.4 µm) copper. Find the IR drop.

**P16.5** Estimate the anti-resonant frequency between an inductive branch of 0.8 nH and a capacitive branch of 1 nF.

**P16.6** *(Approximation failure.)* A 1 µF capacitor has 6 mΩ ESR and 0.6 nH mounted ESL. How many are needed for 2.4 mΩ at 5 MHz? What does the inductive-only shortcut give, and why is it wrong?

**P16.7** *(Imperfect data.)* The capacitor of the previous problem loses half its capacitance at the rail's DC bias. Repeat the count.

**P16.8** *(Inverse design.)* For the network of Worked Example 16.4, what feed resistance gives a quality factor of 0.7?

Chapter 17

# Planes and Return Paths

**Chapter Overview**

Chapter 3 showed that high-frequency return current flows in the plane directly beneath a signal. This chapter follows that return current through the situations that disturb it on real boards: gaps and splits in planes, transitions between layers referenced to different planes, and the plane cavity itself, which resonates. It explains why these disturbances cause signal-integrity, power-integrity, and EMI problems at the same time, and gives the stitching-via and stitching-capacitor rules that preserve the return path.

**Learning Objectives**

1. Explain how a plane split under a trace increases loop inductance, crosstalk, and emissions.
2. Decide when a stitching capacitor can bridge a split, and size it.
3. Trace the return current through a reference-plane change and provide a path for it.
4. Compute cavity resonances of a plane pair and place stitching vias.
5. Evaluate split-plane, partitioned, and solid-plane strategies for mixed-signal boards, and the limits of the 20H rule.

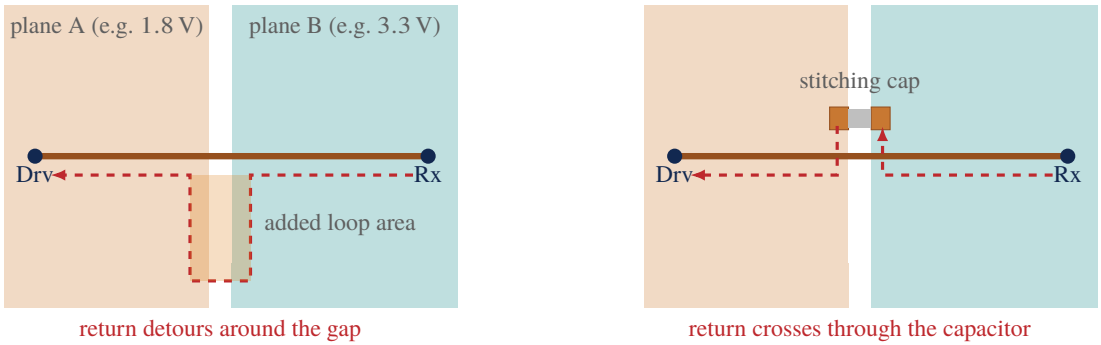
## 17.1 Crossing a plane split

Boards with several supply voltages often divide one plane layer into regions at different potentials. A signal referenced to that layer that crosses the gap between two regions loses its return path at the gap: the return current, unable to cross, detours to the nearest point where the two regions are connected—the end of the gap, a bypass capacitor, or the regulator (Figure 17.1a); Figure 17.1 compares this with a stitched crossing. The consequences are:

- a large added loop inductance in series with the signal, seen as an impedance discontinuity (an inductive bump in TDR) and a slowed edge;
- shared return detours among all signals crossing the same gap, producing strong crosstalk among them;
- a voltage across the gap, which drives the gap as a slot antenna and couples into any cable attached near it [5], [32].

**Design Rule: Never route fast signals across a split**

Route every signal above about 50 MHz, or with edges faster than about 1 ns, entirely over a continuous reference plane. If a crossing cannot be avoided, place a stitching capacitor across the gap within one or two trace widths of the crossing for each signal or pair, or route the signal on a layer referenced to a solid ground plane for that segment.




**(a)** Uncompensated crossing

**(b)** With a stitching capacitor

**Figure 17.1.** A trace crossing a gap between two plane regions. (a) The return current detours to the end of the gap, enclosing a large loop. (b) A stitching capacitor next to the crossing restores a short return path at high frequency.

### 17.1.1 Stitching capacitors

A capacitor across the gap provides a high-frequency return path (Figure 17.1b). It works only where its impedance is small compared with the line's impedance and only if it is placed close to the crossing, because its own mounting inductance and the remaining detour still add loop inductance. An 0402 capacitor with 0.6 nH mounted inductance has $|Z| \approx 1.9\,\Omega$ at 500 MHz—acceptable for a 50 Ω line—but 7.5 Ω at 2 GHz. Stitching capacitors are therefore a repair for moderate-speed signals, not a substitute for a solid reference for multi-gigabit links.

## 17.2 Reference-plane changes

When a signal moves from one layer to another through a via, its return current must move from the reference plane of the first layer to that of the second. How it does so depends on what the two planes are.

**Same plane, both sides**
: If the signal moves between layers on opposite sides of one plane (for example from L1 to L3 around L2), the return current simply flows through the anti-pad clearance from one face of the plane to the other. No extra path is needed.

**Two ground planes**
: The return current flows from one ground plane to the other through the nearest ground via. A ground via placed next to the signal via keeps this loop small (Figure 17.2).

**Ground to power, or two different power planes**
: No DC connection exists near the via. The return current crosses as displacement current through the plane capacitance—spreading across the cavity—or through the nearest decoupling capacitor. This excites the cavity and couples noise between all signals that make similar transitions.

**Common Pitfall: Power planes as references**

Assigning a high-speed stripline layer between a ground plane and a power plane, and assuming that the power plane is "AC ground." It is, but only through the plane capacitance and the decoupling network, both of which are inductive at gigahertz frequencies. The return current in the power-plane half of the stripline must eventually reach ground through those paths; any transition then couples signal noise into

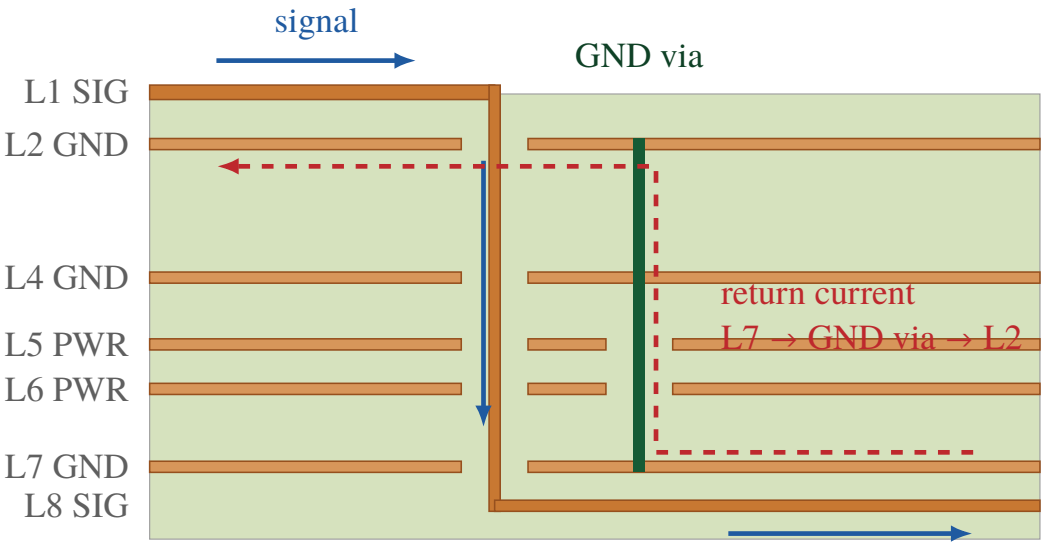



**Figure 17.2.** Return current at a layer transition from L1 (referenced to L2 GND) to L8 (referenced to L7 GND). A ground via adjacent to the signal via carries the return current between the two ground planes.

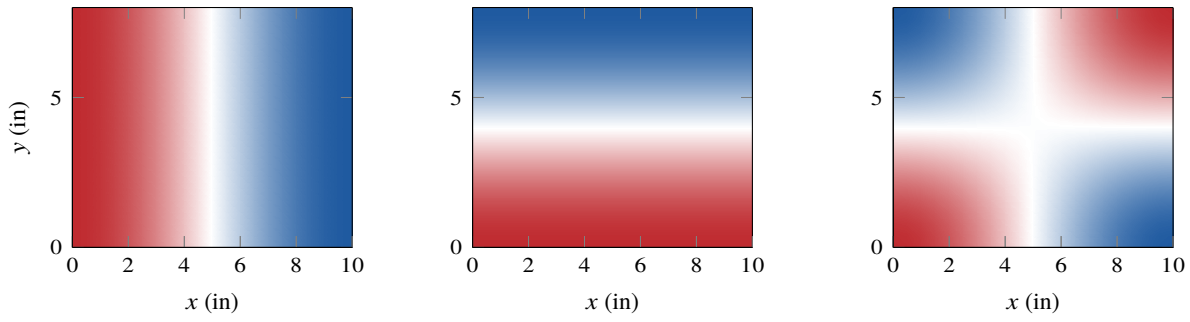
the PDN and PDN noise into the signal. Reference multi-gigabit signals to ground on both sides where possible.

## 17.3 Plane cavity resonance

A pair of planes separated by a dielectric forms a parallel-plate cavity. For a rectangular cavity of dimensions $a \times b$ with open (unloaded) edges, the resonant frequencies are

$$f_{mn} = \frac{c}{2\sqrt{\varepsilon_r}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}, \qquad m, n = 0, 1, 2, \ldots, \ (m, n) \neq (0, 0). \tag{17.1}$$

At these frequencies the voltage between the planes forms standing-wave patterns (Figure 17.3), and the PDN impedance seen at a point depends strongly on where that point lies relative to the nodes and antinodes. Noise injected at one via—by a switching device or by a signal changing reference—couples efficiently to any other via near an antinode, and the cavity edges radiate.




**(a)** $TM_{10}$, 288 MHz **(b)** $TM_{01}$, 360 MHz **(c)** $TM_{11}$, 461 MHz

**Figure 17.3.** Voltage distribution of the first three modes of a 10 in × 8 in plane pair with $\varepsilon_r = 4.2$, from Eq. (17.1). Red and blue denote opposite polarity.

**Worked Example 17.1: Cavity resonances of a board**

Find the first three resonances of a 10 in × 8 in plane pair with $\varepsilon_r = 4.2$.

**Solution.** $c/(2\sqrt{4.2}) = 11.80\,\text{in/ns}/4.099 = 2.88\,\text{GHz}\cdot\text{in}$. Then $f_{10} = 2.88/10 = 288\,\text{MHz}$, $f_{01} = 2.88/8 = 360\,\text{MHz}$, and $f_{11} = 2.88\sqrt{0.01 + 0.0156} = 461\,\text{MHz}$. Every board-size plane pair resonates in the hundreds of megahertz, squarely in the range of clock harmonics.

Cavity resonances are damped by dielectric loss, by decoupling capacitors (which short the cavity locally and shift the resonances up), and by thin dielectrics (which raise the cavity's loss relative to its stored energy). Discrete damping—for example, resistive–capacitive terminations along the board edge—can be used where the PDN impedance at a resonance is critical.

### 17.3.1 Stitching vias

Ground planes on different layers are connected by stitching vias. A dense pattern of stitching vias makes the ground planes behave as one, suppresses the cavity modes between them, and gives every signal transition a nearby return via. Along board edges, a row of ground vias (a "via fence") suppresses edge radiation from the cavities between ground planes.

**Rule of Thumb: Stitching-via spacing**

Space ground stitching vias at no more than about $\lambda/20$ at the highest frequency of concern; at 5 GHz in $\varepsilon_r = 4$ material that is about 60 mil. *Limits:* the factor of 20 is a heuristic; denser spacing is needed near connectors, around sensitive circuits, and along edges where emissions are a concern.

## 17.4 Mixed-signal partitioning

A traditional practice on mixed-signal boards was to split the ground plane into "analog" and "digital" regions joined at a single point near the ADC. With modern fast logic this usually does more harm than good: every signal that crosses between the regions has the problem of Section 17.1, and the single connection point becomes a high-inductance common path. The approach now favored is a *single, solid ground plane* with *partitioned placement*: digital and analog circuits placed in separate regions over the same plane, and routing arranged so that digital return currents, which follow their traces, never pass under the analog region [5]. Splits remain appropriate for safety isolation (for example, under Ethernet magnetics or isolated interfaces), where the crossing signals pass through isolation components rather than copper.

### 17.4.1 The 20H rule

The "20H rule" recommends recessing the edges of power planes by about twenty times the power–ground spacing $H$ from the ground plane's edge, to reduce fringing-field radiation from the plane edges. Measurements and simulations have shown that its benefit is limited and frequency dependent: it can reduce emission at some frequencies but may increase it near cavity resonances, and it reduces plane capacitance [32], [33]. Edge stitching with via fences on ground–ground cavities, thin power–ground dielectrics, and adequate decoupling are more dependable.

**Table 17.1.** Return-path situations and remedies.

| Situation | Problem | Remedy |
|---|---|---|
| Trace over plane split | Detour; crosstalk; slot radiation | Re-route over solid plane; stitching cap at crossing |
| Trace near plane edge | Asymmetric return; edge radiation | Keep $\geq 3H$ to $5H$ from edge |
| Via between two GND planes | Return through cavity | GND via adjacent to signal via |
| Via between GND and PWR | Return through cavity / decoupling | Avoid; else stitching cap at via |
| Anti-pad clusters (BGA, connectors) | Plane perforated | Reduce anti-pads; stagger; back-drill |
| Plane cavity resonance | PDN peaks; coupling; radiation | Thin dielectric, decoupling, stitching, damping |

**Worked Example 17.2: How good is a stitching capacitor?**

A signal crosses a split on a 50 Ω line, with a stitching capacitor of 0.6 nH mounted inductance placed at the crossing. Estimate the reflection it causes at 500 MHz and at 5 GHz, treating it as a series inductance in the return path.

**Solution.** The series impedance is $\omega L = 1.9\,\Omega$ at 500 MHz and $18.8\,\Omega$ at 5 GHz. A small series impedance $Z$ on a line of impedance $Z_0$ reflects roughly $Z/(2Z_0)$: about 2 % at 500 MHz but 19 % at 5 GHz. The capacitor is adequate for a 100 MHz clock with nanosecond edges but not for a multi-gigabit link.

**Key Takeaways**

- A plane split under a trace forces the return current to detour, adding inductance, crosstalk, and slot radiation.
- Stitching capacitors bridge splits only at moderate frequencies; prefer solid references for fast signals.
- At a layer change, provide a ground via next to the signal via; avoid transitions between ground- and power-referenced layers.
- Plane pairs resonate at $f_{mn} = \frac{c}{2\sqrt{\varepsilon_r}}\sqrt{(m/a)^2 + (n/b)^2}$, typically hundreds of megahertz.
- Use a solid ground plane with partitioned placement rather than split grounds; treat the 20H rule with caution.

## Further Reading

Archambeault [32], Ott [5], and Montrose [33] discuss plane splits, reference changes, and cavity resonances from the EMC point of view; Swaminathan and Engin [27] model plane cavities.

## Review Problems

**P17.1** Find $f_{10}, f_{01}$, and $f_{11}$ for a 6 in × 4 in plane pair with $\varepsilon_r = 4.0$.

**P17.2** What stitching-via spacing corresponds to $\lambda/20$ at 10 GHz in $\varepsilon_r = 3.6$ material?

**P17.3** A stitching capacitor has 0.4 nH of mounted inductance. What is its impedance at 2 GHz?

**P17.4** A slot crossing raises a 50 Ω trace's local impedance to 80 Ω. What reflection coefficient results?

**P17.5** By how much would the 20H rule recess a power plane separated from ground by 4 mil?

**P17.6** What is the impedance of 2 nF of plane capacitance at 1 GHz, ignoring inductance?

Part IV

# EMI and EMC

Chapter 18

# Emission Mechanisms and EMC Design

**Chapter Overview**

A product that works perfectly on the bench can still be unsellable if it radiates too much. Radiated and conducted emissions are regulated in every major market, and failing a compliance test late in a program is expensive. This chapter explains the two fundamental emission mechanisms—differential-mode radiation from current loops and common-mode radiation from cables and structures—quantifies them with the standard small-antenna formulas, examines slot and edge radiation from boards, and surveys the design techniques (spread-spectrum clocking, filtering, ferrites, and shielding) used to meet FCC and CISPR limits. It draws heavily on the classic treatments of Ott and Paul [5], [34].

**Learning Objectives**

1. Distinguish differential-mode and common-mode emissions and estimate each.
2. Explain why microamperes of common-mode current on a cable can fail a test.
3. Relate emissions to the spectrum of digital waveforms.
4. Identify slot and edge radiation from boards and prevent them.
5. Apply spread-spectrum clocking, filtering, ferrites, and shielding, and estimate their benefit.
6. Outline the structure of FCC Part 15 and CISPR 32 limits.

## 18.1 Differential-mode radiation

Every signal and its return form a loop (Chapter 3 and Figure 18.1). A loop that is small compared with a wavelength radiates as a magnetic dipole. Its far-field electric field at distance $r$, in free space, is [5], [34]

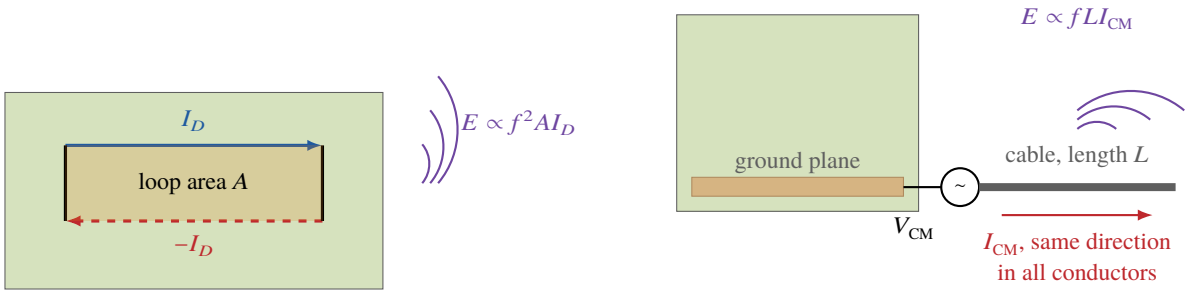
$$E_{\mathrm{DM}} = 131.6 \times 10^{-16} \frac{f^2 A I_D}{r} \quad \mathrm{V/m}, \tag{18.1}$$

with $f$ in hertz, loop area $A$ in square meters, current $I_D$ in amperes, and $r$ in meters. Over a reflecting ground plane, as in a test site, the worst case is about twice this value. Differential-mode emission grows as $f^2$ and in proportion to loop area, so it is controlled by keeping return paths close to their signals—the same rule that minimizes loop inductance.

**Worked Example 18.1: Loop area and emissions**

A 4 in clock trace carries a 10 mA harmonic at 300 MHz. Estimate the field at 3 m if its return is in a plane 5 mil below, and if the return is a wire 1 in away.

**Solution.** With the plane, $A = 4 \times 0.005 = 0.02\,\mathrm{in}^2 = 1.29 \times 10^{-5}\,\mathrm{m}^2$, and Eq. (18.1) gives $E = 131.6 \times 10^{-16} \times (3 \times 10^8)^2 \times 1.29 \times 10^{-5} \times 0.01/3 = 51\,\mu\mathrm{V/m}$, or 34 dBµV/m, below the FCC Class B




**(a)** Differential mode: small current loop

**(b)** Common mode: cable driven as an antenna

**Figure 18.1.** Emission mechanisms. (a) Differential-mode current in a small loop radiates as a magnetic dipole. (b) Common-mode current, driven by a noise voltage between the board's ground and an attached cable, flows in the same direction on all of the cable's conductors and radiates as a monopole.

limit of 46 dBµV/m at that frequency. With the return 1 in away, $A$ is 200 times larger and the field rises by 46 dB, to 80 dBµV/m—a failure by more than 30 dB.

Figure 18.2 extends this example across frequency, using the harmonic envelope of a trapezoidal clock (Eq. (2.8)). Multiplying the current envelope by $f^2$ turns its flat, −20, and −40 dB/decade segments into +40, +20, and 0 dB/decade: differential-mode emissions of a clock *rise* with frequency up to $f_2 = 1/(\pi T_{\text{ramp}})$ and stay high above it.

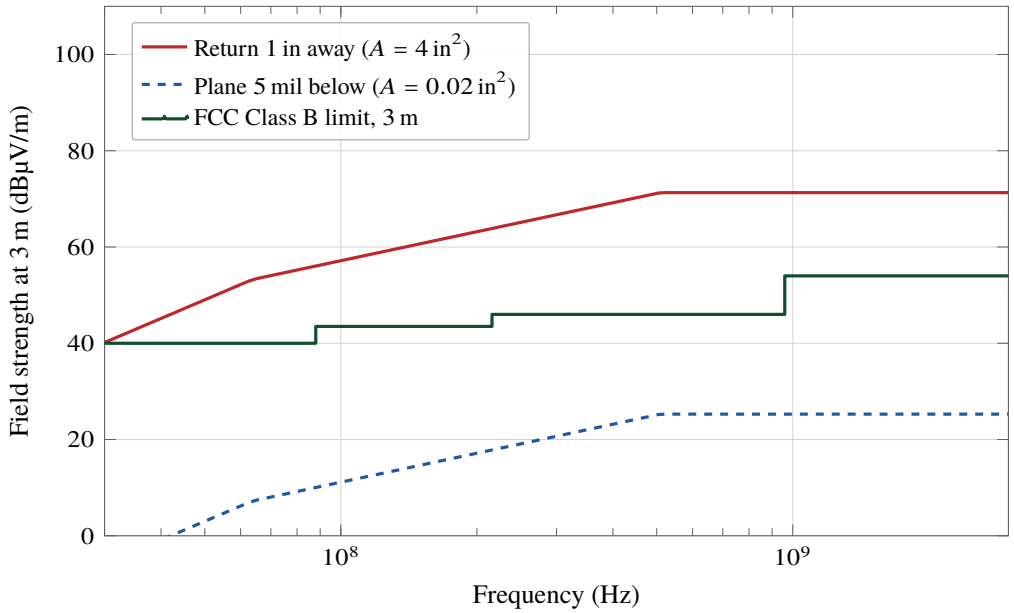



**Figure 18.2.** Envelope of differential-mode emission at 3 m from a 4 in trace carrying the harmonics of a 100 MHz clock ($t_r$ = 0.5 ns, 10 mA harmonic envelope), for two return-path geometries, from Eq. (18.1), compared with the FCC Class B limit. Only the envelope is shown; actual emissions occur at discrete harmonics. Free-space field, RMS; a test site with a ground plane can read up to about 6 dB higher. [Analytical]

## 18.2 Common-mode radiation

Common-mode current flows in the same direction on all conductors of a cable (or on a board's ground structure), returning through stray capacitance to the environment. It is driven by a noise voltage between the board's reference and the cable—ground-plane noise, ground bounce, or voltage across a plane slot. A cable of length $L$ short compared with a quarter wavelength radiates as a monopole:

$$E_{\mathrm{CM}} = 1.257 \times 10^{-6} \frac{f L I_{\mathrm{CM}}}{r} \quad \mathrm{V/m}. \tag{18.2}$$

Equations (18.1) and (18.2) are free-space, far-field estimates for a sinusoidal current: $I$ is the RMS current of one harmonic and $E$ the RMS field. An EMI receiver is calibrated to read the RMS value of a steady sine, so for a stable narrowband harmonic its peak, quasi-peak and average detectors give nearly the same reading; for modulated or intermittent signals they differ, and the detector and bandwidth defined by the regulation apply [35]. A test site with a reflecting ground plane can raise the measured field by up to about 6 dB over the free-space value.

The short-monopole formula assumes the cable is electrically short, roughly $L \lesssim \lambda/10$, with a uniform current along it. Longer cables are not simply "saturated": the current becomes a standing wave whose distribution depends on how the cable is terminated at both ends, the radiation pattern develops lobes, and at resonance the radiated field can be considerably higher than an extrapolation of Eq. (18.2). For cables approaching or exceeding $\lambda/4$, use a numerical model (method of moments) or a measurement of the cable current distribution instead. Even in its valid range, common-mode emission usually dominates: it grows only as $f$, but a cable is a far more efficient antenna than a small board loop.

**Worked Example 18.2: How little common-mode current fails a test**

A 0.25 m cable stub carries 20 µA (RMS) of common-mode current at 100 MHz. Estimate the free-space field at 3 m, and the largest current that meets a 150 µV/m limit. Check that the formula applies.

**Solution.** *Validity:* $\lambda = 3$ m, so $L = \lambda/12$; the short-monopole model is reasonable. Equation (18.2): $E = 1.257\times10^{-6}\times10^{8}\times0.25\times20\times10^{-6}/3 = 210$ µV/m, or 46.4 dBµV/m, about 3 dB over the limit. The maximum allowed current is $150\times10^{-6}\times3/(1.257\times10^{-6}\times10^{8}\times0.25) = 14$ µA. Tens of microamperes of common-mode current are enough to fail, compared with milliamperes of differential-mode current. *What the example does not cover.* A real 1 m USB or Ethernet cable at 100 MHz is longer than $\lambda/4 = 0.75$ m. Multiplying the result by four would be wrong in either direction: the current is no longer uniform, and at a cable resonance the field can exceed that estimate. Such a cable must be modeled or measured.

**Common Pitfall: Treating cables as part of someone else's design**

Testing emissions with the board alone, or with short shielded cables, and discovering at compliance that the system's real cables radiate. Common-mode emission depends on how every cable connects to the board's ground at its connector. Design the connector region—ground plane, filter components, shield termination—as part of the board, and test with representative cables early.

## 18.3 Slot and edge radiation

A slot in a plane, driven by return current that must detour around it (Chapter 17), radiates as a slot antenna. It is most efficient when its length is a half wavelength:

$$L_{\mathrm{slot}} = \frac{\lambda}{2} = \frac{c}{2f} \quad \text{(in air)}; \tag{18.3}$$

a 3 in slot resonates near 2 GHz. Slots inside a board are partly loaded by the dielectric and resonate lower. Plane-pair cavities (Section 17.3) radiate from their edges, most strongly at cavity resonances; ground-via fences along board edges and thin power–ground dielectrics reduce this.

## 18.4 Techniques for reducing emissions

### 18.4.1 Edge rate

The cheapest emission control is to use slower edges. From Eq. (2.8), harmonics above $f_2 = 1/(\pi T_{\text{ramp}})$ (with $T_{\text{ramp}} = t_r/0.8$ for a linear edge) fall by 20 dB/decade faster than below it; doubling $t_r$ halves $f_2$ and reduces every harmonic above the old $f_2$ by 6 dB. Drive-strength and slew-rate settings, and series resistors on clocks, should be set to the slowest value that meets timing.

### 18.4.2 Spread-spectrum clocking

Spread-spectrum clocking (SSC) frequency-modulates a clock by a small amount, typically a triangular modulation of 0.25–0.5 % (usually down-spread so that the maximum frequency is not exceeded) at 30–33 kHz. Each harmonic's energy is spread over a band of width $n\delta f_0$ for the $n$-th harmonic. A receiver measuring in a resolution bandwidth RBW smaller than that band sees a lower peak (Figure 18.3), by roughly

$$\Delta_{\text{SSC}} \approx 10 \log_{10} \frac{n\,\delta f_0}{\text{RBW}} \quad \text{dB}, \tag{18.4}$$

an estimate that is optimistic for the lowest harmonics, where the spread band is only a few RBWs wide. SSC does not reduce the total energy, and the receiving system must tolerate it: PCIe and SATA reference clocks support SSC, but some interfaces (and some PLLs) do not.

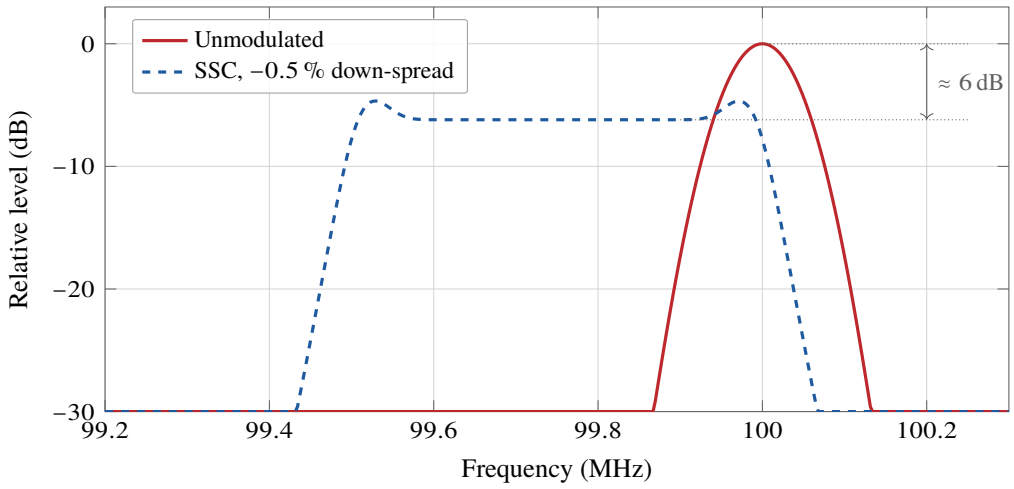



**Figure 18.3.** Effect of −0.5 % down-spread SSC on the fundamental of a 100 MHz clock, as seen by a receiver with a 120 kHz resolution bandwidth (schematic). The peak falls by about 6 dB, the ratio of the 500 kHz spread to the resolution bandwidth expressed in decibels.

**Worked Example 18.3: SSC benefit by harmonic**

A 100 MHz clock uses −0.5 % SSC. Estimate the reduction in a 120 kHz RBW at the fundamental and at the fifth harmonic.

**Solution.** Fundamental: spread 500 kHz, so $10\log_{10}(500/120) = 6.2$ dB. Fifth harmonic: 2.5 MHz, $10\log_{10}(2500/120) = 13.2$ dB. SSC is most effective at the high-order harmonics that are hardest to control by other means.

### 18.4.3 Filtering and ferrites

Signals and power leaving the board through a connector should be filtered so that only the intended bandwidth passes. A series ferrite bead acts as a resistor at high frequency (its impedance becomes predominantly resistive above a few tens of megahertz), absorbing energy rather than reflecting it. A ferrite is effective only against a low source and load impedance; it must be paired with a capacitor to ground on the board side to form a low-pass filter. Common-mode chokes on differential pairs (USB, Ethernet) present high impedance to common-mode current while passing the differential signal. Ferrite beads saturate with DC current and their impedance at rated current may be much lower than the zero-bias curve in the data sheet.

**Design Rule: Connector-region design**

Treat every connector as an antenna port. Provide a solid ground plane under the connector region, bond cable shields to chassis or ground with a low-inductance 360-degree connection at the connector, place filter capacitors and ferrites between the connector and the rest of the board, and keep high-speed traces and clocks away from the connector area unless they leave through it.

### 18.4.4 Shielding

A conductive enclosure attenuates fields by reflection at its surfaces and by absorption in its thickness. The absorption loss of a sheet of thickness $t$ is

$$A_{\text{abs}} = 8.686\,\frac{t}{\delta_{\text{s}}} \quad \text{dB}, \tag{18.5}$$

which for even thin metal at radio frequencies is very large (0.1 mm of copper at 100 MHz gives over 130 dB). In practice, shielding effectiveness is limited not by the metal but by apertures, seams, and penetrating cables: an aperture behaves like a slot antenna, and its leakage depends on its longest dimension, not its area. Board-level shield cans over noisy circuits must be soldered to a ground ring with many vias.

**Table 18.1.** Emission mechanisms and design controls.

| Mechanism | Scaling | Primary controls |
|---|---|---|
| Differential-mode loop | $f^2AI$ | Tight return paths; stripline routing; slower edges |
| Common-mode cable | $fLI_{\text{CM}}$ | Low ground noise; connector filtering; CM chokes; shield bonding |
| Slot in plane | resonant at $\lambda/2$ | No slots under signals; stitching |
| Plane-cavity edge | resonant at $f_{mn}$ | Via fences; decoupling; thin dielectric |
| Periodic clocks | discrete harmonics | SSC; slower edges; short routes |

## 18.5 Regulatory overview

In the United States, unintentional radiators such as digital devices are regulated by FCC Part 15 [35]; Class A covers commercial and industrial equipment and Class B, with tighter limits, residential equipment. Internationally, multimedia equipment emissions are covered by CISPR 32 [36], adopted as EN 55032 in Europe.

Both specify radiated emission limits (typically from 30 MHz to several gigahertz, at 3 m or 10 m) and conducted emission limits on AC mains (150 kHz to 30 MHz). Below 1 GHz, radiated measurements use a quasi-peak detector with a 120 kHz bandwidth; above, peak and average detectors with a 1 MHz bandwidth.

**Specification Note**

Commonly quoted FCC Part 15 Class B radiated limits at 3 m are 100 µV/m (40.0 dBµV/m) from 30–88 MHz, 150 µV/m (43.5 dBµV/m) from 88–216 MHz, 200 µV/m (46.0 dBµV/m) from 216–960 MHz, and 500 µV/m (54.0 dBµV/m) above 960 MHz. CISPR 32 Class B limits at 10 m are commonly quoted as 30 dBµV/m (30–230 MHz) and 37 dBµV/m (230–1000 MHz), quasi-peak. Detector type, measurement distance, upper frequency limit, and the limits themselves must be taken from the current edition of the applicable regulation for the target market.

**Rule of Thumb: Common-mode current budget**

Near 100 MHz, a short cable segment carrying only microamperes to tens of microamperes of common-mode current approaches the FCC Class B limit at 3 m (Worked Example 18.2). Measure cable common-mode current with a clamp-on current probe early in development, and track it relative to a baseline rather than trusting an absolute field prediction. *Limits:* based on the short-monopole formula, valid only for $L \lesssim \lambda/10$; resonant cables can radiate considerably more and must be modeled or measured.

**Key Takeaways**

- Differential-mode emission is $131.6 \times 10^{-16} f^2 AI/r$; keep loop areas small with adjacent return planes.
- Common-mode emission is $1.257 \times 10^{-6} f L I_{\mathrm{CM}}/r$; microamperes on a cable can fail a test.
- Clock harmonics multiplied by $f^2$ make DM emission rise with frequency up to $1/(\pi T_{\mathrm{ramp}})$.
- Slots and plane edges radiate; avoid slots under signals and stitch edges.
- Slow edges, SSC, connector filtering, ferrites, CM chokes, and well-bonded shields are the main controls.

## Further Reading

Ott [5] and Paul [34] are the standard references for emission mechanisms and EMC design; the regulations themselves are FCC Part 15 [35] and CISPR 32 [36].

## Review Problems

**P18.1** Estimate the free-space DM field at 3 m from a 1 $\mathrm{cm}^2$ loop carrying 20 mA at 200 MHz.

**P18.2** Using the short-monopole approximation, find the common-mode current on a 0.15 m cable segment that produces 150 µV/m (free space, RMS) at 3 m and 200 MHz. Would the same calculation be valid for a 1.5 m cable?

**P18.3** What slot length is a half wavelength at 1 GHz in air?

**P18.4** Estimate the SSC reduction at the third harmonic of a 100 MHz clock with 0.5 % spread and 120 kHz RBW.

**P18.5** By how much are harmonics above both corner frequencies reduced when the rise time is doubled from 0.5 ns to 1 ns?

**P18.6** Find the absorption loss of 35 μm copper at 1 GHz.

Part V

# Interfaces in Practice

Chapter 19

# DDR4/DDR5 and LPDDR4/LPDDR5 Memory Interfaces

**Chapter Overview**

Memory interfaces are the densest, most timing-critical buses on most digital boards. They combine source-synchronous data (DQ with strobe DQS), a common-clock command/address bus, many simultaneously switching single-ended lines, and a reference voltage, all running at several gigatransfers per second. This chapter summarizes the electrical characteristics of DDR4, DDR5, LPDDR4, and LPDDR5, explains the fly-by topology and the training mechanisms (write leveling, read and write centering, VREF training) that make it work, derives routing and matching budgets, and lists the failure modes most often encountered at bring-up.

**Learning Objectives**

1. Compare the key electrical parameters of DDR4, DDR5, LPDDR4, and LPDDR5.
2. Explain fly-by and T topologies and why fly-by requires write leveling.
3. Describe pseudo-open-drain signaling, ODT, and VREF training.
4. Derive DQ-to-DQS and CA-to-CK matching budgets from the data rate.
5. Estimate loaded impedance and delay of a fly-by bus.
6. Recognize common DDR failure modes and their causes.

## 19.1 Interface overview

A DDR interface has three groups of signals:

**Data (DQ, DM/DBI) and strobes (DQS)**
Bidirectional, organized in *byte lanes* of eight DQ (or four in some configurations) with one differential DQS pair. Data are transferred on both edges of DQS (source-synchronous, Chapter 12).

**Command/address (CA) and control**
Unidirectional from controller to all DRAMs of a rank, captured by the clock. In DDR4 they transfer once per clock; DDR5 and LPDDR use fewer pins at double data rate on some signals.

**Clock (CK)**
Differential, from controller to all DRAMs, routed with the CA bus.

This three-group picture is exact for DDR4 and DDR5 and for LPDDR4/4X, where a bidirectional DQS strobe times data in both directions. It is *not* adequate for LPDDR5 and LPDDR5X, which separate the data clock from the command clock (Section 19.1.2). Table 19.1 summarizes the four standards.

**Table 19.1.** Representative characteristics of DDR4, DDR5, LPDDR4/4X, and LPDDR5/5X. Values reflect the base JEDEC releases and common practice; see the specification note below.

| | DDR4 [28] | DDR5 [37] | LPDDR4/4X [38] | LPDDR5/5X [39] |
|---|---|---|---|---|
| Data rate (MT/s) | 1600–3200 | 3200–6400+ | up to 4266 | up to 6400 (5X higher) |
| $V_{DDQ}$ | 1.2 V | 1.1 V | 1.1 V / 0.6 V | 0.5 V (typ.) |
| DQ signaling | POD12 | POD11 | LVSTL | LVSTL |
| Channel width | 64 (+8 ECC) | 2 × 32 (+8 ECC) | 16 per channel | 16 per channel |
| DQ VREF | Internal, trained | Internal, trained | Internal, trained | Internal, trained |
| CA VREF | External | Internal | Internal | Internal |
| Rx equalization | — | DFE in DRAM | — | Rx equalization (rate dependent) |
| Topology | Fly-by (CA/CK) | Fly-by (CA/CK) | Point-to-point | Point-to-point |

**Specification Note**

Data rates, supply voltages, and feature sets of JEDEC memory standards are extended in later revisions and vary by speed bin; LPDDR5X in particular supports higher rates than LPDDR5. The values above are for orientation. Design values must come from the current JEDEC document and, above all, from the memory controller vendor's layout guide, which defines the topology, impedance, and matching rules the controller's training was characterized for.

### 19.1.1 Timing groups by family

Board constraints are derived per *timing group*: a set of signals captured by a common clock or strobe. Table 19.2 lists them for each family. The rules that a layout guide imposes—matching, layer, via count—apply within a group, and the training ranges of Section 19.3.2 apply between groups.

**Table 19.2.** Timing groups of the four DRAM families (conceptual; consult the controller documentation for the implemented groups).

| Family | Command/address | Write data | Read data |
|---|---|---|---|
| DDR4 | CA, control with CK (fly-by) | DQ, DM/DBI with DQS (from controller) | DQ with DQS (from DRAM) |
| DDR5 | CA, CS with CK per subchannel (fly-by on DIMMs) | DQ with DQS | DQ with DQS; DRAM Rx has DFE |
| LPDDR4/4X | CA, CS with CK (point-to-point) | DQ, DMI with DQS | DQ with DQS |
| LPDDR5/5X | CA, CS with CK | DQ, DMI with WCK (data clock) | DQ with RDQS where used |

### 19.1.2 LPDDR5 and LPDDR5X: CK, WCK and read strobes

LPDDR5 introduced a separate differential *data clock*, WCK, per byte, sent from the controller to the DRAM and running at a multiple of CK; the slower CK continues to time command and address [39], [40]. Write data on DQ and DMI are captured with WCK. Read data return from the DRAM on DQ, accompanied by

a differential read strobe, RDQS, in configurations that use it (Figure 19.1). Because WCK and CK are separate clocks, the DRAM must align its internal WCK dividers with CK—*WCK-to-CK synchronization*—and training includes a WCK-to-CK leveling step analogous to DDR4 write leveling.

Two consequences matter on the board. First, a DDR4-style rule "match DQ to DQS" cannot be copied: writes are referenced to WCK and reads to RDQS, so each needs its own matching budget, and the CK/CA group has a third. Second, the WCK frequency is higher than the DQS frequency of a comparable DDR4 design, so its intra-pair skew, crosstalk exposure and termination state deserve the same care as the data. The supported WCK:CK ratios, read modes, termination states and delay allocations are implementation specific; a design starts from a controller-specific table listing, for every signal, its direction, termination, timing reference, training adjustment, board-delay allocation and package-delay treatment.

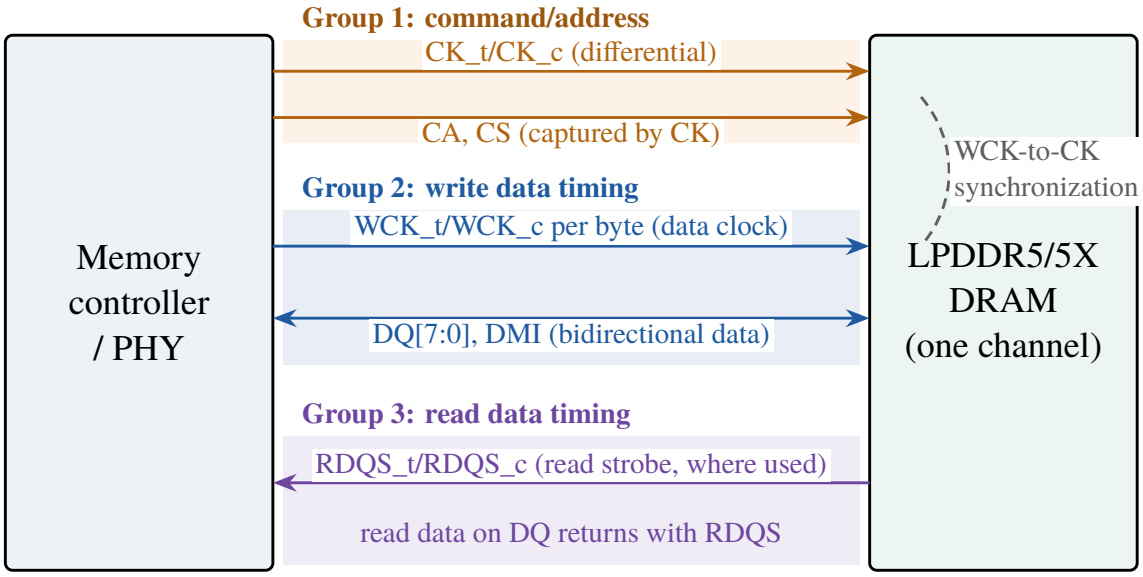



**Figure 19.1.** Logical LPDDR5/5X timing groups of one channel: CK with command/address, WCK with write data, and RDQS with read data where used. Conceptual architecture, not a pinout, termination schematic, clock-ratio specification or training sequence. [Conceptual]

**Common Pitfall: Treating LPDDR5 as fast LPDDR4**

Reusing an LPDDR4 or DDR4 constraint set—one DQ-to-DQS rule per byte—for an LPDDR5 design. The data clock (WCK), the read strobe (RDQS) and the command clock (CK) are separate timing references with separate training steps; derive a constraint for each from the selected controller's documentation.

## 19.2 Signaling and termination

DDR4 and DDR5 data use *pseudo-open-drain* (POD) signaling: the receiver's on-die termination pulls up to $V_{DDQ}$, and the driver pulls the line low through a calibrated pull-down (Figure 19.2). A high level draws no current, which is what makes DBI effective (Chapter 15). Because the signal swings between $V_{DDQ}$ and a level set by the driver/ODT divider, the receiver's threshold is not $V_{DDQ}/2$; it is an internal $V_{REF}$ that training sets to the center of the received eye. LPDDR4 and LPDDR5 use low-voltage swing-terminated logic (LVSTL), terminated to ground, which allows very low $V_{DDQ}$.

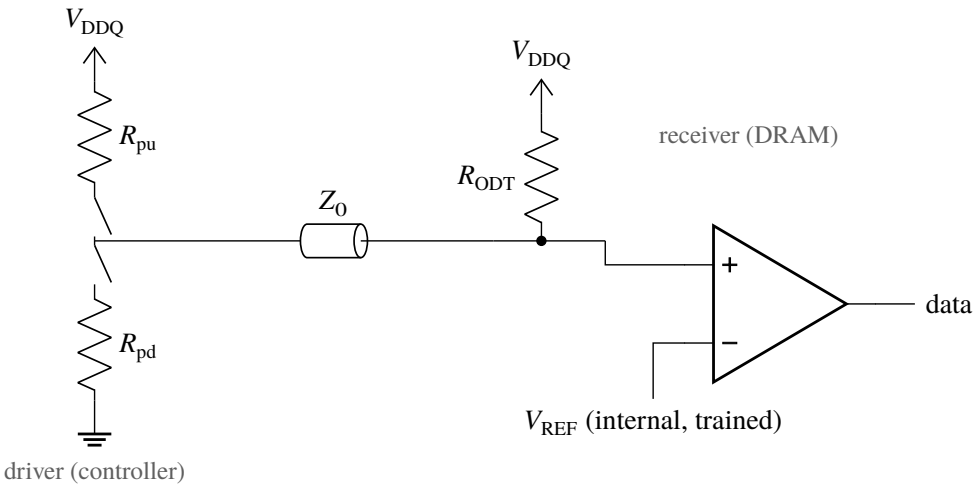



**Figure 19.2.** Pseudo-open-drain DQ signaling: the receiver's on-die termination pulls up to $V_{DDQ}$; the receiver compares the line against an internally generated, trained $V_{REF}$.

Driver output impedance and ODT values are programmable through mode registers and calibrated against an external precision ZQ resistor (240 Ω in DDR4). The controller selects ODT for each rank and each operation (read, write, idle) to optimize the eye in multi-rank systems.

The DDR4 CA bus is terminated at the far end of the fly-by chain to $V_{TT} = V_{DDQ}/2$ through resistors of about the bus's loaded impedance. DDR5 moved CA termination onto the DRAMs.

**Worked Example 19.1: $V_{TT}$ regulator current**

A DDR4 CA bus of 30 lines is terminated with 39 Ω to $V_{TT} = 0.6$ V; the controller drives with 34 Ω. Estimate the current per line and the $V_{TT}$ supply requirement.

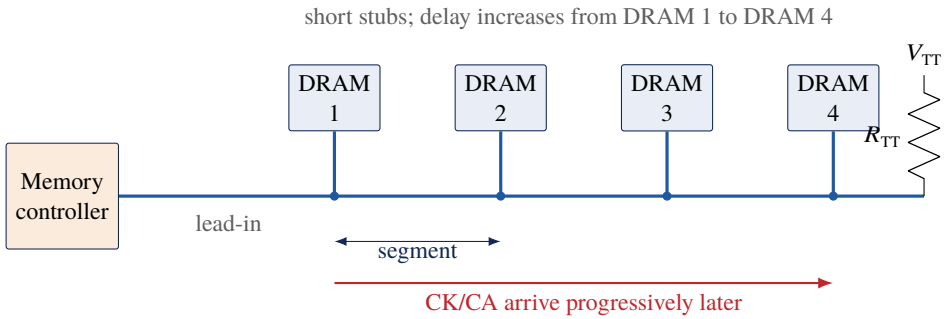
**Solution.** With the driver high (1.2 V), current flows from driver to $V_{TT}$: $I = 0.6/(34 + 39) = 8.2$ mA; driven low, the same current flows from $V_{TT}$. The resistor dissipates $I^2R = 2.6$ mW. With all 30 lines high the $V_{TT}$ supply sinks 247 mA; with all low it sources the same. The $V_{TT}$ regulator must source and sink and must have low impedance at the bus's switching frequencies, so it is decoupled at the termination resistors.

## 19.3 Topologies

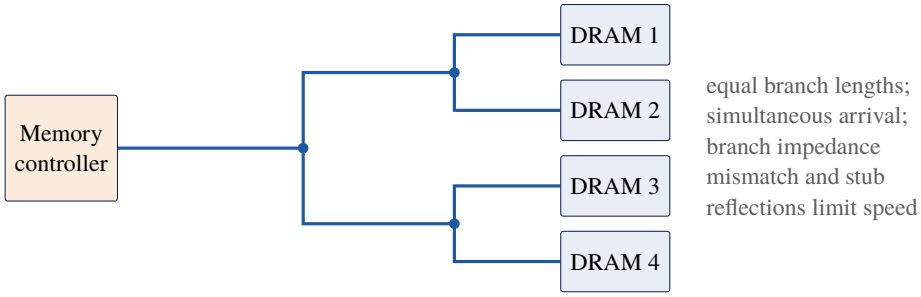
On a DIMM or a board with several DRAM devices per rank, the CA bus and clock must reach every DRAM. Two topologies are possible (Figure 19.3):

**T (tree)**
: The bus splits into balanced branches so that the signal arrives at all DRAMs simultaneously. Each split creates an impedance mismatch and each branch end reflects back into the others; DDR2 used T topologies, but they do not scale to DDR3 rates and beyond.

**Fly-by**
: The bus passes each DRAM in turn, with short stubs to each device, and is terminated at the end. Reflections are small, because the only discontinuities are short stubs and the DRAMs' input capacitance. The signal arrives at each DRAM progressively later.




**(a)** Fly-by (daisy-chain) topology with $V_{TT}$ termination




**(b)** Balanced T (tree) topology

**Figure 19.3.** Command/address and clock topologies. (a) Fly-by: a daisy chain with short stubs, terminated to $V_{TT}$, giving clean signals but staggered arrival times. (b) Balanced T: simultaneous arrival but larger reflections.

### 19.3.1 Loaded impedance and delay

Each DRAM adds an input capacitance $C_d$ (pin, package, and die) at each segment of length $\ell_s$ along the fly-by bus. Distributing it over the segment gives a loaded line with

$$Z_{\text{loaded}} = \frac{Z_0}{\sqrt{1 + C_d/(C'\ell_s)}}, \qquad t_{\text{pd,loaded}} = t_{\text{pd}}\sqrt{1 + \frac{C_d}{C'\ell_s}}, \tag{19.1}$$

valid when the segment delay is short compared with the rise time. The termination resistor is chosen near $Z_{\text{loaded}}$, and the unloaded segments are sometimes routed at a higher impedance so that the loaded impedance matches the lead-in.

**Worked Example 19.2: Loaded fly-by bus**

A 40 Ω stripline ($t_{\text{pd}} = 165$ ps/in) passes four DRAMs spaced 0.65 in apart, each with 1.5 pF of input capacitance. Find the loaded impedance, the loaded delay, and the extra arrival delay at the last DRAM relative to the first.

**Solution.** $C' = t_{\text{pd}}/Z_0 = 4.13$ pF/in, so a segment has 2.68 pF. The loading factor is $\sqrt{1 + 1.5/2.68} = 1.249$, giving $Z_{\text{loaded}} = 32\ \Omega$ and $t_{\text{pd,loaded}} = 206$ ps/in. Three segments add $3 \times 0.65 \times 206 = 402$ ps between the first and the last DRAM—64 % of a 625 ps clock period at DDR4-3200.

### 19.3.2 Write leveling

Because CK reaches each DRAM at a different time, while each byte lane's DQS is routed point-to-point and arrives at its DRAM after its own delay, the DQS-to-CK relationship differs for every byte lane. DDR3

and later solve this with *write leveling*: in a special mode, each DRAM samples CK with the rising edge of DQS and returns the sampled value on DQ. The controller sweeps the DQS delay of each byte lane until the returned value changes from 0 to 1, which marks DQS aligned with CK at that DRAM (Figure 19.4). The required delay range must cover the fly-by skew plus routing differences; the controller's layout guide specifies the maximum CK-to-DQS skew it can level.

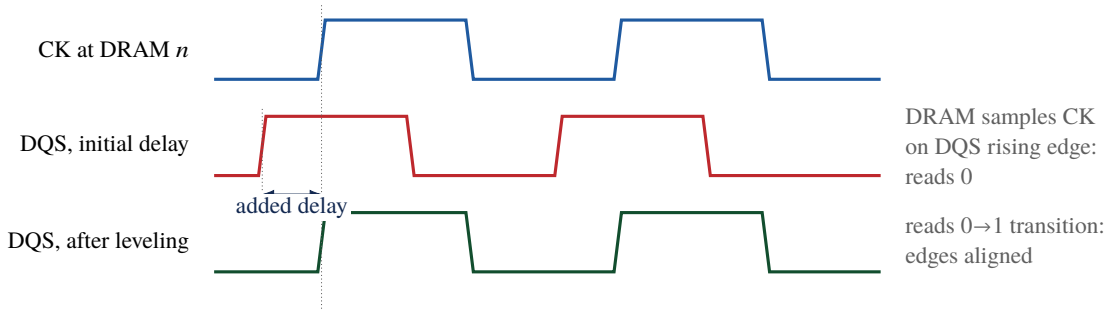



**Figure 19.4.** Write leveling. The controller delays DQS for a byte lane until its rising edge coincides with the CK rising edge at that lane's DRAM, as reported by the DRAM sampling CK with DQS.

### 19.3.3 Read and write centering, and VREF training

After write leveling, the controller performs read and write training: it sweeps the DQ-to-DQS delay (per bit in many controllers) to place the sampling point at the center of each bit's eye, and sweeps the internal $V_{\mathrm{REF}}$ of the DRAM (for writes) and of the controller (for reads) to center the threshold vertically. The result is a two-dimensional eye scan whose margins are an excellent indicator of board quality at bring-up (Chapter 27).

## 19.4 Routing and matching budgets

Training removes static skew within its adjustment range and resolution, but not skew that varies with pattern, temperature, or voltage, and not crosstalk or ISI. Layout guides therefore still impose matching rules, which follow from the UI:

- **DQ to DQS within a byte lane:** DQ and DQS share the same source-synchronous budget. Residual mismatch after per-bit deskew (or all of it, if the controller deskews only per byte) consumes setup/hold margin directly. A few percent of the UI is a typical allocation.
- **DQS to CK:** must be within the write-leveling range.
- **CA to CK:** the CA bus is captured by CK at each DRAM; matched as a group with CK, typically to a larger tolerance than DQ-to-DQS.
- **Intra-pair (CK, DQS):** tight, because pair skew distorts the strobe's crossing point.

**Worked Example 19.3: DQ-to-DQS matching**

For DDR4-3200, allocate 3 % of the UI to static DQ-to-DQS mismatch. What length tolerance is allowed in stripline at 165 ps/in?

**Solution.** UI = 312.5 ps, so the allocation is 9.4 ps, or 57 mil. Layout guides often specify tighter values, such as 10–25 mil, to leave margin for via-count differences, crosstalk-induced delay variation, and measurement uncertainty.

**Design Rule: DDR routing**

Route each byte lane (DQ, DM/DBI, DQS pair) on the same layer, with the same number of vias, referenced to ground, and adjacent to no other byte lane over long parallel runs. Match within the lane to the controller's guide in units of delay. Route CA and CK together in fly-by order with minimal stubs, terminate at the end with resistors decoupled to a low-impedance $V_{TT}$, and keep the total CK-to-DQS skew within the write-leveling range. Decouple $V_{REF}$ (where external) and keep it away from DQ routing.

**Table 19.3.** Typical DDR routing parameters. Controller layout guides take precedence.

| Parameter | Typical practice |
|---|---|
| DQ/DQS/CA single-ended impedance | 36–50 Ω (often 40 Ω) |
| CK and DQS differential impedance | 80–100 Ω |
| DQ-to-DQS matching | 1–5 % of UI, i.e. a few to tens of mil |
| CA-to-CK matching | Looser than DQ-to-DQS; per guide |
| Spacing between byte lanes | $\geq 3H$; more over long parallel runs |
| Fly-by stubs | As short as the package breakout allows |

## 19.5 Common failure modes

**Common Pitfall: DDR bring-up failures**

- *Write-leveling failure* on the last byte lanes: CK-to-DQS skew larger than the leveling range, usually from an unexpectedly long fly-by or a mistake in delay matching across layers.
- *Training passes, but data errors under load*: insufficient margin from SSN, crosstalk between byte lanes, or a noisy $V_{DDQ}/V_{TT}$.
- *Address errors at high temperature*: CA bus reflections from long stubs, or wrong termination value, eroding a small margin.
- *One bit consistently worst*: a via-count mismatch, a split-plane crossing, or a neck-down under the BGA.

Capture the controller's training results and two-dimensional eye margins on the first boards; they localize most of these problems to a byte lane or a bit.

**Rule of Thumb: Loaded delay of a fly-by bus**

DRAM input capacitance typically increases the propagation delay of a fly-by bus by 20–40 % and lowers its effective impedance by the same factor (Eq. (19.1)). *Limits:* depends on device capacitance and segment length; use the controller vendor's loaded-delay values or simulation for write-leveling range checks.

**Key Takeaways**

- DDR data are source-synchronous in byte lanes; CA and CK are common-clock and routed fly-by.
- POD (DDR4/5) and LVSTL (LPDDR4/5) signaling use on-die termination and an internal, trained $V_{REF}$.

- Fly-by gives clean signals but staggered arrival; write leveling aligns DQS to CK per byte lane.
- Loaded impedance and delay follow Eq. (19.1); terminate the CA bus near $Z_{\text{loaded}}$.
- Matching budgets derive from the UI; training removes static skew but not noise, crosstalk, or ISI.

## Further Reading

The JEDEC standards [28], [37]–[39] define the interfaces; memory-controller vendors' layout guides specify the topology and matching rules for their training algorithms.

## Review Problems

**P19.1** Find the UI and Nyquist frequency of LPDDR5-6400.

**P19.2** Allocate 2 % of the UI of DDR5-4800 to DQ-to-DQS mismatch. What is the length tolerance at 165 ps/in?

**P19.3** A 45 Ω bus segment has 2.5 pF of line capacitance and carries a 1.2 pF load. Find the loaded impedance.

**P19.4** A CA bus of 24 lines draws ±8 mA per line from $V_{\text{TT}}$. What source/sink capability is needed?

**P19.5** Five DRAMs are spaced 0.55 in apart on a fly-by bus with a loaded delay of 210 ps/in. Find the CK arrival spread.

**P19.6** An external $V_{\text{REF}}$ must stay within ±1 % of $V_{\text{DDQ}} = 1.2$ V. What is the allowed noise?

**P19.7** A colleague proposes reusing the DDR4 rule "DQ within ±10 mil of DQS" for an LPDDR5 byte. What is wrong, and what information do you need before writing the LPDDR5 rules?

Chapter 20

# PCI Express Gen3 to Gen6

**Chapter Overview**

PCI Express is the reference example of a modern serial link: AC-coupled differential lanes with embedded clocking, adaptive equalization, and a channel specified by its insertion loss at the Nyquist frequency. Each generation has doubled the data rate, from 8 GT/s in Gen3 to 32 GT/s in Gen5 and 64 GT/s with PAM4 in Gen6. This chapter summarizes the electrical layer of PCIe, explains channel loss budgeting and reference-clock architectures, and gives the routing rules that let a board meet its share of the budget.

**Learning Objectives**

1. Tabulate data rate, encoding, Nyquist frequency, and typical loss budget per generation.
2. Allocate a channel insertion-loss budget among packages, board, connectors, and vias.
3. Select and place AC-coupling capacitors.
4. Describe common-clock and separate-reference-clock architectures and SSC.
5. Apply PCIe routing rules for impedance, skew, vias, and stubs.
6. Explain what changes with PAM4 in Gen6.

## 20.1 Electrical overview

A PCIe link consists of one or more lanes (×1, ×4, ×8, ×16), each with a transmit and a receive differential pair. Each pair is AC coupled, so transmitter and receiver may have different common-mode voltages. There is no forwarded clock: the receiver's CDR recovers timing from the data, and lane-to-lane skew is removed by deskew logic in the receiver, so lane matching is loose. Link training negotiates speed and transmitter equalization presets, and Gen3 and later adapt the receiver's CTLE and DFE and the transmitter's FFE coefficients during an equalization phase [41].

**Table 20.1.** PCIe generations. Loss budgets are commonly cited end-to-end (bump-to-bump) values at Nyquist and are given for orientation only.

| Gen | Rate (GT/s) | Encoding | Signaling | $f_{\mathrm{Nyq}}$ (GHz) | Typical loss budget |
|---|---|---|---|---|---|
| 1 | 2.5 | 8b/10b | NRZ | 1.25 | — |
| 2 | 5.0 | 8b/10b | NRZ | 2.5 | — |
| 3 | 8.0 | 128b/130b | NRZ | 4.0 | ≈ 22 dB |
| 4 | 16.0 | 128b/130b | NRZ | 8.0 | ≈ 28 dB |
| 5 | 32.0 | 128b/130b | NRZ | 16.0 | ≈ 36 dB |
| 6 | 64.0 | FLIT + FEC | PAM4 (32 GBd) | 16.0 | ≈ 32 dB |

**Specification Note**

Channel loss budgets, reference-clock jitter limits, AC-coupling capacitance ranges, and equalization presets are defined in the PCI Express Base Specification [41], [42] and, for add-in cards and slots, in the Card Electromechanical (CEM) specification [43]. The loss values in Table 20.1 are those widely quoted in industry literature. Use the current specifications, and the silicon vendor's channel guidelines, for design.

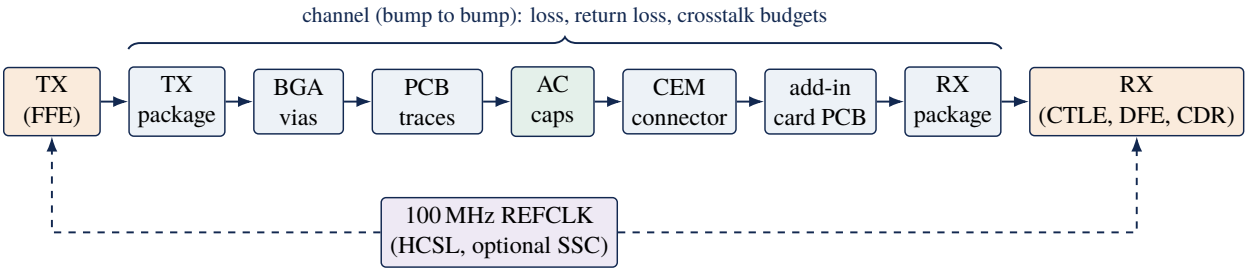



**Figure 20.1.** A PCIe channel from transmitter to receiver, through packages, system-board traces, AC-coupling capacitors, the CEM connector, and an add-in card. The reference clock is distributed to both ends in a common-clock architecture.

## 20.2 Channel loss budgeting

The specification's budget covers the whole channel from transmitter bump to receiver bump (Figure 20.1). Silicon vendors allocate part of it to their packages and to reference add-in-card or cable designs; the remainder is the system board's. A typical allocation is shown in Figure 20.2.

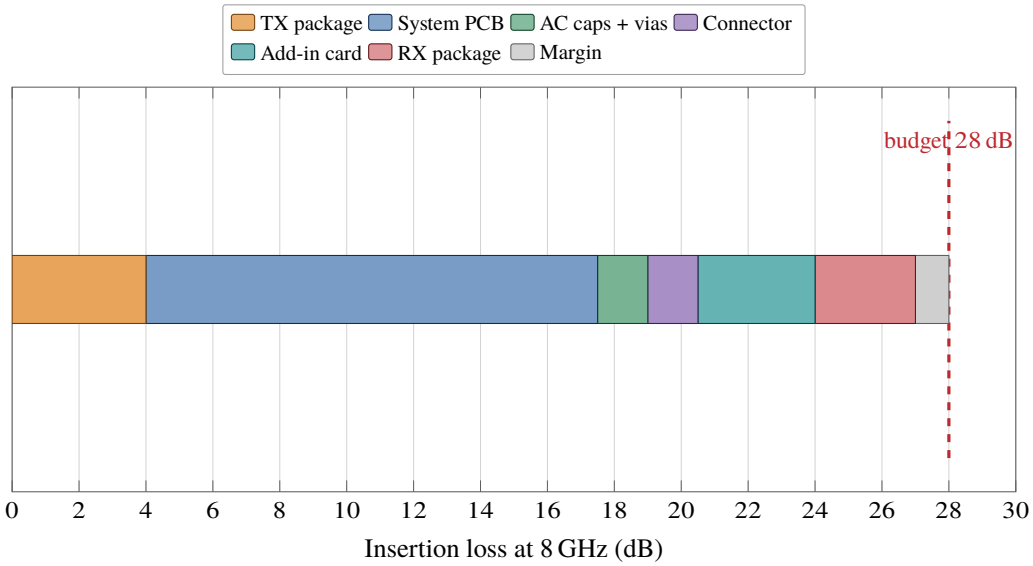



**Figure 20.2.** An illustrative insertion-loss allocation at 8 GHz for a PCIe Gen4 channel with a 28 dB budget. Actual allocations come from the silicon vendor and the CEM specification.

**Worked Example 20.1: Maximum Gen4 trace length**

For the allocation of Figure 20.2, the system board receives 13.5 dB at 8 GHz for its traces. How long may the route be in the mid-loss and low-loss materials of Worked Example 8.2?

**Solution.** In Worked Example 8.2 the HVLP conductor loss of a 5 mil, 50 Ω-equivalent stripline at 8 GHz was 0.72 dB/in. With $D_f = 0.010$, $D_k = 3.7$, dielectric loss is $2.31 \times 8 \times 0.010 \times 1.92 = 0.35$ dB/in; total 1.07 dB/in, allowing 12.6 in. With $D_f = 0.004$, $D_k = 3.5$, the dielectric term falls to 0.14 dB/in and the route may be 15.7 in. Vias, breakouts, and temperature derating reduce these lengths in practice. (An 85 Ω differential pair has somewhat different conductor loss from the single-ended estimate; a field solver should be used for final numbers.)

Insertion loss is not the only requirement. The specification also constrains return loss, insertion-loss deviation (ripple about a smooth fit), and crosstalk, or combines all of these into a channel compliance metric evaluated by simulation with reference transmitter and receiver models. Discontinuities that create reflections or resonances are often what fail a channel that meets its loss target.

## 20.3 AC coupling

Each transmit pair carries a series capacitor in each leg, normally placed near the transmitter (or as required by the CEM specification for slot-based designs). Because PCIe data are scrambled (Gen3 and later) or DC balanced (8b/10b), the low-frequency content is small. Each leg's capacitor is in series between the transmitter's source resistance $R_S$ and the receiver's termination $R_L$, so the high-pass corner is

$$f_{\mathrm{HP}} = \frac{1}{2\pi (R_S + R_L)\, C_{\mathrm{AC}}}, \tag{20.1}$$

which with $R_S = R_L \approx 50\,\Omega$ per leg lies in the kilohertz range. For example, 220 nF gives $1/(2\pi \times 100 \times 220 \times 10^{-9}) = 7.23$ kHz. (Using $R_L$ alone overstates the corner by a factor of two.)

**Baseline wander.** During a run of identical symbols the coupling capacitor charges and the received level droops toward zero with time constant $\tau = (R_S + R_L)C_{\mathrm{AC}} = 22\,\mu$s. For a run of $N$ symbols, the fractional droop is about $N\,\mathrm{UI}/\tau$; a 100-symbol run at 8 GT/s (12.5 ns) droops by 0.06 %. Scrambling keeps runs short, so baseline wander is negligible for PCIe; it matters for unscrambled or poorly balanced data and for much smaller capacitors.[1]

**Design Rule: AC-coupling capacitors**

Use the smallest case size that the assembly process supports (0201 or 0402), place the two capacitors of a pair side by side and symmetric, and keep the footprint on the same layer as the traces to avoid extra vias. If the pads are wider than the trace and the reference is close, they form a capacitive discontinuity; whether to void the reference plane under them (referencing the next ground plane down) depends on pad size, dielectric height and data rate. Decide with a field-solver or 3-D model of the actual footprint, and check that any void does not cut the return path of neighboring traces.

[1] The Base Specification defines an allowed AC-coupling capacitance range for each generation (on the order of 75–265 nF, with a higher minimum for Gen3 and later). Use the values in the current specification.

## 20.4 Reference clocks

PCIe uses a 100 MHz reference clock, usually HCSL, with a frequency tolerance of ±300 ppm. Three architectures are defined:

**Common clock**
The same reference clock (with or without SSC) is distributed to both ends. Jitter common to both ends is partly cancelled by the receiver's CDR, so the jitter limit applies to the difference.

**Separate reference, no SSC (SRNS)**
Each end has its own reference; SSC is not used.

**Separate reference, independent SSC (SRIS)**
Each end has its own reference with SSC; the receiver must absorb the larger frequency offset, which requires extra elasticity in the logic.

SSC down-spreads the reference by up to 0.5 % at 30–33 kHz (Chapter 18). Reference-clock jitter limits are tight and are specified after filtering by the PLL and CDR transfer functions of the link; clock buffers and their layout are discussed in Chapter 24.

## 20.5 Routing rules

**Table 20.2.** Typical PCIe board routing practice.

| Parameter | Typical practice |
|---|---|
| Differential impedance | 85 Ω nominal (many designs use 85 Ω–100 Ω); ±10 % |
| Intra-pair skew | A few mils; compensate at the source of mismatch (Chapter 10) |
| Lane-to-lane skew | Loose (receiver deskew); keep TX lanes similar for EMI |
| TX-to-RX spacing | Route TX and RX on different layers or with large spacing to limit NEXT |
| Pair-to-pair spacing | $\geq 3H$ to $5H$ (stripline), more for long parallel runs |
| Via stubs | Back-drill or route on deep layers at Gen4 and above (Chapter 11) |
| Reference | Continuous ground; no splits; return vias at every layer change |
| Laminate | Mid-loss for Gen3/Gen4; low or very-low loss for Gen5 and Gen6 |

**Common Pitfall: Routing TX and RX pairs side by side**

Interleaving transmit and receive pairs of the same port. A strong transmitter near the start of its channel couples by NEXT into a weak, attenuated receive signal near the end of its channel, where the signal-to-crosstalk ratio is worst. Separate TX and RX groups by layer or by generous spacing, especially near connectors and BGAs.

## 20.6 Gen6 and PAM4

PCIe Gen6 doubles the data rate to 64 GT/s without raising the symbol rate above Gen5's 32 GBd by transmitting two bits per symbol with four-level pulse amplitude modulation (PAM4, Figure 20.4) [42]. The Nyquist frequency, and hence the channel loss per inch, stays at 16 GHz, but each of the three stacked eyes has one-third of the NRZ amplitude, a 9.5 dB penalty in signal-to-noise ratio. Gen6 compensates by allowing a higher raw error rate and correcting it with forward error correction (FEC) in a fixed-size flow-control-unit (FLIT)

mode. For the board designer, the consequences are tighter limits on noise sources—crosstalk, reflections, and power-supply noise—rather than on loss alone.

Figure 20.3 shows the effect under identical conditions: the same symbol rate, the same 2 unit peak-to-peak swing, the same one-pole channel and the same Gaussian noise. At fixed swing, PAM4's adjacent levels are one-third as far apart as NRZ's, so the ideal threshold distance falls from 1 to $1/3$ ($-9.54$ dB). Keep the symbol UI and the bit time distinct: at 32 GBd, the UI is 31.25 ps, the Nyquist frequency is 16 GHz, and uncoded PAM4 carries 64 Gb/s before coding and protocol overhead. The figure is an illustration, not a compliance eye: it has no jitter, CTLE, DFE, CDR, FEC or mask, and a few hundred overlaid symbols say nothing about BER at $10^{-12}$.

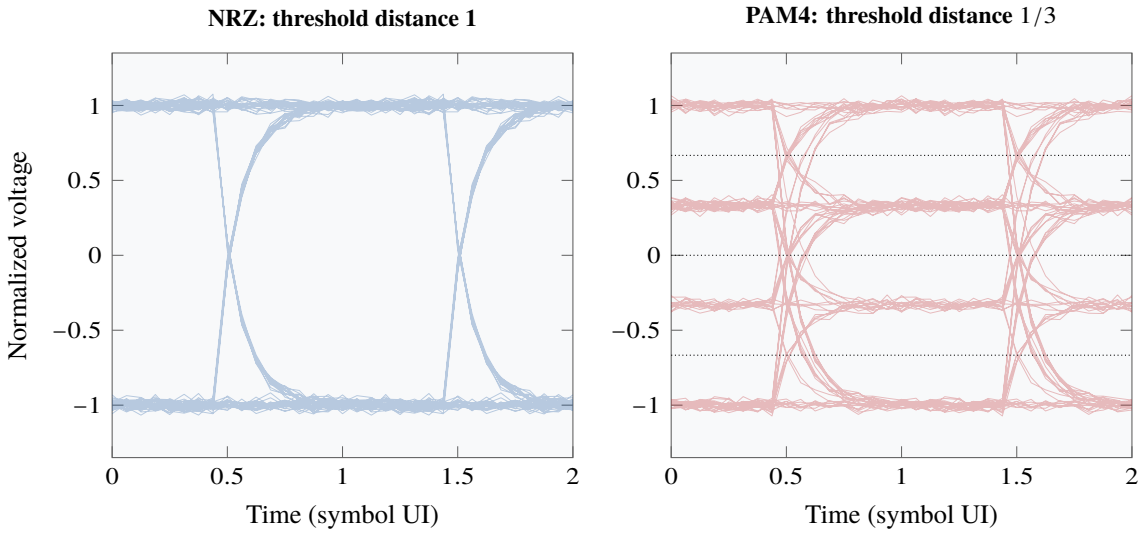



**Figure 20.3.** Synthetic NRZ and PAM4 eyes at the same symbol rate and swing: 1600 random symbols, one-pole channel ($\tau = 0.1$ UI), additive Gaussian noise $\sigma = 0.025$, 80 overlaid two-UI windows, fixed seed. Dotted lines: PAM4 decision thresholds. [Simulation (synthetic)]

### 20.6.1 Raw and post-FEC error rates

With FEC the board no longer has to deliver a raw bit-error ratio of $10^{-12}$. The link tolerates a much higher *raw* (first-burst or pre-FEC) error ratio at the receiver slicer, and the FEC decoder corrects most of those errors. Two cautions follow. First, FEC performance depends on error *statistics*: DFE error propagation and correlated noise produce bursts that a code tuned for random errors corrects less well. Second, FEC changes the reliability interpretation of a margin; it does not restore analog eye height. Channel compliance is therefore judged at the raw error ratio the specification assigns, measured or simulated with the reference receiver.

### 20.6.2 Retimers and channel partitioning

When a channel cannot meet its loss budget—long backplanes, cabled links, or a riser plus an add-in card—a *retimer* splits it into two electrically independent segments. A retimer is a full receiver and transmitter (CDR, equalization, re-launch) that takes part in link training; a *redriver* is only a linear amplifier and equalizer that passes jitter and noise through. Each retimer segment must meet the budget on its own, which makes the partition a board-architecture decision: where the retimer sits determines the two segment lengths, the

number of connectors in each, and the reference-clock architecture. Place it so that both segments have margin, not so that one is minimal and the other marginal.

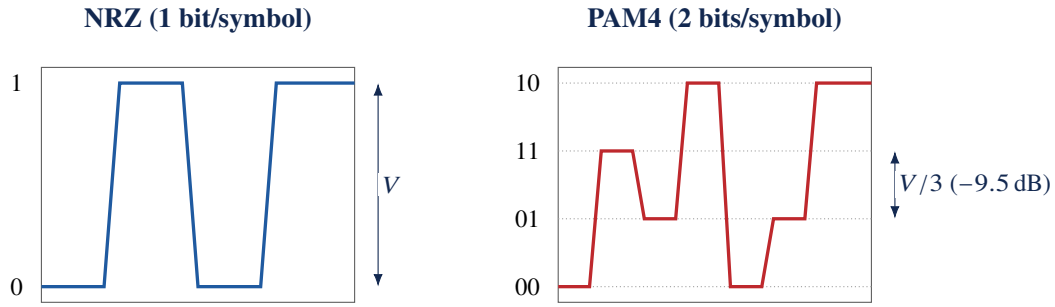



**Figure 20.4.** NRZ and PAM4 at the same peak-to-peak swing. PAM4 carries two bits per symbol with Gray-coded levels; each eye has one-third of the NRZ amplitude.

**Worked Example 20.2: Bandwidth per lane**

What is the payload bandwidth of a Gen4 ×16 link, ignoring protocol overhead above the encoding?

**Solution.** Each lane carries $16 \times 128/130 = 15.75\,\text{Gb/s}$. Sixteen lanes carry $252\,\text{Gb/s}$, or about $31.5\,\text{GB/s}$ in each direction.

**Rule of Thumb: Laminate per generation**

Each doubling of the PCIe data rate roughly doubles the loss per inch at Nyquist for the same material; keeping the same route length usually requires moving one loss class lower (Table 7.3) per generation. *Limits:* Gen6 keeps Gen5's Nyquist frequency, and short routes may stay on a higher-loss material; the budget calculation of Worked Example 20.1 decides.

**Key Takeaways**

- PCIe lanes are AC-coupled, embedded-clock differential pairs; lane-to-lane skew is loose, intra-pair skew tight.
- Each generation doubles the data rate; Gen6 keeps Gen5's Nyquist frequency by using PAM4.
- Budget the channel's loss at Nyquist among packages, board, connectors, and vias; return loss and crosstalk often decide compliance.
- Place AC caps symmetrically; void the reference under the pads only where a 3-D model shows the pad capacitance matters. The AC-coupling corner is $1/[2\pi(R_S + R_L)C]$.
- Choose the reference-clock architecture (common, SRNS, SRIS) early; it constrains SSC and jitter.

## Further Reading

The PCI Express Base and CEM specifications [41]–[43] define the electrical layer; IEEE 370 [44] covers characterization of the board interconnect.

## Review Problems

**P20.1** Find the UI and Nyquist frequency of PCIe Gen5.

**P20.2** What is the payload rate per lane of Gen4 with 128b/130b encoding?

**P20.3** What are the symbol rate and UI of Gen6, and the amplitude penalty of each PAM4 eye relative to NRZ?

**P20.4** Find the high-pass corner of a 100 nF AC-coupling capacitor with 50 Ω source and 50 Ω termination.

**P20.5** A board is allotted 14 dB for traces at Nyquist, and its routing loses 1.1 dB/in. What is the maximum route length?

**P20.6** A 100 MHz reference clock uses 0.5 % down-spread SSC. What is its minimum instantaneous frequency?

Chapter 21

# USB 2.0, USB 3.x, USB4, and Type-C

**Chapter Overview**

USB combines two very different physical layers in one connector: the 480 Mb/s USB 2.0 half-duplex pair and the multi-gigabit, AC-coupled SuperSpeed and USB4 lanes. The Type-C connector added reversibility, configuration through the CC pins, and power delivery up to tens of volts. This chapter summarizes the electrical parameters of each generation, explains how Type-C orientation affects routing, and gives layout guidance for the USB 2.0 pair, the high-speed lanes, ESD protection, and VBUS.

**Learning Objectives**

1. Tabulate signaling rate, encoding, impedance, and coupling for USB 2.0, USB 3.2, and USB4.
2. Explain the Type-C pin map and the routing consequences of reversibility.
3. Route the USB 2.0 pair and its orientation stubs.
4. Place AC-coupling capacitors, orientation muxes, redrivers, and ESD devices on the high-speed lanes.
5. Describe CC-based configuration and size VBUS routing.

## 21.1 Generations and signaling

**Table 21.1.** USB physical layers (representative; see the specification note).

| Mode | Rate per lane | Encoding | Lanes | Coupling |
|---|---|---|---|---|
| USB 2.0 High Speed [45] | 480 Mb/s | NRZI, bit stuffing | 1 pair, half duplex | DC |
| USB 3.2 Gen 1 [46] | 5 Gb/s | 8b/10b | 1 TX + 1 RX | AC (TX caps) |
| USB 3.2 Gen 2 | 10 Gb/s | 128b/132b | 1 TX + 1 RX | AC |
| USB 3.2 Gen 2×2 | 10 Gb/s | 128b/132b | 2 TX + 2 RX (Type-C) | AC |
| USB4 Gen 2/Gen 3 [47] | 10 / 20 Gb/s | 64b/66b, 128b/132b | 2 TX + 2 RX | AC |

USB 2.0 High Speed drives a 17.78 mA current into a 45 Ω-terminated line at each end, producing about 400 mV differential across a 90 Ω pair. It is DC coupled, half duplex, and slow enough that board routing rarely limits it—provided the pair's impedance and symmetry are maintained and the Type-C stubs are short.

SuperSpeed and USB4 lanes are AC-coupled, embedded-clock links like PCIe (Chapter 20), with transmitter de-emphasis and receiver equalization. USB4 version 2.0 extends the rate further with multi-level signaling.

**Specification Note**

Exact impedance targets (90 Ω nominal for USB differential pairs, with tolerances that differ between boards, cables, and generations), AC-coupling capacitance ranges, channel loss allocations, and USB4 version-2 signaling details are defined in the USB-IF specifications [45]–[48]. Check the current revisions and the silicon vendor's guidelines.

## 21.2 The Type-C connector

The Type-C receptacle has 24 contacts in two rows, arranged with 180-degree rotational symmetry so that the plug can be inserted either way up (Figure 21.1) [48]. Its signals are:

- two SuperSpeed/USB4 transmit pairs (TX1, TX2) and two receive pairs (RX1, RX2), of which one TX/RX set is used in a single-lane mode, selected by orientation;
- the USB 2.0 pair, which appears twice (A6/A7 and B6/B7) and is connected to both locations on the host;
- two configuration channel pins (CC1, CC2), through which the host and device detect attachment, orientation, current capability, and power delivery;
- two sideband-use pins (SBU1, SBU2), used by alternate modes;
- four VBUS and four GND contacts.

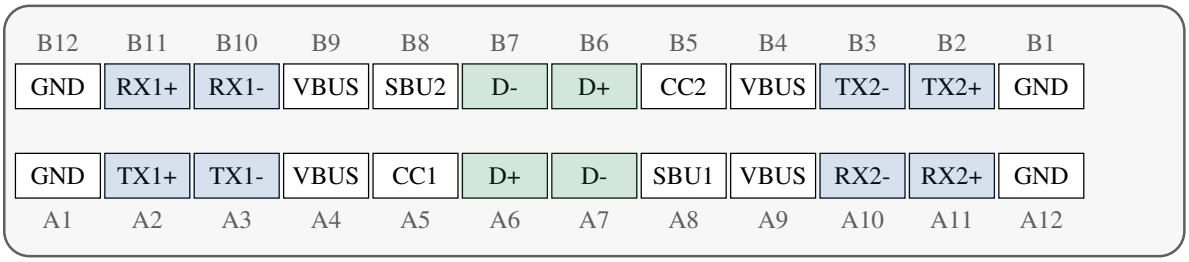



**Figure 21.1.** Type-C receptacle contact map (front view). Row B is rotated by 180 degrees relative to row A, making the connector reversible.

### 21.2.1 Configuration channel

A source presents a pull-up $R_p$ (or current source) on CC; a sink presents a pull-down $R_d$. The voltage on the CC line that has $R_d$ identifies the orientation and advertises the source's current capability (default USB power, 1.5 A, or 3 A). USB Power Delivery then negotiates higher voltages over the same CC wire.[1]

## 21.3 Routing the USB 2.0 pair

On the host side, D+ connects to both A6 and B6 and D− to both A7 and B7. In either orientation, one of each pair of contacts is unused and forms a stub.

[1]The commonly used values are $R_d = 5.1\,\text{k}\Omega$ in a sink and, for a source pulling up to 5 V, $R_p = 56\,\text{k}\Omega$, $22\,\text{k}\Omega$, or $10\,\text{k}\Omega$ for default, 1.5 A, and 3 A advertisement. Confirm against the current Type-C specification, which also defines current-source and 3.3 V pull-up alternatives.

**Worked Example 21.1: Is the orientation stub a problem?**

USB 2.0 High Speed edges are no faster than about 500 ps. What is the critical length on an outer layer with $t_{pd} = 150\,\text{ps/in}$?

**Solution.** $\ell_{crit} = t_r/(6t_{pd}) = 500/900 = 0.56\,\text{in}$. Connecting the two contact rows with a short jog under the connector produces stubs of well under 0.2 in, so the stub is electrically short. A long branch, or tying the contacts together at the far side of a mux, would not be.

**Design Rule: USB 2.0 pair**

Route D+/D− as a 90 Ω pair over a continuous ground plane, tie A6 to B6 and A7 to B7 directly at the receptacle with the shortest possible stubs, place the ESD device and any common-mode choke close to the connector, and keep the pair away from SuperSpeed lanes and clocks.

## 21.4 High-speed lanes

Figure 21.2 shows the typical host-side path. Because the active TX and RX pairs depend on orientation, a host either uses a two-lane PHY that can handle both (as USB4 and Gen 2×2 require) or inserts an orientation multiplexer controlled by the CC logic. Long host routes may need a linear redriver or a retimer; a retimer resets the jitter and loss budgets, while a redriver only amplifies and equalizes.

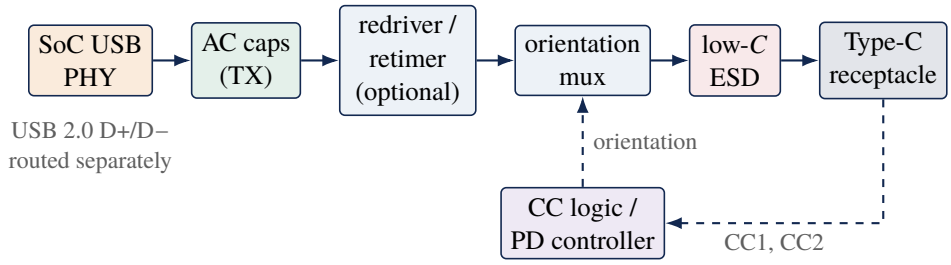



**Figure 21.2.** Host-side signal path for SuperSpeed or USB4 lanes to a Type-C receptacle. The CC logic determines orientation and configures the mux.

### 21.4.1 ESD protection

Every line exposed at the connector needs ESD protection, placed between the connector and everything else so that the discharge current returns through the shortest path to ground. On high-speed lanes the protector's capacitance is a shunt discontinuity. A shunt capacitance $C$ on a line of impedance $Z_0$ reflects

$$|\Gamma| = \frac{x}{\sqrt{4 + x^2}}, \qquad x = \omega C Z_0, \tag{21.1}$$

which follows from Eq. (6.2) with $Z_L = Z_0 \parallel (1/j\omega C)$.

**Worked Example 21.2: ESD capacitance and return loss**

A 10 Gb/s lane ($f_{Nyq} = 5\,\text{GHz}$) uses an ESD device with 0.3 pF per line on a 90 Ω pair (45 Ω per leg in the odd mode). Estimate the reflection at Nyquist, and repeat for 0.5 pF.

**Solution.** $x = 2\pi \times 5 \times 10^9 \times 0.3 \times 10^{-12} \times 45 = 0.424$, so $|\Gamma| = 0.207$ (return loss 13.7 dB). With 0.5 pF, $x = 0.707$ and $|\Gamma| = 0.333$ (9.5 dB). Low-capacitance protectors (well below 0.5 pF), flow-through

packages that the pair passes over without stubs, and a slight local impedance increase to compensate are essential at these rates.

A single capacitance is only a first estimate. A real protector is a package with pin inductance, a die capacitance that depends on bias, and internal routing between the pins of a flow-through package. Vendors of high-speed protectors publish S-parameter models of the device on a test board; use them in the channel simulation of Chapter 25, after de-embedding the vendor's test fixture or confirming that its reference planes match your footprint. Where a model is not available, a TDR of the footprint on a test coupon is the next best evidence.

**Common Pitfall: ESD device far from the connector**

Placing the ESD protector near the SoC "to protect it" or routing its ground through a long trace. The discharge current then travels across the board, coupling into other circuits, and the protector's ground connection adds enough inductance to raise the clamp voltage. Place the protector at the connector with its ground pins via'd directly to the ground plane.

## 21.5 VBUS and power

Type-C sources supply 5 V by default and, with Power Delivery, higher voltages (up to 20 V in standard power range, and up to 48 V with extended power range in recent revisions). VBUS traces and pours must carry up to 3–5 A with acceptable IR drop and temperature rise (IPC-2152), and VBUS needs bulk and local capacitance within the limits the specification imposes on inrush. Connector VBUS pins should be connected with wide copper and multiple vias.

**Worked Example 21.3: VBUS drop**

A 3 A VBUS path runs 2 in in a 100 mil wide, 1 oz trace. Estimate the drop and dissipation.

**Solution.** The trace is 20 squares long, so $R = 20 \times 0.49\,\text{m}\Omega = 9.8\,\text{m}\Omega$; the drop is 29 mV and the dissipation 88 mW. Acceptable for 5 V, but a wider pour or plane would be preferred at 5 A.

**Table 21.2.** USB layout summary.

| Item | Guidance |
|---|---|
| USB 2.0 pair | 90 Ω; short orientation stubs; ESD and CMC at connector |
| SuperSpeed / USB4 pairs | 85 Ω–90 Ω per guide; tight intra-pair skew; TX AC caps |
| Orientation | Two-lane PHY or mux driven by CC logic |
| Redriver / retimer | For routes beyond the host's loss allocation |
| ESD | Sub-picofarad, flow-through, at the connector |
| CC | $R_p$/$R_d$ or PD controller; route as low-speed but keep away from SS lanes |
| VBUS | Wide copper; IR drop and inrush; protection against overvoltage |

**Rule of Thumb: ESD capacitance**

For lanes at 10 Gb/s and above, keep the protector's capacitance below about 0.3 pF per line; by Eq. (21.1) this keeps its reflection near −14 dB or better at 5 GHz. *Limits:* a locally compensated footprint can tolerate more capacitance; higher rates need less.

**Key Takeaways**

- USB combines a DC-coupled 480 Mb/s pair and AC-coupled multi-gigabit lanes in one connector.
- Type-C is reversible; D+/D− appear twice and must be tied with short stubs; high-speed lanes need a two-lane PHY or a mux.
- ESD capacitance is a major return-loss contributor at 10 Gb/s and above.
- CC pins set orientation and power; VBUS can reach tens of volts and several amperes.

## Further Reading

The USB-IF specifications [45]–[48] define the electrical layer and the Type-C connector; silicon vendors' layout guides give host- and device-specific rules.

## Review Problems

**P21.1** Find the UI and Nyquist frequency of USB 3.2 Gen 2.

**P21.2** A USB4 Gen 3 lane runs at 20 Gb/s. Find its UI and Nyquist frequency.

**P21.3** Compute the payload rate of a Gen 1 lane (8b/10b) and a Gen 2 lane (128b/132b).

**P21.4** Find the reflection coefficient at 5 GHz of a 0.5 pF shunt capacitance on a 45 Ω line.

**P21.5** What pull-down resistance identifies a Type-C sink on CC?

**P21.6** Find $\ell_{\text{crit}}$ for a 500 ps edge with $t_{\text{pd}} = 150$ ps/in.

Chapter 22

# Ethernet: 1000BASE-T, 10GBASE-KR, SGMII, and RGMII

**Chapter Overview**

Ethernet on a board involves two very different kinds of interface. On the *media* side, the PHY drives twisted-pair cable through isolation magnetics (1000BASE-T) or a backplane channel (10GBASE-KR). On the *MAC* side, the PHY connects to the MAC in the SoC or switch through a parallel source-synchronous bus (RGMII) or a serial link (SGMII and its relatives). This chapter covers both: the analog front end, isolation, and EMC layout around the magnetics and connector; the backplane channel of 10GBASE-KR; and the timing of RGMII and the routing of SGMII.

**Learning Objectives**

1. Describe 1000BASE-T signaling and the role of the magnetics.
2. Lay out the PHY, magnetics, and RJ45 region with correct isolation and chassis grounding.
3. Explain the Bob Smith termination and its limits.
4. Estimate the channel loss of a 10GBASE-KR backplane.
5. Close RGMII timing with internal or PCB clock delay.
6. Route SGMII as a low-rate SERDES link.

## 22.1 1000BASE-T

1000BASE-T transmits 1 Gb/s over four twisted pairs at 125 MBd per pair, using five-level PAM (PAM5) with trellis coding. Each pair carries traffic in both directions simultaneously; the PHY cancels its own transmitted signal (echo cancellation) and crosstalk from the other three pairs (NEXT and FEXT cancellation) digitally [49]. The board's job on the media side is to deliver a balanced, well-isolated, low-emission connection between the PHY's MDI pins and the cable.

### 22.1.1 Magnetics

Each pair passes through an isolation transformer and usually a common-mode choke (Figure 22.1). The transformer provides galvanic isolation between the cable and the equipment—IEEE 802.3 requires the interface to withstand a specified test voltage[1]—and transfers the differential signal while blocking DC. Its PHY-side center tap is biased or decoupled according to the PHY type (current-mode PHYs need a supply on the center tap; voltage-mode PHYs need only a capacitor). The common-mode choke attenuates common-mode noise traveling in either direction.

[1] Commonly 1500 V rms for 60 s, or an equivalent; see the isolation clause of the current IEEE 802.3 standard.

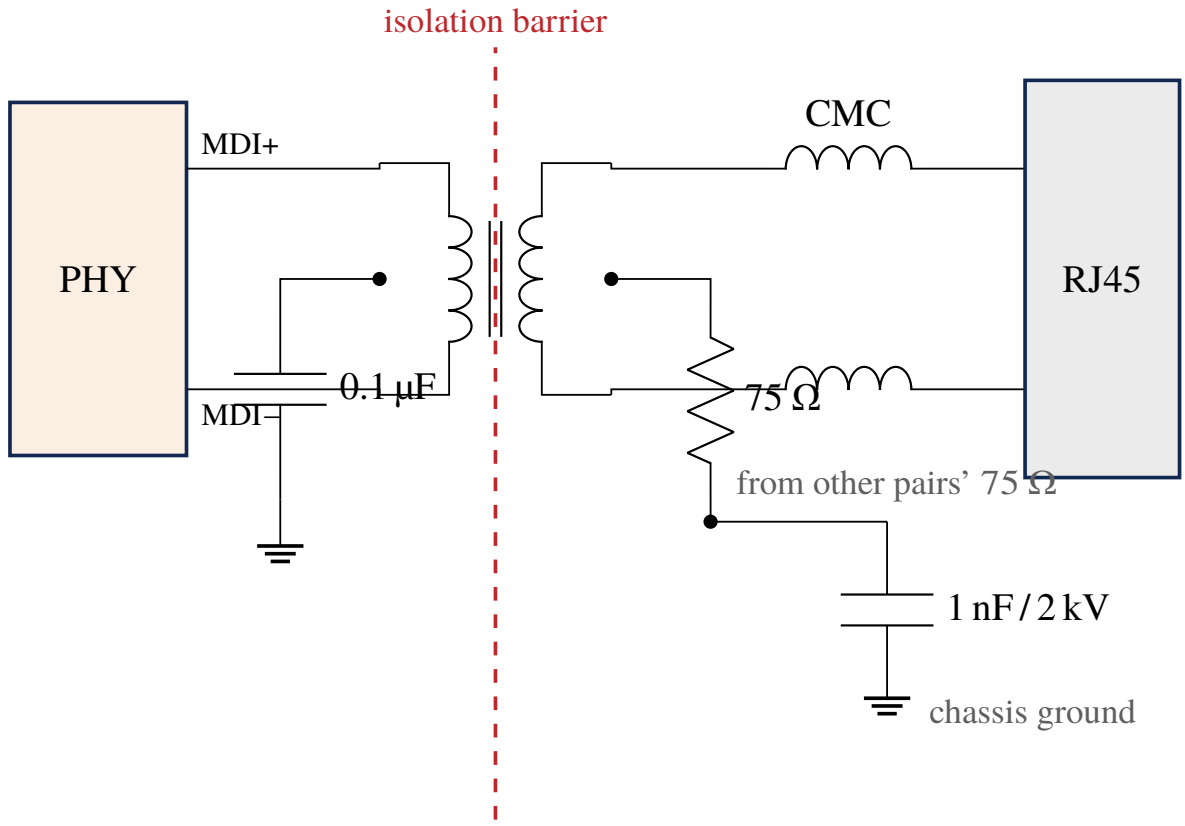



**Figure 22.1.** One pair of a 1000BASE-T interface: PHY, isolation transformer with center taps, common-mode choke (CMC), RJ45, and Bob Smith termination of the cable-side center tap to chassis ground.

### 22.1.2 Bob Smith termination

The cable-side center tap of each pair is connected through a resistor (commonly 75 Ω) to a common node, which is connected to chassis ground through a high-voltage capacitor (often 1 nF, 2 kV). The intent is to terminate the common mode of each pair and the pair-to-pair modes of the cable, reducing common-mode emissions. The 75 Ω value derives from the common-mode impedance of telephone-type cable; whether it is optimal for Ethernet cabling has been debated, and many integrated magnetics modules include their own version of this network. Follow the PHY and magnetics vendors' recommendations.

### 22.1.3 Layout

**Design Rule: Ethernet media-side layout**

Place the magnetics close to the RJ45 (or use an RJ45 with integrated magnetics). Route each MDI pair as a 100 Ω differential pair, matched within the pair, with no vias if possible. Remove all signal-ground and power copper from every layer in the isolation gap under the magnetics (Figure 22.2), keeping creepage and clearance as required by the safety standard. Connect the connector shield and the Bob Smith capacitor to chassis ground, stitched with vias, and keep unrelated traces out of the connector zone.

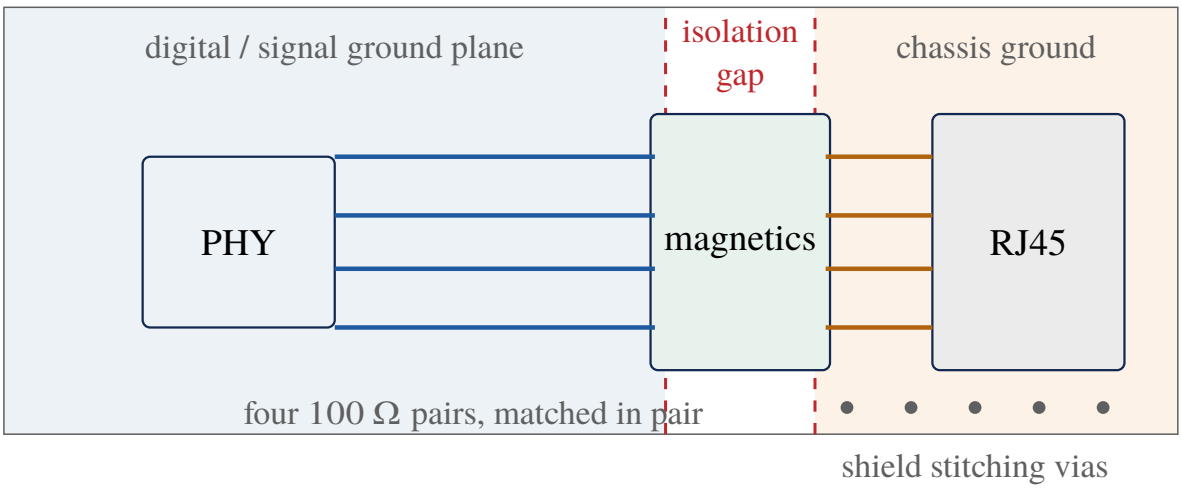



**Figure 22.2.** Layout zones of an Ethernet port: signal ground under the PHY, an isolation gap under the magnetics, and chassis ground under the connector.

**Common Pitfall: Routing other signals through the isolation gap**

Passing an LED line, a clock, or a power trace across the gap between PHY ground and chassis ground, or under the magnetics. The crossing defeats the isolation, couples noise into the cable's common mode, and often appears as an emissions failure at 125 MHz harmonics. LED lines of RJ45s with integrated LEDs should be routed out of the connector on the PHY side of the gap and filtered.

## 22.2 10GBASE-KR

10GBASE-KR runs 10.3125 Gb/s (64b/66b encoding) over a backplane, with a three-tap transmit FFE adapted by a training protocol, and with receiver equalization. The unit interval is 96.97 ps, and the Nyquist frequency is 5.156 GHz. IEEE 802.3 specifies the backplane channel by insertion-loss, return-loss, and crosstalk limits as functions of frequency rather than by a single number [49]; the same electrical framework is reused by many chip-to-chip 10 Gb/s SERDES links.

**Worked Example 22.1: Backplane channel loss**

Estimate the line loss at 5.156 GHz of a 20 in, 6 mil, 50 Ω-equivalent stripline in a mid-loss laminate ($D_f = 0.010$, $D_k = 3.8$) with 1 µm RMS roughness.

**Solution.** $\delta_s = 0.92$ µm and $K_H = 1.65$, so conductor loss is $36\sqrt{5.156}/(6 \times 50) \times 1.65 = 0.45$ dB/in. Dielectric loss is $2.31 \times 5.156 \times 0.010 \times \sqrt{3.8} = 0.23$ dB/in. Total 0.68 dB/in, or 13.6 dB for 20 in, before the two backplane connectors and the line cards' traces and vias.

## 22.3 MAC-to-PHY interfaces

### 22.3.1 RGMII

The Reduced Gigabit Media Independent Interface carries four data bits and a control signal in each direction, clocked on both edges of a 125 MHz clock for gigabit operation (250 Mb/s per data line). It is source synchronous (Chapter 12): the transmitter launches data aligned with its clock edges, and the receiver needs the clock delayed by roughly a quarter of the clock period so that it samples in the middle of the data eye

(Figure 22.3). Early versions of the RGMII specification placed this delay in the PCB routing; later versions allow it to be generated inside the transmitter or receiver ("RGMII-ID").[2]

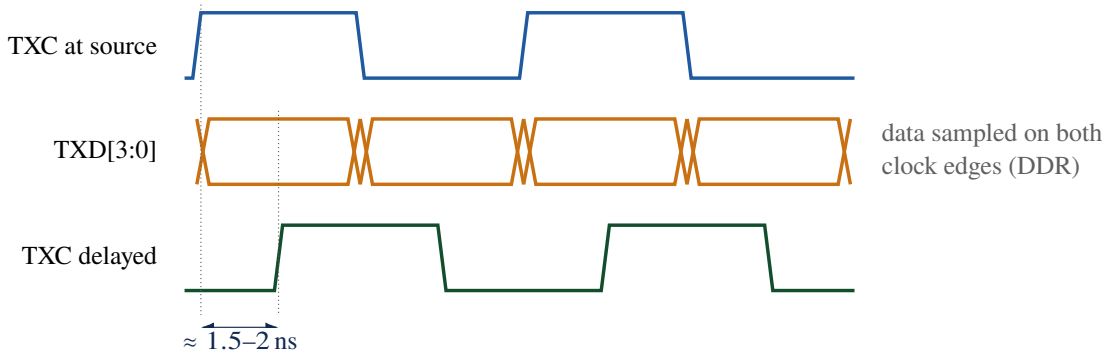



**Figure 22.3.** RGMII transmit timing. Data change at the source clock edges; delaying the clock by about 1.5–2 ns centers each edge in the 4 ns data eye.

**Worked Example 22.2: Clock delay in routing**

If a 2 ns RGMII clock delay were implemented in PCB routing with $t_{pd} = 150\,\text{ps/in}$, how much extra trace would be needed?

**Solution.** $2000/150 = 13.3$ in of extra clock trace—impractical and noisy. Enable the internal delay in exactly one of the two devices (MAC or PHY) for each direction, and match the data lines to their clock to a few tens of picoseconds.

**Worked Example 22.3: A signed RGMII budget**

A receiver needs (illustratively) 1.0 ns setup and 0.8 ns hold, the internal clock delay is 2.0 ns, and the data-to-clock skew at the receiver, $\Delta = t_{data} - t_{clk}$ from transmitter output skew plus board mismatch, ranges from −0.5–0.3 ns. Find the margins.

**Solution.** With a 4 ns data eye and sampling at the delayed clock edge (Eq. (12.3)), setup margin $= 2.0 - 1.0 - \Delta_{max} = 0.7$ ns and hold margin $= 4.0 - 2.0 - 0.8 + \Delta_{min} = 0.7$ ns. A symmetric $\pm 0.5$ ns budget would have reported 0.5 ns of setup margin; the signed range shows that the skew is biased toward early data, so the board can tolerate 0.2 ns more late-data mismatch than a symmetric rule allows. Use the actual minimum and maximum values from the MAC and PHY data sheets.

**Common Pitfall: Double or missing RGMII delay**

Enabling the internal clock delay in both the MAC and the PHY, or in neither. The link may come up at 10 or 100 Mb/s, where the timing budget is enormous, and fail only at 1 Gb/s. Check the delay configuration in both devices' strapping and driver settings at bring-up.

### 22.3.2 SGMII and serial MAC interfaces

SGMII carries gigabit Ethernet over one differential pair in each direction at 1.25 Gb/s (8b/10b). Its signals are low-swing differential (LVDS-like), commonly AC coupled, and routed as 100 Ω pairs with modest intra-

[2]The required delay is usually quoted as about 1.5–2.0 ns; consult the RGMII version used by the MAC and PHY, and their data sheets.

pair matching. Its higher-rate relatives (2500BASE-X, USXGMII, 10GBASE-R interfaces such as XFI/SFI) follow the same SERDES rules as PCIe lanes of similar rate (Chapter 20).

**Table 22.1.** Ethernet interfaces on boards.

| Interface | Rate | Board design notes |
|---|---|---|
| 1000BASE-T MDI | 4 × 125 MBd PAM5 | 100 Ω pairs; magnetics; isolation gap; chassis ground |
| 10GBASE-KR | 10.3125 Gb/s | Backplane SERDES; loss, RL, crosstalk masks |
| RGMII | 4 × 250 Mb/s (DDR) | Source synchronous; one clock delay per direction |
| SGMII | 1.25 Gb/s | 100 Ω pair, usually AC coupled |

**Rule of Thumb: RGMII clock delay**

The RGMII clock delay is about one quarter of the 8 ns clock period, i.e. about 2 ns, which places each clock edge near the center of the 4 ns data bit. *Limits:* the exact window depends on the MAC and PHY timing specifications and on the RGMII version; use the devices' data sheets.

**Key Takeaways**

- 1000BASE-T uses four bidirectional PAM5 pairs at 125 MBd; the PHY cancels echo and crosstalk.
- Magnetics provide isolation and common-mode rejection; keep the isolation gap free of copper and route the connector zone to chassis ground.
- 10GBASE-KR is a backplane SERDES at 10.3125 Gb/s; budget loss at 5.16 GHz.
- RGMII needs exactly one 1.5–2 ns clock delay per direction, preferably internal.

## Further Reading

IEEE 802.3 [49] defines 1000BASE-T and 10GBASE-KR; PHY and magnetics vendors' application notes give layout recommendations for the media-side circuit.

## Review Problems

**P22.1** How many bits per symbol per pair does 1000BASE-T carry to reach 1 Gb/s at 125 MBd on four pairs?

**P22.2** Find the UI and Nyquist frequency of SGMII.

**P22.3** What is the bit rate per RGMII data line at gigabit speed?

**P22.4** Find the UI of 10GBASE-KR.

**P22.5** What trace length gives 1.8 ns of delay at 165 ps/in?

**P22.6** Using Worked Example 22.1, estimate the line loss of a 24 in backplane route.

Chapter 23

# MIPI CSI-2 and DSI: D-PHY and C-PHY

**Chapter Overview**

Cameras and displays in embedded systems connect to the application processor through MIPI CSI-2 (camera serial interface) and DSI (display serial interface). Both protocols run over one of two physical layers: D-PHY, a source-synchronous set of differential lanes with a forwarded clock, or C-PHY, an embedded-clock three-wire "trio" scheme. Both PHYs switch between a high-speed, low-swing, terminated mode and a low-power, large-swing, unterminated mode on the same wires. This chapter explains the two PHYs, their implications for routing on boards and flexible cables, and the common causes of failure.

**Learning Objectives**

1. Describe D-PHY high-speed and low-power modes and their voltage levels.
2. Explain C-PHY three-phase signaling and its throughput.
3. Derive lane-to-clock skew budgets for D-PHY.
4. Route trios and lanes through boards, connectors, and flex cables.
5. Identify failure modes particular to mode switching.

## 23.1 D-PHY

A D-PHY link has one clock lane and one to four data lanes, each a differential pair [50]. In *high-speed* (HS) mode, the transmitter drives a low-swing differential signal (about 200 mV differential on a common mode of about 200 mV) into a 100 Ω differential termination that the receiver switches on only during HS bursts. In *low-power* (LP) mode, each wire is driven single-ended between ground and 1.2 V, unterminated, at a few tens of megabits per second, to carry control sequences and escape-mode data. Each burst starts and ends with an LP sequence (Figure 23.1).

The data lanes are clocked by the forwarded HS clock at half the bit rate (DDR), so D-PHY is source synchronous: the receiver needs the clock-to-data skew within a fraction of a UI. Newer D-PHY versions add deskew calibration bursts that let the receiver compensate static skew at high rates.

**Specification Note**

D-PHY data rates have increased across versions (from 1.5 Gb/s per lane in early versions to several gigabits per second per lane later). HS and LP voltage ranges, termination tolerances, skew limits, and channel insertion-loss requirements are defined by the MIPI Alliance specifications [50], [51], which are available to members. Use the specification version supported by both the processor and the sensor or display.

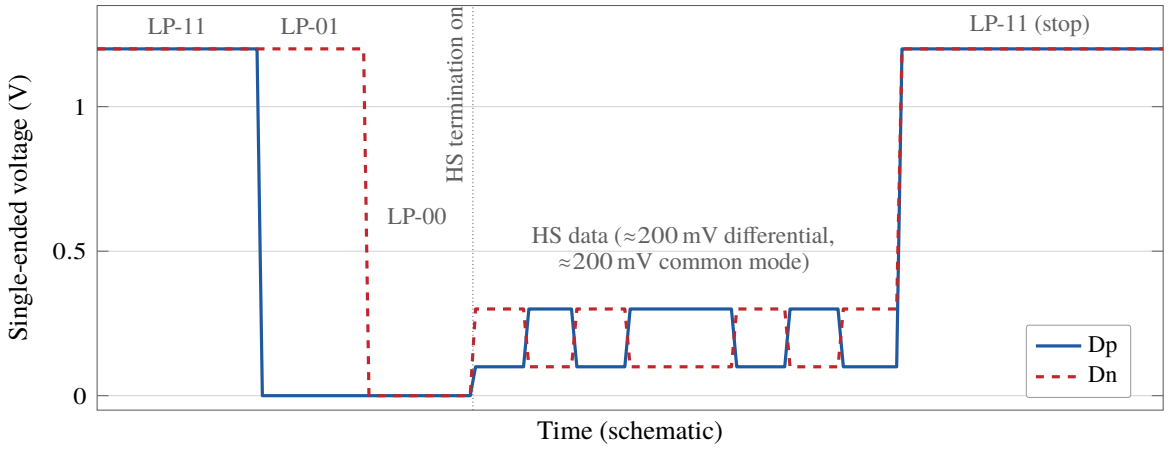



**Figure 23.1.** D-PHY lane voltages during a transition from the low-power stop state (LP-11) through LP-01 and LP-00 into high-speed data and back (schematic).

**Worked Example 23.1: D-PHY lane-to-clock skew**

A camera link runs at 2.5 Gb/s per lane. If 5 % of a UI is allocated to static lane-to-clock routing skew, what length mismatch is allowed at 165 ps/in?

**Solution.** UI = 400 ps; 5 % is 20 ps, or 121 mil. Within each pair, intra-pair skew should be held much tighter, because it distorts the HS differential crossing and the LP single-ended levels are received separately.

## 23.2 C-PHY

C-PHY replaces differential pairs with *trios* of three wires (Figure 23.2) [51]. In each symbol, one wire is driven high, one low, and one to the midpoint; three differential receivers measure A−B, B−C, and C−A. Every symbol differs from the previous one, so each symbol boundary carries a transition and the clock is embedded. The encoding maps 16 bits onto 7 symbols, or about 2.28 bits per symbol.

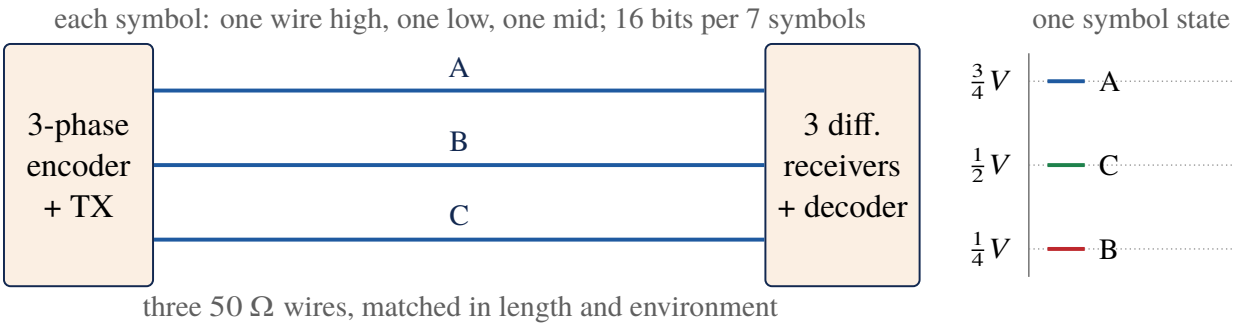



**Figure 23.2.** A C-PHY trio. Three wires each take one of three levels in every symbol; three differential receivers recover the symbol, and the guaranteed transitions carry the clock.

**Worked Example 23.2: C-PHY versus D-PHY throughput**

Compare a three-trio C-PHY link at 2.5 GS/s (symbols per second) with a four-lane D-PHY link at 2.5 Gb/s per lane.

**Solution.** C-PHY: $3 \times 2.5 \times 16/7 = 17.1$ Gb/s on nine wires. D-PHY: $4 \times 2.5 = 10$ Gb/s on ten wires (four data lanes plus the clock lane). C-PHY delivers about 70 % more throughput with fewer wires, at the cost of tighter matching among the three wires of each trio and a more complex PHY.

### 23.2.1 Routing trios

Because the receiver compares wires pairwise, the three wires of a trio must be matched in delay and in their electrical environment: each should have the same impedance to ground (nominally 50 Ω single-ended) and the same coupling to its two partners. Routing the three in a row makes the middle wire couple more strongly than the outer two. Whether that asymmetry is acceptable is a question for an extraction, not a rule of thumb: extract the $3 \times 3$ $\mathbf{L}$ and $\mathbf{C}$ matrices of the actual cross-section (Eq. (9.3)), compare the three wire self-terms and the three mutual terms, and simulate the trio with them. If the imbalance matters, symmetric arrangements (for example, rotating wire positions at intervals, or routing the trio in a triangle across two layers) can equalize it, but they add vias and should be justified by the same extraction rather than applied by default. Different trios are independent (each has its own embedded clock), so trio-to-trio skew is not critical.

## 23.3 Boards, connectors, and flex cables

Camera modules and displays usually connect through board-to-board connectors and flexible printed circuits (FPCs), whose stackups are thin and whose reference planes are often meshed (cross-hatched) for flexibility. A meshed plane raises impedance and loss and lets fields leak; its effect must be simulated, and the mesh pitch kept small compared with the trace width. Connector pin assignments should put ground pins between lanes and keep each pair (or trio) on adjacent pins.

**Design Rule: MIPI routing**

Route D-PHY lanes as 100 Ω differential pairs and C-PHY wires as 50 Ω single-ended lines with equal coupling within each trio. Match data lanes to the clock lane (D-PHY) or wires within each trio (C-PHY) in delay, including the flex and connector. Keep LP-mode aggressors in mind: the 1.2 V single-ended LP swing is six times the HS differential swing, so spacing between lanes must account for LP-to-HS crosstalk. No AC-coupling capacitors are used; both modes are DC coupled.

**Common Pitfall: Treating the flex as "just a cable"**

Designing the main board carefully and leaving the camera or display flex to the module vendor. The flex and its connectors are often the longest and least controlled part of the channel, with meshed references, impedance steps at each connector, and no stitching. Specify impedance, stackup, and ground-pin assignment for the flex, and include it in the channel simulation.

**Table 23.1.** D-PHY and C-PHY compared.

| | D-PHY | C-PHY |
|---|---|---|
| Lane unit | Differential pair | Trio (three wires) |
| Clocking | Forwarded clock lane (source synchronous) | Embedded in symbol transitions |
| Bits per transfer | 1 bit per UI per lane | 16 bits per 7 symbols (2.28 per symbol) |
| Termination | 100 Ω differential, HS only | Per-wire termination, HS only |
| Critical matching | Lane to clock; intra-pair | Within each trio |
| LP mode | 1.2 V single-ended | 1.2 V single-ended |

**Rule of Thumb: LP-mode crosstalk**

The 1.2 V LP swing is about six times the 200 mV HS differential swing, so a coupling level that is harmless between HS lanes may disturb an HS lane when its neighbor is in LP mode. *Limits:* LP and HS phases on adjacent lanes overlap only during transitions in many systems; check whether the interface allows independent lane modes.

**Key Takeaways**

- MIPI CSI-2 and DSI run over D-PHY (pairs plus a clock lane) or C-PHY (embedded-clock trios).
- Both PHYs switch between terminated low-swing HS and unterminated 1.2 V LP modes on the same wires.
- D-PHY needs lane-to-clock matching; C-PHY needs matching within each trio.
- C-PHY carries 2.28 bits per symbol, giving higher throughput per wire.
- Flex cables and connectors are part of the channel and must be specified.

## Further Reading

The MIPI Alliance specifications [50]–[53] define D-PHY, C-PHY, CSI-2, and DSI; they are available to MIPI members.

## Review Problems

**P23.1** Find the UI and Nyquist frequency of a 1.5 Gb/s D-PHY lane.

**P23.2** A C-PHY trio runs at 3.5 GS/s. What is its bit rate?

**P23.3** What is the aggregate throughput and wire count of a four-lane D-PHY link at 2.5 Gb/s per lane?

**P23.4** What symbol rate must three C-PHY trios run to carry 10 Gb/s?

**P23.5** Allocate 5 % of the UI at 4.5 Gb/s to lane-to-clock skew. What length mismatch is allowed at 165 ps/in?

**P23.6** An LP-mode lane (1.2 V) couples 1.5 % into a neighbor. How much noise does the neighbor see?

Chapter 24

# Clock Distribution

**Chapter Overview**

Every synchronous system depends on its clocks, and every serial link's jitter budget begins with its reference clock. Clocks are also the strongest sources of narrowband emissions on most boards. This chapter covers the components of a clock tree—crystals and oscillators, PLLs and jitter cleaners, and fanout buffers—and the board-level practices for routing and terminating clocks. It shows how to design a PLL loop filter for stability and how to convert an oscillator's phase-noise plot into RMS jitter.

**Learning Objectives**

1. Select between XO, TCXO, VCXO, and MEMS oscillators, and size crystal load capacitors.
2. Choose clock output standards (LVCMOS, LVDS, LVPECL, HCSL) and terminate them correctly.
3. Explain the reference- and VCO-noise transfer functions of a PLL and design a second-order loop filter.
4. Integrate phase noise to RMS jitter and combine jitter contributions.
5. Route clocks to minimize jitter pickup and emissions.

## 24.1 Clock sources

**Crystal with on-chip oscillator**
The cheapest source: a quartz crystal and two load capacitors connected to an inverter in the IC. Frequency accuracy depends on the load capacitance matching the crystal's specified $C_L$.

**Crystal oscillator (XO)**
A packaged crystal and oscillator; typical stabilities of $\pm 20$–$\pm 50$ ppm and low phase noise.

**TCXO, OCXO**
Temperature-compensated and oven-controlled oscillators for sub-ppm stability.

**VCXO**
A voltage-controlled crystal oscillator, used inside jitter-cleaner PLLs.

**MEMS oscillators**
Silicon resonators with a PLL; robust and programmable, with phase noise that varies widely among products.

### 24.1.1 Crystal load capacitance

A crystal oscillates at its specified frequency only when loaded with its specified capacitance $C_L$. For the common Pierce circuit with capacitors $C_1$ and $C_2$ from each crystal terminal to ground,

$$C_L = \frac{C_1 C_2}{C_1 + C_2} + C_{\text{stray}}, \tag{24.1}$$

where $C_{\text{stray}}$ includes the IC pins and board traces, typically 2–5 pF.

**Worked Example 24.1: Crystal load capacitors**

A crystal is specified for $C_L = 12$ pF; stray capacitance is estimated at 3 pF. Choose equal load capacitors.

**Solution.** $C_1 C_2/(C_1 + C_2) = C/2 = 12 - 3 = 9$ pF, so $C_1 = C_2 = 18$ pF. Too much load capacitance pulls the frequency low; too little pulls it high, by a few ppm per picofarad.

**Design Rule: Crystal layout**

Place the crystal and its load capacitors next to the IC's oscillator pins, with short traces, a local ground under them connected to the IC's oscillator ground, and no other signals routed under or beside the crystal. Keep the loop of crystal, capacitors, and ground small: it is a sensitive high-impedance circuit.

## 24.2 Clock tree and output standards

A typical clock tree (Figure 24.1) starts from an oscillator, optionally passes through a PLL that multiplies the frequency and cleans the jitter, and fans out through buffers to the loads. Each load receives its own point-to-point output; daisy-chaining clocks to several loads is avoided because of reflections and loading.

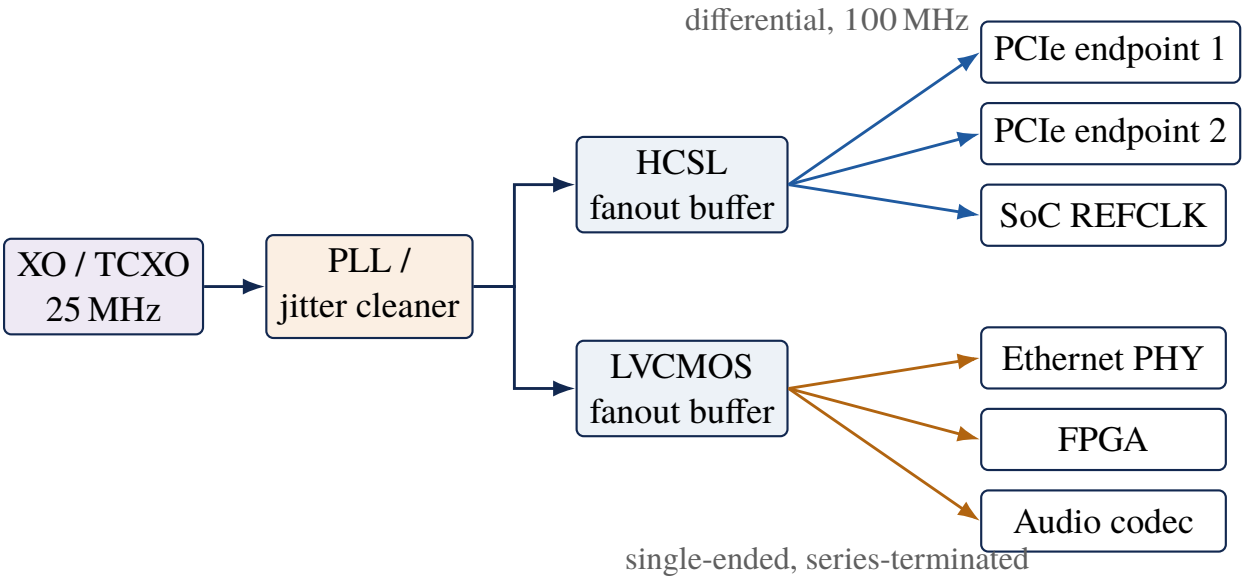



**Figure 24.1.** A clock tree: oscillator, PLL or jitter cleaner, and fanout buffers with point-to-point outputs to each load.

**Table 24.1.** Common clock output standards and their terminations.

| Standard | Type | Swing (typ.) | Termination |
|---|---|---|---|
| LVCMOS | Single-ended | Rail-to-rail | Series resistor at source; one load per output |
| LVDS | Differential | 350 mV | 100 Ω across pair at receiver |
| LVPECL | Differential | 800 mV | 50 Ω to $V_{CC}$ − 2 V (or Thevenin/Y equivalent) |
| HCSL | Differential, current mode | 700 mV | 50 Ω to ground on each leg (source or load end) |
| CML | Differential | 400–800 mV | 50 Ω to $V_{CC}$, internal |

AC coupling between different standards is common for clocks, which are DC balanced; the receiver then needs its own bias. Each standard's data sheet defines its termination precisely; mixing standards without checking common-mode ranges is a frequent source of marginal clocks.

## 24.3 Jitter of clocks

Clock jitter is specified in several ways: *period jitter* (variation of individual periods), *cycle-to-cycle jitter* (difference between adjacent periods), and *phase jitter* or TIE (deviation of edges from an ideal clock), which is what matters for serial links. Oscillators specify their noise as single-sideband phase noise $\mathscr{L}(f)$, in dBc per hertz of offset from the carrier (Figure 24.2). The RMS phase jitter over an offset band $[f_1, f_2]$ is

$$\sigma_t = \frac{1}{2\pi f_0}\sqrt{2\int_{f_1}^{f_2} 10^{\mathscr{L}(f)/10}\,\mathrm{d}f} \quad \text{(seconds, RMS)}, \tag{24.2}$$

where $f_0$ is the carrier frequency and the factor of 2 accounts for both sidebands. The integration band must match the application: 12 kHz to 20 MHz is a conventional band for SONET and Ethernet reference clocks, whereas PCIe defines its own filtered measurement.

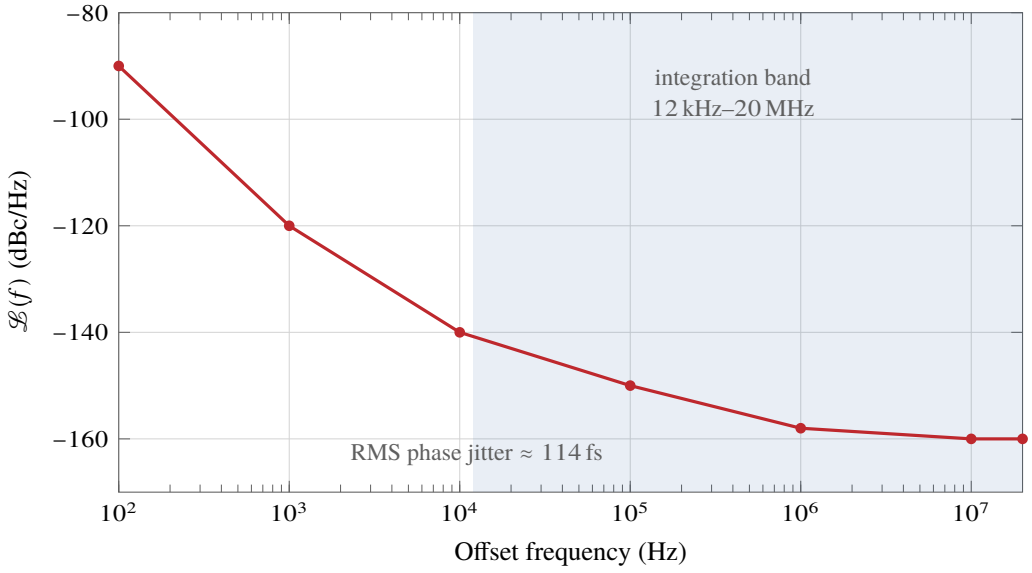



**Figure 24.2.** Representative phase noise of a 100 MHz oscillator. Integrating Eq. (24.2) over the shaded band gives about 114 fs RMS.

**Worked Example 24.2: Phase noise to jitter**

A 100 MHz clock has a flat phase-noise floor of −140 dBc/Hz from 12 kHz to 20 MHz. Find its RMS jitter in that band.

**Solution.** $10^{-14} \times (20 \times 10^{6} - 12 \times 10^{3}) = 1.999 \times 10^{-7}$; doubling and taking the square root gives $6.32 \times 10^{-4}$ rad. Dividing by $2\pi \times 10^{8}$ gives $\sigma_t = 1.0$ ps. A floor 20 dB lower would give 100 fs.

Independent jitter sources add in root-sum-square fashion. A 300 fs reference through a fanout buffer with 100 fs of additive jitter gives $\sqrt{300^2 + 100^2} = 316$ fs.

## 24.4 PLLs and loop filters

A charge-pump PLL (Figure 24.3) compares the divided output with the reference in a phase-frequency detector, converts the phase error into current pulses, filters them into a control voltage, and steers a VCO. Its loop bandwidth determines how noise is shaped:

- reference and phase-detector noise are *low-pass* filtered by the loop and multiplied by $N$ (raised by $20\log_{10} N$ dB) within the bandwidth;
- VCO noise is *high-pass* filtered: the loop suppresses it within the bandwidth and passes it outside.

In transfer-function form, with phase-detector gain $K_{\mathrm{pd}}$ (amperes per radian), loop-filter impedance $Z(s)$, VCO gain $K_{\mathrm{vco}}$ (radians per second per volt) and divider $N$, the open-loop gain is

$$G(s) = \frac{K_{\mathrm{pd}}\, Z(s)\, K_{\mathrm{vco}}}{s\, N}, \tag{24.3}$$

where the $1/s$ is the VCO itself: a VCO integrates frequency into phase, so it adds an integrator to the loop on top of the one in $Z(s)$. Phase noise then reaches the output through

$$\frac{\phi_{\mathrm{out}}}{\phi_{\mathrm{ref}}} = N\,\frac{G}{1+G}, \qquad \frac{\phi_{\mathrm{out}}}{\phi_{\mathrm{vco}}} = \frac{1}{1+G}, \tag{24.4}$$

low-pass and high-pass respectively, with the crossover near the loop bandwidth. The same low-pass response sets the *reference spurs*: charge-pump mismatch and leakage produce ripple at the phase-detector frequency $f_{\mathrm{pd}}$, which modulates the VCO and appears as sidebands at $\pm f_{\mathrm{pd}}$; the loop filter's attenuation at $f_{\mathrm{pd}}$ (the pole $\omega_p$ and any extra filter pole) sets their level.

A jitter cleaner uses a narrow loop bandwidth (tens to hundreds of hertz) with a low-noise VCXO, so that the output follows the reference's frequency but inherits the VCXO's low phase noise. A frequency synthesizer uses a wider bandwidth to suppress the noise of an on-chip VCO.

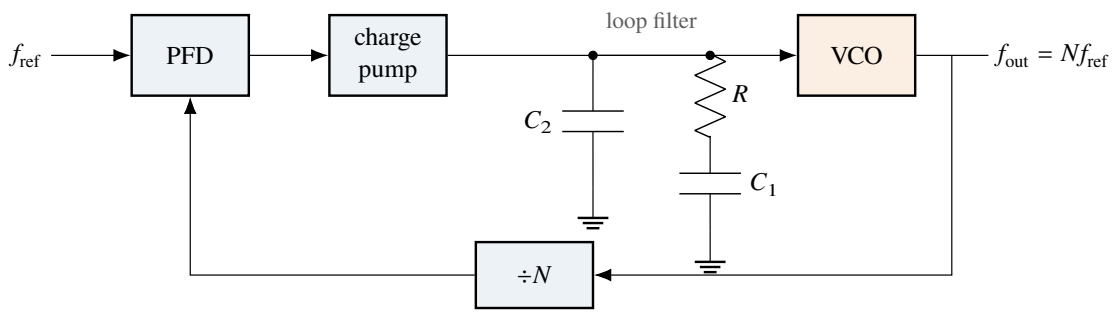



**Figure 24.3.** Charge-pump PLL with a second-order passive loop filter ($R$, $C_1$, $C_2$).

The second-order passive filter of Figure 24.3 has impedance

$$Z(s) = \frac{1 + sRC_1}{s(C_1 + C_2)\left(1 + sR\frac{C_1 C_2}{C_1 + C_2}\right)}, \tag{24.5}$$

with a zero at $\omega_z = 1/(RC_1)$ and a pole at $\omega_p = (C_1 + C_2)/(RC_1C_2)$. Stability requires the open-loop crossover $\omega_c$ to lie between them; the phase margin is maximized at $\omega_c = \sqrt{\omega_z \omega_p}$, where

$$\mathrm{PM}_{\max} = \arctan\sqrt{1 + \frac{C_1}{C_2}} - \arctan\frac{1}{\sqrt{1 + C_1/C_2}}. \tag{24.6}$$

**Worked Example 24.3: Loop-filter phase margin**

A loop filter uses $C_1 = 10C_2$. What is the maximum phase margin?

**Solution.** $\sqrt{11} = 3.317$; $\arctan 3.317 = 73.2°$ and $\arctan(1/3.317) = 16.8°$, so $\mathrm{PM}_{\max} = 56.4°$. Increasing the ratio to $C_1 = 16C_2$ raises it to 62.7°, at the cost of less attenuation of reference spurs above the crossover.

**Rule of Thumb: PLL loop bandwidth**

Set the loop bandwidth near the offset frequency at which the reference's phase noise, multiplied by $N$, crosses the VCO's phase noise; this minimizes integrated jitter. Keep the bandwidth below about one-tenth of the phase-detector frequency for the continuous-time analysis to hold. *Limits:* spur requirements, SSC tracking, and lock time may override the jitter optimum.

## 24.5 Clock routing

**Design Rule: Clock routing**

Route clocks point to point, on inner layers referenced to ground, with at least $3H$ to $5H$ spacing from other signals (and more from other clocks). Terminate every clock so that its edges are monotonic. Keep clock routes short and away from connectors and board edges. Give each clock buffer and PLL its own filtered supply (a ferrite or LDO plus local decoupling); supply noise in a PLL converts directly into jitter.

**Common Pitfall: Supply noise in clock buffers**

Sharing a noisy digital supply with a clock buffer or PLL. A few millivolts of supply ripple at a frequency inside the PLL bandwidth, or at the buffer's output, appears as periodic jitter on every clock in the tree—and often as spurs in the emission spectrum. Measure the clock's phase noise at bring-up with and without the noisy load active.

**Key Takeaways**

- Size crystal load capacitors from $C_L = C_1C_2/(C_1 + C_2) + C_{\text{stray}}$.
- Each clock output standard has a specific termination; use one load per output.
- RMS phase jitter is $\frac{1}{2\pi f_0}\sqrt{2\int \mathscr{L}\, \mathrm{d}f}$ over the application's band; independent contributions add in RSS.
- PLLs low-pass the reference and high-pass the VCO noise; the loop bandwidth trades between them.
- The maximum phase margin of a second-order filter depends only on $C_1/C_2$.

## Further Reading

Li [25] relates phase noise to jitter and discusses PLL jitter transfer; Dally and Poulton [9] cover clock distribution and PLL design for digital systems.

## Review Problems

**P24.1** A crystal circuit uses $C_1 = C_2 = 22\,\text{pF}$ with $4\,\text{pF}$ stray. What load capacitance does the crystal see?

**P24.2** A $156.25\,\text{MHz}$ clock has a flat $-150\,\text{dBc/Hz}$ floor from $12\,\text{kHz}$ to $20\,\text{MHz}$. Find its RMS jitter.

**P24.3** Combine $250\,\text{fs}$, $150\,\text{fs}$, and $100\,\text{fs}$ of independent jitter.

**P24.4** Find the maximum phase margin of a second-order loop filter with $C_1 = 16C_2$.

**P24.5** Find the zero frequency of a loop filter with $R = 1\,\text{k}\Omega$ and $C_1 = 10\,\text{nF}$.

**P24.6** A $25\,\text{MHz}$ oscillator is specified to $\pm 50\,\text{ppm}$. What is its frequency range?

Part VI

# Simulation, Measurement, and Bring-Up

Chapter 25

# Modeling and Simulation

**Chapter Overview**

Closed-form estimates choose the architecture; simulation confirms that it works. This chapter describes the models used in signal- and power-integrity simulation—IBIS and IBIS-AMI buffer models, SPICE models, field-solver extractions, and S-parameters—and the checks (passivity, causality, reciprocity) that keep those models physical. It defines mixed-mode S-parameters and shows how to read them, and lays out a pre-layout and post-layout simulation flow that catches problems while they are still cheap to fix.

**Learning Objectives**

1. Choose between IBIS, IBIS-AMI, and transistor-level models.
2. Select 2-D, 2.5-D, or 3-D field solvers for each part of a channel.
3. Define single-ended and mixed-mode S-parameters and convert between them.
4. Check S-parameter models for passivity, causality, and reciprocity.
5. Run a pre- and post-layout SI/PI flow with appropriate corners.

## 25.1 Buffer models

### 25.1.1 IBIS

The I/O Buffer Information Specification (IBIS) describes a buffer behaviorally, without revealing its circuit [54]. An IBIS model contains:

- current–voltage (V–I) tables for the pull-up and pull-down devices and the power and ground clamps;
- voltage–time (V–t) waveforms of the output switching into specified loads, which the simulator uses to reconstruct the transition;
- the die capacitance (C_comp) and package parasitics (lumped RLC or a package model);
- typical, minimum, and maximum corners for process, voltage, and temperature.

IBIS simulates quickly and is accurate for the reflections, crosstalk, and timing of single-ended and differential buffers. Its traditional form does not capture supply-noise dependence well; newer versions add power-aware features. The fast corner (strongest drive, fastest edges) is usually worst for ringing and crosstalk, the slow corner for timing.

### 25.1.2 IBIS-AMI

Multi-gigabit SERDES contain adaptive equalizers and CDRs that IBIS's analog tables cannot represent. IBIS-AMI (Algorithmic Modeling Interface) attaches executable models of the transmitter's FFE and the receiver's CTLE, DFE, and CDR to the analog IBIS front end. Two simulation modes are used:

**Statistical**
: The channel is characterized by its impulse response, assumed linear and time-invariant, and the eye and bathtub are computed by convolution of the impulse response with the data-pattern statistics. It is fast and gives BER contours down to very low probabilities.

**Time-domain (bit-by-bit)**
: Millions of bits of a pattern are passed through the channel and the equalizer models, including nonlinear and adaptive behavior. It is slower and limited in BER depth but captures adaptation.

A workable AMI flow is: (1) confirm that the TX and RX models match the silicon revision and the operating mode (rate, preset, equalization ranges); (2) build the channel from extracted and checked S-parameters (Section 25.3.4); (3) run a statistical analysis with adaptation enabled to find the optimal TX preset and the equalizer settings the RX converges to; (4) confirm in bit-by-bit mode, with a few million bits, that adaptation converges to similar settings and that the eye agrees with the statistical result at the BER it can reach; (5) sweep corners (silicon, channel tolerance, crosstalk aggressors) and report margins against the specification's reference receiver as well as the vendor's. When the two modes disagree, the difference is a finding, not noise to be averaged (Section 12.4.5).

### 25.1.3 Transistor-level models

Encrypted SPICE models from the silicon vendor are the most accurate but slowest option, and are typically used only to validate IBIS models or to analyze unusual conditions such as power-up or supply collapse.

## 25.2 Interconnect models and field solvers

**Table 25.1.** Field-solver classes and their uses.

| Solver | Geometry | Typical use |
|---|---|---|
| 2-D (cross-section) | Uniform lines | $Z_0$, RLGC matrices, coupling, loss of traces |
| 2.5-D (planar MoM) | Layered planar structures | Planes, PDN impedance, large board regions |
| 3-D full-wave (FEM, FDTD, FIT) | Arbitrary | Vias, connectors, BGA breakouts, packages |
| Quasi-static 3-D | Small structures | Lumped $L$, $C$ of pins and pads |

A complete channel model is usually assembled from pieces: 2-D models of the uniform trace segments, 3-D models of each via and connector transition, and vendor S-parameter models of packages and connectors, cascaded together. The segmentation boundaries must be placed where the fields are again quasi-TEM—a short distance away from each discontinuity—or the cascaded model will be inaccurate.

**Common Pitfall: Using a laminate's nominal $D_k$ and $D_f$**

Building the 2-D models with the laminate family's data-sheet values. Each prepreg and core has its own resin content and therefore its own $D_k$ and $D_f$, which also vary with frequency. Ask the fabricator for the values of the specific glass styles in the stackup, use a causal model (Section 8.3.2), and include roughness.

## 25.3 S-parameters

### 25.3.1 Definitions

An $N$-port network is described at each frequency by the relation between incident waves $a_j$ and reflected waves $b_i$ at its ports, normalized to a reference impedance $Z_{\text{ref}}$ (usually 50 Ω):

$$b_i = \sum_{j=1}^{N} S_{ij}\, a_j, \qquad a_j = \frac{V_j + Z_{\text{ref}} I_j}{2\sqrt{Z_{\text{ref}}}}, \quad b_j = \frac{V_j - Z_{\text{ref}} I_j}{2\sqrt{Z_{\text{ref}}}}. \tag{25.1}$$

$S_{ii}$ is the reflection coefficient at port $i$ with all other ports matched; $S_{ij}$ $(i \neq j)$ is the transmission from port $j$ to port $i$. Insertion loss is $-20\log_{10}|S_{21}|$ and return loss $-20\log_{10}|S_{11}|$. S-parameters are exchanged in Touchstone files (.s2p, .s4p, …).

**Worked Example 25.1: Return loss of an impedance error**

A 55 Ω trace is measured with 50 Ω ports. Find the return loss of the impedance step, and the worst-case $|S_{11}|$ of a long section of it between the two ports.

**Solution.** $\Gamma = 5/105 = 0.0476$, a return loss of 26.4 dB. Reflections from the two ends add in phase when the section is an odd number of quarter-wavelengths long, giving $|S_{11}|_{\max} = 2\Gamma/(1+\Gamma^2) = 0.095$, or −20.4 dB. The return-loss curve therefore ripples between this value and zero, with period $1/(2T_{\text{D}})$.

### 25.3.2 Mixed-mode S-parameters

For differential channels the four single-ended ports are combined into two mixed-mode ports (Figure 25.1). With ports 1 and 2 forming the input pair (P and N) and ports 3 and 4 the output pair, the most used terms are [3], [4]

$$S_{\text{DD21}} = \tfrac{1}{2}\left(S_{31} - S_{32} - S_{41} + S_{42}\right), \qquad S_{\text{DD11}} = \tfrac{1}{2}\left(S_{11} - S_{12} - S_{21} + S_{22}\right), \tag{25.2}$$

$$S_{\text{CD21}} = \tfrac{1}{2}\left(S_{31} - S_{32} + S_{41} - S_{42}\right), \qquad S_{\text{CC21}} = \tfrac{1}{2}\left(S_{31} + S_{32} + S_{41} + S_{42}\right). \tag{25.3}$$

$S_{\text{DD21}}$ is differential insertion loss; $S_{\text{DD11}}$ differential return loss; $S_{\text{CD21}}$ the conversion of an input differential signal to an output common-mode signal, which grows with asymmetry such as skew (Chapter 10). Figure 25.2 shows all three for a representative channel. Port numbering conventions differ between tools (some pair ports 1–3 and 2–4); always check the convention before applying these formulas.

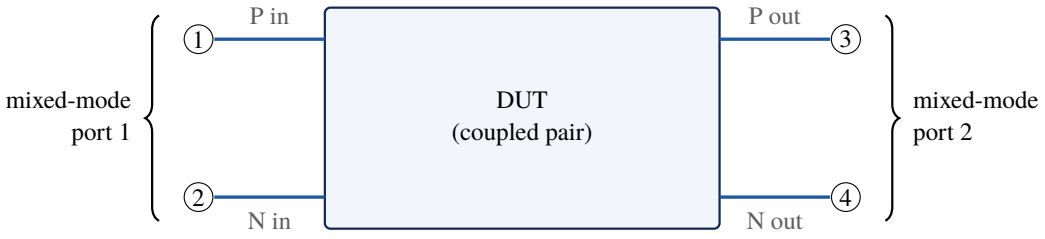



**Figure 25.1.** Port numbering for a four-port differential channel used in this book, and the corresponding mixed-mode ports.

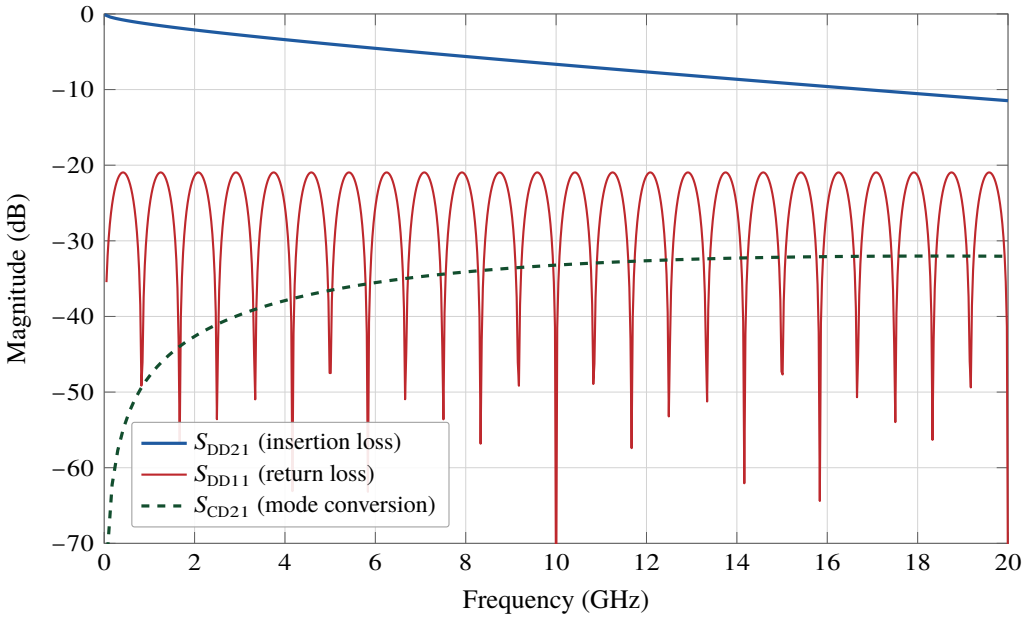



**Figure 25.2.** Mixed-mode S-parameters of a representative 10 in differential channel: insertion loss, return loss from a 93 Ω section ($T_D = 0.6$ ns) between 85 Ω ports, and mode conversion from 3 ps of intra-pair skew.

**Worked Example 25.2: Mode conversion from skew**

A pair has $S_{31} = 0.9$ and $S_{42} = 0.9\angle -10°$ at one frequency, with negligible cross terms. Find $|S_{DD21}|$ and $|S_{CD21}|$.

**Solution.** $S_{DD21} = \frac{1}{2}(0.9 + 0.9\angle -10°)$, magnitude $0.9\cos 5° = 0.897$ (−0.95 dB). $S_{CD21} = \frac{1}{2}(0.9 - 0.9\angle -10°)$, magnitude $0.9\sin 5° = 0.078$ (−22.1 dB). A 10° phase difference—about 2.8 ps at 10 GHz—already produces this conversion. Interpreted correctly (Section 10.4.2), 0.078 is a ratio of power-wave amplitudes: the converted *power* is $0.078^2 = 0.6\,\%$ of the incident differential power, and the common-mode *voltage* defined by Eq. (10.1) is half the ratio, 0.039 of the differential voltage (−28.1 dB). "Eight percent of the signal" would be ambiguous.

### 25.3.3 Physical consistency

Measured or extracted S-parameters must satisfy three physical properties before they are used in time-domain simulation:

**Passivity**

A passive network cannot create energy: all eigenvalues of $S^{H}S$ must be at most one at every frequency. A quick check is that every column satisfies $\sum_i |S_{ij}|^2 \le 1$; this is *necessary but not sufficient*—a matrix can pass every column test and still have an eigenvalue above one when the columns interfere, so the eigenvalue test is the one that decides. Small passivity violations from measurement noise cause time-domain simulations to grow without bound.

**Causality**

The response must not precede the stimulus. Non-causal models (from truncated bandwidth, constant-$D_k$ material models, or interpolation errors) show precursors in the impulse response.

**Reciprocity**

For networks without magnetic materials or active devices, $S_{ij} = S_{ji}$.

S-parameter models must also extend to DC (or have a valid DC extrapolation) and cover a bandwidth of several times the Nyquist frequency, with frequency steps fine enough to resolve the longest delay in the model ($\Delta f \ll 1/T_{\mathrm{D}}$).

### 25.3.4 A Touchstone workflow

S-parameter data are exchanged as Touchstone files: .s2p, .s4p and so on for version 1, or the keyword-structured version 2 format. A reproducible workflow for each file is:

1. *Read the header.* Note the frequency unit, the data format (RI, MA or DB), the reference impedance of every port, and—for version 1 four-ports—the port order, which the file does not state (Figure 25.1).
2. *Inspect the frequency grid.* The lowest point should be close to DC, the highest at least $3f_{\mathrm{Nyq}}$, and the step fine enough for the longest delay.
3. *Check physics.* Run the passivity (eigenvalue), reciprocity and causality checks above, and record the worst violation and its frequency.
4. *Convert and plot.* Compute the mixed-mode terms with the port order confirmed, and plot $S_{\mathrm{DD21}}$, $S_{\mathrm{DD11}}$, $S_{\mathrm{CD21}}$ and the TDR (step response) of each through path.
5. *Sanity-check against expectations.* Compare delay with $\ell\, t_{\mathrm{pd}}$, loss at $f_{\mathrm{Nyq}}$ with the budget of Chapter 8, and TDR impedance with the stackup.
6. *Archive.* Keep the original file, any corrected version, and a note of every correction (passivity enforcement, DC extrapolation) applied.

**Design Rule: S-parameter model hygiene**

Before using any S-parameter file in a channel simulation, check passivity, causality, and reciprocity; confirm the port order and reference impedance; verify that it starts at or near DC and extends to at least three times the Nyquist frequency; and plot its TDR response to confirm that impedances and delays are plausible.

## 25.4 A pre- and post-layout flow

Figure 25.3 shows a practical SI/PI flow. *Pre-layout*, the designer explores topologies, stackups, terminations, and materials with idealized routing, sweeping lengths and corners to find where margins run out; the results become layout constraints. *Post-layout*, the routed board is extracted and every critical net class, the PDN, and the worst crosstalk victims are re-simulated with the real geometry. Corner analysis (fast/slow silicon, voltage, temperature, and fabrication tolerances such as $\pm 10\,\%$ impedance) or Monte Carlo analysis establishes the margin. Finally, simulations are correlated with measurements on the first boards (Chapter 26), which calibrates the models for the next design.

**Rule of Thumb: Simulation bandwidth and resolution**

Extract and simulate channel models to at least three times the Nyquist frequency, with a frequency step no larger than about one-tenth of $1/T_{\mathrm{D}}$ for the longest delay in the model. *Limits:* PAM4 and very low-BER analyses may need more bandwidth; very long channels need finer steps to avoid time-domain aliasing.

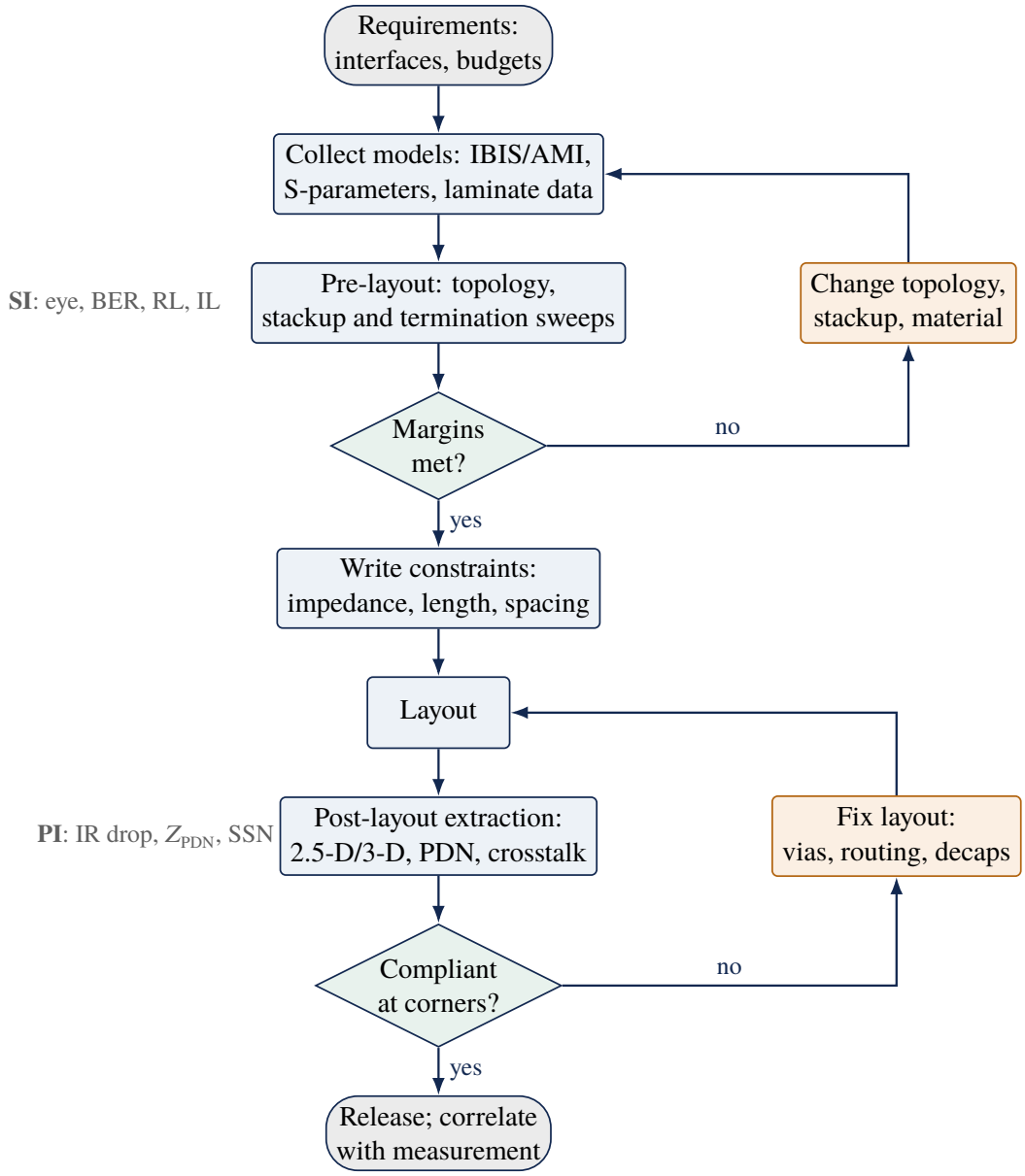



**Figure 25.3.** Pre- and post-layout signal- and power-integrity simulation flow.

**Key Takeaways**

- IBIS models buffers behaviorally from V–I and V–t tables; IBIS-AMI adds executable equalization and CDR models for SERDES.
- Use 2-D solvers for traces, 2.5-D for planes, and 3-D full-wave for vias and connectors; cascade at quasi-TEM boundaries.
- S-parameters describe incident and reflected waves; mixed-mode terms $S_{DD21}$, $S_{DD11}$, $S_{CD21}$ characterize differential channels.
- Check every model for passivity, causality, reciprocity, port order, and bandwidth.
- Simulate pre-layout to set constraints and post-layout to verify them, across corners.

## Further Reading

The IBIS specification [54] defines buffer models; Hall and Heck [4] cover S-parameters, passivity, and causality; IEEE 370 [44] addresses model and fixture quality.

## Review Problems

**P25.1** Find the return loss of a 60 Ω discontinuity measured with 50 Ω ports.

**P25.2** What is the worst-case $|S_{11}|$ of a long 60 Ω line between 50 Ω ports?

**P25.3** A reciprocal two-port has $|S_{11}| = 0.2$ and $|S_{21}| = 0.95$ at one frequency. Does it pass the column-sum passivity check?

**P25.4** With 50 Ω single-ended ports, what are the differential- and common-mode reference impedances?

**P25.5** Find $S_{\mathrm{DD21}}$ if $S_{31} = S_{42} = 0.8$ and $S_{32} = S_{41} = 0.1$ (all real).

**P25.6** How many complex values per frequency point does a .s4p file contain?

Chapter 26

# Measurement: TDR, VNA, Oscilloscopes, and Near-Field Probing

**Chapter Overview**

Simulation predicts; measurement decides. This chapter covers the instruments used to characterize interconnects and to verify signals on working boards: the time-domain reflectometer, the vector network analyzer with its calibration and de-embedding, the oscilloscope and its probes, and the near-field probes used to locate emission sources. For each, it explains what is actually measured, how the measurement setup itself corrupts the result, and how to avoid the most common errors.

**Learning Objectives**

1. Interpret TDR waveforms and extract impedance, excess capacitance, and excess inductance.
2. Estimate TDR spatial resolution from system rise time.
3. Explain VNA calibration and fixture de-embedding.
4. Choose oscilloscope bandwidth and probes, and quantify probe loading and ground-lead effects.
5. Use near-field probes to locate emission sources.

## 26.1 Time-domain reflectometry

A TDR launches a fast step into the device under test through a 50 Ω system and records the sum of the incident and reflected waves. At each instant, the reflection coefficient $\rho(t) = V_r(t)/V_i$ maps to the impedance seen at the corresponding round-trip position:

$$Z(t) = Z_{\text{ref}} \frac{1 + \rho(t)}{1 - \rho(t)}. \tag{26.1}$$

Time on the TDR display is round-trip time, so a feature at time $t$ lies at distance $v_{\text{p}} t/2$ from the reference plane. Capacitive discontinuities appear as dips, inductive ones as bumps, and impedance steps as plateaus (Figure 26.1).

### 26.1.1 Excess capacitance and inductance

For a small discontinuity between uniform lines of impedance $Z_0$, integrating the reflected voltage over the dip or bump gives the excess reactance (Section 6.3):

$$C_{\text{exc}} = -\frac{2}{Z_0 V_i} \int v_r \, \mathrm{d}t, \qquad L_{\text{exc}} = \frac{2 Z_0}{V_i} \int v_r \, \mathrm{d}t. \tag{26.2}$$

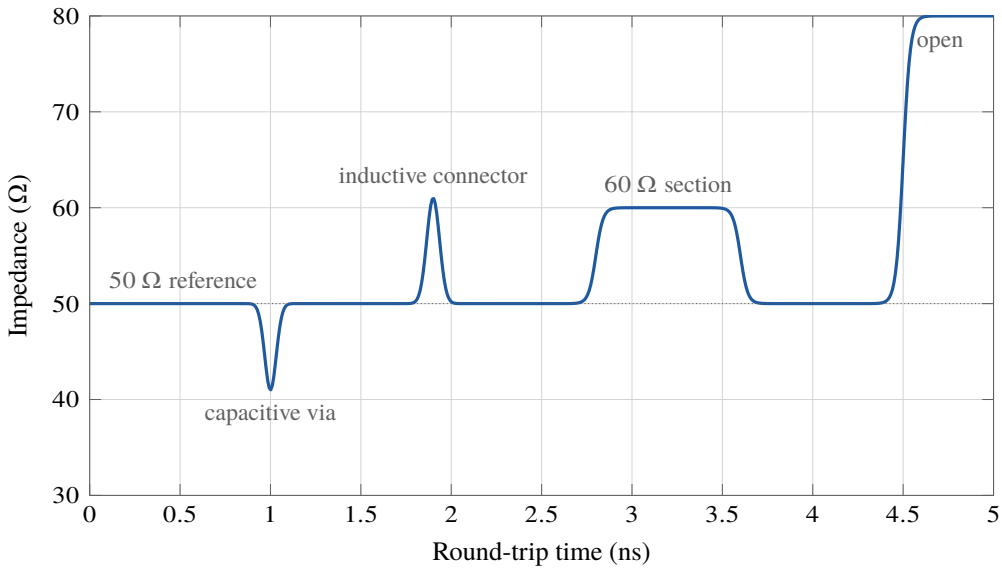



**Figure 26.1.** A TDR impedance profile: a capacitive via (dip), an inductive connector (bump), a 60 Ω trace section, and an open end.

Because the integral is insensitive to the instrument's rise time, this extraction is accurate even when the discontinuity is too small to resolve as a feature.

**Worked Example 26.1: Reading a TDR**

(a) A plateau shows $\rho = -0.10$. What is the impedance? (b) A via produces a triangular dip with peak $\rho = -0.15$ and base width 40 ps. Estimate its excess capacitance on a 50 Ω line.

**Solution.** (a) $Z = 50 \times 0.9/1.1 = 40.9\,\Omega$. (b) The dip's area is $\tfrac{1}{2} \times (-0.15) \times 40\,\text{ps} = -3\,\text{ps}$, so $C_{\text{exc}} = 2 \times 3\,\text{ps}/50\,\Omega = 0.12\,\text{pF}$.

### 26.1.2 Resolution and masking

A TDR displays round-trip time. Two discontinuities a distance $\ell$ apart produce reflections separated by $t = 2\ell/v_{\text{p}}$; they appear as separate features when that separation is at least about the system rise time $t_{\text{r}}$. The spatial resolution is therefore

$$\Delta x \approx \frac{t_{\text{r}}\, v_{\text{p}}}{2}, \tag{26.3}$$

where the factor $\tfrac{1}{2}$ comes from the round trip, not from "half the rise time." A 35 ps system edge and $v_{\text{p}} = 6\,\text{in/ns}$ give $\Delta x \approx 0.105\,\text{in}$. Resolution is not accuracy: a feature shorter than $\Delta x$ is not invisible, but its apparent impedance is a blend of the feature and its surroundings (Figure 26.2). The usual conversion $Z = Z_{\text{ref}}(1+\rho)/(1-\rho)$ then gives an *apparent* impedance, not the local value; reconstructing local geometry requires a model at the same bandwidth.

**Common Pitfall: A slow TDR that makes everything look good**

Accepting a channel because its TDR is flat at 50 Ω when measured with a slow edge. Short discontinuities (pads, via barrels, connector pins) are averaged away by a slow edge but are fully visible to a fast serial link. Specify the TDR edge and the reference plane with every impedance acceptance window, and compare with a model filtered to the same bandwidth.

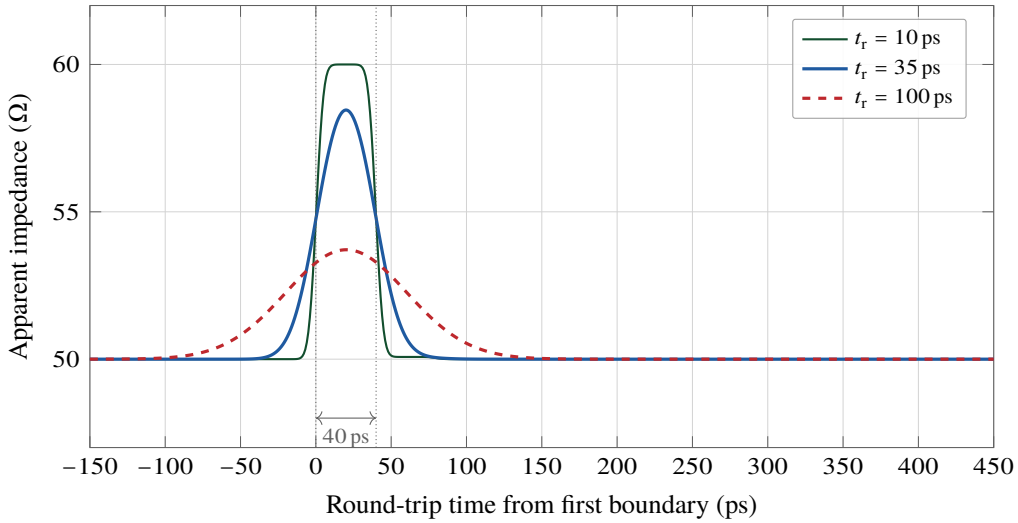



**Figure 26.2.** Simulated TDR of a 60 Ω segment with 20 ps one-way delay between matched 50 Ω lines, for Gaussian-equivalent system edges of 10, 35 and 100 ps. Multiple reflections in the segment are included; no loss or fixture. With the 100 ps edge, the 60 Ω plateau becomes a low bump. [Simulation (analytical lattice)]

Multiple reflections and the loss of the line mask features further down: a large discontinuity near the launch reduces the step that reaches later ones, and a lossy line slows the step so that distant features look smaller and smoother. Time-domain transmission (TDT), measured at the far end, gives delay, skew, and crosstalk; a differential TDR/TDT drives both lines of a pair to measure $Z_{\text{diff}}$ and mode conversion.

## 26.2 Vector network analysis

A VNA measures S-parameters directly in the frequency domain, with far greater dynamic range than a TDR (often 100 dB or more), and its data can be transformed into the time domain. Two steps are essential (Figure 26.3):

**Calibration**

Measuring known standards (short, open, load, thru—SOLT—or thru, reflect, line—TRL) at the ends of the test cables moves the measurement reference plane to the cable connectors and removes the instrument's systematic errors.

**De-embedding**

The DUT on a board is reached through fixtures (launches, probe pads, traces). De-embedding removes their effect mathematically, using models or measurements of the fixtures. The "2×-thru" method measures a structure made of the two fixture halves joined directly, splits it into two halves, and removes them from the DUT measurement. IEEE 370 standardizes fixture design, de-embedding, and quality metrics for board interconnects up to 50 GHz [44].

**Common Pitfall: De-embedding imperfect fixtures**

Assuming that de-embedding removes any fixture. Fixtures with large discontinuities, poor repeatability between the 2×-thru and the DUT fixture, or resonances produce de-embedded results with non-physical ripple and passivity violations. Design fixtures for low reflection and good consistency, and check the de-embedded result with the quality metrics of IEEE 370.

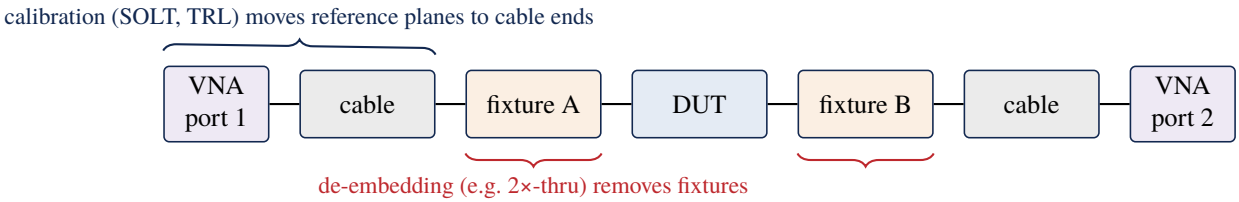



**Figure 26.3.** VNA measurement chain. Calibration places the reference planes at the cable ends; de-embedding removes the test fixtures to isolate the DUT.

### 26.2.1 Checking a de-embedding

A de-embedded result should pass three self-consistency checks before it is trusted: de-embedding the 2×-thru from *itself* should give a perfect thru (0 dB loss, zero phase, negligible reflection) to the accuracy claimed; the fixture halves extracted from the 2×-thru should be passive and causal; and a DUT of known behavior—a short line of the same geometry—should de-embed to the loss per inch measured from two line lengths. Repeating the fixture measurement on a second board or after reconnecting the cables shows how much of the residual is repeatability rather than fixture error, and that spread belongs in the reported uncertainty [44].

## 26.3 Measuring milliohm PDN impedance

A PDN of a few milliohms cannot be measured reliably with a one-port reflection measurement: at $Z \ll 50\,\Omega$ the reflection coefficient is close to $-1$, and the impedance is hidden in its small difference from $-1$, where calibration residuals dominate. The *two-port shunt-through* method connects port 1 and port 2 to the same power–ground location (Figure 26.4). For a shunt impedance $Z$ between two matched 50 Ω ports,

$$S_{21} = \frac{2Z}{Z_0 + 2Z} \quad \Longleftrightarrow \quad Z = \frac{Z_0\, S_{21}}{2(1 - S_{21})} \approx \frac{Z_0}{2} S_{21} \quad (Z \ll Z_0), \tag{26.4}$$

using the complex $S_{21}$ of a calibrated measurement [6]. The transmission is now proportional to $Z$ instead of being a small difference.

**Worked Example 26.2: How hard is a milliohm measurement?**

A purely resistive 5 mΩ PDN is measured shunt-through with 50 Ω ports. Find $|S_{21}|$, and the error contributed by a residual fixture inductance of 10 pH at 100 MHz.

**Solution.** $S_{21} = 0.01/50.01 = 2.0 \times 10^{-4}$, or $-74.0$ dB: the VNA needs dynamic range, source power and IF bandwidth to measure that with low noise. The residual inductance has $|\mathrm{j}\omega L| = 2\pi \times 10^{8} \times 10^{-11} = 6.3\,\mathrm{m}\Omega$ at 100 MHz—larger than the DUT. Unless that inductance is calibrated out or the probe point is defined to include it, the high-frequency result is a measurement of the fixture.

Two further error sources are specific to this method. Above a few hundred hertz, the cable shields of the two ports connect the board ground to the instrument ground and form a loop in parallel with the DUT; at low frequency this ground loop can dominate a milliohm measurement, and it is broken with a common-mode choke (ferrite-loaded cable) or an isolation transformer on one port. And the result is the impedance *at the probe location*, not at the die; for correlation with a simulation, the model must be evaluated at the same port.

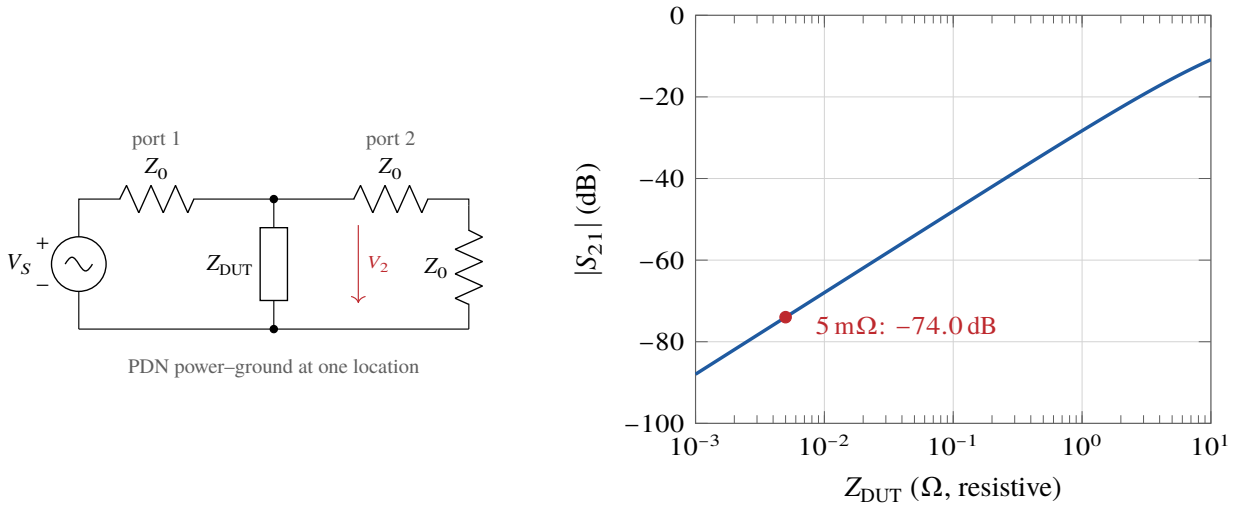



**Figure 26.4.** Left: ideal two-port shunt-through connection; cable shields, calibration standards and ground loops are omitted but matter in the laboratory. Right: $|S_{21}|$ versus a resistive DUT impedance from Eq. (26.4). [Analytical]

**Design Rule: Plan PDN measurement points**

For every critical rail, place a probe point (a pair of coaxial launch footprints or probe pads with adjacent ground) close to the load and record its location in the PDN model. Document the measurement floor (noise and residual inductance) with a shorted fixture before measuring the board.

## 26.4 Oscilloscopes and probing

### 26.4.1 Bandwidth

By Eq. (2.5), an oscilloscope adds its own rise time to the signal's. For a displayed rise time within 5 % of the true value, the scope's rise time must be less than about one-third of the signal's (Worked Example 2.2). For serial-link compliance, specifications usually state a required bandwidth tied to the data rate; at least two to three times the Nyquist frequency is a common minimum. Real-time oscilloscopes capture single events; equivalent-time (sampling) oscilloscopes offer higher bandwidth and lower noise for repetitive signals.

### 26.4.2 Probe loading

Every probe loads the circuit. A passive 10 × probe with 10 pF input capacitance has a reactance of only 32 Ω at 500 MHz; placed on a 50 Ω line, it forms a time constant of $Z_0 C/2 = 250$ ps. Active probes reduce input capacitance to a fraction of a picofarad; differential active probes measure pairs without referencing either side to ground.

### 26.4.3 Ground-lead inductance

The probe's ground lead forms a loop with the probe tip. Its inductance (about 1 nH/mm, so 100–200 nH for a 15 cm clip lead) resonates with the probe's input capacitance and the source impedance, producing ringing that is not on the signal (Figure 26.5).

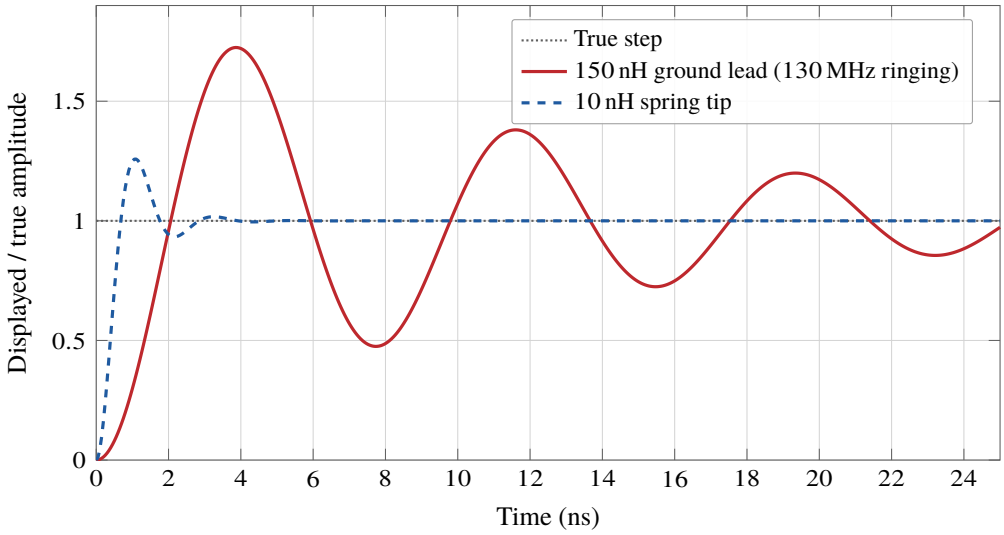



**Figure 26.5.** Step response through a passive probe with 10 pF tip capacitance and a 25 Ω source: a 150 nH ground lead rings at 130 MHz; a 10 nH spring tip resonates near 500 MHz and is well damped.

**Worked Example 26.3: Ground-lead resonance**

Find the resonant frequency of a 150 nH ground lead with a 10 pF probe, and with a 10 nH spring tip.

**Solution.** $f = 1/(2\pi\sqrt{LC})$: 130 MHz and 503 MHz. With a 25 Ω source the damping ratios are 0.10 and 0.40, so the long lead rings for many cycles and overshoots by about 70 %, while the spring tip overshoots by about 25 % and settles quickly.

**Design Rule: Probing fast signals**

Use active or solder-in probes with the shortest possible ground connection, probe at the receiver (where the signal matters), and provide probe points with an adjacent ground pad in the layout. For power-rail ripple, use a 50 Ω coaxial connection (1:1) with the scope input set to 50 Ω and a DC block if needed, rather than a 10 × probe with a clip lead.

## 26.5 Near-field probing

Small magnetic-loop (H-field) and electric-stub (E-field) probes connected to a spectrum analyzer locate emission sources on a board. An H-field probe responds to currents flowing in traces, planes, and cables beneath it and is the more useful for finding loop currents and return-path problems; an E-field probe responds to high-$dV/dt$ nodes such as clock nets and switching regulators. Near-field measurements are relative—they do not predict the far-field result of a compliance test—but comparing amplitudes before and after a change, or across a board, quickly identifies the dominant sources and the cables that carry common-mode current.

**Rule of Thumb: Probe ground leads**

A probe ground lead has roughly 1 nH/mm of inductance. Keep its resonance with the probe's input capacitance at least three to five times above the signal bandwidth. *Limits:* the actual inductance depends on the loop shape; the safest choice is the shortest ground the probe allows.

**Table 26.1.** Measurement instruments and their uses.

| Instrument | Measures | Limitations |
|---|---|---|
| TDR / TDT | Impedance profile, delay, skew, crosstalk | Resolution set by rise time; masking |
| VNA | S-parameters, wide dynamic range | Needs calibration and de-embedding |
| Real-time scope | Waveforms, eyes, jitter, single events | Bandwidth, noise, probe loading |
| Sampling scope | High-bandwidth repetitive waveforms | Needs trigger; repetitive signals only |
| Spectrum analyzer + near-field probe | Emission sources | Relative only |
| BERT | Bit-error ratio, bathtubs | Long test times at low BER |

**Key Takeaways**

- TDR maps reflection coefficient to impedance, $Z = Z_{\text{ref}}(1 + \rho)/(1 - \rho)$; the area of a dip or bump gives excess $C$ or $L$.
- Measure milliohm PDN impedance shunt-through, $Z \approx (Z_0/2)S_{21}$; residual fixture inductance and cable ground loops set the floor.
- TDR resolution is about $t_r v_p/2$ (the $\frac{1}{2}$ is the round trip); shorter features show an apparent, blended impedance, and distant ones are masked.
- VNA measurements require calibration to the cable ends and de-embedding of fixtures; IEEE 370 standardizes the process.
- Probe capacitance and ground-lead inductance corrupt fast measurements; use active probes with short grounds.
- Near-field probes locate emission sources by relative comparison.

## Further Reading

Derickson and Müller [26] describe high-speed measurement instruments and methods; IEEE 370 [44] standardizes de-embedding; Novak and Miller [6] cover PDN measurement.

## Review Problems

**P26.1** A TDR plateau shows $\rho = +0.2$ in a 50 Ω system. What is the impedance?

**P26.2** What reflection coefficient does a 42 Ω section produce in a 50 Ω system?

**P26.3** An inductive bump has $\int v_r \, dt/V_i = 2\,\text{ps}$ on a 50 Ω line. Find the excess inductance.

**P26.4** Find the resonance of a 150 nH ground lead with an 8 pF probe.

**P26.5** Estimate the spatial resolution of a TDR with a 25 ps system rise time on a line with $v_p = 6\,\text{in/ns}$.

**P26.6** What oscilloscope bandwidth displays a 50 ps edge within 5 %?

**P26.7** A shunt-through measurement gives $S_{21} = 4.0 \times 10^{-4} \angle 0°$ at 1 MHz with 50 Ω ports. What is the PDN impedance?

**P26.8** *(Imperfect data.)* A 60 Ω, 20 ps segment measured with a 100 ps TDR edge shows a peak of only about 54 Ω. A colleague concludes the segment is 54 Ω. What would you do?

Chapter 27

# Board Bring-Up and Debug

**Chapter Overview**

The first powered board is where design assumptions meet reality. A disciplined bring-up procedure protects scarce prototypes from damage, isolates faults quickly, and produces the measurements that confirm—or correct—the simulation models. This chapter gives a step-by-step bring-up flow from bare-board inspection to high-speed link qualification, describes boundary scan and the built-in margining features of modern interfaces, and summarizes a root-cause method for the high-speed failures that remain.

**Learning Objectives**

1. Perform pre-power inspection and rail-impedance checks.
2. Power a board safely with current limits and verify sequencing and ripple.
3. Use JTAG and boundary scan to verify connectivity.
4. Interpret DDR training margins and serial-link margining results.
5. Apply a structured root-cause method to high-speed failures.

## 27.1 The bring-up flow

Figure 27.1 outlines the sequence. Its guiding principle is to verify each layer of the system before relying on it: power before clocks, clocks before logic, connectivity before firmware, and simple interfaces before fast ones.

## 27.2 Before power

1. **Visual and X-ray inspection.** Check component orientation, missing or tombstoned parts, solder bridges, and (by X-ray) BGA voiding and bridging, especially on the first article.
2. **Rail-to-ground resistance.** Measure each rail's resistance to ground and compare with a known-good board or with the expected load. Low-voltage, high-current core rails legitimately read low (a fraction of an ohm to a few ohms) because of the load's leakage, so comparison with a reference board is more meaningful than an absolute threshold.
3. **Rail-to-rail checks.** Look for shorts between adjacent rails, which are common under BGAs.

**Worked Example 27.1: Interpreting a low rail resistance**

A 0.8 V core rail measures 0.3 Ω to ground; a reference board measures 8 Ω. If powered at 0.8 V with a 2 A limit, what happens at the fault?

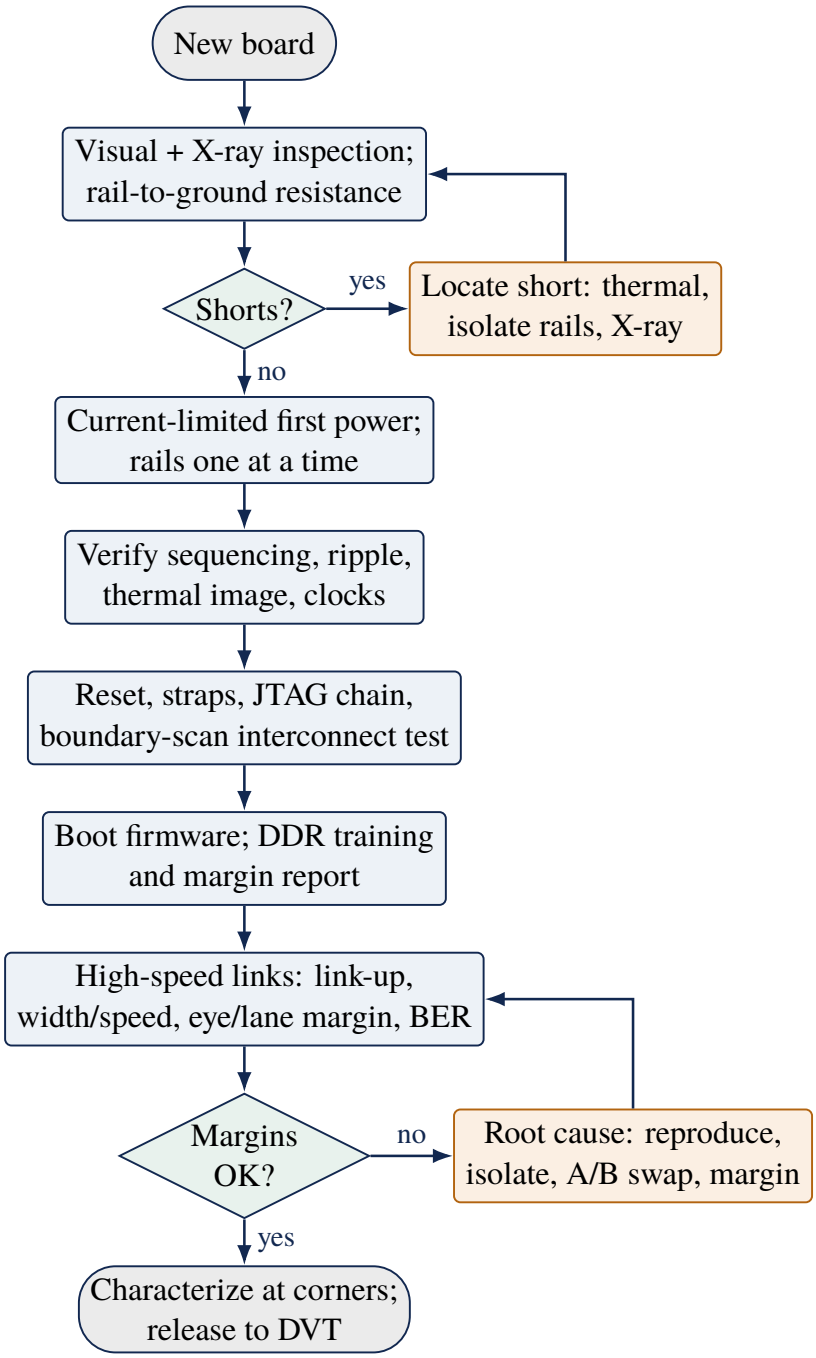



**Figure 27.1.** Board bring-up flow, from inspection to link qualification.

**Solution.** The rail is almost certainly shorted. At the current limit, the fault dissipates $I^2R = 2^2 \times 0.3 = 1.2\,\text{W}$, enough to make it visible on a thermal camera within seconds—which is how such shorts are located. That power is not guaranteed to be harmless: if the short is a small defect (a solder bridge under a BGA, a hairline inner-layer short or a damaged device), 1.2 W concentrated in it can overheat the joint or delaminate the laminate. Start with a lower limit—at 0.5 A the fault dissipates 75 mW—and raise it in steps while watching the thermal image, stopping as soon as the hot spot is located.

## 27.3 First power

Power the board from a bench supply with a current limit a little above the expected idle current, with each regulator enabled separately if the design allows it. For each rail:

- verify the DC voltage at the load (remote-sense point), not only at the regulator;
- verify sequencing and ramp times against the device requirements (Figure 27.2); many SoCs and memories specify the order, maximum ramp time, and maximum delay between rails, and the reset release relative to the last rail;
- measure ripple and transient response with a 50 Ω coaxial connection (Chapter 26), and compare with the PDN budget (Chapter 16);
- image the board thermally to catch hot spots.

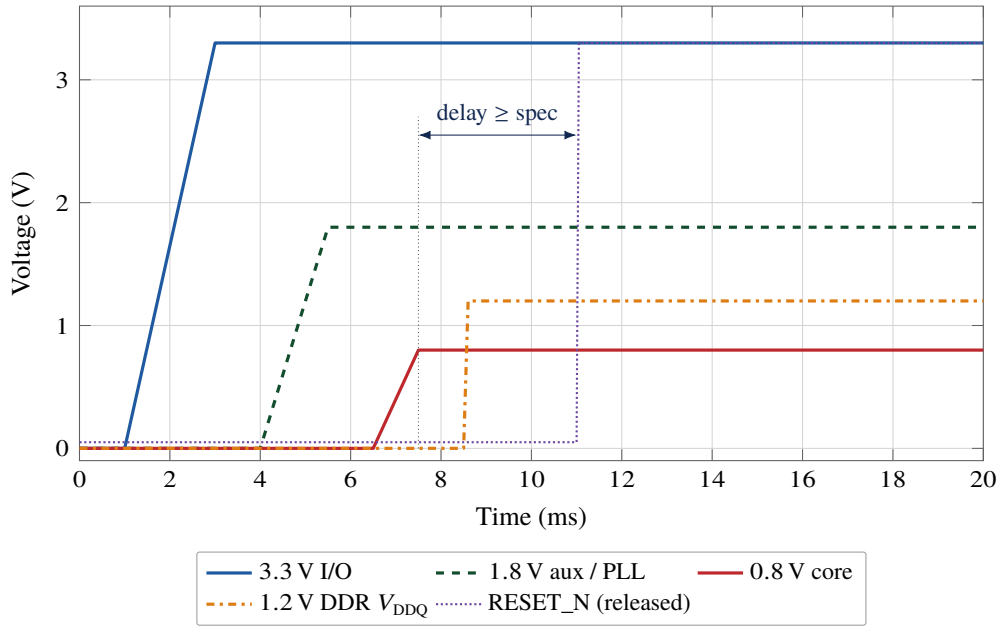



**Figure 27.2.** An example power-up sequence: I/O, auxiliary, core, and memory rails ramp in order, and reset is released after the last rail is stable. The required order and delays come from each device's data sheet.

Then verify clocks: frequency (against the ppm tolerance), amplitude and common mode at the receiver, and, for reference clocks of serial links, phase noise or jitter (Chapter 24).

## 27.4 JTAG and boundary scan

IEEE 1149.1 defines a test access port (TAP)—TCK, TMS, TDI, TDO, and optional TRST—and a boundary-scan register that lets a tester drive and sample every digital pin of a compliant device [55]. With all devices on a board chained TDI-to-TDO, boundary scan verifies:

- the chain itself (reading each device's IDCODE);
- interconnects between boundary-scan devices (opens and shorts);
- connections to non-scan devices that can be exercised indirectly, such as memories (by writing and reading through the controller's pins) and pull resistors.

Boundary scan tests DC connectivity only; it does not exercise signal integrity at speed. It is nevertheless invaluable for BGA-dense boards, where physical probing is impossible.

**Design Rule: Design for JTAG**

Bring the JTAG chain to a standard header, buffer TCK and TMS when they fan out to many devices, terminate TCK (it is a clock), provide a way to bypass or isolate devices that may be unpopulated, and keep the boundary-scan description (BSDL) files and the netlist under revision control for the test team.

## 27.5 Interface bring-up and margining

### 27.5.1 Memory

Memory controllers report training results: write-leveling delays per byte lane, read and write DQ-DQS delays per bit, and $V_{REF}$ settings, often with a two-dimensional eye scan of each bit's timing and voltage margin. Byte lanes whose leveling delays are near the end of the range, or bits whose eyes are markedly smaller than their neighbors', point to layout problems (Section 19.5). Running memory stress tests at elevated temperature and at the supply voltage limits exposes marginal timing.

### 27.5.2 Serial links

Serial links first must train to the intended width and speed; a link that comes up at a lower width or generation indicates a failing lane. PCIe Gen4 and later define *lane margining at the receiver*, in which the receiver reports how far its sampling point can be moved in time and voltage before errors occur, without external equipment. Many SERDES also expose on-die eye monitors. PRBS generators and checkers in the PHY allow long BER tests (Worked Example 12.3).

### 27.5.3 Margins across conditions

A margin measured once, at room temperature and nominal voltage, on one board, says little about production. Plan a small matrix of process, voltage and temperature conditions (Table 27.1) and record the same quantities at each point, so that trends—a margin that shrinks with temperature, or one board that is an outlier—are visible. Keep the raw logs: the controller's training report, the PHY's eye-monitor output, the tool and firmware versions, the board serial number and the instrument settings.

## 27.6 Root-cause method

1. **Reproduce** the failure reliably, and record the board, temperature, voltage, data pattern, and firmware version.

**Table 27.1.** A minimal margining matrix and log template (illustrative).

| Field | Content |
|---|---|
| Conditions | Boards: at least three, from different panels if possible; supply at nominal and both limits; temperature at cold, room and hot limits |
| Memory | Per-bit timing and $V_{REF}$ eye widths; leveling delays per byte; stress-test result and duration |
| Serial links | Trained width and speed; TX preset chosen; lane-margining time and voltage per lane; PRBS error count and duration |
| Power | Rail voltages at the load under idle and stress; ripple and transient captured with stated bandwidth |
| Record | Board serial, firmware and tool versions, instrument and probe settings, date, operator |

**Table 27.2.** Common high-speed symptoms and likely causes.

| Symptom | Likely causes |
|---|---|
| PCIe link trains at reduced width | Swapped or open lane; missing AC cap; lane polarity or reversal not supported |
| PCIe link trains at lower speed | Excess loss; via stubs; reference-clock jitter; marginal equalization |
| DDR write leveling fails on outer lanes | CK-to-DQS skew beyond leveling range |
| DDR errors only when hot | Marginal timing; $V_{REF}$ noise; insufficient decoupling |
| Intermittent lock loss on a SERDES | Supply noise in the PLL; reference-clock noise; SSC mismatch |
| Ethernet works at 100M, fails at 1G | RGMII delay misconfigured; MDI pair swap |
| Emissions at clock harmonics | Return-path discontinuity; unfiltered cable; missing SSC |

2. **Isolate** it: which lane, which bit, which board, which direction. Patterns across boards distinguish design problems from assembly problems.
3. **Measure margin**, not pass/fail: vary voltage, temperature, frequency, or equalization settings and find how far the failure point is from nominal.
4. **Swap** parts or boards (A/B testing) to separate the device, the board, and the partner.
5. **Compare with simulation**: re-simulate the failing net with measured parameters (impedance from TDR, actual stackup from cross-section); a discrepancy between model and measurement is itself a finding.
6. **Confirm the fix** by restoring the margin, not merely by making the failure disappear once.

**Worked Example 27.2: Two failure investigations**

*(a) One PCIe Gen4 lane trains only at Gen3, on two of five boards.* Isolation: the same lane fails on both boards, and swapping the add-in card does not move it, so the board is suspected. Lane margining on the passing boards shows that lane with the smallest time margin. A VNA measurement of the lane (Chapter 26) shows a notch near 40 GHz, below the 48 GHz expected from the back-drill target (Section 11.5.1). A cross-section finds a residual stub of about 37 mil instead of the specified maximum: the back-drill depth on that via row is out of tolerance. *Finding:* a fabrication escape, not a design error; the fix is a drill-depth inspection requirement and coupon, and the design margin is confirmed by re-simulating with the measured stub.

*(b) DDR read errors only above* 70 °C. Isolation: errors are on reads, on all byte lanes, and disappear when the controller's $V_{REF}$ is trained at temperature. A rail measurement with adequate bandwidth shows $V_{DDQ}$ noise peaking at a frequency that matches a PDN anti-resonance predicted by the model but judged acceptable because its peak was "only slightly" above target; the capacitance of the bulk MLCCs falls with temperature and moves the peak. *Finding:* a design margin problem; the fix is damping the anti-resonance (Section 16.4) and re-running the PVT matrix of Table 27.1.

**Common Pitfall: Debugging by changing several things at once**

Swapping a component, changing a firmware setting, and moving a probe at the same time. When the failure disappears, the cause is unknown and the fix may not be real. Change one variable at a time, and record each experiment.

**Worked Example 27.3: Checking a power-up sequence**

A device requires its reset to be held for at least 2 ms after the last supply reaches 95 % of its final value. In the capture of Figure 27.2, the last rail (1.2 V $V_{DDQ}$) is stable at 8.6 ms and reset is released at 11.0 ms. Is the requirement met, and what is the margin?

**Solution.** The hold time is $11.0 - 8.6 = 2.4$ ms, meeting the 2 ms requirement with 0.4 ms of margin. Because regulator ramp times vary with load and temperature, verify the sequence also at the slowest expected ramp.

**Rule of Thumb: First-power current limit**

Set the bench supply's current limit at first power to about 1.2 to 1.5 times the expected idle current. *Limits:* bulk capacitance and regulator soft-start produce inrush currents above the idle value; if the supply folds back during start-up, raise the limit briefly or pre-charge through a resistor.

**Key Takeaways**

- Inspect and measure rail resistances against a reference before power.
- Power with current limits, one rail at a time; verify voltage at the load, sequencing, ripple, temperature, and clocks.
- Boundary scan verifies connectivity of BGA-dense boards without physical access.
- Use DDR training reports, PCIe lane margining, eye monitors, and PRBS BER tests to quantify margin.
- Root-cause by reproducing, isolating, margining, swapping, and comparing with simulation—one change at a time.

## Further Reading

IEEE 1149.1 [55] defines boundary scan; Derickson and Müller [26] discuss BER testing and jitter measurement used in link qualification.

## Review Problems

**P27.1** A 0.8 V rail shows 30 mV peak-to-peak noise against a ±3 % tolerance. Is the dynamic noise within budget, assuming the full tolerance is available?

**P27.2** A 0.3 Ω short carries a 2 A current limit. What power does it dissipate?

**P27.3** Four devices in a JTAG chain have instruction registers of 4, 8, 6, and 5 bits. What is the total instruction-register length?

**P27.4** A boundary-scan chain has 2000 cells and TCK runs at 10 MHz. How long does one full shift take?

**P27.5** A PCIe ×16 link trains at ×8. What fraction of bandwidth is lost?

**P27.6** How long must one 8 GT/s lane run error-free to show BER $\leq 10^{-12}$ at 95 % confidence?

Chapter 28

# Design for Manufacturing and Release

**Chapter Overview**

A high-speed design is only as good as the board that the fabricator builds. The release package must tell the fabricator and the assembler exactly what to build, which tolerances matter, and how compliance will be verified—and the design must be manufacturable within those tolerances at reasonable yield. This chapter covers fabrication notes and drawings, stackup and impedance documentation, impedance coupons, tolerance analysis, design-for-manufacturing and design-for-test rules, the data formats of a release package, and change control.

**Learning Objectives**

1. Write fabrication notes and a stackup/impedance table that a fabricator can build from without ambiguity.
2. Specify impedance coupons and their test method.
3. Estimate the effect of fabrication tolerances and etch on impedance.
4. Apply DFM and DFT rules to a high-speed layout.
5. Assemble a complete release package and control subsequent changes.

## 28.1 Fabrication drawing and notes

The fabrication drawing (Figure 28.1) is the contract with the fabricator. It contains the board outline and dimensions, the stackup and impedance table, the drill table, and the fabrication notes. Typical notes specify:

- the performance class (IPC-6012 Class 2 for most commercial products, Class 3 for high-reliability ones) [56];
- the laminate (by IPC-4101 slash sheet or by named product) and its minimum $T_g$ [22];
- finished thickness and tolerance, copper weights, surface finish, soldermask and legend;
- via treatment (tented, plugged, filled and capped) referenced to IPC-4761 types [57], and any via-in-pad requirements;
- back-drill locations, the layer that must not be cut, and the maximum residual stub (Chapter 11);
- finished-hole tolerances that make physical sense: a tolerance such as "−full" (the hole may plate closed) is a convention for unfilled vias only, and is wrong for filled or capped vias, whose fill needs an open barrel—the drill table of Figure 28.1 uses $\pm 0.05$ mm;
- controlled-impedance requirements, permission (or not) for the fabricator to adjust widths, and the coupon and test method;
- bow and twist, electrical test (to an IPC-D-356 netlist [58]), and marking requirements.

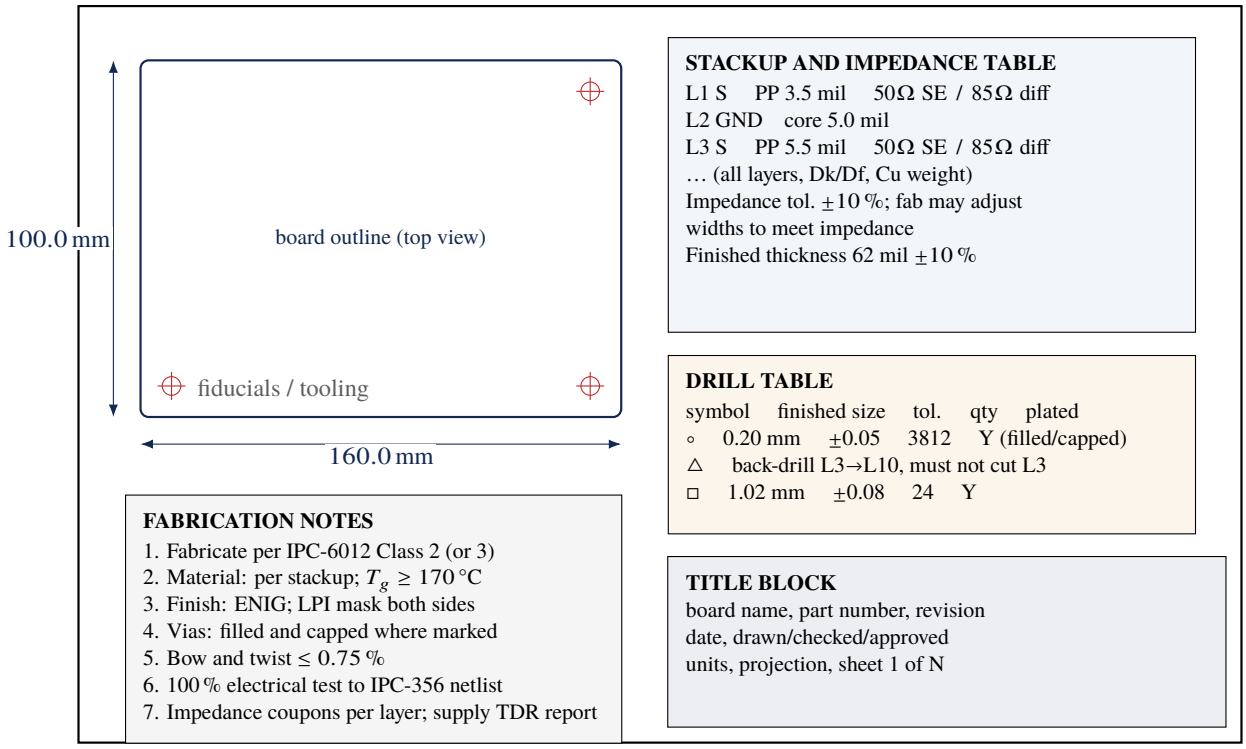



**Figure 28.1.** Organization of a fabrication drawing: outline and dimensions, stackup and impedance table, drill table, fabrication notes, and title block (schematic).

**Design Rule: Let the fabricator tune impedance**

Specify impedance targets and tolerances per layer and per structure, list the designed trace widths and spacings as nominal, and allow the fabricator to adjust widths (within stated limits) and dielectric thicknesses to meet the targets with its materials and process. Ask for the fabricator's stackup proposal before routing, and route to it.

## 28.2 Impedance coupons

Because impedance cannot be measured on the production traces without damaging them, fabricators build *coupons*—test structures on the panel border—with the same layers, widths, and references as the controlled-impedance traces (Figure 28.2). Each coupon is measured by TDR, and the impedance is read from the stable plateau of the trace, away from the launch and the far end. IPC-2141 and the IPC-TM-650 test methods describe coupon design and measurement [17], [59].

**Worked Example 28.1: Coupon length**

A TDR with 35 ps rise time measures a coupon in stripline at 170 ps/in. If the plateau should be at least 1 ns of one-way delay long after excluding launch effects, how long must the coupon be?

**Solution.** 1 ns/170 ps/in = 5.9 in, plus the launch region. This is why coupons are typically 4–6 in or longer, and why they are placed on the panel rather than on the board.

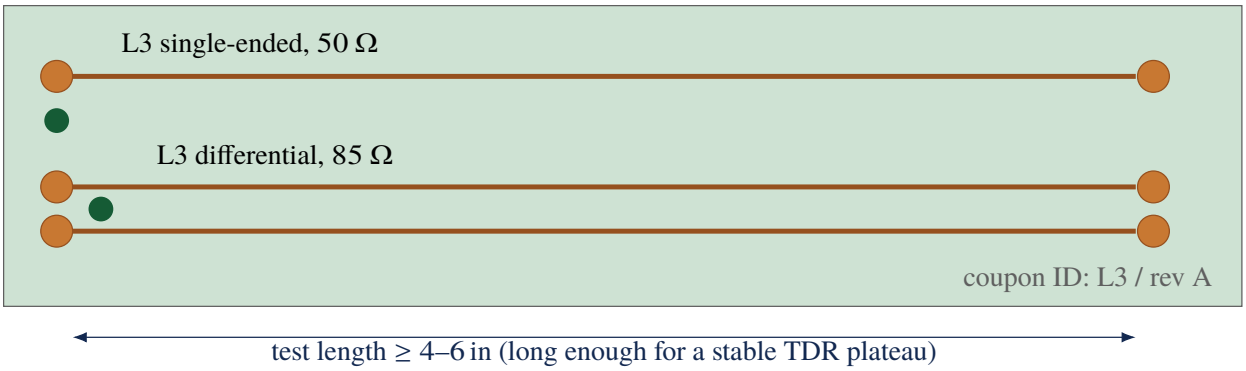



**Figure 28.2.** An impedance coupon with single-ended and differential test lines, ground launch vias (green) next to the probe pads, and layer identification.

### 28.2.1 Acceptance windows

A coupon result is meaningful only with its measurement conditions. The fabrication notes or the purchase specification should state: the TDR edge (rise time at the reference plane, or the filtering applied); the evaluation window along the trace, usually expressed as a fraction of the coupon's delay so that launch and far-end effects are excluded (for example, 30–70 % of the trace); whether the limit applies to the average over the window or to every point within it; whether a loss-induced upward slope is corrected; and the acceptance band itself. Two laboratories with different edges and windows can report different impedances for the same coupon (Section 26.1.2), so a disputed result is settled by repeating the measurement under the stated conditions, not by comparing numbers taken differently.

**Design Rule: Specify how the coupon is judged**

State the TDR rise time, the evaluation window, the statistic (average or all points) and the acceptance band for each controlled-impedance class on the fabrication drawing. Ask for the coupon reports with every lot and keep them with the board's traveler data.

## 28.3 Tolerances

Worked Example 7.4 showed that dielectric thickness, etched width, and $D_k$ together produce a spread of about $\pm 10$ % in impedance. Two manufacturing effects deserve particular attention:

**Etch profile**
: Etching undercuts the top of a trace, leaving a trapezoid narrower at the top than at the bottom. The fabricator compensates by adjusting the artwork width, but the effective width differs from the drawn width.

**Pressed dielectric thickness**
: Prepreg thickness after lamination depends on how much resin flows into the adjacent copper pattern. Sparse copper layers absorb more resin, thinning the dielectric. Balanced copper (with thieving where necessary) keeps thickness predictable.

**Worked Example 28.2: Effect of etch on stripline impedance**

The stripline of Worked Example 7.2 ($W = 4.2$ mil, $T = 0.6$ mil, $B = 10.6$ mil, $D_k = 3.8$) is etched so that its average width is 0.25 mil less than drawn. Estimate the impedance change.

**Solution.** From Eq. (7.5), $\partial Z_0/\partial W = -(60/\sqrt{3.8})\times 0.8/(0.8W+T) = -30.78\times 0.8/3.96 = -6.2\,\Omega/\text{mil}$. A 0.25 mil reduction raises $Z_0$ by about 1.6 Ω, or 3 %—a third of a ±10 % budget from etch alone.

**Rule of Thumb: Bow and twist**

For boards with surface-mount components, bow and twist is usually limited to 0.75 % of the diagonal (IPC-6012 gives the classes' limits). *Limits:* fine-pitch BGAs and large boards may need tighter values; check the assembler's requirement.

## 28.4 Design for manufacturing and test

**Table 28.1.** Key DFM and DFT checks.

| Check | Purpose |
|---|---|
| Minimum trace/space per layer | Within fabricator capability at the copper weight |
| Annular ring | Pad minus finished hole, minus drill wander, ≥ class minimum |
| Copper balance and thieving | Uniform plating and pressed thickness; reduces warp |
| Acid traps, slivers | Avoid acute copper angles and narrow mask/copper features |
| Soldermask dams and clearance | Prevent bridging between fine-pitch pads |
| Via-in-pad | Filled and planarized, or kept out of pads |
| Thermal relief | On plane connections of through-hole parts, for solderability |
| Fiducials and tooling | Global and local fiducials for placement accuracy |
| Component spacing | Rework access, pick-and-place and inspection clearances |
| Test points | One per net for ICT or flying probe, where signal integrity allows |
| JTAG access | Header and chain for boundary scan (Chapter 27) |

**Common Pitfall: Test points on multi-gigabit nets**

Adding a through-hole test pad to every net, including PCIe and USB lanes. Each pad and its via is a stub (Chapter 11). On high-speed nets, omit test points, use small pads directly on the trace with no via, or rely on boundary scan and built-in self-test for connectivity.

## 28.5 The release package

Intelligent formats (ODB++ and IPC-2581) carry the stackup, netlist, drill spans, and component data in one database, reducing the risk of mismatched files; Gerber with X2 attributes adds much of the same information to the traditional format. Whatever format is used, the package should be generated from a single tagged design revision and checked by viewing the output files in an independent viewer before release.

### 28.5.1 Change control

After release, every change follows an engineering change order (ECO or ECN) that records what changed, why, which documents are affected, and which boards in the field or in production are impacted. Board revisions are marked in copper on the board itself, and the BOM revision is tracked separately, since many changes are component-only.

**Table 28.2.** Contents of a PCB release package.

| Item | Content |
|---|---|
| Fabrication data | Gerber (with X2 attributes) [60], ODB++ [61], or IPC-2581 [62] |
| Drill data | NC drill files per drill span, including back-drill files and depths |
| Fabrication drawing | Outline, stackup, impedance table, drill table, notes |
| Netlist | IPC-D-356 for bare-board electrical test |
| Assembly data | Assembly drawings, pick-and-place (centroid) file, paste layers |
| Bill of materials | Manufacturer part numbers, approved alternates, reference designators |
| Stackup document | Materials by layer, thicknesses, $D_k/D_f$, copper weights, impedance targets |
| Readme | File list, units, layer order, revision, contacts |

**Design Rule: Release checklist**

Before release: run DRC with the fabricator's rules; confirm stackup and impedance with the fabricator; verify that back-drill and via-fill layers match the drill table; regenerate all outputs from one tagged revision; inspect every output in a viewer; check the BOM against the placement file; and archive the complete package with the simulation reports that justify it. Appendix C gives a full checklist.

**Key Takeaways**

- The fabrication drawing and notes are the contract: class, material, thickness, finish, vias, back-drill, impedance, and test.
- Specify impedance targets and allow the fabricator to tune widths; verify with TDR on coupons.
- Etch and pressed dielectric thickness dominate impedance tolerance; balance copper.
- Apply DFM and DFT rules, but keep test structures off multi-gigabit nets.
- Release a complete, consistent package from a single revision and control every change with an ECO.

## Further Reading

IPC-6012 [56], IPC-4101 [22], IPC-4761 [57], and IPC-2221 [63] govern board performance, materials, via protection, and design; IPC-2581 [62], ODB++ [61], and the Gerber format [60] define the data exchange.

## Review Problems

**P28.1** What is the bow-and-twist limit at 0.75 % for a 6 in × 4 in board?

**P28.2** A microstrip has $\partial Z_0/\partial W = -5.98\,\Omega/\text{mil}$. What impedance change results from 0.3 mil of etch reduction?

**P28.3** What range corresponds to 85 Ω ±10 %?

**P28.4** What coupon length gives 1 ns of one-way delay at 170 ps/in?

**P28.5** A board has a 10 in × 8 in outline. What is its 0.75 % bow-and-twist limit?

**P28.6** A via has an 18 mil pad and a 10 mil finished hole, and drill positional tolerance is ±3 mil. What is the worst-case annular ring?

# Appendices

Appendix A

# Formula Reference Sheet

This appendix collects the principal equations of the book. Units are SI unless a practical unit is shown; "(mil)" and "(GHz)" indicate the unit of the variable in that formula. Each entry refers to the equation in the text, where its derivation and limits are given.

## Edges, bandwidth, and delay

| Quantity | Formula | Ref. |
|---|---|---|
| RC rise time | $t_r = \tau \ln 9 \approx 2.2\tau$ | Eq. (2.2) |
| Bandwidth (single pole) | $f_{3\mathrm{dB}} \approx 0.35/t_r$ | Eq. (2.3) |
| Knee frequency | $f_{\mathrm{knee}} \approx 0.5/t_r$ | Eq. (2.4) |
| Cascaded rise times | $t_r = \sqrt{\sum t_{r,i}^2}$ | Eq. (2.5) |
| Trapezoid corners | $f_1 = 1/(\pi\tau)$, $f_2 = 1/(\pi T_{\mathrm{ramp}})$, $T_{\mathrm{ramp}} = t_r/0.8$ (linear) | Eq. (2.8) |
| Propagation delay | $t_{\mathrm{pd}} = \sqrt{\varepsilon_{\mathrm{eff}}}/c \approx 84.7\,\mathrm{ps/in}\sqrt{\varepsilon_{\mathrm{eff}}}$ | Eq. (2.10) |
| Critical length | $\ell_{\mathrm{crit}} = t_r/(k\,t_{\mathrm{pd}})$, $k \approx 6$ | Eq. (2.11) |
| Nyquist frequency | $f_{\mathrm{Nyq}} = 1/(2\,\mathrm{UI})$ | Eq. (2.12) |

## Electromagnetics

| Quantity | Formula | Ref. |
|---|---|---|
| Plate capacitance | $C = \varepsilon_0\varepsilon_r A/h$; $\varepsilon_0 = 0.2249\,\mathrm{pF/in}$ | Eq. (3.1) |
| Loop inductance | $L_{\mathrm{loop}} = L_{11} + L_{22} - 2L_{12}$ | Eq. (3.5) |
| Two-wire loop | $L_{\mathrm{loop}} \approx (\mu_0\ell/\pi)\ln(s/r)$ | Eq. (3.8) |
| Sheet inductance | $L_\square = \mu_0 h \approx 32\,\mathrm{pH}$ per square per mil | Eq. (3.9) |
| Return current | $J(x) = \dfrac{I}{\pi H}\dfrac{1}{1+(x/H)^2}$ | Eq. (3.10) |
| Skin depth (Cu) | $\delta_s \approx 66.1\,\mu\mathrm{m}/\sqrt{f/\mathrm{MHz}}$ | Eq. (3.13) |

## Transmission lines and reflections

| Quantity | Formula | Ref. |
|---|---|---|
| Characteristic impedance | $Z_0 = \sqrt{(R' + \mathrm{j}\omega L')/(G' + \mathrm{j}\omega C')} \to \sqrt{L'/C'}$ | Eqs. (4.6) and (4.7) |
| Per-unit-length $L$, $C$ | $C' = t_{\mathrm{pd}}/Z_0$, $L' = Z_0 t_{\mathrm{pd}}$ | Eq. (4.9) |
| Attenuation | $\alpha \approx R'/(2Z_0) + G'Z_0/2$ | Eq. (4.12) |
| Input impedance | $Z_{\mathrm{in}} = Z_0 \dfrac{Z_{\mathrm{L}} + \mathrm{j}Z_0 \tan\theta}{Z_0 + \mathrm{j}Z_{\mathrm{L}} \tan\theta}$ | Eq. (4.16) |
| Reflection coefficient | $\Gamma = (Z_{\mathrm{L}} - Z_0)/(Z_{\mathrm{L}} + Z_0)$, $T = 1 + \Gamma$ | Eq. (6.2) |
| Launch voltage | $V_S Z_0/(Z_0 + R_S)$ | Eq. (6.3) |
| Capacitive load (end, middle) | $\tau = Z_0 C$; $\tau = Z_0 C/2$ | Eqs. (6.6) and (6.7) |
| Shunt-$C$ reflection | $\lvert\Gamma\rvert = x/\sqrt{4 + x^2}$, $x = \omega C Z_0$ | Eq. (21.1) |

## Impedance approximations (see limits in Chapter 7)

| Geometry | Formula | Ref. |
|---|---|---|
| Microstrip (IPC-2141) | $\dfrac{87}{\sqrt{\varepsilon_{\mathrm{r}} + 1.41}} \ln \dfrac{5.98H}{0.8W + T}$ | Eq. (7.1) |
| Stripline (IPC-2141) | $\dfrac{60}{\sqrt{\varepsilon_{\mathrm{r}}}} \ln \dfrac{1.9B}{0.8W + T}$ | Eq. (7.5) |
| Diff. microstrip | $2Z_0[1 - 0.48\mathrm{e}^{-0.96S/H}]$ | Eq. (7.7) |
| Diff. stripline | $2Z_0[1 - 0.347\mathrm{e}^{-2.9S/B}]$ | Eq. (7.8) |
| Odd/even modes | $Z_{\mathrm{odd}} = \sqrt{\frac{L_{11} - L_{12}}{C_g + 2C_m}}$, $Z_{\mathrm{even}} = \sqrt{\frac{L_{11} + L_{12}}{C_g}}$ | Eq. (10.2) |
| Differential, common | $Z_{\mathrm{diff}} = 2Z_{\mathrm{odd}}$, $Z_{\mathrm{cm}} = Z_{\mathrm{even}}/2$ | Eq. (10.3) |

## Loss, crosstalk, skew, vias

| Quantity | Formula | Ref. |
|---|---|---|
| Conductor loss (stripline) | $36\sqrt{f_{\mathrm{GHz}}}/(W_{\mathrm{mil}} Z_0)$ dB/in | Eq. (8.3) |
| Dielectric loss | $2.31 f_{\mathrm{GHz}} D_{\mathrm{f}} \sqrt{D_{\mathrm{k}}}$ dB/in | Eq. (8.6) |
| Hammerstad roughness | $1 + \frac{2}{\pi} \arctan[1.4(\Delta/\delta_{\mathrm{s}})^2]$ | Eq. (8.4) |
| NEXT coefficient | $K_b = \frac{1}{4}(C_m/C + L_m/L)$ | Eq. (9.1) |
| FEXT coefficient | $K_f = \frac{1}{2v_{\mathrm{p}}}(C_m/C - L_m/L)$ ($\lvert\Delta T\rvert \ll t_{\mathrm{r}}$) | Eq. (9.2) |
| FEXT, modal | $V = \frac{1}{2}[a(t - T_e) - a(t - T_o)]$, $K_f \ell = -\Delta T/2$ | Eq. (9.7) |
| Saturation length | $\ell_{\mathrm{sat}} = t_{\mathrm{r}} v_{\mathrm{p}}/2$ | Eq. (9.5) |
| Skew mode conversion | $\lvert V_c/V_d\rvert = \frac{1}{2}\lvert\sin \pi f \Delta t\rvert$ | Eq. (10.4) |
| Via inductance | $5.08h[\ln(4h/d) + 1]$ nH ($h, d$ in inches) | Eq. (11.1) |
| Via pad capacitance | $1.41\varepsilon_{\mathrm{r}} T D_1/(D_2 - D_1)$ pF | Eq. (11.2) |
| Stub resonance | $f = c/(4\ell\sqrt{\varepsilon_{\mathrm{eff}}})$ | Eq. (11.7) |

## Timing, jitter, and clocks

| Quantity | Formula | Ref. |
|---|---|---|
| Setup margin | $T_{\text{clk}} - (t_{\text{co,max}} + t_{\text{flight}} + t_{\text{su}} + t_{\text{skew}} + t_{\text{jit}})$ | Eq. (12.1) |
| Total jitter | $\text{TJ} = \text{DJ}_{\delta\delta} + 2Q_{\text{BER}}\sigma;\ 2Q = 14.07$ at $10^{-12}$ | Eq. (12.5) |
| Two-tap FFE | $\lvert H\rvert = \sqrt{c_0^2 + c_1^2 + 2c_0c_1\cos(2\pi f\ \text{UI})}$ | Eq. (12.7) |
| Phase noise to jitter | $\sigma_t = \frac{1}{2\pi f_0}\sqrt{2\int 10^{\mathscr{L}/10}\,\mathrm{d}f}$ | Eq. (24.2) |
| Crystal load | $C_L = C_1C_2/(C_1 + C_2) + C_{\text{stray}}$ | Eq. (24.1) |

## Power integrity and EMC

| Quantity | Formula | Ref. |
|---|---|---|
| Ground bounce | $V = L_{\text{gnd}}N\ \mathrm{d}i/\mathrm{d}t$ | Eq. (15.1) |
| Parallel pins | $L_{\text{eff}} = [L + (N_g - 1)M]/N_g$ | Eq. (15.3) |
| Target impedance | $Z_{\text{target}} = \Delta V/\Delta I$ | Eq. (16.1) |
| Capacitor SRF | $f_{\text{SRF}} = 1/(2\pi\sqrt{\text{ESL}\,C})$ | Eq. (16.3) |
| Capacitor count | $N(f) \geq \lceil \lvert Z_C(f)\rvert / Z_{\text{target}} \rceil$ (full R–L–C) | Eq. (16.4) |
| Anti-resonance | $f = \frac{1}{2\pi}\sqrt{\frac{C_1+C_2}{(L_1+L_2)C_1C_2}}$, $\lvert Z\rvert \approx X^2/(R_1 + R_2)$ | Eqs. (16.6) and (16.7) |
| Bounce, capacitive load | $\mathrm{d}i/\mathrm{d}t\rvert_{\max} = \frac{\pi^2}{2}C_L\Delta V/T_e^2$ (raised cosine) | Eq. (15.2) |
| Plane resistance | $R_\square = \rho/t$ (0.49 mΩ for 1 oz) | Eq. (16.11) |
| Cavity resonance | $f_{mn} = \frac{c}{2\sqrt{\varepsilon_r}}\sqrt{(m/a)^2 + (n/b)^2}$ | Eq. (17.1) |
| DM emission | $E = 131.6 \times 10^{-16} f^2 AI/r$ | Eq. (18.1) |
| CM emission | $E = 1.257 \times 10^{-6} fLI/r$ | Eq. (18.2) |

## Measurement

| Quantity | Formula | Ref. |
|---|---|---|
| TDR resolution | $\Delta x \approx t_r v_p/2$ (round trip) | Eq. (26.3) |
| Shunt-through PDN | $Z = Z_0S_{21}/[2(1 - S_{21})] \approx (Z_0/2)S_{21}$ | Eq. (26.4) |
| TDR impedance | $Z = Z_{\text{ref}}(1 + \rho)/(1 - \rho)$ | Eq. (26.1) |
| Excess $C$, $L$ | $C = -\frac{2}{Z_0V_i}\int v_r\,\mathrm{d}t;\ L = \frac{2Z_0}{V_i}\int v_r\,\mathrm{d}t$ | Eq. (26.2) |
| Mixed-mode $S_{\text{DD21}}$ | $\frac{1}{2}(S_{31} - S_{32} - S_{41} + S_{42})$ | Eq. (25.2) |

Appendix B

# Material Property Tables

The values in this appendix are representative ranges compiled for orientation. Laminate $D_k$ and $D_f$ depend on resin content, glass style, frequency, temperature, humidity, and the test method used to measure them. Always design with the manufacturer's current data for the specific construction in the stackup, and with the fabricator's recommendation.

## B.1 Laminate classes

**Table B.1.** Representative laminate properties at about 1–10 GHz. Ranges, not guaranteed values; consult product data sheets.

| Laminate class (examples) | $D_k$ | $D_f$ | $T_g$ (°C) | Typical use |
|---|---|---|---|---|
| Standard FR-4, high $T_g$ | 4.0–4.6 | 0.015–0.022 | 150–180 | General digital |
| Mid-loss epoxy/modified epoxy | 3.6–4.1 | 0.008–0.012 | 170–200 | DDR4/5, PCIe Gen3/4 |
| Low-loss (modified PPE, etc.) | 3.3–3.8 | 0.004–0.007 | 180–210 | PCIe Gen4/5, USB4 |
| Very/ultra-low loss | 3.0–3.6 | 0.0015–0.003 | 200+ | PCIe Gen5/6, 56G+ |
| Ceramic-filled hydrocarbon (RF) | 3.0–3.7 | 0.002–0.004 | >280 | RF, mixed builds |
| PTFE (RF) | 2.1–3.0 | 0.0009–0.002 | — | Microwave, antennas |

## B.2 Frequency dependence of a causal model

Table B.2 evaluates the Djordjevic–Sarkar model (Eq. (8.8)) with relaxation limits at 1 kHz and 1 THz, fitted to three laminates at 1 GHz. It illustrates how $D_k$ falls slowly with frequency while $D_f$ stays nearly constant; real materials depart from this ideal, so measured data should be used where available.

**Table B.2.** $D_k$ / $D_f$ versus frequency from a Djordjevic–Sarkar fit at 1 GHz.

| Fit at 1 GHz | 0.1 GHz | 1 GHz | 5 GHz | 10 GHz | 20 GHz | 50 GHz |
|---|---|---|---|---|---|---|
| 4.0 / 0.020 | 4.12 / 0.019 | 4.00 / 0.020 | 3.92 / 0.020 | 3.88 / 0.021 | 3.85 / 0.021 | 3.80 / 0.020 |
| 3.7 / 0.010 | 3.75 / 0.010 | 3.70 / 0.010 | 3.66 / 0.010 | 3.65 / 0.010 | 3.63 / 0.010 | 3.61 / 0.010 |
| 3.5 / 0.004 | 3.52 / 0.004 | 3.50 / 0.004 | 3.49 / 0.004 | 3.48 / 0.004 | 3.47 / 0.004 | 3.47 / 0.004 |

## B.3 Glass styles

**Table B.3.** Common E-glass styles in prepreg (approximate; pressed thickness depends on resin content and adjacent copper).

| Style | Approx. pressed thickness (mil) | Notes |
|---|---|---|
| 106 | 1.5–2.5 | Very thin, open weave, resin rich (lower $D_k$) |
| 1080 | 2.5–3.5 | Thin, open weave; common for thin prepregs |
| 1078 / 1067 / 1035 | 2–3 | Spread (flat) glass; reduced weave skew |
| 2116 | 4–5.5 | Medium, tighter weave |
| 1652 | 4.5–6 | Medium-heavy |
| 7628 | 6.5–8 | Heavy, glass rich (higher $D_k$); not for high-speed layers |

## B.4 Copper

**Table B.4.** Copper foil weights and properties.

| Foil weight | Thickness (µm) | Thickness (mil) | $R_\square$ at 20 °C (mΩ) |
|---|---|---|---|
| 0.5 oz (0.5 oz/ft$^2$) | 17.4 | 0.69 | 0.99 |
| 1 oz | 34.8 | 1.37 | 0.49 |
| 2 oz | 69.6 | 2.74 | 0.25 |

Copper resistivity is $1.724 \times 10^{-8}\ \Omega \cdot \mathrm{m}$ at 20 °C (annealed) with a temperature coefficient of about 0.39 %/°C. Electrodeposited foils are offered in decreasing roughness grades (standard, reverse-treated, VLP, HVLP/ULP); the fabricator or foil supplier provides roughness parameters for each, preferably as fitted Hammerstad or Huray model parameters.

## B.5 Surface finishes

**Table B.5.** PCB surface finishes and their high-speed relevance.

| Finish | Notes |
|---|---|
| ENIG | Flat, good shelf life; ferromagnetic nickel adds loss on outer-layer traces at multi-GHz |
| ENEPIG | As ENIG, with palladium; wire-bondable |
| Immersion silver | Flat, low loss; tarnish sensitive |
| OSP | Lowest loss, flat; limited shelf life and reflow cycles |
| HASL (lead-free) | Inexpensive; not flat enough for fine-pitch BGAs |

Appendix C

# Design-Rule Checklists

These checklists summarize the design rules of the book in review order. They are starting points; adapt them to the interfaces, the silicon vendors' guides, and the fabricator's capabilities for each project.

## C.1 Schematic review

- ☐ Every high-speed net class identified, with driver edge rate and $\ell_{crit}$ computed (Chapter 2).
- ☐ Terminations present and correct (series values, $V_{TT}$, ODT settings, differential terminations) (Chapter 6).
- ☐ AC-coupling capacitors on the correct side of each serial link, with values in the specified range (Chapters 20 and 21).
- ☐ Reference-clock architecture, SSC, and termination standard consistent at both ends (Chapter 24).
- ☐ Power sequencing, reset, and power-good logic match device requirements (Chapter 27).
- ☐ Decoupling quantities and values per rail from the PDN analysis (Chapter 16); filtered supplies for PLLs and clock buffers.
- ☐ ESD protection at every connector, with capacitance suitable for the data rate (Chapter 21).
- ☐ JTAG chain complete; test access for rails and critical signals.
- ☐ RGMII delay, strapping options, and PHY addresses verified.

## C.2 Stackup

- ☐ Each signal layer adjacent to a solid ground plane (Chapter 7).
- ☐ Each power plane within 2–4 mil of a ground plane.
- ☐ Symmetric construction; copper balance reviewed.
- ☐ Laminate chosen from the loss budget at Nyquist; foil roughness specified (Chapter 8).
- ☐ Glass styles for high-speed layers reviewed for weave skew.
- ☐ Fabricator stackup and impedance calculation received and accepted.
- ☐ Via spans, back-drill needs, and HDI structures defined (Chapter 11).

## C.3 Routing

- ☐ Controlled-impedance widths per layer from the fabricator stackup.
- ☐ No high-speed trace over a plane split or near a plane edge (Chapter 17).
- ☐ Ground return via next to every high-speed layer transition.
- ☐ Spacing to aggressors $\geq 2H$–$3H$ (more for long runs); TX and RX groups separated (Chapters 9 and 20).
- ☐ Intra-pair skew compensated near its source; delay matching in time units (Chapter 10).

- ☐ DDR byte lanes on one layer, matched per controller guide, fly-by order correct, write-leveling range respected (Chapter 19).
- ☐ Via stubs within limits (with drill tolerance) or back-drilled; AC-cap pad treatment checked with a 3-D model.
- ☐ LPDDR5/5X: separate constraints for the CK/CA, WCK/write-data and RDQS/read-data groups (Section 19.1.2).
- ☐ Serpentines meet spacing rules; no test points on multi-gigabit nets.
- ☐ Clocks routed point-to-point, terminated, and kept away from connectors.

## C.4 Power distribution

- ☐ Target impedance computed for each rail (Chapter 16).
- ☐ Rail tolerance allocated among set point, DC drop and dynamic noise (Table 16.1); observation port defined.
- ☐ Capacitor counts from full R–L–C impedance with biased capacitance, not from $\omega$ESL alone (Eq. (16.4)).
- ☐ PDN impedance simulated to the package's handoff frequency; anti-resonances reviewed; load-step transient simulated with the specified current profile.
- ☐ PDN measurement points placed and recorded in the model (Section 26.3).
- ☐ Decoupling capacitors mounted with side vias, close to the plane pair.
- ☐ DC IR drop simulated; remote sense connected at the load.
- ☐ Current density in vias and necks within IPC-2152 limits.
- ☐ Each power and ground ball of high-current devices has its own via.

## C.5 EMI/EMC

- ☐ Slowest acceptable edge rates selected; SSC enabled where supported (Chapter 18).
- ☐ Connector region with solid ground, filtering, and shield bonding.
- ☐ No slots in planes under signals; via fences on board edges.
- ☐ Common-mode chokes and ferrites placed at cables where required.
- ☐ Ethernet isolation gap clean; chassis ground connections defined (Chapter 22).

## C.6 Release

- ☐ DRC with fabricator rules clean; all waivers documented (Chapter 28).
- ☐ Post-layout SI/PI simulation reports archived (Chapter 25).
- ☐ Fabrication drawing: class, materials, stackup, impedance table, drill and back-drill tables, via treatment, finish, test requirements.
- ☐ Coupons specified per controlled-impedance layer.
- ☐ Outputs (Gerber/ODB++/IPC-2581, drill, netlist, assembly, centroid, BOM) regenerated from one tagged revision and inspected in a viewer.
- ☐ Revision marked in copper; ECO process in place for changes.

## C.7 Simulation

- ☐ Buffer models (IBIS/IBIS-AMI) at fast, typical, and slow corners from the silicon vendor (Chapter 25).

- ☐ Laminate $D_k/D_f$ per glass style, causal model, roughness parameters.
- ☐ S-parameter models checked for passivity, causality, reciprocity, port order, DC point, and bandwidth ($\geq 3f_{Nyq}$).
- ☐ Channel segmented at quasi-TEM boundaries; 3-D models for vias, connectors, and BGA breakouts.
- ☐ Crosstalk simulated with the two nearest aggressors on each side.
- ☐ PDN impedance and DC IR drop simulated with the routed planes.
- ☐ Margins reported at corners, including $\pm 10\,\%$ impedance.

## C.8 Bring-up

- ☐ Visual and X-ray inspection; rail-to-ground resistance compared with a reference (Chapter 27).
- ☐ Current-limited first power; voltage at the load; sequencing captured.
- ☐ Ripple measured with a $50\,\Omega$ coaxial connection; thermal image.
- ☐ Clocks: frequency, amplitude, common mode, jitter.
- ☐ JTAG chain IDCODEs; boundary-scan interconnect test.
- ☐ DDR training and 2-D eye margins recorded per bit.
- ☐ Serial links: width, speed, lane margining, PRBS BER.
- ☐ TDR of coupons and critical routes compared with fabricator data and models.
- ☐ Near-field scan for emission sources before the first compliance test.

Appendix D

# Unit Conversions and Rules of Thumb

## D.1 Physical constants

**Table D.1.** Physical constants.

| Constant | Value | Board units |
|---|---|---|
| Speed of light $c$ | $2.998 \times 10^{8}$ m/s | 11.80 in/ns; 84.7 ps/in |
| Permittivity $\varepsilon_0$ | $8.854 \times 10^{-12}$ F/m | 0.2249 pF/in |
| Permeability $\mu_0$ | $4\pi \times 10^{-7}$ H/m | 31.9 nH/in; $\mu_0/2\pi$ = 5.08 nH/in |
| Free-space impedance $\eta_0$ | 376.7 Ω | — |
| Copper resistivity $\rho$ | $1.724 \times 10^{-8}$ Ω · m (20 °C) | 0.49 mΩ per square, 1 oz |

## D.2 Length and copper

**Table D.2.** Length conversions.

| Unit | Equivalent |
|---|---|
| 1 mil | 0.001 in = 25.4 µm |
| 1 mm | 39.37 mil |
| 1 in | 25.4 mm = 1000 mil |
| 1 µm | 0.039 37 mil |
| 1 oz copper | 34.8 µm = 1.37 mil |

## D.3 Decibels

## D.4 Rules of thumb and their limits

**Table D.3.** Decibel conversions.

| Quantity | Relation |
|---|---|
| Voltage ratio | $\mathrm{dB} = 20\log_{10}(V_2/V_1)$; 6.02 dB = 2×, 20 dB = 10× |
| Power ratio | $\mathrm{dB} = 10\log_{10}(P_2/P_1)$ |
| Nepers | 1 Np = 8.686 dB |
| dBm | $10\log_{10}(P/1\,\mathrm{mW})$; 0 dBm = 224 mV rms in 50 Ω |
| dBµV | $20\log_{10}(V/1\,\mu\mathrm{V})$; 0 dBm = 107 dBµV in 50 Ω |
| Field | 1 µV/m = 0 dBµV/m; 100 µV/m = 40 dBµV/m |

**Table D.4.** Rules of thumb used in this book, with their limits.

| Rule | Statement | Limits |
|---|---|---|
| Bandwidth | $f_{3\mathrm{dB}} \approx 0.35/t_\mathrm{r}$ | Exact for single pole; 0.34–0.45 for others |
| Knee frequency | $f_\mathrm{knee} \approx 0.5/t_\mathrm{r}$ | Heuristic; content above it still matters for EMI |
| Propagation | ≈ 6 in/ns, 170 ps/in in FR-4 stripline | $D_\mathrm{k}$ varies with resin, frequency |
| Critical length | $\ell_\mathrm{crit} = t_\mathrm{r}/(6t_\mathrm{pd})$ | $k$ = 4–10 depending on tolerance to ringing |
| 50 Ω line | 3.4 pF/in, 8.5 nH/in | FR-4 stripline; scale with $Z_0$ and $t_\mathrm{pd}$ |
| Wire inductance | ≈ 1 nH/mm | Logarithmic in $\ell/r$; order of magnitude only |
| Sheet inductance | 32 pH per square per mil | Plane pairs, uniform current |
| Return current | 80 % within $\pm 3H$ | Thin trace approximation |
| Small reflection | $\Gamma \approx \Delta Z/(2Z_0)$ | $\lvert\Delta Z\rvert \lesssim 0.2Z_0$ |
| Dielectric loss | $2.31 f_\mathrm{GHz} D_\mathrm{f}\sqrt{D_\mathrm{k}}$ dB/in | Stripline; less in microstrip |
| Conductor loss | $36\sqrt{f_\mathrm{GHz}}/(WZ_0)$ dB/in | Smooth Cu, stripline; times $K_\mathrm{rough}$ |
| 3W rule | $D \geq 3W$ for low coupling | Only when $W \approx 2H$; specify in $H$ |
| Stub length | $f_{\lambda/4} \geq 3f_\mathrm{Nyq}$ | Heuristic; include drill and $\varepsilon_\mathrm{eff}$ tolerance; verify by simulation |
| Lumped stub | Within 5 % below $\lambda/16$ | $\lambda/10$ already gives 14 % error |
| Edge duration | $T_\mathrm{ramp} = t_\mathrm{r}/0.8$ | Linear ramps only |
| Stitching vias | $\leq \lambda/20$ spacing | Heuristic; denser near connectors |
| Ground bounce | $\propto N/t_\mathrm{r}$ (line-driven); $\propto N/t_\mathrm{r}^2$ (lumped $C$) | Doubling $t_\mathrm{r}$ gives 6 dB or 12 dB |
| Guard-via spacing | $\leq t_\mathrm{r} v_\mathrm{p}/10$ | $\lambda/10$ at $f_\mathrm{knee}$ is twice as wide |
| TDR resolution | $t_\mathrm{r} v_\mathrm{p}/2$ | Resolution, not accuracy |

Appendix E

# Python Scripts

The three scripts below implement calculations from the text. They use only the Python standard library (Python 3.8 or later) and can be run directly, e.g. python3 impedance.py. Each script's __main__ block reproduces a worked example; the printed output is shown after each listing. The source files are in appendices/code/. A separate script, tools/gen_figdata.py, regenerates the data-driven figures of the book.

## E.1 Microstrip and stripline impedance

**Listing E.1.** impedance.py — closed-form impedance estimates (Chapter 7).

```
#!/usr/bin/env python3
"""impedance.py - closed-form PCB impedance estimates (Chapter 7).

All dimensions in mils. Results are approximations; see the accuracy
limits stated in Chapter 7 and confirm with a field solver.
"""
import math


def microstrip_ipc(w, h, t, er):
    """IPC-2141 microstrip. Valid for 0.1 < w/h < 2.0 and 1 < er < 15."""
    if not 0.1 < w / h < 2.0:
        print("  warning: w/h outside IPC-2141 range")
    return 87.0 / math.sqrt(er + 1.41) * math.log(5.98 * h / (0.8 * w + t))


def microstrip_hammerstad(w, h, er, t=0.0):
    """Simplified Hammerstad-type closed forms of Chapter 7 with optional
    Wheeler thickness correction. Not the full Hammerstad-Jensen model.

    Returns (Z0, e_eff).
    """
    if t > 0:
        w = w + (t / math.pi) * (1.0 + math.log(2.0 * h / t))
    u = w / h
    a = (er + 1.0) / 2.0
    b = (er - 1.0) / 2.0
    if u <= 1.0:
        eeff = a + b * ((1.0 + 12.0 / u) ** -0.5 + 0.04 * (1.0 - u) ** 2)
        z0 = 60.0 / math.sqrt(eeff) * math.log(8.0 / u + u / 4.0)
    else:
        eeff = a + b * (1.0 + 12.0 / u) ** -0.5
```

```
        z0 = 120.0 * math.pi / (math.sqrt(eeff) * (u + 1.393 + 0.667 * math.log(u + 1.444)))
    return z0, eeff


def microstrip_hj_full(w, h, er, t=0.0):
    """Full Hammerstad-Jensen (1980) quasi-static microstrip, with the
    Hammerstad-Jensen thickness correction. Returns (Z0, e_eff).
    Idealized geometry: rectangular strip, no soldermask, infinite ground."""
    eta0 = 376.73
    u = w / h

    def z01(x):
        f = 6.0 + (2.0 * math.pi - 6.0) * math.exp(-(30.666 / x) ** 0.7528)
        return eta0 / (2.0 * math.pi) * math.log(f / x + math.sqrt(1.0 + 4.0 / x ** 2))

    def ee(x):
        a = (1.0 + math.log((x ** 4 + (x / 52.0) ** 2) / (x ** 4 + 0.432)) / 49.0
             + math.log(1.0 + (x / 18.1) ** 3) / 18.7)
        b = 0.564 * ((er - 0.9) / (er + 3.0)) ** 0.053
        return (er + 1.0) / 2.0 + (er - 1.0) / 2.0 * (1.0 + 10.0 / x) ** (-a * b)

    if t > 0:
        th = t / h
        du1 = th / math.pi * math.log(1.0 + 4.0 * math.e /
                                      (th / math.tanh(math.sqrt(6.517 * u)) ** 2))
        dur = 0.5 * (1.0 + 1.0 / math.cosh(math.sqrt(er - 1.0))) * du1
        u1, ur = u + du1, u + dur
    else:
        u1 = ur = u
    eeff = ee(ur) * (z01(u1) / z01(ur)) ** 2
    return z01(ur) / math.sqrt(ee(ur)), eeff


def stripline_ipc(w, b, t, er, warn=True):
    """IPC-2141 symmetric stripline; b = plane-to-plane spacing."""
    if warn and (w / (b - t) >= 0.35 or t / b >= 0.25):
        print("  warning: outside IPC-2141 stripline range")
    return 60.0 / math.sqrt(er) * math.log(1.9 * b / (0.8 * w + t))


def stripline_wheeler(w, b, t, er):
    """Wheeler's centered stripline with finite thickness (Chapter 7)."""
    if t > 0:
        m = 2.0 / (1.0 + 2.0 / 3.0 * t / (b - t))
        dw = t / math.pi * (1.0 - 0.5 * math.log((t / (2 * b - t)) ** 2
                                                 + (0.0796 * t / (w + 1.1 * t)) ** m))
    else:
        dw = 0.0
    x = (b - t) / (w + dw)
    return 30.0 / math.sqrt(er) * math.log(
        1.0 + 4.0 / math.pi * x * (8.0 / math.pi * x
                                   + math.sqrt((8.0 / math.pi * x) ** 2 + 6.27)))

```

```

def stripline_exact_t0(w, b, er):
    """Exact zero-thickness stripline (conformal mapping), via the AGM."""
    def ellipk(k):
        a, g = 1.0, math.sqrt(1.0 - k * k)
        for _ in range(30):
            a, g = 0.5 * (a + g), math.sqrt(a * g)
        return math.pi / (2.0 * a)
    k = 1.0 / math.cosh(math.pi * w / (2.0 * b))
    kp = math.sqrt(1.0 - k * k)
    return 30.0 * math.pi / math.sqrt(er) * ellipk(k) / ellipk(kp)


def zdiff_microstrip(z0, s, h):
    """IPC-2141 edge-coupled microstrip differential impedance."""
    return 2.0 * z0 * (1.0 - 0.48 * math.exp(-0.96 * s / h))


def zdiff_stripline(z0, s, b):
    """IPC-2141 edge-coupled stripline differential impedance."""
    return 2.0 * z0 * (1.0 - 0.347 * math.exp(-2.9 * s / b))


def tpd_ps_per_inch(eeff):
    """Propagation delay per inch for an effective permittivity."""
    return 84.73 * math.sqrt(eeff)


def solve_width(func, target, lo=0.5, hi=40.0):
    """Bisection for the width that gives a target impedance (Z falls with W)."""
    for _ in range(100):
        mid = 0.5 * (lo + hi)
        if func(mid) > target:
            lo = mid
        else:
            hi = mid
    return 0.5 * (lo + hi)


if __name__ == "__main__":
    # Worked Example 5.1: 4-mil trace on 3.5-mil prepreg, Dk = 3.8, T = 1.9 mil
    w, h, t, er = 4.0, 3.5, 1.9, 3.8
    z_ipc = microstrip_ipc(w, h, t, er)
    z_s0, ee0 = microstrip_hammerstad(w, h, er)
    z_s, ee = microstrip_hammerstad(w, h, er, t)
    z_f, eef = microstrip_hj_full(w, h, er, t)
    print("Microstrip W=%.1f H=%.1f T=%.1f Dk=%.2f" % (w, h, t, er))
    print("  IPC-2141                    Z0 = %.1f ohm" % z_ipc)
    print("  simplified set, T=0         Z0 = %.1f ohm, e_eff = %.2f" % (z_s0, ee0))
    print("  simplified + Wheeler dW     Z0 = %.1f ohm, e_eff = %.2f" % (z_s, ee))
    print("  full Hammerstad-Jensen, T   Z0 = %.1f ohm, e_eff = %.2f, tpd = %.0f
    ps/in"
          % (z_f, eef, tpd_ps_per_inch(eef)))

    # Worked Example 5.2: stripline, B = 10.6 mil, T = 0.6 mil
    b, t2 = 10.6, 0.6
```

```
    w50 = solve_width(lambda x: stripline_ipc(x, b, t2, er, warn=False), 50.0)
    print("Stripline B=%.1f T=%.1f Dk=%.2f" % (b, t2, er))
    print("  IPC width for 50 ohm  W = %.2f mil  (W/(B-T) = %.3f)" % (w50, w50 / (b
    - t2)))
    stripline_ipc(w50, b, t2, er)          # prints the range warning
    w50w = solve_width(lambda x: stripline_wheeler(x, b, t2, er), 50.0)
    print("  Wheeler width for 50 ohm W = %.2f mil" % w50w)
    print("  check T=0, W=4.2: Wheeler %.2f ohm, exact %.2f ohm"
          % (stripline_wheeler(4.2, b, 0.0, er), stripline_exact_t0(4.2, b, er)))
    print("  Zdiff (S = 6 mil) = %.1f ohm" % zdiff_stripline(50.0, 6.0, b))
```

Output:

```
Microstrip W=4.0 H=3.5 T=1.9 Dk=3.80
  IPC-2141                    Z0 = 53.8 ohm
  simplified set, T=0         Z0 = 70.9 ohm, e_eff = 2.81
  simplified + Wheeler dW     Z0 = 60.7 ohm, e_eff = 2.87
  full Hammerstad-Jensen, T   Z0 = 61.4 ohm, e_eff = 2.57, tpd = 136 ps/in
Stripline B=10.6 T=0.6 Dk=3.80
  IPC width for 50 ohm  W = 4.21 mil  (W/(B-T) = 0.421)
  warning: outside IPC-2141 stripline range
  Wheeler width for 50 ohm W = 4.42 mil
  check T=0, W=4.2: Wheeler 58.17 ohm, exact 58.20 ohm
  Zdiff (S = 6 mil) = 93.3 ohm
```

## E.2 PDN target impedance and capacitor sizing

**Listing E.2.** pdn_target.py — target impedance, capacitor impedance, anti-resonance, and IR drop (Chapter 16).

```
#!/usr/bin/env python3
"""pdn_target.py - PDN target impedance and capacitor bank sizing (Chapter 16)."""
import math


def target_impedance(vdd, ripple_fraction, i_transient):
    """Z_target = Vdd * ripple / I_transient (ohms)."""
    return vdd * ripple_fraction / i_transient


def cap_impedance(f, c, esr, esl):
    """|Z| of a capacitor with series ESR and ESL at frequency f (Hz)."""
    w = 2.0 * math.pi * f
    return math.hypot(esr, w * esl - 1.0 / (w * c))


def srf(c, esl):
    """Self-resonant frequency (Hz)."""
    return 1.0 / (2.0 * math.pi * math.sqrt(esl * c))


def caps_needed(z_target, f, c, esr, esl):
    """Pointwise lower bound on the number of identical, independent capacitors
    whose parallel bank meets z_target at frequency f: ceil(|Z_branch|/Z_target).
    Uses the full series R-L-C branch; ignores shared mounting inductance."""
```

```
    return math.ceil(cap_impedance(f, c, esr, esl) / z_target)


def caps_needed_inductive_only(z_target, f, esl):
    """The common shortcut ceil(w*ESL/Z_target); valid only well above the SRF."""
    return math.ceil(2.0 * math.pi * f * esl / z_target)


def branch(f, c, esr, esl):
    """Complex impedance of one series R-L-C branch."""
    w = 2.0 * math.pi * f
    return complex(esr, w * esl - 1.0 / (w * c))


def parallel_peak(branches, f_lo, f_hi, n=200000):
    """Frequency (Hz) and magnitude (ohm) of the largest |Z| of parallel
    R-L-C branches between f_lo and f_hi (log-spaced brute-force sweep)."""
    best = (0.0, 0.0)
    for i in range(n + 1):
        f = f_lo * (f_hi / f_lo) ** (i / n)
        y = sum(1.0 / branch(f, *b) for b in branches)
        z = abs(1.0 / y)
        if z > best[1]:
            best = (f, z)
    return best


def antiresonance_lossless(c1, l1, c2, l2):
    """Admittance zero of two lossless series L-C branches in parallel (Hz)."""
    return math.sqrt((c1 + c2) / ((l1 + l2) * c1 * c2)) / (2.0 * math.pi)


def ir_drop(current, squares, copper_um=34.8, rho=1.724e-8):
    """DC drop across a plane path of a given number of squares."""
    return current * squares * rho / (copper_um * 1e-6)


if __name__ == "__main__":
    zt = target_impedance(0.8, 0.03, 10.0)
    print("Target impedance (0.8 V, 3 %%, 10 A): %.2f mOhm" % (zt * 1e3))
    print("SRF of 100 nF with 0.5 nH: %.1f MHz" % (srf(100e-9, 0.5e-9) / 1e6))
    print("|Z| of that capacitor at 100 MHz: %.3f ohm"
          % cap_impedance(100e6, 100e-9, 0.01, 0.5e-9))
    zb = cap_impedance(20e6, 100e-9, 0.010, 0.5e-9)
    print("100 nF / 10 mOhm / 0.5 nH at 20 MHz: |Z| = %.2f mOhm" % (zb * 1e3))
    print("  capacitors for Z_target (full R-L-C):   %d"
          % caps_needed(zt, 20e6, 100e-9, 0.010, 0.5e-9))
    print("  capacitors (inductive-only shortcut):   %d   <- invalid below SRF"
          % caps_needed_inductive_only(zt, 20e6, 0.5e-9))
    fp, zp = parallel_peak([(100e-9, 0.020, 0.5e-9), (10e-9, 0.030, 0.4e-9)], 10e6,
    200e6)
    print("Anti-resonance 100 nF/20 mOhm/0.5 nH || 10 nF/30 mOhm/0.4 nH:")
    print("  full model peak %.2f MHz, %.3f ohm" % (fp / 1e6, zp))
    print("  lossless admittance zero %.2f MHz"
          % (antiresonance_lossless(100e-9, 0.5e-9, 10e-9, 0.4e-9) / 1e6))
    print("IR drop, 15 A over 3 squares of 1 oz: %.1f mV" % (ir_drop(15, 3) * 1e3))
```

```
    print()
    print("  f (MHz)   |Z| 10uF   |Z| 1uF   |Z| 100nF  (ohm)")
    for fmhz in (0.1, 1, 3, 10, 30, 100, 300):
        f = fmhz * 1e6
        print("  %7.1f   %8.4f   %8.4f   %8.4f" % (
            fmhz,
            cap_impedance(f, 10e-6, 0.003, 0.8e-9),
            cap_impedance(f, 1e-6, 0.005, 0.6e-9),
            cap_impedance(f, 100e-9, 0.010, 0.5e-9)))
```

Output:

```
Target impedance (0.8 V, 3 %, 10 A): 2.40 mOhm
SRF of 100 nF with 0.5 nH: 22.5 MHz
|Z| of that capacitor at 100 MHz: 0.298 ohm
100 nF / 10 mOhm / 0.5 nH at 20 MHz: |Z| = 19.50 mOhm
  capacitors for Z_target (full R-L-C):   9
  capacitors (inductive-only shortcut):   27   <- invalid below SRF
Anti-resonance 100 nF/20 mOhm/0.5 nH || 10 nF/30 mOhm/0.4 nH:
  full model peak 55.16 MHz, 0.443 ohm
  lossless admittance zero 55.64 MHz
IR drop, 15 A over 3 squares of 1 oz: 22.3 mV

  f (MHz)   |Z| 10uF   |Z| 1uF   |Z| 100nF  (ohm)
      0.1     0.1587     1.5912    15.9152
      1.0     0.0113     0.1555     1.5884
      3.0     0.0102     0.0420     0.5212
     10.0     0.0488     0.0224     0.1281
     30.0     0.1503     0.1079     0.0424
    100.0     0.5025     0.3754     0.2984
    300.0     1.5079     1.1305     0.9372
```

The capacitor count (27) is the rounded-up value of the 26.2 estimated in Worked Example 16.2.

## E.3 Reflection and bounce-diagram calculation

**Listing E.3.** bounce.py — lattice-diagram waves and end voltages (Chapter 6).

```
#!/usr/bin/env python3
"""bounce.py - lattice (bounce) diagram for a resistive source and load (Chapter 6).

Prints each traveling wave and the cumulative voltage at the source and load
ends, for a step source of amplitude vs, source resistance rs, line
impedance z0 and one-way delay td, and a load resistance rl (use
float('inf') for an open circuit).
"""


def gamma(z, z0):
    """Reflection coefficient of impedance z on a line of impedance z0."""
    if z == float("inf"):
        return 1.0
    return (z - z0) / (z + z0)


def lattice(vs, rs, z0, rl, td, n_trips=6):
```

```
    gs, gl = gamma(rs, z0), gamma(rl, z0)
    wave = vs * z0 / (z0 + rs)              # launched wave
    v_src, v_load = wave, 0.0
    events = [(0.0, "source", "launch", wave, v_src)]
    t = 0.0
    for _ in range(n_trips):
        t += td                              # wave reaches the load
        refl = wave * gl
        v_load += wave + refl
        events.append((t, "load", "%+.4f in, %+.4f out" % (wave, refl), refl, v_load))
        wave = refl
        t += td                              # wave reaches the source
        refl = wave * gs
        v_src += wave + refl
        events.append((t, "source", "%+.4f in, %+.4f out" % (wave, refl), refl,
    v_src))
        wave = refl
    v_final = vs if rl == float("inf") else vs * rl / (rs + rl)
    return gs, gl, events, v_final


if __name__ == "__main__":
    # Worked Example 4.1: 3.3 V, Rs = 10 ohm, Z0 = 50 ohm, open load, TD = 1 ns
    gs, gl, events, vf = lattice(3.3, 10.0, 50.0, float("inf"), 1.0, n_trips=4)
    print("Gamma_S = %+.4f, Gamma_L = %+.4f, final value = %.3f V" % (gs, gl, vf))
    print("  t (ns)   end      waves                          V (V)")
    for t, end, desc, _, v in events:
        print("  %5.1f    %-7s  %-26s  %7.3f" % (t, end, desc, v))
```

Output:

```
Gamma_S = -0.6667, Gamma_L = +1.0000, final value = 3.300 V
  t (ns)   end      waves                          V (V)
     0.0   source   launch                           2.750
     1.0   load     +2.7500 in, +2.7500 out          5.500
     2.0   source   +2.7500 in, -1.8333 out          3.667
     3.0   load     -1.8333 in, -1.8333 out          1.833
     4.0   source   -1.8333 in, +1.2222 out          3.056
     5.0   load     +1.2222 in, +1.2222 out          4.278
     6.0   source   +1.2222 in, -0.8148 out          3.463
     7.0   load     -0.8148 in, -0.8148 out          2.648
     8.0   source   -0.8148 in, +0.5432 out          3.191
```

These values match Figure 6.1 and Worked Example 6.1.

Appendix F

# Case Study: A 10-Layer SoC Single-Board Computer

This appendix applies the methods of the book to one design from requirements to bring-up: a compact single-board computer built around an application SoC with LPDDR4X memory (Figure F.1), an M.2 NVMe slot on PCI Express Gen3 ×4, a USB Type-C port, gigabit Ethernet, and a four-lane MIPI CSI-2 camera input. The SoC and all interface parameters are generic; on a real project they would come from the SoC vendor's hardware design guide, which always takes precedence over the estimates shown here.

## F.1 Requirements

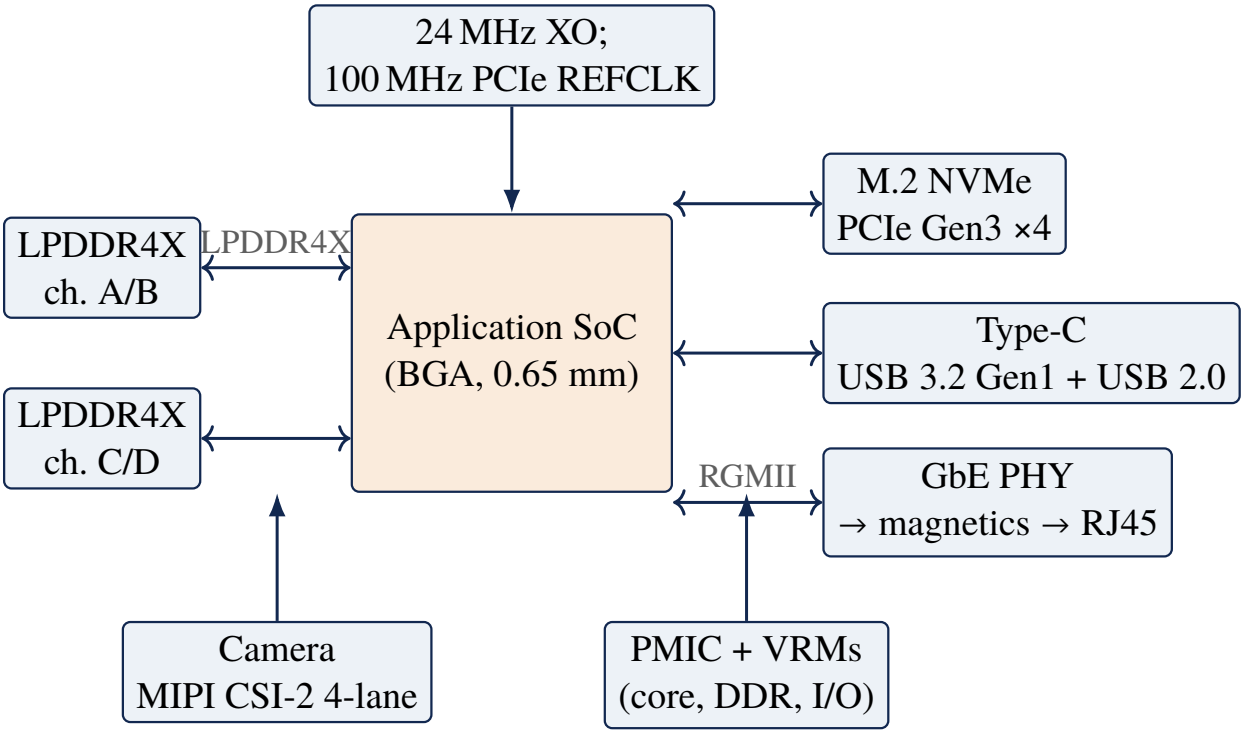



**Figure F.1.** Block diagram of the case-study board.

The fastest single-ended edges are the LPDDR4X data lines; the highest Nyquist frequency is PCIe Gen3 at 4 GHz. With edges of about 60 ps on the PCIe lanes, $f_{\text{knee}} \approx 8$ GHz, so channel models must be valid to at least 12 GHz (three times Nyquist).

**Table F.1.** Interface requirements of the case-study board (illustrative).

| Interface | Rate | UI | $f_{\mathrm{Nyq}}$ | Impedance | Notes |
|---|---|---|---|---|---|
| LPDDR4X | 4266 MT/s | 234 ps | 2.13 GHz | 40 Ω SE / 80 Ω diff | Point-to-point, 4 channels |
| PCIe Gen3 ×4 | 8 GT/s | 125 ps | 4 GHz | 85 Ω diff | M.2 socket, AC caps on TX |
| USB 3.2 Gen1 | 5 Gb/s | 200 ps | 2.5 GHz | 90 Ω diff | Type-C, mux, ESD |
| USB 2.0 | 480 Mb/s | 2.08 ns | 240 MHz | 90 Ω diff | Short orientation stubs |
| MIPI CSI-2 (D-PHY) | 2.5 Gb/s | 400 ps | 1.25 GHz | 100 Ω diff | 4 lanes + clock, FPC |
| RGMII | 250 Mb/s | 4 ns | 125 MHz | 50 Ω SE | Internal delay in PHY |
| 1000BASE-T | 125 MBd | 8 ns | 62.5 MHz | 100 Ω diff | Magnetics, chassis zone |

### F.1.1 Requirements ledger

Each requirement is carried through the design in a *ledger* (Table F.2) that records where it came from, the budget derived from it, how it will be verified, and the evidence behind it. A row is closed only when its verification has been done on the real board, not when a simulation passes. Keeping the ledger from the start is what makes the bring-up plan of Appendix F.8 possible.

**Table F.2.** Requirements ledger for the case-study board (excerpt; values illustrative).

| Requirement | Source | Budget or constraint | Verification | Evidence |
|---|---|---|---|---|
| PCIe Gen3 channel loss | Base spec. and SoC guide | Board share at 4 GHz (Worked Example F.1) | Post-layout extraction; TDR/VNA of one board | Sim. then meas. |
| LPDDR4X DQ–DQS skew | Controller guide | 3 % UI static (Worked Example F.2) | Training report, 2-D eye scan per bit | Meas. |
| Core rail ±3 % | SoC data sheet | 3.2 mΩ dynamic target (Worked Example F.3) | PDN sweep and load step; shunt-through at probe point | Sim. then meas. |
| Controlled impedance | Stackup plan | 85 Ω/90 Ω/100 Ω ±10 % | Coupons with stated edge and window (Section 28.2.1) | Meas. |
| Radiated emissions | FCC/CISPR Class B | Margin target at 3 m | Pre-scan with cables attached | Meas. |

## F.2 Floorplan

The floorplan (Figure F.2) follows three principles from the text: the memory sits immediately beside the SoC's memory ball field, so that DQ byte lanes are short and matched (Chapter 19); the connectors for fast interfaces share one edge, with their ESD protection and AC caps between connector and SoC (Chapters 20 and 21); and the Ethernet port forms its own corner with an isolation gap and chassis ground (Chapter 22). The PMIC and core regulator sit as close to the SoC as thermal design allows, to keep the IR drop and the inductance of the high-current path low (Chapter 16).

## F.3 Stackup and impedance plan

The 10-layer, 62.6 mil stackup of Figure 7.3 fits this board:

- L1 and L10 (microstrip): BGA breakout, short escapes, connectors, and components;

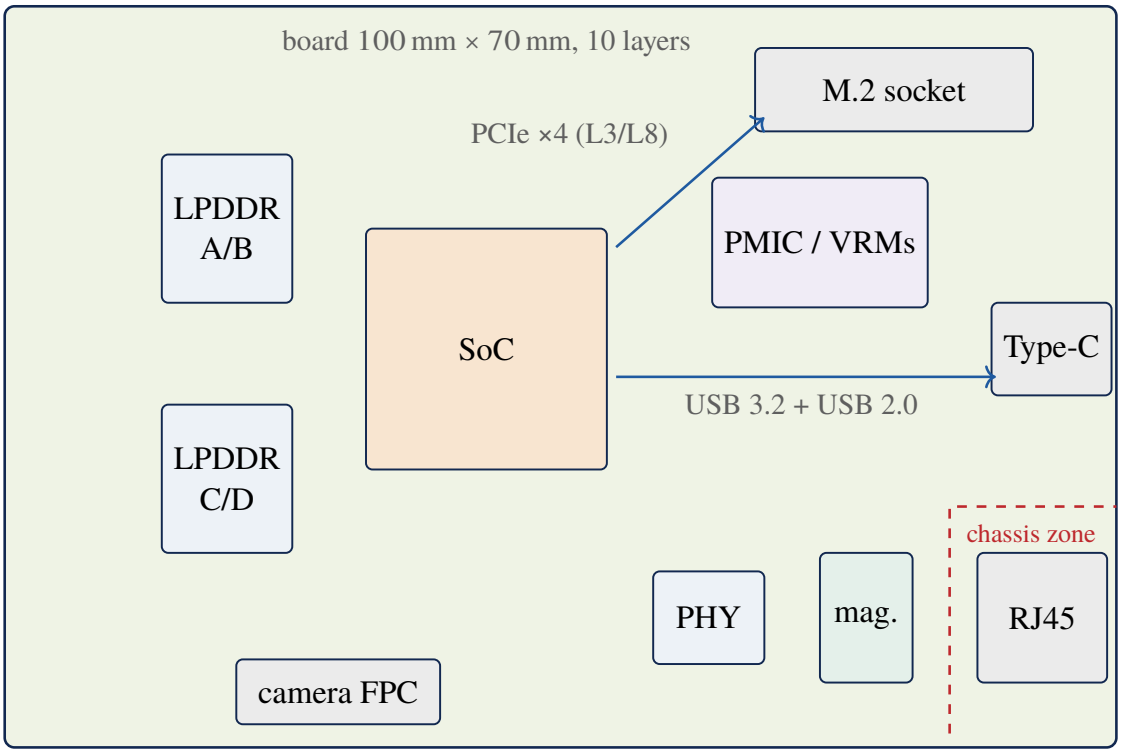



**Figure F.2.** Floorplan of the case-study board (schematic). The memory is placed beside the SoC's DDR ball field, high-speed connectors along one edge, and the Ethernet magnetics and RJ45 in a corner chassis zone.

- L3 and L8 (stripline, ground on both sides): PCIe, USB SuperSpeed, MIPI, and the longest DDR routes;
- L5 (core rail) and L6 (memory and I/O rails), each coupled to ground through a 3 mil core;
- L2, L4, L7, L9: ground.

A mid-loss laminate ($D_k \approx 3.7$, $D_f \approx 0.010$ at a few gigahertz) with HVLP foil on L3 and L8 is chosen from the loss budget below. The fabricator is asked to confirm widths for 40 Ω and 50 Ω single-ended and 80 Ω, 85 Ω, 90 Ω, and 100 Ω differential on each signal layer, with $\pm 10$ % tolerance and permission to adjust widths. Starting values from Worked Examples 7.1 and 7.2 are about 4.2 mil for 50 Ω on L3/L8; differential geometries are taken directly from the fabricator's solver.

## F.4 Loss budgets

**Worked Example F.1: PCIe Gen3 route to the M.2 socket**

The PCIe lanes run 4 in on L3 from the SoC to the M.2 socket. Estimate the board's insertion loss at 4 GHz and compare it with a Gen3 budget of about 22 dB end to end.

**Solution.** With $W \approx 5$ mil, 50 Ω-equivalent, HVLP ($\Delta = 1$ µm): $\delta_s = 1.045$ µm, $K_H = 1.58$, conductor loss $0.288 \times 1.58 = 0.45$ dB/in. Dielectric loss $2.3 \times 4 \times 0.010 \times \sqrt{3.7} = 0.18$ dB/in. Total 0.63 dB/in, or 2.5 dB for 4 in. Adding about 0.5 dB for vias and AC caps and 1 dB for the M.2 connector, the board's share is about 4 dB, leaving most of the budget for the SoC package, the SSD's package and traces, and margin. Via stubs from L3 run to L10 (about 50 mil), resonating near 30 GHz (Worked Example 11.4)—well above $3f_{\text{Nyq}} = 12$ GHz, so back-drilling is not needed for Gen3.

The USB 3.2 Gen1 lanes (5 in to the Type-C mux and connector) lose about 0.47 dB/in at 2.5 GHz, or 2.4 dB—small compared with what the PHY can equalize, so the dominant risks are the mux, the ESD device, and the connector's return loss rather than trace loss. The MIPI lanes are short on the board; their channel is dominated by the camera FPC and its two connectors, which must be specified with the module vendor (Chapter 23).

## F.5 Timing and matching

**Worked Example F.2: LPDDR4X byte-lane matching**

Allocate 3 % of the LPDDR4X-4266 UI to static DQ-to-DQS mismatch. What length tolerance results in stripline at 165 ps/in?

**Solution.** UI = 234 ps; 3 % is 7.0 ps, or 43 mil. The controller guide's tighter rule (for example 10–20 mil within a byte) is used for layout; the computed value shows how much of the timing budget that rule represents.

For the MIPI CSI-2 lanes at 2.5 Gb/s (UI 400 ps), 5 % of a UI corresponds to about 120 mil of lane-to-clock mismatch (Worked Example 23.1); intra-pair matching is held to a few mils. The PCIe lanes need tight intra-pair matching (compensated near each bend) but no lane-to-lane matching. The RGMII interface uses the PHY's internal clock delay, with data matched to clock within about 100 mil.

## F.6 Power distribution

**Worked Example F.3: Core-rail decoupling plan**

The SoC core rail is 0.8 V with a ±3 % allowance for dynamic noise and a 7.5 A transient. Find the target impedance and the number of 100 nF capacitors (ESR 10 mΩ, mounted ESL 0.4 nH) needed at 20 MHz.

**Solution.** $Z_{\text{target}} = 0.024/7.5 = 3.2$ mΩ. The capacitor's SRF is 25.2 MHz, so at 20 MHz it is still capacitive: $X_L = 50.3$ mΩ, $X_C = 79.6$ mΩ, and $|Z_C| = 31.0$ mΩ. By Eq. (16.4), $N \geq \lceil 31.0/3.2 \rceil = 10$ (the inductive-only shortcut would give 16). The same bank needs 8 at 30 MHz but 42 at 10 MHz and 30 at 50 MHz: one value cannot cover the band, so 1 µF and 10 µF banks take the lower frequencies and the planes and package the higher ones. The plan uses at least 10 of the 100 nF parts with side-via mounting on L10 directly below the core balls, connected through short vias to the L5–L4/L6 plane region, and the count is confirmed by a sweep of the whole network and a load-step simulation (Section 16.6). Bulk polymer capacitors near the regulator cover the range below 1 MHz; the die–package anti-resonance

above that is left to the SoC's package design, and the board's contribution is checked by PDN simulation (Figure 16.2).

The DC drop from the regulator to the SoC across the L5 plane is estimated with Eq. (16.11): with the regulator about two squares away and 15 A maximum, 1 oz copper gives about 15 mV. The regulator's remote sense is connected at the SoC's core balls to remove this static error.

## F.7 EMC and protection

- All fast nets are routed on L3/L8 between ground planes where possible; microstrip segments are limited to breakouts.
- The Type-C connector has low-capacitance ESD protection on SuperSpeed and USB 2.0 lines, a common-mode choke option on USB 2.0, and VBUS protection.
- The Ethernet corner has an isolation gap under the magnetics, a Bob Smith network, and a chassis-ground connection to the RJ45 shield.
- The PCIe reference clock and the SoC's memory clocks run with SSC where supported; clock buffers have filtered supplies.
- A via fence along the board edge stitches the ground planes at about 100 mil spacing, tighter near the connectors.

## F.8 Bring-up plan

The bring-up follows Figure 27.1: rail resistances against a reference board, current-limited first power with the PMIC's sequence captured, clock checks, JTAG chain and boundary-scan connectivity, then firmware. The DDR training report and two-dimensional eye margins are recorded for every byte lane on the first five boards; PCIe link width and speed, and lane margining (where the endpoint supports it), are logged for the M.2 slot; USB is tested in both plug orientations; and the camera link is tested with PRBS patterns if the sensor supports them. TDR measurements of the impedance coupons and of one board's PCIe and USB routes are compared with the fabricator's report and with the post-layout models, closing the loop for the next design.

**Table F.3.** Summary of the case-study design decisions.

| Decision | Basis |
|---|---|
| 10 layers, 4 signal + 6 planes | Every signal layer referenced to ground; two P/G pairs (Chapter 7) |
| Mid-loss laminate, HVLP on L3/L8 | PCIe Gen3 loss budget with margin (Worked Example F.1) |
| No back-drilling | Stub resonance $\gg 3f_{\mathrm{Nyq}}$ at Gen3 (Chapter 11) |
| DQ–DQS matching per controller guide | $\approx 3\,\%$ of UI equals 43 mil (Worked Example F.2) |
| $\geq$10 100 nF plus 1 µF/10 µF banks | $Z_{\mathrm{target}} = 3.2\,\mathrm{m\Omega}$; full R–L–C sizing and sweep (Worked Example F.3) |
| Remote sense at SoC | 15 mV static plane drop (Chapter 16) |
| RGMII internal delay in PHY | Avoids 13 in of delay routing (Worked Example 22.2) |

Appendix G

# Interface Quick-Reference Sheets

Each sheet condenses the board-level rules for one interface into a single page for use during schematic and layout review. The values are typical starting points drawn from the chapters of Part V; the controlling documents are the interface specification and the silicon vendor’s design guide, which must be checked for every project.

## G.1 DDR4 / DDR5 / LPDDR4 / LPDDR5

| | |
|---|---|
| **Signals** | DQ, DM/DBI, DQS (byte lanes); CA, control; CK. LPDDR5/5X: WCK per byte and RDQS where used |
| **Clocking** | DDR4/5, LPDDR4/4X: DQ source synchronous to DQS; CA to CK. LPDDR5/5X: writes to WCK, reads to RDQS, CA to CK (Table 19.2) |
| **Topology** | Fly-by for CA/CK on multi-device ranks; point-to-point for LPDDR |
| **Impedance** | 36–50 Ω single-ended (often 40 Ω); 80–100 Ω diff. CK/DQS |
| **Termination** | ODT on DQ; $V_{TT}$ resistors at fly-by end (DDR4 CA); on-die CA termination (DDR5) |
| **Matching** | Per timing group, per controller guide; CK–DQS (or WCK–CK) within the leveling range |
| **Layers** | Byte lane on one layer, same via count, ground referenced |
| **Spacing** | $\geq 3H$ between byte lanes; more for long parallel runs |
| **VREF** | Internal and trained for DQ; external CA VREF (DDR4) decoupled and isolated |
| **PDN** | Low-impedance $V_{DDQ}$; $V_{TT}$ that sources and sinks |
| **Bring-up** | Capture training results and 2-D eye margins per bit |
| **Common failures** | Leveling range exceeded; byte-lane crosstalk; $V_{TT}/V_{DDQ}$ noise |
| **References** | Chapter 19; JEDEC [28], [37]–[39] |

## G.2 PCI Express

| | |
|---|---|
| **Signals** | TX and RX differential pairs per lane; REFCLK (HCSL); PERST#, CLKREQ#, WAKE# |
| **Rates** | Gen3 8, Gen4 16, Gen5 32 GT/s NRZ; Gen6 64 GT/s PAM4 |
| **Impedance** | 85 Ω differential nominal; ±10 % |
| **Coupling** | AC caps on TX, symmetric; void reference under pads only where a 3-D model shows it helps; $f_{\mathrm{HP}} = 1/[2\pi(R_S + R_L)C]$ |
| **Skew** | Intra-pair tight and compensated near bends; lane-to-lane loose |
| **Loss** | Budget at Nyquist across packages, board, connector, card |
| **Vias** | Back-drill or deep-layer routing at Gen4+; return via at each transition |
| **Crosstalk** | Separate TX and RX groups; ≥ $3H$–$5H$ pair spacing |
| **REFCLK** | 100 MHz, CC/SRNS/SRIS architecture chosen early; SSC compatibility |
| **Laminate** | Mid-loss to Gen4; low or very-low loss at Gen5/Gen6 |
| **Bring-up** | Link width and speed; lane margining; eye monitors |
| **References** | Chapter 20; PCI-SIG [41]–[43] |

## G.3 USB and Type-C

| | |
|---|---|
| **Signals** | USB 2.0 D+/D−; SuperSpeed/USB4 TX/RX pairs; CC1/CC2; SBU; VBUS |
| **Impedance** | 90 Ω differential nominal (check per generation) |
| **USB 2.0** | Tie A6/B6 and A7/B7 at the receptacle with short stubs |
| **SuperSpeed** | TX AC caps; orientation mux or two-lane PHY; redriver/retimer if needed |
| **ESD** | At connector; sub-picofarad on high-speed lanes; short ground |
| **CC** | $R_p/R_d$ or PD controller; $R_d = 5.1\,\text{k}\Omega$ in sinks |
| **VBUS** | Wide copper; inrush limits; overvoltage protection |
| **EMI** | CM choke option on USB 2.0; shield bonded to chassis at connector |
| **Bring-up** | Test both plug orientations; all supported speeds |
| **References** | Chapter 21; USB-IF [45]–[48] |

## G.4 Ethernet

| | |
|---|---|
| **Media side** | Four 100 Ω MDI pairs; magnetics; RJ45 |
| **Isolation** | Gap under magnetics free of copper on all layers; creepage per safety standard |
| **Termination** | Center taps per PHY type; Bob Smith network to chassis |
| **Layout** | Magnetics close to RJ45; pairs matched within pair; minimal vias |
| **RGMII** | One 1.5–2 ns clock delay per direction (internal preferred) |
| **SGMII** | 1.25 Gb/s, 100 Ω, usually AC coupled |
| **10GBASE-KR** | 10.3125 Gb/s; loss/RL/crosstalk per IEEE 802.3 |
| **EMI** | LED and other lines routed on PHY side of the gap |
| **Bring-up** | Link at all speeds; RGMII delay settings in both devices |
| **References** | Chapter 22; IEEE 802.3 [49] |

## G.5 MIPI CSI-2 / DSI

| | |
|---|---|
| **PHYs** | D-PHY (pairs + clock lane) or C-PHY (trios, embedded clock) |
| **Impedance** | D-PHY 100 Ω differential; C-PHY 50 Ω per wire |
| **Modes** | HS (terminated, low swing) and LP (1.2 V, unterminated) on same wires |
| **Matching** | D-PHY lane-to-clock and intra-pair; C-PHY within each trio |
| **Coupling** | DC coupled; no AC caps |
| **Crosstalk** | Spacing for LP-to-HS coupling (LP swing ≈ 6× HS) |
| **Flex** | Specify FPC stackup, impedance, ground pins, and hatch pitch |
| **Bring-up** | PRBS or test patterns where supported; check both modes |
| **References** | Chapter 23; MIPI [50]–[53] |

## G.6 Clocks

| | |
|---|---|
| **Sources** | XO, TCXO, VCXO, MEMS; crystals with $C_L$-matched load caps |
| **Standards** | LVCMOS (series term.), LVDS, LVPECL, HCSL, CML—terminate per data sheet |
| **Topology** | Point to point; one load per output; fanout buffers |
| **Jitter** | Integrate phase noise over the application's band; RSS independent terms |
| **PLL** | Loop bandwidth at crossover of scaled reference and VCO noise; phase margin from $C_1/C_2$ |
| **Supplies** | Filtered supply (ferrite or LDO) for each PLL and buffer |
| **Routing** | Inner layers, ground referenced, $\geq 3H$–$5H$ from other nets |
| **EMI** | SSC where allowed; slowest acceptable edges; keep away from connectors |
| **References** | Chapter 24 |

Appendix H

# Glossary

**AC coupling** Series capacitors that block DC between transmitter and receiver while passing the signal (Chapter 20).

**Aggressor** The net whose switching induces crosstalk in another (the victim) (Chapter 9).

**Anti-pad** Clearance hole in a plane around a via that is not connected to it (Chapter 11).

**Anti-resonance** Impedance peak formed by an inductive branch in parallel with a capacitive branch of a PDN (Chapter 16).

**Back-drilling** Removing the unused portion of a plated via barrel with an oversized drill to shorten its stub (Chapter 11).

**Baseline wander** Drift of the received level of an AC-coupled link during long runs of identical symbols (Section 20.3).

**Bathtub curve** Plot of BER versus sampling phase across a unit interval (Chapter 12).

**BER** Bit-error ratio: the fraction of received bits in error.

**Bob Smith termination** Resistor network from Ethernet cable-side center taps to chassis through a high-voltage capacitor (Chapter 22).

**Boundary scan** IEEE 1149.1 test method that drives and samples device pins through a serial chain (Chapter 27).

**Causality** Property of a physical system that its response does not precede its stimulus; required of dielectric and S-parameter models (Chapters 8 and 25).

**CDR** Clock and data recovery: a receiver circuit that extracts timing from the data transitions (Chapter 12).

**Common mode** The average of the voltages on the two conductors of a pair (Chapter 10).

**Critical length** Length beyond which an interconnect must be treated as a transmission line, $\ell_{\text{crit}} = t_r/(k t_{\text{pd}})$ (Chapter 2).

**CTLE** Continuous-time linear equalizer: an analog peaking filter in a receiver (Chapter 12).

**Cursor** One sample per UI of the channel's pulse response; the main cursor is the largest, with precursors before and postcursors after it (Section 12.4.3).

**DBI** Data-bus inversion: optional inversion of a byte to reduce simultaneous switching (Chapter 15).

**De-embedding** Mathematical removal of test fixtures from a measurement (Chapter 26).

**DFE** Decision-feedback equalizer: cancels post-cursor ISI using previous decisions (Chapter 12).

**Dk, Df** Dielectric constant (relative permittivity) and dissipation factor (loss tangent) of a laminate (Chapters 7 and 8).

**Dual-Dirac model** Jitter model representing DJ by two delta functions convolved with Gaussian RJ (Chapter 12).

**ESL, ESR** Equivalent series inductance and resistance of a capacitor (Chapter 16).

**Evidence label** The bracketed tag at the end of each figure caption stating whether the figure is analytical, simulated, extracted, measured or conceptual (Section 1.5).

**Eye diagram** Overlay of many unit intervals of a waveform showing amplitude and timing margin (Chapter 12).

**FEXT** Far-end crosstalk: forward-traveling coupled noise arriving at the victim's far end (Chapter 9).

**FFE** Feed-forward equalizer: a finite-impulse-response filter, usually in the transmitter (Chapter 12).

**Fly-by topology** Daisy-chained routing of DDR command, address, and clock past each memory device (Chapter 19).

**Glass-weave skew** Skew caused by local $D_k$ differences between glass and resin in a laminate (Chapter 7).

**Ground bounce** Shift of a die's ground relative to the board due to $L\,\mathrm{d}i/\mathrm{d}t$ in shared return paths (Chapter 15).

**IBIS** I/O Buffer Information Specification: behavioral buffer models (Chapter 25).

**Insertion loss** $-20\log_{10}|S_{21}|$: attenuation of a signal through a channel (Chapter 8).

**ISI** Inter-symbol interference: residual energy from one bit corrupting others, mainly due to loss and reflections (Chapters 8 and 12).

**Knee frequency** $f_{\text{knee}} \approx 0.5/t_r$, the upper limit of a digital edge's significant spectrum (Chapter 2).

**Lattice diagram** Distance–time diagram of reflections on a line (Chapter 6).

**Loop inductance** Inductance of a closed current path, including its return (Chapter 3).

**Microvia** Laser-drilled blind via spanning one thin dielectric layer (Chapter 11).

**Mode conversion** Transfer of energy between differential and common modes due to asymmetry (Chapter 10).

**NEXT** Near-end crosstalk: backward-traveling coupled noise at the victim's near end (Chapter 9).

**ODT** On-die termination (Chapters 6 and 19).

**PAM4** Four-level pulse amplitude modulation, two bits per symbol (Chapter 20).

**Passivity** Property that a network does not generate energy; required of S-parameter models (Chapter 25).

**PDN** Power distribution network from regulator to die (Chapter 16).

**Phase noise** Single-sideband noise power density of an oscillator relative to its carrier, in dBc/Hz (Chapter 24).

**PSIJ** Power-supply-induced jitter: edge-timing variation caused by supply noise acting on driver delay (Chapter 15).

**Pulse response** A channel's received response to a single one-UI pulse; sampled once per UI it gives the cursors used for equalization analysis (Section 12.4.3).

**RDQS** LPDDR5/5X differential read data strobe, returned by the DRAM with read data in configurations that use it (Section 19.1.2).

**Retimer** A protocol-aware device that recovers and re-transmits a serial signal, splitting a channel into two electrically independent segments (Chapter 20).

**Return loss** $-20\log_{10}|S_{11}|$: a measure of reflection (Chapter 25).

**Rise time** Time for an edge to go from 10 % to 90 % of its swing (Chapter 2).

**S-parameters** Scattering parameters relating incident and reflected waves of a network (Chapter 25).

**Saturation length** Coupled length beyond which NEXT stops growing, $t_r v_p/2$ (Chapter 9).

**Shunt-through measurement** Two-port VNA method for milliohm impedances, with both ports connected to the same node; $Z \approx (Z_0/2)S_{21}$ (Section 26.3).

**Skin depth** Depth at which current density falls by $1/e$ in a conductor at high frequency (Chapter 3).

**SSC** Spread-spectrum clocking (Chapter 18).

**SSN** Simultaneous switching noise (Chapter 15).

**Stitching capacitor** Capacitor that provides a high-frequency return path across a plane split or between planes (Chapter 17).

**Stitching via** Via connecting ground planes on different layers (Chapter 17).

**Stub** Unused branch of a signal path, such as the barrel of a via below its last connection (Chapter 11).

**T network** Differential termination that matches both modes: $Z_{odd}$ from each line to a center node and $(Z_{even} - Z_{odd})/2$ to ground (Chapters 6 and 10).

**Target impedance** Maximum PDN impedance that keeps supply noise within tolerance, $\Delta V/\Delta I$ (Chapter 16).

**TDR** Time-domain reflectometry (Chapter 26).

**Unit interval (UI)** Duration of one symbol (Chapter 2).

**Victim** The net receiving crosstalk (Chapter 9).

**WCK** LPDDR5/5X differential data (write) clock, separate from the command clock CK (Section 19.1.2).

**Write leveling** DDR training that aligns each byte lane's DQS to CK at its memory device (Chapter 19).

Appendix I

# Study Paths and Integrated Exercises

The chapters are written so that each can be read on its own, but design problems cross chapter boundaries. This appendix gives two things: study paths through the book for readers with different goals, and integrated exercises that carry one problem from the interconnect to a decision. The exercises use illustrative requirements and make no compliance claims; each states what its model still lacks.

## I.1 Study paths

Table I.1 groups the material into a core path and six advanced modules. The core path is enough to design and review a typical ten-layer board; each module adds the analysis needed for one class of hard problem. Readers already fluent in the foundations can take the diagnostic problems marked *(Approximation failure)* in Chapters 2, 4 and 6 as a self-test before skipping ahead.

**Table I.1.** Core path and advanced modules.

| Path | Sections | Integrated exercise |
|---|---|---|
| Core | Chapters 1 and 2; Chapter 4; Chapters 5 and 6; Chapter 13; Chapter 7; Section 16.1.1; Appendix F | Case-study ledger (Table F.2) |
| A: Channel modeling | Sections 4.6.1, 8.1, 25.3.4 and 26.2.1 | — |
| B: SERDES and PAM4 | Sections 12.4.3, 12.4.5 and 20.6 | Appendix I.2 |
| C: Source-synchronous timing | Sections 12.1, 19.1.1 and 19.1.2 | — |
| D: PDN co-design | Chapter 14; Sections 16.1.1, 16.4, 16.6 and 26.3 | Appendix I.3 |
| E: Tolerance and validation | Sections 7.7, 11.5.1, 26.1.2 and 28.2.1 | Appendix I.4 |
| F: EMC diagnosis | Sections 2.3.1 and 18.2; Chapter 17 | — |

Three kinds of review problem recur in the chapters and are labeled in the text: *approximation failure* problems ask where a quick formula breaks; *inverse design* problems start from a requirement and solve for a dimension or a tolerance; and *imperfect data* problems give a measurement or a coupon set that does not, by itself, support the obvious conclusion.

## I.2 Exercise A: from pulse response to receiver margin

*Purpose:* connect the interconnect chapters, the equalization of Chapter 12 and the simulation workflow of Chapter 25. Synthetic sampled channel; no compliance claim.

**Table I.2.** Design inputs for Exercise A.

| Input | Value or condition |
|---|---|
| Symbol rate and signaling | 16 GBd NRZ; UI = 62.5 ps; levels $\pm 1$ |
| Sampled channel | $h_0 = 1$; $h_{-1} = -0.08$; $h_1, h_2, h_3 = 0.25, 0.12, 0.05$ |
| Noise at the slicer | Independent Gaussian, $\sigma = 0.03$ (cursor units) |
| Initial receiver | Threshold 0; fixed sampling phase; no jitter or CDR error |
| Candidate receiver | Three-tap DFE with exact taps and correct prior decisions |

**Step 1: amplitude budget.** Without feedback the worst-case ISI is $0.08 + 0.25 + 0.12 + 0.05 = 0.50$, so the minimum distance to threshold is 0.50 ($16.7\sigma$). With the ideal DFE only the precursor remains: 0.92 ($30.7\sigma$). These are static amplitude distances, not BER values (Worked Example 12.5).

**Step 2: a plausible failure.** Add a reflection postcursor of 0.15 at $k = 8$ (500 ps). It lies outside the tap span; the half-height falls to 0.77. Find the via, connector or package discontinuity whose round-trip delay is 500 ps ($\approx 1.5$ in of stripline one way), and compare two remedies: a longer-span or floating-tap DFE, and a board change that removes the echo.

**Step 3: what the model lacks.** Sampling-phase error and CDR response, noise correlation, tap quantization and adaptation limits, transmitter swing limits, and decision errors (one wrong decision on the first tap costs 0.50). The correct-decision subtraction is an upper bound; an adaptive bit-by-bit simulation should report convergence time and error propagation, and a statistical analysis should state its assumptions (Section 12.4.5).

**Required submission.** Cursor table, tap values, residual-ISI budget, before-and-after eyes at the same reference plane, and a short sensitivity sweep. The decision must name the impairment that controls the margin and argue whether the board or the receiver is the more robust place to fix it.

**Extension.** Repeat with PAM4 levels $\pm 1$, $\pm \frac{1}{3}$ and three thresholds; track the three eye margins and level-dependent distortion, and explain why FEC changes the reliability interpretation without restoring analog eye height (Section 20.6).

## I.3 Exercise B: from a rail tolerance to a validated PDN

*Purpose:* turn the target-impedance calculation of Chapter 16 into a design process. Allocations are illustrative requirements for the exercise.

**Step 1: define the observation port.** Board load pads or die bumps? State whether the load current is board or die current, include the package model and on-die capacitance when predicting die voltage, and reserve measurement uncertainty inside the allocation before using it as an acceptance threshold.

**Step 2: full model and meaningful corners.** Use each capacitor's biased capacitance, ESR and mounted ESL (Eq. (16.4)); include shared vias, spreading inductance, regulator output impedance and the package. Sweep frequency and port location, and check the corners and the correlations that the mounting geometry creates (Figures 16.4 and 16.6).

**Table I.3.** Rail budget for Exercise B.

| Item | Allocation |
|---|---|
| Nominal and allowed deviation | 0.8 V ±3 % = ±24 mV |
| Set point and regulator error | ±4 mV |
| Uncompensated DC path drop | 6 mV maximum |
| Dynamic allocation | 14 mV for the specified worst-direction transient |
| Load change | 10 A increase, defined ramp and dwell |
| Initial dynamic target | 14 mV / 10 A = 1.4 mΩ at the chosen port |

**Step 3: excite with the real current.** Apply the specified rise time, amplitude, repetition and dwell; a periodic load can excite a narrow resonance that an isolated step does not (Section 16.6). Compare peak droop, overshoot and settling with the asymmetric budget.

**Step 4: plan correlation before fabrication.** Place a shunt-through measurement point (Section 26.3), document the measurement floor, and compare simulated and measured impedance at the same port. Record temperature, probe setup and bias. Explain residual differences rather than forcing a fit by changing several model parameters at once.

**Required submission.** Allocation table, network diagram, biased capacitor inventory, frequency and transient plots, corner table, measurement plan, and one paragraph defending the remaining margin.

## I.4 Exercise C: from a coupon report to a production decision

*Purpose:* practice reasoning from imperfect data (Chapters 7, 26 and 28).

A first lot of boards arrives with coupon reports for the 85 Ω differential class: ten coupons from two panels, mean 88.9 Ω, all within ±10 %. The reports state a 35 ps TDR edge but no evaluation window. A PCIe Gen5 link on the board shows less margin than simulated.

1. List what the coupon data can and cannot establish: within-lot versus lot-to-lot variation; coupon versus board geometry; average versus local impedance; the effect of the unstated window (Section 28.2.1).
2. Estimate how much of the margin loss a 4.6 % impedance offset could explain, using the return-loss rule of thumb of Chapter 6, and what else could explain it (stub length after back-drill, Section 11.5.1; de-embedding error, Section 26.2.1).
3. Specify the additional measurements—TDR of a production trace with a stated edge and window, a back-drill cross-section, a fixture self-check—and the decision each could change.

*What a good answer avoids:* concluding from the coupon mean alone that the fabrication is either the cause or not the cause.

# Answers to Review Problems

Numerical answers are given to two or three significant figures, which is the accuracy justified by the approximations used in most problems. Where a problem asks for a design choice, a brief justification is given instead.

### Chapter 1: The PCB as an Electrical System

**P1.1** Pressed dielectric thickness (impedance rises with thickness); etched width and trapezoid (narrower traces raise impedance); dielectric permittivity of the chosen glass and resin (higher $D_k$ lowers impedance). Plated thickness and roughness are also acceptable answers.

**P1.2** $T_D = 290\,\text{ps}$. With 1 ns: $T_D/t_r = 0.29$, between lumped and distributed, marginal. With 80 ps: $T_D/t_r = 3.6$, a transmission line.

**P1.3** $R = 1.724 \times 10^{-8} \times 0.254/(203 \times 10^{-6} \times 34.8 \times 10^{-6}) \approx 0.62\,\Omega$.

**P1.4** $C_{tot} = T_D/Z_0 = 10\,\text{pF}$; $L_{tot} = Z_0 T_D = 25\,\text{nH}$.

**P1.5** The display is round-trip: $T_D = 0.6\,\text{ns}$, so the distance is $0.6/0.170 \approx 3.5\,\text{in}$.

**P1.6** Slot: discontinuity and crosstalk from shared detour; noise coupled into the plane cavity; loop and slot radiation. Unstitched reference change: return-path discontinuity; noise injected between the planes; excitation of plane-pair resonances and radiation from the board edge.

### Chapter 2: What Makes a Design "High-Speed"

**P2.1** $f_{3dB} \approx 10\,\text{GHz}$; $f_{knee} \approx 14.3\,\text{GHz}$.

**P2.2** $T_D = 580\,\text{ps}$; $\ell_{crit} = 2.30\,\text{in}$; the 4 in line is distributed ($T_D/t_r = 0.29 > 1/6$).

**P2.3** Scope $t_r = 43.75\,\text{ps}$; displayed $\approx 74.3\,\text{ps}$.

**P2.4** $t_{pd} \approx 156\,\text{ps/in}$; $T_D \approx 1.09\,\text{ns}$.

**P2.5** $f_1 = 15.9\,\text{MHz}$; $T_{ramp} = 1.25\,\text{ns}$, so $f_2 = 255\,\text{MHz}$; $f_{knee} = 500\,\text{MHz}$.

**P2.6** UI = 62.5 ps; $f_{Nyq} = 8\,\text{GHz}$; $t_r = 25\,\text{ps}$, so $f_{knee} = 20\,\text{GHz}$.

### Chapter 3: Electromagnetic Fundamentals

**P3.1** 6.61 µm and 0.661 µm.

**P3.2** $C \approx 10.8\,\text{nF}$.

**P3.3** 70.5 %; 87.4 %.

**P3.4** $L_\square = \mu_0 h \approx 128\,\text{pH}$ per square.

**P3.5** $L_{loop} \approx 0.94\,\text{nH}$.

**P3.6** $v = L\,\text{d}i/\text{d}t = 0.4\,\text{V}$.

**P3.7** $R \approx 2.48\,\Omega$.

### Chapter 4: Transmission Line Theory

**P4.1** $Z_0 = 50\,\Omega$; $t_{pd} = 180\,\text{ps/in}$; $\varepsilon_r \approx 4.51$.

**P4.2** $t_{pd} = 158.5\,\text{ps/in}$; $C' = 3.52\,\text{pF/in}$; $L' = 7.13\,\text{nH/in}$.

**P4.3** $\alpha_d \approx 0.140\,\text{dB/in}$.

**P4.4** $\sqrt{50 \times 100} = 70.7\,\Omega$.

**P4.5** $\theta = 1.068\,\text{rad}$; $Z_{in} = \text{j}90.95\,\Omega$, i.e. 14.47 nH, versus $L'\ell = 8.50\,\text{nH}$: a ratio of $\tan\theta/\theta = 1.70$ (Eq. (4.17)).

**P4.6** $f = R'/(2\pi L') \approx 4.0\,\text{MHz}$.

**P4.7** $\theta = 0.387$, so $T_D \le 0.387/(2\pi \times 8 \times 10^9) = 7.7\,\text{ps}$ (about 45 mil of stripline). Exact: $1 - 0.387/\tan 0.387 = 5.0\,\%$.

**P4.8** $T_D = Z_0 C = 22.5\,\text{ps}$, i.e. 0.13 in (132 mil).

### Chapter 5: What Is Signal Integrity?

**P5.1** The received signal is on the correct side of the decision threshold by more than the receiver's sensitivity, within its timing window, with margin for noise and variation, so that the bit-error ratio meets the requirement.

**P5.2** $\Gamma = (40 - 50)/90 = -0.111$; $-111\,\text{mV}$; 19.1 dB.

**P5.3** (a) ISI from loss; (b) crosstalk; (c) overshoot from reflection/ringing.

**P5.4** $10^{-12/20} = 0.25$; yes, by the rule of thumb of this chapter an unequalized NRZ eye would be small or closed.

**P5.5** A TDR for location (impedance versus round-trip time); a VNA for $S_{11}$ at 8 GHz.

**P5.6** $16 - 1 - 1.5 - 3 = 10.5\,\text{dB}$, so $10.5/0.9 \approx 11.7\,\text{in}$.

**P5.7** Supply noise modulates driver delay and PLL phase, producing jitter; simultaneous switching noise shifts the ground reference and closes the eye vertically.

### Chapter 6: Reflections and Termination

**P6.1** 75 Ω: $\Gamma = 0.2$, $T = 1.2$. 25 Ω: $\Gamma = -0.333$, $T = 0.667$.

**P6.2** $V_{launch} = 0.8\,\text{V}$; $\Gamma_S = -1/3$.

**P6.3** 1.333, 0.889 and 1.037 V.

**P6.4** $R_S = 38\,\Omega$ ideal; use 39 Ω.

**P6.5** $\tau = Z_0 C/2 = 100\,\text{ps}$; added rise time $\approx 220\,\text{ps}$.

**P6.6** $C \ge 96\,\text{pF}$; choose 100 pF.

**P6.7** $R_1 = 45\,\Omega$ to the center node and $R_2 = 7.5\,\Omega$ from it to ground. The split termination gives $\Gamma_e = (45 - 60)/(45 + 60) = -0.143$.

**P6.8** Only the launch and first arrival are reasonably approximated; later $\Gamma_S$ values depend on the instantaneous operating point. Use the IBIS $V$–$I$ and $V$–$t$ tables in a nonlinear simulator (Chapter 25).

### Chapter 7: PCB Stackup and Impedance Control

**P7.1** $W \approx 4.7\,\text{mil}$.

**P7.2** $W \approx 7.5\,\text{mil}$; the zero-thickness formula ignores the large plated thickness, which widens the effective strip, and the two approximations differ intrinsically.

**P7.3** $Z_{diff} \approx 88\,\Omega$.

**P7.4** IPC-2141 gives $Z_0 \approx 51.3\,\Omega$, but $W/(B-T) = 0.40 > 0.35$, so this is an extrapolation; Wheeler's formula gives $52.4\,\Omega$.

**P7.5** $\varepsilon'_{\mathrm{r}} \approx 3.63$; $Z_0 \approx 44.5\,\Omega$.

**P7.6** $\approx 17.6$ ps (2.2 ps/in).

**P7.7** $\Delta H$ gives $1.9\,\Omega$, $\Delta D_{\mathrm{k}}$ about $0.6\,\Omega$; the remaining $2.9\,\Omega$ allows $\Delta W = 2.9/5.98 \approx \pm 0.48$ mil.

**P7.8** No. They show within-panel variation only (mean $52.5\,\Omega$, $s \approx 0.4\,\Omega$); lot-to-lot dielectric thickness and $D_{\mathrm{k}}$ variation, which dominate, are absent. Collect coupons across panels and lots.

### Chapter 8: Losses

**P8.1** $\approx 0.57$ dB/in.

**P8.2** $\approx 0.127$ dB/in.

**P8.3** $f_{\times} \approx 2.3$ GHz.

**P8.4** $\delta_{\mathrm{s}} = 0.935\,\mu$m; $K_H \approx 1.90$.

**P8.5** $K \to 1 + 1.5 \times 1.2 = 2.8$.

**P8.6** Conductor $\approx 1.08$ dB/in, dielectric $\approx 0.28$ dB/in; total $\approx 8.1$ dB.

### Chapter 9: Crosstalk

**P9.1** $K_b = 0.0325$.

**P9.2** $\ell_{\mathrm{sat}} = 0.15$ in.

**P9.3** $2T_{\mathrm{D}} = 33.3$ ps; NEXT $\approx 0.0325 \times 33.3/50 = 21.7$ mV.

**P9.4** $\approx -115$ mV.

**P9.5** $K_b = 0.025$, so NEXT = 25 mV (saturated); FEXT = 0.

**P9.6** From 0.100 to 0.027, a factor of 3.7 (11.4 dB).

### Chapter 10: Differential Signaling

**P10.1** $Z_{\mathrm{diff}} = 96\,\Omega$; $Z_{\mathrm{cm}} = 30\,\Omega$.

**P10.2** $R_1 = 48\,\Omega$ on each trace; $R_2 = 6\,\Omega$ to ground.

**P10.3** $\Delta t = 3.4$ ps.

**P10.4** 0.050, i.e. −26 dB.

**P10.5** $\Delta\ell = 75.4$ mil; $\Delta t = 12.8$ ps.

**P10.6** $\Delta t = 3.75$ ps; $\Delta\ell \approx 22$ mil.

### Chapter 11: Vias and Discontinuities

**P11.1** $L \approx 2.1$ nH.

**P11.2** $C \approx 0.50$ pF.

**P11.3** $f \approx 22.2$ GHz.

**P11.4** $3f_{\mathrm{Nyq}} = 24$ GHz; $\ell \le 63$ mil.

**P11.5** $Z_{\mathrm{via}} \approx 41.6\,\Omega$.

**P11.6** $3/0.8 = 3.75$ mil; specify 4 mil.

**P11.7** Clearance $(0.80 - 0.50)/2 = 0.15$ mm; $Z_{\mathrm{via}} = (60/\sqrt{3.6})\ln(0.80/0.25) \approx 36.8\,\Omega$.

**P11.8** $0.25\,e^{50\sqrt{3.6}/60} \approx 1.22$ mm; no—the maximum on a 1.0 mm grid is 0.85 mm.

### Chapter 12: Timing, Jitter, and Eye Diagrams

**P12.1** Setup margin = −0.05 ns (fails); hold margin = 0.6 ns.

**P12.2** TJ ≈ 23.3 ps.

**P12.3** $3.0 \times 10^{12}$ bits; 187 s.

**P12.4** $20\log_{10}(1.0/0.6) = 4.4$ dB.

**P12.5** $20\log_{10}(8/2) = 12$ dB.

**P12.6** $2b_1 = 0.50$ of the cursor. The half-height drops from 0.92 to 0.42: still positive without noise, but at $14\sigma$ rather than $30.7\sigma$, and a further error on tap 2 (0.24) would leave 0.18.

**P12.7** $1 - 0.1 - |h_3| \geq 0.6$, so $|h_3| \leq 0.30$.

**P12.8** At 40 MHz: $|\mathrm{j}10/(1 + \mathrm{j}10)| = 0.995$ (practically not attenuated). At 400 kHz: 0.0995, about −20 dB.

### Chapter 13: What Is Power Integrity?

**P13.1** The ability of the PDN to keep each supply within its tolerance, under all operating conditions, at the point the device specifies (package balls or die).

**P13.2** $L\,\mathrm{d}I/\mathrm{d}t = 50$ mV during the ramp; $IR = 10$ mV afterward.

**P13.3** $\Delta V = 18$ mV; $Z_{\text{target}} = 3\,\text{m}\Omega$.

**P13.4** $C = L/Z_c^2 = 10^{-9}/(25 \times 10^{-6}) = 40\,\mu$F; droop about $10 \times 5 = 50$ mV plus IR drop.

**P13.5** It forms a resonance at about 5 MHz with $\sqrt{L/C} = 32\,\text{m}\Omega$; the ramp, about half a resonant period long, excites it and the droop grows (Figure 13.3).

**P13.6** (a) Oscilloscope with a rail probe; (b) VNA shunt-through impedance; (c) oscilloscope at the load during the workload, or an injected load step.

**P13.7** The noise modulates the VCO frequency and phase; the resulting clock jitter transfers to the transmitted data or the receiver's sampling clock, shrinking the eye width and raising the BER.

### Chapter 14: Transient Current, Inductance, and the PDN Response

**P14.1** $\mathrm{d}I/\mathrm{d}t = 2 \times 10^8$ A/s; 100 mV and 400 mV.

**P14.2** $6 \times 2 \times 10^{-6}/470 \times 10^{-6} \approx 25.5$ mV.

**P14.3** $f_0 \approx 240$ kHz; $Z_c \approx 3.0\,\text{m}\Omega$; $\zeta = 0.0015\sqrt{1.1 \times 10^5} \approx 0.50$.

**P14.4** +0.2 V for 20 ns, zero for 50 ns, then −0.4 V for 10 ns.

**P14.5** The capacitor supplies the fast current, so the path current does not change at $10^8$ A/s; the simulated droop is 75 mV.

**P14.6** $L = Z_c^2 C = 4 \times 10^{-6} \times 10^{-4} = 0.4$ nH.

**P14.7** The third harmonic (6.9 MHz); a square wave has strong odd harmonics, so this component, about one third of the fundamental, is multiplied by 35 mΩ instead of a few milliohms.

**P14.8** The first droop (die–package). Board capacitors cannot reach it through the package inductance; it is set by on-die and package capacitance. Also confirm the probe bandwidth and grounding before trusting a nanosecond feature.

### Chapter 15: Ground Bounce and SSN

**P15.1** $V_{\text{gb}} = 0.32$ V.

**P15.2** 0.30 nH.

**P15.3** $(1.2 + 3 \times 0.2)/4 = 0.45$ nH.

**P15.4** $N \leq 15$.

**P15.5** $f \approx 113\,\text{MHz}$.

**P15.6** Line-driven: by a factor of two, 6 dB. Lumped capacitive loads (Eq. (15.2)): by a factor of four, 12 dB.

### Chapter 16: Power Distribution Network Design

**P16.1** $Z_{\text{target}} = 10\,\text{m}\Omega$.

**P16.2** $f_{\text{SRF}} \approx 6.5\,\text{MHz}$.

**P16.3** $N \geq 2\pi f\,\text{ESL}/Z_{\text{target}} = 15.1$; use 16. The inductive-only formula is acceptable only because the problem states that $f \gg f_{\text{SRF}}$.

**P16.4** $R_{\square} = 0.99\,\text{m}\Omega$; $\Delta V \approx 40\,\text{mV}$.

**P16.5** $f_{\text{ar}} \approx 178\,\text{MHz}$.

**P16.6** $f_{\text{SRF}} = 6.5\,\text{MHz}$; $|Z_C| = 14.3\,\text{m}\Omega$, so $N = 6$. The shortcut gives $\lceil 7.85 \rceil = 8$ but ignores the capacitive reactance below the SRF, and would give an even larger error further below it.

**P16.7** With 0.5 µF, $|Z_C| = 45.2\,\text{m}\Omega$ and $N = 19$: the DC-bias derating, not the ESL, now dominates. Always use biased capacitance.

**P16.8** $R = \sqrt{L/C}/Q = 7.07/0.7 \approx 10\,\text{m}\Omega$; the DC drop is then 10 mV per ampere.

### Chapter 17: Planes and Return Paths

**P17.1** 492, 738 and 887 MHz.

**P17.2** $\lambda = 0.622\,\text{in}$; spacing $\approx 31$ mil.

**P17.3** $\approx 5.0\,\Omega$.

**P17.4** $\Gamma = 0.23$.

**P17.5** 80 mil.

**P17.6** $1/(\omega C) \approx 80\,\text{m}\Omega$.

### Chapter 18: Emission Mechanisms and EMC Design

**P18.1** $E \approx 351\,\mu\text{V/m}$ (50.9 dBµV/m).

**P18.2** $\lambda = 1.5\,\text{m}$, so 0.15 m is $\lambda/10$ (just valid); $I_{\text{CM}} \approx 12\,\mu\text{A}$. A 1.5 m cable is a full wavelength long, four times $\lambda/4 = 0.375\,\text{m}$: the short-monopole formula does not apply and the cable must be modeled or measured.

**P18.3** $\approx 5.9\,\text{in}$ (150 mm).

**P18.4** $10\log_{10}(1.5/0.12) \approx 11\,\text{dB}$.

**P18.5** By 6 dB.

**P18.6** $\delta_s = 2.09\,\mu\text{m}$; $A \approx 145\,\text{dB}$.

### Chapter 19: DDR4/DDR5 and LPDDR4/LPDDR5

**P19.1** UI = 156.25 ps; $f_{\text{Nyq}} = 3.2\,\text{GHz}$.

**P19.2** 4.2 ps; about 25 mil.

**P19.3** $Z_{\text{loaded}} \approx 37\,\Omega$.

**P19.4** $\pm 192\,\text{mA}$.

**P19.5** $4 \times 0.55 \times 210 = 462\,\text{ps}$.

**P19.6** $\pm 12$ mV.

**P19.7** LPDDR5 writes are referenced to WCK and reads to RDQS (where used), not a single bidirectional DQS. You need the controller's list of timing groups, clock ratio, read mode, training ranges and the package delays to derive separate write, read and CK/CA budgets.

### Chapter 20: PCI Express

**P20.1** UI = 31.25 ps; $f_{\text{Nyq}} = 16$ GHz.

**P20.2** 15.75 Gb/s.

**P20.3** 32 GBd; UI = 31.25 ps; one-third, $-9.5$ dB.

**P20.4** $f_{\text{HP}} = 1/[2\pi(50 + 50) \times 100 \times 10^{-9}] \approx 15.9$ kHz.

**P20.5** 12.7 in.

**P20.6** 99.5 MHz.

### Chapter 21: USB and Type-C

**P21.1** UI = 100 ps; $f_{\text{Nyq}} = 5$ GHz.

**P21.2** UI = 50 ps; $f_{\text{Nyq}} = 10$ GHz.

**P21.3** 4 Gb/s and 9.70 Gb/s.

**P21.4** $|\Gamma| = 0.333$ (9.5 dB return loss).

**P21.5** $R_d = 5.1$ kΩ.

**P21.6** $\approx 0.56$ in.

### Chapter 22: Ethernet

**P22.1** Two bits per symbol per pair ($4 \times 2 \times 125 = 1000$).

**P22.2** UI = 800 ps; $f_{\text{Nyq}} = 625$ MHz.

**P22.3** 250 Mb/s (DDR at 125 MHz).

**P22.4** 96.97 ps.

**P22.5** $\approx 10.9$ in.

**P22.6** $\approx 16.4$ dB.

### Chapter 23: MIPI D-PHY and C-PHY

**P23.1** UI = 667 ps; $f_{\text{Nyq}} = 750$ MHz.

**P23.2** $3.5 \times 16/7 = 8.0$ Gb/s.

**P23.3** 10 Gb/s on ten wires.

**P23.4** $\approx 1.46$ GS/s.

**P23.5** 11.1 ps; $\approx 67$ mil.

**P23.6** 18 mV, almost 10 % of a 200 mV HS swing.

### Chapter 24: Clock Distribution

**P24.1** $C_L = 15$ pF.

**P24.2** $\approx 204$ fs.

**P24.3** $\approx 308$ fs.

**P24.4** $\approx 62.7°$.

**P24.5** $f_z = 15.9\,\text{kHz}$.

**P24.6** $25\,\text{MHz} \pm 1.25\,\text{kHz}$.

### Chapter 25: Modeling and Simulation

**P25.1** $\Gamma = 0.091$; return loss 20.8 dB.

**P25.2** $2\Gamma/(1+\Gamma^2) = 0.180$ (−14.9 dB).

**P25.3** Yes: $0.04 + 0.9025 = 0.9425 \le 1$.

**P25.4** 100 Ω and 25 Ω.

**P25.5** 0.7 (−3.1 dB).

**P25.6** 16.

### Chapter 26: Measurement

**P26.1** 75 Ω.

**P26.2** $\rho = -0.087$.

**P26.3** $L = 2 \times 50 \times 2\,\text{ps} = 200\,\text{pH}$.

**P26.4** ≈ 145 MHz.

**P26.5** ≈ 75 mil.

**P26.6** Scope rise time ≤ 16 ps; bandwidth ≳ 22 GHz.

**P26.7** $Z \approx 25 \times 4.0 \times 10^{-4} = 10\,\text{m}\Omega$ (exact expression: 10.004 mΩ).

**P26.8** The feature is shorter than $\Delta x$ for that edge, so 54 Ω is an apparent, blended value (Figure 26.2). Re-measure with a faster edge, or compare with a model of the expected geometry filtered to the same 100 ps edge.

### Chapter 27: Board Bring-Up and Debug

**P27.1** ±3 % is ±24 mV (48 mV peak-to-peak); 30 mV fits, but leaves little for DC error.

**P27.2** 1.2 W.

**P27.3** 23 bits.

**P27.4** About 200 µs.

**P27.5** Half.

**P27.6** $3.0 \times 10^{12}$ bits; about 374 s.

### Chapter 28: Design for Manufacturing and Release

**P28.1** Diagonal 7.21 in; limit ≈ 0.054 in.

**P28.2** ≈ +1.8 Ω.

**P28.3** 76.5–93.5 Ω.

**P28.4** ≈ 5.9 in.

**P28.5** ≈ 0.096 in.

**P28.6** Nominal 4 mil; worst case 1 mil.

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# Index


1000BASE-T, 195
10GBASE-KR, 197
20H rule, 158
3W rule, vii, 79

AC coupling, 183
anti-pad, 91
anti-resonance, 148
attenuation, 31
attenuation constant, 30

back-drilling, 96
bandwidth
    rise-time relation, 12
bathtub curve, 106
BER, 105
BGA breakout, 99
bit-error ratio, 105
blind via, 98
Bob Smith termination, 196
bounce diagram, 50
boundary scan, 232
buried via, 98

capacitance, 21
capacitive load, 53
causality, 72, 216
cavity resonance, 157
characteristic impedance, 30
CISPR 32, 167
clock and data recovery, 105
common mode, 83
common-clock timing, 103
common-mode radiation, 165
conductor loss, 31, 69
configuration channel, 190
connectivity versus geometry, 4
coplanar waveguide, 59
copper roughness, 66
    Hammerstad model, 70
    Huray model, 70
coupling
    tight vs. loose, 84
critical length, 16
crosstalk, 75
crystal oscillator, 205
CTLE, 108
cursor, 108

data-bus inversion, 141
DBI, *see* data-bus inversion
DDR4, 173
DDR5, 173
de-embedding, 223
de-emphasis, 108
DFE, 109
DFM, 240
DFT, 240
dielectric loss, 31, 71
differential mode, 83
differential-mode radiation, 163
dispersion, 35
displacement current, 22
distributed model, 16
Djordjevic–Sarkar model, 72
dog-bone, 99
dual-Dirac model, 105

ECO, 240
edge radiation, 165
effective dielectric constant, 15
embedded clock, 105
EMC, v
EMI filter, 167

equalization, 108
ESD protection, 191
ESL, 146
ESR, 146
even mode, 83
eye diagram, 106
eye mask, 106

fabrication drawing, 237
fanout buffer, 206
far-end crosstalk, *see* FEXT
FCC Part 15, 167
FEC, 185
ferrite bead, 167
FEXT, 77
FFE, 108
fiber-weave effect, *see* glass-weave skew
field solver, 214
first droop, 131
fly-by topology, 176
FR-4, 65

Gerber, 240
glass-weave skew, 65
ground bounce, 137
ground lead inductance, 225
guard trace, 80

HCSL, 206
HDI, 98
hold time, 103

IBIS, 213
IBIS-AMI, 213
impedance
    closed-form, 60
    common-mode, 83
    differential, 83
    tolerance, 67
impedance coupon, 238
inductance, 22
    internal, 24
    loop, 22
    mutual, 23, 139
    partial, 23
input impedance, 33
insertion loss, 69
inter-symbol interference, 69
IPC-2141, 60
IPC-2581, 240
IR drop, 151

jitter, 105
    deterministic, 105
    phase, 207
    power-supply-induced, 140
    random, 105
JTAG, 232

knee frequency, 12

laminate, 65
lane margining, 232
lattice diagram, 50
length matching, 86
loaded impedance, 177
loop filter, 208
loss budget, 182
loss tangent, 71
LPDDR4, 173
LPDDR5, 173
    WCK, 174
lumped model, 16
LVDS, 206
LVPECL, 206

magnetics, 195
microstrip, 59
microvia, 98
MIPI C-PHY, 202
MIPI D-PHY, 201
mixed-signal layout, 158
mode conversion, 85
mounted inductance, 147
mutual capacitance, 75
mutual inductance, 75

near-end crosstalk, *see* NEXT
near-field probe, 226
NEXT, 76
notation, 8
Nyquist frequency, 17

ODB++, 240
odd mode, 83

ODT, *see* on-die termination
on-die termination, 57
oscilloscope, 225
overshoot, 52

PAM4, 17, 184
passivity, 216
PCB
    anatomy, 3
PCI Express, 181
PCIe, *see* PCI Express
PDN, 143
    elements, 116
    measurement, 224
    transient response, 150
phase constant, 30
phase noise, 207
plane capacitance, 22
plane split, 155
PLL, 208
POD, *see* pseudo-open drain
power integrity, v
    definition, 115
power sequencing, 231
pre-emphasis, 108
probe, 225
propagation constant, 30
propagation delay, 15
pseudo-open drain, 175
pulse response, 108

quasi-TEM, 35

RDQS, 174
reciprocity, 216
reference clock, 184
reference plane change, 156
reflection coefficient, 49
retimer, 185
return current, 21
    frequency dependence, 25
return via, 99
RGMII, 197
ringing, 52
rise time, 11
    10–90 percent, 11
    10–90 versus full ramp, 14
    20–80 percent, 11
    cascaded, 13
RLGC model, 29

S-parameters, 214
    mixed-mode, 215
saturation length, 76
SCD21, 215
SDD21, 215
self-resonant frequency, 146
serpentine, 86
setup time, 103
SGMII, 198
sheet inductance, 24
shielding, 167
shunt-through measurement, 224
signal integrity, v
    definition, 39
simultaneous switching noise, 137
skew
    inter-pair, 86
    intra-pair, 85
skin depth, 25, 26
skin effect, 25
slot antenna, 165
source-synchronous timing, 104
spatial extent of the edge, 16
spread-spectrum clocking, 166
spreading inductance, 150
SSN, *see* simultaneous switching noise
stackup, 63
stitching capacitor, 156
stitching via, 99, 158
stripline, 59

T topology, 176
target impedance, 119, 143
TDR, 221
telegrapher's equations, 30
TEM mode, 35
termination, 55
    AC, 56
    differential, 57, 84
    diode, 57
    parallel, 56
    series, 55
    Thevenin, 56

Touchstone, 217
transmission line
    lossless, 30
trapezoidal waveform
    spectrum, 13

unit interval, 17
USB, 189
USB Type-C, 190

via, 89
    impedance, 90
via stub, 96
VNA, 223
VREF training, 178
VTT termination, 56

WCK, 174
write leveling, 177